



Honeywell

Fail Safe Control Software Manual

Version 500

Revision 01 (01/98)

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Total Plant



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REFERENCES

FSC Documentation:

Publication Title	Publication Number
FSC Safety Manual	PM.MAN.8047
FSC Hardware Manual	PM.MAN.8048
FSC Software Manual	PM.MAN.8025

FSC-SM Documentation:

Publication Title	Publication Number	Binder Title	Binder Number
FSC Safety Manager Installation Guide	FS20-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Implementation Guidelines	FS11-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Control Functions	FS09-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Parameter Reference Dictionary	FS09-550	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Configuration Forms	FS88-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Service Manual	FS13-500	Implementation FSC Safety Manager	TPS 3076





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Fail Safe Control

Section 1: FSC Primer

Version 500

Revision 01 (01/98)

Total Plant



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1.1 Introduction

The Fail Safe Control (FSC) system is a safety system for applications in the process industries. The FSC system is a microprocessor-based, modular, software-programmable system. This allows it to be adapted easily to a broad spectrum of applications.

Through dedicate software and hardware, the FSC system can always guarantee its ability to lead the process to a safe state in case of any potential hazard, which results in:

Proven safety

In redundant FSC configurations, all system parts are continuously and actively involved in the execution of the system tasks and are continuously subject to self-tests. Thus, it is guaranteed that in case of failure of one of the redundant Central Parts, the other Central Part is able to continue the process control, which results in:

Proven availability

The first part of the FSC Primer provides a bird's-eye view of the FSC system. The second part describes the basic information you need to know in order to properly work with the FSC Navigator software.

For more detailed information, refer to the following documents:

- Safety manual (PM.MAN.8047) : FSC system safety aspects,
- Hardware manual (PM.MAN.8048) : System hardware component documentation, and
- Software manual (PM.MAN.8025) : FSC Navigator user manual.

1.2 Fail Safe Controller configurations

The Fail Safe Controller supports two basic configurations (single and redundant) which can be combined into four different configurations (see Table 1-1).

Table 1-1 FSC configurations

Configuration	Processor	I/O	I/O racks	Remarks
S + s	Single	Single	Multiple	
R + s	Redundant	Single	Multiple	processor with inter processor communication.
R + r	Redundant	Redundant	Multiple	processor with inter processor communication, and redundancy for availability.
R + r/s	Redundant	Redundant and Single	Multiple	processor with inter processor communication, redundancy for availability, and single I/O for non-critical functions (e.g. lamps).

All FSC configurations can be used for safety applications. The preferred configuration depends on the safety requirements.

The FSC configurations defined in Table 1-1 are discussed in more detailed below.



1.2.1 Single processor, single I/O

This FSC configuration has a single processor (Central Part) and single input and output (I/O) modules (see Figure 1-1). The I/O modules are controlled via the Vertical Bus Driver (VBD), which is located in the Central Part, and the Vertical bus (V-Bus), which controls up to 10 I/O racks. Each I/O rack is controlled via the Horizontal Bus Driver (HBD). No redundancy is present except as built into those modules where redundancy is required for safety (memory and watchdog).

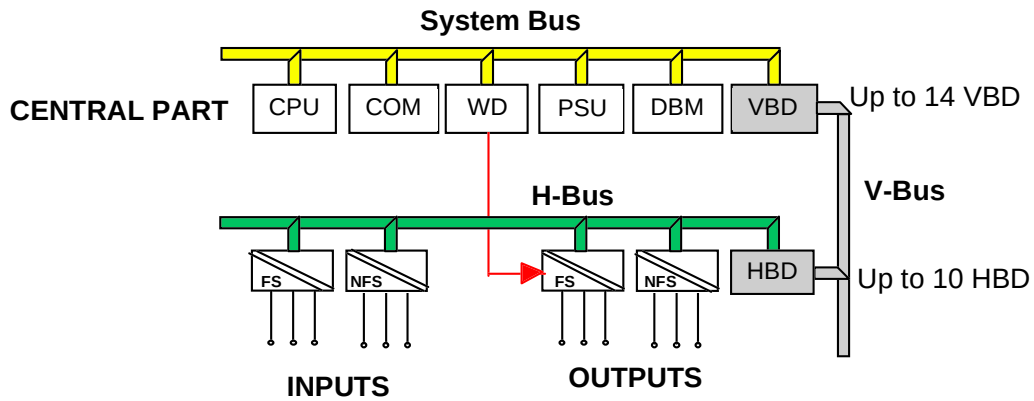


Figure 1-1 Single processor, single I/O configuration

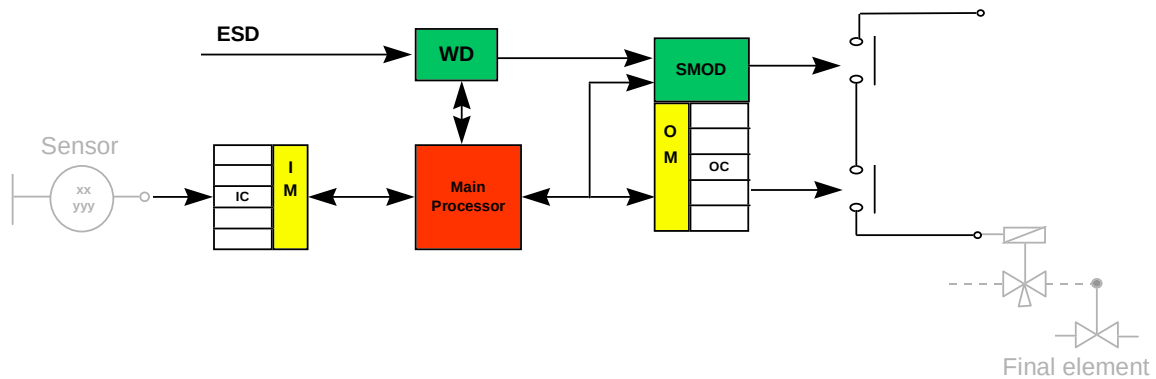


Figure 1-2 Functional diagram: single processor, single I/O

1.2.2 Redundant processor, single I/O

This FSC configuration has a redundant processor (Central Parts) and single input and output (I/O) modules (see Figure 1-3 and Figure 1-4). The I/O modules are controlled via the VBDs, which are located in each Central Part, and the V-Bus, which controls up to 10 I/O racks. Each I/O rack is controlled via the HBD. The processor is fully redundant, which allows continuous operation and bumpless (zero-delay) transfer in case of a processor failure.

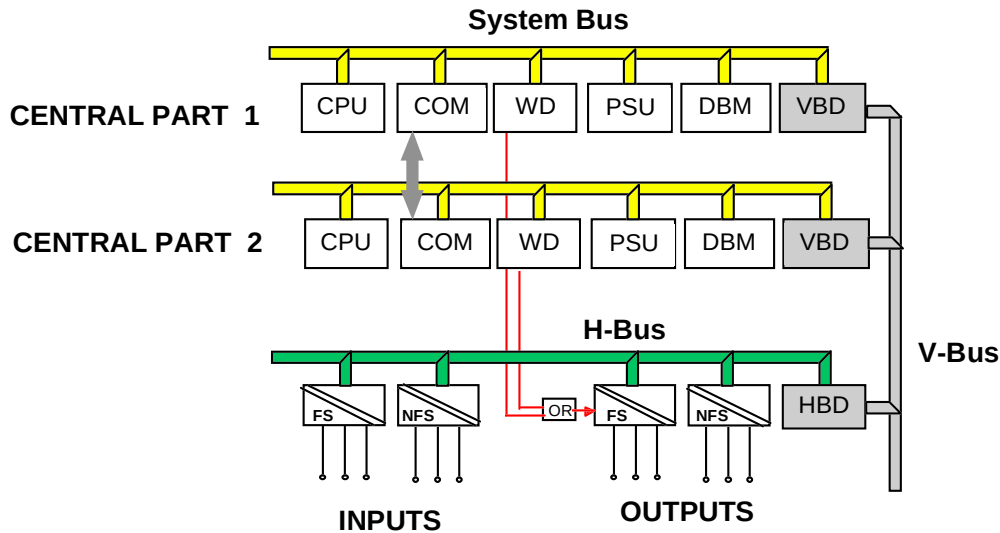


Figure 1-3 Redundant processor, single I/O configuration

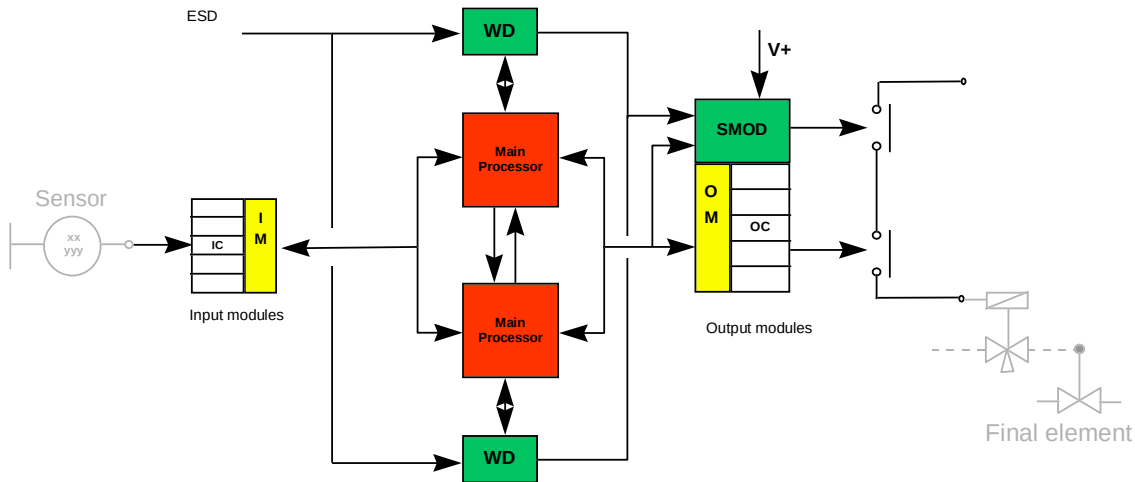


Figure 1-4 Functional diagram: redundant processor, single I/O



1.2.3 Redundant processor, redundant I/O

This FSC configuration has a redundant processor (Central Parts) and redundant input and output (I/O) modules (OR function on outputs) (see Figure 1-5 and Figure 1-6). The I/O modules are controlled via the VBDs, which are located in each Central Part and the V-Bus, which controls up to 10 I/O racks. Each I/O rack is controlled via the HBD. The processor and I/O are fully redundant, which allows continuous operation and bumpless (zero-delay) transfer in case of a processor or I/O failure.

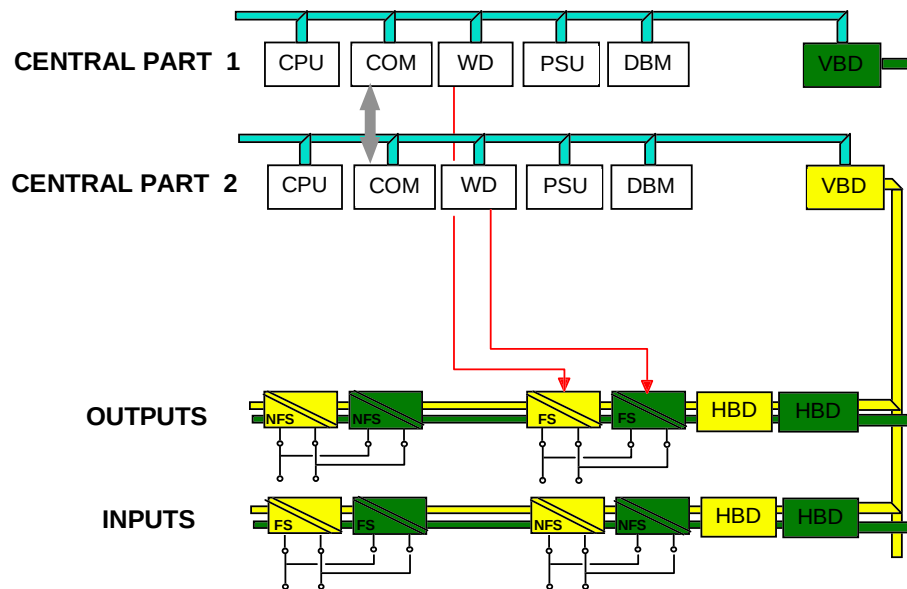


Figure 1-5 Redundant processor, redundant I/O configuration

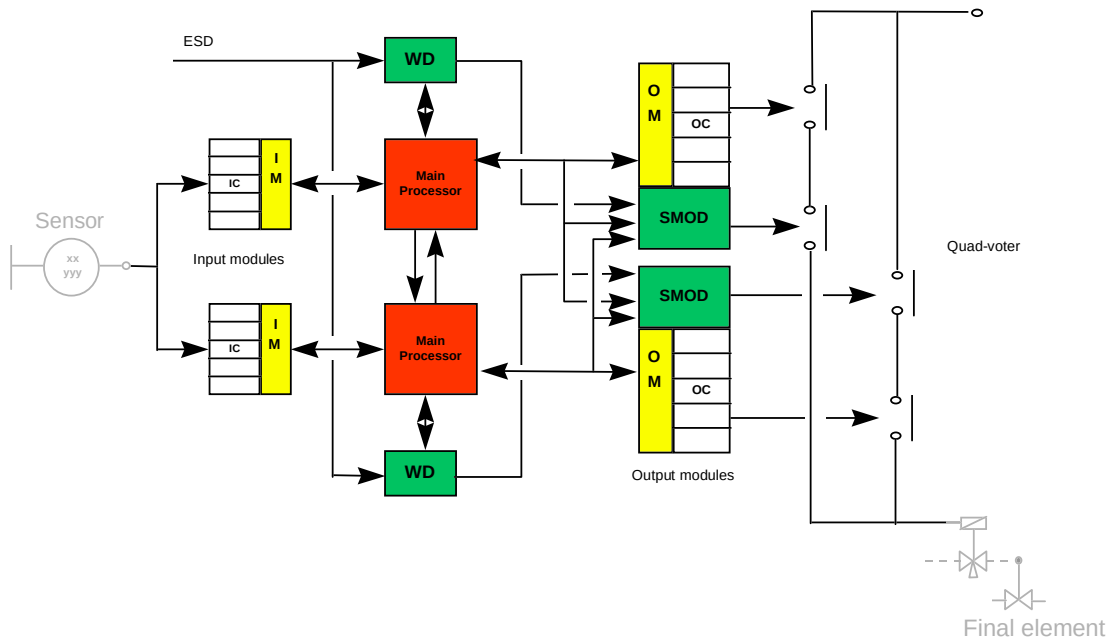


Figure 1-6 Functional diagram: redundant processor, redundant I/O

1.2.4 Redundant processor with redundant and single I/O

This FSC configuration has a redundant processor (Central Part) and redundant input and output (I/O) modules (OR function on outputs) combined with single input and output modules (see Figure 1-7 and Figure 1-8). The I/O modules are controlled via the VBDs, which are located in each Central Part, and the V-Bus, which controls up to 10 I/O racks. Each I/O rack is controlled via the HBD. The processor and I/O are fully redundant, which allows continuous operation and bumpless (zero-delay) transfer in case of a processor or I/O failure of the redundant I/O modules.

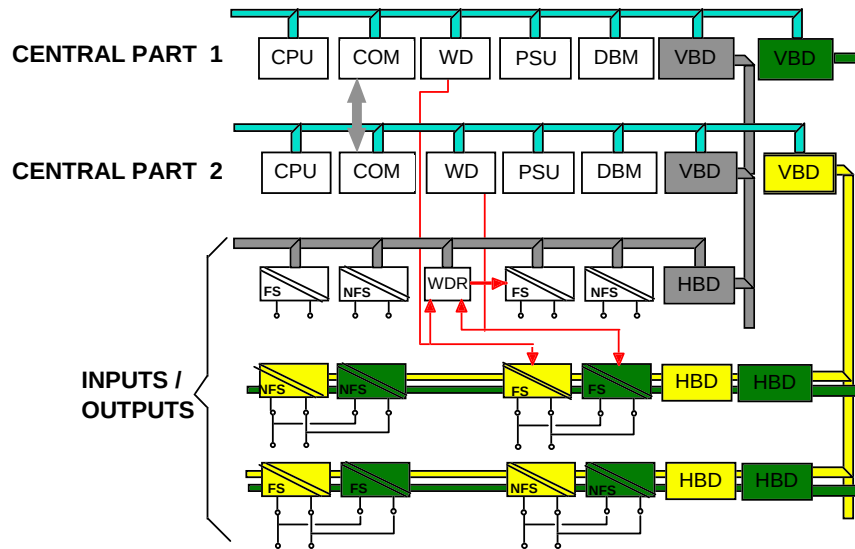


Figure 1-7 Redundant processor with redundant and single I/O configuration

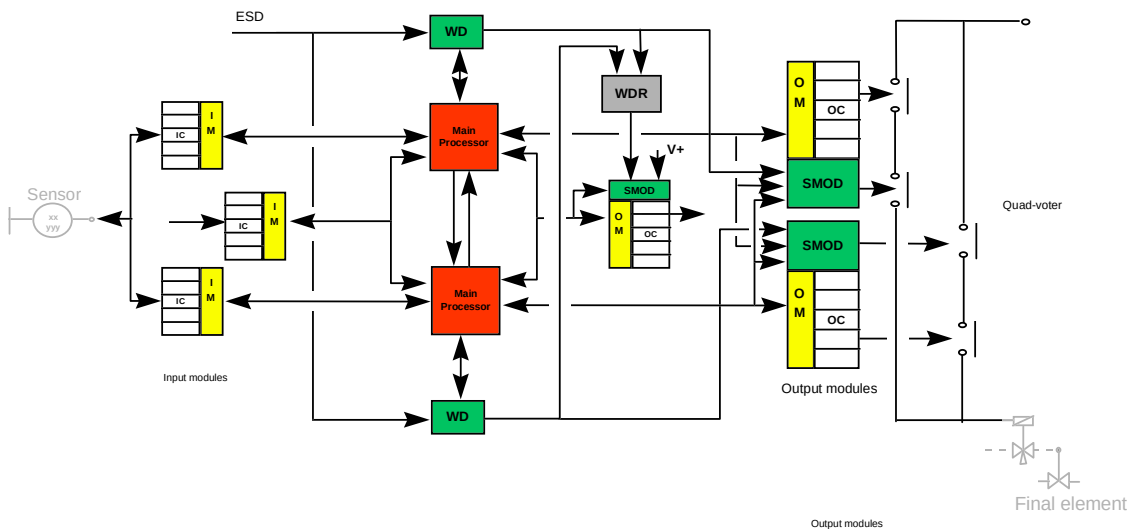


Figure 1-8 Functional diagram: redundant processor with redundant and single I/O



1.3 FSC configuration and programming

The specific tasks of the FSC system within the process environment are determined by the system configuration and the application program.

During the configuration stage, the FSC system type, specification, system parameters and the field instrumentation parameters are defined (see Figure 1-9).

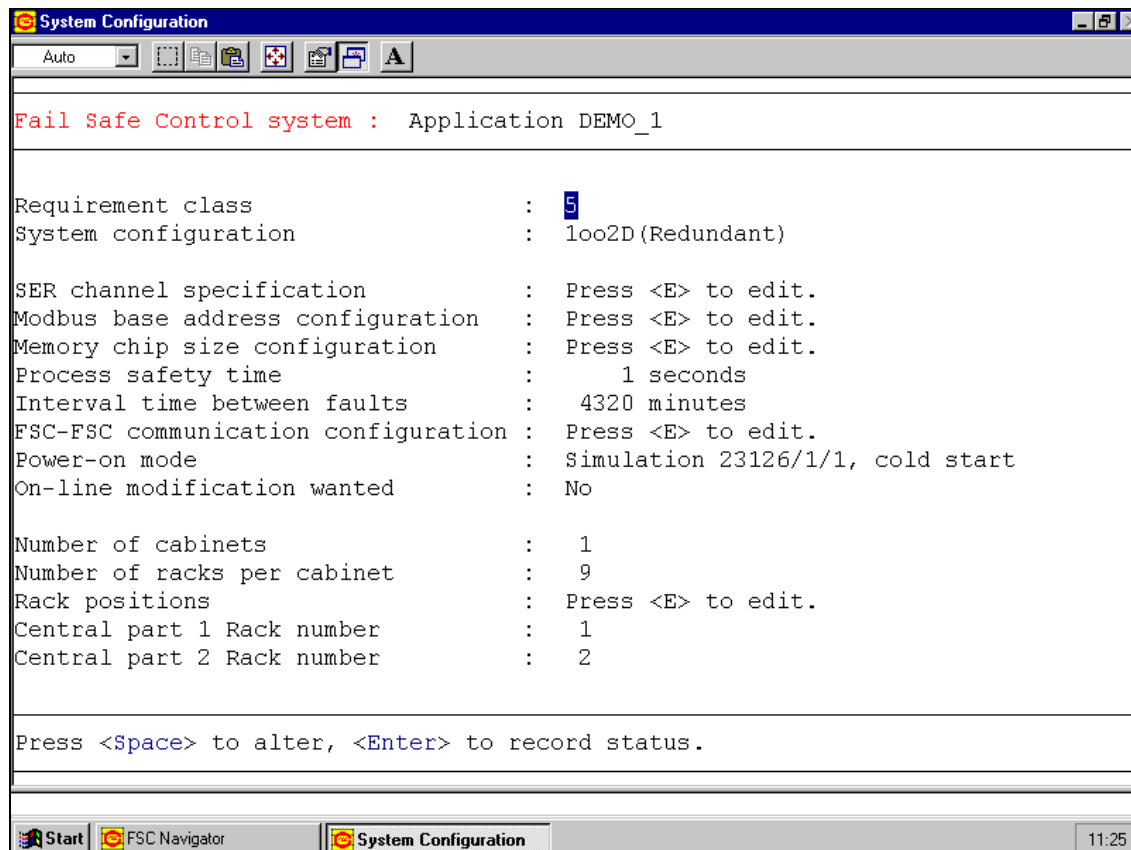
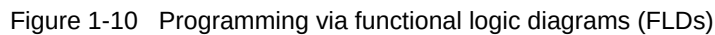


Figure 1-9 Instrumentation parameter configuration

During the application design stage, the application program is created by means of functional logic diagrams (FLDs), which define the relationships between inputs and outputs. Figure 1-10 shows an example of a functional logic diagram.



The FSC Navigator software provides support for the tasks that must be carried out during the configuration and design stages. With FSC Navigator, project documentation can be generated automatically and is readily available.



1.4 FSC features

1.4.1 FSC hardware modules

The set of FSC modules includes:

Digital input modules:	Fail-safe:	24 Vdc, 48 Vdc, 60 Vdc
	Non fail-safe:	24 Vdc
		Ex-i
Analog input modules:	Fail-safe:	0(4)-20 mA, 0(1)-5 Vdc, 0(2)-10 Vdc
Digital output modules:	Fail-safe:	24 Vdc, 48 Vdc, 60 Vdc, 110 Vdc, 220 Vdc
	Non fail-safe:	24 Vdc
		Ex-i
Analog output modules:	Fail-safe:	0(4)-20 mA
Miscellaneous:		Central Processor Unit (CPU)
		Watchdog (WD) module
		Memory (MEM) module
		Communication (COM) module
		Diagnostic and battery module (DBM)
		I/O bus driver modules
		Power supply units (PSU)
		Watchdog repeater (WDR) module.

Refer to the FSC Hardware Manual for details on the FSC modules.

1.4.2 FSC standard functions

The FSC system has a number of several standard functions which are commonly used in process control. They include:

- Alarm processing,
- PID control, and
- Sequence of event recording (SER).

Note:

Data collected by the sequence-of-event recording (SER) feature is used by the FSCSOE program, which is a separate Windows application.

The function parameters, e.g. alarm type, are defined during system configuration. Interaction with the application program is realized via the functional logic diagrams (FLDs).

Refer to the appendices of the FSC Software Manual for details on the standard FSC functions.



1.4.3 FSC communication facilities

Multiple FSC systems can be interconnected to form a communication network. The systems in such a network can be programmed to perform a joint task. Also, process information can be gathered at a central point within the network.

The FSC system also has facilities for communication with:

- the Universal Control Network (UCN) of Honeywell's TotalPlant Solution (TPS),
- Distributed Control Systems (DCSs), via Modbus (RTU mode) or RKE3964R protocol,
- FSCSOE data collecting devices,
- Hartmann & Braun, and
- FSC user station running FSC Navigator (using the FSC Development System protocol).

Available interfaces include:

- RS-232,
- RS-422/RS-485,
- Fiber optic, and
- Current loop.

The communication configuration of the FSC system is specified during the system configuration. The functional logic diagrams are used to realize the interaction of communication data with the FSC application.

For details on the FSC system's communication facilities refer to Appendix F of the FSC Software Manual ("FSC Communication Manual").

1.4.4 On-line modification

The FSC system supports on-line modification in redundant Central Part configurations. This allows changes to be made in the functional logic diagrams and the system software configuration while process operation continues.

During the on-line modification of the application, the FSC system performs compatibility checks to guarantee a safe changeover from the old to the new application. The sheet numbers of changed FLDs are reported, which allows easy validation of the changes that have been implemented in the application program.

If the FSC system is running Release 310 or later, the FSC system software of release 500 can also be upgraded on-line.

For details on on-line modification refer to Appendix D of the FSC Software Manual.



1.4.5 Power-on modes of the FSC system

The power-on mode defines the conditions for the start-up of the FSC system. Table 1-2 below lists the power-on modes that are available.

Two start-up modes are possible: cold and warm. If the system starts up for the first time, a cold start is performed, which means that the power-on values of the variables (as configured in the variable database) will be used as default variable values in the system. Configure a cold start power-on mode if the system must always start up with the power-on values as laid down in the variable database. Configure a warm start power-on mode if the system must start up with the current system power-on values.

Notes:

1. If the system is started up after a shutdown that was caused by a fault, a cold start-up will always be executed.
 2. Using the warm start option in combination with on-line modification of the application program may result in spurious diagnostic messages and Central Part shutdown.
-

Table 1-2 FSC power-on modes

Power-on mode	Description
Run, cold start	The FSC system is completely reset and the application EPROMs are used.
Run, warm start	The FSC system is partly reset and the application EPROMs are used.
RAM operation, cold start	The FSC system is completely reset and the application is downloaded for use.
RAM operation, warm start	The FSC system is partly reset and the application is downloaded for use.
Simulation, 23125/1/1, cold start	The FSC system is completely reset and the application is downloaded for test purposes (Training unit 23125/1/1 only; single configuration).
Simulation, 23125/1/1, warm start	The FSC system is partly reset and the application is downloaded for test purposes (Training unit 23125/1/1 only; single configuration).
Simulation, 23126/1/1, cold start	The FSC system is completely reset and the application is downloaded for test purposes (Training unit 23126/1/1 only; single configuration in one rack).
Simulation, 23126/1/1, warm start	The FSC system is partly reset and the application is downloaded for test purposes (Training unit 23126/1/1 only; single configuration in one rack).
Simulation, 23125/R/1, cold start	The FSC system is completely reset and the application is downloaded for test purposes (Training unit 23125/R/1 only; redundant configuration).



Table 1-2 FSC power-on modes (continued)

Simulation, 23125/R/1, warm start	The FSC system is partly reset and the application is downloaded for test purposes (Training unit 23125/R/1 only; redundant configuration).
Simulation, 23250/1/1, cold start	The FSC system is completely reset and the application is downloaded for test purposes (Training unit 23250/1/1 only; redundant configuration).
Simulation, 23250/1/1, warm start	The FSC system is partly reset and the application is downloaded for test purposes (Training unit 23250/1/1 only; redundant configuration).
Simulation, 23260/1/1, cold start	The FSC system is completely reset and the application is downloaded for test purposes (Training unit 23260/1/1 only, single configuration in one rack).
Simulation, 23260/1/1, warm start	The FSC system is partly reset and the application is downloaded for test purposes (Training unit 23260/1/1 only, single configuration in one rack).
Simulation, 23251/1/1, cold start	The FSC system is completely reset and the application is downloaded for test purposes (Training unit 23251/1/1 only, redundant configuration).
Simulation, 23251/1/1, warm start	The FSC system is partly reset and the application is downloaded for test purposes (Training unit 23251/1/1 only, redundant configuration).



1.4.6 Variable description

This subsection discusses the attributes and parameters of the variable types that are supported by the FSC system. Variable attributes and parameters are defined during FSC system configuration. Table 1-3 below lists the variable types that are available in the FSC system.

Table 1-3 FSC variable types

Type	Description	Type	Description
I (DI)	Digital input	XO	Output multiplexer
AI	Analog input	M	Marker
BI	Multiple input	C	Counter
XI	Input multiplexer	T	Timer
O (DO)	Digital output	R	Register
AO	Analog output	P	PID control function
BO	Multiple output	A	Alarm

Tag number

The program allows the variable type and tag number to be combined only once.

A typical tag number for I/O (I, AI, BI, XI, O, AO, BO, XO, A, P) consists of three parts:

Tag number = <Prefix>'_'<Description>'_'<Loop number>

The underscore sign ('_') may also be replaced by a hyphen ('-'). Valid characters of tag numbers are all characters, except space (' '), semicolon(';'), comma(',') and question mark('?'). The tag number is sometimes followed by a sequence of characters that provide further information, e.g.:

01-HLS-320HH (HLS for high low switch, HH for high high)

Tag numbers of internal variables are automatically created from the program block number and a sequence number. These tag numbers cannot be changed.

Examples: 32_1 First timer in program block 32.
10_12_3 Third reference between program blocks 10 and 12.

Service

The service field provides a description about the tag number, e.g. 'Level glycol tower'.

Qualification

The qualification field provides information about the one (or active) status of boolean variables. If the qualification is TRUE or 1, then the status of the variable is TRUE or 1. It is only used for variables of type I, O, M, C, T.



Location

The location field gives information about the location of the sensor or the actuator. The location is specified as an abbreviation of three characters. It is only used for I/O variables (I, AI, BI, XI, O, AO, BO, XO and A). The FSC system has a number of reserved locations, which are listed in Table 1-4.

Table 1-4 Reserved locations in FSC system

Location	Description
ANN	Alarm function (I, O, A only).
COM	Variable that can be altered or read via a communication link (I, BI, O, BO only).
MUX	Multiplexer variable (BI, BO, XI, XO only).
SYS	System variable (reserved for use by the FSC programs).
FSC	Variable that can be altered or read by another FSC system (I, BI, O, BO only).

Unit

This field contains the name of the unit for which this variable is used in the application program, e.g. F-210 (Furnace 210).

Subunit

This field contains the name of the subunit for which this variable is used in the application program, e.g. B-210 (Burner on furnace 210).

FLD

This is the FLD where the variable is placed in the functional logic diagrams.

Address

This field contains the address of the variable if an address has been defined for the variable type.

Relative address

Some I/O variables will be allocated to an address within a memory area that is reserved for a certain application (e.g. a communication link). The relative address is the address within such a reserved area.

Safety-related

This field specifies the safety relation of the variable. For I/O variables, this field determines the action to be taken in case of a hardware fault of the I/O module which the variable is connected to.



Force enable

This field specifies for each variable whether it is possible to force the variable using the 'On-Line Environment' options of FSC Navigator. It is possible to force I/O variables and on-sheet references. The value which is forced will be valid until the force of the variable is disabled. All forces are controlled by a force enable input. Forces are only allowed if the force enable input is high and the force enable flag is true. Forces will be cleared if the force enable input is low.

Write enable

This field specifies for each variable whether it is possible to write a variable using the 'On-Line Environment' options of FSC Navigator. The value which is assigned to the variable will be overwritten in the next program cycle.

SER enable

This field specifies whether sequence-of-event recording (SER) should take place for the variable. If an SER sequence range has been specified and the variable is SER-enabled, an SER sequence is displayed. This field can be used to specify a unique number for numerical SER format.

Table 1-5 provides an overview of the characteristics of the various variable types.

Table 1-5 Characteristics of FSC variables

Variable type	Location	Safety-related	Force enable	Write enable	SER enable
I		X	X	N	X
I	COM	N	X	Y	X
I	FSC	X	X	N	X
I	SYS	Y/X ¹⁾	N	N	X
I	ANN	N	N	N	X
AI		X	X	N	X
BI		X	N	N	N
BI	COM	N	N	Y	N
BI	FSC	X	N	N	N
BI	MUX	N	N	N	N
XI		N	N	N	N

N = No

Y = Yes

X = Can be modified

Note:

1) = Safety-related except for the COM I/O input and the clock synchronization input. Safety relation of the fault reset and force-enable variables can be modified.

Table 1-5 Characteristics of FSC variables (continued)

Variable type	Location	Safety-related	Force enable	Write enable	SER enable
O		X	X	N	X
O	COM	N	N	N	X
O	FSC	X	N	N	X
O	SYS	N	N	N	Y/N ⁵⁾
O	ANN	N	X ⁴⁾	N	X ⁷⁾
AO		X	X	N	X
BO		X	N	N	N
BO	COM	N	N	N	X
BO	FSC	X	N	N	X
BO	MUX	N	N	N	N
XO		N	N	N	N
M		Y/N ²⁾	N ³⁾	X ⁶⁾	X
C		Y/N ²⁾	N	X	X
T		Y/N ²⁾	N	X	X
R		Y/N ²⁾	N	X ⁶⁾	N
P		N	N	N	N
A		N	N	N	N

N = No

Y = Yes

X = Can be modified

Notes:

2) = Calculated by the compiler.

3) = Can only be modified for digital on/off-sheet references.

4) = Only for annunciator variables with a hardware location.

5) = Depends on the function it performs for FSC applications:

RKE triggers: N

BO SER triggers: Y

6) = Not allowed for on/off-sheet references (write enable = N)

7) = Only for first-up alarm indication.

Transmitter alarm setpoint low

The minimum value for each analog input must be entered. If the analog input goes below the minimum setpoint, the system will give an analog sensor break alarm.



Transmitter alarm setpoint high

The maximum value for each analog input must be entered. If the analog input goes above the maximum setpoint, the system will give an analog sensor break alarm.

SER setpoint low

If the value which is read or written for an analog I/O is lower than SER setpoint low, sequence-of-event recording takes place (providing the SER enable flag is set to 'Yes'). The SER setpoint low should be higher than, or equal to, the bottom scale, and lower than the SER setpoint high.

SER setpoint high

If the value which is read or written for an analog I/O is higher than the SER setpoint high, sequence-of-event recording takes place (providing the SER enable flag is set to 'Yes'). The SER setpoint high should be lower than, or equal to, the top scale, and higher than the SER setpoint low.

Power-up value and Power-up status

These fields contain the value which the variable will get after a power-up sequence. The power-up value can be specified for multiple inputs (BI) with location COM, FSC or MUX, multiple outputs (BO), analog outputs (AO), counters (C) and registers (R). The power-up status can be specified for digital inputs (I) with location COM or FSC, digital outputs (O) and markers (M).

Sensor is fail-safe

This field specifies if the sensor is fail-safe, i.e. goes to a defined state (0 or OFF) in case of a sensor failure. It is only used for safety-related digital and analog inputs.

Sensor is redundant

When a sensor is not fail-safe and the digital input or the analog input is safety-related, a redundancy type should be selected. Redundancy is used to increase the reliability of the data read. Depending on the redundancy type, more channels should be allocated to this variable. Refer to Appendix C ("Safety-related inputs with non-fail-safe sensors") of the FSC Software Manual for further details.

Signal type

The signal type is used for analog I/O. It should match the hardware configuration used. The following signal types are possible:

- 0 .. 20 mA
- 4 .. 20 mA
- 0 .. 5 V
- 1 .. 5 V
- 0 .. 10 V
- 2 .. 10 V



Engineering units

The engineering units are used with SER-enabled analog I/O, PIDs and BOs. The engineering units are defined as a string with a maximum length of five characters of the unit which is read or written with the variable, e.g. °C, Psi, %.

Bottom scale

The bottom scale of an analog variable should contain a value in engineering units which is read or written on the bottom value of the defined signal type. If, for example, a temperature of 20°C is measured and the voltage of the sensor is 2 Volt with a signal type of 2-10 V, you should enter 20 as the bottom scale value. Refer to section 4 ("System Configuration") of the FSC Software Manual for further details.

Top scale

The top scale of an analog variable should contain a value in engineering units which is read or written on the top scale value of the defined signal type.

Register type

The register type defines the format of the data stored. There are four different register types. The first three are integer values, the last (Float) is a real value which consists of an exponent and a mantissa.

Table 1-6 FSC register types

Register type	Integer/Real	Minimum value	Maximum value
Byte (8-bit)	Integer	-128	+127
Word (16-bit)	Integer	-32,768	+32,767
Long (32-bit)	Integer	-2,147,483,648	+2,147,483,647
Float (32-bit)	Real	-10e38	+10e38

MUX tag number

A BI variable is 'connected' to the XI variable via the MUX tag number. The tag number of the accessory multiplexer is the MUX tag number. A BO variable will be 'connected' to an XO variable. Refer to Appendix G ("FSC Multiplexers") of the FSC Software Manual for further details.

Number of input bits

This field specifies the number of input bits of a BI (with location <> COM, <> FSC) or an XI, and has a range of 1 to 32 bits. All bits of the variable will be allocated to sequential channels. If the number of bits does not fit on the first module, a second module must be defined. On this next module, the allocation starts at channel one. It may be necessary to specify as much as four modules.



Selection output rack

The selection output rack is the rack where the selection outputs of the multiplexed variable will be allocated. Refer to Appendix G ("FSC Multiplexers") of the FSC Software Manual for further details.

Selection output position

The selection output position is the position in the rack where the multiplexed variable will be allocated. Remember to allocate a BI variable with location MUX to a digital output! All the selection outputs must be allocated. The selection output channel is the first channel of the first module which the multiplexed variable will be allocated to. Refer to Appendix G ("FSC Multiplexers") of the FSC Software Manual for further details.

Sel OUT active time

The select output active time is the time in which the select output should be true with multiplexed I/O. Select output active time should be in the range from 1 to 127 ms. Select output active time is used for multiplexed I/O (XI and XO variables). For more information see appendix 'Multiplexers'.

Sel OUT active status

The select output active status defines the value which is used for the select outputs of multiplexed I/O. When select output active status is On, the data of the multiplexed I/O is read or written to the data path (defined with the connected XI or XO) when the select output is HIGH. Refer to Appendix G ("FSC Multiplexers") of the FSC Software Manual for further details.

Number of output bits

This field specifies the number of output bits of a BO (with location $\neq$ COM, $\neq$ FSC) or an XO, and has a range of 1 to 32 bits. All bits of the variable will be allocated on sequential channels. If the number of bits does not fit on the first module, a second module must be defined. On this next module, the allocation starts at channel one and continues with the next channels. It may be necessary to specify as much as four modules.

DCS address

This field specifies a unique identification of a variable in the FSC system. It is used by a DCS to access the attributes of this variable.

1.4.7 Safety relation

The safety relation of the I/O variables is specified in a tree structure. Each leaf of the tree can be safety-related **only** if the **complete** path from root to leaf is safety-related. An I/O signal can only be safety-related if the I/O module is safety-related. An I/O module can only be safety-related if the HBD module in the same rack is safety-related.

Safety-related variables can only be allocated to safety-related I/O. The safety-related outputs in the application program should depend only on safety-related inputs. The compiler will check for conflicts of the safety relation in the application program. **I/O modules can only be safety-related if the hardware is fail-safe.**

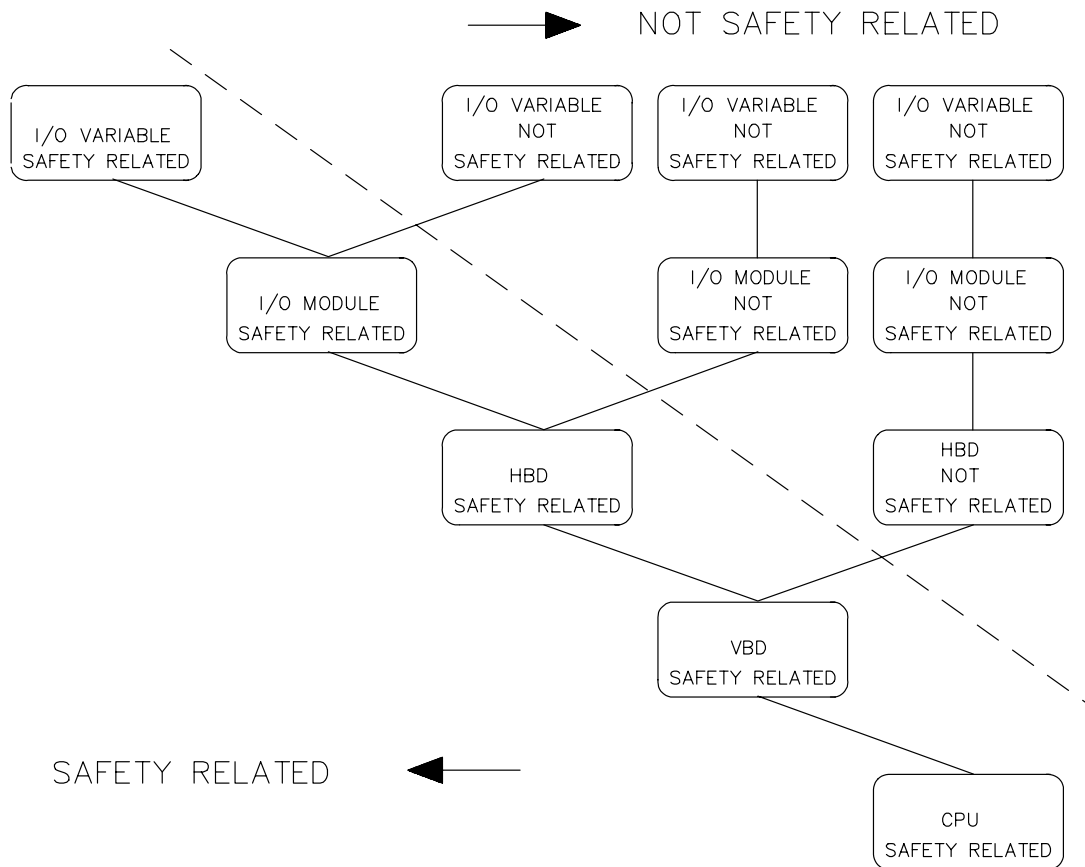


Figure 1-11 Safety relation of I/O variables



1.4.8 Voting

The FSC system is available in single and redundant mode, both for Central Part and I/O, in several combinations. For details on the various FSC configurations refer to subsection 1.2.

If the Central Part and I/O are operating in single mode, it is obvious what will happen in case a fault is detected: the Central Part or I/O will go to the safe (i.e. non-operational) state. For redundant Central Parts and/or I/O, this is less obvious, and users may want to define the system response in case a fault is detected in one part of the redundant components. This is the reason that voting has been incorporated into the system, which allows the users to optimize the system response to his safety needs.

1.4.8.1 Single components

For all single components in the FSC system, two voting schemes are available depending on the hardware that is being used. The table below lists the various options.

Table 1-7 Voting schemes for single FSC components

Voting scheme	Used for hardware modules...	Fault results in...
1001D	diagnostics capabilities (e.g. 10101/./ digital input modules)	switch-off
1001	without diagnostic capabilities (e.g. 10104/./ digital input modules)	incorrect operation or switch-off

1.4.8.2 Redundant components

Redundant components have more voting schemes to choose from, depending on the hardware that is being used and on the primary action in case a fault is detected: switch-off or continue. Table 1-8 and Table 1-9 below list the various options.

Table 1-8 Voting schemes for redundant components

	Hardware	
Primary action at fault	Fail-safe	Non fail-safe
Safety (switch-off)	1002D	1002
Availability (continue)	2002D	2002

Table 1-9 Redundancy voting schemes

Voting scheme	Used for hardware modules...	Primary action directed at...	Response to faults
1oo2	without diagnostics capabilities (e.g. 10104/./ digital input modules)	safety (switch-off)	The first fault may result in switch-off as the faulty module may overrule the correct one.
2oo2	without diagnostics capabilities (e.g. 10104/./ digital input modules)	availability (continue)	The first fault may result in incorrect operation as the faulty module may overrule the correct one.
1oo2D	with diagnostics capabilities (e.g. 10101/./ digital input modules)	safety (switch-off)	For detected faults, operation continues as desired. A fault that cannot be detected by the diagnostics (probability = $1 - \text{diagnostic coverage}$) may result in switch-off as the faulty module may overrule the correct one.
2oo2D	with diagnostics capabilities (e.g. 10101/./ digital input modules)	availability (continue)	For detected faults, operation continues as desired. A fault that cannot be detected by the diagnostics (probability = $1 - \text{diagnostic coverage}$) may result in incorrect operation as the faulty module may overrule the correct one.



1.4.9 Default application variables

Whenever a new project is created, 29 system variables are stored in the variable database. These variables can be used in the application software.

Pulse generators:

- I Flasher - 0.5 Hz
- I Flasher - 1 Hz
- I Flasher - 2 Hz
- I Flasher - 5 Hz (only for single configurations).

Function: Pulses of different frequencies. These variables are not safety-related.

Real-time clock functions (only available if a DBM module has been placed):

- BI MONTH (byte)
- BI DATE (byte)
- BI DAY (byte)
- BI HOURS (byte)
- BI MINUTES (byte)
- BI SECONDS (byte)
- BI YEAR (word)

Function: Indicate the date and time as available in the FSC system.
These variables are not safety-related.

System inputs and outputs:

- I COM I/O
- O COM I/O

Only used for fully redundant configurations. These variables are not safety-related.

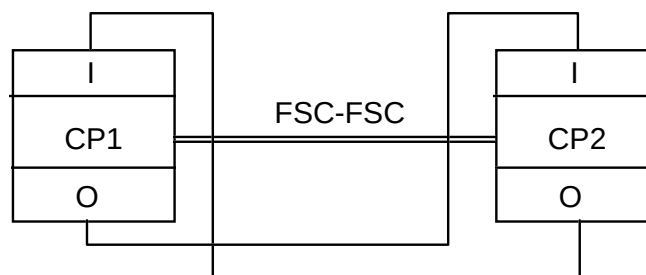


Figure 1-12 COM I/O variables

If FSC communication fails, the Central Part can use these COM I/O variables to see if the other Central Part is still running. This will be the case if the input is high (the connected output is also high).

**System reset:**

- I <tag number> Default: RESET.

Function: All errors detected and stored in memory will be cleared. This hardware input is edge-triggered. The variable safety relation can be modified (default 'Yes').

Important!

Always read diagnostic data prior to a system fault reset. Otherwise diagnostic data information will be lost.

Force enable:

- I <tag number> Default: ENABLE.

If this hardware input is high, forces to other variables can be given. The variable safety relation can be modified (default 'Yes'). This input must also be high if you want to download the application file to the FSC system.

Clock synchronization input:

- I <tag number> Default: CLOCK_SYNC

This variable is not safety-related. The hardware input is used to synchronize the real-time clock of the FSC system. It can be used in combination with the clock set option via the Modbus or RKE3964R protocol. When the input is asserted, the clock value (set by DCS) is accepted by the FSC system. If the input is asserted while no clock set command was initiated by the DCS, the FSC system will set its clock at 03.00.00 (a.m.). This allows simple adaptation to daylight saving time.

Secondary switch-off:

- O <tag number> Default: SEC.SWITCH-OFF

In normal operation the output is low. If a condition occurs which requires a Central Part to deactivate the outputs of the other Central Part, the output is set high and an emergency shutdown is effected on the watchdog module of the other Central Part. The shutdown becomes active when allocated to a hardware output channel (10201/1/1 or 10216/1/1). The output may be used in the application program to initiate a shutdown at a user-defined condition.

System markers:

The system markers are safety-related variables.

- I CENTR.PART-FAULT
Normally high, but if a fault is detected in a module connected to this Central Part, this input will go low.
- I INPUT-FAILURE
Normally high, but if a fault is detected in an input test routine, this input will go low. If a subsequent fault on inputs is detected, it will give a pulse.
- I OUTPUT-FAILURE
Normally high, but if a fault is detected in an output test routine, this input will go low. If a subsequent fault on inputs is detected, it will give a pulse.



- I TEMP.PRE-ALARM
Normally high, but if the temperature of the DBM is too high or too low, this input will go low. The default setting is 55°C (131°F).
- I IO-FORCED
Normally high, but if any input, output or sheet reference has been forced, this input will go low. If a subsequent variable is forced, it will give a pulse.
- I DEVICE-COM.FLT
Normally high, but if a device fails (e.g. Modbus link, FSC SER PC), this input will go low. If a subsequent device communication error occurs, it will give a pulse.
- I RED.INPUT FAULT
Normally high, but if a fault has been detected in redundant input sensors (e.g. discrepancy) , this input will go low. If another fault of this kind is detected, it will pulse. It is only used for safety-related inputs with non-fail-safe sensors.
- I EXT.COMMUNIC.FLT
Normally high, but if an error in the external communication is detected, this input will go low. If another fault of this kind is detected, it will give a pulse.
- I INT.COMMUNIC.FLT
Normally high, but if a Central Part has shut down, this input will go low.
- I TRANSMIT-FAULT
Normally high, but if an analog input exceeds the transmitter setpoint alarms, this input will go low. If another fault of this kind is detected, it will give a pulse.
- I FSC-SYSTEM-FAULT
Normally high, but if any fault is detected in the FSC system (see above), this input will go low. If another fault of this kind is detected, it will give a pulse.
- I IO-COMPARE
Normally high, but if the status of an input or output variable in Central Part 1 differs from the status of that variable in Central Part 2 and the discrepancy cannot be attributed to an existing hardware fault, this input will go low. If another fault of this kind is detected, it will give a pulse. This variable is not available in FSC systems with only one Central Part.



Left blank intentionally.



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Helping You Control Your World



Fail Safe Control

Section 2: Installation Guide

Version 500

Revision 01 (01/98)



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2.1 Installation requirements

Hardware requirements

FSC Navigator and its installation program are both Windows applications, so you must already have Windows 95 running on the target computer to install FSC Navigator. The installation program creates folders as needed and copies files from the distribution disks or CD-ROM to your hard disk.

The target PC should meet the following hardware requirements:

- Pentium 166-MHz processor or higher (recommended)*
- 16 MB of RAM (32 MB recommended)
- PC serial interface module 07177/1/. installed**
(see subsection 2.2)
- 15 MB of hard disk space for the FSC Navigator application files
- Sufficient free hard disk space to hold all project files. The exact disk space required depends on the number and complexity of the projects, but an average of 2 MB per project should be sufficient.

Notes:

* If the PC has a Pentium processor below 166 MHz, the maximum communication baud rate may be limited to 19k2. FSC communication at 125 kbaud requires a Pentium 166-MHz processor or higher.

** The PC serial interface module 07177/1/. is only required if you want to program applications to EPROMs.

Software requirements

FSC Navigator was specifically developed to run under Windows 95. It is **not** supported by Windows NT or Windows 3.11. Using FSC Navigator under Windows NT or Windows 3.11 is considered improper use of the software. Honeywell Safety Management Systems cannot be held liable in any way for any damages that result from using FSC Navigator under Windows NT or Windows 3.11.



Printer requirements FSC Navigator can output hardcopy to a printer or plotter. Table 2-1 below lists the output devices that are supported:

Table 2-1 Supported output devices

Matrix printers		Laser printers	Plotters
Epson MX-80	Epson FX-1000	HP LaserJet series II	HP 7475
Epson RX-80	Epson FX-800	HP LaserJet series III	HP 7550
Epson FX-80	NEC P6	Canon LBP-8II	
Epson FX-85	NEC P7	Brother HL-8e	
Epson FX-100			
Epson FX-105			
Epson LX-800			

If you intend to use the HP LaserJet series III with full graphics, make sure that the printer has at least 2 Mbytes of memory (and preferably more) to avoid memory overflow problems when printing complex functional logic diagrams.

Also, enable page protection for the paper size that you expect to use most (using the printer's front panel menu). If page protection is set, additional memory is reserved for page image processing, which allows the printer to create the entire page image in memory before outputting to paper.



2.2 Serial interface module 07177/1/.

Introduction

The serial interface module 07177/1/. (see Figure 2-1) is only required if you want to program applications to EPROMs. The interface module is used to connect the EPROM programmer (07170/1/1) to the FSC user station.

Since the Demo and Contractor versions of FSC Navigator do not allow programming applications to EPROMs, you do not need the serial interface module 07177/1/. for these versions of FSC Navigator.

Note:

For technical details on the serial interface module 07177/1/. and the EPROM programmer 07170/1/1 refer to the FSC Hardware Manual.

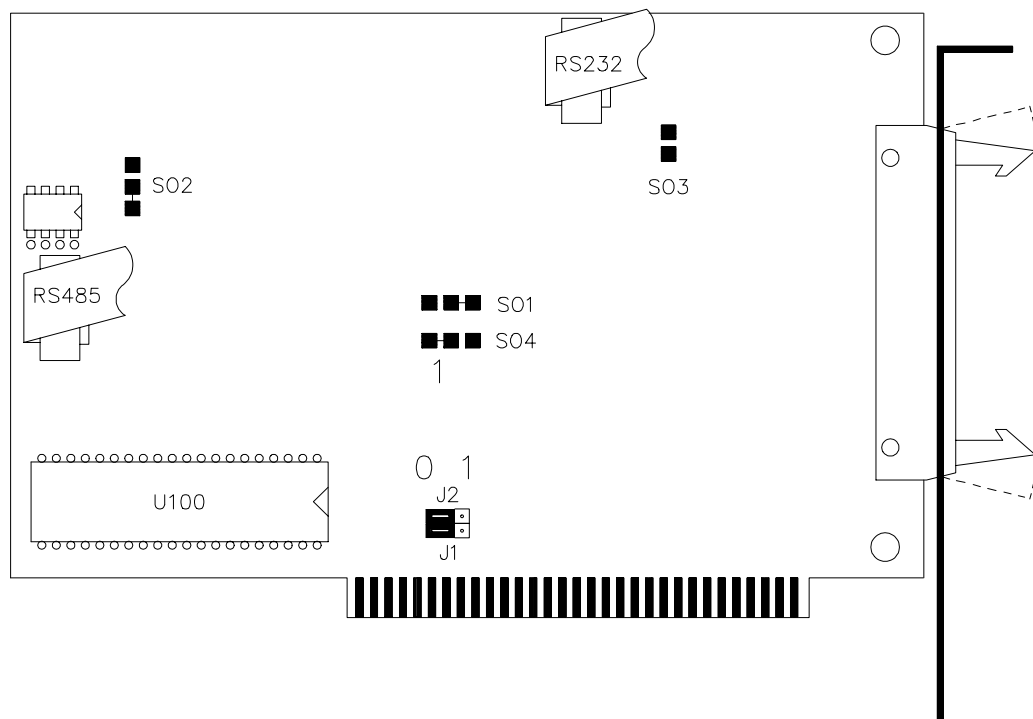


Figure 2-1 07177/1/. serial interface module



- Installing the module** To install the serial interface module 07177/1/., do the following:
1. Turn off your PC.
 2. Remove the cover from your PC.
 3. Select an unused expansion slot and remove its bracket.
 4. Verify that the address setting of the serial interface module 07177/1/. (see below) does not interfere with any other modules (e.g. network interface modules).
 5. Hold the serial interface module by its top edges and press it firmly into an unused expansion slot. After positioning the module, tighten the screw to secure the module bracket.
 6. Put the PC cover back into place.
 7. Turn on your PC.
-

Module address

The FSC software uses interrupt level 5 (IRQ5), DMA channel 3 and hexadecimal I/O address 300h for the serial interface module 07177/1/.

The interrupt level and the DMA channel cannot be changed. The address can, however, be changed by modifying the setting of jumpers J1 and J2 on the serial interface module (see Figure 2-1). Table 2-2 below shows the jumper settings for the various module addresses.

Table 2-2 07177/1/. module addresses

Jumper settings		Base address
J1	J2	
0	0	300 (default)
1	0	320
0	1	340

FSC Navigator assumes hexadecimal module address 300h. If you wish to use a different address, you can modify it using the 'Program EPROMs' option of FSC Navigator. The address is stored in the FSC.INI initialization file, which is located in the FSC Navigator application folder (C:\FSC by default).

For details on programming applications in EPROMs refer to Section 10 of the FSC Software Manual ("Programming EPROMs").



2.3 FSC Navigator versions

FSC Navigator versions

FSC Navigator is available in four versions. The customer will always purchase a particular version. A unique license number is then issued which allows the customer to install and use the appropriate version. The four FSC Navigator versions are the following:

- **Demo**

This version has limited functionality and is primarily for demonstration purposes.

- **Standard**

This version offers all FSC Navigator functions.

- **Contractor**

This version has limited functionality and allows FSC contractors to design a system.

- **Hartmann & Braun**

This version is functionally identical to the Standard version, but it also supports the Hartmann & Braun (H&B) communication protocol.

Table 2-3 below lists the differences between the various software versions.

Table 2-3 FSC Navigator versions

Function	FSC Navigator versions			
	Demo	Contractor	Standard	H&B
System configuration	✓	✓	✓	✓
Print	✓	✓	✓	✓
Design FLDs	✓	✓	✓	✓
Max. number of FLDs in project	10	999	999	999
View log files	✓	✓	✓	✓
Translate application		✓	✓	✓
Program EPROMs			✓	✓
Edit SER formats			✓	✓
Verify application			✓	✓
Monitor system and view process status			✓	✓
FSC-SMM support	✓	✓	✓	✓
Hartmann & Braun comm. protocol + P-Bus				✓



2.4 Installing FSC Navigator

The FSC Navigator software is supplied on 3.5" diskettes or CD-ROM. If you wish to install a version that allows programming of EPROMs (Standard and H&B versions), make sure that the serial interface module (07177/1/.) has been properly installed in your PC (see subsection 2.2).

Note:

The installation procedure as described below assumes that you use the English version of Windows 95 and that you choose English as the Setup language. However, depending on the language version of your Windows 95 and the setup language that you select, the messages may also be displayed in a different language than English.

Previous installations of FSC user software

Setup automatically detects if other FSC user software (FSC Navigator or FSC Development System) has been installed on your PC. If that is the case, Setup will provide you with the option of uninstalling the "old" software (if this is possible) or installing the "new" software in an alternative folder.

Note:

If you previously installed the FSC Development System (FSC-DS) software on your PC and you want to install FSC Navigator in the same folder (C:\FSC by default), you may want to uninstall the FSC Development System software before starting the installation procedure.

To uninstall FSC-DS, start its Install program and choose the "Uninstall FSC from hard disk" option. Do not just delete the contents of the FSC-DS folder as this will decrease the number of available installations by one. Please note that the FSC-DS uninstall program will not remove the project files that were created using FSC-DS, which means that you can use your old projects in FSC Navigator.



Installation procedure To install FSC Navigator, do the following:

- Starting Setup**
1. Put diskette 1 (Installation disk) in the disk drive or the CD-ROM in the CD-ROM drive.
 2. Close all Windows applications that are open.
 3. From the **Start** menu, choose the **Run** command. This will open the **Run** dialog box. Click the **Browse** button, select the disk drive (usually A:) or CD-ROM drive and double-click SETUP.EXE to close the **Browse** dialog box.
 4. Click the **OK** button to close the **Run** dialog box and to start the Setup program.
- Setup language**
5. You will first be asked to choose which language should be used for the setup process (see Figure 2-2). Select either English or German from the drop-down list and click the **OK** button to continue.

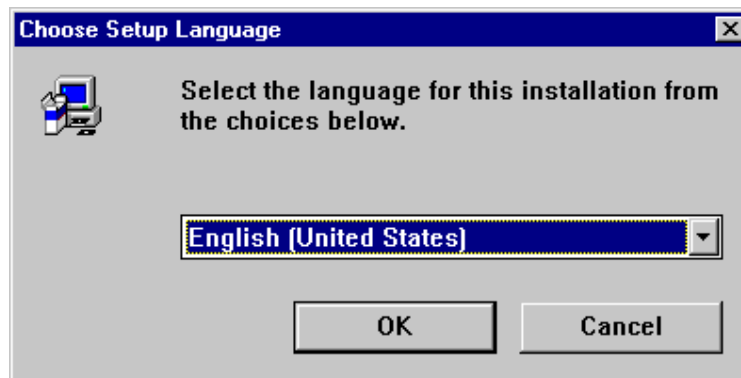
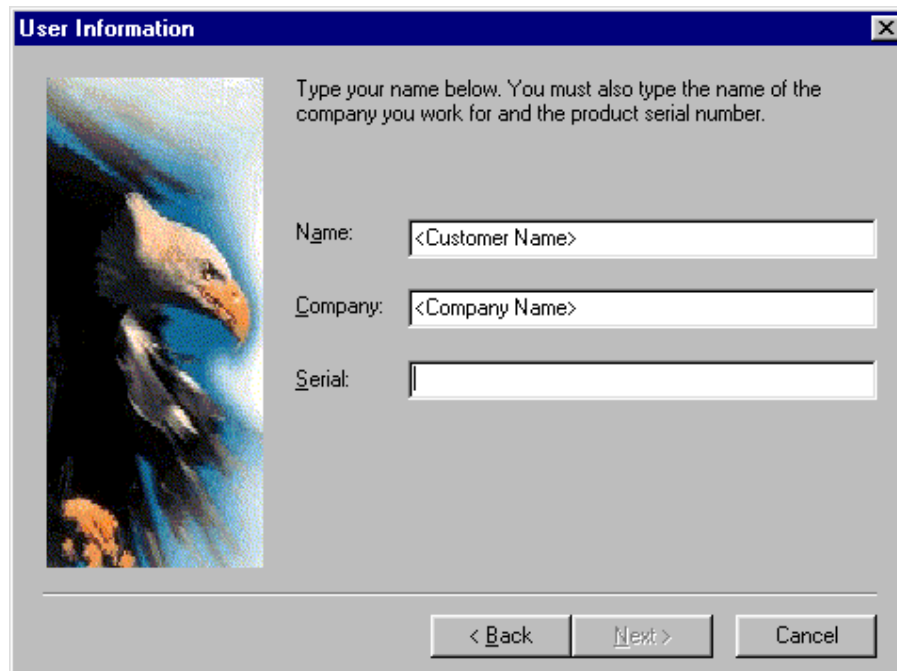


Figure 2-2 Choosing the setup language

- User information**
6. A Welcome window will appear. Click the **Next** button to open the **User Information** dialog box (see Figure 2-3).
- Serial number**
- Enter the user name and the company name. In the **Serial** text box, enter the serial number that was provided with the installation diskettes or CD-ROM. Please note that any letters in the serial number are case-sensitive (i.e. 'x' is not the same as 'X'). You **must** enter a valid serial number before you can install FSC Navigator. After you have successfully completed the **User Information** dialog box, click the **Next** button.



The 'User Information' dialog box features a blue title bar with a close button. On the left is a vertical image of an eagle's head. The main area contains instructional text and three input fields. The 'Name' field is pre-filled with '<Customer Name>', the 'Company' field with '<Company Name>', and the 'Serial' field is empty. At the bottom are three buttons: '< Back', 'Next >', and 'Cancel'.

Type your name below. You must also type the name of the company you work for and the product serial number.

Name:

Company:

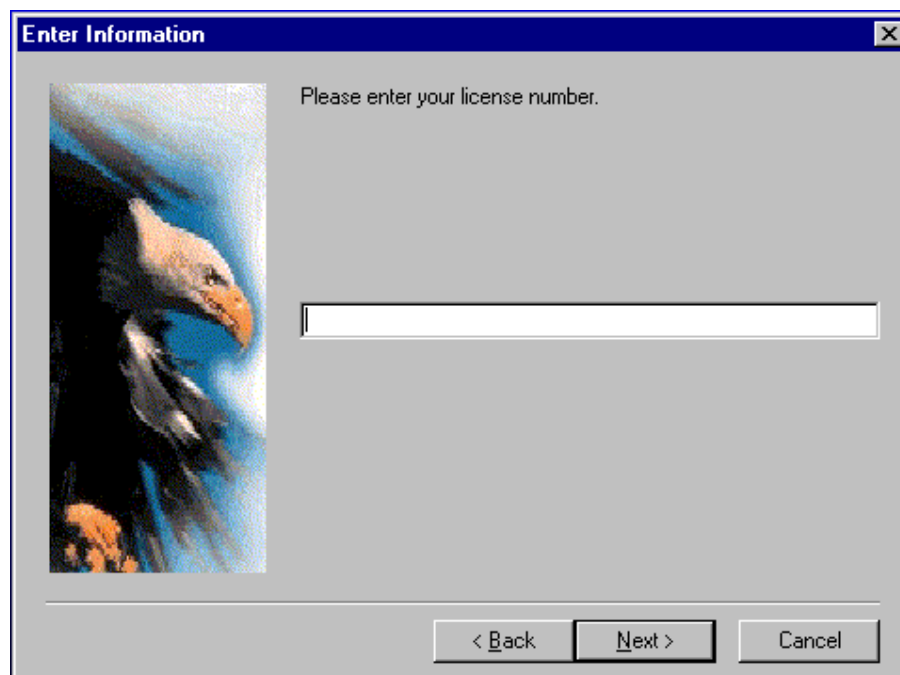
Serial:

< Back Next > Cancel

Figure 2-3 Entering user information

License number

7. A dialog box will appear which prompts you to enter the license number that was supplied with the installation diskettes or CD-ROM (see Figure 2-4).



The 'Enter Information' dialog box has a blue title bar and a close button. It includes the same eagle image on the left. The text prompts the user to enter their license number. A single input field is provided for this purpose. The bottom buttons are '< Back', 'Next >', and 'Cancel'.

Please enter your license number.

< Back Next > Cancel

Figure 2-4 Entering license number



The license number determines which version of FSC Navigator you are entitled to use. The Setup program will automatically install the correct version.

Please note that any letters in the serial number are case-sensitive (i.e. 'x' is not the same as 'X'). You **must** enter a valid license number before you can proceed.

After you have entered a valid license number, click the **Next** button.

Software license

8. This will open the **Software License Agreement** window (see Figure 2-5). Carefully read the license agreement before proceeding. You **must** accept the license agreement before you can continue. To accept the license agreement and continue with the installation, click the **Yes** button. If you decline the license agreement, click the **No** button, and the setup procedure will be aborted.

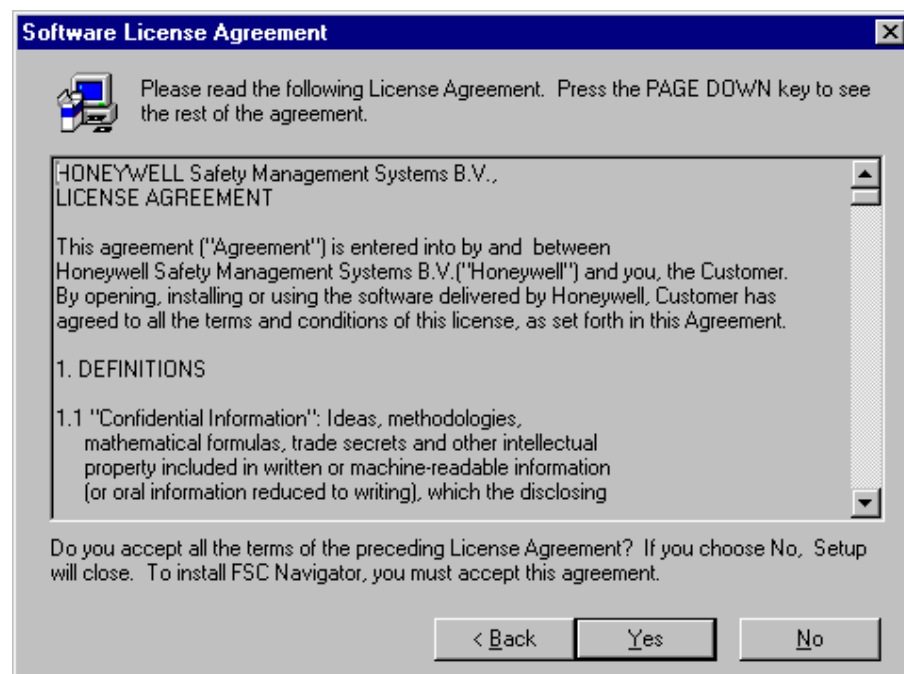


Figure 2-5 Software License Agreement window

Destination folder

9. The window as shown in Figure 2-6 will appear. This window prompts for the folder on the hard disk where the FSC Navigator application files should be installed. You can accept the default suggestion (C:\FSC), or you can use the **Browse** button to choose a different folder. When you are done, click the **Next** button.

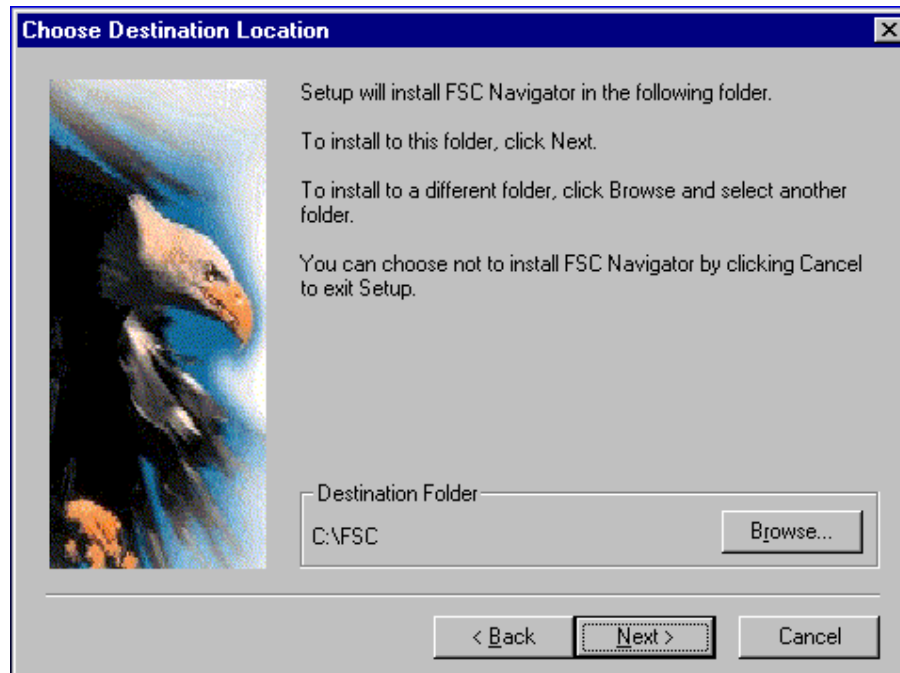


Figure 2-6 Choosing the destination folder

Another version

At this point, Setup may detect that another version of the FSC user software (FSC Navigator or FSC Development System) has already been installed in the destination folder. If this version can be uninstalled, the window as shown in Figure 2-7 will appear. If no other FSC software version is detected, the step below will be skipped and Setup will continue with step 10.

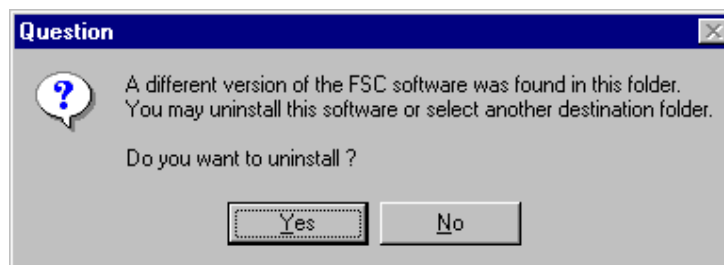


Figure 2-7 Uninstalling a previously installed version

If you choose **Yes**, the previously installed version of the FSC user software will be uninstalled. If you choose **No**, this version will not be uninstalled. You must then choose a different destination folder for FSC Navigator.



If the existing FSC software in the destination folder cannot be uninstalled, the following window will appear:

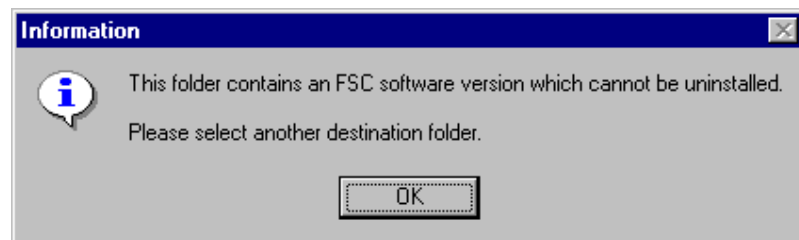


Figure 2-8 Previously installed version cannot be uninstalled

Note:

The FSC Development System software (< v500) can never be uninstalled by Setup. You must then choose a different destination folder.

Alternatively, you may choose to exit Setup, uninstall the FSC-DS software (using the FSC-DS Install program) and run Setup again. If you choose to uninstall FSC-DS, the project subfolders will not be removed, which means that you can use your old project files in FSC Navigator.

Program folder 10. The window as shown in Figure 2-9 will appear.

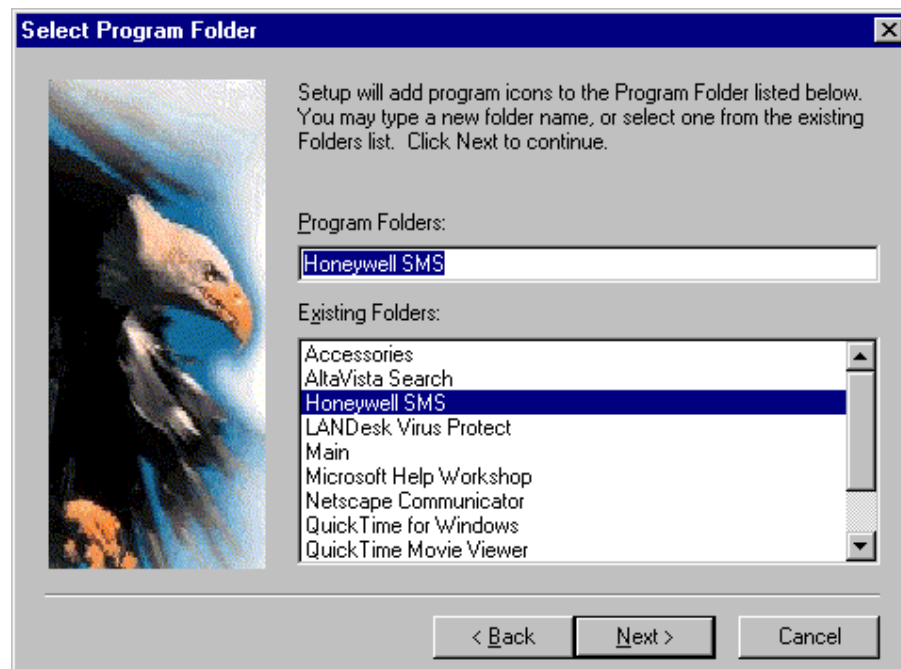


Figure 2-9 Selecting a program folder

This window prompts for the name of the program folder that will contain the shortcut to FSC Navigator. You can accept the default suggestion (Honeywell SMS), or you can enter a different program folder. When you are done, click the **Next** button.

Note:

Please note that the program group of all other FSC-related software applications will also default to 'Honeywell SMS'.

AUTOEXEC.BAT

11. If, at this point, Setup detects that your AUTOEXEC.BAT file requires modification, the window as shown in Figure 2-10 will appear. If no changes are required, this step will be skipped.

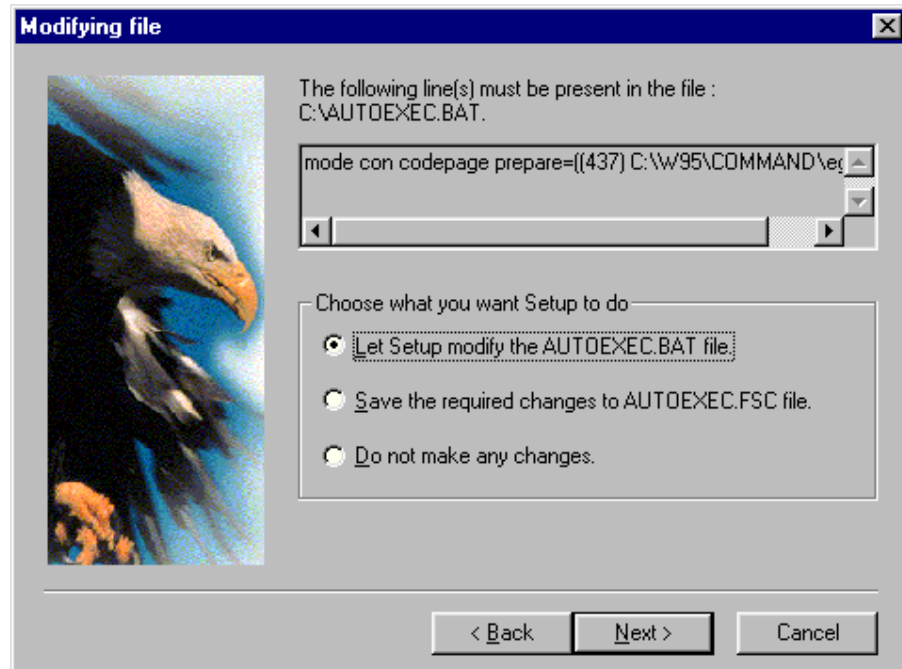


Figure 2-10 Modifying AUTOEXEC.BAT (if required)

It is strongly recommended that you let Setup modify the AUTOEXEC.BAT file for you.

Note:

FSC Navigator can **only** operate correctly if the indicated modifications to the AUTOEXEC.BAT file are carried out.



CONFIG.SYS

12. If, at this point, Setup detects that your CONFIG.SYS file requires modification, the window as shown in Figure 2-11 will appear. If no changes are required, this step will be skipped. It is strongly recommended that you let Setup modify the CONFIG.SYS file for you.

Note:

FSC Navigator can **only** operate correctly if the indicated modifications to the CONFIG.SYS file are carried out.

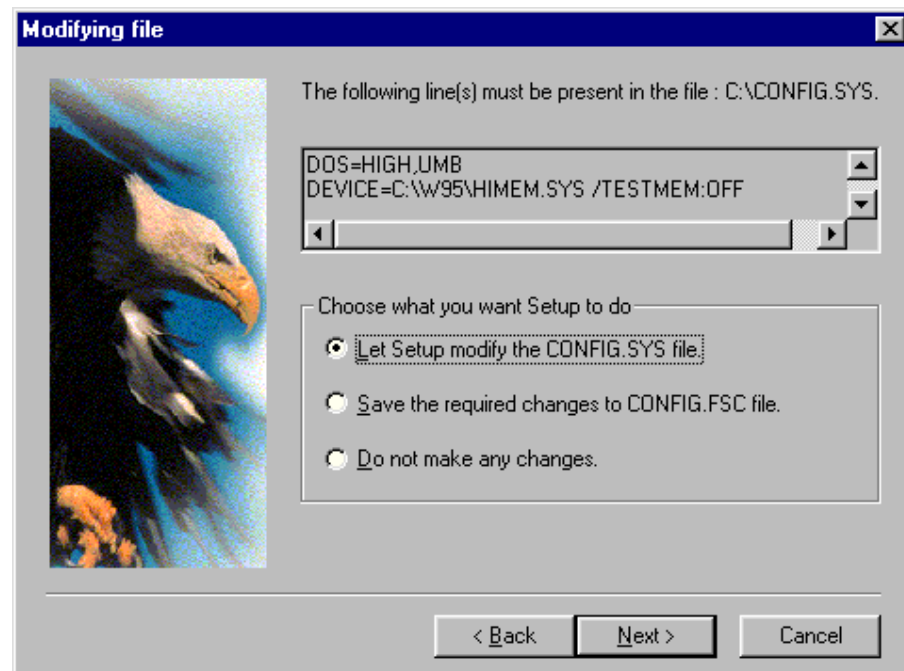


Figure 2-11 Modifying CONFIG.SYS (if required)

Summary window

13. The window as shown in Figure 2-12 will appear.

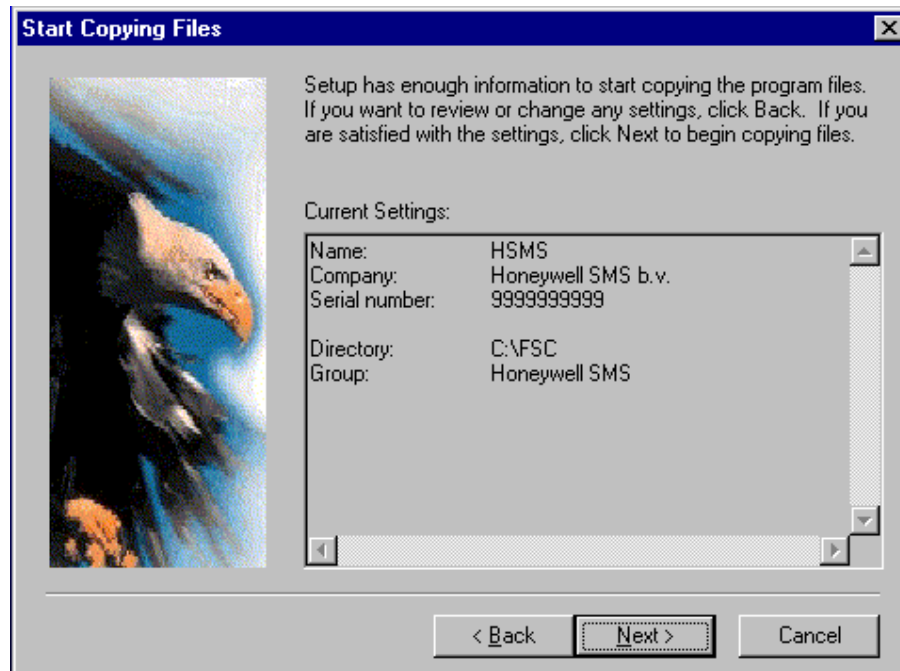


Figure 2-12 Installation summary window

This window presents a brief summary of the installation parameters before the setup program will start copying the files to your hard disk. If you want to change any of the parameters, use the **Back** button to move to the appropriate window and change the parameter. Then use the **Next** button(s) to return to this window. If you are ready to install FSC Navigator, click the **Next** button in this window. The files will now be copied to hard disk.

Finalizing Setup

14. If any of your system files were modified, you will be prompted to restart your computer. Click the **Finish** button to complete the setup process and restart your computer (if required).

Note:

If you are prompted to restart your system, FSC Navigator will only operate correctly *after* you reboot your computer.



Starting FSC Navigator

If the FSC Navigator application has been properly installed on the FSC user station, two items have been added to the Windows 95 user interface:

- A shortcut on the Windows 95 desktop called 'FSC Navigator 500' (see Figure 2-13), and
- A program group in the Programs menu under the Start button called 'Honeywell SMS'.



Figure 2-13 FSC Navigator shortcut on desktop

You can start FSC Navigator by double-clicking the desktop shortcut, or by moving to the 'Honeywell SMS' program group and choosing the 'FSC Navigator 500' option.



2.5 Error messages

Error messages

The following error messages may appear while installing the FSC Navigator software on your system.

English: A fatal error has occurred during setup. Error: <code>.

German: Fataler Fehler während des Setup. Fehler: <Kode>.

This message appears if an error occurs that prevents Setup from continuing the installation procedure. The setup process will be terminated. <Code> is an internal error code that helps HSMS engineers determine what went wrong.

English: An error occurred during the move data process. Error: <code>.

German: Fehler während der Übertragung der Daten. Fehler: <Kode>.

This message appears if an error occurs while data is being copied from the distribution disks or CD-ROM to hard disk. <Code> is an internal error code that helps HSMS engineers determine what went wrong.

English: An error occurred while uninstalling the FSC software. Please select another destination folder.

German: Fehler beim Deinstallieren der FSC-Software. Wählen Sie bitte einen anderen Zielordner.

This message appears if an error occurs while the previously installed FSC software is being uninstalled from your system. You can only continue with the setup process if you select a different destination folder for FSC Navigator.

English: FSC Navigator requires Windows 95. Setup will be aborted.

German: FSC Navigator erfordert Windows 95. Setup wird abgebrochen.

This message appears if you attempt to install FSC Navigator on a computer which contains a different Windows version than Windows 95. The setup process will be terminated.



English: Invalid license number entered.

German: Die eingegebene Lizenznummer ist ungültig.

This message appears if you enter an incorrect license number. Make sure that the license number that you enter is **exactly** as it was provided to you. (Please note that the letters are case-sensitive, i.e. 'x' is not the same as 'X'.) This message may also appear if the **serial** number that you entered earlier is not correct.

English: Invalid serial number entered.

German: Die eingegebene Seriennummer ist ungültig.

This message appears if you enter an incorrect serial number. Make sure that the serial number that you enter is **exactly** as it was provided to you. (Please note that the letters are case-sensitive, i.e. 'x' is not the same as 'X'.)

English: This program requires VGA or better resolution.

German: Dieses Programm benötigt VGA- oder bessere Auflösung.

This message appears if you attempt to install FSC Navigator on a computer that does not have sufficient graphic capabilities. Make sure that your computer has a VGA-compatible video card.

English: Unable to modify AUTOEXEC.BAT. Error: <code>, <code>.

German: Änderung nicht möglich: AUTOEXEC.BAT. Fehler: <Kode>, <Kode>.

This message appears if Setup cannot carry out the modifications to your AUTOEXEC.BAT file which are required for FSC Navigator to work properly.

English: Unable to modify CONFIG.SYS. Error: <code>, <code>.

German: Änderung nicht möglich: CONFIG.SYS. Fehler: <Kode>, <Kode>.

This message appears if Setup cannot carry out the modifications to your CONFIG.SYS file which are required for FSC Navigator to work properly.

English: Unable to modify system files. Error: <code>, <code>, <code>, <code>.

German: Änderung der Systemdateien nicht möglich. Fehler: <Kode>, <Kode>, <Kode>, <Kode>.

This message appears if Setup cannot carry out the modifications to your AUTOEXEC.BAT and/or CONFIG.SYS files which are required for FSC Navigator to work properly.



English: Unable to modify SYSTEM.INI. Error: <code>, <code>.

**German: Änderung nicht möglich: SYSTEM.INI. Fehler:
<Kode>, <Kode>.**

This message appears if Setup cannot carry out the modifications to your SYSTEM.INI file which are required for FSC Navigator to work properly.

English: Unable to write to registry.

German: Schreiben in die Registrierdatei ist nicht möglich.

This message appears if Setup cannot carry out the modifications to the Windows 95 registry which are required for FSC Navigator to work properly.

English: Uninstaller setup failed to initialize. You may not be able to uninstall the program.

German: Setup-Deinstallationsprogramm konnte nicht initialisiert werden. Sie können das Programm nicht deinstallieren.

This message appears if Setup cannot initialize the program component that logs the FSC Navigator installation. This log allows FSC Navigator to be uninstalled later, and if it is compromised in any way, uninstalling may not be possible.



2.6 Uninstalling FSC Navigator

FSC Navigator has an uninstall facility which allows it to be completely removed from your system.

Uninstall procedure

To uninstall FSC Navigator, do the following:

1. Double-click the **Add/Remove Programs** applet from **Control Panel**. (Please note that in Windows 95 versions other than English these program components are called differently.) The following window will appear:

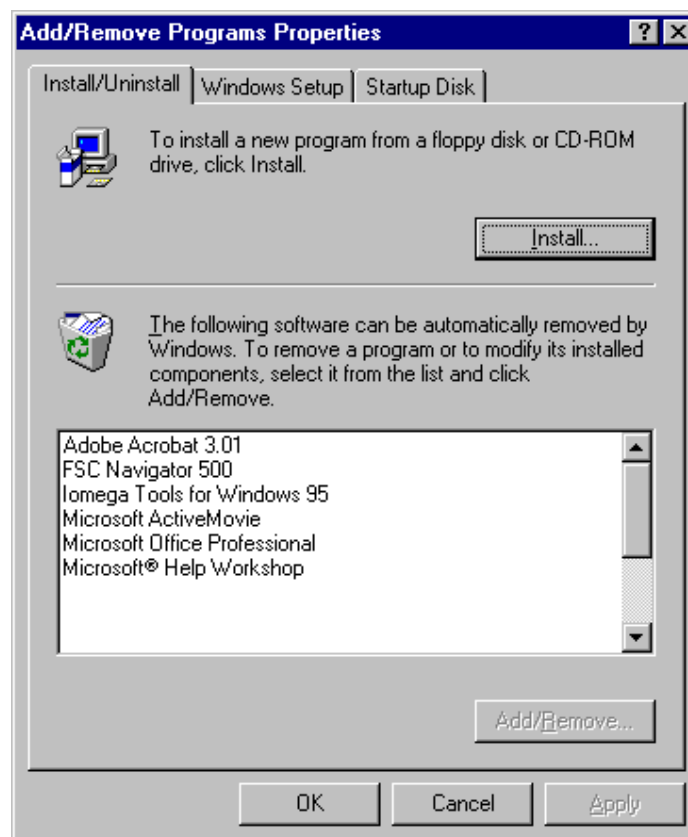


Figure 2-14 Uninstalling FSC Navigator

2. From the list of applications that can be automatically removed by Windows 95, select **FSC Navigator 500** and click the **Add/Remove** button. The window as shown in Figure 2-15 will appear.

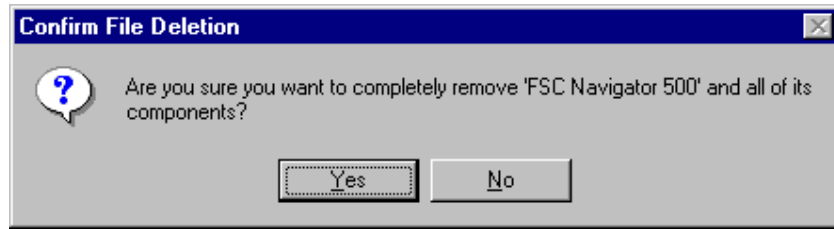


Figure 2-15 Confirming uninstall

If you want to remove FSC Navigator from your system, choose **Yes**. All FSC Navigator files will now be removed.

Any files and registry entries that were created after FSC Navigator was installed on your system will not be removed. This means that the project files in the project subfolders will not be deleted from your hard disk. If you want to remove them, you should delete them manually (e.g. using Explorer).

Please note that the modifications to the AUTOEXEC.BAT and CONFIG.SYS files will not be undone.



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Helping You Control Your World



Fail Safe Control

Section 3: Using FSC Navigator

Version 500

Revision 01 (01/98)



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Version 500
Revision 01 (01/98)

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3.1 Program overview

Note:

It is assumed that you have a good understanding of Windows basics and are comfortable using the Windows environment. If this is not the case, please refer to the Windows documentation for more information on how to use Microsoft Windows.

Program overview

FSC Navigator provides a Windows-based user interface with the FSC system. It is a powerful tool which supports the user in performing a number of design and maintenance tasks. FSC Navigator can be used to:

- configure the FSC system,
- design the application program,
- generate application documentation, and
- monitor the FSC system.

The interface between the FSC user station and the FSC system is established through a serial communication link. FSC Navigator uses this link to communicate with the FSC system.

For installation issues refer to Section 2 ("Installation Guide") of the FSC Software Manual.

Note:

It is strongly recommended that you use the FSC user station with FSC Navigator *exclusively* for FSC-related design and maintenance tasks. Other tasks may affect the stability and integrity of the Windows 95 environment.



Starting FSC Navigator

If the FSC Navigator application has been properly installed on the FSC user station, two items have been added to the Windows 95 user interface:

- A shortcut on the Windows 95 desktop called 'FSC Navigator 500' (see Figure 3-1), and
- A program group in the Programs menu under the Start button called 'Honeywell SMS'.



Figure 3-1 FSC Navigator shortcut on desktop

You can start FSC Navigator by double-clicking the desktop shortcut, or by moving to the 'Honeywell SMS' program group and choosing the 'FSC Navigator 500' option.

It is not possible to run two instances of FSC Navigator. If you attempt to start FSC Navigator for a second time, the instance that was started last will be closed automatically.

Quitting FSC Navigator

FSC Navigator can be closed in any of the following four ways:

- From the **File** menu on FSC Navigator's menu bar, choose **Exit**.
- From the Control menu (which is called up by clicking the box in the upper left corner of the application window), choose **Close**.
- With the main application window active, hold down the ALT key and press *F4*.
- Click the  icon in the upper right corner of the application window.



Main screen

After FSC Navigator has started up successfully, the main screen will be displayed (see Figure 3-2). This screen contains the following areas:

- the title bar, which contains the project name and system number (if an FSC project has been selected),
- the menu bar (see subsection 3.2), which can be used for selecting the functions of FSC Navigator,
- the button bar (see subsection 3.4), with speed buttons for quick access to certain functions, and
- the navigation area (see subsection 3.5), which provides links to various parts of FSC Navigator and provides a good "guideline" for the design and maintenance cycle.

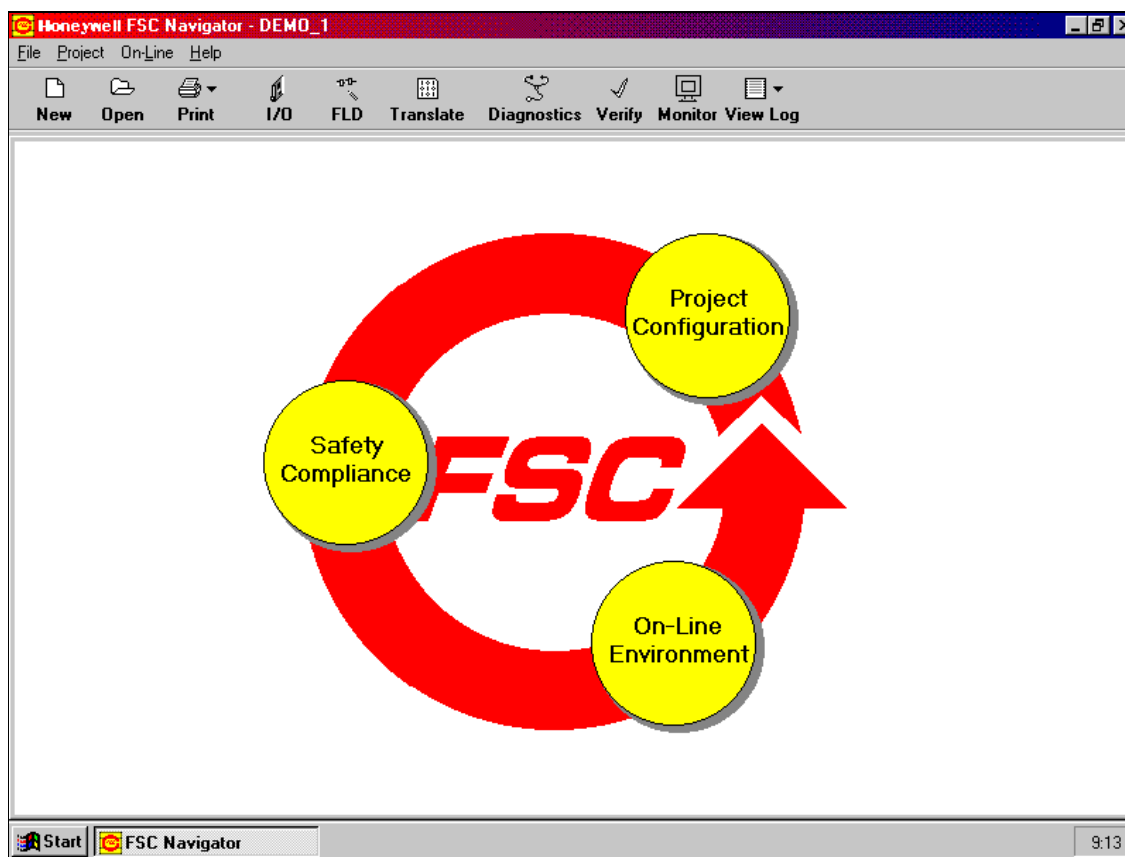


Figure 3-2 FSC Navigator main screen



3.2 FSC Navigator versions

FSC Navigator versions

FSC Navigator is available in four versions. The customer will always purchase a particular version. A unique license number is then issued which allows the customer to install and use the appropriate version. The four FSC Navigator versions are the following:

- **Demo**
This version has limited functionality and is primarily for demonstration purposes.
- **Standard**
This version offers all FSC Navigator functions.
- **Contractor**
This version has limited functionality and allows FSC contractors to design a system.
- **Hartmann & Braun**
This version is functionally identical to the Standard version, but it also supports the Hartmann & Braun (H&B) communication protocol.

Table 3-1 below lists the differences between the various software versions.

Table 3-1 FSC Navigator versions

Function	FSC Navigator versions			
	Demo	Contractor	Standard	H&B
System configuration	✓	✓	✓	✓
Print	✓	✓	✓	✓
Design FLDs	✓	✓	✓	✓
Max. number of FLDs in project	10*	999**	999**	999**
View log files	✓	✓	✓	✓
Translate application		✓	✓	✓
Program EPROMs			✓	✓
Edit SER formats			✓	✓
Verify application			✓	✓
FSC-SMM support	✓	✓	✓	✓
Monitor system and view process status			✓	✓
Hartmann & Braun comm. protocol + P-Bus				✓

* Function blocks can be defined in FLD 5 and higher.

** Function blocks can be defined in FLD 500 and higher.



3.3 Menu bar

Menu bar

The FSC Navigator application provides a menu bar for easy access to all program functions. Figure 3-4 on the next page shows an overview of the program's menu structure.

As shown in Figure 3-2, the menu bar is made up of a number of drop-down menus, which can be easily accessed by clicking on them. The following drop-down menus can be selected from the menu bar.

- **F**ile
- **P**roject
- **O**n-**L**ine
- **H**elp

Each drop-down menu or menu item contains an underlined shortcut character, which allows direct access to that menu or menu item by holding down the *ALT* key on the keyboard and pressing the shortcut character. For example, the **F**ile drop-down menu can be called up by pressing *ALT + F* (or *f*).

Figure 3-4 on the next page shows a diagram with all available menu options.

Menu availability

If no FSC project has been selected, the **P**roject menu and **O**n-**L**ine menu (and all their related buttons and shortcuts) are not available. Menu options will only be available if they are valid options. If, for example, certain log files are not available for viewing, the corresponding options on the **V**iew **L**og submenu will be disabled:

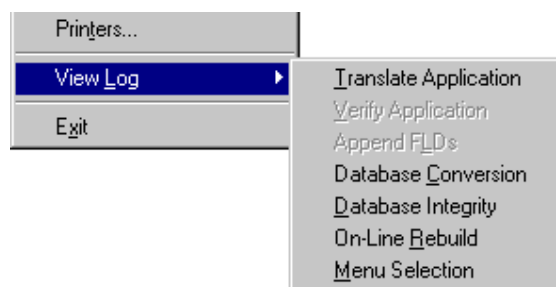


Figure 3-3 Unavailable menu options

Some menu options may not be available depending on the FSC Navigator version (see subsection 3.2). Menu options will also not be available if there are any errors, e.g. if the corresponding application component does not exist or is corrupted.

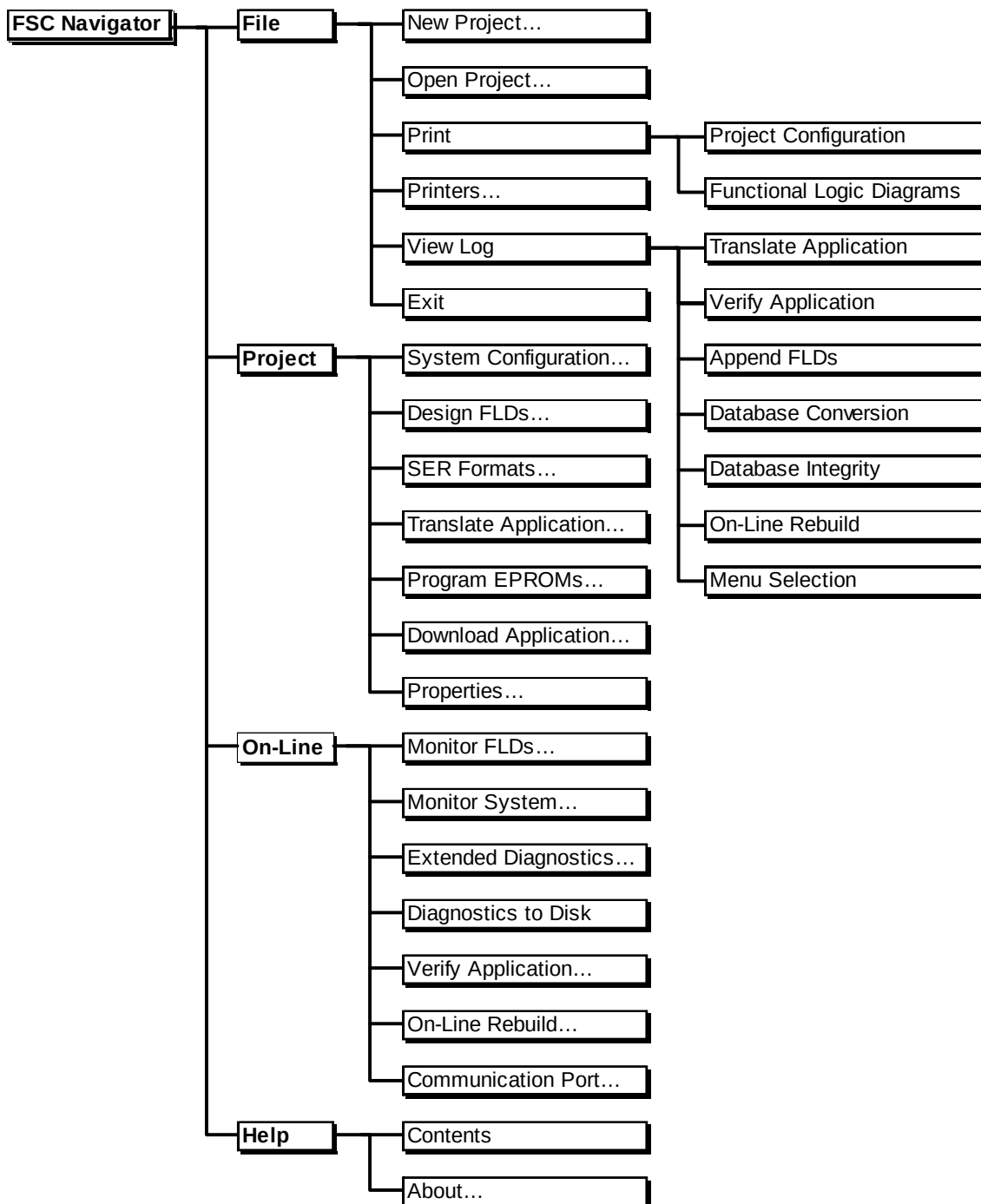


Figure 3-4 FSC Navigator menu structure



File drop-down menu

The **File** drop-down menu has the following menu items:

- **New Project...**
Use this option to create a new FSC project. (For details refer to subsection 3.10 on page 28.)
 - **Open Project...**
Use this option to open an existing FSC project. (For details refer to subsection 3.10 on page 28.)
 - **Print** ▶
Use this option to create hardcopy of the system configuration or functional logic diagrams of the current FSC project. This menu option has a submenu, which allows you to choose what you wish to print. (For details refer to Section 7 of the FSC Software Manual.)
 - **Printers...**
Use this option to define and configure the output device(s) used for creating hardcopy. (For details refer to subsection 3.8 on page 25.)
 - **View Log** ▶
Use this option to view the various log files that are available in FSC Navigator. This menu option has a submenu, which allows you to choose the specific log file you wish to view. (For details refer to Section 9 of the FSC Software Manual.)
 - **Exit**
Use this option to quit the FSC Navigator program.
-

Project drop-down menu

The **Project** drop-down menu will only be available if an FSC project has been selected. It has the following menu items:

- **System Configuration...**
Use this option to configure and maintain the system in the current FSC project. (For details refer to Section 4 of the FSC Software Manual.)
- **Design FLDs...**
Use this option to design the functional logic diagrams (FLDs) which are used to create the application logic. (For details refer to Section 6 of the FSC Software Manual.)
- **SER Formats...**
Use this option to define the layout of the sequence-of-event (SER) messages and reports. (For details refer to Section 5 of the FSC Software Manual.)



Project drop-down menu (continued)

- **Translate Application...**
Use this option to convert the project function specification (designed using FLDs) into application code that the FSC system can use. (For details refer to Section 8 of the FSC Software Manual.)
 - **Program EPROMs...**
Use this option to program application EPROMs or communication EPROMs. (For details refer to Section 10 of the FSC Software Manual.)
 - **Download Application...**
Use this option to load an application program from your hard disk into the FSC system. (For details refer to Section 12 of the FSC Software Manual.)
 - **Properties...**
Use this option to set the symbol library that will be used for the current FSC project. (For details refer to subsection 3.11 on page 31.)
-

On-Line drop-down menu

The **On-Line** drop-down menu will only be available if an FSC project has been selected. It has the following menu items:

- **Monitor FLDs...**
Use this option to view the actual status of the variables in the FLDs of the current FSC project. (For details refer to Section 12 of the FSC Software Manual.)
- **Monitor System...**
Use this option to monitor the FSC system and view the process status. (For details refer to Section 12 of the FSC Software Manual.)
- **Extended Diagnostics...**
Use this option to view and store all diagnostics of the Central Part(s) in the FSC system. (For details refer to Section 12 of the FSC Software Manual.)
- **Diagnostics to Disk...**
Use this option to back up the extended diagnostic database to hard disk or retrieve a previously stored database. (For details refer to Section 12 of the FSC Software Manual.)
- **Verify Application...**
Use this option to verify the integrity of an FSC application by comparing the application in the FSC system to the project database in FSC Navigator. (For details refer to Section 11 of the FSC Software Manual.)



On-Line drop-down menu (continued)

- **On-Line Rebuild...**
Use this option to rebuild the FSC project databases. (For details refer to Section 4 of the FSC Software Manual.)
 - **Communication Port...**
Use this option to set the communication port that FSC Navigator uses to communicate with the FSC system. (For details refer to subsection 3.7 on page 24.)
-

Help drop-down menu

Use this menu option to access the help information of FSC Navigator. The **Help** drop-down menu has the following menu items:

- **Contents**
Use this option to call up the Windows help facility that comes with FSC Navigator.
 - **About...**
Use this option to display a window which contains information about the application (including version numbers). (For details refer to subsection 3.9 on page 27.)
-

3.4 Button bar

FSC Navigator's main screen contains a button bar with a number of speed buttons that provide quick access to certain program functions (see Figure 3-5). If any program options are unavailable, the corresponding speed buttons will be disabled.



Figure 3-5 Button bar

If there is a down arrow (▼) to the right of the button icon, clicking the speed button will open a submenu (see Figure 3-6).

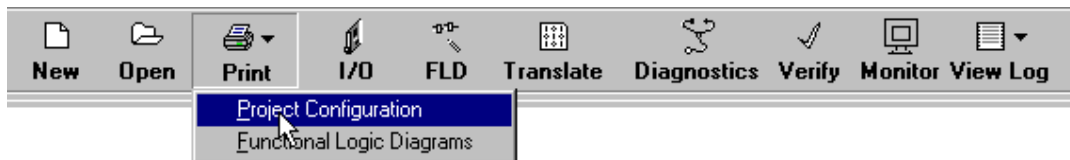


Figure 3-6 Button bar with submenu

The speed buttons basically act as "shortcuts" to menu items on the menu bar. Table 3-2 shows the links between the speed buttons and the menu bar. For details on the menu bar refer to subsection 3.3 on page 5.

Table 3-2 Links between speed buttons and menu items

Speed button...	Links to menu item...
New	File / New Project
Open	File / Open Project
Print	File / Print
I/O	Project / System Configuration
FLD	Project / Design FLDs
Translate	Project / Translate Application
Diagnostics	On-Line / Extended Diagnostics
Verify	On-Line / Verify Application
Monitor	On-Line / Monitor System
View Log	File / View Log



3.5 Navigation area

The navigation area in FSC Navigator's main screen contains a chronological representation of the FSC design and maintenance cycle (see Figure 3-7).

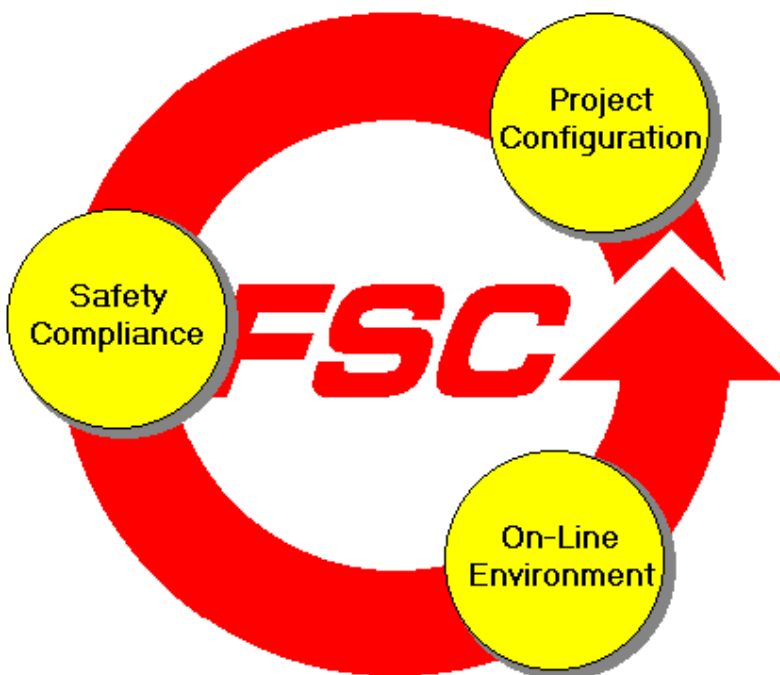


Figure 3-7 FSC design and maintenance cycle

This visualization provides a good guideline when designing and maintaining FSC systems. The route has been subdivided into three main components (represented by three yellow circles):

- **Project Configuration,**
- **Safety Compliance, and**
- **On-Line Environment.**

Each component represents a stage in the design and maintenance cycle, with a number of distinct tasks.

Clicking any of the three circles in the navigation area will take you to the corresponding screen (see Figure 3-8, Figure 3-9 and Figure 3-10).

3.5.1 FSC Project Configuration screen

Project configuration The FSC Project Configuration screen is the starting point for each new FSC project. It consists of a number of stages:

- creation of the FSC project,
- configuration of the FSC project,
- definition of the FSC project logic,
- translation of the FSC project logic into application code, and
- transfer of the FSC application to EPROMs and/or to memory.

If you click the 'Project Configuration' circle in the navigation area, the FSC Project Configuration screen will appear (see Figure 3-8).

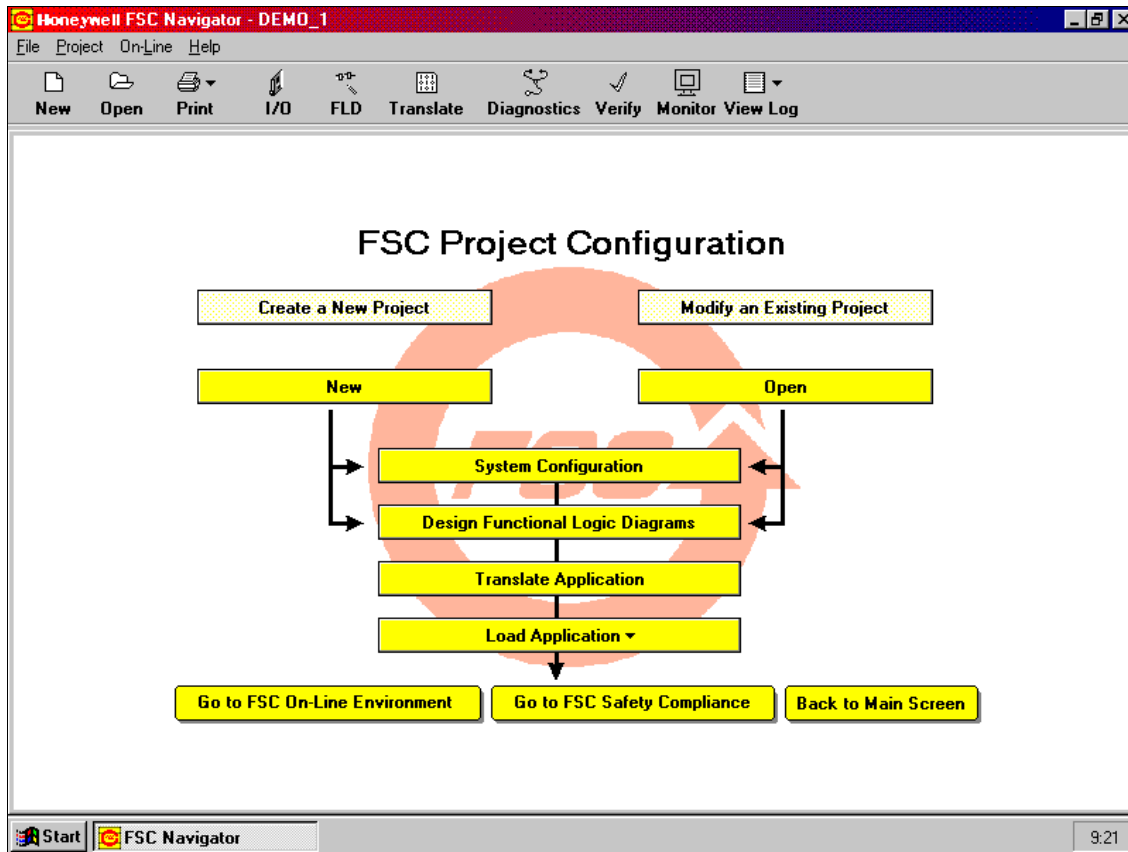


Figure 3-8 FSC Project Configuration screen



This screen presents a chronological sequence of tasks that must be accomplished to successfully create and implement an FSC project. Clicking any of the yellow buttons will take you to the corresponding FSC Navigator option. If there is a down arrow (▼) to the right of the button text, clicking the button will open a submenu (see Figure 3-9).

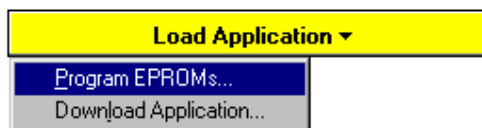


Figure 3-9 Button in navigation area with submenu

If any program options are unavailable, the corresponding buttons or menu options in the navigation area will be disabled. The buttons at the bottom of the screen allow you to go back to the main screen or move forward to the FSC Safety Compliance screen or FSC On-Line Environment screen.

Options

The FSC Project Configuration screen contains the following buttons:

- **New**
Use this option to create a new FSC project. (For details refer to subsection 3.10 on page 28.)
- **Open**
Use this option to open an existing FSC project. (For details refer to subsection 3.10 on page 28.)
- **System Configuration**
Use this option to configure and maintain the system(s) in the current FSC project. (For details refer to Section 4 of the FSC Software Manual.)
- **Design Functional Logic Diagrams**
Use this option to design the functional logic diagrams (FLDs) which are used to create the application logic. (For details refer to Section 6 of the FSC Software Manual.)
- **Translate Application**
Use this option to convert the application function specification (designed using FLDs) into program code that the FSC system can use. (For details refer to Section 8 of the FSC Software Manual.)



- **Load Application / Program EPROMs**

Use this option to load an application program from your hard disk into the FSC system. (For details refer to Section 12 of the FSC Software Manual.)

- **Load Application / Download Application**

Use this option to load an application program from your hard disk into the FSC system. (For details refer to Section 12 of the FSC Software Manual.)

The buttons and submenu options in the FSC Project Configuration screen basically act as "shortcuts" to menu items on the menu bar. Table 3-3 shows the links between the buttons and the menu bar. For details on the menu bar refer to subsection 3.2 on page 4.

Table 3-3 Links between buttons and menu items
(FSC Project Configuration screen)

Button...	Links to menu item...
New	File / New Project
Open	File / Open Project
System Configuration	Project / System Configuration
Design Functional Logic Diagrams	Project / Design FLDs
Translate Application	Project / Translate Application
Load Application / Program EPROMs	Project / Program EPROMs
Load Application / Download Application	Project / Download Application



3.5.2 FSC Safety Compliance screen

Safety compliance

The FSC Safety Compliance screen provides a number of links that allow you to:

- view the log files that are available in FSC Navigator,
- print the system configuration or FLDs, and
- verify the application that was created in the FSC Project Configuration screen.

If you click the 'Safety Compliance' circle in the navigation area, the FSC Safety Compliance screen will appear (see Figure 3-10).

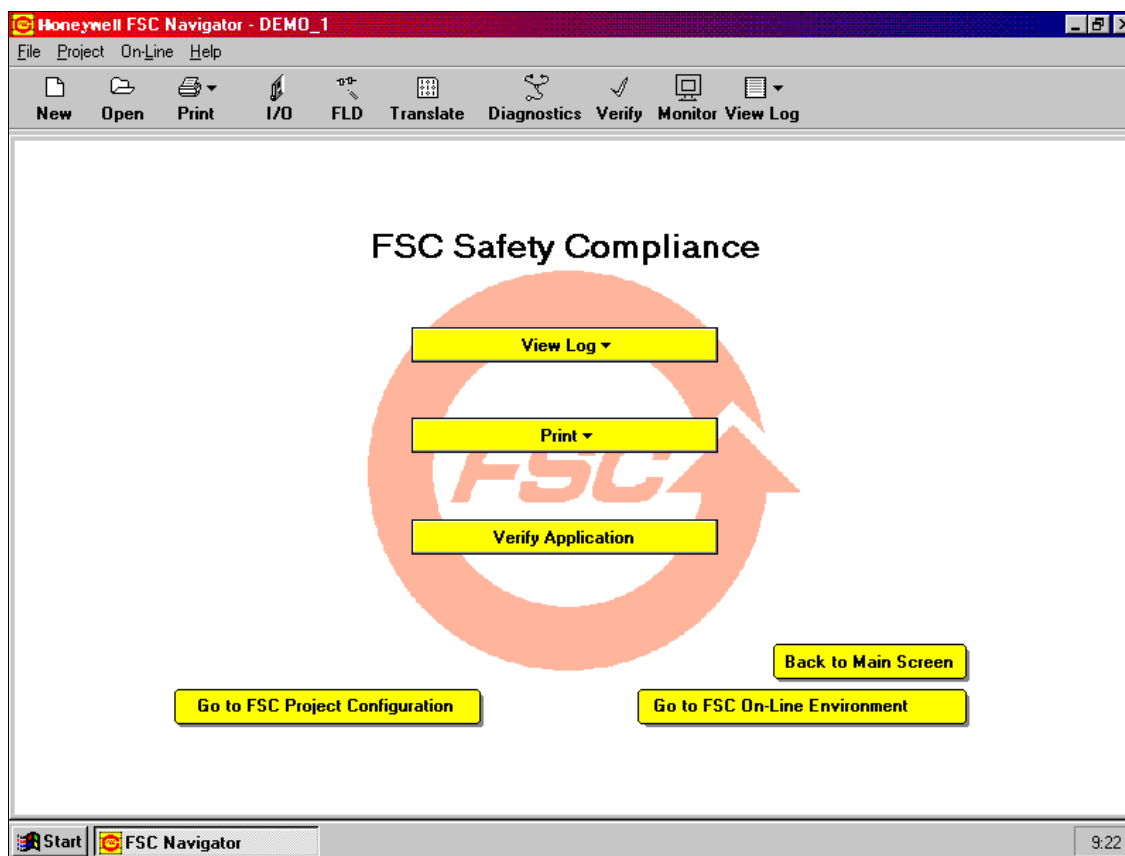


Figure 3-10 FSC Safety Compliance screen

This screen presents a number of individual tasks that allow you to verify the integrity of an application and create output of system information (on screen or to a printer).

Clicking any of the yellow buttons will take you to the corresponding FSC Navigator option. If there is a down arrow (▼) to the right of the button text, clicking the button will open a submenu (see Figure 3-11).

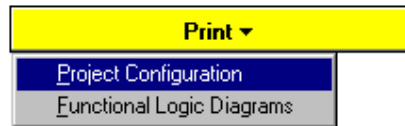


Figure 3-11 Button in navigation area with submenu

If any program options are unavailable, the corresponding buttons or menu options in the navigation area will be disabled.

The buttons at the bottom of the screen allow you to go back to the main screen or the FSC Project Configuration screen, or move forward to the FSC On-Line Environment screen.

Options

The FSC Safety Compliance screen contains the following buttons and options:

- **View Log**

Use this option to view the various log files that are available in FSC Navigator. This menu option has a submenu, which allows you to choose the specific log file you wish to view. (For details refer to Section 9 of the FSC Software Manual.)

- **Print**

Use this option to create hardcopy of the system configuration or functional logic diagrams of the current FSC project. This menu option has a submenu, which allows you to choose what you wish to print. (For details refer to Section 7 of the FSC Software Manual.)

- **Verify Application**

Use this option to verify the integrity of an FSC application by comparing the application program in the FSC system to the application database in FSC Navigator.

This is a very important option. It verifies if the FSC application software as present in the FSC system is consistent with the FSC database and the functional logic diagrams as maintained by FSC Navigator on the user station. (For details refer to Section 11 of the FSC Software Manual.)



The buttons and submenu options in the FSC Safety Compliance screen basically act as "shortcuts" to menu items on the menu bar. Table 3-4 shows the links between the buttons and the menu bar. For details on the menu bar refer to subsection 3.2 on page 4.

Table 3-4 Links between buttons and menu items
(FSC Safety Compliance screen)

Button...	Links to menu item...
View Log (with submenu)	File / View Log (with submenu)
Print (with submenu)	File / Print (with submenu)
Verify Application	On-Line / Verify Application

3.5.3 FSC On-Line Environment screen

On-line environment The FSC On-Line Environment screen provides a number of links that allow you to:

- monitor the FSC system,
- view the status of I/O signals, system variables and FLDs,
- access and process extensive diagnostic information.

If you click the 'On-Line Environment' circle in the navigation area, the FSC On-Line Environment screen will appear (see Figure 3-12).

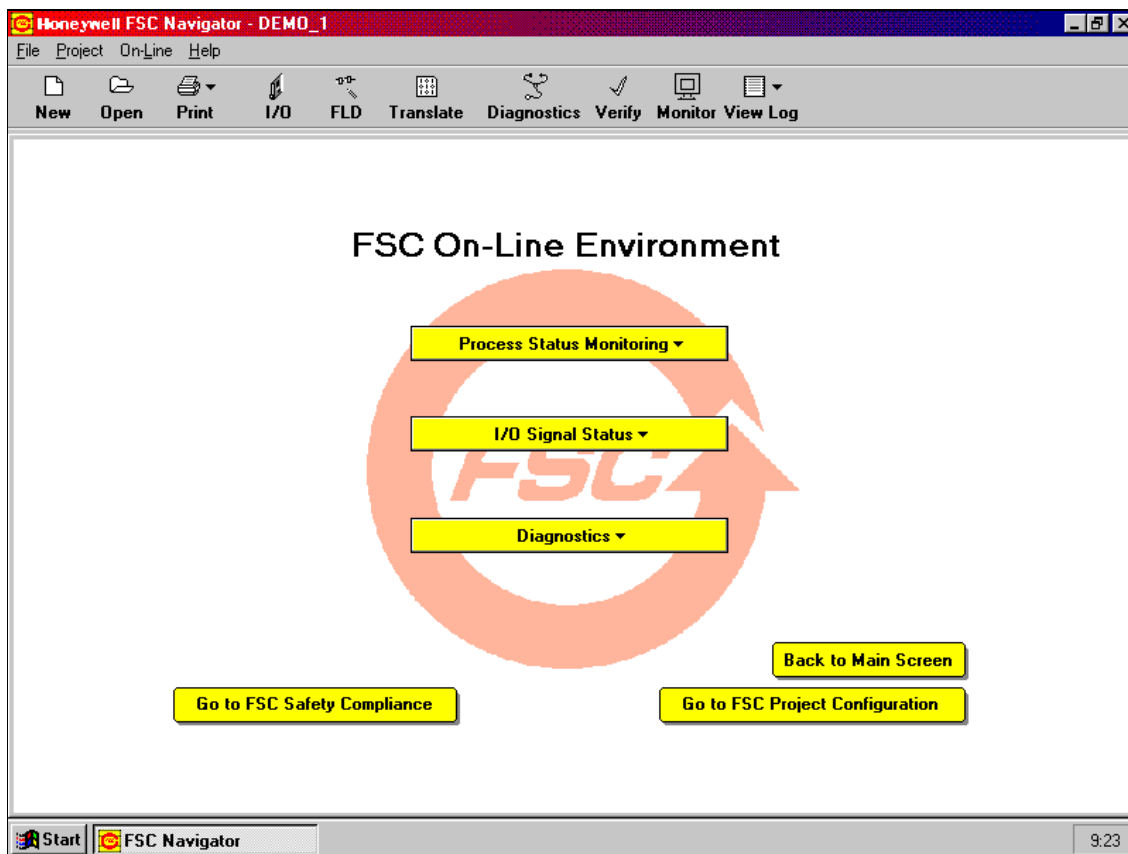


Figure 3-12 FSC On-Line Environment screen



This screen presents a number of individual tasks that allow you to monitor the FSC system and access diagnostic information. Clicking any of the yellow buttons will take you to the corresponding FSC Navigator option. If there is a down arrow (▼) to the right of the button text, clicking the button will open a submenu (see Figure 3-13).

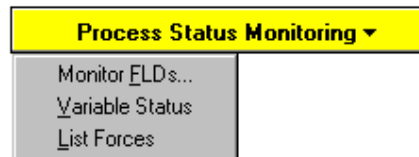


Figure 3-13 Button in navigation area with submenu

If any program options are unavailable, the corresponding buttons or menu options in the navigation area will be disabled. The buttons at the bottom of the screen allow you to go back to the main screen, the FSC Safety Compliance screen or the FSC Project Configuration screen.

Options

The FSC On-Line Environment screen contains the following buttons:

- **Process Status Monitoring / Monitor FLDs**
Use this option to view the actual status of the variables in the FLDs of the current FSC project. (For details refer to Section 12 of the FSC Software Manual.)
- **Process Status Monitoring / Variable Status**
Use this option to view the actual status of the variables in the FLDs of the current FSC project. (For details refer to Section 12 of the FSC Software Manual.)
- **Process Status Monitoring / List Forces**
Use this option to view a list of all forced variables in the FSC system. (For details refer to Section 12 of the FSC Software Manual.)
- **I/O Signal Status**
Use this option to view the status of the input/output loops. This menu option has a submenu, which allows you to choose which loops you wish to view. (For details refer to Section 12 of the FSC Software Manual.)



- **Diagnostics / Extended Diagnostics**

Use this option to view and store all diagnostics of the Central Part(s) in the FSC system. (For details refer to Section 12 of the FSC Software Manual.)

- **Diagnostics / Diagnostics to Disk**

Use this option to back up the extended diagnostic database to hard disk or retrieve a previously stored database. (For details refer to Section 12 of the FSC Software Manual.)

- **Diagnostics / I/O Loop Diagnostics**

Use this option to view all diagnostics of the I/O loops in the FSC system. (For details refer to Section 12 of the FSC Software Manual.)

- **Diagnostics / System Information**

Use this option to view important system parameters of the FSC system. (For details refer to Section 12 of the FSC Software Manual.)

Some of the buttons and submenu options in the FSC On-Line Environment screen act as "shortcuts" to menu items on the menu bar. Table 3-5 shows the links between the buttons and the menu bar. For details on the menu bar refer to subsection 3.2 on page 4.

Table 3-5 Links between buttons and menu items
(FSC On-Line Environment screen)

Button...	Links to menu item...
Process Status Monitoring / Monitor FLDs	On-Line / Monitor FLDs
Diagnostics / Extended Diagnostics	On-Line / Extended Diagnostics
Diagnostics / Diagnostics to Disk	On-Line / Diagnostics to Disk



3.6 Password protection

Password protection

Some options of FSC Navigator are password-protected to prevent unauthorized access to these options. Enabling and disabling the use of passwords is accomplished through the System Configuration option, which can be called up in three ways:

- Clicking the System Configuration button in the FSC Project Configuration screen,
- Clicking the I/O speed button on the button bar, and
- Choosing the System Configuration option from the Project menu.

To set the passwords, choose Install / Password. If you choose the Password option, you will first be asked to enter the supervisor password. This is a special password that prevents unauthorized users from changing and/or deleting passwords. In the application's initial state, the supervisor password is 'SUPER' (in capital letters) and all other passwords are disabled. If you have entered the correct supervisor password, the following screen will appear:

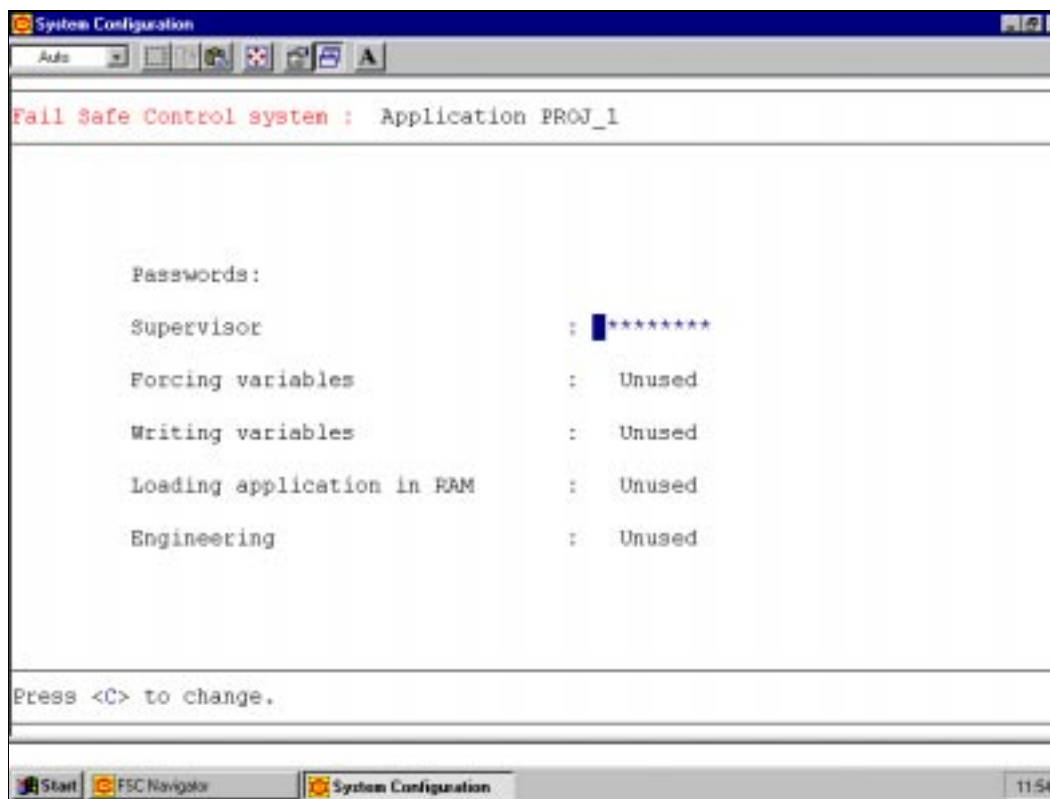


Figure 3-14 Password definition screen



If a password is enabled, it is identified by eight asterisks ('*'), which are displayed to the right of the password. A disabled password is identified by the text 'Unused'. All passwords can be disabled, except for the supervisor password.

Password maintenance

Setting a password

To enable a password, move the cursor to the line of the password you wish to set, and press 'C'. Enter a password of four to eight alphanumeric characters. You will be asked to re-enter the password to confirm. The text to the right of the selected password will change to eight asterisks ('*').

Please note that the passwords are case-sensitive (i.e. 'PASSWORD' is a different password than 'password' or 'PassWord').

Changing a password

If a password is already enabled, there will be eight asterisks ('*') to its right. To modify a password, move the cursor to the appropriate line, and press 'C'. If you want to change the supervisor password, you will first have to enter the old supervisor password. Enter a new password of 4 to 8 alphanumeric characters. You will be asked to re-enter the password to confirm.

Disabling a password

Only enabled passwords can be disabled (except the supervisor password). To disable a password, move the cursor to the line of the password you wish to disable, and press 'D'. You will be asked to confirm the action. The eight asterisks ('*') to the right of the selected password will change to 'Unused'.

Using passwords

Passwords are used to prevent unauthorized users from accessing certain options in FSC Navigator. If an option is password-protected and the password has been enabled, you will be asked to enter the corresponding password. You have three attempts to enter the correct password before being denied access to the password-protected program option.

If you enter a password, any valid (i.e. alphanumeric) key stroke is echoed on screen as an asterisk (*). Typing errors can be corrected using the BACKSPACE key.

Whenever a password is required, the supervisor password will be accepted as a correct password.



Passwords

Forcing variables

The 'Forcing variables' password secures the access to the 'Force' options in the On-Line Environment option of FSC Navigator. If the password has been disabled, no password is required.

Writing variables

The 'Writing variables' password secures the access to the 'Write' options in the On-Line Environment option of FSC Navigator. If the password has been disabled, no password is required.

Loading application in RAM

The 'Loading application in RAM' password secures the 'Load' option in the On-Line Environment option of FSC Navigator. If the password has been disabled, no password is required.

Engineering

The 'Engineering' password secures the access to the following FSC Navigator options:

Menu bar:

- Project / System Configuration
- Project / Design FLDs
- Project / SER Format
- Project / Translate Application
- On-Line / On-Line Rebuild

Speed buttons:

- I/O
- FLD
- Translate

Navigation area: In FSC Project Configuration screen:

- System Configuration button
- Design Functional Logic Diagrams button
- Translate Application

If the password has been disabled, no password is required.

Passwords in converted projects

Existing projects that were created with FSC software prior to R500 are automatically converted to R500 when they are opened in FSC Navigator. If these projects include password protection, all passwords in the converted project are set to their defaults:

Supervisor:	SUPER (in capital letters)
Forcing variables:	empty (i.e. not used)
Writing variables:	empty (i.e. not used)
Loading application in RAM:	empty (i.e. not used)
Engineering:	empty (i.e. not used)

3.7 Communication port

FSC Navigator uses a serial communication link to interface with the FSC system. The program can only communicate with the FSC system if the communication port has been set correctly.

Setting the communication port

To set the communication link, do the following:

1. From the **On-Line** menu, choose **Communication Port**.
2. The dialog window as shown in Figure 3-15 will appear.

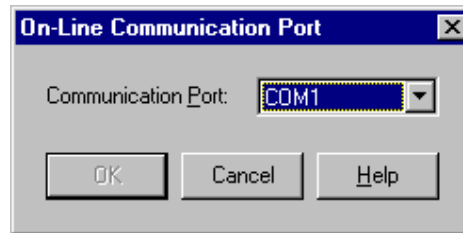


Figure 3-15 Setting the communication port

3. Select the appropriate communication port from the drop-down list (see below).
4. Choose the **OK** button to confirm or **Cancel** to abort the operation. Any changes will become effective immediately.

Supported communication ports

The following communication ports are supported:

- 07177 - RS232 (serial port on the HSMS-proprietary PC serial interface module 07177/1/1)
- 07177 - RS485 (serial port on the HSMS-proprietary PC serial interface module 07177/1/1)
- COM1 (standard PC serial port)
- COM2 (standard PC serial port)
- COM3 (serial port on a third-party serial interface module)
- COM4 (serial port on a third-party serial interface module)

The default FSC communication port is COM1.



3.8 Printers

FSC Navigator allows you to output system information and functional logic diagrams (FLDs) to a printer or plotter. The program can only generate output if one or more printers have been defined and set correctly.

Setting a printer

To set a printer, do the following:

1. From the **File** menu, choose **Printers**.
2. The dialog window as shown in Figure 3-16 will appear.

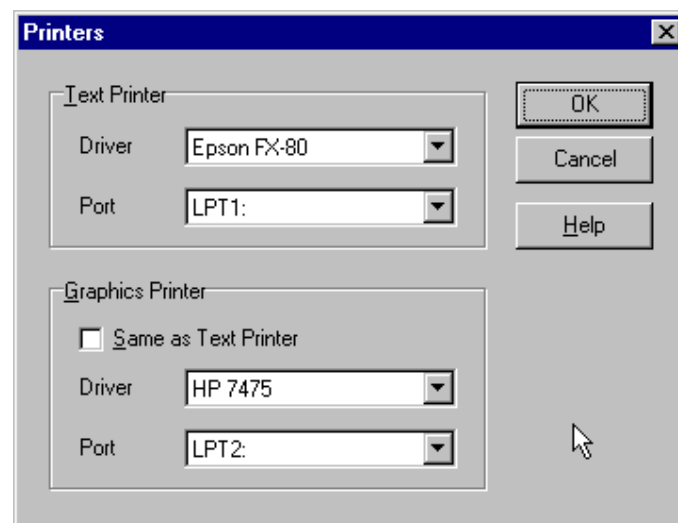


Figure 3-16 Setting a printer

3. Select the appropriate printer driver(s) and port(s) from the drop-down lists (see below).
4. Choose the **OK** button to confirm or **Cancel** to abort the operation. Any changes will become effective immediately.

Printer drivers

Table 3-6 below lists the supported printer drivers.

Table 3-6 Supported printer drivers

	Matrix printers	Laser printers	Plotters
Text printer	Epson MX-80 Epson RX-80 Epson FX-80 Epson FX-85 Epson FX-100 Epson FX-105 Epson LX-800 Epson FX-1000 NEC P6 NEC P7 Epson FX-800	HP LaserJet series II HP LaserJet series III Canon LBP-8II Brother HL-8e	—
Graphics printer	—	HP LaserJet series II HP LaserJet series III Canon LBP-8II Brother HL-8e	HP 7475 HP 7550

The default text printer is Epson FX-80.

The default graphics printer is HP 7475 (plotter).

If your printer is not in the list above, choose a printer that it can emulate. Most matrix printers can emulate an Epson printer, and most laser printers a Hewlett-Packard LaserJet.

Printer ports

The drop-down list shows the printer ports of all configured Windows 95 printers (as defined in **Control Panel**).

The default printer port is the printer port of the default Windows 95 printer.

To (re)configure the printer port of a Windows 95 printer, go to **Control Panel** and choose the **Printers** applet.

Text and graphics printers

The **Same as Text Printer** check box allows you to set the same output device for both text and graphics.

As shown in Table 3-6, a matrix printer cannot be used as a graphics printer. If you select a matrix printer as the text printer, you cannot use it as the graphics printer as well. The **Same as Text Printer** check box will then be disabled and not available.



3.9 User and version information

User information

FSC Navigator is licensed to a particular "user" (which may also be a company). To view the license information, choose the **About** option from the **Help** menu. This will open the window as shown in Figure 3-17. This window shows the user that FSC Navigator is licensed to, as well as the application's serial number.

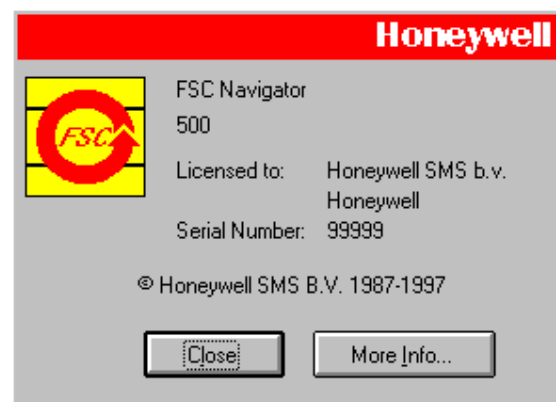


Figure 3-17 User and version information

Version information

The **About** box has an extra button called **More Info**. If you choose this button, a new window will open (see Figure 3-18), which provides detailed information about the versions of the various application components of FSC Navigator.

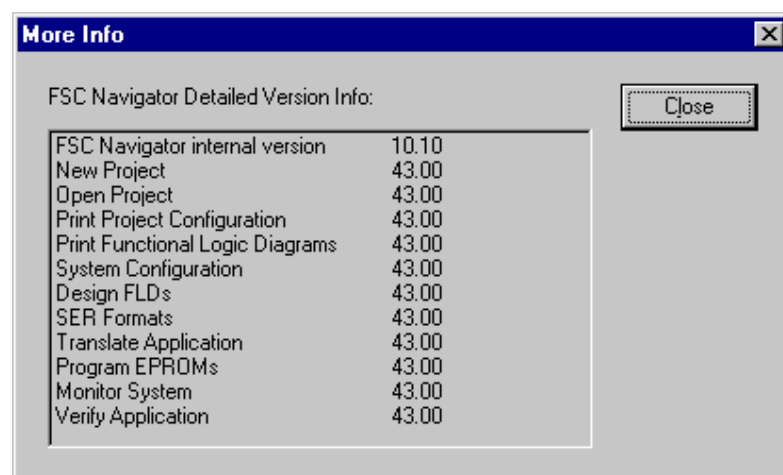


Figure 3-18 Version information

3.10 Working with projects

Creating a new project

To create a new project, do any of the following:

- From the **File** menu on FSC Navigator's menu bar, choose **New Project**.
- On the speed button bar, choose the **New** button.
- In the FSC Project Configuration screen, choose the **New** button.

This will open the following window:

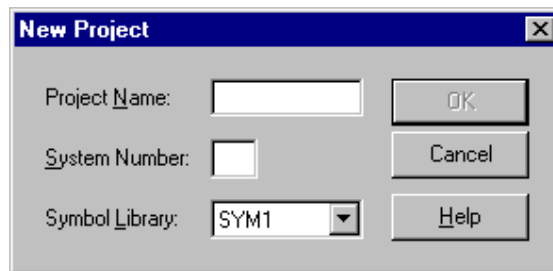


Figure 3-19 Creating a new project

The **OK** button will initially be disabled and cannot be selected. It will become available as soon as a valid project name and system number have been entered. A valid project name has a maximum length of five or six alphanumeric characters (e.g. PROJ1), depending on the system number. The system number must be between 1 and 63. If you use a double-digit system number (≥ 10), the length of the project name is limited to a maximum of five characters.

If you enter the name of an existing project, the **OK** button will remain disabled. To open an existing project, use the Open option (see below).

The **Project Name** box accepts alphanumeric input (i.e. letters and numbers) and a number of other characters (e.g. &, +, #).

The **System Number** box only accepts numeric input (between 1 and 63).

When creating a new project, you must also select a symbol library from the **Symbol Library** drop-down list. For details on symbol libraries refer to subsection 3.11 on page 31.

After you have created a new project, its name will be shown in the main screen's title bar (e.g. PROJ1_1).



Opening an existing project

To open an existing project, do any of the following:

- From the **File** menu on FSC Navigator's menu bar, choose **Open Project**.
- On the speed button bar, choose the **Open** button.
- In the FSC Project Configuration screen, choose the **Open** button.

This will open the following (resizeable) window:

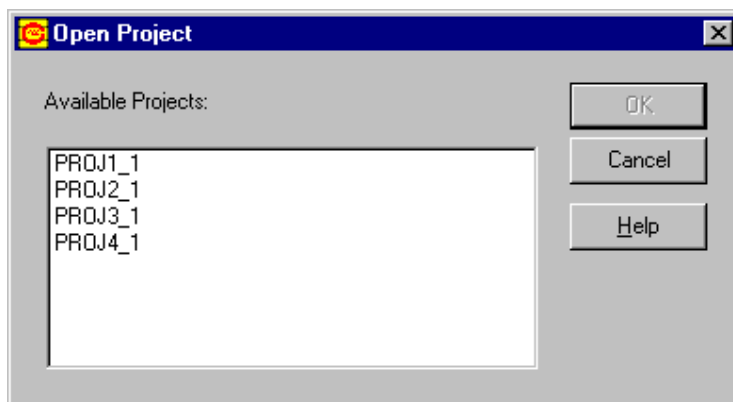


Figure 3-20 Opening an existing project

To open a project, select a project from the list of available projects and choose the **OK** button.

The **OK** button will initially be disabled and cannot be selected. It will become available as soon as an existing project has been selected from the list. (If the selected project is the currently open project, the **OK** button will remain disabled.)

After you have opened an existing project, its name will be shown in the main screen's title bar (e.g. PROJ1_1). When you quit FSC Navigator with a project opened, the program remembers which project was open. It will then automatically load this project the next time you start FSC Navigator, unless the project's integrity has been compromised, e.g. if one of the project files (see 'Project files' below) has become corrupt. In that case no project will be loaded.



Project conversion

If you open an existing project that was created with an earlier version of the FSC user software, you will be asked whether the project should be converted. If you choose **Yes**, the project will be converted and opened. If you choose **No**, the project will not be opened. If the project cannot be converted, a message will be displayed on screen.

After the conversion, a log file is created which can be viewed using the **View Log** option of the **File** menu. (For details on log files refer to Section 9 of the FSC Software Manual.)

Note:

A subfolder 'BACKUP.TMP' is used to store a temporary backup of the application being converted. If this folder already exists, conversion of the application will be aborted. In that case you should remove this folder and convert the application again.



3.11 Project files

Project files

For every project, FSC Navigator creates a number of files and stores them in a subfolder of the main application folder (C:\FSC by default). The subfolder is created automatically upon creation of the project, and will have the same name as the project name (e.g. PROJ1). The project files are identified by their project name and system number, which are separated by an underscore sign (e.g. PROJ1_1). The various file types are identified by their file extension.

The project files are essential for the project, and if any of them are corrupted or deleted, the project integrity is compromised. This may mean that the project can no longer be opened and will not be shown in the list of available projects. In some cases, the project can be opened, but the error will become apparent as soon as a particular program option is called which needs the corrupted or deleted file.

The project subfolder also contains an initialization (INI) file, which stores which symbol library is used for the project (SYM1 by default). If this file is corrupted or deleted and the project is opened, SYM1 will be assumed as the symbol library.

3.12 Symbol libraries

New projects

When creating a new project, you must also select the symbol library that will be used with the new project. A symbol library contains the symbols that can be used to create the functional logic diagrams (FLDs). Table 3-7 below lists the available symbol libraries and their characteristics. The default symbol library is SYM1.

Table 3-7 Available symbol libraries

Library name	Contents	Orientation
SYM1	DIN standard symbols	Landscape
SYM2	DIN standard symbols	Portrait
SYM3	DIN standard symbols	Portrait
SYM4	DIN standard symbols	Portrait

Existing projects

It is possible to change the symbol library of an existing project. To change the symbol library, do the following:

1. Make sure that the current project is the project whose symbol library you wish to change.
2. From the **Project** menu, choose **Properties**.
3. The dialog window as shown in Figure 3-21 will appear.

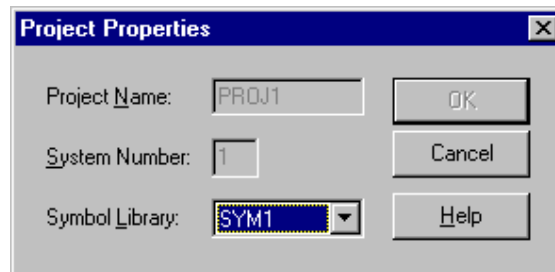


Figure 3-21 Project properties

The **Project Name** box and **System Number** box are disabled and cannot be changed.

To change the symbol library, select the new symbol library from the drop-down list and choose the **OK** button. The **OK** button will initially be disabled and cannot be selected. It will become available as



soon as a new symbol library has been selected from the list. Choose the **OK** button to confirm the change or **Cancel** to abort the operation.

Exchangeability of symbol libraries

There are some limitations as to the "exchangeability" of symbol libraries. Table 3-8 below shows which symbol libraries can be exchanged in which situations.

Table 3-8 Exchangeability of symbol libraries

Current symbol library	Any FLDs in project?	Symbol library can be changed to...
SYM1	No	SYM2, SYM3, SYM4
	Yes	SYM2
SYM2	No	SYM1, SYM3, SYM4
	Yes	no changes possible*
SYM3	No	SYM1, SYM2, SYM4
	Yes	SYM4
SYM4	No	SYM1, SYM2, SYM3
	Yes	no changes possible*

* The **Properties** option of the **Project** menu will be disabled altogether.

Note:

Within an FSC network, the application of the FSC systems that are interconnected via an FSC-FSC communication link must be designed using the same symbol library.



3.13 Basic operations in program components

Mouse support

Most of the program components of FSC Navigator are character-oriented. Besides the main program, the only graphical components are the program options that involve functional logic diagrams (FLDs), e.g. the Design FLDs tool.

Mouse support is only available in the graphical components of FSC Navigator. In character-oriented screens, the mouse cursor will be visible but cannot be used in the screens.

If mouse support is available, clicking the left mouse button is then the same as pressing the ENTER key on the keyboard. It can, for example, be used to accept the input or to acknowledge an on-screen message. Clicking the right mouse button is the same as pressing the ESC key on the keyboard. It can, for example, be used to abort an input action and return to the menu.

Moving through menus

You can use keys on the keyboard to move through the menus of the program components. If mouse support is available (in graphical screens), you can also use the mouse.

Mouse

If the graphical screen shows a menu, you can move the mouse to select the desired menu item. A selected menu item is shown highlighted. If you wish to select the menu item, press the left mouse button. Either a submenu will be shown or, if there are no more submenus, the program will execute the selected menu item. The right mouse button brings you back to the next higher menu level.

Keyboard

Use the left and right cursor keys to move left and right in the menu. A selected menu item will be shown highlighted. If you wish to select the menu item, press ENTER. Either a submenu will be shown or, if there are no more submenus, the program will execute the selected menu item.

Every menu item has a shortcut key which is highlighted. Pressing the shortcut key is an easy and fast way to select a menu item. The rightmost menu item 'Main' can be used to return to the next higher menu level. If you are on the highest menu level, the rightmost menu item will be 'eXit'. Selecting this menu item will quit the program component and return to FSC Navigator's main screen.



The following keys can be used to move through the menus in the FSC Navigator program components: PGUP, PGDN, TAB, ESC, ←, →, ↑, ↓ and ENTER.

Intelligent menus

Sometimes a different menu may be shown than you might expect, with menu items added or deleted from the menu. The reason for this is that "intelligent menu building" has been used throughout the application software. If, for example, the FSC system configuration is not known yet, the hardware specification of variables cannot be edited. The 'Hardware specs' menu option will therefore only be shown if the FSC system configuration has been defined.

Keyboard input

When entering strings (text or values), you can use all standard ASCII characters with ASCII values between 32 and 127 (letters, digits and common symbols) as well as non-graphical extended ASCII characters (e.g. é, Å and Ö). Table 3-9 below lists the various shortcut keys for keyboard input in the character-oriented FSC Navigator program components.

Table 3-9 Shortcut keys for data input

Shortcut key	Action
<Ctrl> + <A>	Moves the cursor to the first character of the string.
<Home>	Moves the cursor to the first character of the string.
<Ctrl> + <F>	Moves the cursor to the last character of the string.
<End>	Moves the cursor to the last character of the string.
<Ctrl> + <V>	Toggles between insert and overwrite mode.
<Insert>	Toggles between insert and overwrite mode.
<Ctrl> + <G>	Deletes the character under the cursor.
<Delete>	Deletes the character under the cursor.
<Ctrl> + <H>	Deletes the character left of the cursor.
<Backspace>	Deletes the character left of the cursor.
<Ctrl> + <T>	Deletes all characters after the cursor position.
<Ctrl> + <Y>	Deletes the complete string.
<←/→>	Moves the cursor left or right through the entered string.



Function keys

In the character-oriented program components, the following function keys are available for on-line help:

- <F1>: A window will appear which provides specific information about the item that the cursor is on.
 - <F2>: A window will appear which provides specific information about variable allocations (only available in the 'System Configuration' program component).
-



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Helping You Control Your World



Fail Safe Control

Section 4: System Configuration

Version 500

Revision 01 (01/98)



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4.1 Introduction

The 'System Configuration' option of FSC Navigator is used to view and maintain the databases which are involved in the configuration of the FSC system, the description of the variables and the allocation of I/O-related variables.

All information that is entered is checked for consistency. When a new project is created, the following information must be entered in the following order:

- Application description,
- I/O variables specification,
- Function specification (using 'Design FLDs' option of FSC Navigator),
- Type of configuration of the FSC system and related information,
- Positions of racks in the cabinet(s),
- Positions of hardware modules in the racks,
- Allocation of I/O variables to channels of I/O modules or COM modules.

The 'System Configuration' option of FSC Navigator will only be available if a project is open. If no project is open, you must either create a new project or open an existing one (for details refer to Section 3 of the FSC Software Manual).

The 'System Configuration' option can be called in three ways:

- **Menu bar:** Project / System Configuration
- **Button bar:** I/O
- **Navigation area:** FSC Project Configuration / System Configuration

Note:

If the Engineering password is active, you must enter the correct password to start up the 'System Configuration' option of FSC Navigator. (For details on passwords refer to Section 3 of the FSC Software Manual or the FSC Navigator Help file.)

4.2 Getting started

4.2.1 "System Configuration" main menu

The 'System Configuration' option of FSC Navigator has the following main menu:

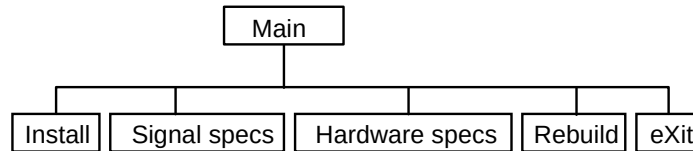


Figure 4-1 Structure of 'System Configuration' main menu

The main menu consists of the following options:

Install	: Entry and modification of the FSC application specification.
Signal specs	: Entry and modification of input and output variables.
Hardware specs	: Hardware specification of variables
Rebuild	: Rebuild databases index files and address allocation.
eXit	: Return to the FSC main menu.

Note:

The option 'Hardware specs' will only appear after the configuration has already been defined.

4.2.1.1 'Install' menu

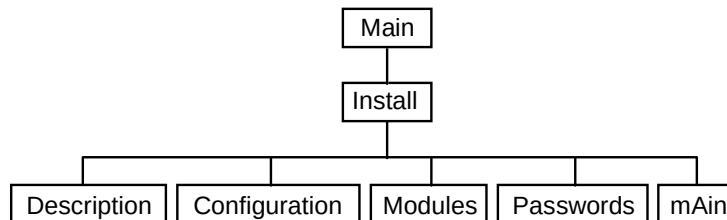


Figure 4-2 'Install' menu

The Install menu consists of the following options:

Description	: Enter application information as system description.
Configuration	: Define the configuration, e.g. system type, rack numbers.
Modules	: Entry and modification of the placement of the modules (only available if the configuration has been defined).
Passwords	: Define passwords for protected options in the FSC Navigator software.
Main	: Return to the previous menu.



4.2.1.1.1 'Modules' menu

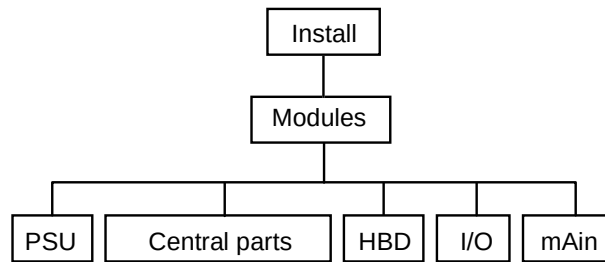


Figure 4-3 'Modules' menu

You can choose to install the following module types:

PSU	: Power supply unit.
Central Parts	: Submenu for Central Part definition and Central Part modules.
HBD	: Horizontal bus driver.
I/O	: I/O modules (only if one or more HBDs have been placed).
mAin	: Return to the previous menu.

4.2.1.1.2 'Central Parts' menu

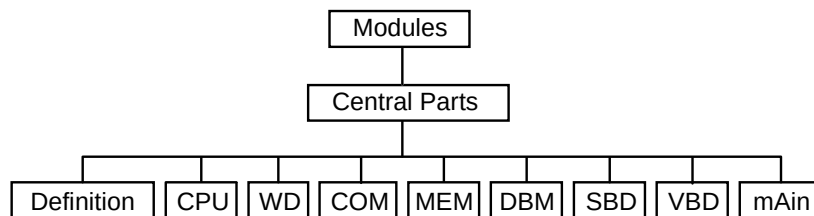


Figure 4-4 'Central Parts' menu

You can choose between entering/editing the following module types:

Definition	: Definition of Central Part modules
CPU	: Central processing unit (10002/1/.)
WD	: Watchdog module (10005/1/1)
COM	: Communication module (10004/./.)
MEM	: Memory module (10003/1/1)
DBM	: Diagnostic and battery module (10006/./.)
VBD	: Vertical bus driver (10001/./1)
SBD	: Single bus driver (10007/1/1)

Note:

The options MEM, DBM, VBD and SBD will only be available if the corresponding modules have been defined.

4.2.1.2 Signal specification

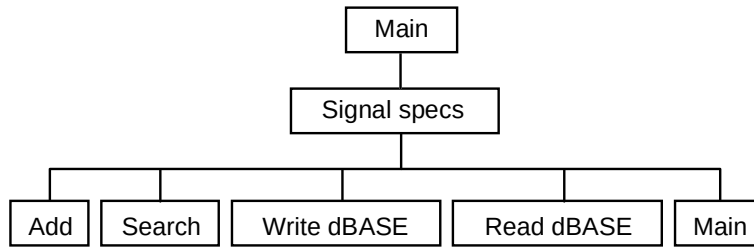


Figure 4-5 'I/O signal specs' menu

The Signal Specification option is used for entry and modification of the signal related part of the variables. The signal specification menu contains the following options:

Add	: Add variables to the variable database.
Search	: Search for a variable on tag number or a part of the tag number.
Main	: Return to the System Configuration main menu.
Write dBASE	: Write the I/O variables of the variable database to a dBASE 3 or 4 compatible file
Read dBASE	: Read I/O variables from a dBASE 3 or 4 compatible file.

Note:

The 'Write dBASE' option will only be available if at least one additional variable has been created.

4.2.1.3 Hardware specification

The Hardware Specification option is used for entry and modification of the hardware-related part of the variables.

4.2.1.4 Rebuild

The 'Rebuild' option is used to create new index files in case one or more index files are corrupted (e.g. due to a powerdown failure of your PC). It is also used to change the address allocation in case you want to optimize the memory usage.

For details on the Rebuild option refer to subsection 4.2.1.4.

4.2.1.5 Exit

If you choose eXit, you will return to the FSC Navigator main screen.



4.3 Install

4.3.1 Description

If you create a new application, the following default application information is stored. These texts are retrieved from the message file (FSC_INIT.MSG) and can be customized by the user.

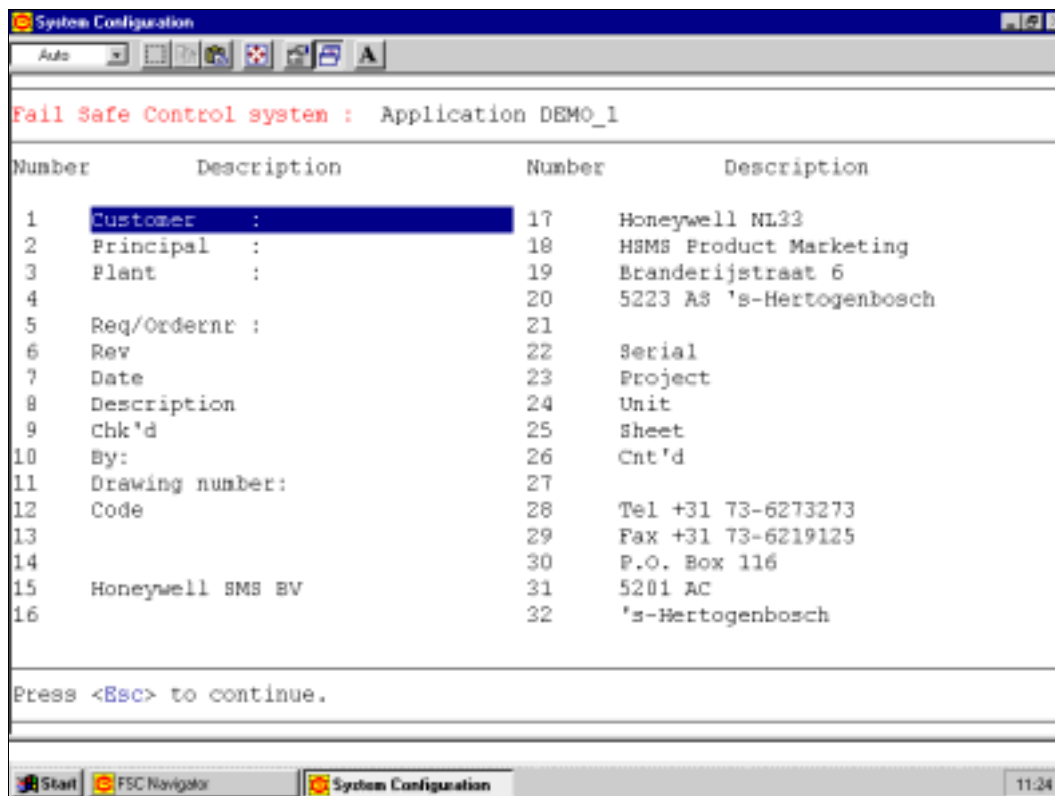


Figure 4-6 Install: Description

After changing the default descriptions, the program asks if changes must be saved to the FSC_INIT.MSG file:

Default descriptions changed. Save new defaults? (Y/N)

Confirmation is requested. Every new application you create will use these new default descriptions.

4.3.2 Configuration

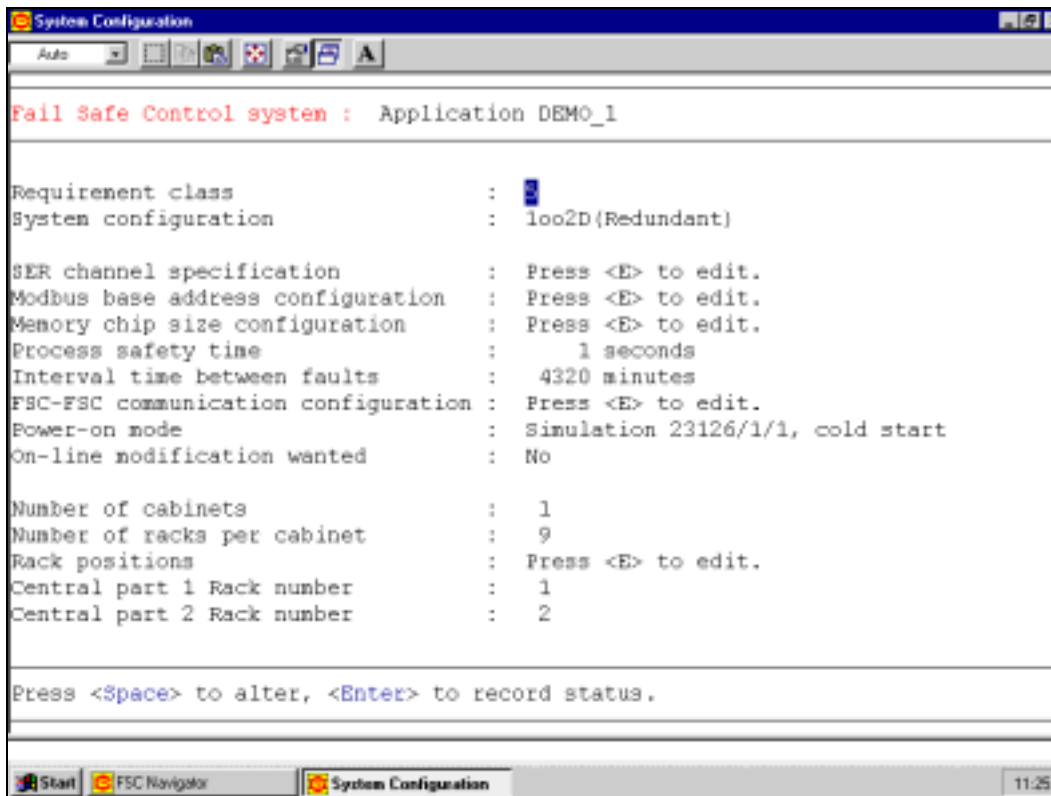


Figure 4-7 Install: Configuration menu

This option is used to define the configuration of the FSC system.

Requirement class

In accordance with DIN V 19250, six requirement classes (safety classes) can be selected (class 1 to 6). Class 1 is the lowest requirement class (low safety level), class 6 is the highest requirement class (high safety level). To select a requirement class the user must toggle between the different classes.

Requirement class 4 is the default class.

The table on the next page shows the relation between FSC configurations and requirement classes and availability degrees, respectively.



Table 4-1 Relation between FSC configurations and requirement classes AK1-6, according to DIN V 19250

AVAILABILITY	SAFETY		
	AK 1-4	AK 1-5	AK 1-6
	NORMAL	1001D S + s	
	INCREASED	1001D S + s	R + s
OPTIMAL	2002D/1002D R + r	R + r	

where:

AK = Anforderungsklasse = requirement class

R = Redundant processor with inter processor communication

S = Single processor

r = redundant I/O

s = single I/O

1001D = 1-out-of-1 voting with diagnostics

1002D = 1-out-of-2 voting with diagnostics

2002D = 2-out-of-2 voting with diagnostics

For more information on voting refer to the FSC Safety Manual

If a new requirement class is selected and the current FSC configuration cannot meet the demands of the new requirement class, the user is notified that the FSC configuration cannot be used with this new requirement class:

'FSC configuration not possible for new requirement class. Continue? (Y/N)'

Confirmation is requested. If you do not want to continue, the new requirement class is ignored. Otherwise the user has the possibility to change the FSC configuration.

System configuration

Selection of an FSC system type depends on the requirement class (see Table 4-1). Toggle to select an FSC system type.

If you change the FSC system type of a configured FSC system, module placement is cleared and variable allocations are deallocated. Confirmation is requested:

'Changing clears module placement and I/O allocations'

'Change FSC system type? (Y/N)'

If the system type has been defined, the following information can be entered:

- SER channel specification
- Modbus base address configuration
- Memory chip size configuration
- Process safety time
- Interval time between faults
- FSC-FSC communication configuration
- Power-on mode
- On-line modification wanted
- Number of cabinets
- Number of racks per cabinet
- Rack positions
- Central Part rack numbers

SER channel specification (option in current system only configurable if COM has been defined)

It is possible to concentrate event collection at a single communication port (see appendix B 'Sequence of event recording'). The SER channel may be located in any system within a network of FSC systems. The selected channel may be configured at:

- the printer protocol, for direct connection to a printer
- a Development System protocol in which case it can be used with the FSCSOE application, which is optional to the FSC software.
- a Modbus channel (numerical SER only).

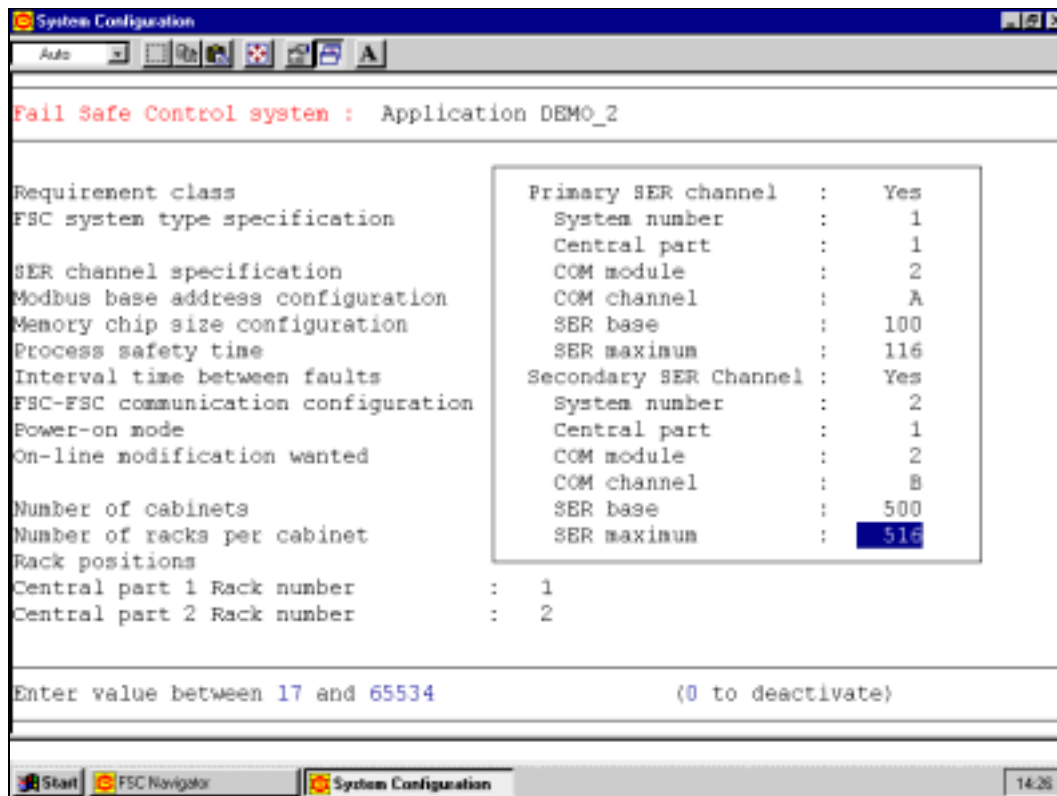


Figure 4-8 SER channel specification



SER may be collected at two different communication channels of the FSC system (dual SER). This allows two independent SER PCs (or printers) to be connected, each of them receiving all SER data. Should one PC (or printer) fail, the other will still be able to receive and process the events generated by the system.

The secondary SER channel is optional. It can only be configured if a valid primary SER channel has been defined. There are some restrictions that must be taken into account when configuring a secondary channel:

- If the primary SER channel has been configured in the current system, the primary SER channel must be configured at a 'lower' channel than the secondary SER channel. For example, if the primary SER channel is at Central Part 1 / COM module 2 / channel B, the secondary SER channel cannot be configured at Central Part 1 / COM module 1 / channel B, but can be configured at Central Part 1 / COM module 3 / channel A or Central Part 2 / COM module 2 / channel B.
- When configuring redundant SER (with connected Central Parts), the two SER channels per Central Parts must be allocated to two different COM modules. If you do not, the failure of one SER link will block the other link for a period of more than 4 hours after a complete buffer of events has been generated.
- The secondary channel can only be configured in the current system. Therefore, the system number of the secondary SER channel cannot be modified (always equal to the current system).

If the allocation of the primary SER channel is changed, the secondary SER channel will be deallocated if the above restrictions are violated.

An SER channel is configured by toggling the 'SER channel' field to 'Yes', otherwise the SER channel cannot be allocated. The value of the system number can now be entered. If the system number is not specified (when it is left 0) the SER channel cannot be allocated (1-63 are valid system numbers). If the system is an external system number, a range of 18 channels can be chosen: from (1, 1, A) to (2, 4, B). For redundant systems, however, the internal FSC-FSC link cannot be selected: (1, 1, A), (2, 1, A), (2, 1, A) and (2, 1, B).

If the system number is identical to the system's own number, then all variable communication channels which have an undefined protocol, a Development System protocol or a printer protocol can be selected. The primary SER channel allocation will be removed if:

- The SER channel field is toggled to No.
- The system number is changed.
- The Central Part number is changed (COM module and COM channel become zero).

If the primary SER channel is edited and the restrictions mentioned are violated, the secondary SER channel allocation will be removed.

The secondary SER channel can only be allocated if the primary SER channel has been allocated and the secondary SER channel field is toggled to Yes by the user.

The allocation of the secondary SER channel is based on the same principle as the allocation of the primary SER channel.

Numerical SER ranges

You can enter a range of numeric SER values (from 1 to 65534). In the SER base field you can enter the start of this range of SER values. The end of the range can be entered in the SER maximum field. The first 16 SER values are reserved by the system. The 'System Configuration' option of FSC Navigator does not check for overlapping numerical SER ranges, but the 'Translate Application' option (the compiler) does.

Example:

Suppose the entered range is 100 to 200. Now you have $(200-100-16)+1 = 85$ available numeric SER values. If a range of 1000 to 2000 has been entered in system 10, then a range of 1500 to 2500 can be entered in system 9.

Entering zero for the SER base or the SER maximum leads to deallocation of all SER values. Numerical SER will be disabled. If the secondary SER range has been configured, it will be deactivated as well. The secondary SER range can be configured as soon as the primary SER range has been configured. Both SER ranges must be identical or disjunct. The first 16 SER values of the secondary SER range are also reserved.

Modbus base address configuration

Pressing <E> in this field will open a window in which you can specify the base addresses of the Modbus addresses for import/export to dBASE files and for the base address used by the options 'Print FSC system configuration' and 'Hardcopy of functional logic diagrams' (see Figure 4-9).

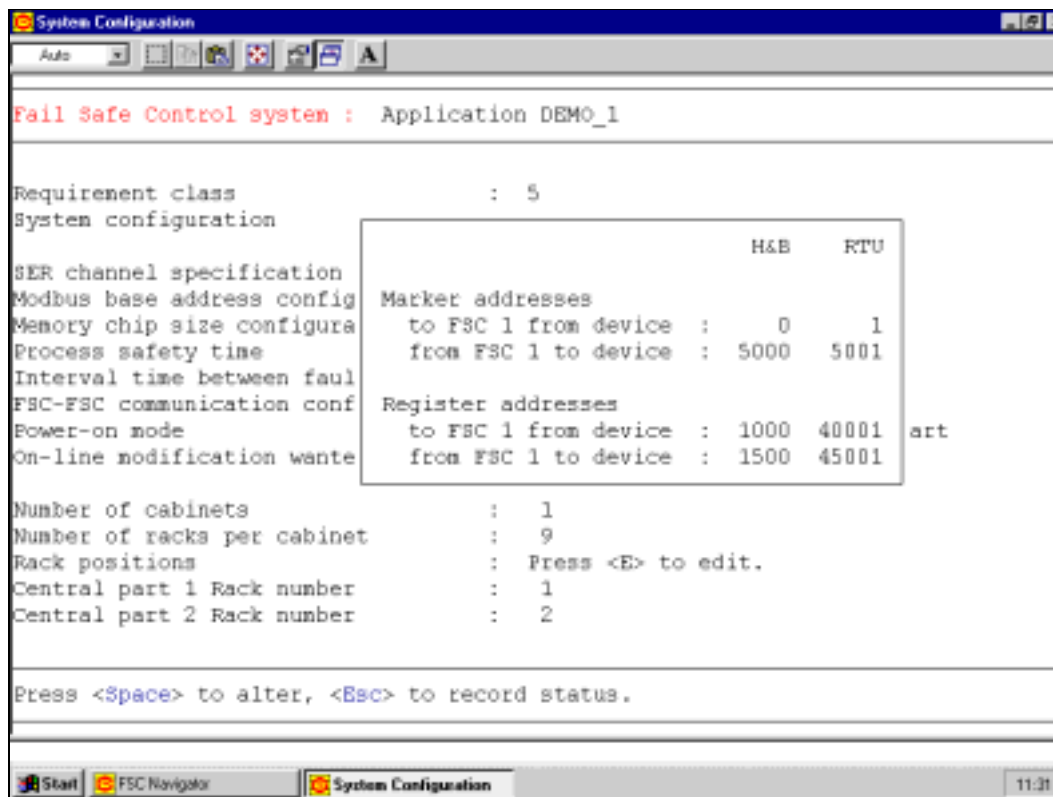


Figure 4-9 Modbus base address configuration

Memory chip size configuration

Pressing <E> in this field will open a window in which you can specify the size of the COM and CPU memory chips (RAM, EPROM) (see Figure 4-10).

The 'Translate application' option of FSC Navigator will generate files of the correct length according to the chosen memory chip size. For communication chips, you can select 512 KBit, 1 MBit, 2 MBit or Variable.



If you select 'Variable', the required memory chip size depends on the compiler output. The smallest possible memory chip size will be selected.

The selection of the CPU memory chip size depends on the configuration. You can only select the CPU chip size if the application is on the CPU (see Central Part definition screen). If the power-on mode is 'Run mode', you can select the chip types: 512K, 1M, 2M, 4M and variable. In RAM mode or simulation mode you can select the chip types: 512K, 1M and 4M.

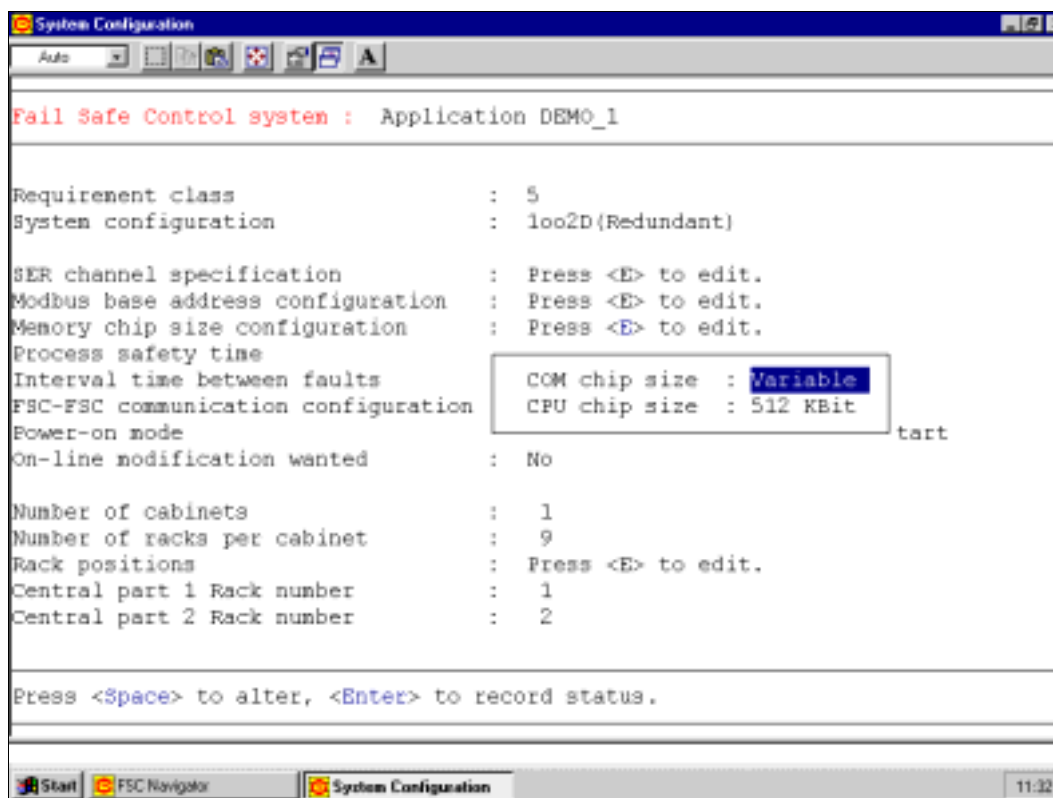


Figure 4-10 Memory chip size configuration

Process safety time

The process safety time is the time that a fault may be present in the safety system, without possible danger for an installation or the environment. In the FSC system it specifies the period in which a complete self test will be executed. The process safety time has a range of 1 to 20 seconds.

Interval time between faults

During operation, each Central Part of the FSC system performs self-tests on itself and allocated I/O modules.

If a fault is detected during self-testing, the Central Part will report the failure and take action to guarantee a safe operational result. If possible, the failure will be isolated and Central Part operation continues. If continuation of fail-safe operation cannot be guaranteed, the Central Part shuts down.

For a restricted number of failures, isolation of the failure is possible, but safe operation can only be guaranteed as long as no additional faults occur which, in correlation with the first failure, may lead to unsafe operation.

Therefore, when continuing operation, a certain risk is present that such an additional, correlating fault occurs. The longer the Central Part operates, the larger the risk becomes.

In order to keep the risk within acceptable limits, a time interval must be defined which reflects the maximum period of time the Central Part is allowed to operate after the first failure has occurred (the "interval time between faults"). When the interval time between faults expires, the Central Part will shut down.

The interval time between faults can be defined between 0 minutes (direct Central Part shutdown) and 22 days, or it can be completely deactivated. The default values for the various requirement classes are as follows: Requirement class 1-5: 72 hours (4320 minutes)

Requirement class 6: 1 hour (60 minutes).

For more information on FSC system response to failures, refer to the FSC Safety Manual.

FSC-FSC communication configuration

Pressing <E> in this field will open a window in which you can enter the FSC-FSC communication timeout and the application names for each connected system (see Figure 4-11).

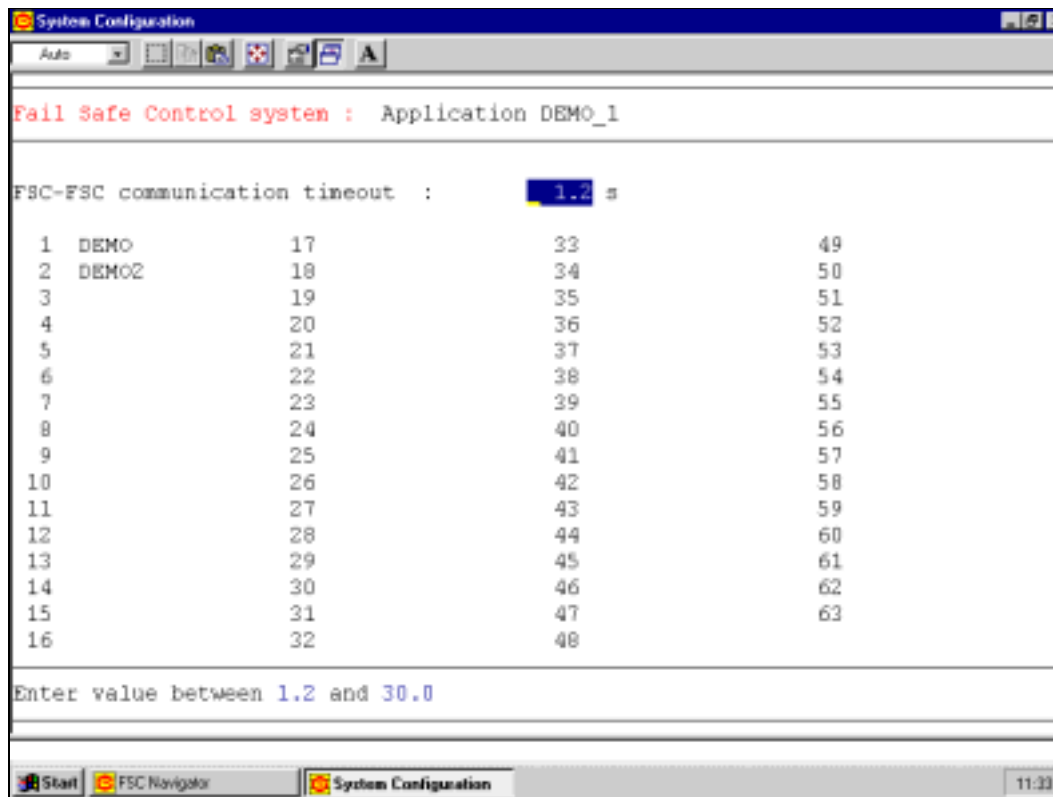


Figure 4-11 FSC-FSC communication configuration

The communication timeout is the total timeout for scanning all connected FSC systems. In other words, the FSC master expects all FSC slaves in a network to respond to a command within the given time frame. The timeout can be set to a value between 1.2 and 30 seconds (in steps of 0.1 second). The default value is 1.2 seconds. All FSC systems in an FSC network must have the same communication timeout. If they do not, communication errors may occur and the compiler will report differences.



The system numbers for FSC-FSC communication must be unique. The correct application name must be entered for each connected system number.

The configuration is used by the 'Translate application' option of FSC Navigator (the compiler) to verify the correctness of the FSC communication network.

Power-on mode

The power-on mode provides the conditions for the start-up of the FSC system.

– Run, cold start	The FSC system is completely reset and the application in EPROM is used.
– Run, warm start	The FSC system is partly reset and the application in EPROM is used.
– Ram operation, cold start	The FSC system is completely reset and the application is downloaded for use.
– Ram operation, warm start	The FSC system is partly reset and the application is downloaded for use.
– Simulation, 23125/1/1 cold start	The FSC system is completely reset and the application is downloaded for test purposes. (Training unit 23125/1/1 only; single configuration)
– Simulation, 23125/1/1 warm start	The FSC system is partly reset and the application is downloaded for test purposes. (Training unit 23125/1/1 only; single configuration)
– Simulation, 23126/1/1 cold start	The FSC system is completely reset and the application is downloaded for test purposes. (Training unit 23126/1/1 only; single configuration in one rack)
– Simulation, 23126/1/1 warm start	The FSC system is partly reset and the application is downloaded for test purposes. (Training unit 23126/1/1 only, single configuration in one rack)
– Simulation, 23125/R/1 cold start	The FSC system is completely reset and the application is downloaded for test purposes. (Training unit 23125/R/1 only, redundant configuration)
– Simulation, 23125/R/1 warm start	The FSC system is partly reset and the application is downloaded for test purposes. (Training unit 23125/R/1 only, redundant configuration)
– Simulation, 23250/1/1 cold start	The FSC system is completely reset and the application is downloaded for test purposes. (Training unit 23250/1/1 only, single configuration)
– Simulation, 23250/1/1 warm start	The FSC system is partly reset and the application is downloaded for test purposes. (Training unit 23250/1/1 only, single configuration)
– Simulation, 23260/1/1 cold start	The FSC system is completely reset and the application is downloaded for test purposes. (Training unit 23260/1/1 only, single configuration in one rack)
– Simulation, 23260/1/1 warm start	The FSC system is partly reset and the application is downloaded for test purposes. (Training unit 23260/1/1 only, single configuration in one rack)
– Simulation, 23251/1/1 cold start	The FSC system is completely reset and the application is downloaded for test purposes. (Training unit 23251/1/1 only, redundant configuration)
– Simulation, 23251/1/1 warm start	The FSC system is partly reset and the application is downloaded for test purposes. (Training unit 23251/1/1 only, redundant configuration)



On-line modification wanted

The on-line modification option allows you to exchange software (i.e. EPROMs) in an operational FSC system without shutting down the process.

If you have chosen a redundant system configuration, you may set the on-line modification option active.

Use <Space> to select and press <Enter> to record the status.

For more information refer to appendix D of the FSC Software Manual ("On-line modification").

Number of cabinets and number of racks per cabinet

You can enter the size of the FSC system:

- Number of cabinets
- Number of racks per cabinet

The modules of an FSC system can be accommodated in several racks. A maximum of 10 racks can be accommodated in one cabinet. The entire system can be accommodated in more cabinets. The default number of cabinets is 1. This can be increased to a maximum of 11 cabinets. The default number of racks per cabinet is 9, and has the maximum of 10 racks per cabinet. The maximum number of racks is 99.

These numbers can be altered. If these numbers are decremented the program checks if no racks were defined in the part that is deleted.

For single-rack configurations the number of cabinets is fixed at 1.

Rack positions

After entering the number of cabinets and the number of racks per cabinet, you should define the position of the racks in the cabinets in the 'Rack positions' option. This is done by giving an unique number to every rack position that is to be used.

The program uses these numbers to check the validity of rack numbers which are entered in the remaining part of the program.

If a rack number is entered that is already placed elsewhere, the program asks if it has to replace the rack. If a rack number is altered in which modules are already placed or which is defined as a central rack, the program asks if it has to renumber the rack. If so, the rack is renumbered in the database. This means that the placement information of all modules placed in that rack is converted, including I/O variable allocations to these modules.

Central Part 'X' rack number

After defining the racks in the cabinets, the Central Part(s) rack number(s) should be defined. If there is more than one Central Part, they have to be accommodated in identical racks, or in racks on consecutive positions. If the rack number of a Central Part is not defined, no modules can be placed. When modules are placed in the Central Part(s), the rack number(s) cannot be altered.



4.3.3 Modules

This option is only available after you have defined the configuration. The module option is used to place the modules in the racks, and to configure the hardware (e.g. safety relation) if necessary. It has the following menu structure:

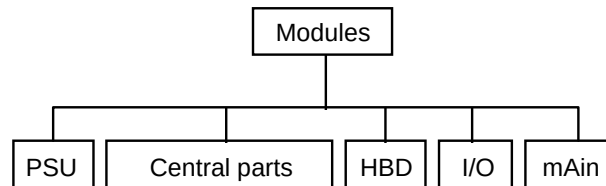


Figure 4-12 'Modules' menu

Module types

Table 4-2 to Table 4-5 list the available module types. (Refer to the FSC Hardware Manual for technical specifications.)

The following module types belong to the Central Part(s):

Table 4-2 Central Part modules

Module type	Abbreviation	Description
10001/1/1	VBD	Vertical bus driver module
10001/R/1	VBD	Vertical bus driver module with relay
10002/1/1	CPU	Central processor unit
10002/1/2	CPU	Central processor unit
10003/1/1	MEM	Memory module
10004/./.	COM	Communication module
10005/1/1	WD	Watchdog module
10006/1/1	DBM	Diagnostic and battery module
10006/2/1	DBM	Diagnostic and battery module (RTC)
10006/2/2	DBM	Diagnostic and battery module (DCF)
10007/1/1	SBD	Single bus driver module
10008/2/U	FSC-SMM	FSC-SMM communication module
10008/3/P	P-BUS	FSC to P-bus communication module

The following modules have a special function:

Table 4-3 Special modules

Module type	Abbreviation	Description
10100/1/1	HBD	Horizontal bus driver module
10100/2/1	HBD	Horizontal bus driver module
10300/1/1	PSU	24 Vdc to 5 Vdc/12 A converter
10302/1/1	WDR	Watchdog repeater module
10302/2/1	WDR	Watchdog repeater module
10303/1/1	PSD	Power supply distribution module
10305/1/1	AIC	0-20 mA to 0-5 V analog input converter (16 channels)

The following I/O modules are available:

Table 4-4 FSC input modules

Module type	Description	Tested	Fail-safe	Single configurations		Redundant configurations	
				Suited?	Voting	Suited?	Voting
10101/1/1	Fail-safe digital input module (24 Vdc, 16 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10101/1/2	Fail-safe digital input module (60 Vdc, 16 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10101/1/3	Fail-safe digital input module (48 Vdc, 16 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10101/2/1	Fail-safe digital input module (24 Vdc, 16 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10101/2/2	Fail-safe digital input module (60 Vdc, 16 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10101/2/3	Fail-safe digital input module (60 Vdc, 16 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10102/1/1	Fail-safe analog input module (4 channels)	Yes	Yes	Yes	1oo1D	No	— —
10102/1/2	Fail-safe analog input module (4 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10102/2/1	Fail-safe analog input module (4 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D
10103/1/1	Intrinsically safe input module (4 channels)	No	No	Yes	1oo1	No	—
10103/1/2	Intrinsically safe input module (4 channels)	No	No	Yes	1oo1	No	—
10104/1/1	Digital input module (24 Vdc, 16 channels)	No	No	Yes	1oo1	Yes	2oo2
10104/2/1	Digital input module (24 Vdc, 16 channels)	No	No	Yes	1oo1	Yes	2oo2
10105/2/1	Fail-safe high-density analog input module (24 Vdc, 16 channels)	Yes	Yes	Yes	1oo1D	Yes	2oo2D 1oo2D

Note:

The default voting schemes for input modules that support different voting schemes are as follows:

- New projects: **1oo2D**
- Existing projects with requirement class AK1-4, after conversion from an FSC version that does not have voting (< 500) to FSC R5xx: **2oo2D**
- Existing projects with requirement class AK5-6, after conversion from an FSC version that does not have voting (< 500) to FSC R5xx: **1oo2D**



Table 4-5 FSC output modules

Module type	Description	Tested	Fail-safe	Single configurations		Redundant configurations	
				Suited?	Voting	Suited?	Voting
10201/1/1	Fail-safe digital output module (24 Vdc, 0.55 A, 8 channels)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10201/2/1	Fail-safe digital output module (24 Vdc, 0.55 A, 8 channels)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10203/1/2	Fail-safe digital output module with double switch-off (24 Vdc, 0.9 A, 4 channels)	Yes	Yes	Yes	1001D	No	– –
10205/1/1	Fail-safe analog output module (0(4)-20 mA, 2 ch.)	Yes	Yes	Yes	1001D	No	–
10205/2/1	Fail-safe analog output module (0(4)-20 mA, 2 ch.)	Yes/No	Yes	Yes	1001D	No	–
10206/1/1	Digital output module (24 Vdc, 0.55 A, 12 channels)	No	No	Yes	1001	Yes	2002
10206/2/1	Digital output module (24 Vdc, 0.55 A, 12 channels)	No	No	Yes	1001	Yes	2002
10207/1/1	Intrinsically safe optocoupler output module (8 channels)	No	No	Yes	1001	No	–
10208/1/1	Relay output module (contacts, 12 channels)	No	No	Yes	1001	Yes	2002
10208/2/1	Relay output module (contacts, 10 channels)	No	No	Yes	1001	Yes	2002
10209/1/1	Digital output module (24 Vdc, 0.1 A, 16 channels)	No	No	Yes	1001	Yes	2002
10209/2/1	Digital output module (24 Vdc, 0.1 A, 16 channels)	No	No	Yes	1001	Yes	2002
10212/1/1	Digital output module (24 Vdc, 0.9 A, 8 channels)	Yes	Yes/No	Yes	1001D	Yes	2002D 1002D
10213/1/1	Fail-safe digital output module (110 Vdc, 0.32 A, 4 ch.)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10213/1/2	Fail-safe digital output module (60 Vdc, 0.67 A, 4 ch.)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10213/1/3	Fail-safe digital output module (48 Vdc, 0.75 A, 4 ch.)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10213/2/1	Fail-safe digital output module (110 Vdc, 0.32 A, 4 ch.)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10213/2/2	Fail-safe digital output module (60 Vdc, 0.67 A, 4 ch.)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10213/2/3	Fail-safe digital output module (48 Vdc, 0.75 A, 4 ch.)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10214/1/2	Fail-safe digital output module (220 Vdc, 0.25 A, 3 ch.)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10215/1/1	Fail-safe digital output module (24 Vdc, 2 A, 4 channels)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10215/2/1	Fail-safe digital output module (24 Vdc, 2 A, 4 channels)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10216/1/1	Fail-safe loop-monitored digital output module (24 Vdc, 1 A, 4 channels)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10216/2/1	Fail-safe loop-monitored digital output module (24 Vdc, 1 A, 4 channels)	Yes	Yes	Yes	1001D	Yes	2002D 1002D
10216/2/3	Fail-safe loop-monitored digital output module (48 Vdc, 0.5 A, 4 channels)	Yes	Yes	Yes	1001D	Yes	2002D 1002D

Note:

The default voting schemes for output modules that support different voting schemes are as follows:

- New projects: **1002D**
- Existing projects with requirement class AK1-4, after conversion from an FSC version that does not have voting (< 500) to FSC R5xx: **2002D**
- Existing projects with requirement class AK5-6, after conversion from an FSC version that does not have voting (< 500) to FSC R5xx: **2002D** (*except* 10216/./ modules, which are set to **1002D**)



Placing the modules

If the FSC system type or rack number(s) of the Central Part(s) are not defined, no modules can be placed. A distinction is made between modules that are placed in the Central Part (PSU, PSD, CPU, WD, SBD, VBD, COM, MEM, DBM), and modules that are not placed in the Central Parts (PSU, PSD, HBD, I/O). For the Central Part modules you only need to define the desired number of modules of every type; the modules are automatically placed. For the other modules the rack and position need to be defined. Please note that the PSU (10300/1/1), COM and DBM modules occupy two positions, all other modules one. The PSU, CPU, WD, COM, MEM, SBD, VBD and DBM are always safety-related. The same is true for HBD 1 of VBD 1. The safety relation for the other HBDs and the I/O modules can be defined by the user.

Note:

By editing the Central Part definition and PSUs, the Central Part module positions can change even though no changes have been made (due to different specifications in the past).

4.3.3.1 Power supply unit

Power supply units can be configured in the PSU screen. The 5 Vdc power supplies and the Delta/External power supplies use different screens. Use <PgUp>, <PgDn> to step through these screens. You can do the following:

- Add and delete power supply units,
- Assign a rack and position for a power supply unit,
- Alter the type of the power supply unit,
- Configure a PSD for a 5 Vdc power supply unit (if there is a VBD that controls single I/O in a redundant FSC system), and
- Reserve the space of a PSD of a 5 Vdc power supply unit.

The program defaults to the 5 Vdc power supply screen. Use the <PgUp> and <PgDn> keys to step through the power supply screens. Every time a new screen of Delta power supply units must be displayed, the next multiple of 11 Delta units appear until the last multiple has been reached (89-99).



Editing the 5 Vdc power supply

The 5 Vdc power supply unit is the 10300/1/1 (uses two positions in the rack).

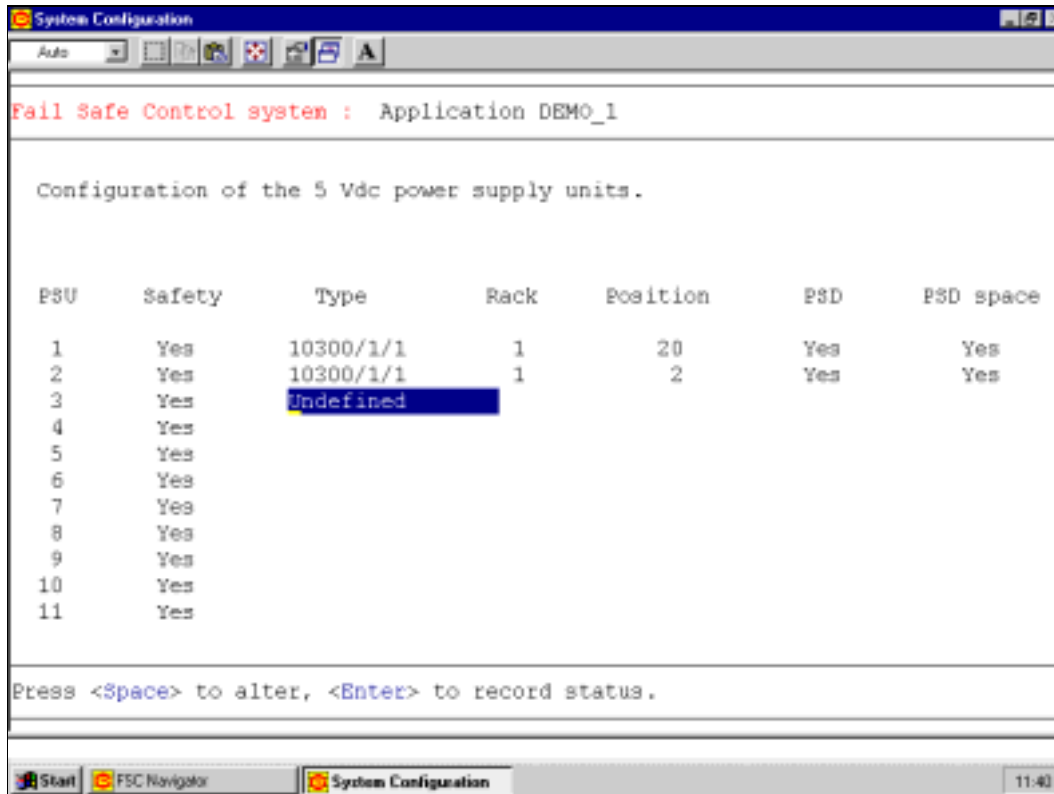


Figure 4-13 Power supply unit

The first PSU was configured automatically when the user configured the FSC configuration. This is the Central Part PSU. The rack and position of this PSU cannot be changed. The 10300/1/1 module will be the default PSU module. The PSD will not be placed and the PSD space will not be reserved unless an HBD has been placed in a redundant system which controls single I/O.

PSUs must be configured in ascending order. It is not possible to configure, for example, PSU 10 if PSU 6 has not been configured.

The PSUs can only be deleted in descending order: You can only delete the PSU module that was added last (highest number). A PSU can be deleted by setting the PSU type to 'Undefined' or by deallocating the PSU. The Central Part PSU cannot be removed.

Possible error messages are:

– If you try to remove a PSU that is not the last PSU the following error will be displayed

Only last module can be deleted. Restore? (Y/N)



- If, in a redundant configuration, you tried to delete the last PSU, a PSD must be allocated for the PSU that becomes the last module. If this is not possible, the following message will be displayed:

Cannot allocate the PSD to the last PSU.

After entering the type you can enter the rack number for the PSU.

Possible error messages are:

- If you tried to select an undefined rack:

Rack does not exist.

- If you tried to select the Central Part rack of a configuration using a SBD:

SBD placed, only one PSU allowed in the central part rack.

- You can configure PSUs in the Central Part racks. Whenever, only Central Part 1 can be selected. Otherwise the following error message will be displayed:

Not allowed to define PSUs in other central part than the first.

- If two Central Parts are configured in one rack, extra PSUs cannot be configured in the Central Part rack:

More than one central part in central part rack. Not allowed to place PSU.

If you change the rack number, the position of the PSU will be reset to zero. You can enter the position of the PSU in the rack.

Possible error messages are:

- If you tried to assign the PSU to a position in the Central Part:

Position in use for central part.

- If the position is an impossible position:

Position out of range.

- If the position has already been allocated to another module:

Position allocated to <module>

- If the position for the PSD has already been allocated to another module:

Position for PSD allocated to <module>



If the configuration allows it, a PSD space of a PSU can be reserved for future expansions. The PSD space field must be toggled to 'Yes'. By default, the PSD space field will not be reserved.

Possible error messages are:

- If the PSD cannot be placed because the position has already been allocated to another module:

Position for possible PSD already allocated to <Module>

- If the position for the PSD is out of range:

Position for PSD out of range.

If the configuration allows it (VBD which controls single I/O), a PSD can be allocated for a PSU. The PSD field must be toggled to 'Yes'.

Possible error messages:

- If the PSD cannot be placed because the position has already been allocated to another module:

Position for PSD has already been allocated to <Module>

- If the position for the PSD is out of range:

Position for PSD out of range.

The PSD will be configured by the program automatically if the FSC configuration is a redundant system with single I/O and the current PSU is the last one. You cannot delete this PSD.

Editing the Delta/External power supply

Possible 24 Vdc power supply units:

- 24-20HE (uses the entire rack)
- 24-12HE (uses half the rack)

Possible 48 Vdc power supply units:

- 48-10HE (uses the entire rack)
- 48-05HE (uses half the rack)

Possible 60 Vdc power supply units:

- 60-05HE (uses half the rack)

No power supplies are available in the 110 and 220 Vdc voltage classes.

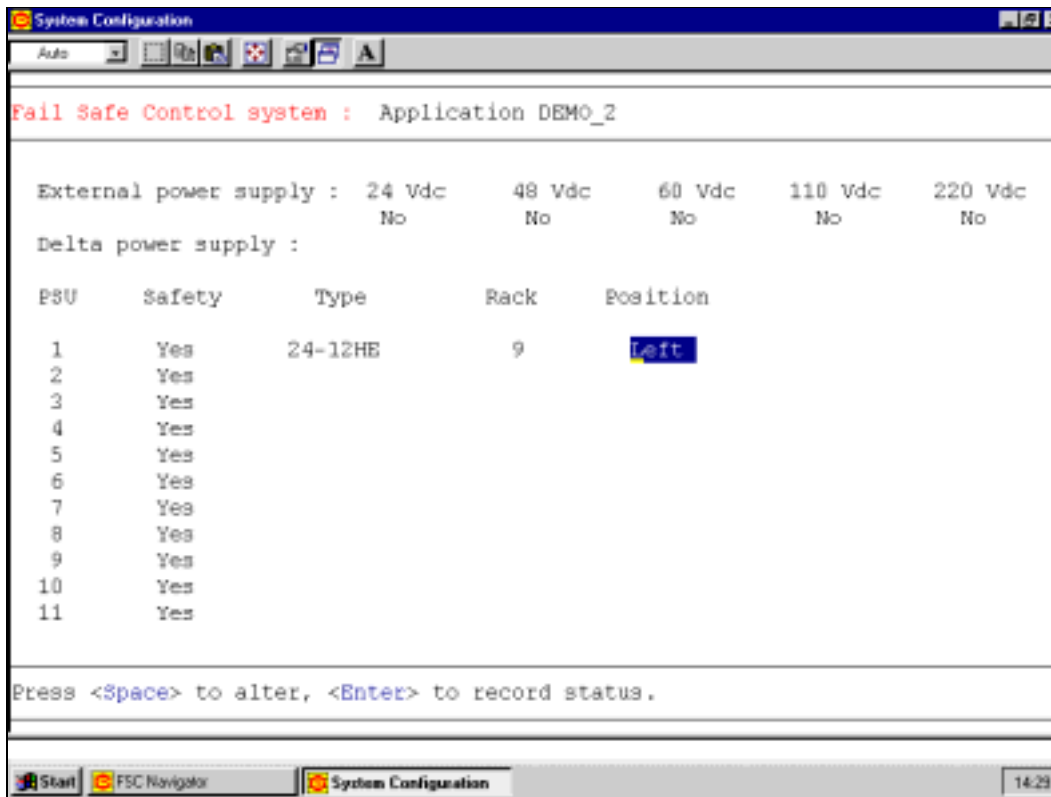


Figure 4-14 Power supply units

An external power supply can be selected for each voltage class (except the 5 Vdc class). Toggle the external power supply fields of the voltage classes to change. If you select the external power supply in a specific voltage class and Deltas of this voltage class are already configured, the program asks:

Selecting the external power supply will delete Deltas of this voltage class. Change? (Y/N)

Confirmation is requested. If the external power supply of a voltage class has been configured, it is not possible to configure Delta power supply units of this voltage class.

The user can change the Delta type by toggling. If changing is not possible, the following message will be displayed:

Changing the type will deallocate the Delta unit. Change? (Y/N)

Confirmation is requested. This could, for example, occur if the type changes from 24-12HE to 24-20 HE and the entire rack cannot be allocated for the 24-20HE.



After entering the Delta type you can enter the rack number of the Delta. Possible error messages are:

- If the rack has not been configured:

Rack number does not exist.

- If you tried to select a physical rack that is within the range of the Central Part and I/O racks:

Mixing <Delta type>, Central Part and I/O racks is not allowed.

- If the rack is already in use by modules other than Delta and 5 Vdc PSU modules:

<Delta type> can only be configured in racks that are empty or contain PSUs.

- If the Delta power supply cannot be allocated:

Rack already in use by other power supply unit(s).

The position of the Delta can only be changed for Delta power supplies using only half a rack. Toggle the position field to select the left (default) or right part of the rack. If the left or right part of the rack has already been allocated to another Delta unit, the position cannot be changed. The Delta will be configured in the empty part.

4.3.3.2 Central Part modules

After defining the required number of modules of every type, the module positions in the Central Part rack(s) can be viewed by selecting a specific module from the menu.

For VBD, COM and DBM modules, the extra hardware configuration data can be defined.

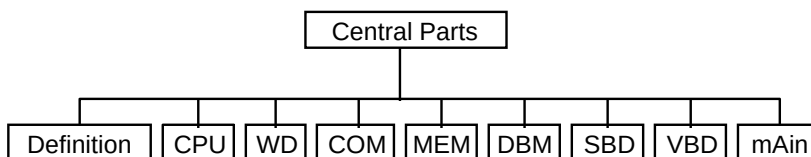


Figure 4-15 'Central Parts' menu

4.3.3.2.1 Definition

Note:

By editing the Central Part definition and PSUs, the Central Part module positions can change even though no changes have been made (due to different specifications in the past).

In the Central Part definition screen, the definition of the Central Parts can be changed:

- The size of the Central Part bus
- The processor (CPU type)
- The location of application program
- The type of the COM module(s)
- The number of Central Part modules: CPU (and WD), MEM, COM, VBD, SBD, DBM.

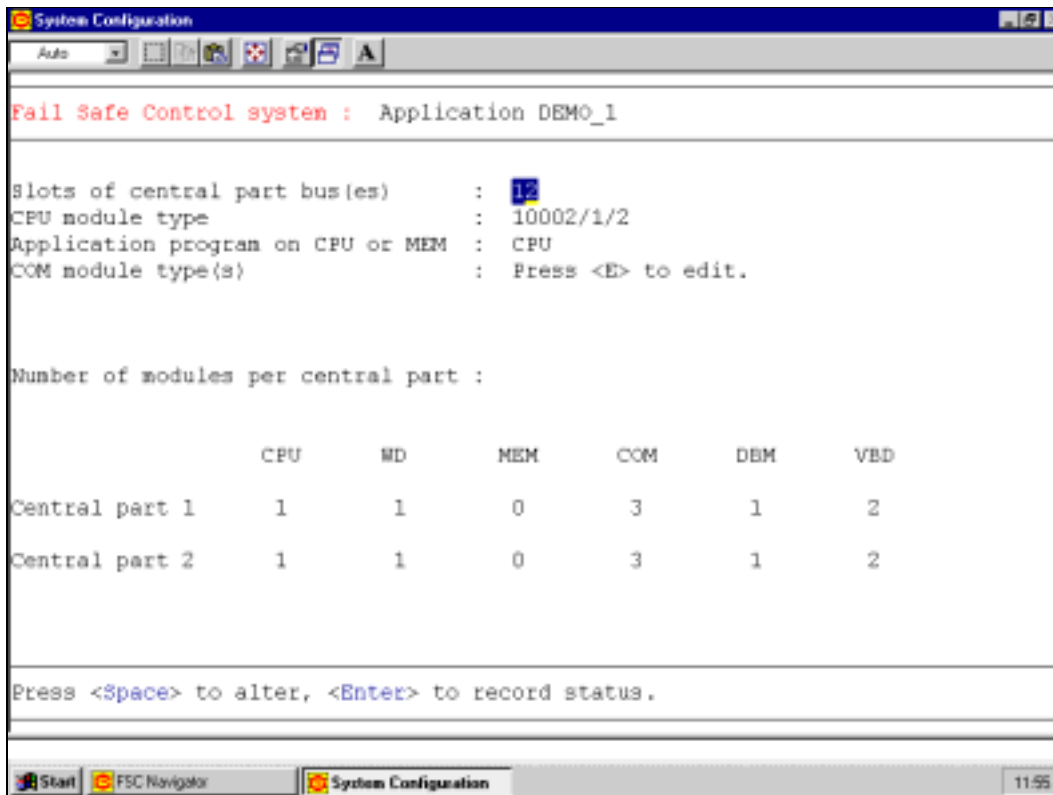


Figure 4-16 Central Part definition

The default numbers for the various module types are the following:

VBD: 1	MEM: 0
SBD: 0	CPU: 1
COM: 0	WD: 1
DBM: 1	

If the 'Definition' menu item of the Central Part menu has been selected, the cursor will be on the 'Slots of Central Part bus(es)' item. Use the cursor keys and the <Enter> key to step through the screen. Screen items that cannot be edited are skipped.

By pressing <Esc> the program stores the new Central Part definition and returns to the Central Part menu.

Slots of central part bus(es):

The first item in this option is the size of the system bus. There are 5-, 7-, 12- and 17-slot system buses. If the FSC configuration has two Central Parts in one rack, only a 5-slot or a 7-slot bus can be selected.

CPU module type:

Select the CPU module type. All CPUs in the application are of the same type. Possible CPU module types are:

- 10002/1/1 Standard 12 MHz CPU module (obsolete). The application must be stored on MEM.
- 10002/1/2 Improved 10002/1/1 12 MHz. CPU module. Application can be stored on CPU or MEM.

**Application program on CPU or MEM:**

For the 10002/1/2 CPU module, the user may configure whether to store the application program on the CPU (default) or on (a) MEM module(s). If 'Application program on MEM' is selected, automatically a minimum of 1 MEM module will be configured. The field cannot be altered if the 10002/1/1 CPU is configured. In this case, the application is always located on MEM.

COM module type(s):

With this option you can define the communication module types. By default, the communication module types of all modules will be configured as 10004/./.. If desired, the COM module with the highest number may be changed to 10008/2/U (UCN protocol) or 10008/3/P.

Number of modules per central part:

With this option you can define how many MEM, COM, DBM and VBD modules you want to place in every Central Part. Central Part 2 is always equal to Central Part 1, except for the number of COM modules: this number can be one less for Central Part 2.

When leaving the option by pressing <Esc>, the modules you selected are placed in the Central Parts.

An SBD can be placed configurations with a 5-, 7- or 12-slot bus, one Central Part per rack and only one PSU per Central Part. If an SBD is placed, only one CPU and one VBD are possible.

If placement is not possible because you selected more modules than the number of slots of the system bus allows, you are asked whether the situation on entering the option should be restored or not.

The position of the Central Part modules within the rack is allocated automatically and cannot be modified.

4.3.3.2.2 Central processor unit

The positions in the Central Part rack, of the central processor unit (CPU) modules are listed. No additional hardware configuration necessary.

4.3.3.2.3 Watchdog

The positions in the Central Part rack, of the watchdog (WD) modules are listed. No additional hardware configuration necessary.

4.3.3.2.4 Communication module

For every Central Part you can configure the channels A and B of each communication module (COM) (see Figure 4-17). Select a Central Part and a channel by pressing the spacebar. Depending on the protocol you choose, some fields have a fixed value, others can be altered. Use the arrow keys to select a field.

In redundant configurations, channel A of COM module 1 in Central Part 1 and 2 always has protocol FSC-FSC. Listed below are the fields that can be defined.

- **Protocol**
 - Possible values:
 - Undefined
 - FSC-FSC
 - Development system
 - Printer
 - Modbus RTU
 - Modbus H&B (Optional)
 - RKE3964R (Optional)
 - DS (Modem)

- **Redundant**
Define a redundant link by changing the redundant field to 'Yes'. The link definition will be copied to the second Central Part automatically. You are only allowed to change the configuration of Central Part 1. A redundant link can also be configured for a single configuration with redundant communication links. The configuration of channel A will be copied to channel B automatically. You are only allowed to change the configuration of channel A.

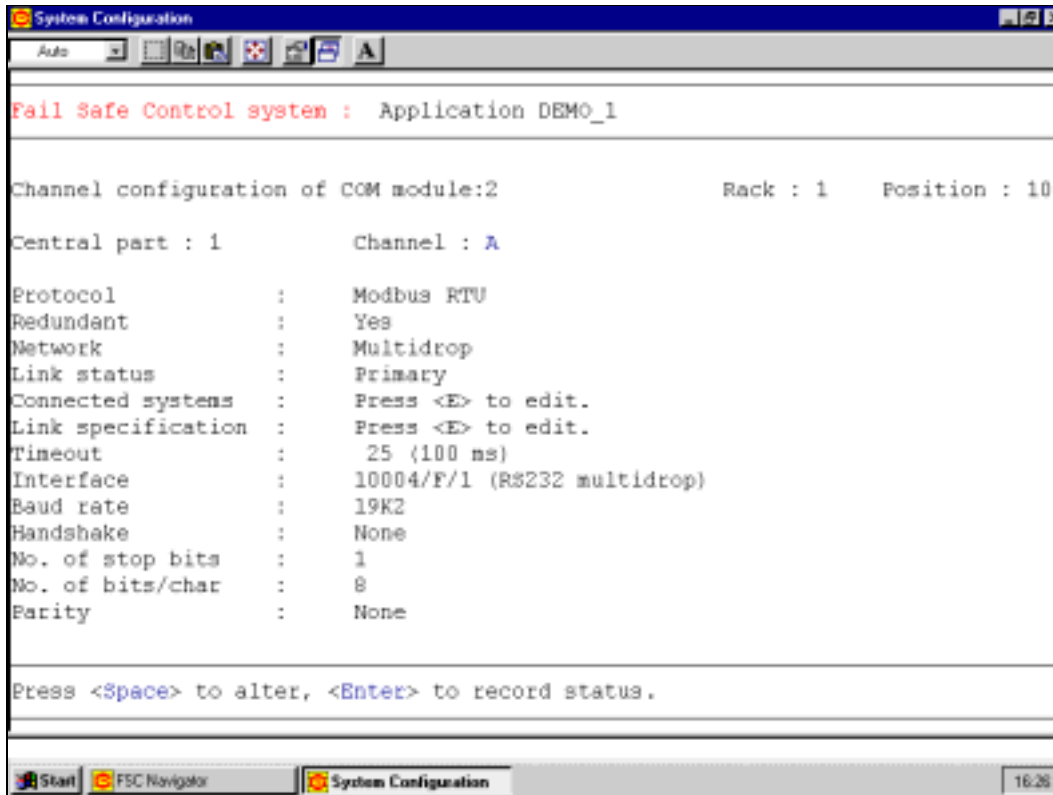


Figure 4-17 Defining the channel configuration

- **Network**
Possible values
 - Point to point
 - Multidrop
- **Link status**
Only defined for the protocols FSC-FSC, Development System, Modbus RTU, Modbus H&B and RKE3964R.
If the protocol is FSC-FSC, the link status can be defined as master (communication server) or slave. Multiple FSC systems may cooperate to accomplish a specific task. In such a configuration one FSC system, the master, controls the operation of one or more other systems, the slaves. For the other protocols the link status can be defined as primary or secondary. For primary protocols a memory area can be reserved to be used for the communication. Secondary links use the same memory area as the primary link, they serve as a kind of back-up, e.g. to be used if the primary channel fails.
For details on communication refer to Appendix F of the FSC Software Manual ("Communication Manual").



- **Connected systems**
Only defined for multidrop protocols and for point-to-point FSC-FSC protocols (Figure 4-17) All available system numbers are shown in a popup window. If a connection to a system already exists, the system is shown highlighted. Use the cursor keys to select a system. Press <S> to establish a connection with a system, press <D> to remove a connection with a system. If the protocol is FSC-FSC, the systems can be selected which communicate with the system, via the communication channel.

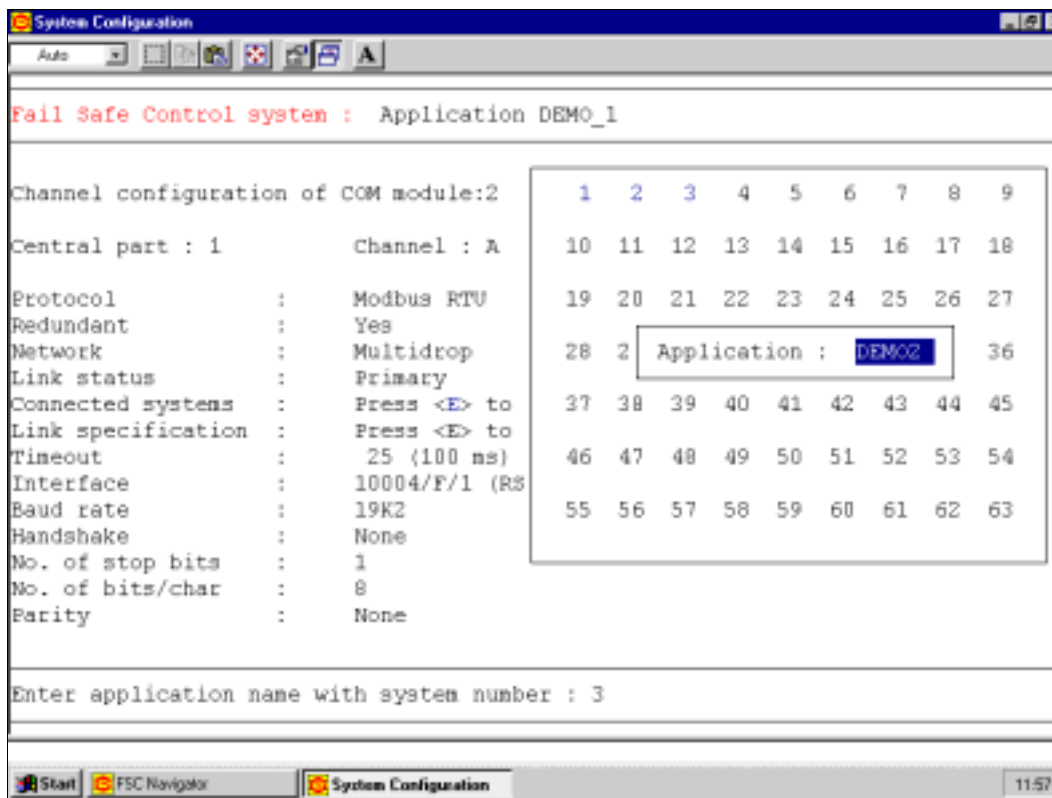


Figure 4-18 Editing the connected systems

- **Link specification**
Only defined for protocol FSC-FSC, Development system, Modbus RTU, Modbus H&B, RKE3964R and printer.
 - **FSC-FSC link specification**
If the protocol is FSC-FSC, the amount of memory (marker bytes and register bytes) that is used for the communication with every connected system has to be defined. The program automatically allocates the required amount, and all inputs and outputs that are to be used for the communication with a particular system are allocated within the reserved area. (See chapter for the allocation of inputs and outputs to COM modules.)

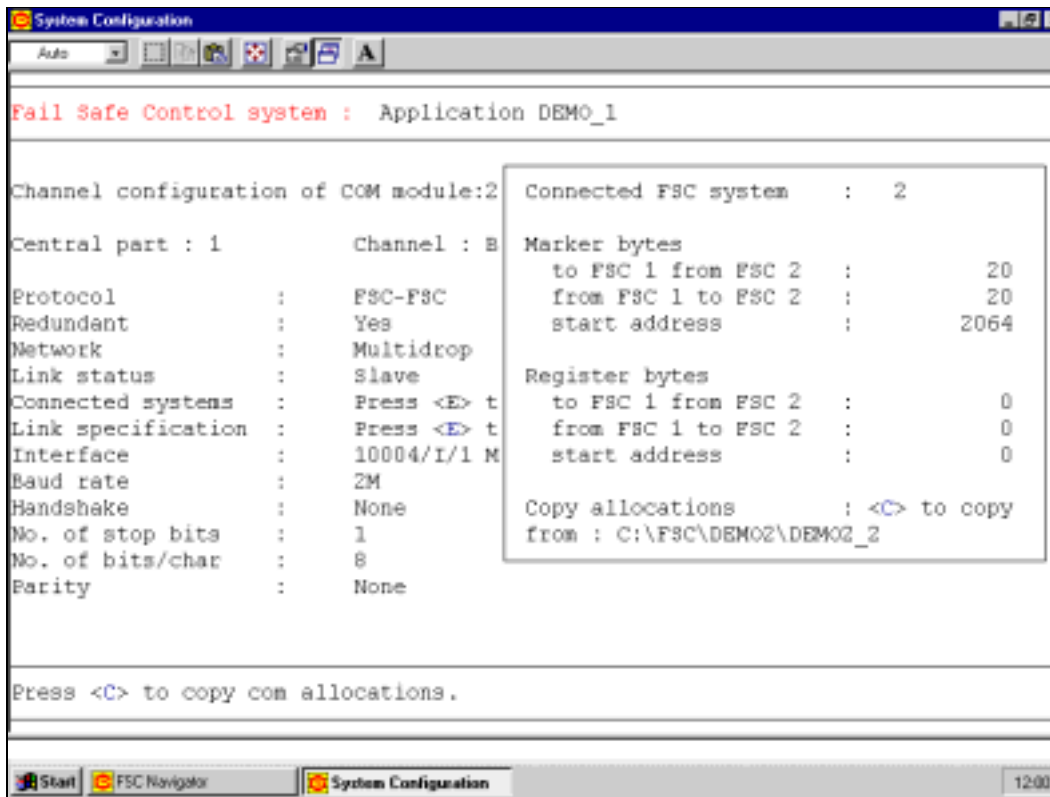


Figure 4-19 Editing the link specification for an FSC-FSC protocol

The start addresses of the allocated marker and register areas are displayed.

– Copy allocations

In an FSC-FSC network configuration, the bytes reserved for communication between two FSC systems must be consistent. This means that the number of bytes for written to system X must be the same as the number of bytes read by system X. The tag numbers must also be the same. Example:

Table 4-6 Link specification

Application →	TEST_1	TEST_2
Marker bytes	To FSC1 from FSC2:4 From FSC1 to FSC2:0	To FSC2 from FSC1:0 From FSC2 to FSC1:4
Tag number	I test loc. FSC connected from system 2. Relative address 3.	O test loc. FSC connected to system 1. Relative address 3.

This is a correctly defined link for transferring digital information from system 2 to system 1 via the variable 'test'. If you have defined a link specification and variables are already allocated in system 2, you can do a copy allocations, which first checks if the defined link in your own database as well as in the other database are correct. If so, it will copy the bytes defined in the other database with the accessory variables (tag number, service, qualification, unit and sub unit), in such way that the link is correct. Variables which are allocated incorrectly, will be



deallocated.

If you have a link to more than one system (e.g. a master or communication server to more slaves), you also have an option to copy the information of all slaves at once.

Make sure that the application names of the connected system are defined (FSC-FSC communication configuration window). In all the other applications the name of the application you are configuring at the moment must be defined.

- **FSC device link specification**

If the protocol is 'Device', the amount of memory (marker bytes and register bytes) that is used for the communication with the device has to be defined. The program automatically allocates the required amount, and all inputs and outputs that are to be used for the communication are allocated within the reserved area. (see chapter for the allocation of the inputs and outputs to COM modules)

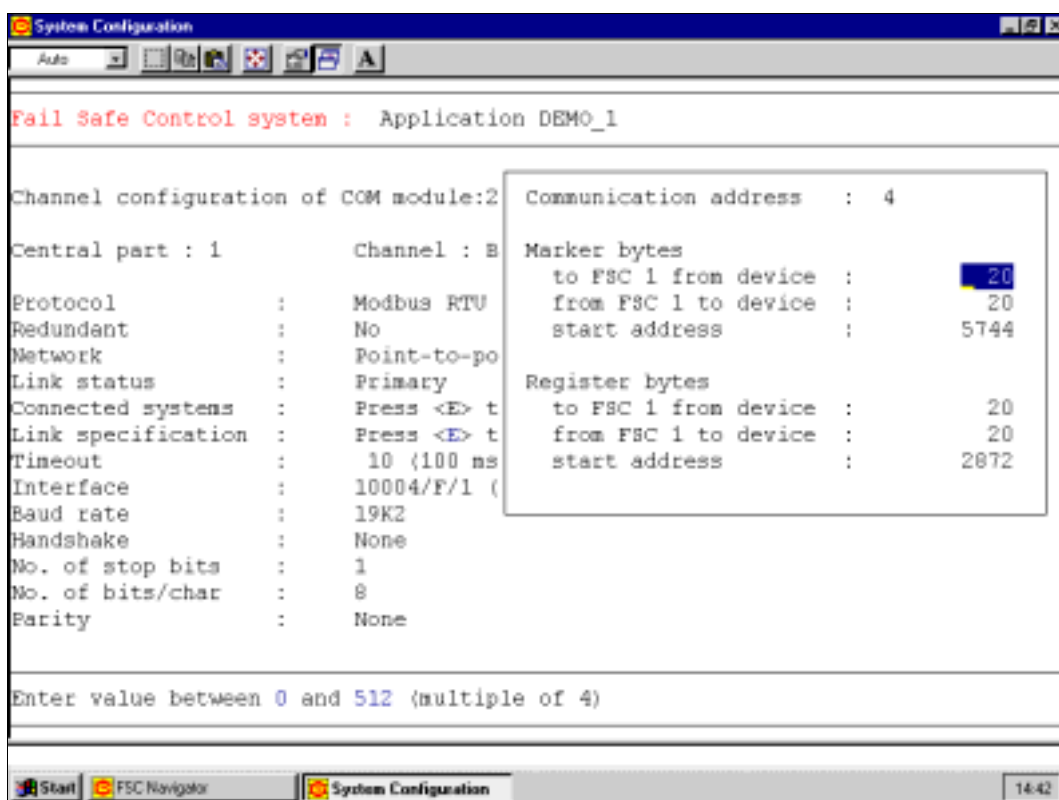


Figure 4-20 Editing the link specifications for a Modbus protocol

If the protocol is Development system or Modbus and the link status is primary, the amount of memory used for communication with the device has to be defined. If the protocol is printer and the printer channel is the SER channel, a number of output bytes can be defined (BO SER). If the SER event of one of these BOs occurs, the value of the BO will be printed.

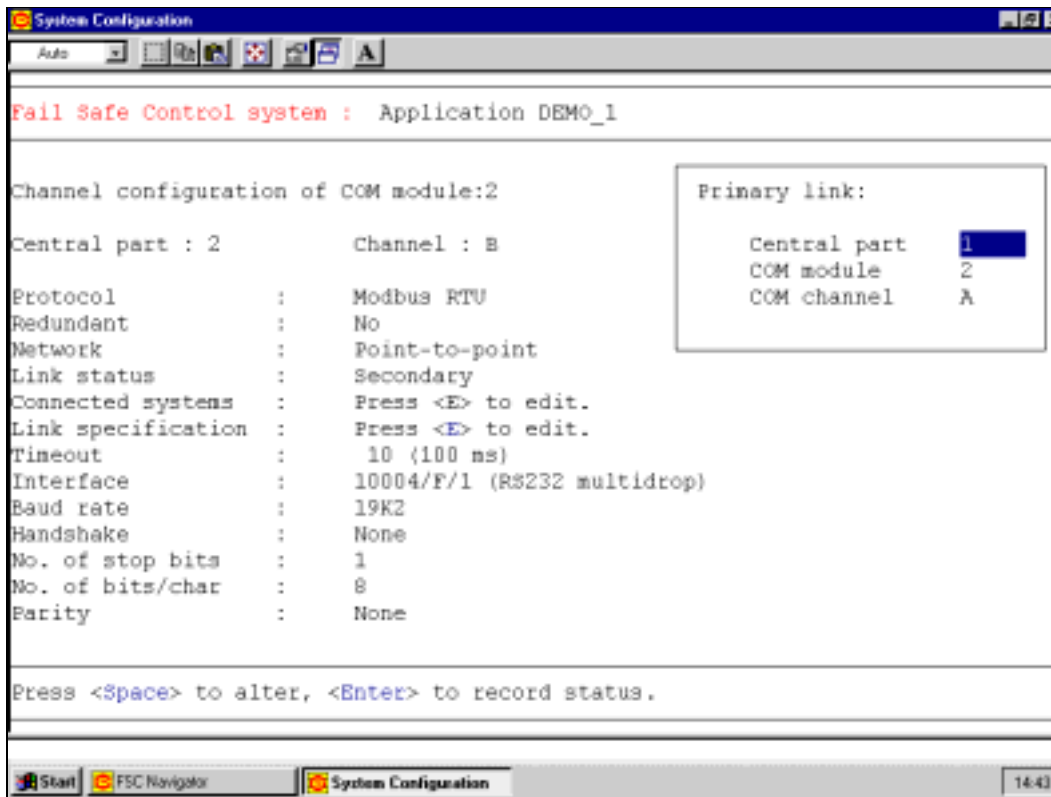


Figure 4-21 Editing the link for a secondary Development system protocol

Channel B on module 2 in Central Part 1 will be the primary Development system protocol. If the link is secondary, the primary link has to be defined.

Note:

If you change the configuration of a channel to which inputs or outputs are allocated (protocols FSC-FSC, Development system, Modbus RTU, Modbus H&B and RKE3964R) and these changes affect the allocation of these variables, you will be asked for a confirmation.

– Timeout

Only defined for the Modbus RTU and Modbus H&B protocols. The timeout can be any value in the range of 10 to 250 (x 100 ms)

– Interface

Possible values	– 10004/B/1	RS232C
	– 10004/E/1	Current loop
	– 10004/F/1	RS232C multidrop
	– 10004/G/1 Manchester	Fiber optic
	– 10004/G/1 FM0	Fiber optic
	– 10004/G/1 FM1	Fiber optic
	– 10004/H/1	Between Central Parts
	– 10004/I/1 No encoding	RS485 / RS422
	– 10004/I/1 Manchester	RS485 / RS422
	– 10004/I/1 FM0	RS485 / RS422
	– 10004/I/1 FM1	RS485 / RS422



The values that can be selected depend on the chosen protocol and network.

- **Baud rate**
 - Possible values
 - 2M
 - 1M
 - 125K
 - 50K
 - 38K4
 - 19K2
 - 9600
 - 4800
 - 2400
 - 1200
 - 600
 - 300

The values that can be selected depend on the chosen protocol and the interface.

- **Handshake**
 - Possible values
 - Hardware
 - Xon - Xoff
 - None
- **Number of stop bits**
 - Possible values
 - 1
 - 2
- **Number of bits/char**
 - Possible values
 - 7
 - 8
- **Parity**
 - Possible values
 - Even
 - Odd
 - None

4.3.3.2.5 Safety Manager Module

The FSC Safety Manager Module (FSC-SMM) is configured by selecting the UCN communication protocol.

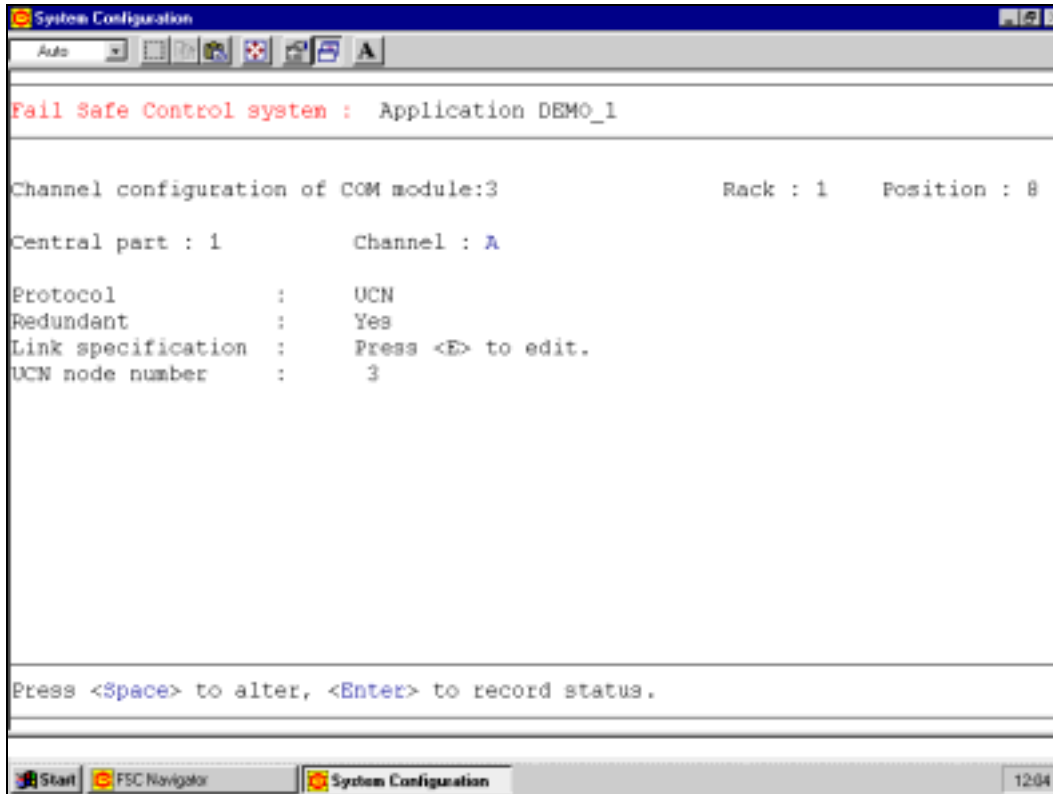


Figure 4-22 Protocol definition for communication with the TPS system

- **Protocol**
UCN.
- **Redundant**
Not configurable.
Redundant = yes, if the Safety Manager Module is configured in an FSC system with redundant Central Parts.
Redundant = no, if the Safety Manager Module is configured in an FSC system with single Central Parts.
- **Link specification**
The amount of memory (marker bytes and register bytes) that is used for the communication with the TotalPlant Solution (TPS) system has to be defined. The program automatically allocates the required amount, and all inputs and outputs that are to be used for the communication are allocated within the reserved area (see chapter for the allocation of the inputs and outputs to COM modules).

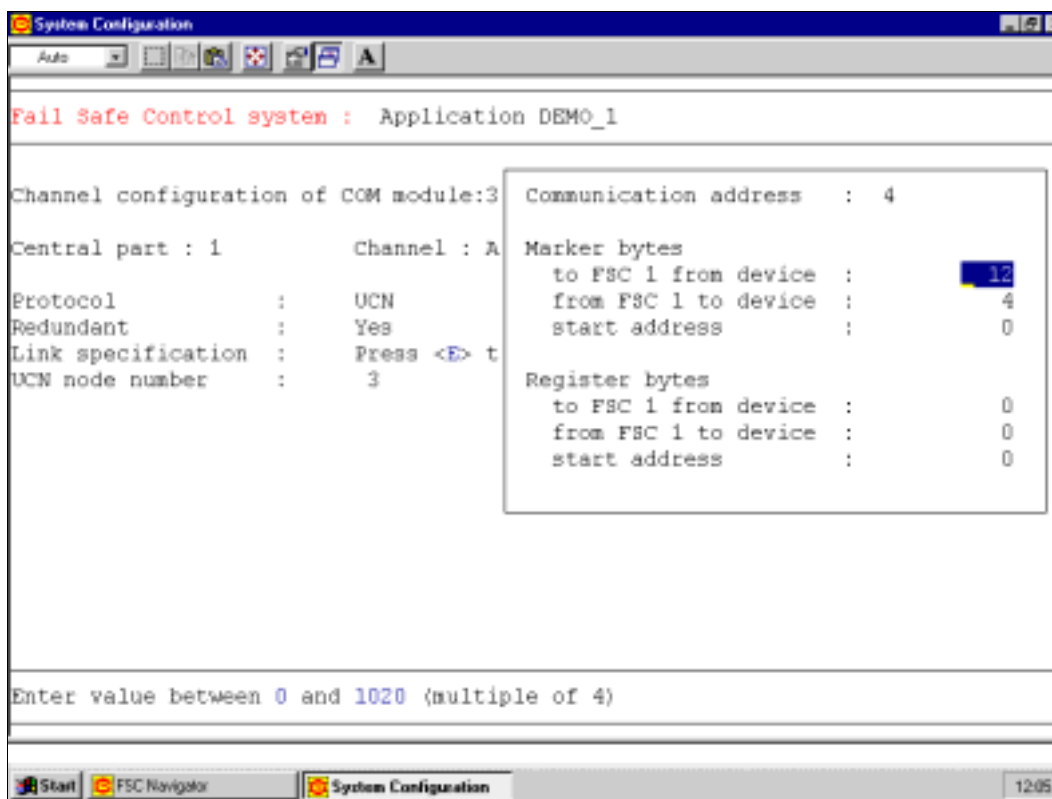


Figure 4-23 Editing the link specifications for the FSC-SMM

- **UCN node number**
Enter the UCN node number of the FSC-SMM.

4.3.3.2.6 Memory module

The positions in the Central Part racks of the memory modules are listed. No additional hardware configuration is necessary.

4.3.3.2.7 Diagnostic and battery module (DBM)

The user is able to modify the following DBM characteristics:

- Module type
- Temperature settings :
 - Degree type (Celsius, Fahrenheit or Kelvin)
 - Low temperature shutdown
 - Low temperature alarm
 - High temperature alarm
 - High temperature shutdown

DBM module type

The DBM module type can be specified as:

- 10006/1/1 (RTC)
- 10006/2/1 (RTC)
- 10006/2/2 (DCF)

The 10006/2/2 DBM module has the so-called DCF capability. It is capable of receiving atomic clock information from the digital clock Frankfurt within the radius around Frankfurt indicated in Figure 4-24. Atomic clock information is received on a continuous basis.



Figure 4-24 Receiving radius of DCF signal

DBM temperature settings

Table 4-7 DBM values

Description	Celsius		Fahrenheit		Kelvin	
	Default	Range	Default	Range	Default	Range
Low temperature shutdown	0	0 to 20	32	32 to 68	273	273 to 293
Low temperature alarm	5	5 to 25	41	41 to 77	278	278 to 298
High temperature alarm	55	35 to 55	131	95 to 131	328	308 to 328
High temperature shutdown	60	40 to 60	140	104 to 140	333	313 to 333

The minimum difference between low temperature shutdown and low temperature alarm must be 5 degrees Celsius, 5 degrees Kelvin or 9 degrees Fahrenheit. The same holds for high temperature alarm and high temperature shutdown.

4.3.3.2.8 Single bus driver

The position in the Central Part racks of the single bus driver (SBD) is listed. No additional hardware configuration is needed.



4.3.3.2.9 Vertical bus driver

The following fields are specified for a vertical bus driver (VBD): Function, Module type and CPU access. The following considerations apply to the values the fields can have:

Function

The VBD function can be changed by pressing the spacebar. Table 4-8 below shows the possible VBD functions.

Table 4-8 Possible VBD functions per FSC system type

FSC system type	VBD functions
Single Redundant	Single Redundant + Single

If HBDs have already been connected to the VBD, the VBD function cannot be changed anymore. Consecutive VBDs must always have the same VBD function. It is not allowed to mix VBD functions, as shown in the examples below.

Correct:

VBD no.:	1	2	3	4	5	6
VBD function:	R	R	R	R	S	S

Incorrect:

VBD no.:	1	2	3	4	5	6
VBD function:	S	R	S	S	R	R

Module type

The module type can be toggled to either 10001/1/1 or 10001/R/1. Module type 10001/R/1 means that the VBD is equipped with a relay to isolate the VBD from the vertical bus in case of a VBD fault. If there are one or more VBDs which control single I/O in a redundant system, then all VBDs are of the type 10001/R/1.

Note:

For new applications only select 10001/R/1.

CPU access

Access by CPU1 is always allowed. This field cannot be changed.

4.3.3.3 Horizontal bus driver

For every placed VBD, 10 horizontal bus drivers (HBD) can be placed.

The safety relation can be altered.

Exception: – HBDs to which safety-related I/O is connected, cannot be made non-safety-related.

Safety-related I/O modules can only be connected to a safety-related HBD.

The number of the rack in which the HBD is to be placed, must be entered. The following limitations apply:

- If I/O has been placed for the HBD already, the rack number cannot be changed anymore.
- The rack number must have been defined in the configuration option, but cannot be a Central Part rack number.
- The rack may not be controlled by another HBD already.
- An HBD can be removed by entering zero, but always start removing with the highest numbered HBD.
- An HBD cannot be placed in a rack that is above the Central Part rack(s).

- I/O racks and Delta power supply racks cannot be mixed.
- The position(s) where the HBD is placed depends on the function of the VBD:
- single I/O: position 21.
 - redundant I/O: position 20 for Central Part 1, 21 for Central Part 2.

If the VBD controls redundant I/O, the HBD can control one extra rack. For the rack number the following limitations apply:

- The rack number cannot be changed if I/O has been placed in the controlled rack.
The rack number must have been defined in the configuration option, but cannot be a central rack.
- The numbers for the rack in which the HBD is placed, and for the extra controlled racks must be consecutive.
- The extra controlled racks must be in the same cabinet as in which the HBD is placed, and on consecutive positions.
- The rack may not be controlled by another HBD.

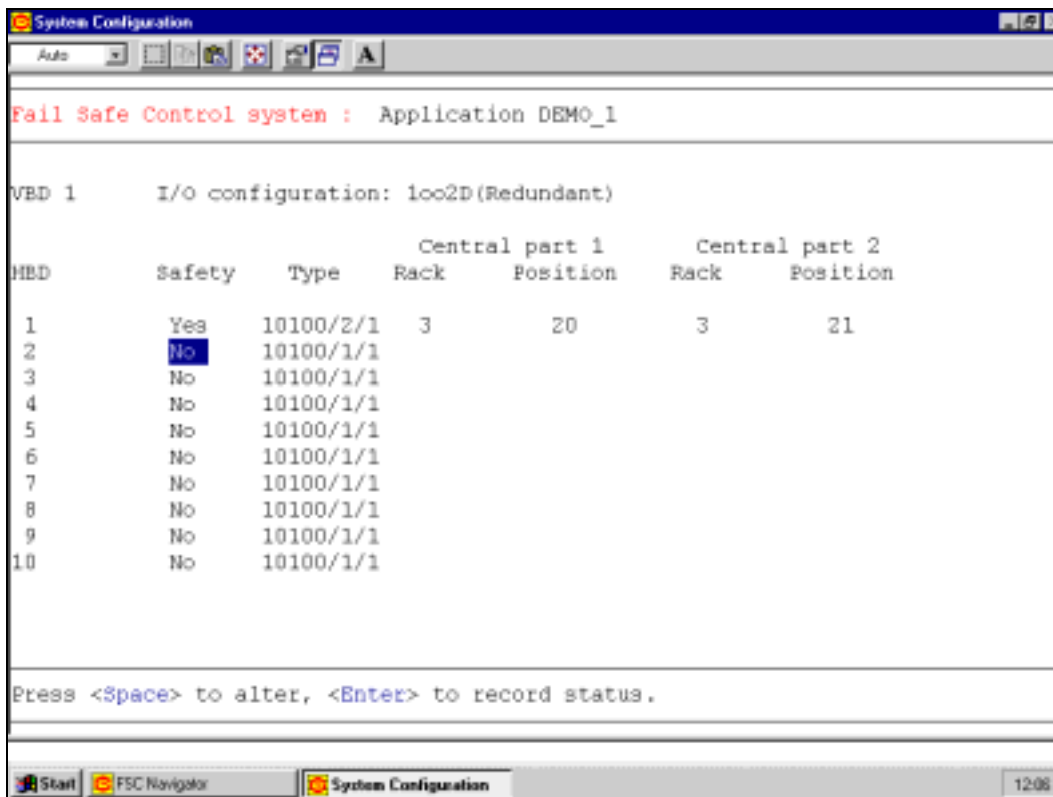


Figure 4-25 Select the type of HBD

Note:

Selecting a specific HBD type can limit the selection of I/O module types. For example, selecting the 10100/2/1 limits the set of selectable I/O modules to the modules identified as type 2.



4.3.3.4 I/O modules

I/O modules can be placed in every rack in which an SBD or an HBD is placed or which is controlled by an HBD. Use <PgUp> and <PgDn> to select a rack.

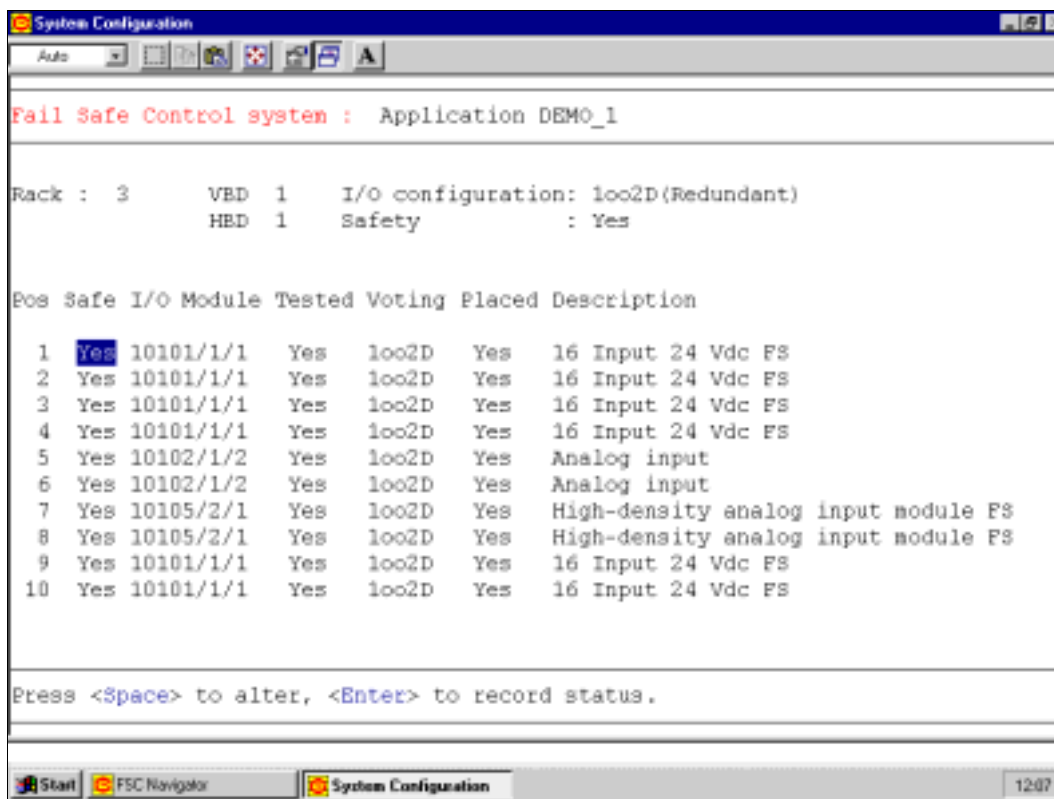


Figure 4-26 Placing the I/O modules

Safe

If the HBD is non safety-related, the I/O modules are also non safety-related. Otherwise the safety relation can be modified. Choose between Yes and No.

- Exceptions:
- Modules to which safety-related signals are connected cannot be made non safety-related.
 - If the module is non fail-safe, it can only be non safety-related.

Tested

Normally, fail-safe I/O modules are always tested by the FSC system. However, you may disable the test for the fail-safe modules. The possible values are Yes (tested) and No (not tested). Disabling the test for fail-safe modules is only possible for analog output modules. Non fail-safe I/O modules are always non safety-related and they are not tested.

Voting

The table below shows which voting types are possible for each module.

Table 4-9 Voting type per I/O module

Module	VBD controls redundant I/O	VBD controls single I/O
10101/./.	2oo2D/1oo2D	1oo1D
10104/./.	2oo2	1oo1
10103/1/.	2oo2	1oo1
10102/./.	1oo2D	1oo1D
10105/2/.	1oo2D	1oo1D
10201/./.	2oo2D	1oo1D
10205/./.	–	1oo1D
10206/./.	2oo2	1oo1
10207/1/1	2oo2	1oo1
10208/./.	2oo2	1oo1
10209/./.	2oo2	1oo1
10212/1/.	2oo2D 1oo2D	1oo1D
10213/./.	2oo2D 1oo2D	1oo1D
10214/1/2	2oo2D 1oo2D	1oo1D
10215/./.	2oo2D 1oo2D	1oo1D
10216/./.	2oo2D 1oo2D	1oo1D
10100/./.	–	–

If the chosen rack is a Central Part rack with an SBD, all positions not in use for the Central Parts can be selected for I/O modules.

If a module is placed for a VBD which controls redundant I/O, then the module is automatically placed at the adjacent positions for the higher Central Parts.

Keyboard commands

Use the following keys to select or to delete a module:

- <D>: Deletes the module placed at the position indicated by the cursor.
- <C>: Copies the module that has been placed last in the selected rack.
- <S>: Shows a window with all the modules that can be placed at the selected position:
 - If the module is safety-related only fail-safe modules can be placed.
 - Depending on the VBD function, only certain modules can be placed.
 - Depending on the HBD type, types of modules will be excluded



- The program checks that at the low-numbered positions in the rack only Exi modules can be placed and at the high numbered positions only non Exi modules. There has to be at least one free position between the two groups.
- The program checks that at the low numbered positions in the rack only high voltage modules (10101/. /2, 10101/. /3, 10213/. /., and 10214/1/2) can be placed and at the high numbered positions the other (low voltage) modules. There has to be at least one free position between the groups.
- High-voltage modules cannot be placed in the same rack as the Exi modules.

Use the arrow keys and <PgUp> and <PgDn> to move around in the window. By pressing <Enter> a highlighted module can be selected for placement. With <Esc> the window can be left without selecting a module.

Note:

High-voltage modules are all modules operating at voltages higher than 50 Vdc.

When a module is deleted or replaced by another module, the program checks if any signals are connected to the module. If so, you are asked if the module has to be deleted anyway. If you answer 'Yes', the signals are deallocated. The following modules are exchanged without the need to deallocate the signals:

- 10101/1/1 ↔ 10101/2/1
- 10101/1/2 ↔ 10101/2/2
- 10101/1/3 ↔ 10101/2/3
- 10104/1/1 ↔ 10104/2/1
- 10102/1/2 ↔ 10102/2/1
- 10201/1/1 ↔ 10201/2/1
- 10205/1/1 ↔ 10205/2/1
- 10206/1/1 ↔ 10206/2/1
- 10209/1/1 ↔ 10209/2/1
- 10213/1/1 ↔ 10213/2/1
- 10213/1/2 ↔ 10213/2/2
- 10213/1/3 ↔ 10213/2/3
- 10215/1/1 ↔ 10215/2/1
- 10216/1/1 ↔ 10216/2/1

Other modules cannot be exchanged due to different functionality between the /1/ and /2/ versions or due to different voltage classes.

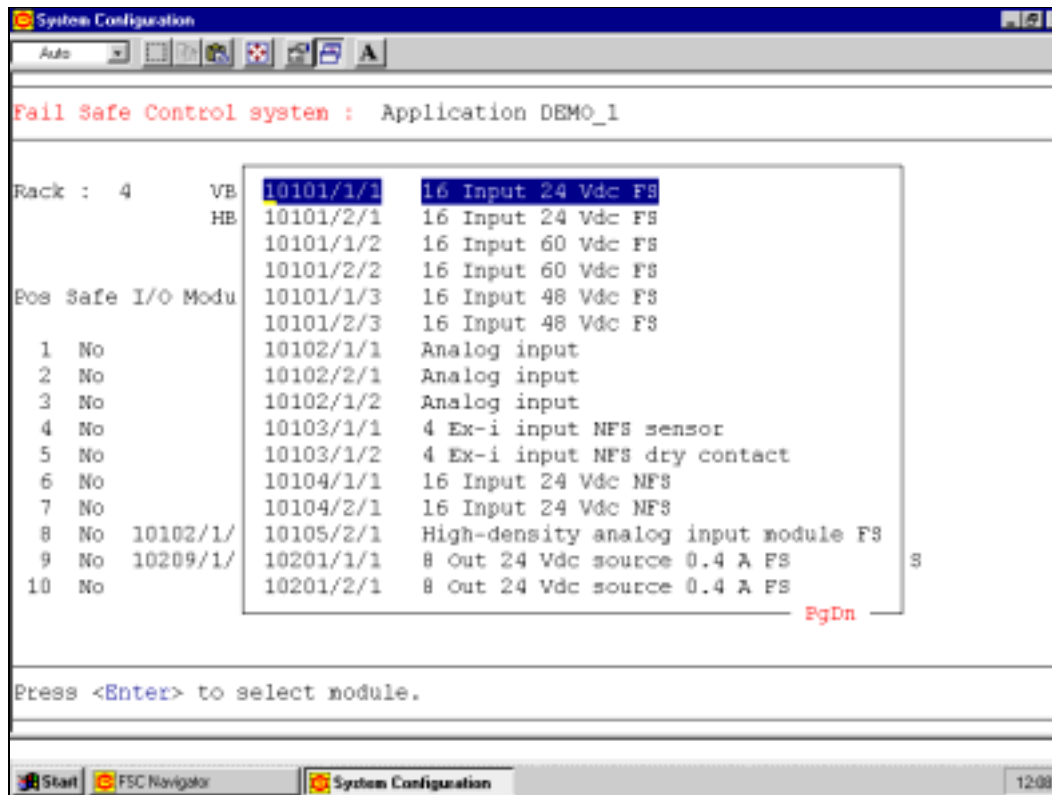


Figure 4-27 Selection of a module

After selecting a module it can be defined to be placed or not placed. Choose placed if the module indeed will be used in the system. Choose not placed if the module will not be used yet, but in future extensions of the system. I/O variables can only be allocated to placed modules.

The range of an analog input value can be determined from the bottom and top scale values. These values contain the bottom and top scale values. If the bottom scale value is smaller than the top scale value, the scaling is positive, else the scaling is negative. So, the bottom scale value does not always contain the lowest value and the top scale does not always contain the highest value!

4.3.4 Passwords

Some options of FSC Navigator are password-protected to prevent unauthorized access to certain features. For details on passwords refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or FSC Navigator's Help file.



4.4 Signal specs

V	Tag number	Service	Qualification	Loc	Unit	Subunit
I	HS-155A	MAIN VALVE	NOT HIGH	FLD		

I/O specs: Add variable (I,AI,BI,XI,O,AO,BO,XO,A or P).

Figure 4-28 I/O signal specs

Seven fields are defined:

- V : Variable type
- Tag number : Variable tag number
- Service : The description of the variable (e.g. level in glycol tower)
- Qualification : Information about the one (or active) status of the variable in case of boolean variables (e.g. HIGH).
- Loc : Information about the location of the sensor, actuator (e.g. FLD). Some locations are reserved.
- Unit : This field contains the name of the unit for which the variable is used in the application program (e.g. furnace 210).
- Subunit : This field contains the name of the subunit for which the variable is used in the application program (e.g. Burner 2 on furnace 210).

Main menu options:

- Add : Enter add mode
- Search : Enter search mode
- Write dBASE : Write the I/O variables from the variable database to a dBASE 3 or 4 compatible file.
- Read dBASE : Read I/O variables from a dBASE 3 or 4 compatible file.
- Main : Return to the 'System Configuration' main menu.



4.4.1 Add

You can enter the variable type and the tag number. If this combination already exists the program does not accept it. Otherwise the variable is added to the database. Now you can edit other fields such as service, qualification, location (only for I/O), unit and subunit. Note that for some variable types certain fields cannot be edited. System variables (variables with location SYS), markers, counters, registers and timers cannot be added.

4.4.1.1 Tag number restrictions

The user can enter a tag number. The following restrictions should be accounted for:

- Empty tag numbers are not allowed. The error message 'Tag number cannot be an empty string' will be generated if you try to add a variable with an empty tag number.
- Tag numbers cannot contain a comma, semicolon or question mark. The error message 'Tag number contains a comma, semicolon or question mark' will be generated if you try to enter a variable using a tag number that contains these reserved characters.
- Tag numbers can only contain trailing spaces. Tag numbers like 'HRDW INPUT ONE ' and 'HLS 34' are not allowed. The error message 'Tag number contains non trailing spaces' will be generated if you try to enter such tag numbers.
- Duplicate tag numbers of the same variable type are not allowed. All tag numbers of a certain variable type must be unique. The error message 'Variable type + tag number is not unique' will be generated if you try to add a variable which type and tag number already exists in the variable database.
- The tag numbers of default system variables are reserved. The error message 'Tag number is reserved' will be generated if you try to add a variable with one of the following tag numbers:

AI	VM r/p/17 ¹⁾	I	FSC-SYSTEM-FAULT
BI	DATE	I	INPUT-FAILURE
BI	DAY	I	INT.COMMUNIC.FLT
BI	HOURS	I	IO-COMPARE
BI	MINUTES	I	IO-FORCEDI OUTPUT-FAILURE
BI	MONTH	I	RED.INPUT-FAULT
BI	SECONDS	I	TEMP.PRE-ALARM
BI	YEAR	I	TRANSMIT.-FAULT
I	CENTR.PART-FAULT	I	WD r/p/s ²⁾
I	COM-I/O	O	BO SER triggers ³⁾
I	DEVICE-COM.FLT	O	COM-I/O
I	EXT.COMMUNIC.FAULT	O	E cp_com_ch_blk ⁴⁾
I	FLASHER-0.5Hz	O	SEC.SWITCH-OFF
I	FLASHER-1Hz		
I	FLASHER-2Hz		
I	FLASHER-5Hz		

Notes:

- 1) r/p/17 = rack / position, e.g. VM 3/8/17
- 2) r/p/s = rack / position / sequence no., e.g. WD 3/7/1
- 3) depends on corresponding BO
- 4) RKE block triggers: cp_com_ch_blk = central part_COM module_channel_block no., e.g. E 1_1_B_23



4.4.1.2 Variable field restrictions

The next table shows the variable fields that can be changed. If the variable can be deleted, the Del field is set to 'Yes'.

Table 4-10 Variable field restrictions

Type	Tag number	Service	Qualification	Location	Unit	Subunit	Del
I I	Y N ¹⁾	Y Y	Y Y	Y ⁴⁾ N (SYS)	Y N	Y N	Y N
O O	Y N	Y Y	Y Y	Y N (SYS)	Y N ⁶⁾	Y N	Y N
AI AI	Y N	Y N	N N	Y ⁵⁾ N (SYS)	Y N	Y N	Y N
AO	Y	Y	N	Y ⁵⁾	Y	Y	Y
BI BI	Y N	Y Y	N N	Y N (SYS)	Y N	Y N	Y N
BO	Y	Y	N	Y	Y	Y	Y
XI ³⁾	Y	Y	N	N (MUX)	Y	Y	Y
XO ³⁾	Y	Y	N	N (MUX)	Y	Y	Y
A ³⁾	Y	Y	N	N (ANN)	Y	Y	Y
P ³⁾	Y	N	N	N	Y	Y	Y
M	N	N ²⁾	N	N	N	N	N
R	N	N ²⁾	N	N	N	N	N
T	N	Y	N	N	N	N	N
C	N	Y	N	N	N	N	N

Notes:

- 1) Tag number can be edited if the system input is the fault reset input, force enable input or the clock synchronization input.
- 2) The service of function block tag numbers can be changed.
- 3) The amount of XI, XO, A and P are bound to be specific maximum. The maximum number of XI and XO variables is 16. The maximum number of A and P are variables is 64. If this number is reached, the program cannot be add more variables of this type.
- 4) If the location of a variable is SYS or MUX, the user cannot change it to another location type: These locations are reserved and cannot be assigned to variables either.
- 5) Location cannot be changed to a reserved location: COM, FSC, MUX, SYS or ANN.
- 6) For BO SER triggers, this field show the engineering units of the BO.



4.4.2 Search

You can enter a variable type and a tag number or a part of a tag number and the location. All variables with matching type, tag number and location are displayed on the screen. If a variable record was already on the screen the line on which it was displayed is deleted. The display can be stopped by pressing <Esc> and be paused and restarted by pressing <F2>.

In the add and search modes you can use the cursor control keys to select the field of a variable on the screen for editing. The variable type can never be changed. A variable can be deleted by pressing <Ctrl> + <U>. If the variable is allocated you will be asked for a confirmation. System variables cannot be deleted.

Use <PgUp> and <PgDn> to step through the database page by page.

4.4.3 dBASE interface

The FSC software uses an interface with dBASE via 'Write dBASE' and 'Read dBASE' options in the 'Signal Specs' menu. These options allow you to create and/or read dBASE III and IV compatible files.

4.4.3.1 dBASE file format

The following dBASE file format is used:

Table 4-11 dBASE file format

FIELD	LENGTH	TYPE	USE	USE
TYPE	2	Character	Type	General variable information Type and Tag number are a unique combination.
TAGNUMBER	16	Character	Tag number	
SERVICE	19	Character	Service	
QUALIFICAT	16	Character	Qualification	
UNIT	8	Character	Unit	
SUB_UNIT	8	Character	Subunit	
LOC	3	Character	Location	
SHEET	3	Numeric	Sheet 1	Sheet number(s) of the variable if they are allocated on a sheet. (These fields are not used yet).
SHT 2	3	Numeric	Sheet 2	
SHT 3	3	Numeric	Sheet 3	
SHT 4	3	Numeric	Sheet 4	
RACK	2	Numeric	Rack	Hardware allocation
POS	2	Numeric	Position	
CHAN	2	Numeric	Channel	
SAFETY	1	Character	Safety relation	Communication allocation for variables with location COM
FORCE	1	Character	Force enable	
WRITE	1	Character	Write enable	
SER	1	Character	SER enable	
SERSEQ	6	Numeric	SER enable	
CNTPT	1	Numeric	Central Part	Communication allocation for variables with location COM
COM	1	Numeric	COM module	
CHANNEL	1	Character	COM channel	
RKEBLOCK	3	Numeric	RKE3964R block number	



Table 4-11 dBASE file format (continued)

FIELD	LENGTH	TYPE	USE	USE
SYSTEM	2	Numeric	System number	Communication allocation for variables with location FSC
ADDRESS	5	Numeric	Relative address (Modbus address ¹)	The relative address (or Modbus address) of a variable with location COM or FSC
DCS_ADDR	5	Numeric	DCS address	Alias address for FSC system used for data exchange with process computers. In the case of TPS, it is the counterpart of the PLCADDR parameter of a TPS point allocated to the FSC UCN interface.
AENGUNIT ANTOP ANBOTTOM	5 19 19	Character Numeric ² Numeric ²	Analog engineering unit AN top scale AN bottom scale	Scaling range of an analog input/output engineering unit indication.

Notes:

¹ See appendix F for a detailed specification of the Modbus addresses.

² With decimal point at 10th position (9 digits in front of decimal point and 9 digital after)



4.4.3.2 Write dBASE

This option allows you to write variables from the variable database to a dBASE file. If you have selected this option, you are prompted for the dBASE file name. The extension '.DBF' (dBASE file extension) is automatically added. If an invalid file name has been entered, an error message will be displayed. If you have selected an existing file, the program will ask if the file may be overwritten. If the file does not exist, the program will ask if the file must be created.

If a 'Write dBASE' command is performed, variables with the following variable type will be written to the dBASE file:

I	Location FLD (hardware), COM, FSC, ANN
O	Location FLD (hardware), COM, FSC, ANN
BI	Location FLD (hardware), COM, FSC, ANN
BO	Location FLD (hardware), COM, FSC, ANN
AI	
AO	

The system records (location SYS) that are used for the FSC system are also written to the dBASE file, but are ignored if the 'Read dBASE' command is used. They are only added for inspection, for example to generate a complete list of SER sequence numbers in dBASE (SER sequence is also allowed for, for example, the system markers).

4.4.3.3 Read dBASE

With this option you can import variables from a dBASE 3 or 4 compatible file. You are prompted for the file name. If the file does not exist, an error message is generated. The program lists the format of the dBASE file. While reading the contents of the dBASE file, the type and tag number of new or changed variables are listed. The data read from the dBASE file is checked for consistency (see tables shown below).

Examples: A hardware variable that is safety-related cannot be allocated to a NFS I/O module. A hardware variable cannot be allocated to a non existing or an already allocated hardware position.

dBASE files must be of one of the variable types that can be written to the dBASE file. All other variable types are ignored. If you added variables with location MUX or SYS (reserved locations) or if you changed the location of a variable to MUX or SYS, the location is reset to location FLD.

Note:

The maximum number of fields in the dBASE file is 40.



Table 4-12 Field interpretation for dBASE imports

Type	Loc.	Service / qualification	Unit / subunit	Safety	Rack / position / channel	Force enable	Write enable	Eng. units	Top scale	Bott. scale
I	COM	Copy	Copy	No	Ignored	Y=Yes Other=No	Yes	Copy	Ignored	Ignored
I	FSC	Copy	Copy	Y=Yes N=No Other=Undef.	Ignored	Y=Yes Other=No	No	Copy	Ignored	Ignored
I	ANN	Copy	Copy	No	Copy (*1)	No	No	Copy	Ignored	Ignored
I	FLD	Copy	Copy	Y=Yes N=No Other=Undef.	Copy (*1)	Y=Yes Other=No	No	Copy	Ignored	Ignored
AI	<>	Copy	Copy	Y=Yes N=No Other=Undef.	Copy (*1)	Y=Yes Other=No	No	Copy	Copy (if <>0) (*9-13, 15)	Copy (if <>0) (*9-13, 15)
BI	COM	Copy	Copy	No	Ignored	No	Yes	Copy	Ignored	Ignored
BI	FSC	Copy	Copy	Y=Yes N=No Other=Undef.	Ignored	No	No	Copy	Ignored	Ignored
BI	FLD	Copy	Copy	Y=Yes N=No Other=Undef.	Copy (*1)	No	No	Copy	Ignored	Ignored
O	COM	Copy	Copy	No	Ignored	No	No	Copy	Ignored	Ignored
O	FSC	Copy	Copy	Y=Yes N=No Other=Undef.	Ignored	No	No	Copy	Ignored	Ignored
O	ANN	Copy	Copy	No	Copy (*1)	Y=Yes (*2) Other=Nodbase1.tbl	No	Copy	Ignored	Ignored
O	FLD	Copy	Copy	Y=Yes N=No Other=Undef.	Copy (*1)	Y=Yes Other=No	No	Copy	Ignored	Ignored
AO	<>	Copy	Copy	Y=Yes N=No Other=Undef.	Copy (*1)	Y=Yes Other=No	No	Copy	Copy (if <>0) (*9-14)	Copy (if <>0) (*9-14)
BO	COM	Copy	Copy	No	Ignored	No	No	Copy	Ignored	Ignored
BO	FSC	Copy	Copy	Y=Yes N=No Other=Undef.	Ignored	No	No	Copy	Ignored	Ignored
BO	FLD	Copy	Copy	Y=Yes N=No Other=Undef.	Copy (*1)	No	No	Copy	Ignored	Ignored

Notes:

- (*1) If the hardware requirements for the allocation of a hardware channel are met. Otherwise the location is reset.
- (*2) If the variable is allocated. Otherwise No.
- (*3) If AENGUNIT field is missing in the dBASE file, the engineering units field contents of the variable in the FSC database is left unchanged. If it concerns a new variable, then the initial contents of the engineering units field will be 5 spaces.
- (*4) ANTOP, ANBOTTOM: if both are NOT available, then the top and bottom field contents of the variable in the FSC database is left unchanged. If it concerns a new variable, then the initial contents will be 0. If ANTOP available and ANBOTTOM is not (or vice versa) then the program will NOT import the contents of just 1 field (both fields must be available).

- (*5) If illegal situations occur, i.e. $|top - bottom| < 0.001$, the values of bottom and top as specified in the dBase file will not be accepted: the top and bottom field contents of the variable in the FSC database is left unchanged. If it concerns a new variable, then the initial contents of the top and bottom value will be 0.
- (*6) If the ANBOTTOM or ANTOP values are $> 10^{-6} - 1$ (maximum value) or $< -10^{-5} + 1$ (minimum value), then these new value settings are not imported nor clamped to the maximum values. If it concerns a new variable, then the initial contents of the top and bottom value will be 0.
- (*7) If bottom and top have been imported, the SER setpoints can have values outside the range of the new top and bottom scale values! SER setpoints will be re-scaling if necessary.
- (*8) (Analog Output only): If necessary, the program sets the power-on value of an AO to the outer boundary (if it becomes out of range).
- (*9) (Analog Input only): Note that if the analog range setting changes, the maximum discrepancy value of the AI will be recalculated by the program.

Table 4-12 Field interpretation for dBASE imports (continued)

Type	SER / SER sequence	Central Part / COM module channel / RKE3964 block (location COM)	System (location FSC)	Relative address (Modbus address (*7))	DCS address
I	Y=Yes Other=No Copy SER sequence (*10)	Copy (*11)	Ignored	Copy (*13)	Copy (*15)
I	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Copy (*12)	Copy (*13)	Copy (*15)
I	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Ignored	Ignored	Copy (*15)
I	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Ignored	Ignored	Copy (*15)
AI	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Ignored	Ignored	Copy (*15)
BI	No	Copy (*4)	Ignored	Copy (*13)	Copy (*15)
BI	No	Ignored	Copy (*12)	Copy (*13)	Copy (*15)
BI	No	Ignored	Ignored	Ignored	Copy (*15)
O	Y=Yes Other=No Copy SER sequence (*10)	Copy (*4)	Ignored	Copy (*13)	Copy (*15)
O	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Copy (*12)	Copy (*13)	Copy (*15)
O	No	Ignored	Ignored	Ignored	Copy (*15)
O	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Ignored	Ignored	Copy (*15)
AO	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Ignored	Ignored	Copy (*15)
BO	Y=Yes Other=No Copy SER sequence (*10)	Copy (*4)	Ignored	Copy (*13)	Copy (*15)
BO	Y=Yes Other=No Copy SER sequence (*10)	Ignored	Copy (*12)	Copy (*13)	Copy (*15)
BO	Ignored	Ignored	Ignored	Ignored	Copy (*15)

**Notes:**

- (*10) If the requirements for the SER and allocation of an SER sequence number are met.
- (*11) If a communication channel exists with a device's protocols, the RKE block number must exist also. Otherwise the allocation is reset.
- (*12) If the connection with this system is already made. Otherwise the allocation is reset.
- (*13) If the relative address is in the range and can be located. Otherwise the allocation is reset.
- (*14) See appendix F for a specification of Modbus addresses.
- (*15) If the DCS address is in the range and can be allocated. Otherwise the allocation is reset.

4.4.3.4 Using dBASE

In dBASE the possibility exists to change the specification of existing variables. It is also possible to add variables to the database. You cannot delete variables from the variable database by simply removing them from the dBASE file. These variables remain unchanged after a read dBASE command.

Examples of dBASE usage:

1. Creation of lots of tag numbers using a small dBASE program. Signal specs does not provide options to create lots of variables automatically. By writing a small dBASE program you could, for example, create tag numbers like 01-HLS-XXXHH, where XXX is a number from 000 to 999.
2. Allocation of hardware or communication (FSC or COM) variables. Hardware specs does not provide options to allocate variables to hardware automatically. After writing the variables to a dBASE file, variables with location FLD can be allocated easily by using a simple dBASE program.
3. Updating of variable allocations if lots of variables are moved from, for example, on rack to another or from one communication channel to another. Hardware specs does not provide options to reallocate hardware allocated variables automatically. If for some reason multiple I/O modules must be moved from rack X, position Q to rack Y, position R, a simple dBASE program can be used to reallocate the variables.
4. Updating the SER sequence numbers of variables. The program does not provide an option to, for example, automatically allocate SER sequence numbers for all the variables that do not have an SER sequence number yet. By using a simple dBASE program, these SER sequence numbers can be generated quickly.
5. Creating tables containing, for example, all SER sequence numbers with accompanying tag numbers. By using a simple dBASE program all variables with an SER sequence number can be filtered and stored in a different dBASE file.

4.5 Hardware specs

This option allows you to define or edit the hardware specifications of a variable. Every variable type has its own kind of hardware specification.

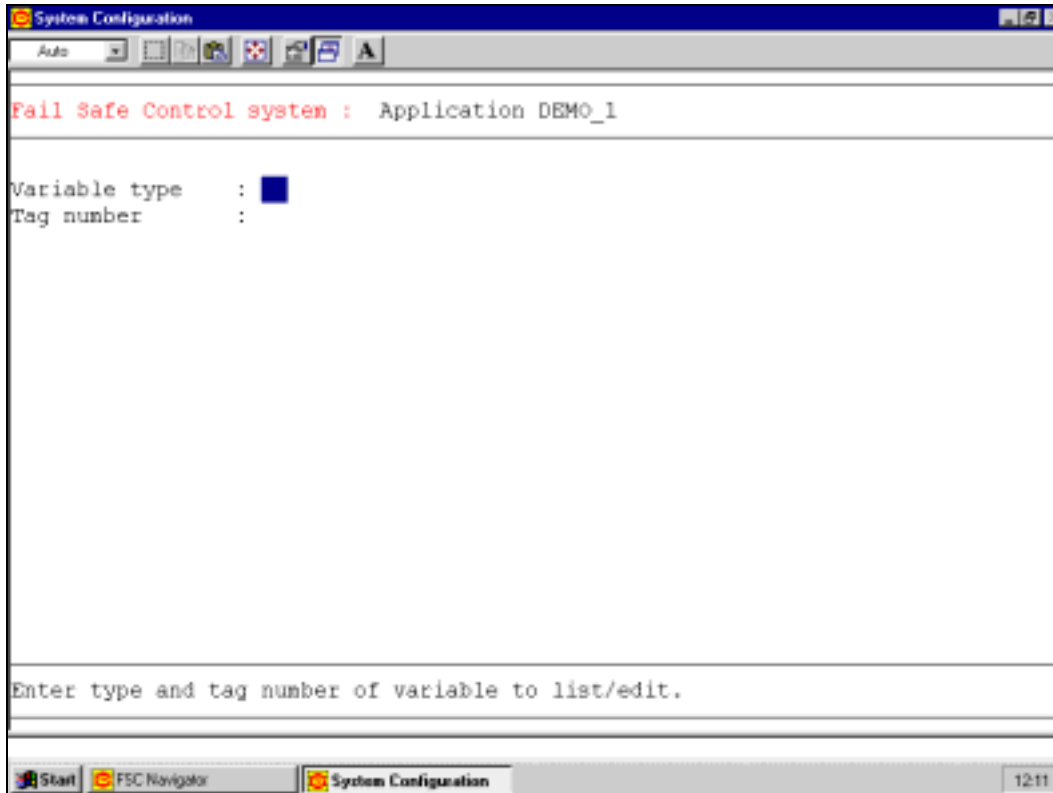


Figure 4-29 Hardware specs

- Variable type : Enter the variable type
If you enter 'NA', the program will show the unallocated variables.
- Tag number : Enter the tag number of the variable you want to examine
If you enter an incomplete tag number, the program will show you all variables that match the partial tag number.

4.5.1 Allocation of I/O variables

All I/O variables have to be allocated. Variables with allocation 'COM' or 'FSC' have to be allocated to the channel of a COM module. Other variables have to be allocated to the channels of I/O modules.

4.5.1.1 Allocation to I/O modules

Inputs and outputs which are not used for communication (location not 'COM' or 'FSC') have to be allocated to channels of I/O modules. (Digital inputs and outputs with location 'ANN' do not have to be allocated, except for outputs whose alarm type is 'Alarm').



Therefore a rack number, position and channel have to be defined per variable. For safety-related variables of type I and AI with non fail-safe input sensors, one or two redundant channels can be allocated (see subsection 4.5.1.1.3). For multiple inputs and outputs (BI, XI, BO, XO), channels must be allocated for every input or output (see subsection 4.5.1.1.4). A useful tool to see the allocations of variables is the <F2> function key. This will show the hardware allocations.

4.5.1.1.1 Rack

This field specifies the number of the rack, in which the module is placed to which the variable can be allocated. The program checks if the rack number is defined and if it is a Central Part rack with a SBD, or a rack that is controlled by a VBD.

Entering zero or changing the rack number will deallocate all the channels that are allocated to the variable.

After entering a valid rack number, a picture in the lower left corner of the screen shows you which modules are placed in the rack. Modules that can be selected for the variable are highlighted. If there are one or more modules highlighted, the position can be entered, otherwise the position field is skipped.

4.5.1.1.2 Position

This field specifies the position in the selected rack, at which the module is placed to which the variable can be allocated. Enter the position of a module that is highlighted in the lower left corner of the screen. A module can be selected if:

- Its position is within the set of possible positions.
This set is:
 - for Central Part rack with a SBD, all positions left of the Central Part.
 - 1..18 if the VBD controls single I/O
 - 1, 3, 5, 7, 9, 11, 13, 15, 17 if the VBD controls redundant I/O.
- It is of a type suited for the variable. For example, an analog input can only be connected to an analog input module.

Table 4-12 I/O modules suited for the different variable types

Inputs		Outputs		
I, BI, XI	AI	O, BO, XO		AO
10101/1/1	10102/1/1	10201/1/1	10213/1/2	10205/1/1
10101/1/2	10102/1/2	10201/2/1	10213/1/3	10205/2/1
10101/1/3	10102/1/3	10203/1/2	10213/2/1	
10101/2/1	10102/1/4	10206/1/1	10213/2/2	
10101/2/2	10102/2/1	10206/2/1	10213/2/3	
10101/2/3	10105/2/1	10207/1/1	10214/1/2	
10103/1/1		10208/1/1	10215/1/1	
10103/1/2		10208/2/1	10215/2/1	
10104/1/1		10209/1/1	10216/1/1	
10104/2/1		10209/2/1	10216/2/1	
		10212/1/1	10216/2/3	
		10213/1/1		

- The module is safety-related if the variable is also safety-related.
- There are channels free on the module.
- The module is defined as being placed

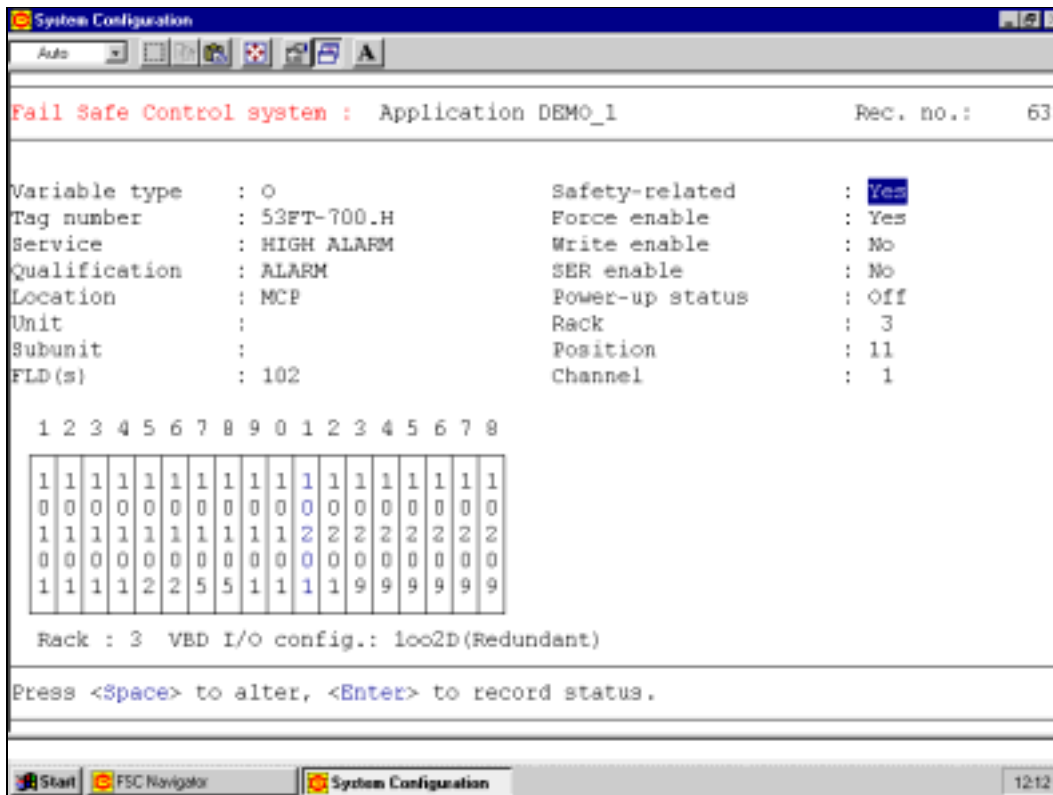


Figure 4-30 Position

The modules at position 1 to 10 cannot be chosen because they are input modules. The modules at position 13 to 18 are of the correct type but cannot be chosen because they are, for example, not safety-related whereas the signal is, or there are no free channels left on the module. The 10201 modules remain, but only position 11 can be selected, because the VBD that controls the rack controls redundant I/O.

Entering zero or changing the position will deallocate all channels that were already allocated. After entering a valid position, a picture in the lower left corner of the screen shows you which channels are still free on the module.

4.5.1.1.3 Channel

This field specifies the channel of the selected module, to which the variable can be allocated. The channels that are still free and can be selected are shown in the lower left corner of the screen. After entering a value the program checks if:

- The entered value is not greater than the number of channels on the module.
- It is tried to connect a safety-related variable to channel 5.8 of a 10212/1/1 module, as these channels can only be used for not safety-related variables.
- The channel is still free. If it is not, the message line will tell you to which variable the channel is allocated.

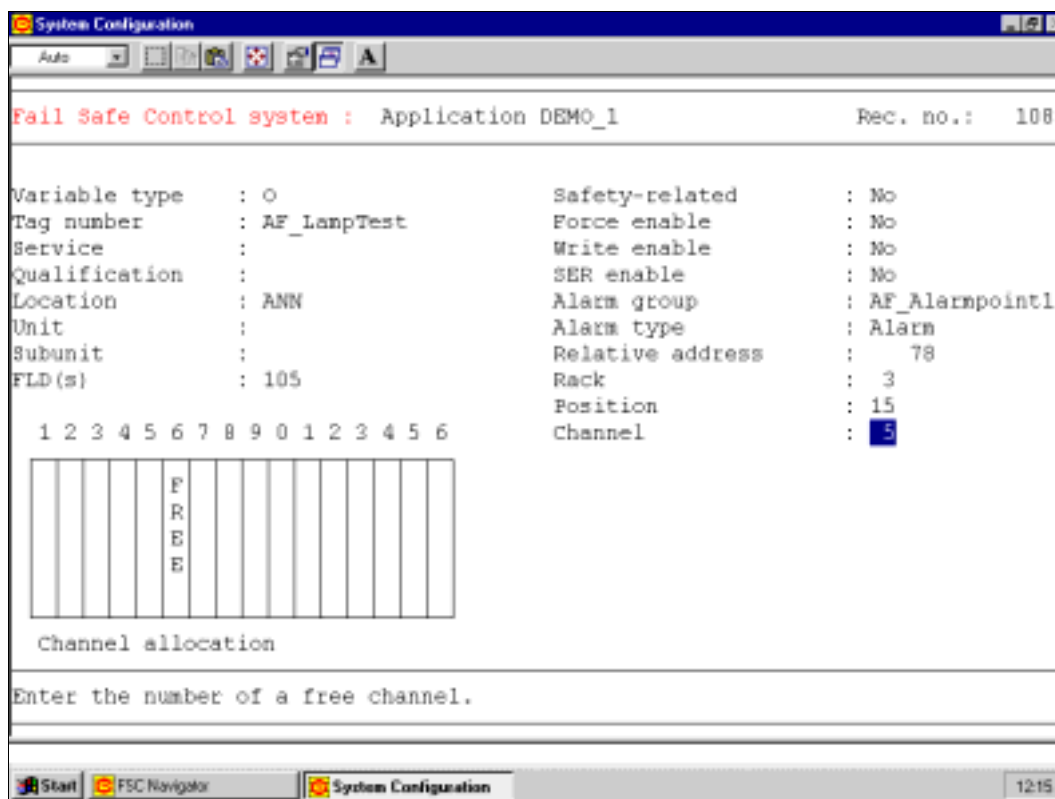


Figure 4-31 Channel

The picture tells that channel 6 is still free and can be selected to allocate the variable to. Channels 1 to 5 and 7 to 16 have already been allocated.

Entering zero or changing the channel will deallocate a channel that was already allocated.

4.5.1.1.4 Multiple inputs and outputs

For variables of type BI, XI, BO and XO channels, have to be allocated for the total number of bits. The fields rack, position and channel now define where the allocation starts. Starting with the defined channel, either enough consecutive channels have to be free to allocate all the bits, or all the channels until the end of the module have to be free. In the later case extra modules can be used to allocate the remaining bits: up to three extra modules for the BIs and BOs and for the XIs and XO.

Apart from the considerations above, the following applies:

- All the channels needed have to be allocated at once.
Changing the allocation will deallocate all channels that were already allocated. Pressing <Esc> aborts the allocation.
- For BI's and BO's with location 'MUX', and for XI and XO, only single I/O is allowed. This means the VBD controlling the rack cannot control redundant I/O.
- The extra racks must be controlled by the same VBD as the main rack.
- When extra modules are used, allocation always starts with the first channel. Enough consecutive channels have to be free to allocate all the remaining unallocated bits, or all channels have to be free.
- The extra modules have to be of the same type as the main module

4.5.1.2 Allocation to COM modules

Inputs and outputs which are used for communication (location is 'COM' or 'FSC') have to be allocated to a channel of a COM module.

4.5.1.2.1 Location 'FSC'

Inputs and outputs with location 'FSC' have to be allocated to a channel with protocol FSC-FSC.

For inputs with location 'FSC' only a system number has to be defined.

If you press <F2>, in a record with location FSC, you can see all allocated variables.

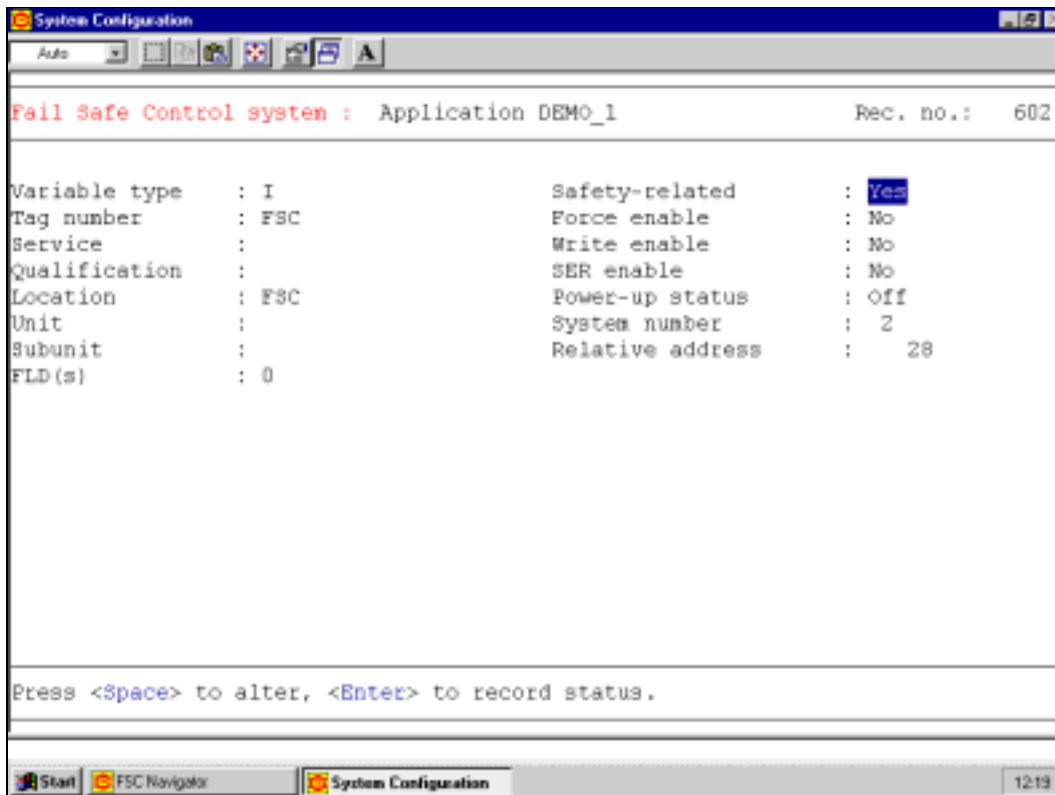


Figure 4-32 Allocation to a COM module (Location 'FSC')

4.5.1.2.1.1 System number

Variables with location 'FSC' are used for the communication with another FSC system.

Variables are allocated to an FSC-FSC link by selecting the number of the target FSC system. The program will show you the available FSC systems, which can be selected. This means that the link must be defined in the communication channel definition. The allocated system is highlighted in the window. You can (re)allocate a system by entering <S>.

You can deallocate a system by pressing <D> on the spot of the highlighted system number.

The program will ask you to allocate the variable automatically. If you confirm the variable will be allocated to the first free address. Otherwise you can enter a free address manually.



4.5.1.2.1.2 Relative address

In general a variable is automatically allocated to a free address in the memory area reserved for the system number. The address displayed is relative to the start address of the reserved memory area. This address should be the same as the address of the corresponding variable in the connected FSC system.

4.5.1.2.2 Location 'COM'

Inputs and outputs with location 'COM' have to be allocated to a channel with protocol UCN, Modbus RTU, Modbus H&B, development system or RKE3964R.

For inputs with location COM, a Central Part, communication module and channel have to be defined per input. If you press <F2>, in a record with location COM, you can see all allocated variables.

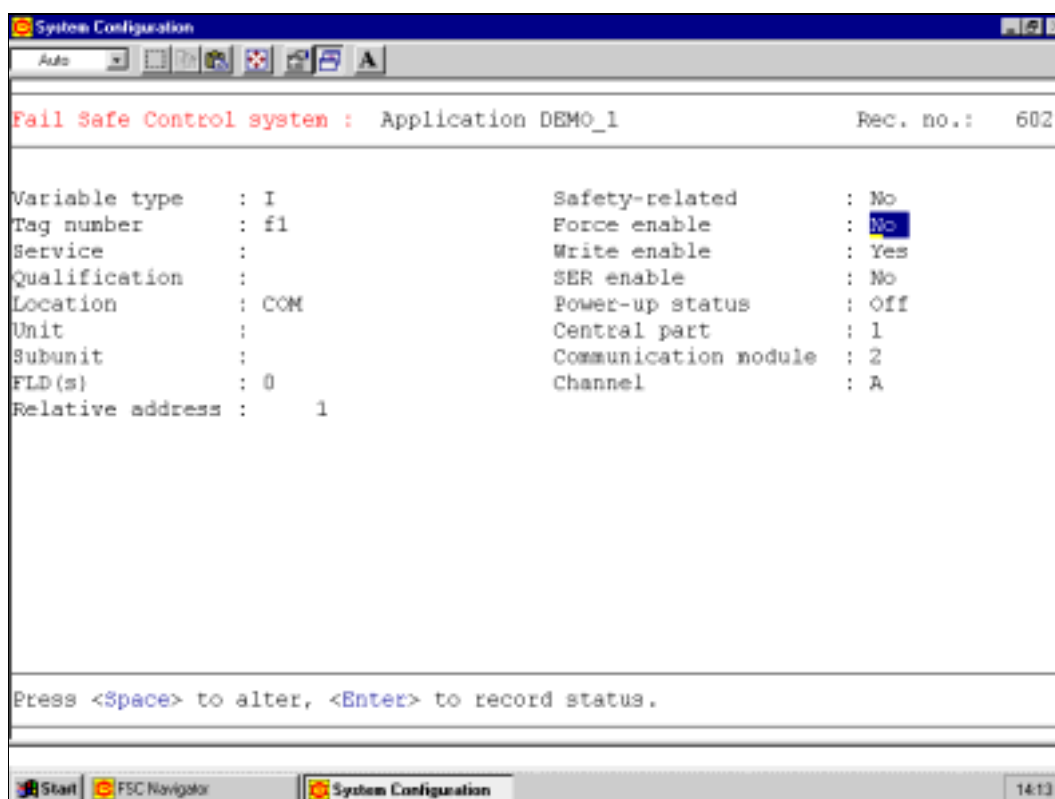


Figure 4-33 Allocation to COM module (Location COM)

4.5.1.2.3 Central Part

Select a Central Part by toggling. Possible values depend on the FSC system type. Selecting 'Undef' removes an existing allocation.



4.5.1.2.4 Communication module

After selecting a Central Part, a communication module placed in that Central Part can be selected by toggling.

Note:

Selecting 'Undef' removes an existing allocation.

4.5.1.2.5 Channel

Select a channel (A or B) on the specified communication module. Selecting 'Undef' removes an existing allocation.

Variables with location 'COM' can only be allocated to channels with protocol Development system, UCN, Modbus RTU, Modbus H&B, RKE3964R or printer (for BO SER variables only).

If the amount of memory available allows it, the variable is allocated to the selected channel. The address, relative to the start of the memory area of the COM channel, is displayed at the bottom of the left column on the screen.

4.5.1.2.6 Relative address

In general a variable is automatically allocated to a free address in the memory area reserved for the channel. The address displayed is relative to the start address of the reserved memory area.

4.5.1.3 Automatic allocation of variables

After allocating a variable you will get the question:

Do you want to step through the database to copy this allocation? (Y/N)

Answering 'Yes' gives you the possibility to copy the allocation you just made to other variables of the same type and with the same location which have not been allocated yet.

The next question after answering 'Yes' will be:

Do you want to confirm the allocation of every variable? (Y/N)

If you answer 'No', then the allocation will automatically be copied to every suitable unallocated variable found.

If your answer is 'Yes', the program will display every unallocated variable found and will ask: 'Copy allocation? (Y/N)'

For variables of type BI and BO you may change the register type of the variables selected for copying, because the address depends on the register type.

For variables connected to FSC-FSC channels, each relative address can be entered manually.

The program will ask: 'Allocate all variables automatically? (Y/N)'

If you answer with 'Yes', the address is allocated automatically. Otherwise you can edit the relative address of each variable manually.

The copying process can be interrupted at any time by pressing <Esc>.



4.5.1.4 Automatic allocation of SER sequence numbers

If you activate numerical SER, SER sequence numbers must be allocated for variables that are SER enabled. A possibility exists to allocate SER sequence numbers for variables that are SER enabled, but do not have an SER sequence number yet.

This option can be used, e.g.:

- If the SER sequence range changes. SER sequence numbers falling out of the range will then be deallocated.
- If SER has been enabled for a lot of variables and the user activates numerical SER.

The option is activated if the SER sequence number is assigned for a variable. The program asks 'Do you want to step through the database to allocate sequence numbers? (Y/N)' and confirmation is requested. If you want to allocate sequence numbers, the program scans the variable database for variables that are SER enabled, but do not have a sequence number yet. If such variables are found, the program asks: 'Allocate all sequence numbers automatically? (Y/N)' and confirmation is requested. If such variables are not found, the message 'No more unallocated SER sequence numbers' will be displayed.

If automatic allocation is requested, the program runs through the variable database to allocate SER sequence numbers. If the area for SER sequence numbers is full, automatic allocation is aborted with the message: 'Area for SER sequence numbers is full'. If the automatic SER sequence allocation finished, the message: 'No more unallocated SER sequence numbers' is displayed.

If no automatic allocation is requested, the program searches for the first unallocated SER sequence number and allows you to enter a sequence number manually. A free sequence number is already displayed by default. The user can accept this sequence number or enter another (not used) sequence number. If a sequence number has been entered, the program searches for the next unallocated SER sequence number. If no unallocated SER sequence numbers remain, the message: 'No more unallocated SER sequence numbers' will be displayed.

The program always returns to the variable where the automatic allocation of SER sequence numbers was started.

<Esc> can be used to abort the allocation of SER sequence numbers at any time.

4.5.1.5 Scaling values

The range of an analog input value can be determined from the bottom and top scale values. These values contain the bottom and top scale values. If the bottom scale value is smaller than the top scale value, the scaling is positive, else the scaling is negative. So, the bottom scale value does not always contain the lowest value and the top scale does not always contain the highest value.

4.5.2 Alarm configuration

For more information refer to Appendix A of the FSC Software Manual ("Alarm functions").

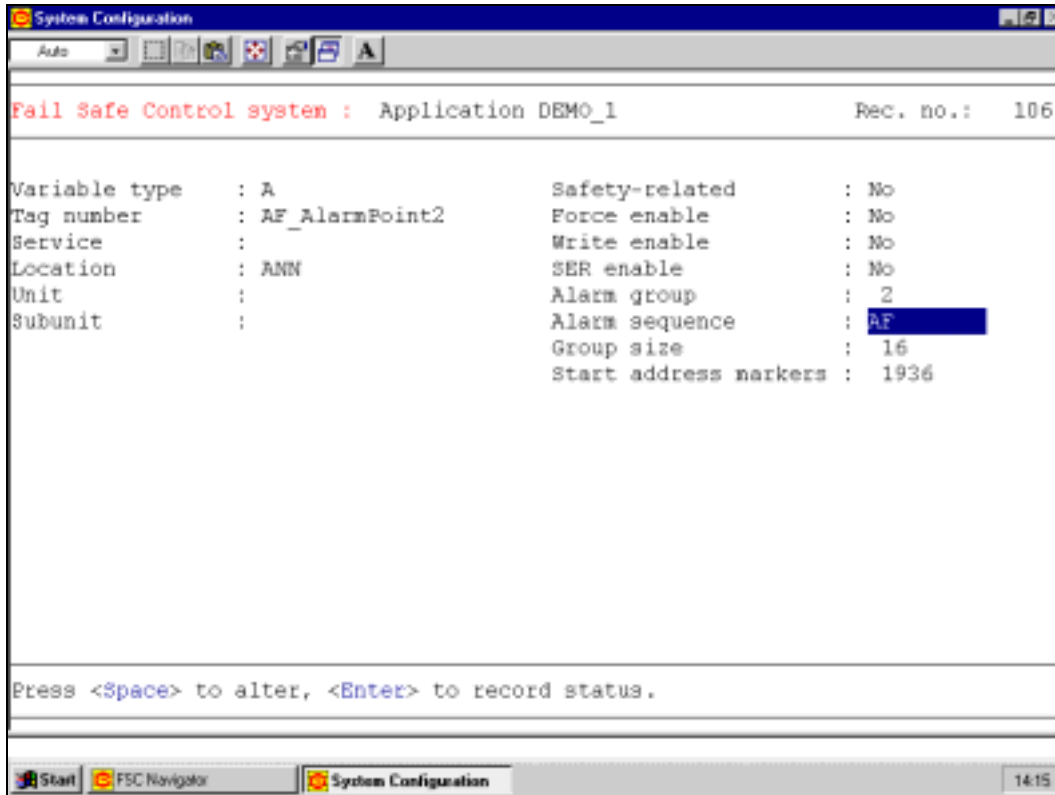


Figure 4-34 Hardware specs: A

Safety relation	: Cannot be modified. Always No.
Force enable	: Cannot be modified. Always No.
Write enable	: Cannot be modified. Always No.
SER enable	: Cannot be modified. Always No.
Alarm group	: Cannot be modified. The alarm group number is generated by the program. Maximum number of alarm groups: 64.

**Alarm sequence**

: Possible values :

- AF	Basic flashing
- AM	Manual lamp reset
- FR	Flasher reset
- FRM	Flasher / lamp reset
- AR	Ringback
- ARR	Double audible ringback
- TFS	First out indication
- TFS_COM1	First out indication transmitter FSC system.
- TFS_COM2	First out indication receiver FSC system.

Press <Space> to select alarm sequence.

Group size

: Specifies the number of digital outputs with location 'ANN' and the alarm type 'Alarm' that can be connected to the alarm. Minimum group size is 8. Maximum group size is 256.

Start address markers

: For each alarm group, memory is allocated in the marker area. Inputs and outputs connected to the alarm are allocated within this area. This field gives the start address of the allocated area. It is generated by the program and cannot be altered. The number of markers allocated depends on the selected alarm sequence and the alarm group size.

4.5.3 Digital input signal (DI)

There are several kinds of inputs, depending on the location field:

- Location = 'COM' : The input is received via a non-FSC communication channel.
- Location = 'FSC' : The input is received from another FSC system.
- Location = 'ANN' : The input is an annunciator control input
- Location = 'SYS' : The input is a system variable.
- Other locations : The input is a normal digital input.

4.5.3.1 Location = 'COM'

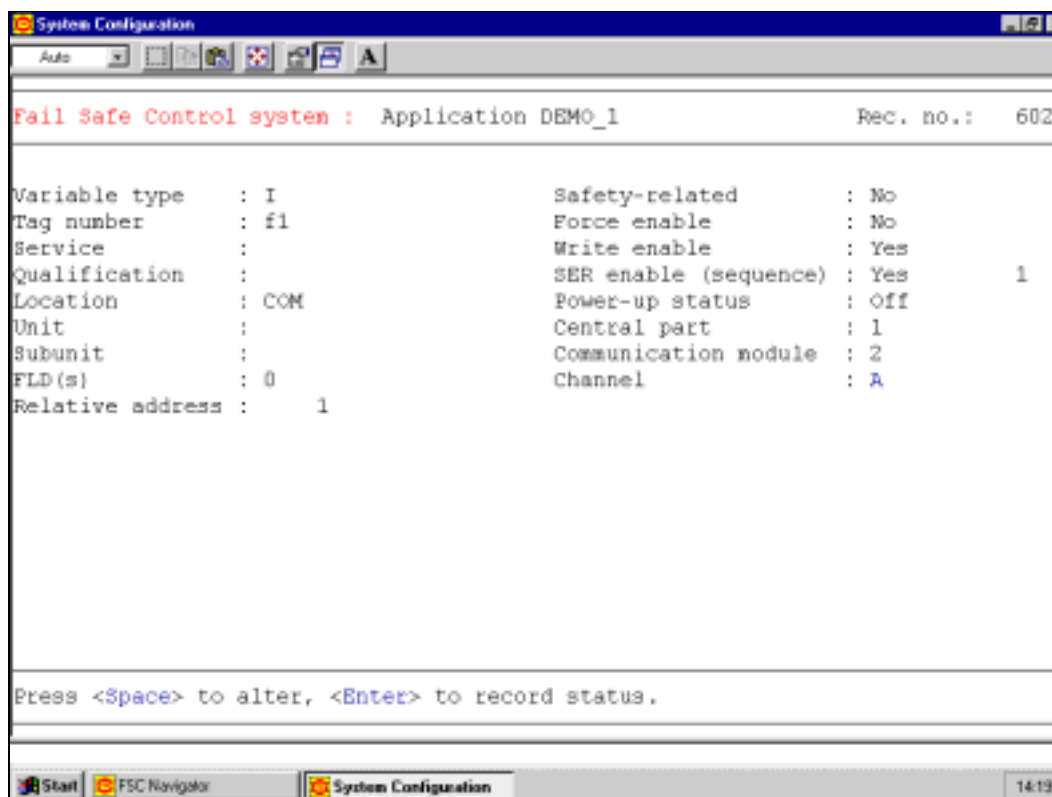


Figure 4-35 Hardware specs: I : Location = 'COM'

Safety-related : Cannot be modified. Always No.

Force enable : Press <Space> to alter.

Write enable : Cannot be modified. Always Yes.

SER enable (sequence) : Press <Space> to alter.
 If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER



enable and SER sequence number.

Power-up status	: If this field is On, the 'input' is set to 1 after power-up. If it is Off the 'input' is set to 0 after power-up. The default value is Off. Press <Space> to alter the power-up status.
Central Part	: Select a Central Part by toggling.
Communication module	: Select a module by toggling
Channel	: Select a channel by toggling.
Relative address	: If you allocate automatically, the address is calculated by the program, otherwise the relative address has to be entered by the user.
DCS address	: Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.

4.5.3.2 Location = 'FSC'

For more information see Appendix F of the FSC Software Manual ("FSC communication manual").

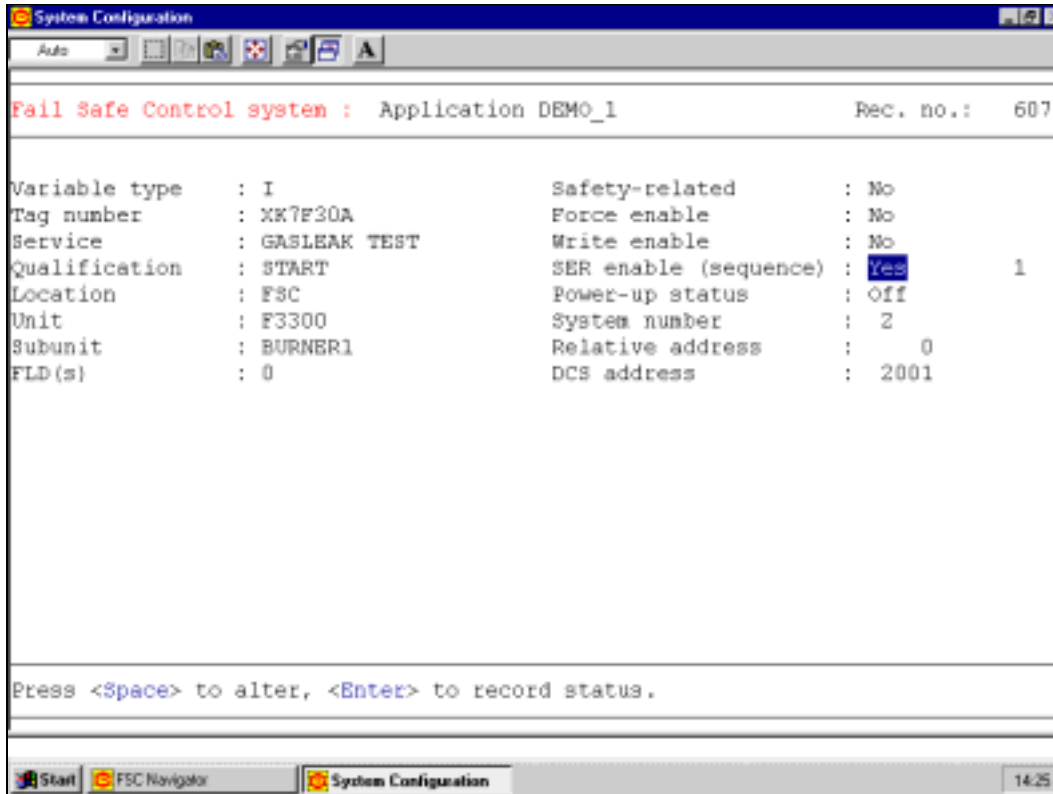


Figure 4-36 Hardware specs: I : Location = 'FSC'

Safety-related	: Press <Space> to alter.
Force enable	: Press <Space> to alter.
Write enable	: Cannot be modified. Always No.
SER enable (sequence)	: Press <Space> to alter. If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
Power-up status	: If this field is On, the 'input' is set to 1 after power-up. If it is Off the 'input' is set to 0 after power-up. The default value is Off. Press <Space> to alter the power-up status.



- System number** : Select the system number in the window that appears on the screen. The highlighted number is the selected system number. Press <D> on the spot of the highlighted number to deallocate the system, press <S> to select a system.
- Relative address** : If you allocate automatically, the address is calculated by the program, otherwise this can be entered by the user.
- DCS address** : Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.

4.5.3.3 Location = 'ANN'

For more information refer to Appendix A of the FSC Software Manual ("Alarm functions").

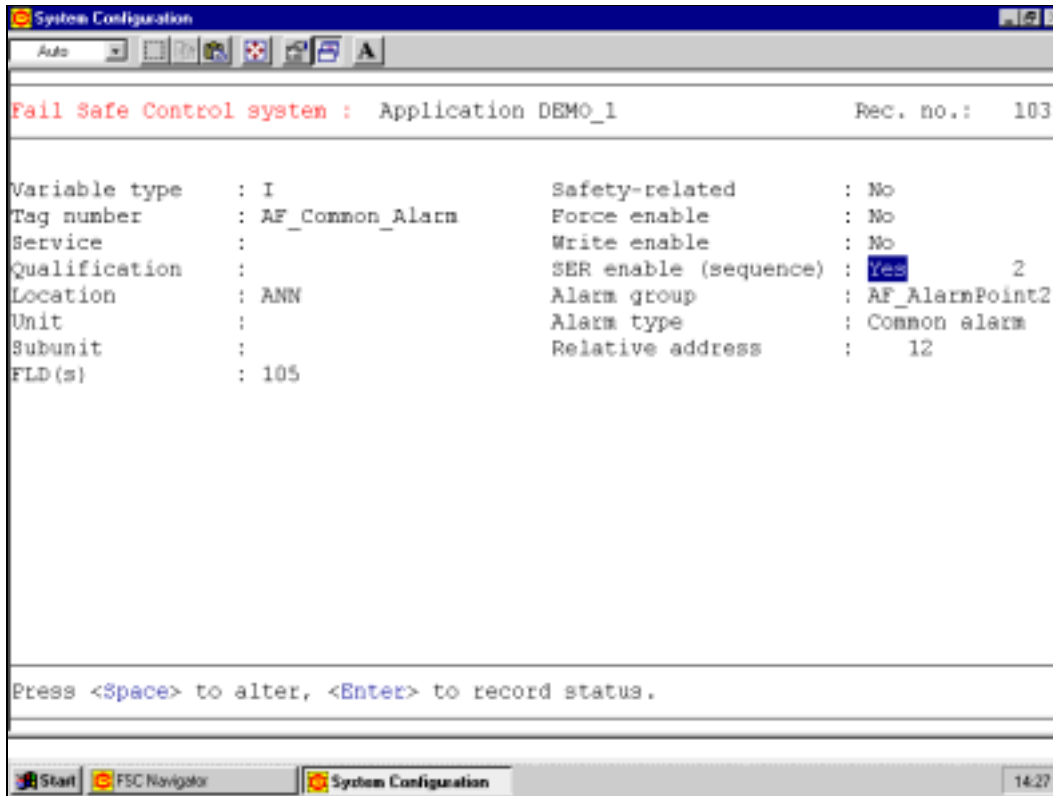


Figure 4-37 Hardware specs: I : Location = 'ANN'

- Safety-related** : Cannot be modified. Always No.
- Force enable** : Cannot be modified. Always No.
- Write enable** : Cannot be modified. Always No.
- SER enable (sequence)** : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
- Alarm group** : Press <S> to select the alarm group. <D> will deallocate the variable from the alarm group.



- Alarm type** : Press <Space> to alter the alarm type.
Possible values depend on the alarm sequence of the alarm group the input is connected to. Every alarm type can only be selected once per alarm group.
- Relative address** : Always calculated automatically.

4.5.3.4 Location = 'SYS'

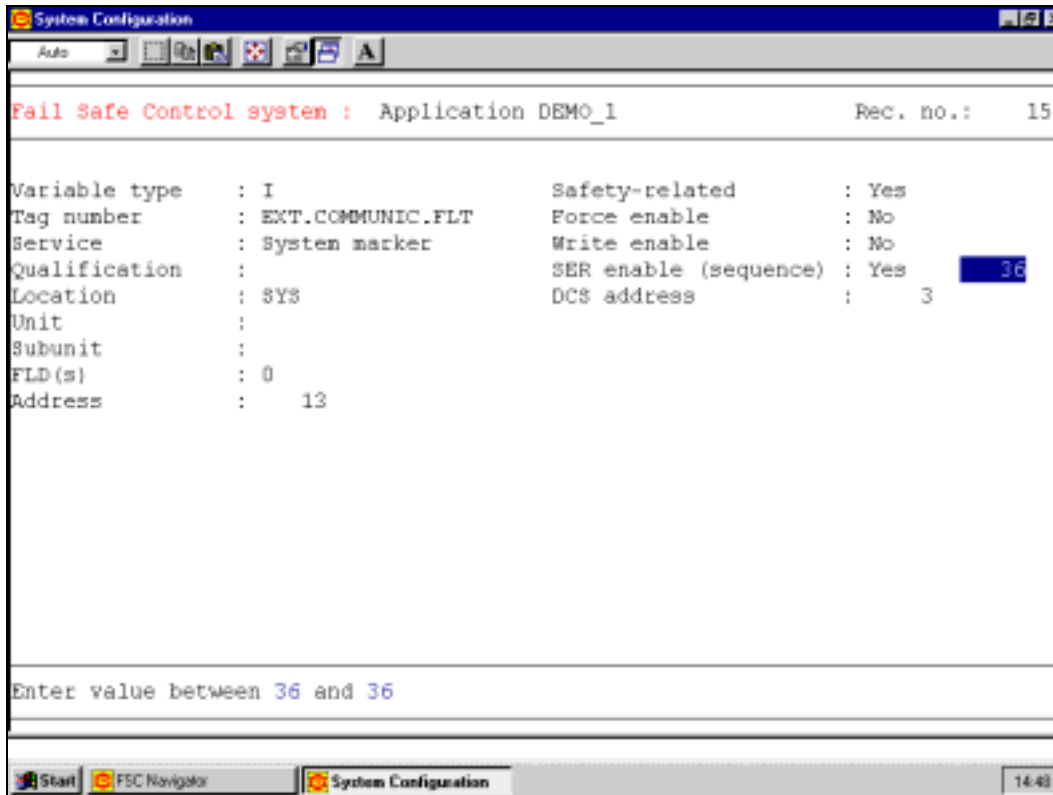


Figure 4-38 Hardware specs: I : Location = 'SYS'

- Safety-related** : Cannot be modified. Always Yes.
- Force enable** : Cannot be modified. Always No.
- Write enable** : Cannot be modified. Always No.
- SER enable (sequence)** : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
- DCS address** : Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.3.5 Other locations

System Configuration

Auto

Fail Safe Control system : Application DEMO_1 Rec. no.: 501

Variable type	: I	Safety-related	: Yes
Tag number	: 91XA-651A	Force enable	: No
Service	: Door switch	Write enable	: No
Qualification	: Close	SER enable (sequence)	: Yes 0
Location	: AH	Rack	: 3
Unit	: 5000	Position	: 9
Subunit	: 91UZ-650	Channel	: 13
FLD(s)	: 0	DCS address	: 2002
		Sensor is fail-safe	: No
		Sensor is redundant	: Redund. 202
		Max. discrepancy time	: 10 s
		Maximum on-time	: 28800 s
		Second input rack	: 3
		Second input position	: 9
		Second input channel	: 14

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	2	5	5	1	1	1	1	9	9	9	9	9	9	9

Rack : 3 VBD Func.: Redundant

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 4/27

Figure 4-39 Hardware specs: I : Other locations

Safety-related : Press <Space> to alter.

Force enable : Press <Space> to alter.

Write enable : Cannot be modified. Always No.

SER enable (sequence) : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.

Rack : Enter the rack number where the variable will be allocated.

Position : Enter the position in the rack where the variable will be allocated.

Channel : Enter the channel number of the module where the variable will be allocated.



Sensor is fail-safe : Press <Space> to define the fail-safe status of the sensor. Can only be changed if the variable is safety-related.

If the sensor is NOT fail-safe, define the type of redundancy required. For more information refer to Appendix C of the FSC Software Manual ("Safety-related inputs with non-fail-safe sensors").

Sensor is redundant : Press <Space> to select the redundancy type.

Max discrepancy time : Enter the maximum discrepancy time. The value must be in the range 1 to 2047 seconds, The default value is 10 seconds.

Maximum on time : Enter the value of the maximum on time or deactivate the maximum on time. The value must be in the range of 10 - 122820 seconds (2047 minutes). The default value is 28800 seconds (= 8 hours). Deactivate the maximum on time by pressing <D>.

Second and third input rack, position and channel : Enter the second and third input rack, position and channel (depends on the redundancy type).

DCS address : Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.4 Analog input signal (AI)

System Configuration

Fail Safe Control system : Application DEMO_1 Rec. no.: 90

Variable type	: AI	Safety-related	: Yes
Tag number	: PRESSURE	Force enable	: Yes
Service	:	Write enable	: No
Location	: FLD	SER enable	: No
Unit	:	Signal type	: 4..20 mA
Subunit	:	Trm. alarm setp. low	: 3 mA
FLD(s)	: 105		20.93 mA
		Engineering units	: %
		Bottom scale	: 3 %
		Top scale	: 97 %
		SER setpoint low	: 0 %
		SER setpoint high	: 0 %
		Rack	: 3
		Position	: 5
		Channel	: 3
		DCS address	: -1
		Sensor is fail-safe	: Yes

1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	2	2	5	1	1	1	1	9	9	9	9	9	9	9

Rack : 3 VBD I/O config.: 1002B(Redund)

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 14:33

Figure 4-40 Hardware specs: AI

Safety-related : Press <Space> to alter.

Force enable : Press <Space> to alter.

Write enable : Cannot be modified. Always No.

SER enable (sequence) : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.

Signal type : Press <Space> to select the signal type which matches with your configuration.



Trm. alarm setp. low and Trm. alarm setp. high	: The minimum and the maximum alarm value must be entered. If the analog input goes below the minimum setpoint or the analog input goes above the maximum setpoint, the system will give an analog sensor break alarm. You can press <D> to deactivate each.
Engineering units	: Enter the units which are measured with this variable. You can enter 5 characters.
Bottom scale	: Enter the lowest possible value (in engineering units) of the variable which is measured.
Top scale	: Enter the highest possible value (in engineering units) of the variable which is measured.
SER setpoint low	: If SER enable is Yes, a value can be entered to generate an SER message as soon as the analog input goes below this value. Default is the bottom scale.
SER setpoint high	: If SER enable is Yes, a value can be entered to generate an SER message as soon as the analog input goes above this value. Default is the top scale.
Rack	: Enter the rack number where the variable will be allocated.
Position	: Enter the position in the rack where the variable will be allocated.
Channel	: Enter the channel number of the module where the variable will be allocated.
Sensor is fail-safe	: Press <Space> to define the fail-safe status of the sensor. Can only be changed if the variable is safety-related.
If the sensor is NOT fail-safe, define the type of redundancy required. For more information refer to Appendix C of the FSC Software Manual ("Safety-related inputs with non-fail-safe sensors").	
Sensor is redundant	: Press <Space> to select the redundancy type. Depending on the redundancy type, more channels may need to be allocated to this variable.
Max discrepancy value	: Enter the maximum discrepancy value between redundant analog inputs. This is the maximum difference between all redundant input sensors of an analog variable. (Limited to 10 % of the span of the analog input as maximum and 0.1 % of the span for the minimum).
DCS address	: Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.4.1 Location = 'SYS' (for 10105/2/1 module only)

If one or more high-density analog input modules 10105/2/1 have been defined, you need to define the voltage monitoring settings for each 10105/2/1 module. This is done in the screen as shown in Figure 4-41. Please note that the tag number and allocation are predefined. The tag number consists of 'VM'+rack+position+channel. The rack, position and channel are automatically derived from allocation data that was entered earlier for the 10105/2/1 module.

Variable type	Tag number	Service	Location	Unit	Subunit	FLD(s)	Safety-related	Force enable	Write enable	SER enable (sequence)	Signal type	Trim. alarm setp. low	Trim. alarm setp. high	Engineering units	Bottom scale	Top scale	SER setpoint low	SER setpoint high	Rack	Position	Channel	DCS address
AI	VM 03/05/17		SYS			0	No	No	No	Yes	0..5 V	1 V	4 V	V	0 V	40 V	0 V	40 V	3	5	17	1000

Rack : 3 VBD I/O config.: 1002D(Redund)

Press <Space> to alter, <Enter> to record status.

Figure 4-41 Hardware specs: AI (voltage monitoring for 10105/2/1)

- Safety-related** : Cannot be modified. Always No.
- Force enable** : Cannot be modified. Always No.
- Write enable** : Cannot be modified. Always No.
- SER enable (sequence)** : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
- Signal type** : Cannot be modified. Always '0..5 V'.



Trm. alarm setp. low and Trm. alarm setp. high	: The minimum and the maximum alarm value must be entered. If the analog input goes below the minimum setpoint or the analog input goes above the maximum setpoint, the system will give an analog sensor break alarm. You can press <D> to deactivate each.
Engineering units	: Enter the units which are measured with this variable. You can enter 5 characters.
Bottom scale	: Enter the lowest possible value (in engineering units) of the variable which is measured.
Top scale	: Enter the highest possible value (in engineering units) of the variable which is measured.
SER setpoint low	: If SER enable is Yes, a value can be entered to generate an SER message as soon as the analog input goes below this value. Default is the bottom scale.
SER setpoint high	: If SER enable is Yes, a value can be entered to generate an SER message as soon as the analog input goes above this value. Default is the top scale.
Rack	: Cannot be modified.
Position	: Cannot be modified.
Channel	: Cannot be modified.
DCS address	: Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.5 Multiple input signal (BI)

The variable type BI can have four different kinds of hardware specifications, depending on the location field:

- Location = 'COM' : The BI variable is received via a non-FSC communication channel.
- Location = 'FSC' : The BI variable is received from another FSC system.
- Location = 'MUX' : The BI variable is a part of multiplexed I/O.
- Other locations : The BI variable is a multiple input, but is no part of multiplexed I/O.

4.5.5.1 Location = 'COM'

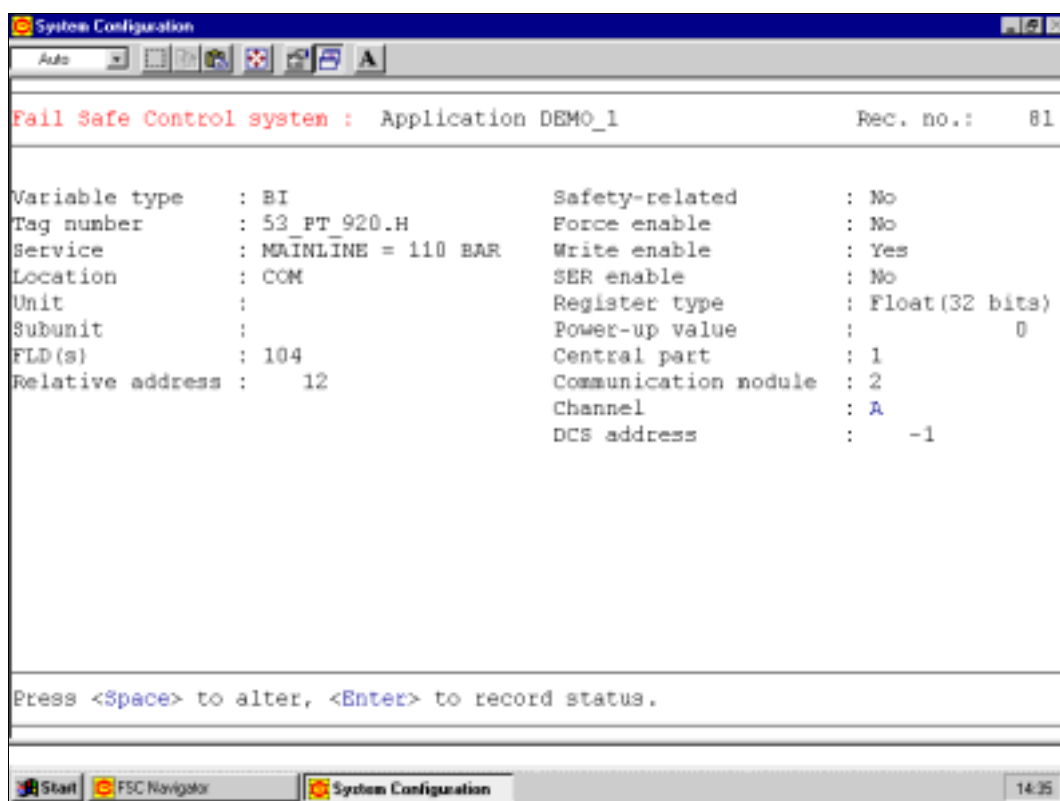


Figure 4-42 Hardware specs: BI : Location = 'COM'

- | | |
|-----------------------|---|
| Safety-related | : Cannot be modified. Always No. |
| Force enable | : Cannot be modified. Always No. |
| Write enable | : Cannot be modified. Always Yes. |
| SER enable | : Cannot be modified. Always No. |
| Register type | : Press <Space> to change the register type (Byte, Word, Long, Float).
If the register type is changed, the power-up value may be adjusted to the maximum value of that register type. |



Power-up value	: The value of the register when the FSC system is switched on. The range of this value is depending on the register type. The default value is 0.
Central Part	: Select a Central Part by toggling.
Communication module	: Select a module by toggling.
Channel	: Select a channel by toggling.
Relative address	: If you allocate automatically, the address is calculated by the program, otherwise it has to be entered by the user.
DCS address	: The DCS address can only be configured if the register type is 'Float'. Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.5.2 Location = 'FSC'

For more information refer to Appendix F of the FSC Software Manual ("FSC communication manual").

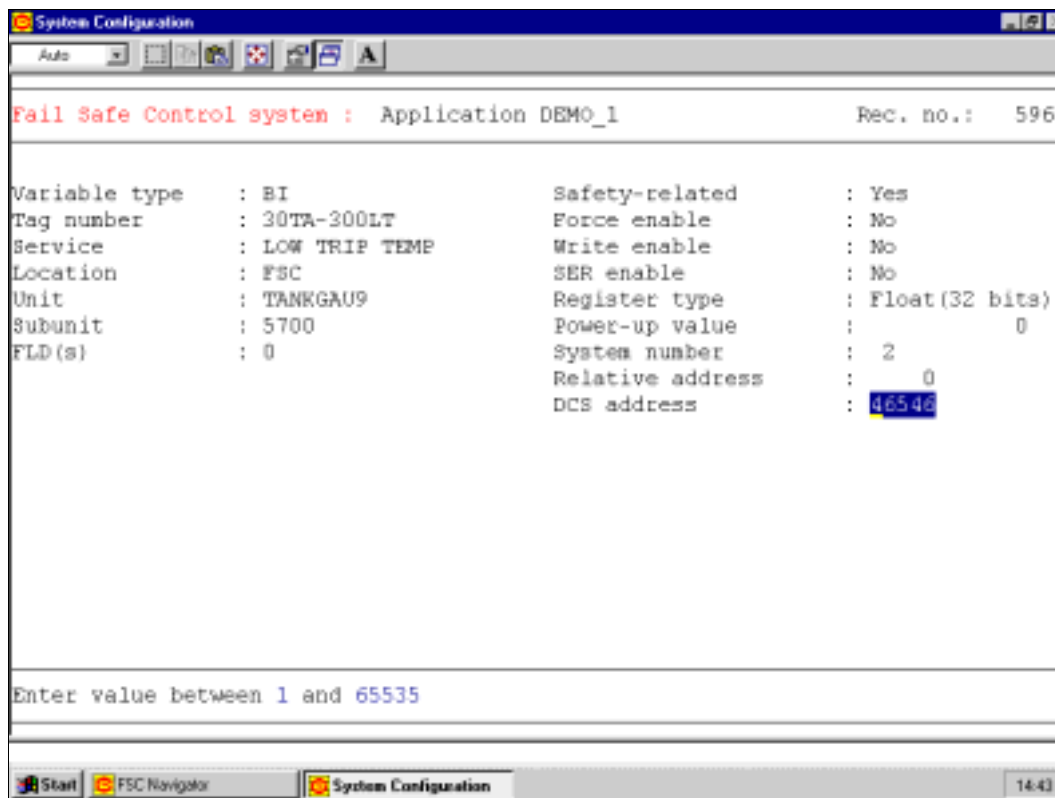


Figure 4-43 Hardware specs: BI : Location = 'FSC'

Safety-related	: Press <Space> to alter.
Force enable	: Cannot be modified. Always No.
Write enable	: Cannot be modified. Always No.
SER enable	: Cannot be modified. Always No.
Register type	: Press <Space> to change the register type (Byte, Word, Long, Float). If the register type is changed, the power-up value may be adjusted to the maximum value of that register type.
Power-up value	: The value of the register when the FSC system is switched on. The range of this value is depending on the register type. The default value is 0.
System number	: Select the system number in the window that appears on the screen. The highlighted number is the selected system number. Press <D> on the spot of the highlighted number to deallocate the system. Press <S> to select a system.



Relative address	: If you allocate automatically, the address is calculated by the program, otherwise it has to be entered by the user.
DCS address	: The DCS address can only be configured if the register type is 'Float'. Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.5.3 Location = 'MUX'

For more information see Appendix G of the FSC Software Manual ("FSC multiplexers").

The screenshot shows the 'System Configuration' window for 'Application DEMO_1' (Rec. no.: 576). It displays various configuration parameters for a variable of type BI. The parameters are organized into two columns. The first column includes: Variable type (BI), Tag number (WHEELSET 1), Service (ROT. COUNT), Location (MUX), Unit (3200), Subunit (32C0-10), FLD(s) (0), and Address (2512). The second column includes: Safety-related (No), Force enable (No), Write enable (No), SER enable (No), MUX tag number (KE-576-D1LOCAL), Data representation (Binary), Number of input bits (4), Register type (Byte (8 bits)), and Power-up value (0). Below these parameters is a table with 16 columns (labeled 1-16) and 4 rows of data. The data is as follows:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
								1	1	1	1	1	1	1	1
								0	0	0	0	0	0	0	0
								1	2	2	2	2	2	2	3
								0	0	0	1	1	0	0	0
								2	9	5	5	6	9	9	2

Below the table, it says 'Rack : 4 VBD I/O config.: 1001b(Single)'. At the bottom, it says 'Press <S> to (re-)select, <D> to deselect.' The taskbar at the bottom shows 'Start', 'FSC Navigator', and 'System Configuration'.

Figure 4-44 Hardware specs: BI : Location = 'MUX'

Safety-related	: Cannot be modified. Always No.
Force enable	: Cannot be modified. Always No.
Write enable	: Cannot be modified. Always No.
SER enable	: Cannot be modified. Always No.
Mux tag number	: Press <S> to select the XI variable. <D> will deallocate the BI.
Number of input bits	: Enter the number of bits of the variable. This value should be in the range 1 to 32 bits.
Data representation	: Press <Space> to select a Binary representation or a BCD representation.
Register type	: Cannot be changed, depends on the number of input bits.
Power-up value	: The value of the register when the FSC system is switched on. The range of this value depends on the register type. The default value is 0.

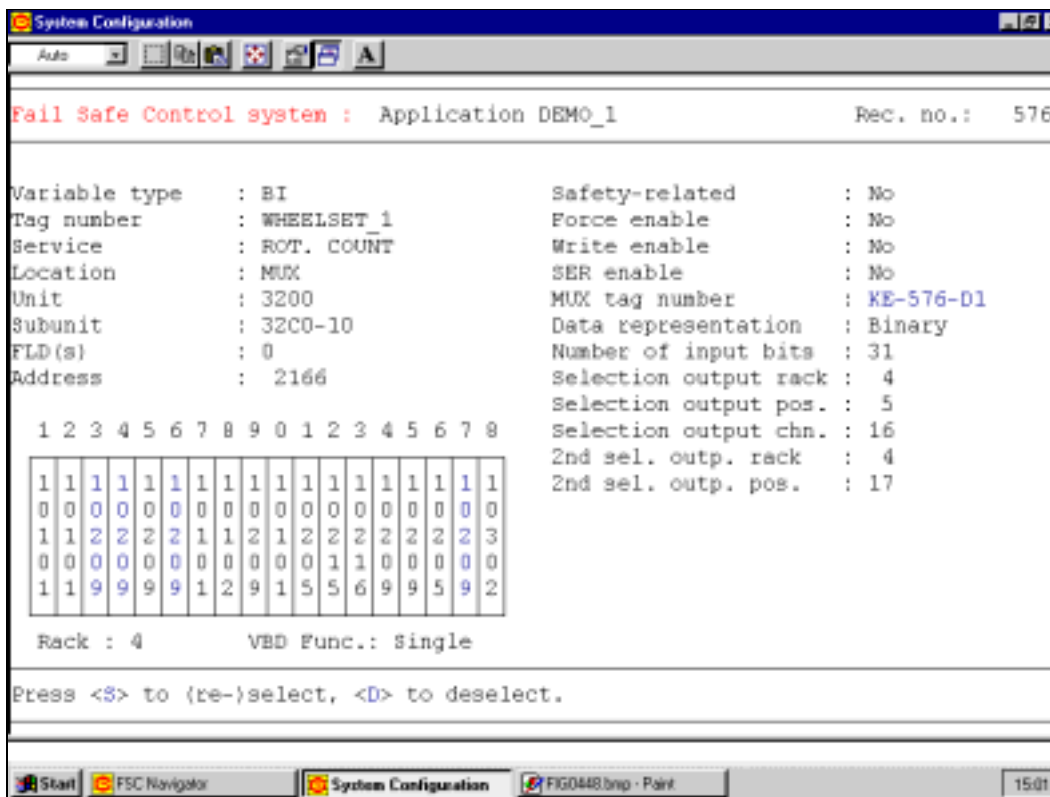


Figure 4-45 Hardware specs: BI : Location 'MUX'

- Selection output rack** : Enter the rack number where the variable will be allocated.
- Selection output pos.** : Enter the position in the rack where the variable will be allocated.
Remember to allocate a BI variable with Location = 'MUX' to output modules.
- Selection output chn.** : Enter the first channel number of the module where the variable will be allocated.
- 2nd, 3rd and 4th Sel. output rack and output position** : If it is not possible to allocate selection output bits on the module defined by Rack and position starting with channel, then one or more extra modules have to be defined to allocate the remaining bits.



4.5.5.4 Other locations

System Configuration

Fail Safe Control system : Application DEMO_1 Rec. no.: 80

Variable type	: BI	Safety-related	: No
Tag number	: 53_PT_920.L	Force enable	: No
Service	: MAINLINE = 75 BAR	Write enable	: No
Location	: FLD	SER enable	: No
Unit	:	Data representation	: Binary
Subunit	:	Number of input bits	: 2
FLD(s)	: 104	Register type	: Byte (8 bits)
Address	: 2258		

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	2	2	5	5	1	1	1	9	9	9	9	9	9	9

Rack : 3 VBD I/O config.: 1002D(Redund)

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 15:08

Figure 4-46 Hardware specs: BI : Other locations

Safety-related	: Press <Space > to alter.
Force enable	: Cannot be modified. Always No.
Write enable	: Cannot be modified. Always No.
SER enable	: Cannot be modified. Always No.
Data representation	: Press <Space> to select a Binary representation or a BCD representation.
Number of input bits	: Enter the number of bits of the variable. This value should be in the range 1 to 32 bits.
Register type	: Cannot be changed, depends on the number of input bits.

System Configuration

Fail Safe Control system : Application DEMO_1 Rec. no.: 80

Variable type	: BI	Safety-related	: No
Tag number	: 53_PT 920.L	Force enable	: No
Service	: MAINLINE = 75 BAR	Write enable	: No
Location	: FLD	SER enable	: No
Unit	:	Data representation	: Binary
Subunit	:	Number of input bits	: 2
FLD(s)	: 104	Rack	: 3
Address	: 2258	Position	: 9
		Channel	: 1

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	2	2	5	5	1	1	1	1	9	9	9	9	9	9

Rack : 3 VBD I/O config.: 1oo2D(Redund)

Enter the number of the rack where to allocate the variable.

Start FSC Navigator System Configuration 15:08

Figure 4-47 Hardware specs: BI : Other locations

Rack : Enter the rack number where the variable will be allocated.

Position : Enter the position in the rack where the variable will be allocated.

Channel : Enter the first channel number of the module where the variable will be allocated.

Second, third and fourth input rack and input position : If it is not possible to allocate all input bits on the module defined by Rack and position starting with channel, then one or more extra modules have to be defined to allocate the remaining input bits.



4.5.6 Input multiplexer (XI)

For more information refer to Appendix G of the FSC Software Manual ("FSC multiplexers").

System Configuration

Fail Safe Control system : Application DEMO_1 Rec. no.: 572

Variable type	: XI	Safety-related	: No
Tag number	: KE-576-D1	Force enable	: No
Service	: DATA SELECT	Write enable	: No
Location	: MUX	SER enable	: No
Unit	: 5600	MUX tag number	: 1
Subunit	: 56-10	Number of input bits	: 3
		Scan time	: 500 ms
		Debounce	: Yes
		Sel OUT active time	: 23 ms
		Data hold time	: 0 ms
		Sel OUT active status	: On
		MUX data bus status	: Inverted

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	2	2	2	2	1	1	2	1	2	2	2	2	2	2	2	3
0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0
1	1	9	9	9	9	1	2	9	1	5	5	6	9	9	5	9	2

Rack : 4 VBD Func.: Single

Enter value between 2 and 100

Start FSC Navigator System Configuration 15:05

Figure 4-48 Hardware specs: XI

- | | |
|-----------------------------|---|
| Safety-related | : Cannot be modified. Always No. |
| Force enable | : Cannot be modified. Always No. |
| Write enable | : Cannot be modified. Always No. |
| SER enable | : Cannot be modified. Always No. |
| Mux tag number | : The MUX tag number is automatically allocated by the program. |
| Number of input bits | : Enter the number of bits of the variable. This value should be in the range 1 to 32 bits. |
| Scan time | : This time should be in the range from 100 to 60,000 ms The scan time is the time in which all 'connected' BI variables should be read. Enter the scan time. The minimum must be greater than the application time. If an application is not yet translated, the minimum scan time will be 100 ms for systems with only one Central Part and 200 ms for systems with multiple Central Parts. The maximum scan time is 60,000 ms. |



- Debounce** : Press <Space> to alter the debounce value. When the debounce value is yes, the value which is read, will be compared with the last value read. If these values are equal, the last value will be transferred to the application program.
- Sel OUT active time** : Enter the select output active time. The select output active time is the time which the select output should be true before the data is valid. The select OUT active time must be greater then twice the delay time of the inputs on the module on which the XI variable has been allocated. If the XI has not yet been allocated, the minimum select OUT active time is 2 ms The maximum time is 100 ms The delay times of the different I input modules are:
- 10101/1/1: 20 ms
 - 10101/1/2: 20 ms
 - 10101/1/3: 20 ms
 - 10101/2/1: 20 ms
 - 10101/2/2: 20 ms
 - 10101/2/3: 20 ms
 - 10104/1/1: 1 ms
 - 10104/2/1: 1 ms
 - 10103/1/1: 1 ms
 - 10103/1/2: 1 ms
- Data hold time** : Enter the data hold time. This time should be in the range from 0 to 100 ms The data hold time is the time in which the data stabilizes on the input multiplexer data bus.
- Sel OUT active status** : Press <Space> to alter the select output active status.
- MUX data bus status** : Press <Space> to alter the MUX data bus status. The MUX data bus status can be used to invert the data read from the input multiplexer data bus. (normal = no inversion)
- Rack** : Enter the rack number where the variable will be allocated.
- Position** : Enter the position in the rack where the variable will be allocated.
- Channel** : Enter the first channel number of the module where the variable will be allocated.
- Second, third and fourth input rack and input position** : If it is not possible to allocate all input bits on the module defined by Rack and position starting with channel, then one or more extra modules have to be defined to allocate the remaining input bits.



4.5.7 Digital output signal (DO)

There are five different kinds of outputs, depending on the location field:

- Location = 'COM' : The output is sent via a non-FSC communication channel.
- Location = 'FSC' : The output is sent to another FSC system.
- Location = 'ANN' : The output is an annunciator output
- Location = 'SYS' : The output is a system variable.
- Other locations : The output is a normal digital output

4.5.7.1 Location = 'COM'

For more information refer to Appendix F of the FSC Software Manual ("FSC communication manual").

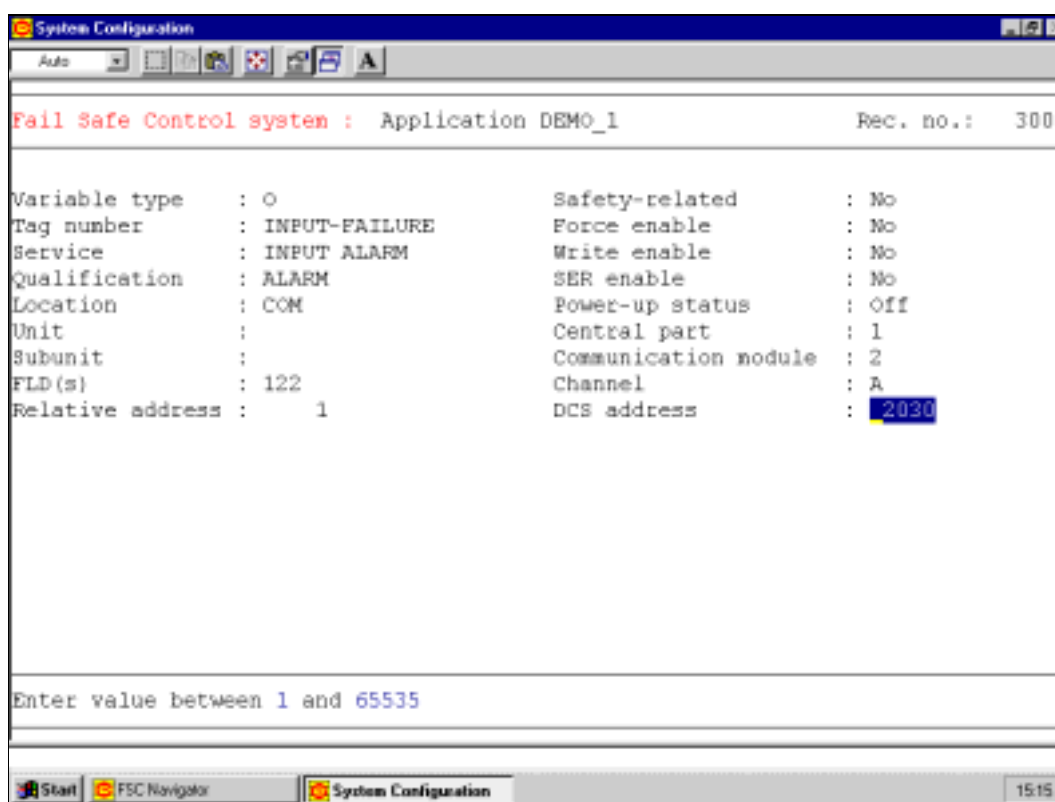


Figure 4-49 Hardware specs: O : Location = 'COM'

Safety-related : Cannot be modified. Always No.

Force enable : Cannot be modified. Always No.

Write enable : Cannot be modified. Always No.



SER enable (sequence)	: Press <Space> to alter. If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
Power-up status	: If this field is On the 'output' is set to 1 after power-up. If it is Off the 'output' is set to 0 after power-up. The default status is Off. Press <Space> to alter the power-up status.
Central Part	: Select a Central Part by toggling.
Communication module	: Select a module by toggling.
Channel	: Select a channel by toggling.
Relative address	: If you allocate automatically, the address is calculated by the program, otherwise it has to be entered by the user.
DCS address	: Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.7.2 Location = 'FSC'

For more information refer to Appendix F of the FSC Software Manual ("FSC communication manual").

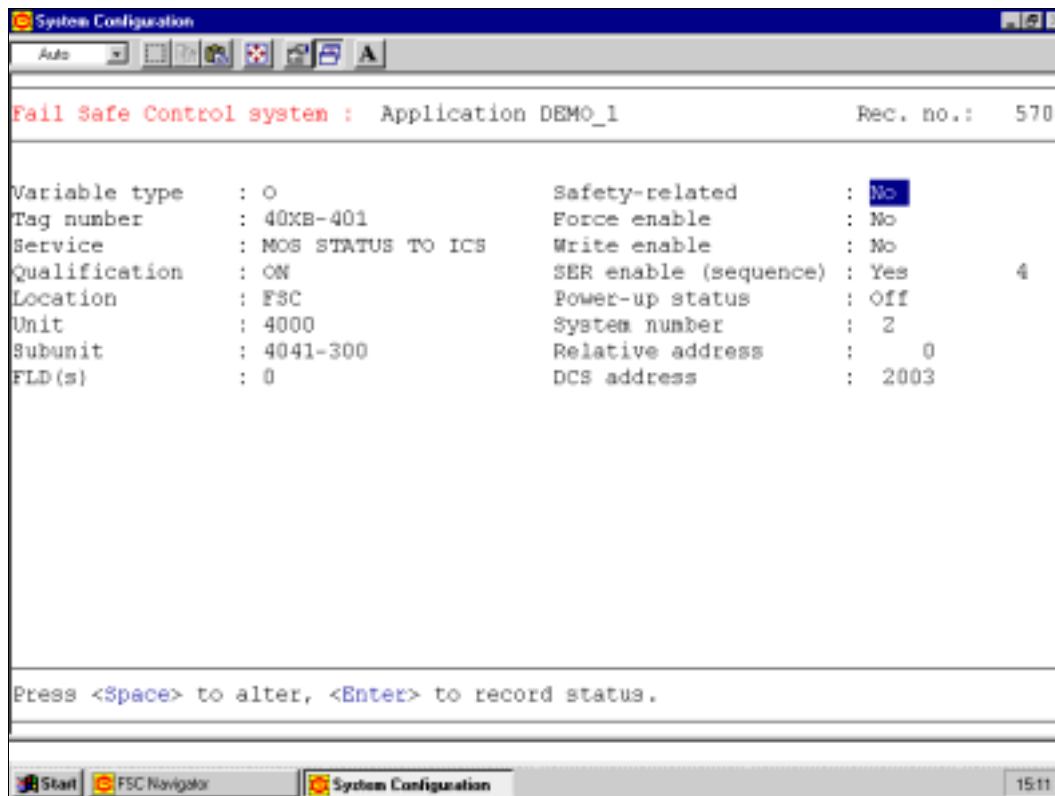


Figure 4-50 Hardware specs: O : Location = 'FSC'

Safety-related : Press <Space> to alter.

Force enable : Cannot be modified. Always No.

Write enable : Cannot be modified. Always No.

SER enable (sequence) : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.

Power-up status : If this field is On the 'output' is set to 1 after power-up. If it is Off the 'output' is set to 0 after power-up. The default status is Off. Press <Space> to alter the power-up status.



-
- System number** : Select the system number in the window that appears on the screen. The highlighted number is the selected system number. Press <D> on the spot of the highlighted number to deallocate the system. Press <S> to (re)select a system.
- Relative address** : If you allocate automatically, the address is calculated by the program, otherwise it has to be entered by the user.
- DCS address** : Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.7.3 Location = 'ANN'

For more information refer to Appendix A of the FSC Software Manual ("Alarm functions").

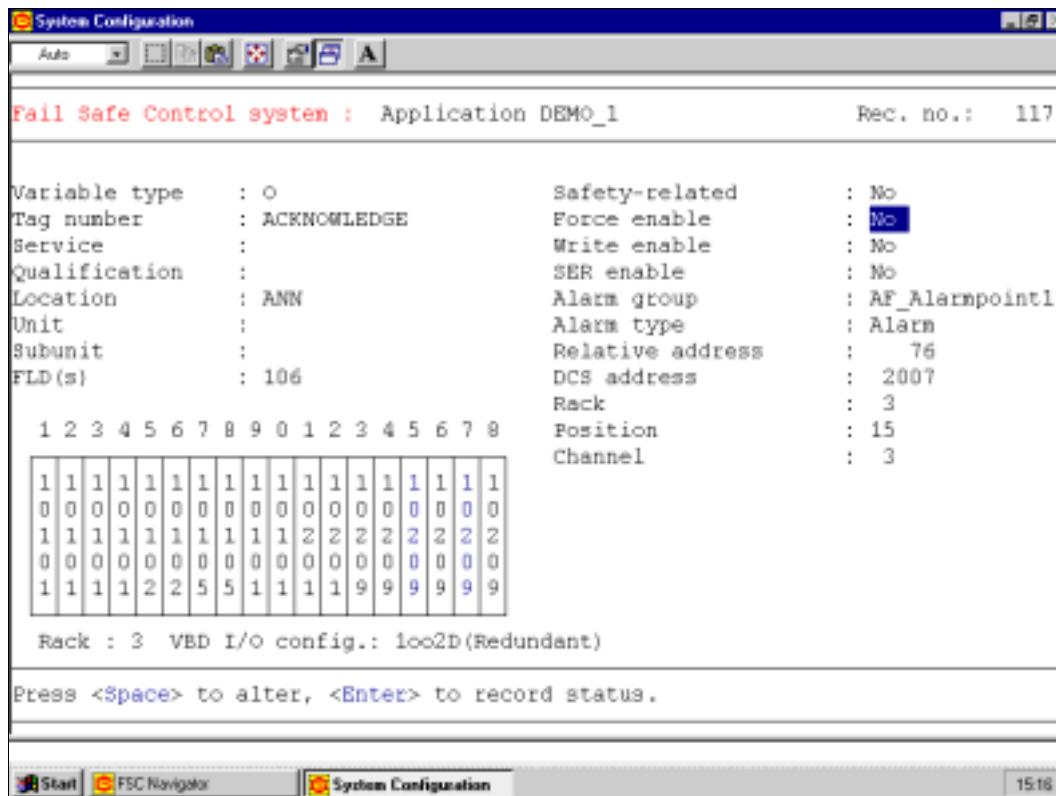


Figure 4-51 Hardware specs: O : Location = 'ANN'

- Safety-related** : Cannot be modified. Always No.
- Force enable** : Press <Space> to alter (Control outputs always No).
- Write enable** : Cannot be modified. Always No.
- SER enable** : Cannot be modified. Always No.
- Alarm group** : Press <S> to select the alarm group. <D> will deallocate the variable from the alarm group.
- Alarm type** : Press <Space> to alter the alarm type. Possible values depend on the alarm sequence of the alarm group the input is connected to. Every alarm type can only be selected once per alarm group. The exception is alarm type 'Alarm' which is used for the actual alarm outputs. This alarm type can be selected as many times as the group size defined for the alarm function. Only for this alarm type a rack, position and channel have to be defined.



Relative address : If you allocate automatically, this is calculated by the program, otherwise this has to be entered by the user.

If the alarm type is 'Alarm' the hardware allocation must be defined.

Rack : Enter the rack number where the variable will be allocated.

Position : Enter the position in the rack where the variable will be allocated.

Channel : Enter the channel number where the variable will be allocated.

DCS address : Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



4.5.7.4 Location = 'SYS'

There are two kinds of digital outputs with location 'SYS':

- The RKE event triggers for transmitting an RKE block.
- The SER event trigger for printing the value of a BO with location COM or FSC.

Figure 4-52 shows an example of a SER event trigger of a BO variable with location COM/FSC.

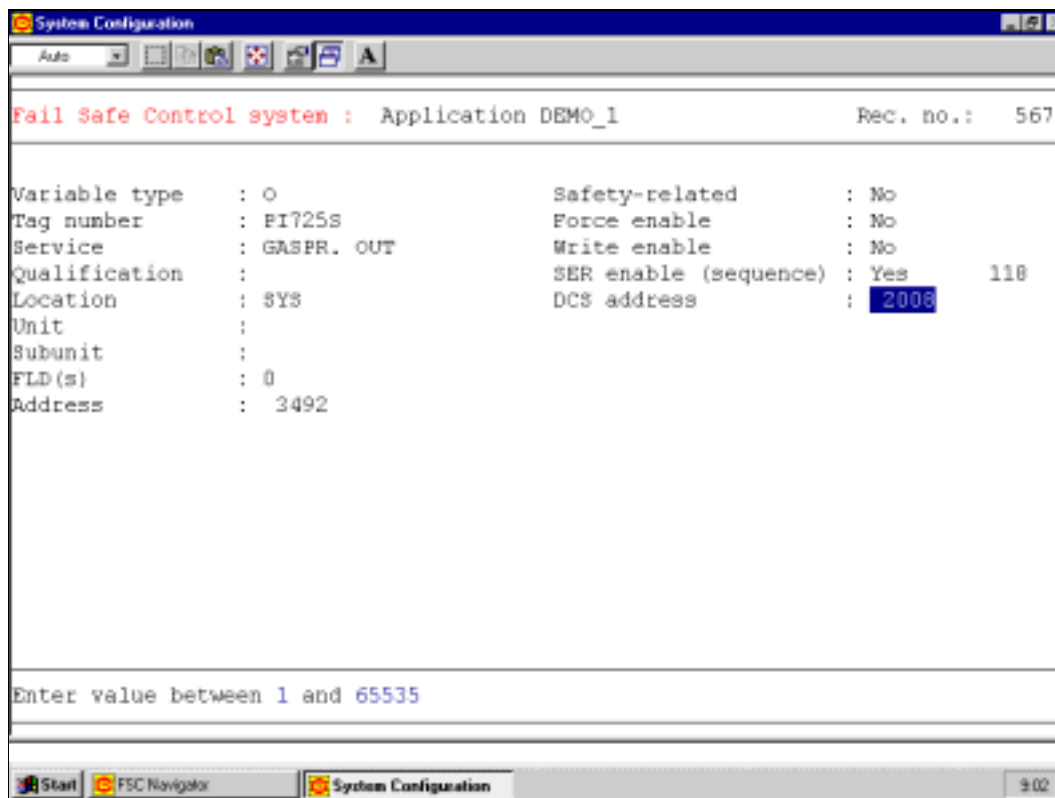
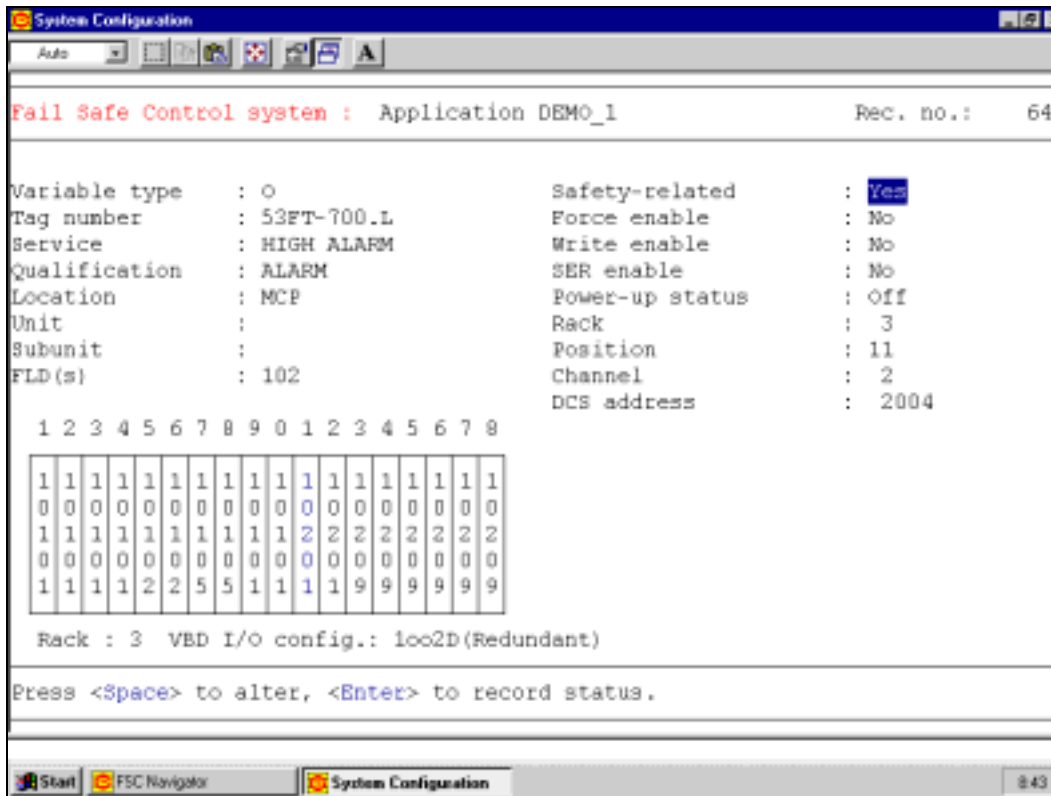


Figure 4-52 SER event trigger of a BO variable with location COM/FSC

Nothing can be edited for digital outputs with location SYS. They are controlled by the application program.

4.5.7.5 Other locations



System Configuration

Fail Safe Control system : Application DEMO_1 Rec. no.: 64

Variable type	: O	Safety-related	: Yes
Tag number	: 53FT-700.L	Force enable	: No
Service	: HIGH ALARM	Write enable	: No
Qualification	: ALARM	SER enable	: No
Location	: MCP	Power-up status	: Off
Unit	:	Rack	: 3
Subunit	:	Position	: 11
FLD(s)	: 102	Channel	: 2
		DCS address	: 2004

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	2	2	5	5	1	1	1	9	9	9	9	9	9	9	9

Rack : 3 VBD I/O config.: 1oo2D(Redundant)

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 843

Figure 4-53 Hardware specs: O : Other locations

Safety-related : Press <Space> to alter.

Force enable : Press <Space> to alter.

Write enable : Cannot be modified. Always No.

SER enable (sequence) : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.

Power-up status : The power-up status is defined after the FSC system has been reset (e.g. after power-up). The default status is Off. Press <Space> to change the power-up status.

Rack : Enter the rack number where the variable will be allocated.



- Position** : Enter the position in the rack where the variable will be allocated.
- Channel** : Enter the channel number of the module where the variable will be allocated.
- DCS address** : Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.



Top scale	: Enter the highest possible value (in engineering units) of the controlling variable.
SER setpoint low	: If SER enable is Yes, a value can be entered to generate an SER message as soon as the analog output goes below this value. The default value is the bottom scale.
SER setpoint high	: If SER enable is Yes, a value can be entered to generate an SER message as soon as the analog output goes above this value. The default value is the top scale.
Power-up value	: The power-up value is defined after the FSC system has been reset (e.g. after power-up). The default value is the bottom scale.
Rack	: Enter the rack number where the variable will be allocated.
Position	: Enter the position in the rack where the variable will be allocated.
Channel	: Enter the channel number of the module where the variable will be allocated.
DCS address	: Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.

4.5.9 Multiple output signal (BO)

The variable type BO can have four different kinds of hardware specifications, depending on the location field:

- Location = 'COM' : The BO variable is sent via a non-FSC communication channel.
- Location = 'FSC' : The BO variable is sent to another FSC system.
- Location = 'MUX' : The BO variable is a part of multiplexed I/O.
- Other locations : The BO variable is a multiple output, but is no part of multiplexed I/O.

4.5.9.1 Location = 'COM'

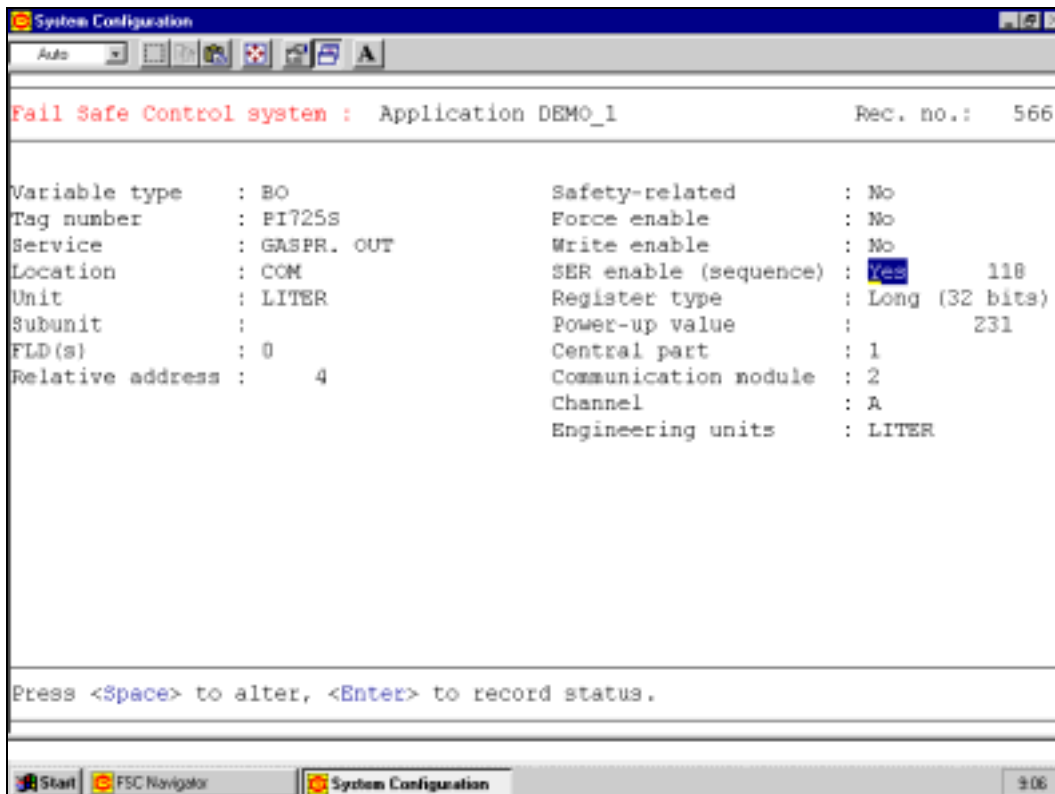


Figure 4-55 Hardware specs: BO : Location = 'COM'

- Safety-related** : Cannot be modified. Always No.
- Force enable** : Cannot be modified. Always No.
- Write enable** : Cannot be modified. Always No.
- SER enable (sequence)** : Press <Space> to alter.
If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.



Register type	: Press <Space> to alter the register type.
Power-up value	: The power-up value is specified after the FSC system is reset (e.g. after power-up). The default value is 0.
Central Part	: Select a Central Part by toggling
Communication module	: Select a module by toggling
Channel	: Select a channel by toggling.
Relative address	: If you allocate automatically, this is calculated by the program. Otherwise it has to be entered by the user.
Engineering units	: If the SER enable = Yes, the engineering units field can be entered as a string of 5 characters. These units can be printed in the SER message.
DCS address	: The DCS address can only be configured if the register type is 'Float'. Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.

4.5.9.2 Location = 'FSC'

For more information refer to Appendix F of the FSC Software Manual ("FSC communication manual").

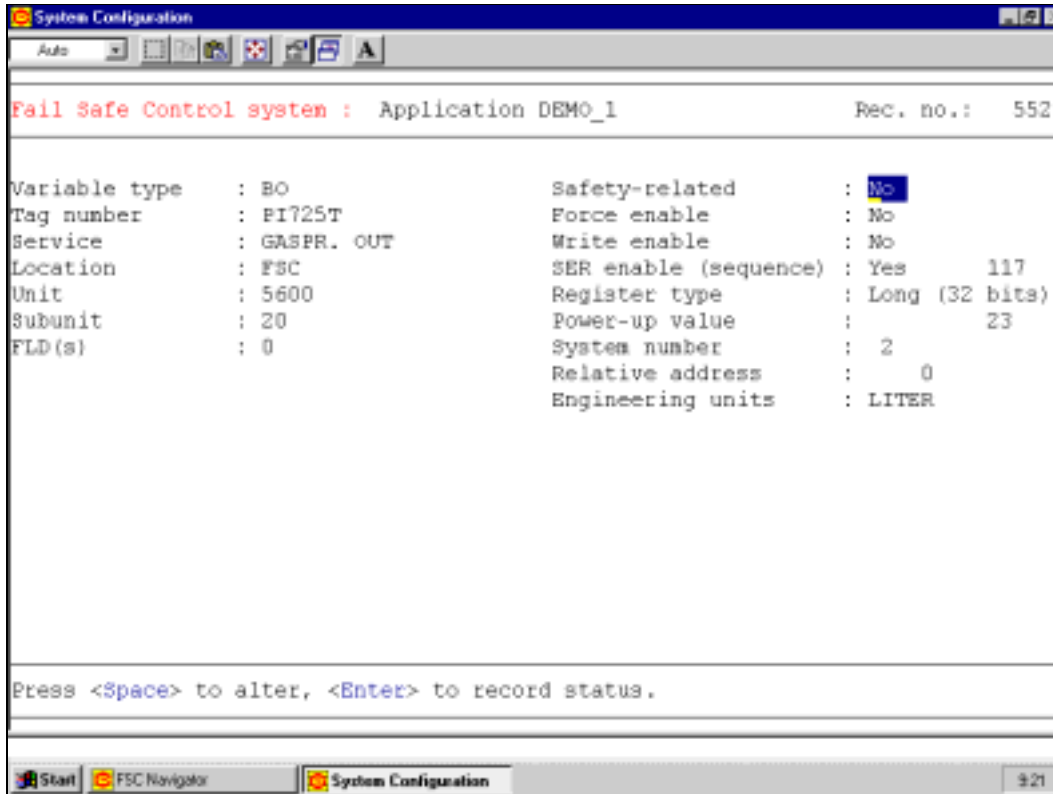


Figure 4-56 Hardware specs: BO : Location = 'FSC'

Safety-related	: Press <Space> to alter.
Force enable	: Cannot be modified. Always No.
Write enable	: Cannot be modified. Always No.
SER enable (sequence)	: Press <Space> to alter. If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
Register type	: Press <Space> to alter the register type.
Power-up value	: The power-up value is specified after the FSC system is reset (e.g. after power-up). The default value is 0.



- System number** : Select the system number in the window that appears on the screen. The highlighted number is the selected system. Press <D> on the spot of the highlighted number to deallocate the system. Press <S> to (re) select a system.
- Relative address** : If you allocate automatically this is calculated by the program. Otherwise it has to be entered by the user.
- Engineering units** : If SER enable = Yes, the engineering units field can be entered as a string of 5 characters. These units can be printed in the SER message.
- DCS address** : The DCS address can only be configured if the register type is 'Float'. Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.

4.5.9.3 Location = 'MUX'

For more information refer to Appendix G of the FSC Software Manual ("FSC multiplexers").

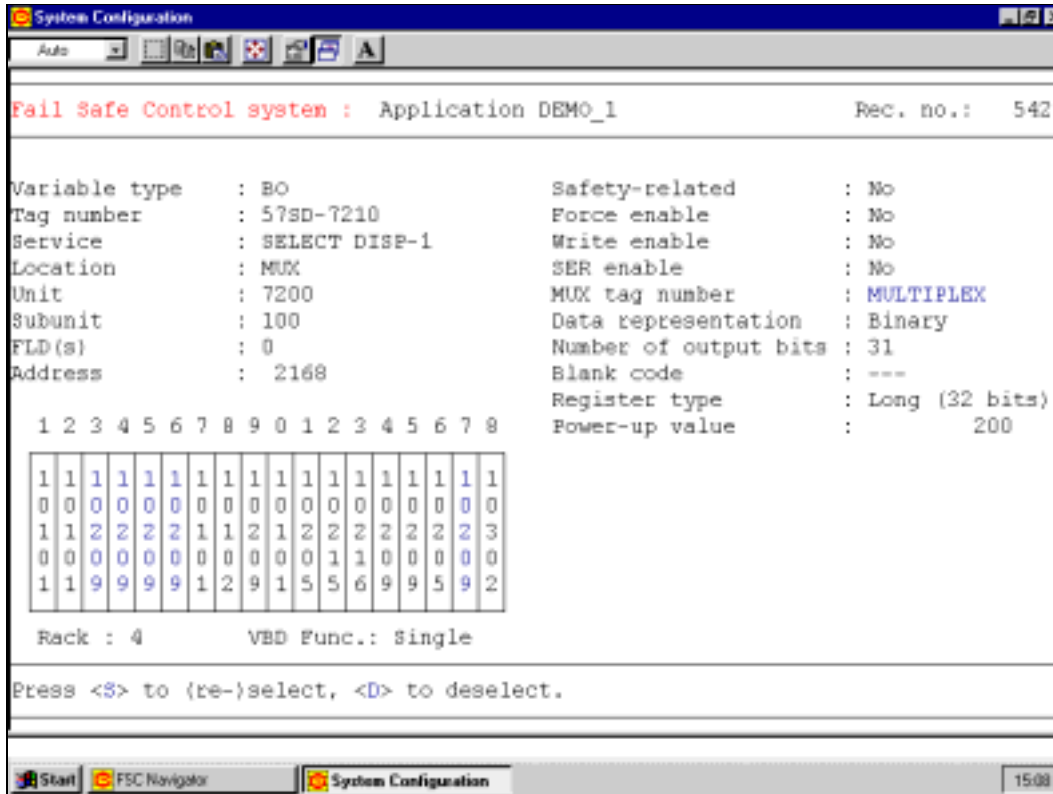


Figure 4-57 Hardware specs: BO : Location = 'MUX'

- Safety-related** : Cannot be modified. Always No.
- Force enable** : Cannot be modified. Always No.
- Write enable** : Cannot be modified. Always No.
- SER enable** : Cannot be modified. Always No.
- Mux tag number** : Press <S> to select the XO variable. <D> will deallocate the BO.
- Data representation** : Press <Space> to select a Binary representation or a BCD representation.
- Number of output bits** : Enter the number of bits of the variable. This value should be in the range 1 to 32 bits.
- Blank code** : This value can be entered if BCD representation is selected. This value is used to blank BCD display in case of loading zeros. The default 0 means no blanking.



Register type	: Cannot be modified if the number of output bits > 0.
Power-up value	: The power-up value is defined after the FSC system has been reset (e.g. after power-up) The default value is 0.

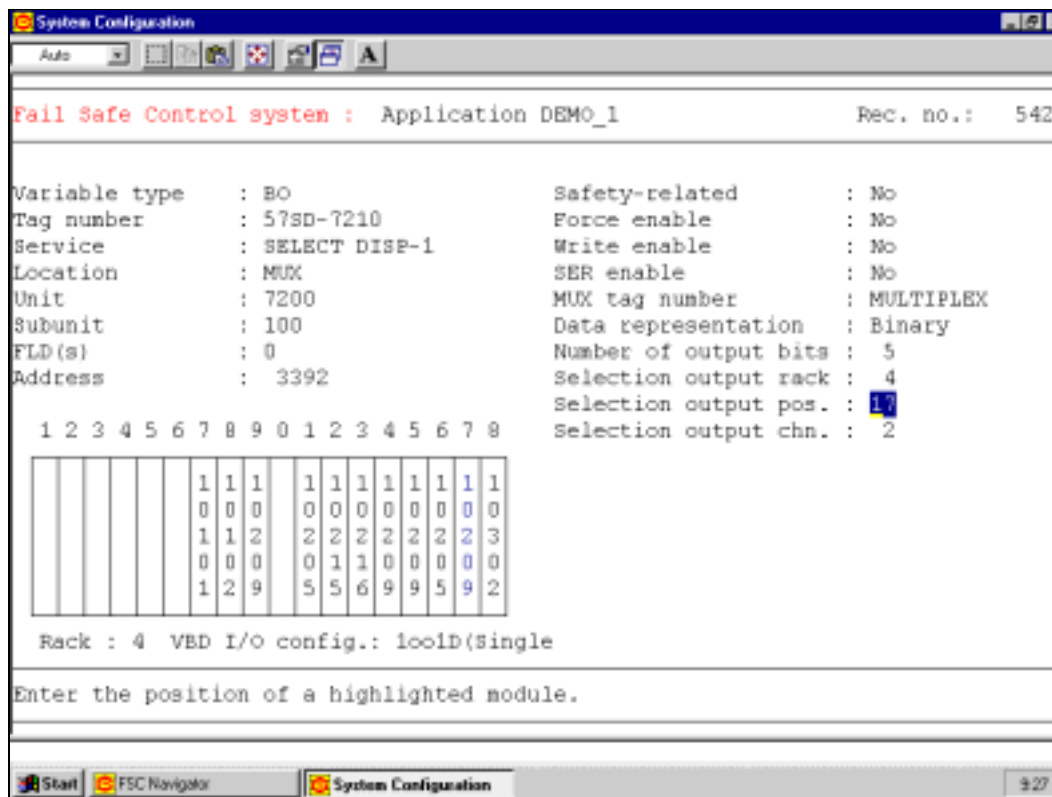


Figure 4-58 Hardware specs: BO : Location = 'MUX'

- Selection output rack** : Enter the rack number where the variable will be allocated.
- Selection output pos.** : Enter the position in the rack where the variable will be allocated.
- Selection output chn.** : Enter the first channel number of the module where the variable will be allocated.
- 2nd, 3rd and 4th Sel. outp. rack and outp. position** : If it is not possible to allocate all output selection bits on the module defined by rack and position starting with channel, then one or more extra modules have to be defined to allocate the remaining output selection bits.



4.5.9.4 Other locations

The screenshot shows the 'System Configuration' window with the following details:

- Fail Safe Control system : Application DEMO_1
- Rec. no.: 542
- Variable type : BO
- Tag number : 57SD-7210
- Service : SELECT DISP-1
- Location : MUX
- Unit : 7200
- Subunit : 100
- FLD(s) : 0
- Address : 3392
- Safety-related : No
- Force enable : No
- Write enable : No
- SER enable : No
- MUX tag number : MULTIPLEX
- Data representation : Binary
- Number of output bits : 5
- Blank code : ---
- Register type : Byte (8 bits)
- Power-up value : 20

Below the specifications is a table with 16 columns and 4 rows of data:

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8
					1	1	1			1	1	1	1	1	1	1	1
					0	0	0			0	0	0	0	0	0	0	0
					1	1	2			2	2	2	2	2	2	2	3
					0	0	0			0	1	1	0	0	0	0	0
					1	2	9			5	5	6	9	9	5	9	2

Rack : 4 VBD I/O config.: 1001D(Single)

Press <S> to (re-)select, <D> to deselect.

Figure 4-59 Hardware specs: BO : Other locations

- Safety-related** : Press <Space> to alter.
- Force enable** : Cannot be modified. Always No.
- Write enable** : Cannot be modified. Always No.
- SER enable** : Cannot be modified. Always No.
- Data representation** : Press <Space> to select a binary representation or a BCD representation.
- Number of output bits** : Enter the number of bits of the variable. This value should be in the range 1 to 32 bits.
- Blank code** : This value can be entered if BCD representation is selected. This value is used to clear a part of the variable on, for example, BCD display. The default 0 means no blanking.
- Register type** : Cannot be modified. Depends on the number of output bits.
- Power-up value** : The power-up value is defined after the FSC system has been reset (e.g. after power-up). The default value is 0.

System Configuration

Fail Safe Control system : Application DEMO_1 Rec. no.: 536

Variable type	: BO	Safety-related	: No
Tag number	: XI-210-25	Force enable	: No
Service	: MIXING TIME	Write enable	: No
Location	: CAB	SER enable	: No
Unit	: 2100	Data representation	: Binary
Subunit	: 2102	Number of output bits	: 31
FLD(s)	: 101	Rack	: 4
Address	: 2172	Position	: 17
		Channel	: 15
		Second output rack	: 4
		Second output position	: 3
		Third output rack	: 4
		Third output position	: 4

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	2	2	2	1	1	2	1	2	2	2	2	2	2	2	2	3
0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
1	1	9	9	9	1	2	9	1	5	5	6	9	9	5	9	2	

Rack : 4 VBD Func.: Single

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 15:14

Figure 4-60 Hardware specs: BO : Other locations

- Rack** : Enter the rack number where the variable will be allocated.
- Position** : Enter the position in the rack where the variable will be allocated.
- Channel** : Enter the first channel number of the module where the variable will be allocated.
- Second, third and fourth output rack and output position** : If it is not possible to allocate all outputs bits on the module defined by rack and position starting with channel, then one or more extra modules have to be defined to allocate the remaining output bits.



4.5.10 Output multiplexer (XO)

For more information refer to Appendix G of the FSC Software Manual ("FSC multiplexers").

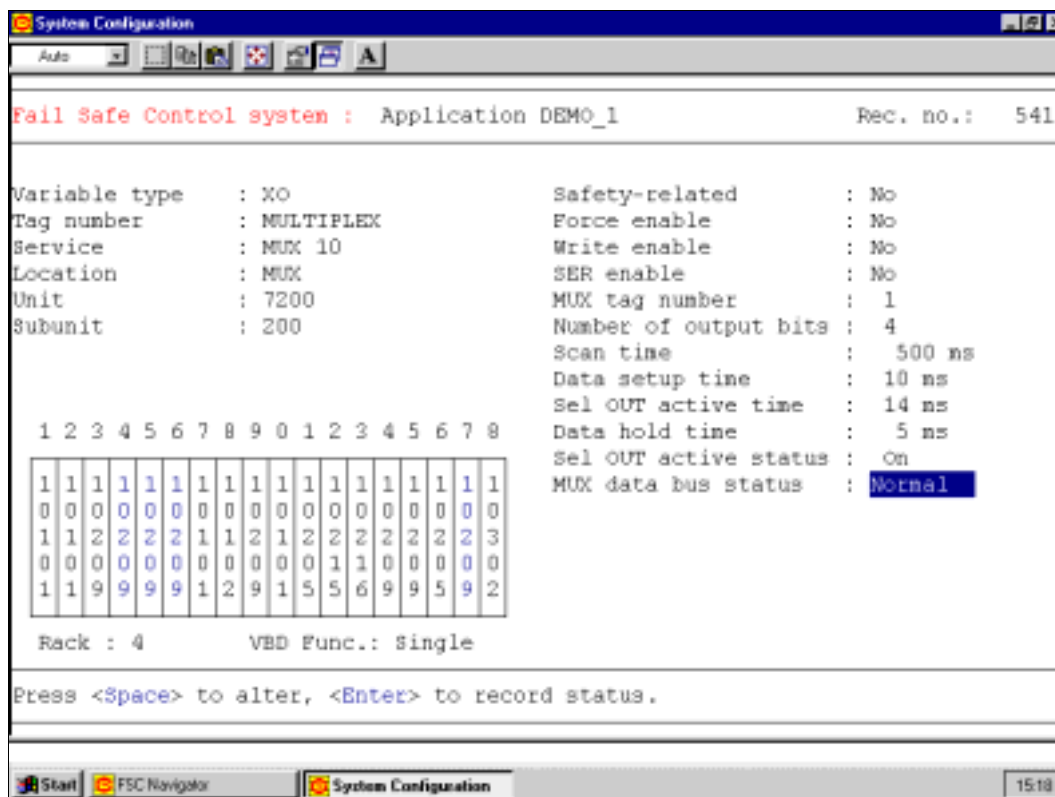


Figure 4-61 Hardware specs: XO

Safety-related	: Cannot be modified. Always No.
Force enable	: Cannot be modified. Always No.
Write enable	: Cannot be modified. Always No.
SER enable	: Cannot be modified. Always No.
Mux number	: Cannot be modified. Automatically allocated by the program.
Number of output bits	: Enter the number of bits of the variable. This value should be in the range 1 to 32 bits.
Scan time	: Enter the scan time. This time should be in the range from 100 to 60,000 ms. The scan time is the time in which all 'connected' BO variables should be written.
Data setup time	: Enter the data setup time. This time should be in the range from 0 to 100 ms. The data setup time is the time which the data lines should have the correct data status before the select output will be activated.



Sel OUT active time	: Enter the select output active time. This time should be in the range from 1 to 100 ms. The select output time is the time which the select output should be active.
Data hold time	: Enter the data hold time. This time should be in the range from 1 to 100 ms. The data hold time is the time in which the data has to stay active after the select output is de-activated.
Sel OUT active status	: Press <Space> to alter the select output active status.
MUX data bus status	: Press <Space> to alter the MUX data bus status. The MUX data bus status can be used to invert the data (normal = no inversion).
Rack	: Enter the rack number where the output multiplexer data bus will be allocated.
Position	: Enter the position in the rack where the output multiplexer data bus will be allocated.
Channel	: Enter the first channel number of the module where the output multiplexer data bus will be allocated.
Second, third and fourth output rack and output position	: If it is not possible to allocate all outputs bits on the module defined by rack and position starting with channel, then one or more extra modules have to be defined to allocate the remaining output bits.



4.5.11 Marker (M)

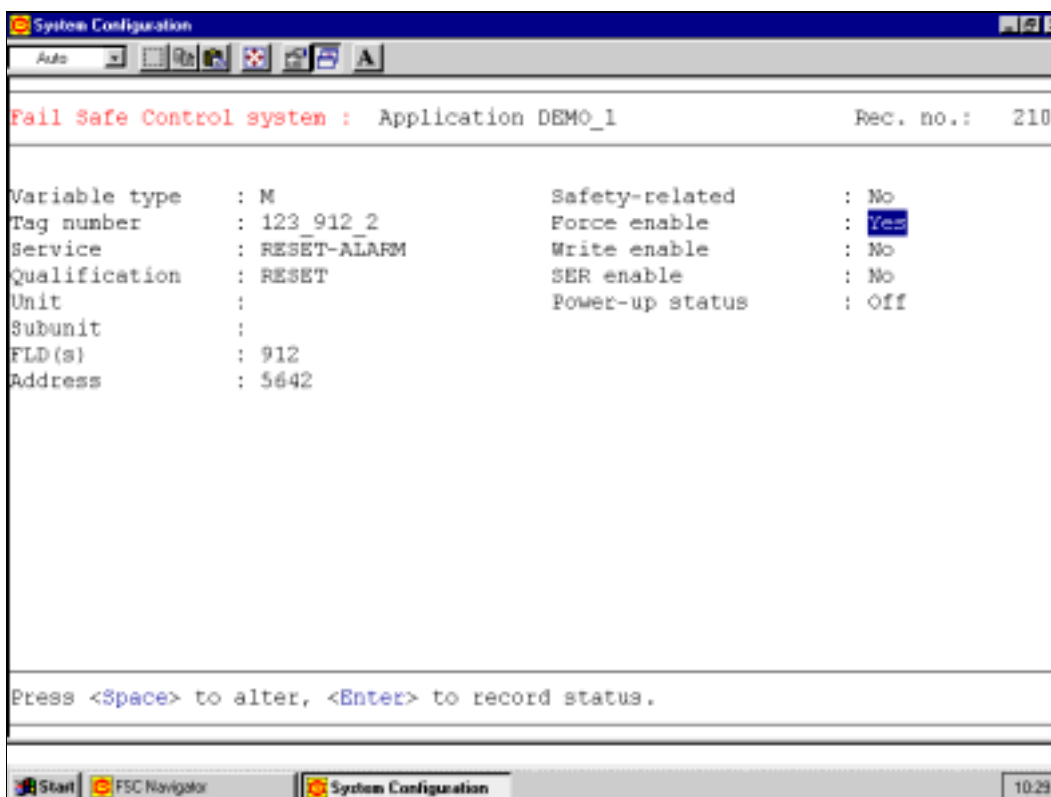


Figure 4-62 Hardware specs: M

Safety-related	: Cannot be modified; will be determined by the 'Translate application' option of FSC Navigator (the compiler).
Force enable	: Markers can only be forced for on-sheet references. Markers used in a sheet or off-sheet references cannot be forced
Write enable	: Press <Space> to alter.
SER enable (sequence)	: Press <Space> to alter. If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
Power-up status	: If this field is On the marker is set to 1 after power-up. If it is Off the marker is set to 0 after power-up. The default status is Off. Press <Space> to alter the power-up status.
FLD(s)	: This field lists the destination FLD of a reference between two or more FLDs. This field cannot be modified.

4.5.12 Counter (C)

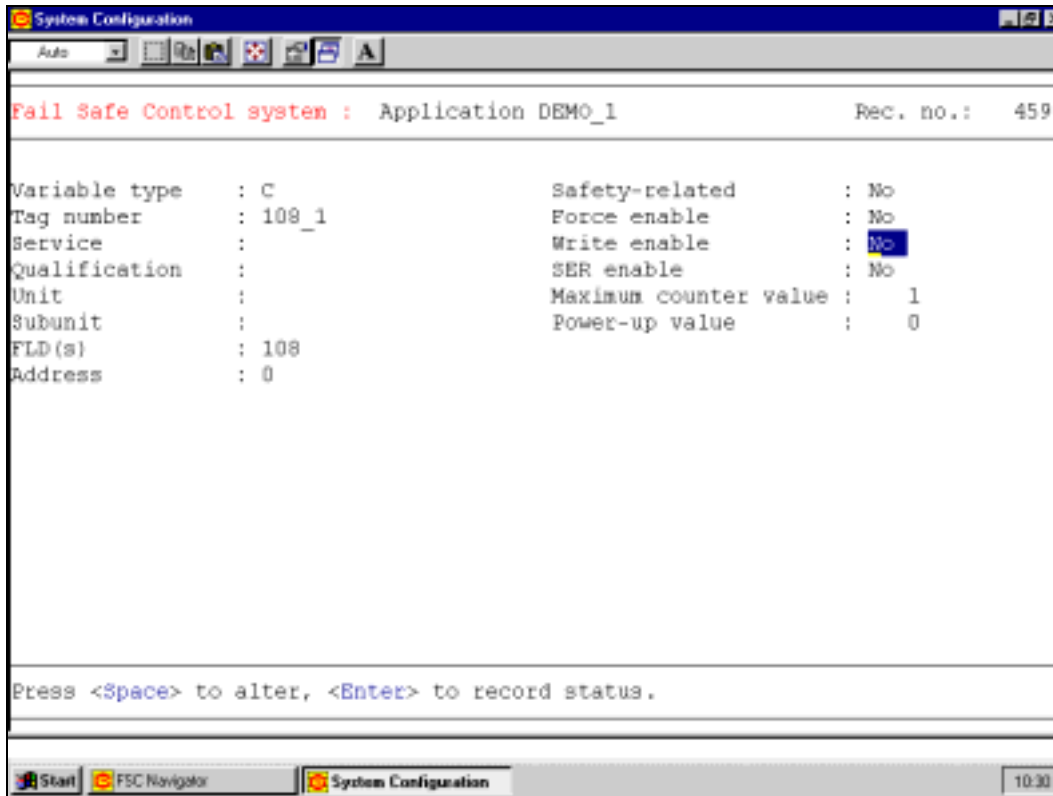


Figure 4-63 Hardware specs: C

Safety-related	: Cannot be modified; will be determined by the 'Translate application' option of FSC Navigator (the compiler).
Force enable	: Cannot be modified. Always No.
Write enable	: Press <Space> to alter.
SER enable (sequence)	: Press <Space> to alter. If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
Maximum counter value	: This is the maximum value that the counter can reach after subsequent count up pulses. After this value the counter resets automatically to 0.
Power-up value	: The value of the counter when the FSC system is reset (e.g. after power-up). The value should be in the range from 0 to the maximum counter value. The default value is 0.
FLD(s)	: This is the FLD where the counter is placed.



4.5.13 Timer (T)

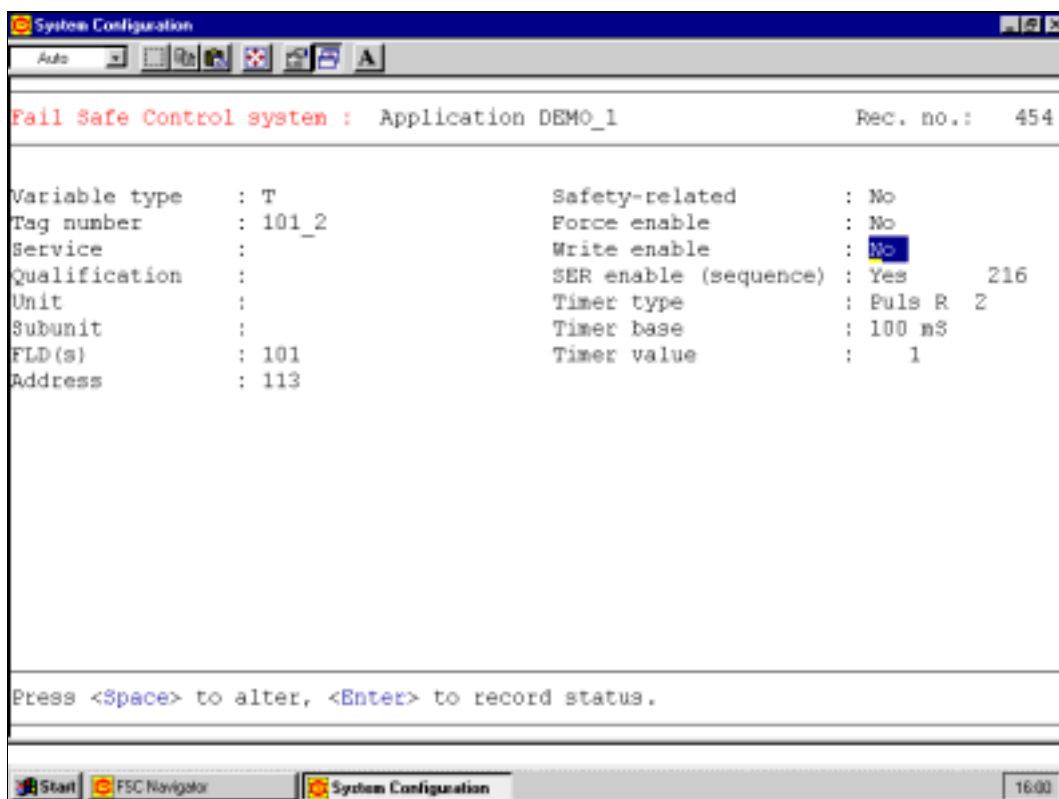


Figure 4-64 Hardware specs: T

Safety-related	: Cannot be modified; will be determined by the 'Translate application' option of FSC Navigator (the compiler).
Force enable	: Cannot be modified. Always No.
Write enable	: Press <Space> to alter.
SER enable (sequence)	: Press <Space> to alter. If SER is enabled and an SER sequence range has been specified, an SER sequence number must be entered. For the SER sequence a unique number between SER base and SER maximum of the primary or secondary SER channel can be entered. The default is the first free sequence number. Use right/left arrow keys to switch between the SER enable and SER sequence number.
Timer type	: The timer function type. Timers can be divided into: 1. Timers with timer value dependent on the application program. 2. Timers with constant time value defined in the Design functional logic diagrams. The second type of timers has an extension of 2 to the timer type (see screen above).



The following different timer types are possible:

- **None** : No type specified
- **Pulse** : Pulse time function on rising edge of the timer input.
- **Pulse R** : Pulse time function on rising edge of the timer input, the time is restarted after every rising edge.
- **Del ON** : Delay On time function on rising edge of the timer input.
- **Del ON M** : Delay On time function on rising edge of the timer input. The time continues to count even if the time input goes to zero state.
- **Del OFF** : Delay Off function on falling edge of the timer input.

Timer base : This can be 10 milliseconds (10 ms), 100 milliseconds (100 ms), 1 second (1 S) or 1 minute (1 M). The timer base is the period in which the timer value is decremented once.

Timer value : Constant timer value. This value has a range of 0 to 2047.

DCS address : Enter the unique identification used for accessing the attributes of this variable. A value between 1 and 65535 can be entered. Enter (-1) to deallocate the address.

The timer type, timer base and timer value cannot be changed using the 'System Configuration' option of FSC Navigator. Use the 'Design FLDs' option to change the timer type, timer base or timer value.



4.5.14 Register (R)

The screenshot shows a 'System Configuration' window with a menu bar (Auto, icons, A) and a title bar. The main area displays the following information:

Fail Safe Control system : Application DEMO_1 Rec. no.: 233

Variable type	: R	Safety-related	: No
Tag number	: 101_102_1	Force enable	: No
Service	: MAIN LINE FLOW	Write enable	: No
Unit	:	SER enable	: No
Subunit	:	Register type	: Byte (8 bits)
FLD(s)	: 102	Power-up value	: 0
Address	: 2819		

Enter value between -128 and 127

The taskbar at the bottom shows 'Start', 'FSC Navigator', and 'System Configuration' (active) with a clock showing 16:02.

Figure 4-65 Hardware specs: R

Safety-related	: Cannot be modified; will be determined by the 'Translate application' option of FSC Navigator (the compiler).
Force enable	: Cannot be modified. Always No.
Write enable	: Press <Space> to alter.
SER enable	: Cannot be modified. Always No.
Register type	: The type of the register. The register type cannot be modified by this program.
Power-up value	: The value of the register when the FSC system is switched on. The range of this value depends on the register type. The default value is 0. This field can be modified if the address is ≤ 7000 .

Note:

Registers are also used for value sheet references.

4.5.15 PID (P)

For more information refer to Appendix E of the FSC Software Manual ("PID controllers").

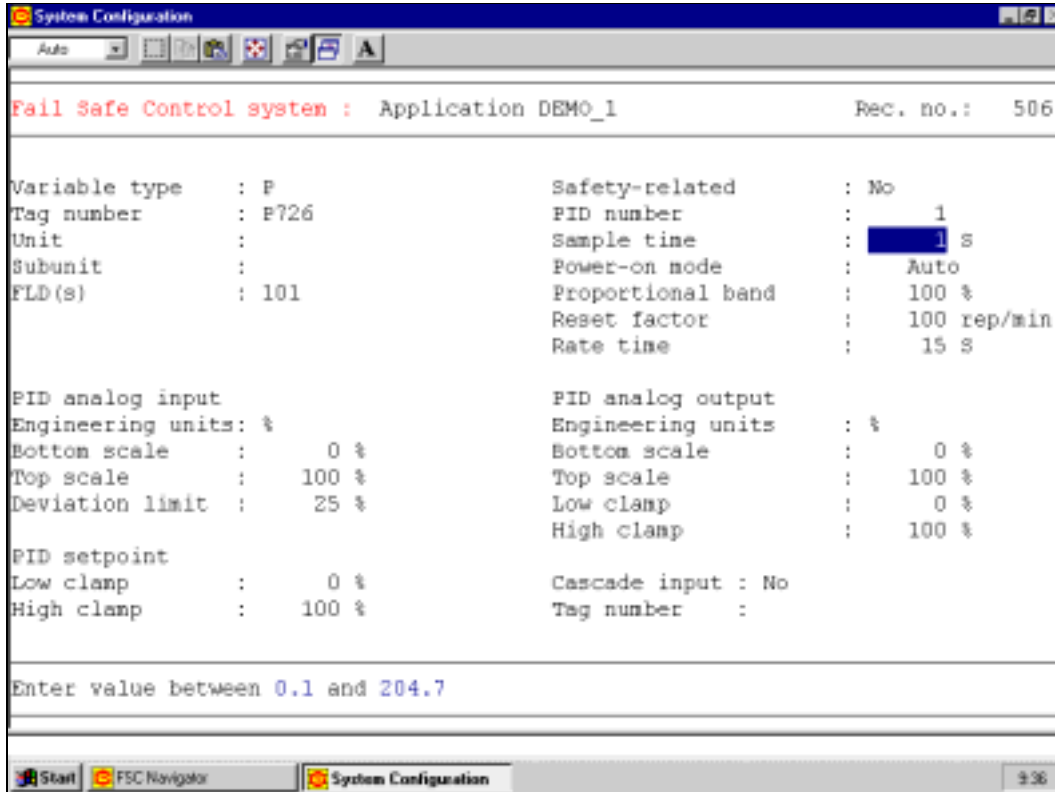


Figure 4-66 Hardware specs: P

Safety-related	: Cannot be modified. Always No.
PID number	: Cannot be modified. The PID number is generated by the program and is for internal use only. The maximum number of PID functions is 64.
Sample time	: The sample time is the time between two samples and has a range of 0.1 to 204.7 seconds. Enter the sample time.
Power-on mode	: The power-on mode which is used for PIDs in the FSC system. There are four options: – Auto : The output is calculated each sample timeout of the difference between the input and the setpoint, according to the setup of the parameters. – Manual : Only the output can be entered. This value is copied to the PID output and is not a calculated value. – Test : In this mode, the input (a virtual input) and the setpoint (virtual setpoint) can be entered. The output is calculated using these values. This option can be used to check the PID function. – Cascade : The PID will act as a slave, controlled by another master slave (chain) or a master PID in Auto mode.



Proportional band	: This indicates the change (as a percentage) in the difference setpoint - input for doubling the output. Enter the Proportional band (<> 0).
Reset factor	: This indicates the times per minute in which the integration is calculated. The Reset factor has a range of ≥ 0 rep/Min. Enter the reset factor.
Rate time	: The rate time is the time section over which the differentiations are discreted, and has a range of 0 to 3276.7 seconds. Enter the rate time.
Engineering units	: Enter the units of the input and of the output of the PID (two fields). You can enter 5 characters.
Bottom scale	: Enter the lowest possible value (in engineering units) of the input and of the output of the PID (Two fields).
Top scale	: Enter the highest possible value (in engineering units) of the input and of the output of the PID (two fields).
PID deviation limit	: This is the high clamp for the difference between PID input and PID analog setpoint. Take care: there will be no correct controlling if the parameter is zero. Enter the PID deviation limit in engineering input units. The deviation limit must exceed 1% of the input range.
PID analog output low clamp	: The analog output will never become smaller than this value. Enter the PID analog output low clamp in engineering units.
PID analog output high clamp	: The analog output will never exceed this value. Enter the PID analog output high clamp in engineering units.
PID setpoint low clamp	: The setpoint will never become smaller than this value. Enter the PID setpoint low clamp in engineering units.
PID setpoint high clamp	: The setpoint will never exceed this value. Enter the PID setpoint high clamp in engineering units.
Cascade input	: If PID operation mode is cascade and this PID is the slave then enter Yes else enter No.
Tag number	: If in cascade enter the tag number of the master PID.



4.6 Rebuild

If you choose this option you are asked to confirm that you want to rebuild the databases:

Do you want to rebuild the databases? (Y/N)

If your answer is 'Yes' and on-line modification is on 'No', the next question will be asked:

Do you want to change the address allocation? (Y/N)

If you answer 'No', or on-line modification is on 'Yes' the address allocation remains the same and only the index files are rebuilt. The next question is:

Are you sure? (Y/N)

Answering 'Yes' once again will start the rebuilding process.

4.6.1 On-line rebuild

If 'on-line modification wanted' is configured, it is possible to rebuild on-line. With this option the allocation of variables can be checked/recovered out of the FSC system.

There is a shortcut to the 'On-Line Rebuild' option. If you choose the 'On-Line Rebuild' command on the On-Line menu of FSC Navigator, the first two questions will be skipped (assuming the answer 'Yes') and you will immediately be taken to the last question, which asks you to confirm the on-line rebuild.

If you choose the rebuild option, you are asked to confirm that you want to rebuild the databases:

Do you want to rebuild the databases ? (Y/N)

If your answer is 'Yes', the next question will be asked:

Do you want to rebuild on line ? (Y/N)

If you answer 'No', the address allocation remains the same and only the index files are rebuilt.

If you answer 'Yes' the allocation will be checked/restored.

Hereafter you are asked for a confirmation.

Are you sure ? (Y/N)

If you answer 'No', the rebuild function is aborted. If you answer 'Yes', the development system will start reading the allocation of the variables from the FSC system via the communication link. The gathered information is put into temporary files which are used by the on-line rebuild function.

The on-line rebuild will act on the following situations:

- A tag number has been changed after translation and programming EPROMs and the allocation is still the same.
Action : The variable will be deallocated or set to undefined.
- The allocation of a variable has been changed during on-line modification.
Action : The variable will be reallocated to its previous allocation.



- An I/O module has been deleted and all variables on it have been deallocated.
Action : The module as well as the accessory variables will be reallocated to the rack and position they had before.
- The rack of a HBD has been changed (after deleting the I/O modules).
Action : The HBD will be reallocated to the rack, it had before.

See also Appendix D of the FSC Software Manual ("On-line modification").



Left blank intentionally.



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Helping You Control Your World



Fail Safe Control

Section 5: Editing Reports and SER Format

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5.1 Introduction

The 'Edit reports and SER Format' option of FSC Navigator can be called by choosing the **SER Formats** command in the **Project** menu. The windows as shown in Figure 5-1 will appear.

This option of FSC Navigator serves two purposes:

1. It can be used to define the journal of the SER messages.
2. It can be used to create the layout for up to 99 reports and to define the conditions under which each of the reports must be generated by the FSC system. For creating the layouts, the program provides a screen editor and a number of other editing features.

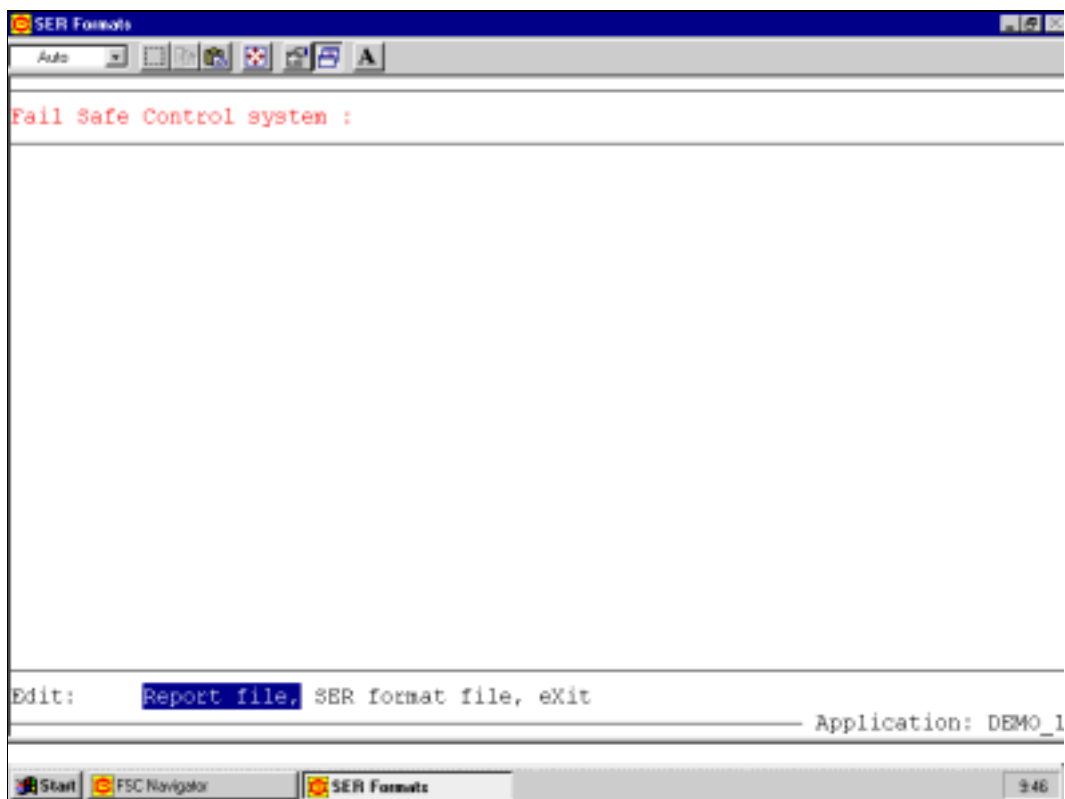


Figure 5-1 Edit reports and SER format: start-up screen

On start-up of the program, you can choose between editing a report, editing the SER format file, deleting files (if there are any files that can be deleted), or returning to the FSC Navigator main screen (eXit). The SER format messages for an application are stored in a file called <application name>.FMT. If you want to edit the SER format file and this file does not exist, a default file will be created. This default file has the same layout as the default format file used by the 'Translate Application' option (compiler) of FSC Navigator and the FSCSOE application. The difference is the value of the qualification of markers. The compiler default file generates no marker qualification table, the user format file will.



5.2 Basic editing commands

For defining the journal of the SER messages and for creating a report, the program provides an editor. This editor, of course, allows you to enter text and to manipulate this text, but also to set graphic attributes as bold printing, the background color, etc. It is also possible to indicate where, for example, in the text a date or a page number should be included in the current SER message or report generated by the FSC system.

Note:

An empty line is indicated by a dash ('-'). This character is not inserted into the actual text.

Below you will find an overview of the available editing commands. They are activated by pressing the key or key combination shown between the < > brackets.

<Ins> or <Ctrl>+<V>

Use this key to switch between insert and overwrite mode. The current mode is displayed in the upper right corner of the screen.

Arrow keys

Use the arrow keys to move around on the screen.

<Enter>

If the total number of lines is fixed (report header in the SER format file), pressing <Enter> puts the cursor at the beginning of the next line. If the current line is the last line, the cursor will be put at the beginning of the last line. If the number of lines is report text, pressing <Enter> inserts a new line below the current line. The text starting at the cursor position will be moved to the new line. The cursor will be put at the beginning of the new line.

<Ctrl> + <→>

Brings the cursor to the beginning of the next word, or to the end of the line if there are no more words on the line.

<Ctrl> + <←>

Brings the cursor to the beginning of the word indicated by the cursor. If the cursor is already on the beginning of the word, or the cursor indicates a space, then the cursor is moved to the beginning of the previous word.

<Home>

Brings the cursor to the beginning of the line.

<End>

Positions the cursor after the last character on the line.

Deletes the character at the current cursor position.

<Backspace>

Deletes the character left of the cursor.

**<Ctrl> + <T>**

Deletes a word. If the cursor indicates a space, then all the spaces until the next word are deleted. If the cursor indicates a word, then all characters until the beginning of the next word are deleted.

<Ctrl> + <Y>

Deletes a complete line.

5.3 Edit report file

If you want to edit a report, you are asked to enter the number of the report, which must be in the range 1 to 99.

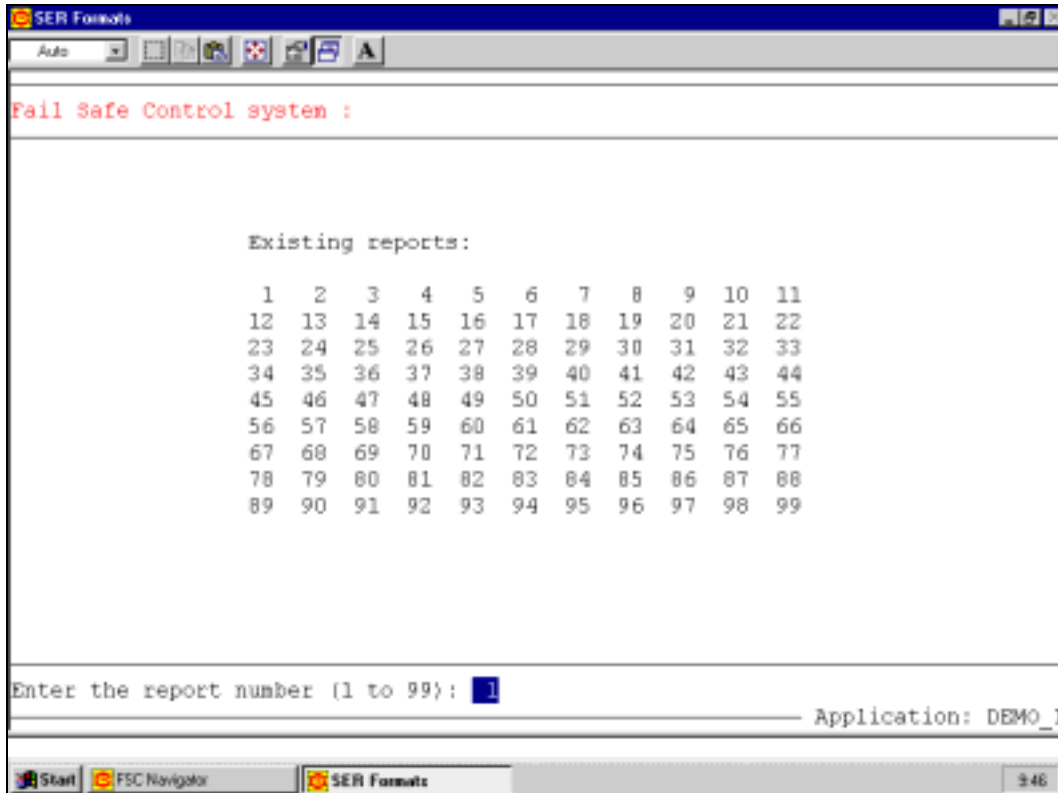


Figure 5-2 Selecting a report file for editing

The existing reports are displayed highlighted. A report is stored in the disk file <Application name>.R<report number>, so a report file for the application TEST_1 could be TEST_1.R05. If you enter the number of a report that does not exist, a new empty report is automatically created.

After selecting a report file, the data is read and checked for invalid data. When the file is corrupted, the program prompts:

Report is corrupted. Do you want to delete the file? <Y/N>

After entering Yes or No, you will return to the main menu.
If the report file is correct or empty, the following screen will be displayed.

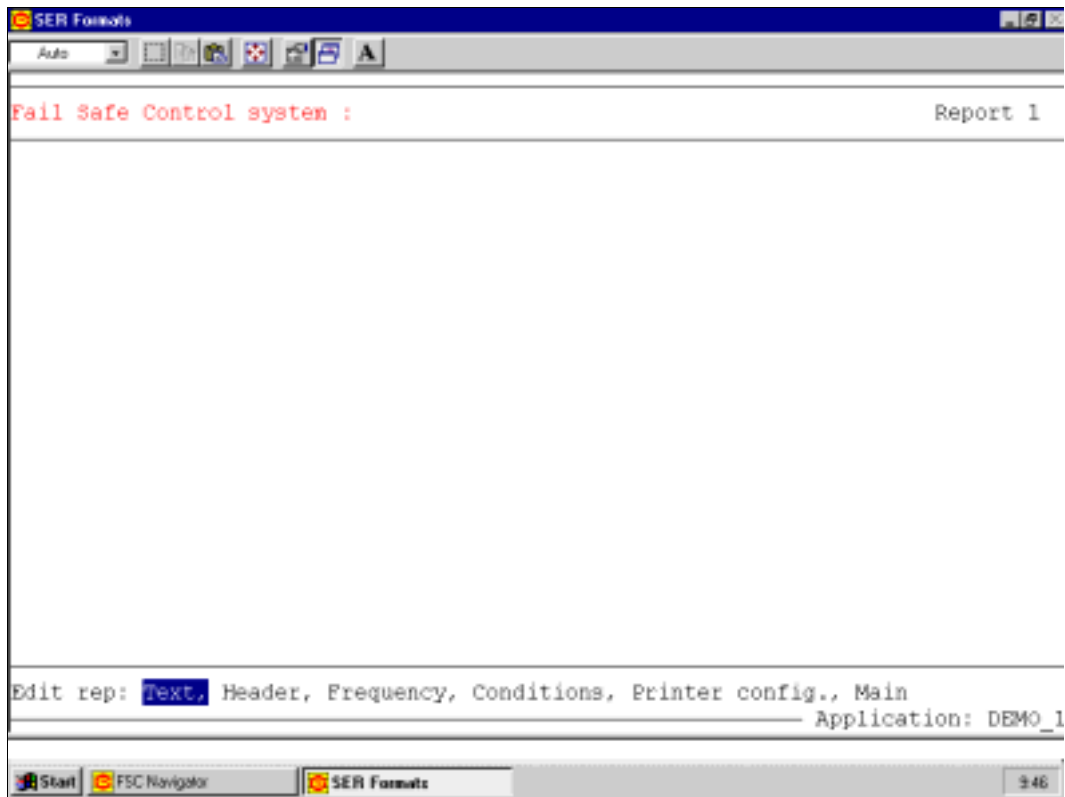


Figure 5-3 Edit menu for report file

5.3.1 Report text

You can enter the report text. There are a number of special functions for placing tag numbers, services, values, etc. in the report text. They are:

<Alt> + <T>

By pressing this key combination you can indicate where in the report you want a tag number to be shown. The program will ask you for the variable type and the tag number.

<Alt> + <S>

By pressing this key combination you can indicate where in the report you want the service of a variable to be shown. The program will ask you for the variable type and the tag number.

<Alt> + <Q>

By pressing this key combination you can indicate where in the report you want the qualification of a variable to be shown. The program will ask you for the variable type and the tag number.

<Alt> + <V>

By pressing this key combination you can indicate where in the report you want the current value of a variable to be shown. The program will ask you for the variable type and the tag number.

While inserting text, you can press <Esc> to call the following editing menu.

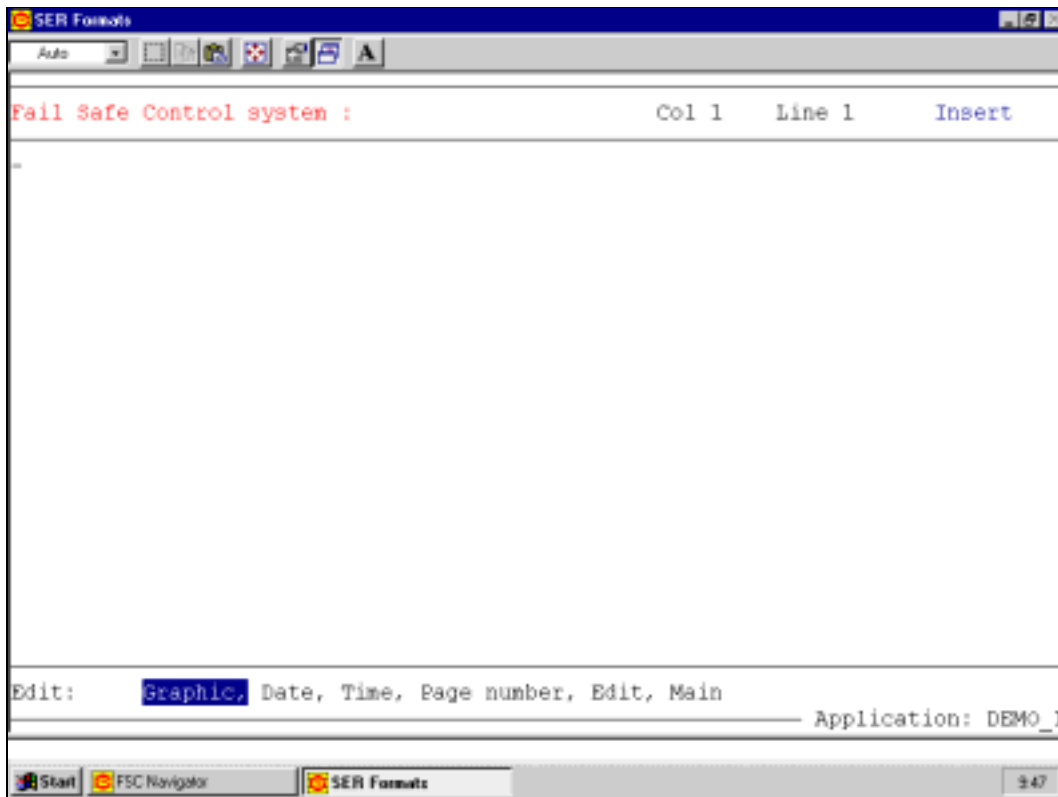


Figure 5-4 Edit menu for report text

- Graphic : Gives access to a submenu used to set graphic attributes of the text. Attributes are valid from the current cursor position until the end of the line.
By choosing the attribute once more it is switched off again.
Only the valid attributes will be displayed.
The possible attributes are :
 - Bold The text will be shown bold.
 - Underline The text will be underlined, only for printer output.
- Color : Gives access to a sub menu to set foreground colors. Only available for color printers.
- Date : Choosing this option inserts an indication in the text that tells where in the text the current date must be shown. This is shown by inserting the text '-- date -----' in the line. You cannot edit this text and the cursor can only be positioned on the first character of the text. Pressing when the cursor is on the first character removes the date indication and text.
- Time : This option has a similar function as the 'Date' function, but it inserts the current time.
- Page number : This option inserts the page number indication.
- Edit : Brings you back to the edit screen.
- Main : Brings you back to the report editor main menu.



5.3.2 Report header

You can enter the report header text. While inserting text, you can press <Esc> to enter the following editing menu.

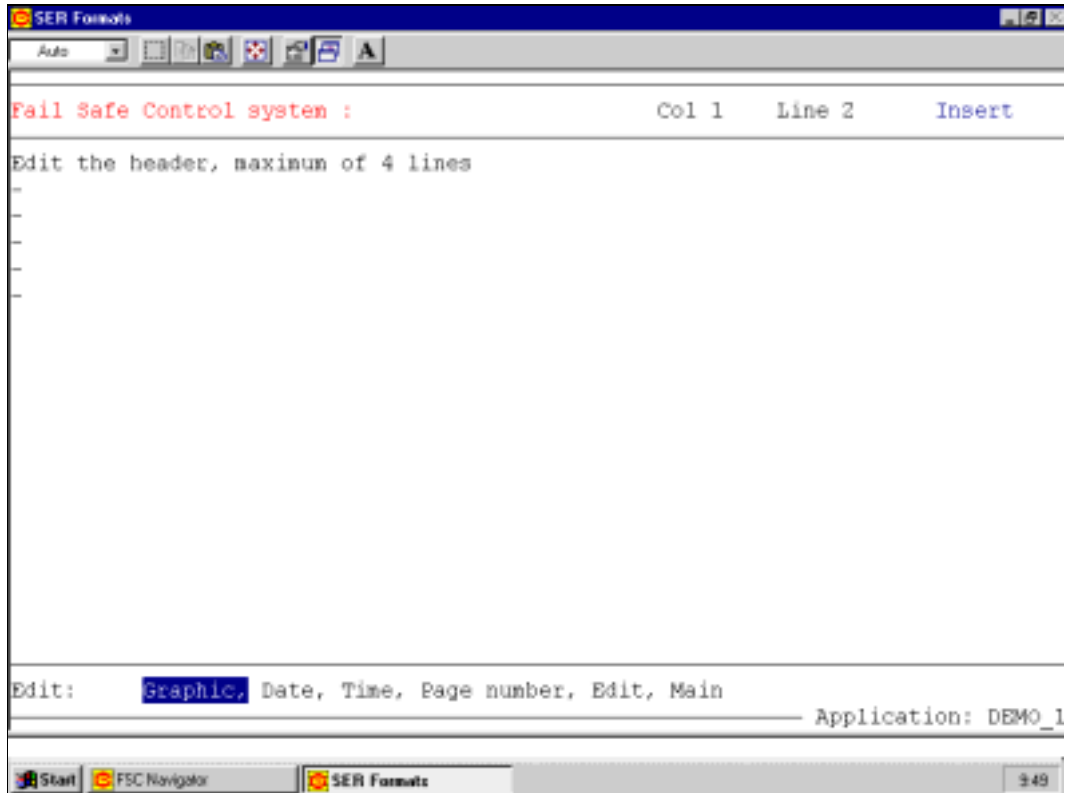


Figure 5-5 Edit menu for report header

- Graphic** : Gives access to a sub menu used to set graphic attributes of the text. Attributes are valid from the current cursor position until the end of the line. By choosing the attribute once more it is switched off again. Only the valid attributes will be displayed. The possible attributes are :
- Bold The text will be shown bold.
 - Underline The text will be underlined, only for printer output.
- Color** : Gives access to a sub menu to set foreground colors. Only available for color printers.
- Date** : Choosing this option inserts an indication in the text that tells where in the text the current date must be shown. This is shown by inserting the text '-- date -----' in the line. You cannot edit this text and the cursor can only be positioned on the first character of the text. Pressing when the cursor is on the first character removes the date indication and text.
- Time** : This option has a similar function as the 'Date' function ; it inserts the current time.

Page number : This option inserts the page number indication.

Edit : Brings you back to the edit screen.

Main : Brings you back to the report editor main menu.

5.3.3 Time-based report generation

If you select the 'frequency' option, you are asked whether or not you want to generate the report on a time base (only possible if you have placed a DBM module in the central part). If your answer is 'Yes', you must define the frequency with which the report must be generated. The frequency is determined by the values given to Month, Date, Day, Hour and Minute.

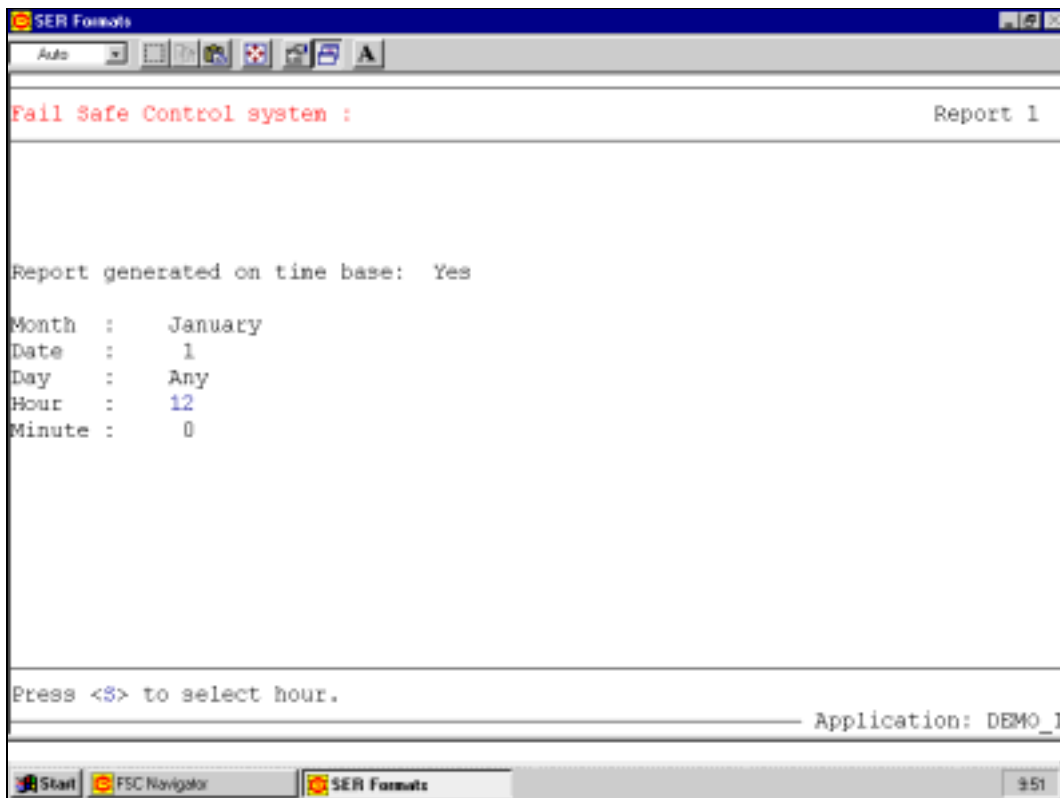


Figure 5-6 Editing the report frequency



If you want a yearly report, you only have to select a specific month. The report is generated when the system timer matches the selected month. If, for example, you wish to report on the first day of the year at 12 o'clock, you should select the month 'January', the date '1' and the hour '12'. The day should be specified as 'Any'. The minute may be specified as '0' or as 'Any'. Use <Space> to select a month or <?> to get a list of all possible options.

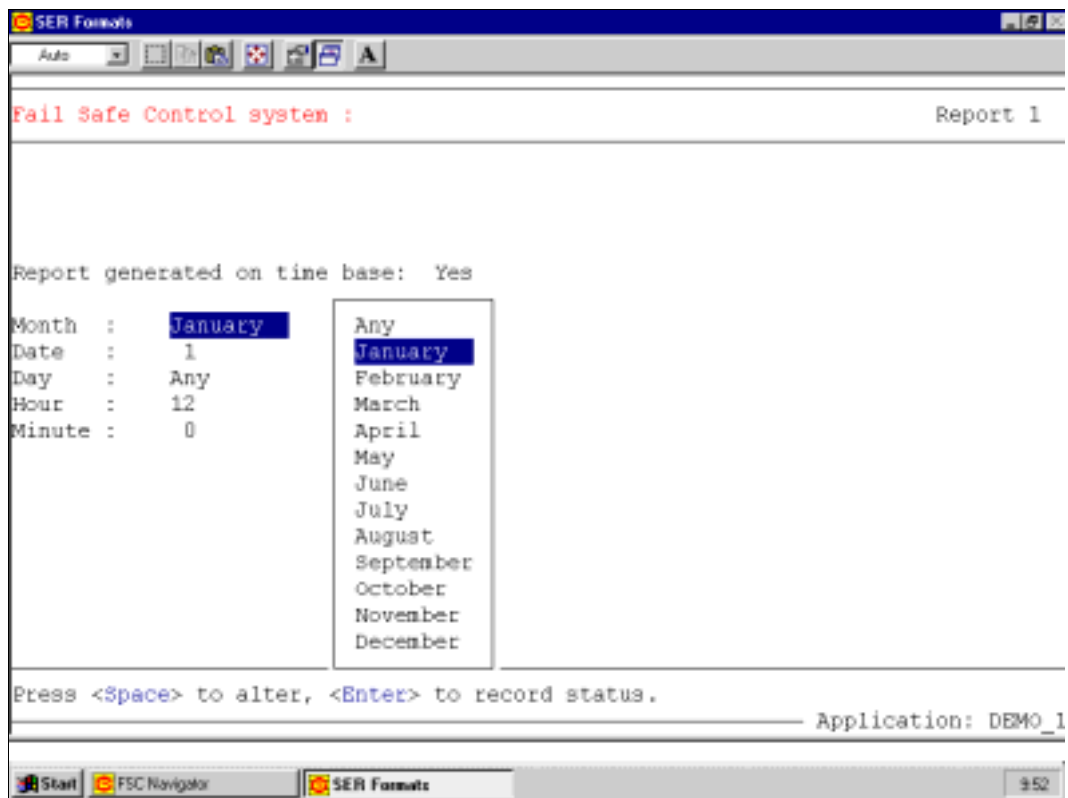


Figure 5-7 Selecting a month

If you want a monthly report, specify the month as 'Any' and specify a date of the month. Use <S> to get the selection window. Use the cursor keys to choose the date and press <S> to select this date. The selected date will be highlighted. If you want to specify 'Any' for the date, press <A>. After selecting a date or 'Any', press <Esc> to return to the selection of month, date, etc.

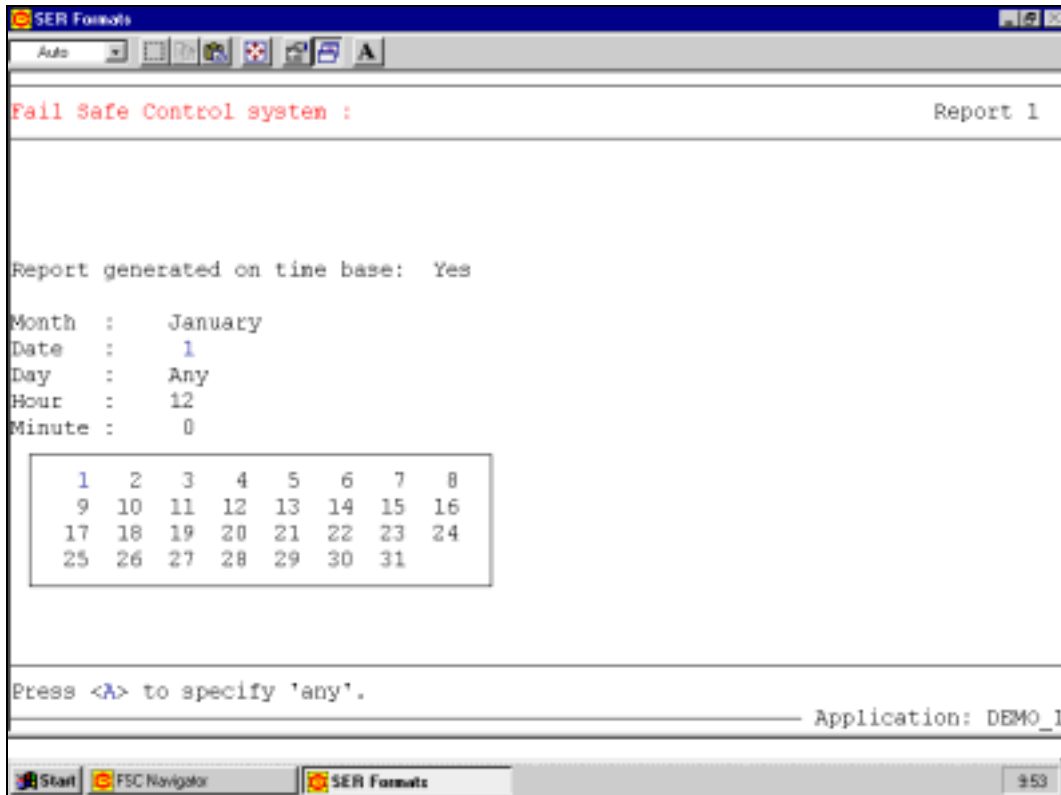


Figure 5-8 Selecting a date



If you want a weekly report, specify both month and date as 'Any' and select a specific day of the week. Use <Space> to select a day of the week or use <?> to get a list of all possible options.

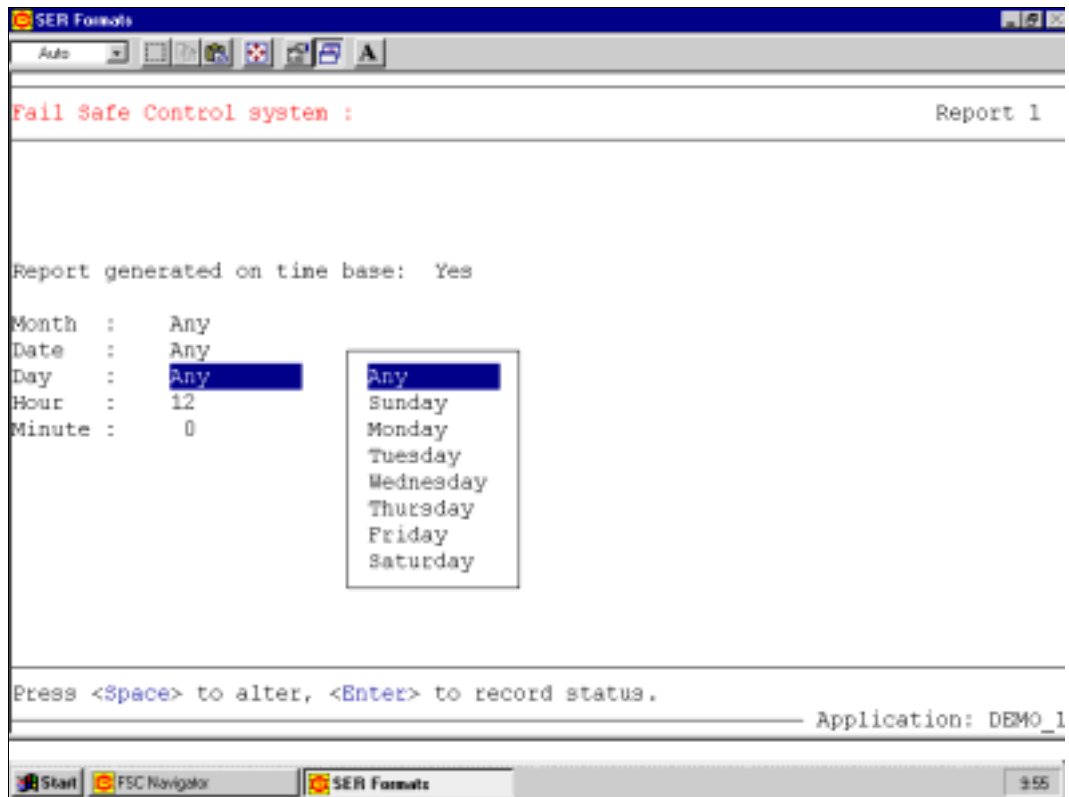


Figure 5-9 Selecting a day

If you want a daily report, specify the month, the date and the day as 'Any' and select a specific hour. Use <S> to get the selection window. Use the cursor keys to choose the hour and press <S> to select this hour. The selected hour will be highlighted. If you want to specify 'Any' for the hour, press <A>. After selecting an hour or 'Any', press <Esc> to return to the selection of month, date, etc.

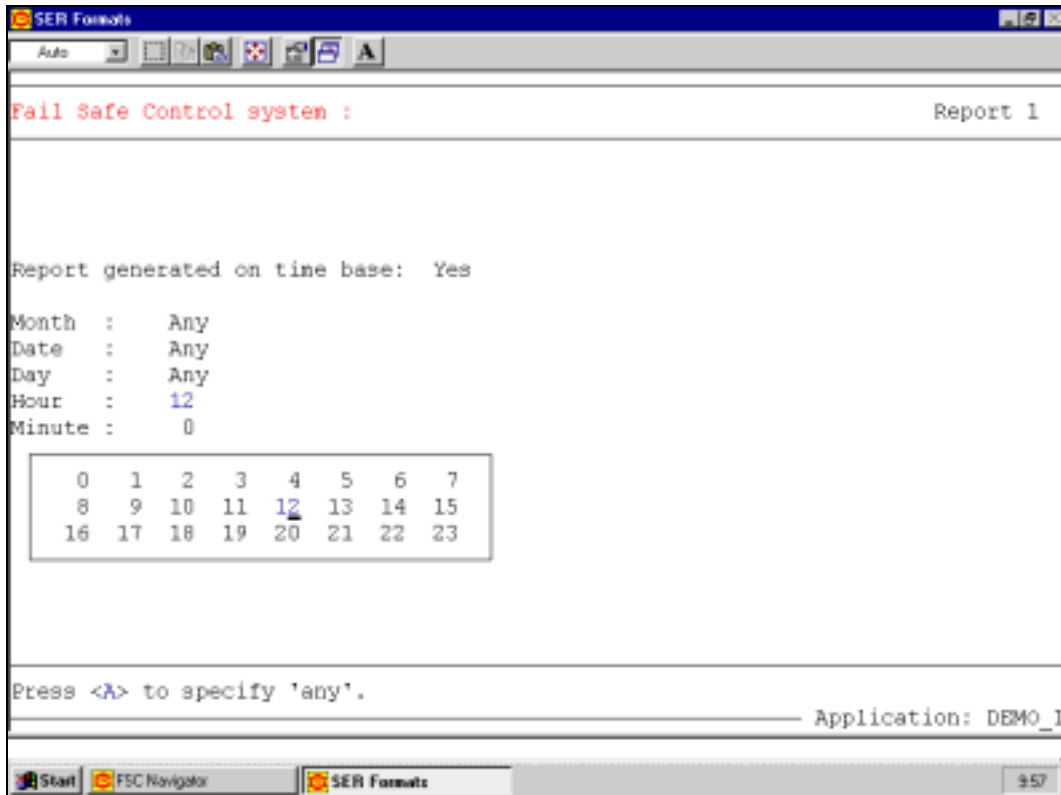


Figure 5-10 Selecting an hour



If you want a report every hour, specify the month, the date, the day and the hour as 'Any' and select a specific minute. Use <S> to get the selection window. Use the cursor keys to choose a minute and press <S> to select this minute. The selected minute will be highlighted. If you want to specify 'Any' for the minute, press <A>. After selecting a minute or 'Any', press <Esc> to return to the selection of month, date, etc.

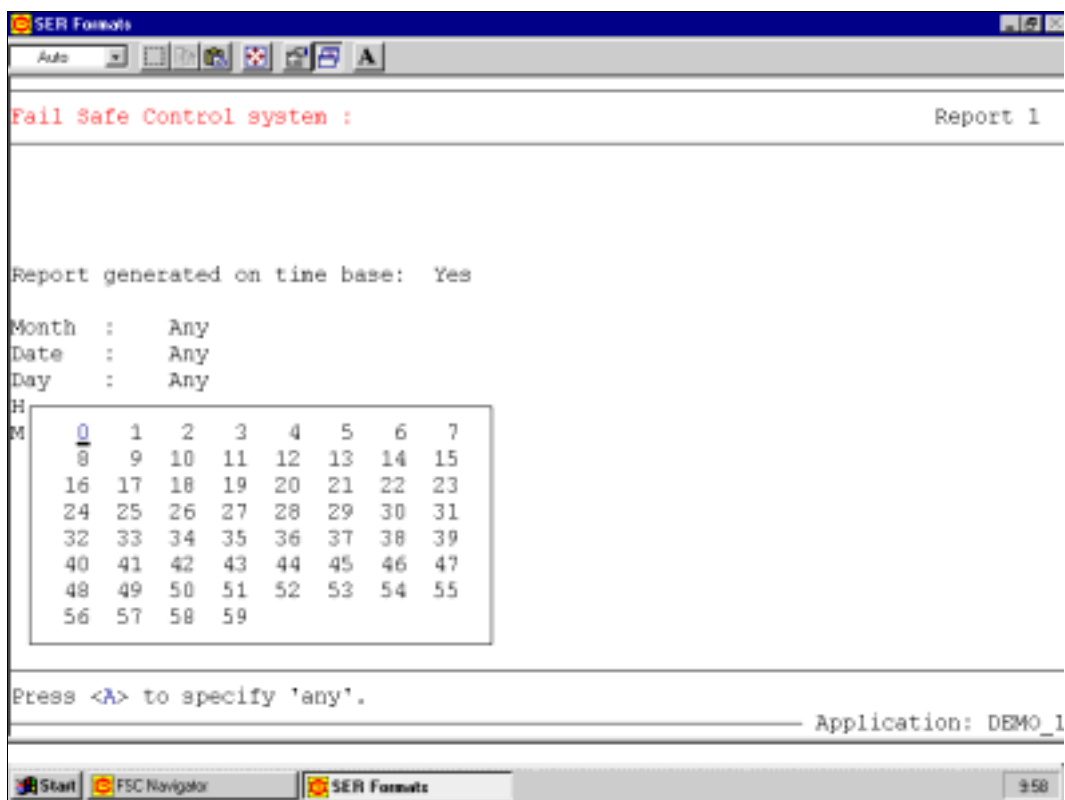


Figure 5-11 Selecting a minute

5.3.4 Conditional report generation

Conditional variables are variables that cause a report to be generated when a certain condition is met. You can specify up to three types of conditions.

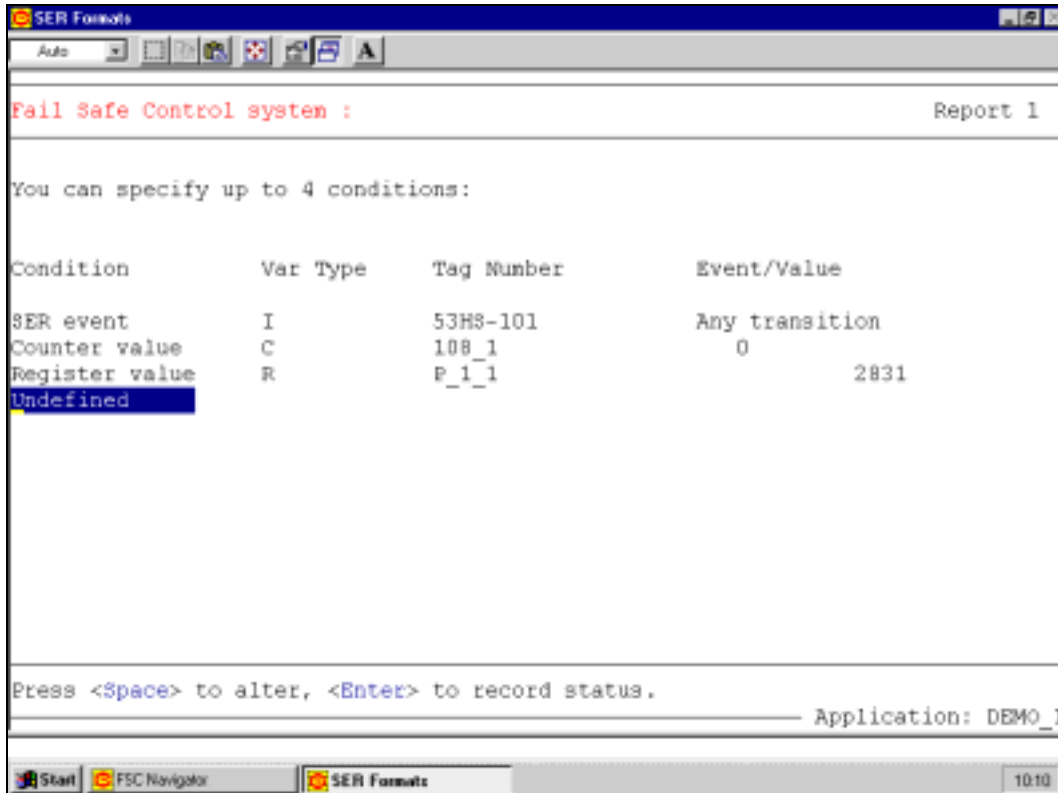


Figure 5-12 Editing the variables for conditional report generation

First select the type of condition under which to generate a report. After selecting the condition, you must name the variable for which the condition will be tested, and you must specify the condition.

There are three possible types of conditions:

- **SER event**
A report is generated if the specified SER event takes place for the conditional variable.
- **Counter value**
A report is generated if the specified counter reaches the specified value.
- **Register value**
A report is generated if the specified register reaches the specified value.

After selecting the condition, you must name a variable in the database for which the condition will be tested. If the condition is an SER event, the variable type must be entered. All variable types that can be made SER-enabled are allowed. Next, the tag number of the variable must be entered. The program checks if the variable is present in the database. If it is, and the condition is SER Event but the variable is not SER-enabled, you are asked whether to make it SER-enabled or not. If your answer is 'Yes', the



database is adapted and the variable is accepted. Otherwise the condition is ignored. SER events are not possible with numerical SER format.

Finally, the condition must be specified. For the conditions 'Counter value' and 'Register value', a value must be entered. For the condition 'SER event', the condition can be specified by toggling. If the variable type is I (digital input), O (digital output), M (marker), T (timer) or C (counter), the possible SER events are 'Any transition', 'Transition to low' and 'Transition to high'. If the variable type is AI (analog input) or AO (analog output), the possible SER events are 'Exceed setpoint', 'Exceed setpoint low' and 'Exceed setpoint high'.

5.3.5 Report printer configuration

For every report and for the SER messages, you can define the configuration of the printer that is used for outputting the report or the SER messages.

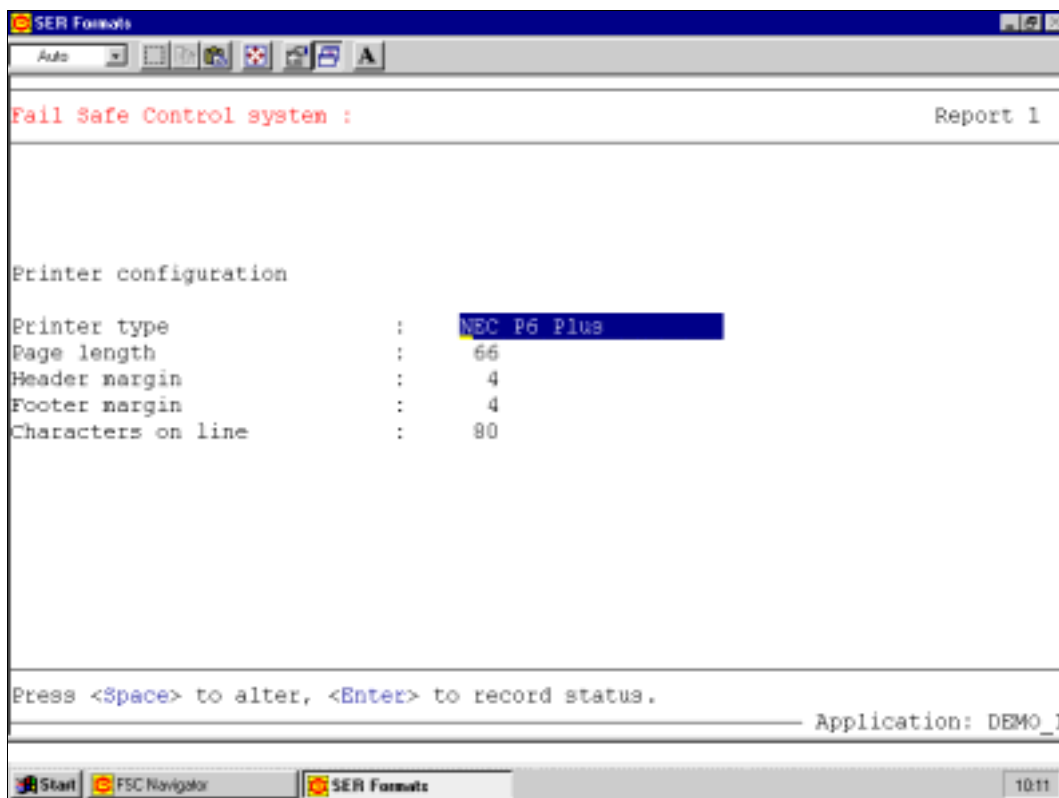


Figure 5-13 Defining the printer configuration

The following items define the printer configuration:

Printer type

Select the desired printer by toggling with <Space>.

Possible printer types are:

- | | | | |
|----------------|---------------|-------------------|------------------|
| – NEC P6 Plus | – EPSON FX80 | – EPSON FX1000 | – SiemensPT89-11 |
| – NEC P7 Plus | – EPSON FX80+ | – EPSON RX80 | |
| – NEC P6 Color | – EPSON FX100 | – Dummy printer | |
| – NEC P7 Color | – EPSON FX800 | – Siemens PT88-11 | |



A special option is the dummy printer. If you select this option, all attributes you used in the report text will be ignored.

Page length

Select one of the values 66 (11" paper) or 72 (12" paper) by toggling with <Space>.

Header margin

The number of empty lines on top of each page.

Footer margin

The number of empty lines at the bottom of each page.

Characters on line

The number of characters on one line. Select a value by toggling. The possible values depend on the selected printer type.

5.3.6 Main

The program asks if you want to save your changes. After you have answered the question, the program returns to the 'Edit reports and SER format' main edit menu.



5.4 SER format file

If you choose the 'SER format file' option, the program checks if a format file already exists. If there is a format file, the file is read and checked for invalid contents. If the file is corrupted, the following message is displayed:

SER message file corrupted.

The file is then ignored. If you want to keep this version, do not save the file when you leave the SER message editing. If the SER format file is correct, the following screen will be displayed:

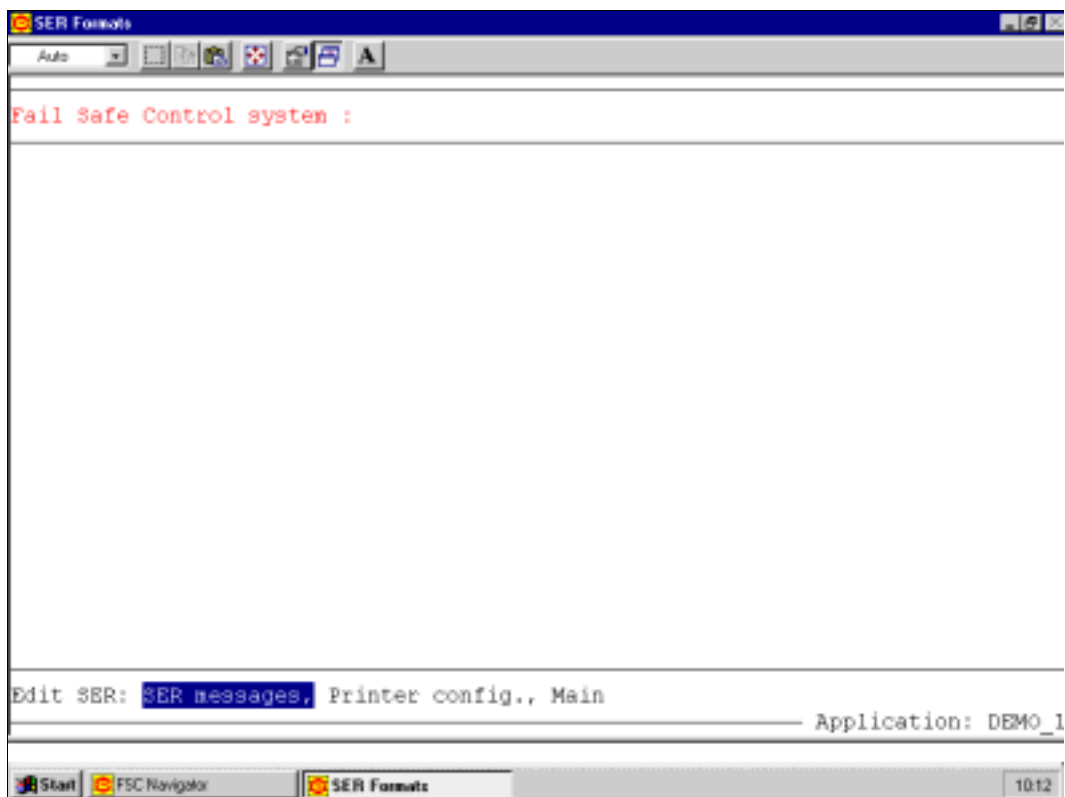


Figure 5-14 Edit menu for SER format file

Note:

The 'Edit reports and SER format' option of FSC Navigator can be used to change the default format to include service and qualification for these variables (see SER printer specification). As the service and qualification are stored in the COM module EPROMs, inclusion is only possible as long as the <Application>.Cxx files do not exceed the capacity of 2 Mbit EPROMs.

5.4.1 SER messages

If you choose the 'SER messages' option from the menu, the following screen will be displayed:

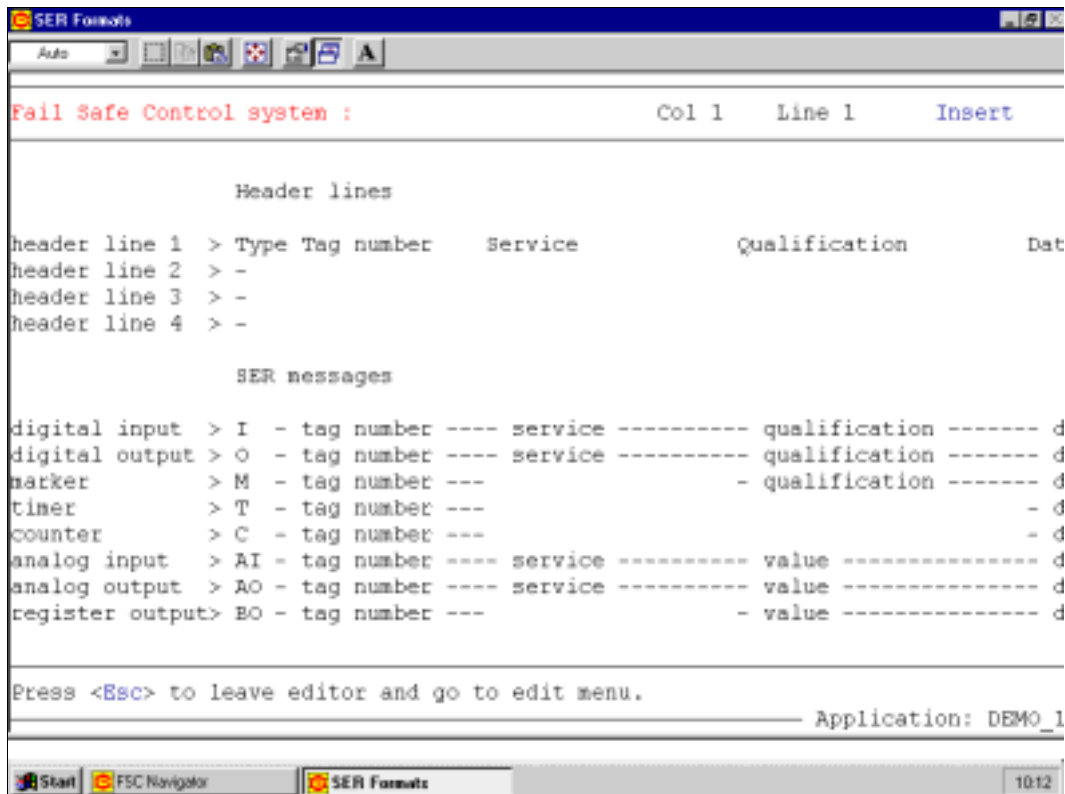


Figure 5-15 SER messages

Special functions to place tag numbers, services, values, etc. in the SER messages.

<Alt> + <T>

By pressing this key combination you can indicate where in the current SER message you want the tag number of the variable to be shown. As a result, 16 positions in the message are reserved for the tag number. This is indicated by inserting the text — tag number — in the line. You cannot edit this text and the cursor can only be positioned on the first character of the text. Pressing the key when the cursor is on the first character, removes the tag number indication and text.

<Alt> + <S>

By pressing this key combination you can indicate where in the current SER message you want the service of the variable to be shown. The service message is not defined for register outputs (BO).

<Alt> + <Q>

By pressing this key combination you can indicate where in the current SER message you want the qualification of the variable to be shown. This can only be done when editing the SER message for digital inputs, digital outputs and markers. For the other variable types the qualification is not defined.



<Alt> + <V>

By pressing this key combination you can indicate where in the current SER message you want the value of the variable to be shown. This can be done for variable types AI, AO and BO.

Note:

In future releases, SER for timers, counters and markers will not be supported anymore.

If you press <Esc>, the following menu will be displayed:

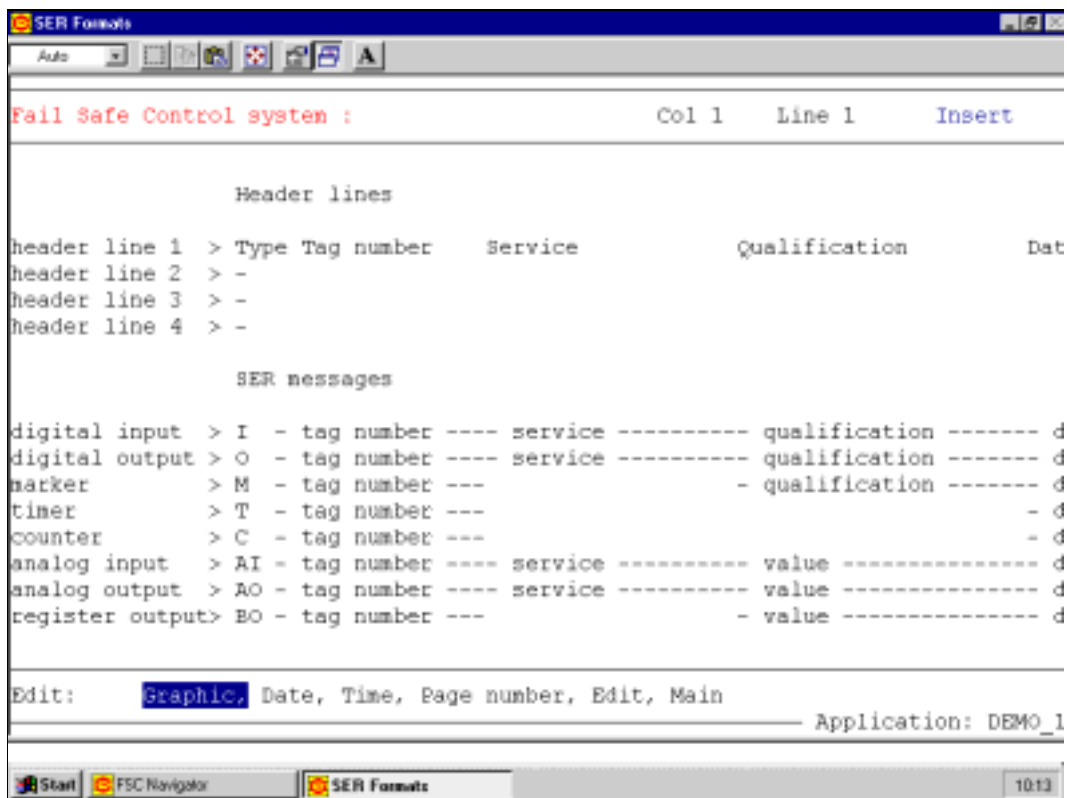


Figure 5-16 Edit menu for SER messages

- Graphic** : Gives access to a sub menu used to set graphic attributes of the text. Attributes are valid from the current cursor position until the end of the line. By choosing the attribute once more it is switched off again. Only the valid attributes will be displayed. The possible attributes are :
- Bold The text will be shown bold.
 - Underline The text will be underlined, only for printer output.
- Color** : Gives access to a sub menu to set foreground colors. Only available for color printers.
- Date** : Choosing this option inserts an indication in the text that tells where in the text the current date must be shown. This is shown by insert in the text '-- date ----' in the line. You cannot edit this text and the cursor can only be positioned on the first character of the text. Pressing when the cursor is on the first character removes the date indication.



Time	: This option has a similar function as the 'Date' function, but it inserts the current time.
Page number	: This option inserts the page number indication. The page number option is only available when editing the header of the SER format file.
Edit	: Brings you back to the edit screen.
Main	: Brings you back to the edit menu.



5.4.2 SER printer configuration

You can define the configuration of the printer on which the SER messages are to be printed.

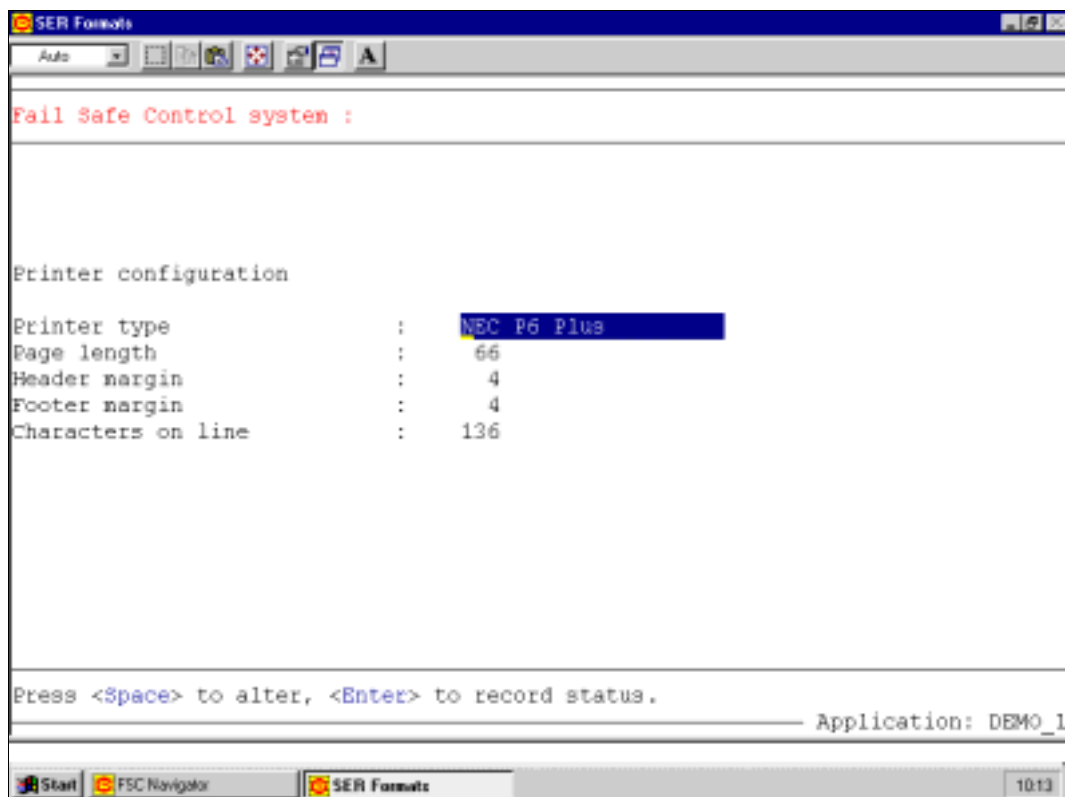


Figure 5-17 Defining the printer configuration

The following items define the printer configuration:

Printer type

Select the desired printer by toggling with <Space>.

The following printer types can be selected:

- | | | | |
|----------------|---------------|------------------|------------------------|
| – NEC P6 Plus | – EPSON FX80 | – EPSON FX1000 | – Numerical SER format |
| – NEC P7 Plus | – EPSON FX80+ | – EPSON RX80 | – Siemens PT88-11 |
| – NEC P6 Color | – EPSON FX100 | – Dummy printer | – Siemens PT89-11 |
| – NEC P7 Color | – EPSON FX800 | – FSC SER format | |

Special options are the dummy printer, FSC SER format and numerical SER.

Notes:

If numerical SER ranges are defined in the 'System Configuration' option of FSC Navigator, the numerical SER format must be used.

The format that FSCSER used remains supported. However, from FSC Release 500 onward, the FSCSER option has been replaced by FSCSOE, which is a separate Windows application. SER printing to directly connected printers is still supported.

If the FSC SER format has been selected, the marker service and marker qualification may be available in the SER. To specify marker services in the SER, toggle the 'Marker service in EPROM' option to 'Yes'. To specify the marker qualification in the SER, toggle the 'Marker qualification in EPROM' option to 'Yes'.

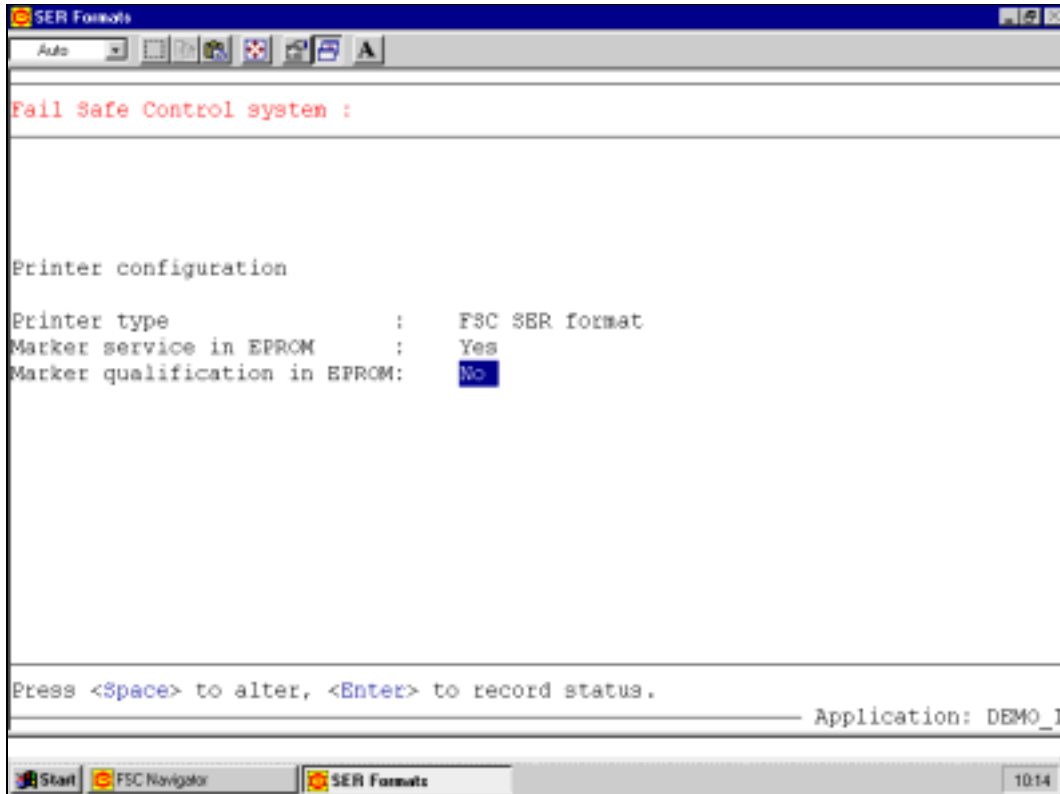


Figure 5-18 FSC SER format

If a printer device is selected, the following parameters can be set:

Page length

Select one of the values 66 (11" paper) or 72 (12" paper) by toggling with <Space>.

Header margin

The number of empty lines on top of each page.

Footer margin

The number of empty lines at the bottom of each page,

Characters on line

The number of characters on a line. Select a value by toggling. The possible values depend on the selected printer type.

5.4.3 Main

The program asks if you want to save your changes. After you have answered the question, the program returns to the main edit menu.



5.5 Deleting files

If you choose the 'Delete files' option, the following screen will be displayed:

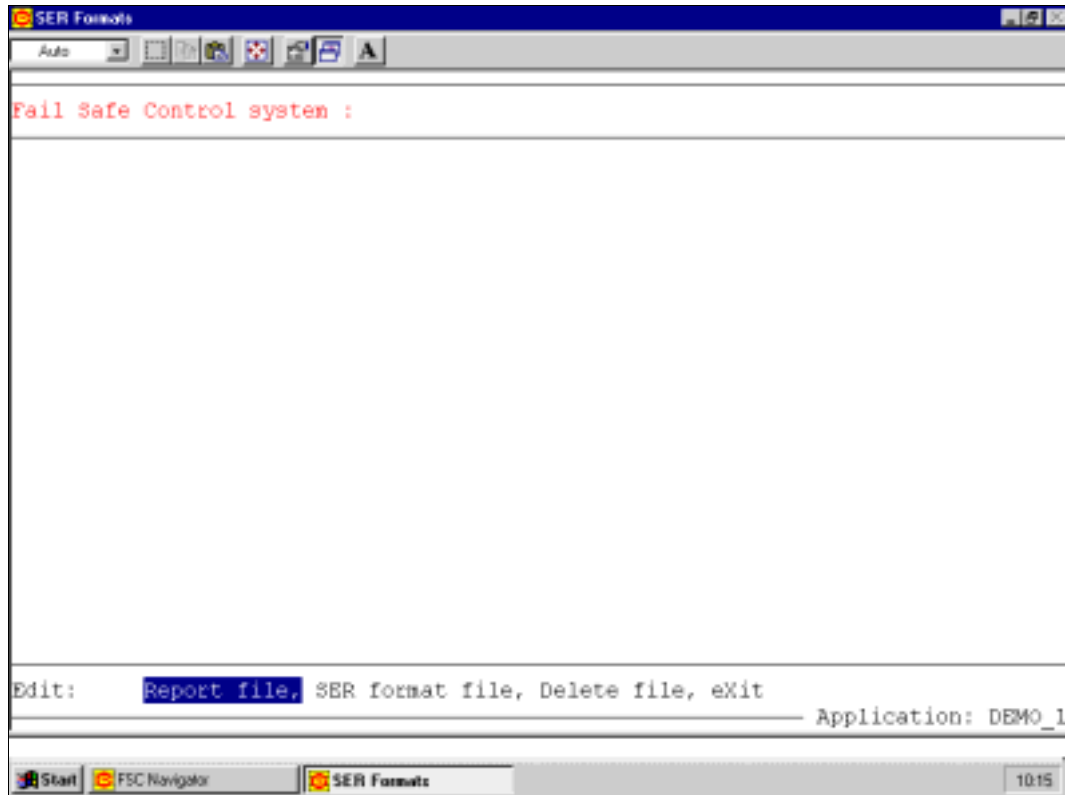


Figure 5-19 Delete menu

To delete a report file, choose 'Report file' from the delete menu. The following screen will be displayed:

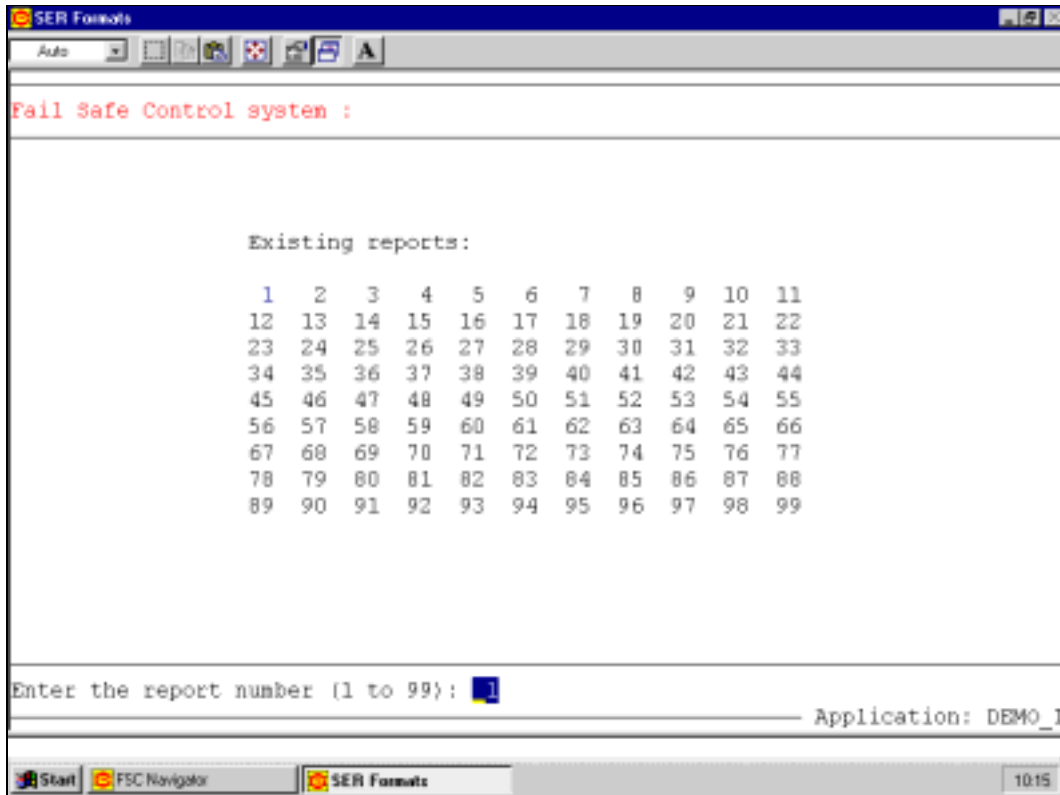


Figure 5-20 Deleting a report file

Enter the number of the report file you want to delete. The following question will be asked:

Do you want to delete the report file? (Y/N)

If you answer 'Y', the report file will be deleted.

To delete the SER format file, select 'SER format file' from the delete menu. The following question will be asked:

Do you want to delete the SER format file? (Y/N)

If you answer 'Y', the SER format file will be deleted.



5.6 Default SER format specification

This subsection describes the default format of the SER reports. The default SER format is used in combination with FSCSOE, which is a separate Windows application.

A distinction is made between event reports for process variables and SER messages. The SER header and variable event formats can be changed using the 'SER Format' option of FSC Navigator. The format of the SER messages is fixed.

5.6.1 Default SER header

The SER header is printed at the top of every new page. It is a fixed text string of 87 characters: 'Type', <space>, 'Tag number', <4 spaces>, 'Service', <12 spaces>, 'Qualification', <9 spaces>, 'Date', <19 spaces>, 'Time'.

The SER header is preceded by a printer control sequence, consisting of an ASCII form-feed character and four CR-LF character combinations. The header is terminated by two CR-LF character combinations.

5.6.2 Default variable event formats

An identical format is used for all variable types. The format consists of 104 ASCII printable characters. It contains the following fields:

Field	Description	Length
margin	line margin of 8 spaces	8
type	type identification	3
tag number	variable tag number	16
service	variable service	19
qualification	variable qualification	22
date	date stamp	23
time	time stamp	11
terminator	carriage return + line feed	2

total length		104

5.6.2.1 Type

The type identification field consists of a fixed string of three characters. Depending on the variable type, the string is:

- 'I ' for digital inputs,
- 'O ' for digital outputs,
- 'M ' for markers,
- 'T ' for timers,
- 'C ' for counters,
- 'AI ' for analog inputs,
- 'AO ' for analog outputs,
- 'BO ' for register outputs.



5.6.2.2 Tag number

The tag number field contains the tag number of the variable as assigned in the system configuration or FLDs. If the tag number is less than 16 characters, the remaining part of the field is filled with spaces.

5.6.2.3 Service

The service field contains the service of the variable as assigned in the system configuration or FLDs. If the service is less than 19 characters, the remaining part of the field is filled with spaces. The service is only printed for variable types I, O, AI, AO and optionally for M. For variable types BO, T and C, the field is completely filled with spaces.

5.6.2.4 Qualification

The format of the qualification field depends on the variable type.

Variable types I, O and M

For types I, O and M, the field consists of two parts. The first part is 6 characters long and is used to indicate the current status of the variable. If the status is high, the field is filled with spaces. If the status is low, a text of five characters plus a trailing space is printed. The text is derived from the compiler message file and depends on the language. For the English version, the text is 'NOT'.

The second part of the qualification field contains the qualification of the variable as assigned in the system configuration or FLDs. If the qualification is less than 16 characters, the remaining part of the field is filled with spaces.

Notes:

1. At this moment, the qualification is not printed for markers. If you wish to include the marker qualification in the event report, use the 'Edit reports and SER format' option of FSC Navigator.
 2. It is quite common that the qualification represents a negation, e.g. 'not low'. In this case, when the variable status is low, both the 'not' in the first part of the qualification field and the 'not' text within the qualification itself are suppressed, i.e. the string that is printed is 'low' instead of 'not not low'. The remaining part of the field is filled with spaces.
-

Variable types T and C

For variable types T and C, the field is completely filled with spaces.

Variable types AI and AO

For the variable types AI and AO, the qualification field is used to report the type of event. Three different texts can appear to indicate that the analog value exceeded the configured SER high value, became less than the SER low value, or re-entered the healthy range between SER low and SER high. The messages are derived from the compiler message file and depend on the language. For the English version, the texts are 'Maximum', 'Minimum' and 'healthy', respectively. A number of 20 characters is reserved for the texts. The last two characters of the qualification field are always filled with spaces.



Variable type BO

For type BO, the qualification field contains the value of the BO variable followed by the configured engineering units.

Force events

If a variable in the FSC system is forced to a specific value, or a previous force is cleared, an event report is generated. In this case, the qualification field is used to indicate the type of force event. The texts that are printed are derived from the compiler message file and depend on the language. For the English version, the texts are 'force set' and 'force cleared', respectively. The force texts may be 22 characters long. The remaining characters are filled with spaces.

5.6.2.5 Date

The date field consists of the following parts: weekday, date and month. The weekday and date parts are separated by a comma and a space. The date and month parts are separated by a dash.

The weekday part contains a text that is derived from the compiler message file and depends on the language. For the English version the following texts are used: 'Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday' and 'Saturday'. A number of 16 characters is reserved for the weekday. The weekday text is always immediately followed by a comma and a space character. If the text is less than 16 characters, the remaining characters of the date field are filled with spaces.

The date and month parts are printed as a two-character decimal value. If the value of the date or month is less than 10, a space is printed at the first character position.

5.6.2.6 Time

The time field consists of the following parts: hours, minutes, seconds and hundredths of seconds. The hours, minutes and seconds parts are separated by a dot. The seconds and hundredths of seconds parts are separated by a colon.

All parts of the time field are printed as two-character decimal values. If the value of the hours is less than 10, a space is printed at the first character position. If any of the other values is less than 10, a zero is printed at the first character position.



5.6.3 Standard SER messages

The FSC system contains several standard SER messages. The messages cannot be suppressed. The texts are derived from the compiler message file and depend on the language.

5.6.3.1 System fault message

The system fault message is printed if a fault is detected within the FSC system. The message indicates the system number, rack and position where the fault occurred and the date and time. The format of the message is adapted to the formats for variable events. The default format consists of fixed text and five variable fields:

- 12 leading spaces,
- text: 'System fault ',
- system number field,
- rack number field,
- position number field,
- date field,
- time field.
- carriage return, line feed.

The system, rack and position number fields are separated by a comma and a space character. The total length of the message is equal to the length of the variable event report, i.e. 104 characters for the default format.

System number field

The system number field consists of the text 'System:' and the system number in decimal representation. The system number field is aligned with the service field of the variable event formats as described above. The length of the field depends on the system number. If the system number is less than 10, the field is 9 characters long. If the system number is 10 or higher, the field is 10 characters. If the service field is not configured in the variable SER format, the system number field is not printed.

Rack number field

The rack number field consists of the text 'Rack:' and the rack number in decimal representation. The rack number value always occupies two character positions. If the rack number is less than 10, a space is printed at the first character position. The rack number field is aligned with the qualification field of the variable event formats as described above. If the qualification field is not configured in the variable SER format, the rack number field is not printed.

Position number field

The position number field consists of the text 'Position:' and the position number in decimal representation. The position number value always occupies two character positions. If the position number is less than 10, a space is printed at the first character position. The position number field is printed directly after the rack number field. If the qualification field is not configured in the variable SER format, the position number field is not printed.



Date and Time

The format of the date and time fields is identical to the format used in the variable event reports. The fields are aligned to the corresponding fields in the variable event formats.

5.6.3.2 All forces cleared message

This message is generated if a command was issued from the FSC Development System to clear all forces. The message consists of the text 'All forces cleared' and the date and time.

The text is aligned with the service field of the variable event formats as described above. If no service field is configured, the text is aligned with the qualification field. If both the service and qualification field are not configured, no text is printed.

The format of the date and time fields is identical to the format used in the variable event reports. The fields are aligned to the corresponding fields in the variable event formats.

5.6.3.3 SER message error

The SER message error text is printed if an invalid SER message is selected. The message consists of a text 'SER message error'. The text is aligned with the service field of the variable event formats as described above. If no service field is configured, the text is aligned with the qualification field. If both the service and qualification field are not configured, no text is printed.

This SER message error should never appear. If it does, please contact Honeywell Safety Management Systems.



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Helping You Control Your World



Fail Safe Control

Section 6: Functional Logic Diagrams

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6.1 Introduction

The application task of the FSC system is specified by means of functional logic diagrams (FLDs). These FLDs can be created using the 'Design FLDs' option of FSC Navigator. This option starts the FLD design editor, which allows you to use the screen as a drawing sheet and the cursor as a pencil.

The 'Design FLDs' option can be called in three ways:

- **Menu bar:** Project / Design FLDs
- **Buttons bar:** FLD
- **Navigation area:** FSC Project Configuration / Design Functional Logic Diagrams

This section explains how to use design functional logic diagrams:

- Starting up the FLD design editor
- Menu control
- Input of variable data

Your functional logic diagrams (FLDs) are limited to a predefined set of symbols. When designing your functional logic diagrams, you can select a symbol from a special graphics sheet (symbol selection sheet) (see subsection 6.3.2). Certain symbols need additional data like symbol height, width or I/O tag number. If required, you will be asked to enter the data on the bottom line of the screen. In a number of cases you can select data like tag numbers, sheet references or FLD numbers from a pop-up window. The FLD design editor will assist in selecting a position for a symbol, by means of a grid and a snap-to-grid function, which automatically selects the nearest grid position or half grid position of the cursor.

6.2 Getting started

6.2.1 Starting the FLD design editor

If you activate the 'Design FLDs' option of FSC Navigator, the start-up screen appears:

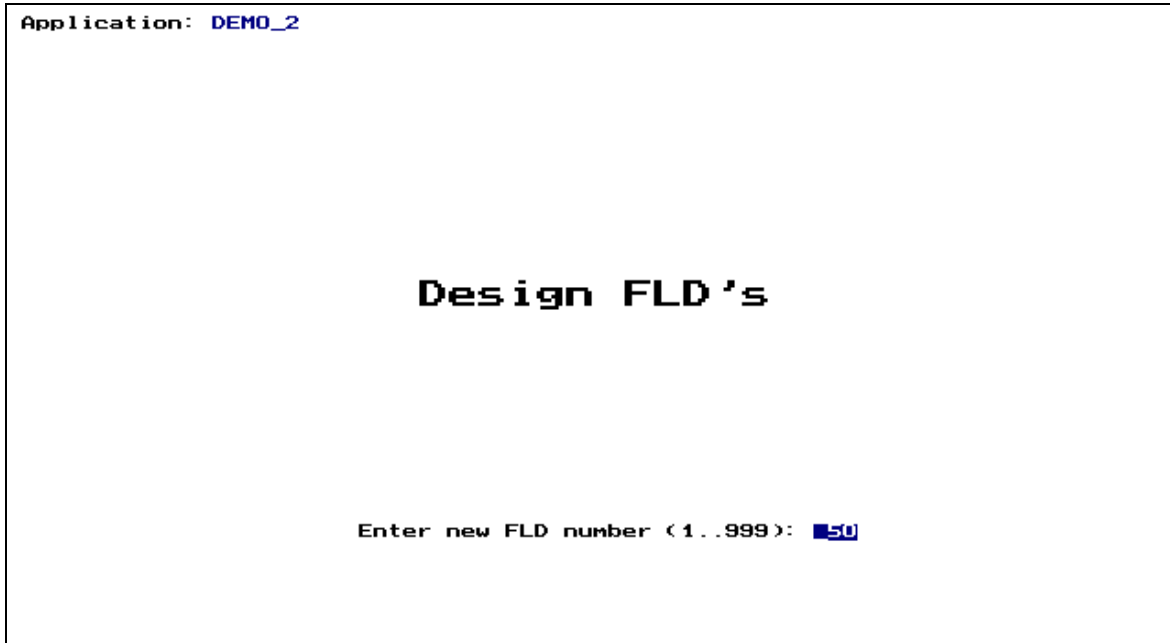


Figure 6-1 FLD design editor: start-up screen

To open a functional logic diagram, enter its number (between 1 to 999). You can also press <?> to open a popup window that lists the available FLD numbers with their description. Move the cursor bar to the FLD you wish to open, and press <Enter>.

If the FLD you entered already exists, a message appears on screen which says that the symbol selection sheet is being loaded.

If the FLD you entered does not exist, the title block appears on the screen (see Figure 6-2). Here you must enter data which is needed during the design of your FLD: block type, unit, subunit, revision, engineer, design date and a maximum of 9 descriptive texts. Most of this data is also used in the hard copy of the FLD.

You can leave the title block at the bottom in several ways:

- <Enter>
- <PgDn>
- <↓> or
- <Esc>

If you leave the title block immediately at the top by pressing <Esc> without entering any data, you will return to the previous situation.



```

FLD number      : 110
Block type      : Program block

Unit            :
Subunit         :
Revision        :
                  No Date      Description
                  0  11-6-1997  FIRST ISSUE
                  A
                  B
                  C
                  D
                  E
Engineer         : RK NL33
Design date      : 11-6-1997
Text1           : FUNCTIONAL LOGIC DIAGRAMS
Text2           :
Text3           :
Text4           :
Text5           :
Text6           :
Text7           :
Text8           :
Text9           :

```

Figure 6-2 Title block

After you have entered the correct data in the title block, the design sheet is displayed.

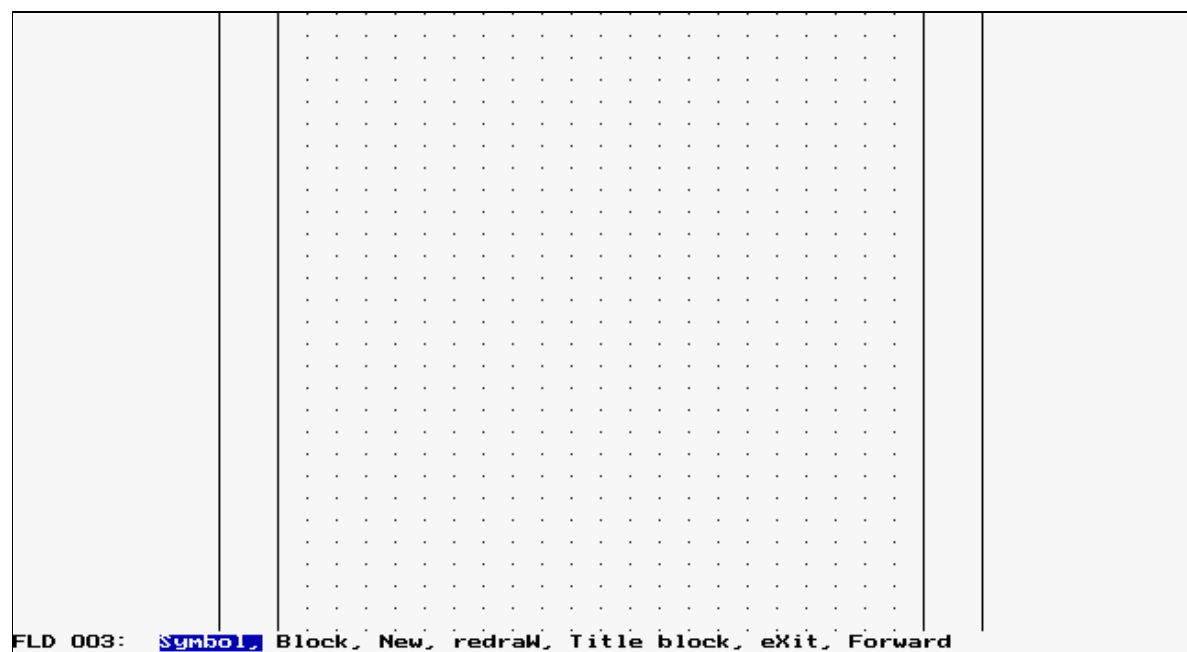


Figure 6-3 Design sheet (main menu)

The FLD number to the left of the menu bar is the number of the functional logic diagram that is being edited.



6.2.2 Using the FLD design editor

In order to be able to work with the FLD design editor, you must first know what you have at your disposal for making your design.

– Design sheets

You can use 999 FLDs for your design (1..999). Each of these FLDs can have the size as specified in your symbol library. The screen displays the current FLD. You can design six types of FLDs:

- Program blocks Normal logics.
- Function blocks Logics used as a subroutine function in other FLDs (program blocks or function blocks)
- Equation blocks Contains an equation approximation, which can be used as a macro function on other logic diagrams
- Comment FLDs containing descriptive texts and/or symbols which are not placed in the variable database (e.g. cover sheet, legend of symbols). These FLDs will be ignored by the 'Translate Application' option of FSC Navigator (the compiler).
- Sheet index FLD which only contains a title block and no logics. When this FLD is present in the range of FLDs to be plotted or printed, all non-empty FLDs, which are present in the application, are registered in a sheet index listing containing the FLD number, revision number and description text 4 of the title block. If it does not fit on a single FLD, it is continued on the next FLD(s). You must therefore take care that enough empty FLDs are reserved after the sheet index. Index sheets are ignored by the 'Translate application' option of FSC Navigator (the compiler).
- Tag number index This is similar to a sheet index, with the difference that now all tag numbers of placed inputs and outputs are registered in an index listing. For each tag number the service is recorded as well as the FLD(s) on which the signal is used. If the service of an on-sheet reference symbol is an I/O tag number, this FLD is also recorded.

– Message line

The bottom line of the screen is used to display messages, to ask for input or to display the menus. The same line will be used by the user for entering and editing input information. Sometimes the complete screen is used when a block of strings must be edited (i.e. title block)

The service of the menus and the available edit functions are described in section 6.2.3 and 6.2.4. Input can be given with the mouse or the keyboard. An error message is displayed for two seconds unless you press any key (keyboard or mouse) or move the mouse to another position.

– Selection sheets and the symbol library

A selection sheet is stored in memory which shows all graphic representations of the available symbols in the symbol library (see subsection 6.3.2). The selection sheet for a function block is different from the selection sheet of a program block (see title block or function block). A function block has no system outputs or off-sheet transfers, but it has other symbols which cannot be used on program blocks: function block inputs and function block outputs. When you want to select a symbol from the symbol library, the selection sheet and the design sheet are swapped and swapped back after the selection.



– Cursor

If you are selecting a screen position, a cross cursor is displayed at the current screen position. If a menu option must be chosen, a block cursor marks the current option. When string input must be entered, a text cursor (underscore) is available. The cursors can be controlled with the mouse or the arrow keys of the keyboard.

– Grid

Raster drawn on the design sheet to help the user with choosing a position on the screen. Only half grid and grid positions can be chosen.

In order to be able to draw with the FLD design editor some basic operations must be familiar to the user.

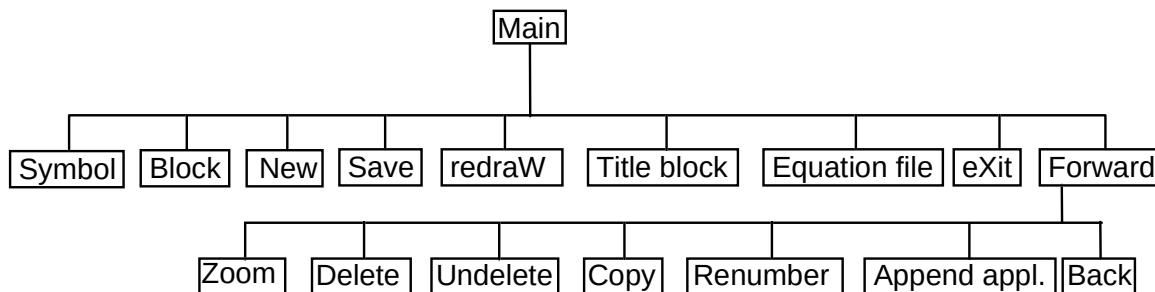
– Selection of a command from the menu

The commands can be entered by choosing an option of one of the menus of the FLD design editor. How you can choose an option from the menu is described in subsection 6.2.3 ("Menu control"). The FLD design editor has three menus:

- MAIN
- SYMBOL
- BLOCK

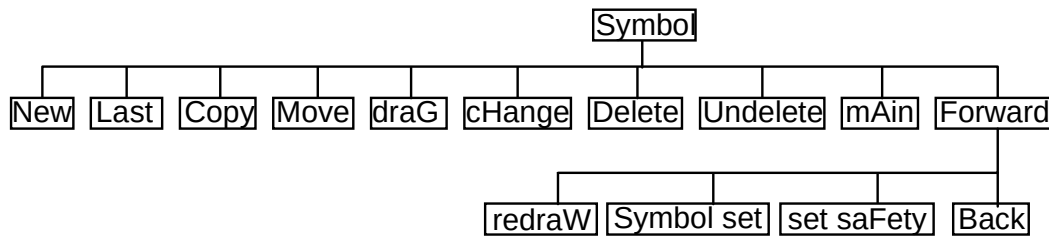
SYMBOL and BLOCK are submenus of MAIN which can be reached by choosing the **Symbol** or **Block** option.

In certain situations some options are not recorded in the menu because they have no effect. For example, If an FLD is empty, you cannot select the options change, move, drag, delete or copy a symbol because there are no symbols. You cannot choose undelete when you did not recently use the delete option. Beside SYMBOL and BLOCK, MAIN contains commands which concern the whole FLD:



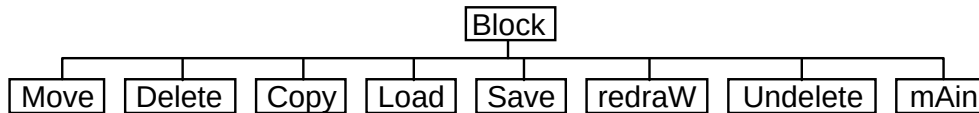
Refer to subsections 6.4.1 and 6.5.1.1 on details on using the MAIN menu and commands.

The SYMBOL menu contains options which concern operations on single symbols:



Refer to subsections 6.4.2 and 6.5.1.2 for details on using the SYMBOL menu and commands.

The BLOCK menu contains options which concern a group of symbols:



Refer to subsections 6.4.3 and 6.5.1.3 for details on using the BLOCK menu and commands.

If you choose one of the options Move, Delete, Copy or Save, the FLD design editor will prompt you to enter two corner positions of a rectangle which is the block to be manipulated. The block is marked with a dashed rectangle.

The MAIN menu and the SYMBOL menu are split into two parts. The **Forward** option will take you to the second part of the menu, and the **Back** option will take you back to the first part. Every command can be canceled at every step where input is asked, by pressing <Esc>. In that case the FLD design editor returns to the situation before the start of the canceled command. For details on using the menus refer to subsection 6.2.3.

– Selection of a symbol from the screen

The first thing you will do when you entered a new FLD is placing a new symbol. To select a symbol you must choose one from the selection sheet which is displayed after you choose the SYMBOL command **New**. You can select a symbol by moving the cursor into the inner area of a symbol and pressing <Enter>, or the left mouse button. You must also select a symbol when you choose one of the symbol options move, drag, change, copy or delete. In that case the procedure is the same but you must select an already placed symbol from the design sheet.

– Selection of a position on the screen

When you want to place a symbol or when you want to move or copy a block or a symbol, a (new) position must be chosen on the design sheet. This position can be chosen by moving the symbol or block to a position on the screen and pressing <Enter>.

In the inner sheet area, grid is drawn to help the user with choosing a position. If the symbol overlaps with another symbol a beep is generated and you must select a legal position or you must cancel the operation with <Esc>.



– Automatic connection

If you draw a line at the screen and define a starting point or an end point inside a symbol, the FLD design editor makes a connection with that symbol. The line is automatically cut at the outer border of that symbol. You will be asked for connection points again until the last drawn line is connected at both ends. This means that you can continue drawing lines until you press <Esc> or connect to the point that was chosen last. A line is also cut at the borders of the inner sheet. This cutting point is regarded as a connection point (later an I/O or transfer can be connected there). If you try to make an illegal connection, a beep will be generated. The connection will not be made, and the line that was drawn last will be undone. Illegal connections are connections of wrong signals to a certain symbol i.e. a timer only can have a boolean input signal at its set input. An analog output can only be connected to a binary signal. Calculation functions can only have binary inputs and/or outputs. Inputs and outputs of a function block are drawn as a part of the function block symbol to help the user with making legal connections. The same is true for registers and counters.

For logic functions, calculations, inputs, outputs, constants and timers, the FLD design editor shifts connections that are made at the output side of the symbol, to half height of the symbol. If a line is drawn across a symbol, the largest of the line parts on the input side or the output side of the symbol is accepted as a connection, the other part is erased from the screen.



6.2.3 Menu control

The FLD design editor is completely menu-driven. The menus are placed on the bottom line of the screen. The menus can be operated with both the keyboard keys or a mouse.

In each menu you can move a block cursor from left to right (or right to left) by pressing the right and left arrow keys on the keyboard or by moving the mouse from left to right (or right to left). A menu option can be chosen by moving the block cursor to a particular menu option and pressing <Enter> or by pressing the left mouse button. A choice can also be made by pressing the uppercase letter of an option without the need of moving the cursor to a particular option first. The character may be uppercase or lowercase. The MAIN menu and the SYMBOL menu are split up in two parts. The **Forward** option will take you to the second part of the menu, and the **Back** option will take you back to the first part.

6.2.4 The string editor

If the program needs information from the user, a message is displayed on the last line of the screen. The string editor, is activated from that moment on until a termination character is entered. It can happen that the program changes the characters that you entered into uppercase characters (i.e. when location for input is entered).

If the input must be integer numbers, the editor will check whether this condition is satisfied after each character input. As soon as a non-integer character is entered, a beep will be generated and the character is not accepted. The FLD design editor generates a default value for most input, which is either an empty string or a value that was previously entered. If you wish to edit the default input, the following key commands are available:

- | | | | |
|----------------|----|--------------|--|
| – <←> | or | <Ctrl> + <S> | Move left. |
| – <→> | or | <Ctrl> + <D> | Move right. |
| – <Backspace> | or | <Ctrl> + <H> | Move left and delete character under new position. |
| – <Home> | or | <Ctrl> + <A> | Move to the begin of the string. |
| – <End> | or | <Ctrl> + <F> | Move to the end of the string. |
| – | or | <Ctrl> + <G> | Delete character under cursor. |
| – <Ins> | or | <Ctrl> + <V> | Switches insert mode on or off. If it is off a new character will overwrite an old one (default off). |
| – <Ctrl> + <P> | | | Makes a printout of the displayed screen. You can go on with editing after the printout is completed (Epson MX-80 only). |
| – <Ctrl> + <Y> | | | Deletes from the cursor position to the end of the string. |
| – <PgUp> | | | Makes the cursor step twice as big. The maximum value is 22 x grid. |
| – <PgDn> | | | Makes the cursor step twice as small. The minimum value is 0.5 grid. |



The termination characters which can be used in most cases are:

- <Enter> End of the input (and possible go to the next line).
- <Esc> Interrupt string input and return to the situation just after the last successful command. (In most cases the program returns to the menu, but when entering an FLD number for a multiple sheet transfer, the references which are entered before the current one, are accepted and placement goes on).

When a block of strings is edited:

- <←> or <Ctrl> + <S> Move left. If the cursor is at the begin of the string, the cursor will move to the last character of the previous string.
- <→> or <Ctrl> + <D> Move right. If the cursor is at the end of a string, the cursor will move to the first character of the next string.
- <↑> or <Ctrl> + <E> Line up
- <↓> or <Ctrl> + <X> Line down
- <PgUp> or <Ctrl> + <R> Page up. Go to the first string.
- <PgDn> or <Ctrl> + <C> Page down. Go to the last string.

The editor only accepts a certain number of characters which depends upon the application of the input. When you try to enter more characters, the beeper is sounded. When not all characters are entered, the tail of the string is trailed with space characters up to the full length of the string. If illegal characters are entered the beeper sounds.

If you have to enter predefined possibilities you can toggle between them by using <Space>. If it is possible to toggle the following message will be displayed on screen:
'Press <Space> to alter'.

6.2.5 The revision editor

Every time you change something on an FLD, the FLD design editor displays a revision window when you leave that FLD.

The revision window will be displayed if:

- the application was compiled before, and
- you did not make any changes on the FLD on the same day.

This means that there will only be one revision per day.

If you are manipulating a sheet transfer, you can also edit a revision upgrade of a source or destination FLD.

Note:

Watch the FLD number in the header of the revision window.

When you enter an empty date or description this revision will not be adapted in the revision list.

You may not enter an empty revision number.

6.2.6 Selection lists

In several situations the FLD design editor can help you with selecting data to be inserted by means of a selection list containing legal input data. You can ask for these listings by pressing <?> during editing.

You can select a component of this listing by pressing <Enter>.

If you must enter a string which is non-numerical (i.e. a tag number), you can also enter some characters first and then enter <?>. In this case only the strings beginning with these characters are listed.

You can leave a listing and return to editing by pressing <Esc>.

You can change your choice by moving the cursor.

You can get a new page by pressing <PgDn>.

You can get the previous page by pressing <PgUp>.

You can select data by pressing <Enter>.

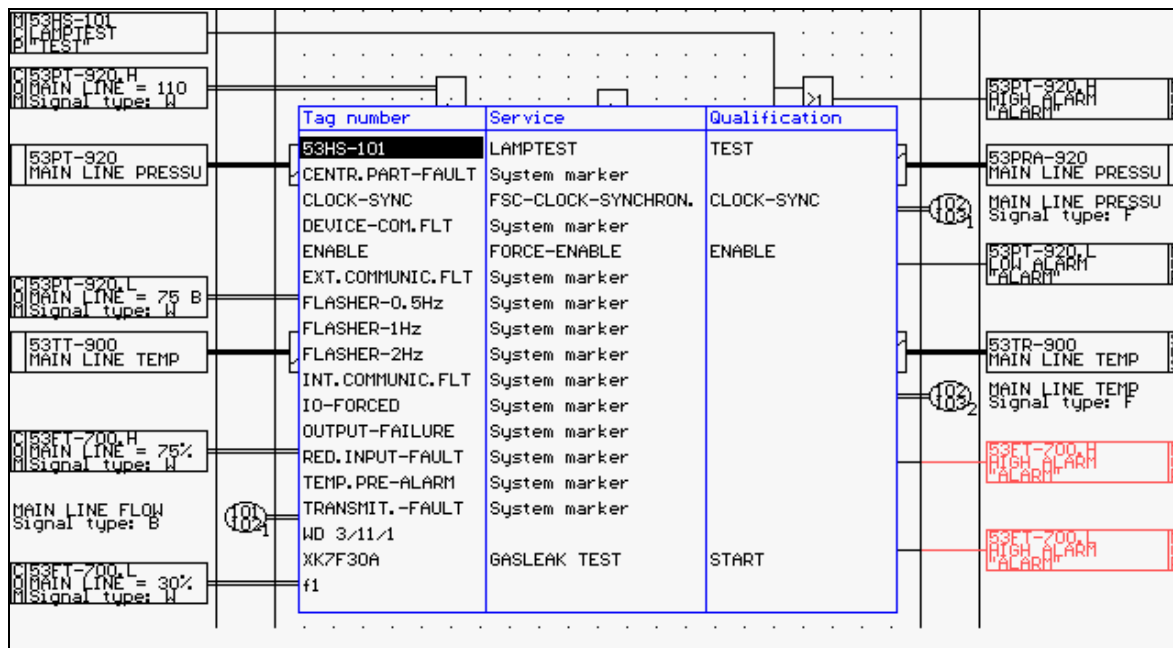


Figure 6-4 Example of a selection list



These listings can be asked in the following cases:

- Input of tag numbers of inputs or outputs.
Service, qualification and FLD number are recorded in the listing.
- Input of source FLD or destination FLD for the selection of a sheet reference.
All unlinked sheet references are recorded in the list as well as the service text and the FLD number.
Also listed are all off-sheet references which do not contain the maximum allowed number of sheet references.
- Input of sequence number for selection of a sheet reference
All unlinked sheet references on the destination FLD entered before are listed with the service.
- During input of service of a sheet reference, the tag number of an input or an output that is directly connected to the reference symbol on the source FLD of the sheet reference. Only the inputs and outputs that are placed on the source FLD are recorded.
- Input of a tag number of a PID. Also the PID number is recorded in the list.
- Input of the sheet number of a function block or equation block when you want to select a function block or equation block. The fourth descriptive text of the title block is recorded and the FLD number of the function block or the equation block.
- Input of an FLD number in the option "New". All FLD numbers of existing FLDs and the **fourth** descriptive text of the title block will be listed.

6.3 Symbol set

The FLD design editor uses a symbol set which is divided into 11 subsets:

1. Non-functional symbols
2. Connection lines
3. Logical functions, calculations, and compare functions
4. Registers, counters, constants, signal conversion
5. Function blocks, PIDs, equation blocks
6. Function block inputs and function block outputs
7. Timers
8. Inputs and outputs (I/O)
9. Markers: sheet transfers
10. Markers: flip-flops
11. Texts

Each of these symbols is explained below.

6.3.1 Non-functional symbols

The non-functional symbols are:

– Basic sheet symbol

The basic sheet is a symbol which divides the FLD into an I/O area and the internal area.

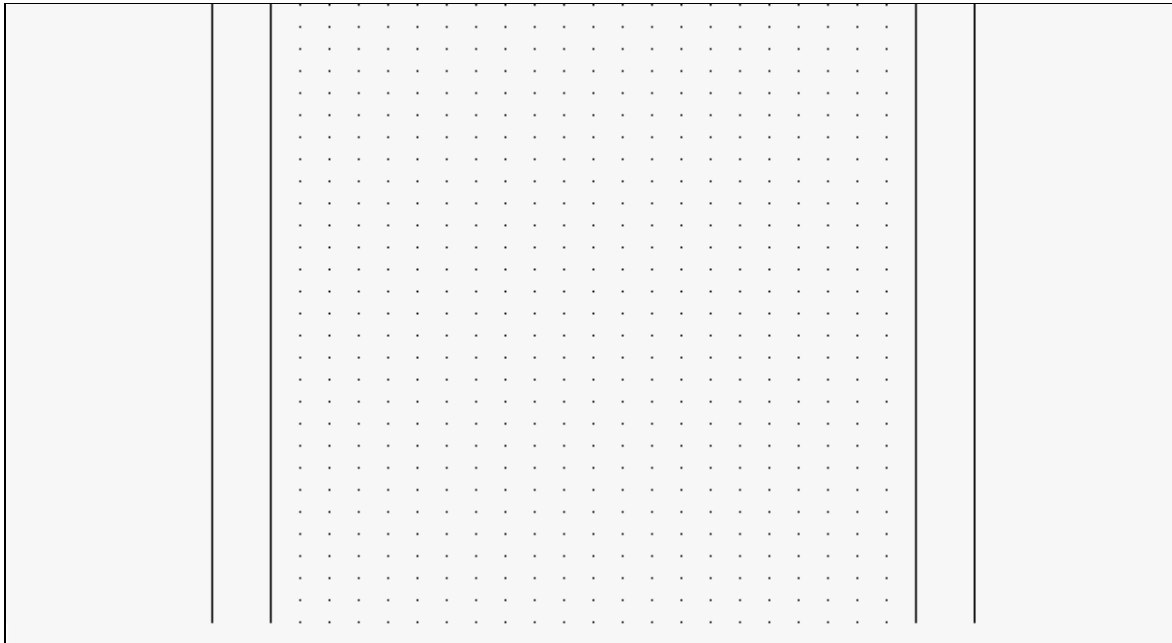


Figure 6-5 Example of basic sheet symbol as defined in symbol library SYM1 (screen view)



– Sheet identification symbol (title block)

This symbol contains all sheet-dependent data

- Unit
- Subunit
- Revision: RevNo RevDate RevDescription
- Engineer
- Design date
- A maximum of 9 comment texts (e.g. the fourth text is the sheet description)

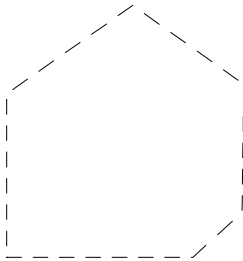
This data must be entered when a new FLD is selected. When an FLD already exists, the data is displayed and can be changed by means of the title block option in the MAIN menu of the FLD design editor. The sheet identification symbol is not visible for the user, but when you print the FLD, this symbol is drawn as an information block under the FLD. The information block also contains data from the description fields which were defined using the 'System Configuration' option of FSC Navigator (Installation \ Description).

E				Customer: Text5	Text1	Text2					
D				Principal: Text6	Text3	Text4					
C				Plant 1: Text7	SAFETY MANAGEMENT SYSTEMS	Tel.: +31 73-6273273	Date:		By:		
B				Plant 2: Text8		Fax: +31 73-6219125	Drawing number:				
A				Req/Order no.: Text9		P.O. Box 116					
0						5201 AC	Serial	Project	Unit	Sheet	Cont'd
Rev.	Date	Description	Chk'd			's-Hertogenbosch	Code		Code		

Figure 6-6 Example of sheet identification symbol as defined in the symbol library SYM1

The basic sheet symbol and sheet identification symbol are automatically placed when an FLD is entered.

– Revision cloud(s)



This symbol consists of dashed lines which during drawing are automatically connected by the program until the first point is connected to the last point or <Esc> is pressed. Clouds are used to mark parts of the logics which have been revised.

6.3.2 Functional symbols

If you entered a design FLD, you can start drawing your functional logics. If you select 'New' in the symbol menu, a selection sheet will appear on your screen. Depending on the block type you entered in the title block, the selection sheet for a program block or the selection sheet for a function block will appear.

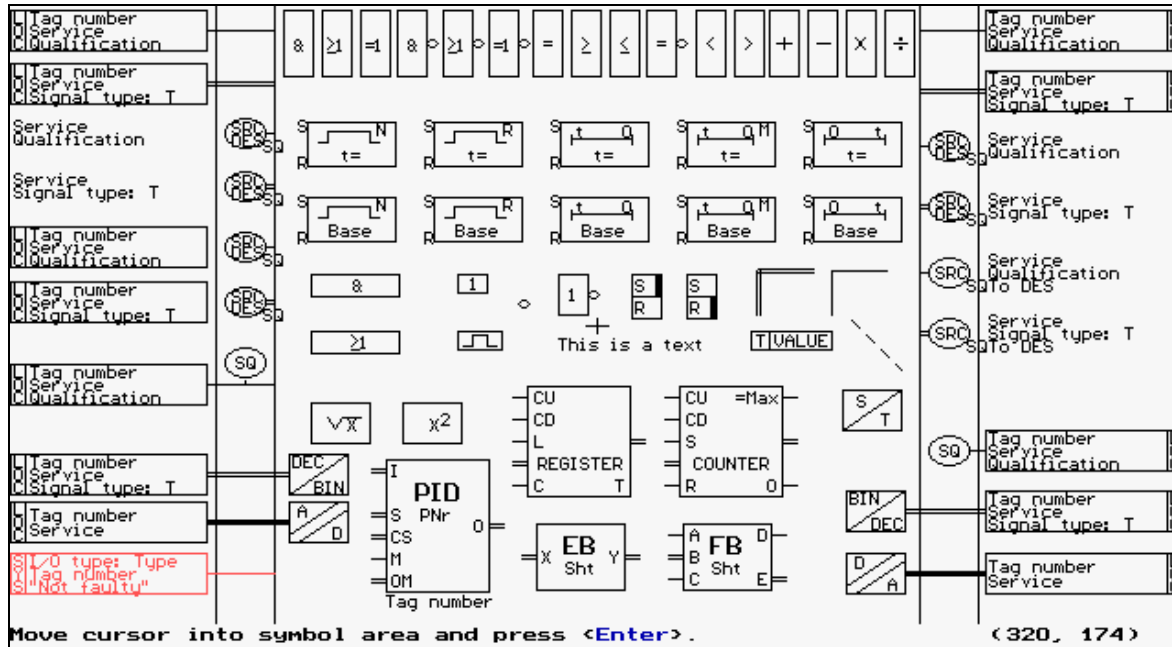


Figure 6-7 Selection sheet of a program block as defined in SYM1

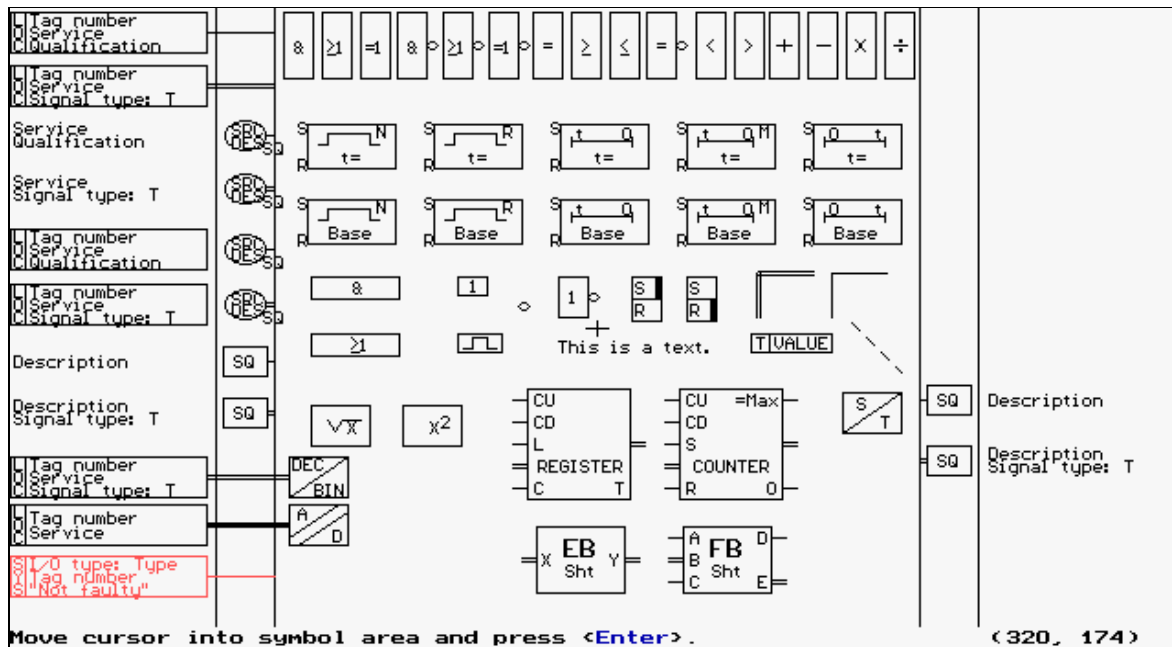


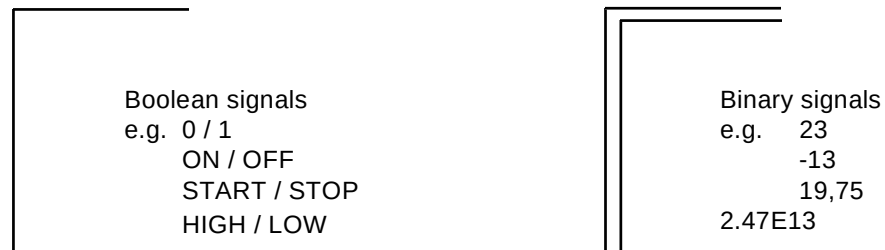
Figure 6-8 Selection sheet of a function block as defined in SYM1



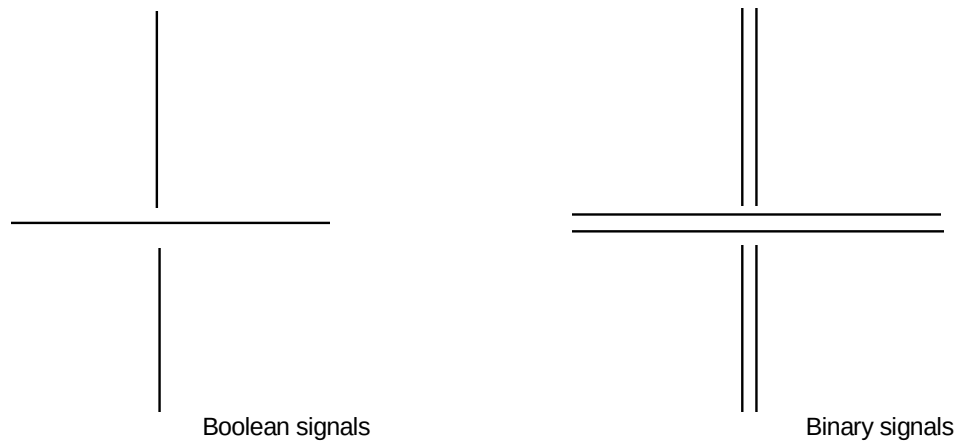
6.3.2.1 Connection lines

Lines represent the connection between two components. Two types of lines (signals) can be drawn vertical or horizontal but the direction has no functional meaning. When lines cross each other they are not connected (the program automatically makes a break in the vertical line). When lines intersect, they are connected and the line is split into two lines at the connection point, unless a connection is made at an end point of a line.

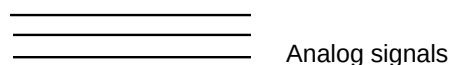
Single lines and double lines



Crossing of signals



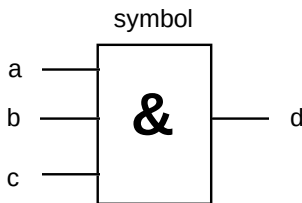
A third line type (thick line) is used for analog signals as represented in the Analog I/O converter symbols.



6.3.2.2 Logical functions, calculations and compare functions

Calculations on invalid numbers (e.g. divide by 0 or square root of negative numbers) or results out of range of the chosen signal type (e.g. 100 + 73 is out of range for signal type byte) will result in a total system shutdown.
If an overflow occurs for a binary code decimal (BCD) signal, it is clamped to its maximum value.

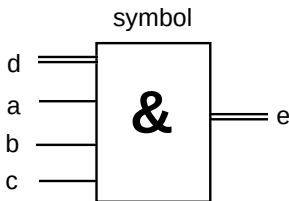
- AND gate



The symbol height of the AND gate can be varied to connect multiple input signals or to upgrade the layout of the drawing.
The maximum number of boolean inputs is 26.

Function : $d = a \cdot b \cdot c$ $\cdot = \text{AND}$

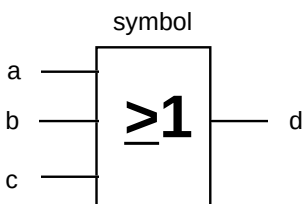
- AND filter



The symbol height of the AND filter can be varied to correct multiple input signals or to upgrade the layout of the drawing.
The maximum number of boolean inputs is 26.
The number of binary inputs can only be one. The position of this input can be chosen at the input side of the symbol.

Function : $a \cdot b \cdot c = 1 \Rightarrow e = d$
 $a \cdot b \cdot c = 0 \Rightarrow e = 0$

- OR gate

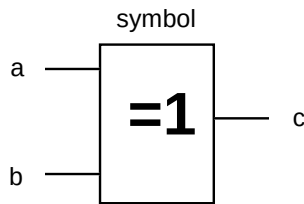


The symbol height of the OR gate can be varied to connect multiple input signals or to upgrade the layout of the drawing.
The maximum number of boolean inputs is 26.

Function : $d = a + b + c$ $+ = \text{OR}$



- XOR gate



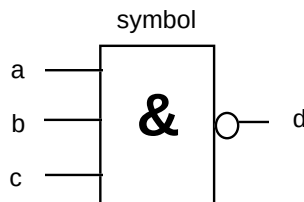
There are only two inputs allowed to the XOR gate.
The symbol height cannot be varied.

Function :

a	b	c
0	0	0
0	1	1
1	0	1
1	1	0

$c = (-a).b + a.(-b)$
(-) = Inversion

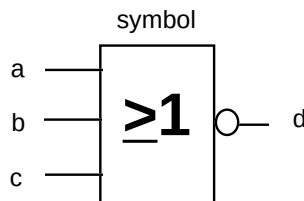
- NAND gate



The symbol height of the NAND gate can be varied to connect multiple input signals or to upgrade the layout of the drawing.
The maximum number of boolean inputs is 26.

Function : $d = -(a . b . c)$. = AND

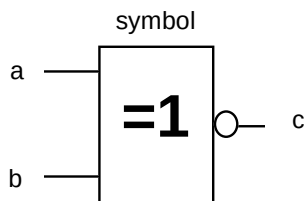
- NOR gate



The symbol height of the NOR gate can be varied to connect multiple input signals or to upgrade the layout of the drawing.
The maximum number of boolean inputs is 26.

Function : $d = -(a + b + c)$ + = OR

- XNOR gate



There are only two inputs allowed to the XNOR gate.
The symbol height cannot be varied.

Function :

a	b	c
0	0	1
0	1	0
1	0	0
1	1	1

$c = ((-a).b).(a.(-b))$

- INVERTER

symbol

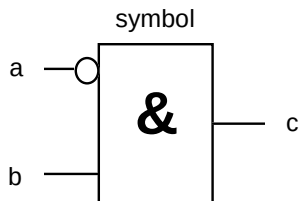
a  b

The inverter can be placed:

- at the front of logical functions
- at the output side of a pulse
- at the end of a timer or a flip-flop
- at the set and reset input of a timer or a flip flop
- at the front or the end of rotated logical functions

Function : $b = (-a)$

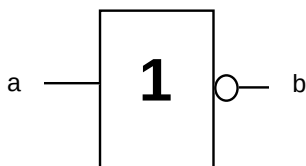
Example:



Function : $c = (-a) . b$

- BUFFER INVERTER

symbol

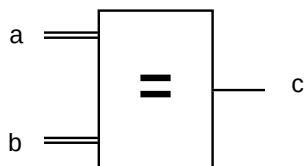


There is only input allowed to the buffer inverter.
The symbol height cannot be varied.

Function : $b = (-a)$ $(-) = \text{Inversion}$

- EQUAL gate (EQ)

symbol

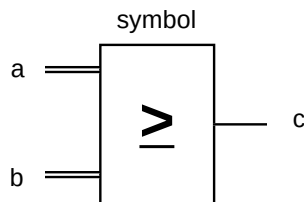


There are only two inputs allowed to the EQUAL gate.
The symbol height cannot be varied.

Function : $a = b : c = 1$ $== \text{EQUAL}$
 $a <> b : c = 0$



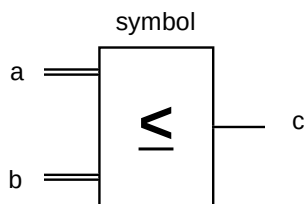
- GREATER THAN OR EQUAL (GTE) gate



There are only two inputs allowed to the GREATER THAN OR EQUAL gate. The symbol height cannot be varied.

Function : $a \geq b$: $c = 1$ $\geq$ = GREATER THAN OR EQUAL
 $a < b$: $c = 0$

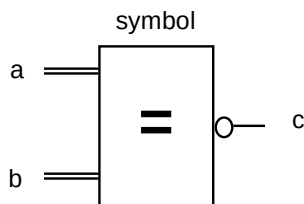
- LESS THAN OR EQUAL (LTE) gate



There are only two inputs allowed to the LESS THAN OR EQUAL gate. The symbol height cannot be varied.

Function : $a \leq b$: $c = 1$ $\leq$ = LESS THAN OR EQUAL
 $a > b$: $c = 0$

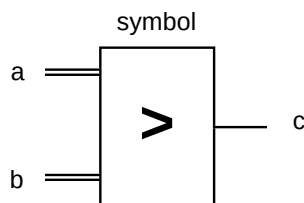
- NOT EQUAL (NEQ) gate



There are only two inputs allowed to the NOT EQUAL gate. The symbol height cannot be varied.

Function : $a \neq b$: $c = 1$ $\neq$ = NOT EQUAL
 $a = b$: $c = 0$ $=$ = EQUAL

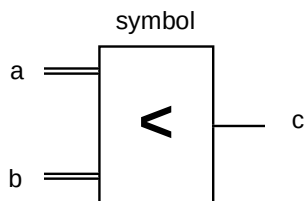
- GREATER THAN (GT) gate



There are only two inputs allowed to the GREATER THAN gate. The symbol height cannot be varied.

Function : $a > b$: $c = 1$ $>$ = GREATER THAN
 $a \leq b$: $c = 0$

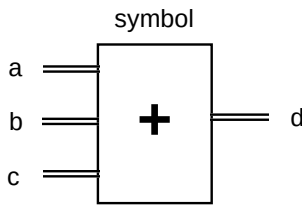
- LESS THAN (LT) gate



There are only two inputs allowed to the LESS THAN gate. The symbol height cannot be varied.

Function : $a < b$: $c = 1$ $<$ = LESS THAN
 $a \geq b$: $c = 0$

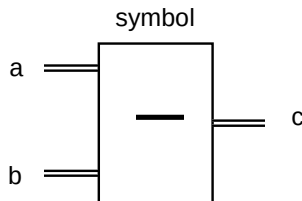
- ADDITION (ADD) gate



The symbol height of the ADDITION gate can be varied to connect multiple input signals or to upgrade the layout of the drawing. The maximum number of binary inputs is 26.

Function : $d = a + b + c$ $+$ = ADDITION

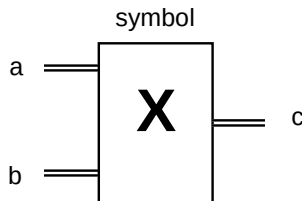
- SUBTRACT (SUB) gate



There are only two inputs allowed to the SUBTRACT gate. The symbol height cannot be varied.

Function : $c = a - b$ $-$ = SUBTRACT

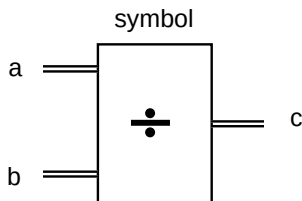
- MULTIPLY (MUL) gate



There are only two inputs allowed to the MULTIPLY gate. The symbol height cannot be varied.

Function : $c = a \times b$ $\times$ = MULTIPLY

- DIVIDE (DIV) gate



There are only two inputs allowed to the MULTIPLY gate. The symbol height cannot be varied.

Function : $a \div b$ $\div$ = DIVIDE

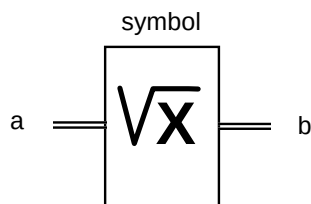
For floating point numbers a real divide is taken. For byte/word/ long numbers a DIV function is taken = divide with trunc function e.g. $16 \div 5 = 3$.

Note:

If $b = 0$, the system will shut down.



- SQUARE ROOT (FSQRT) gate



There is only one input allowed to the SQUARE ROOT gate.
The symbol height cannot be varied.

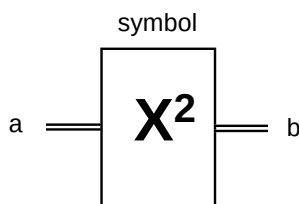
Function : $b := \sqrt{a}$ $\sqrt{x}$ = SQUARE ROOT

SQUARE ROOT always gives a floating point result.

Note:

If $a < 0$, the system will shut down.

- SQUARE (SQR) gate

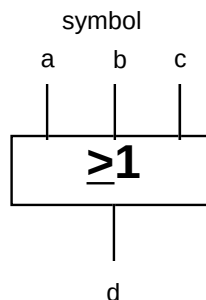


There is only one input allowed to the SQUARE gate. The
symbol height cannot be varied.

Function : $b := a^2 = a \times a$ x = MULTIPLY

A square root function gives always a floating point result.

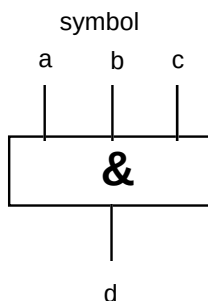
- rotated OR gate



The symbol width of the OR gate can be varied to connect
multiple input signals or to upgrade the layout of the
drawing. The maximum number of boolean inputs is 26.

Function : $d := a + b + c$ $+$ = OR

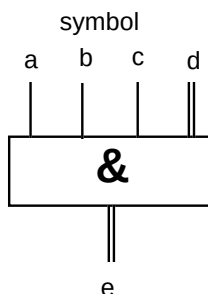
- rotated AND gate



The symbol width of the AND gate can be varied to connect multiple input signals or to upgrade the layout of the drawing. The maximum number of boolean inputs is 26.

Function : $d := a . b . c$ $. = \text{AND}$

- rotated AND filter



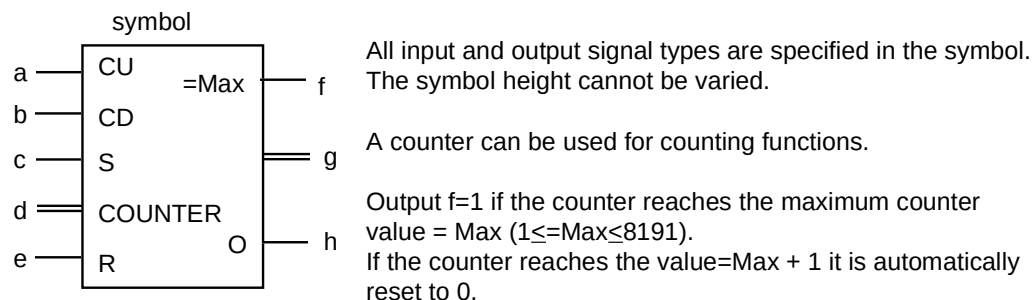
The maximum number of boolean inputs is 25.
The number of binary inputs can only be one. The position of this input can be chosen at the input side of the symbol.

Function : $a . b . c = 1 \Rightarrow e = d$
 $a . b . c = 0 \Rightarrow e = 0$



6.3.2.3 Registers, counters, constants and signal conversions

- COUNTER



a = Count up
b = Count down
c = Set
d = Value in
e = Reset

f = Counter = maximum
g = Value out
h = Counter = 0

Function	a	b	c	d	e	f	g	h
Reset	*	*	*	*	1	0	0	1
Set	*	*	1	y(*4)	0	*(1)	y	*(2)
Count down	0	1	0	*	0	0	g-1	0
Count up	1	0	0	*	0	0	g+1	0
Unchanged	*	*	0	*	0	f	g	*(3)

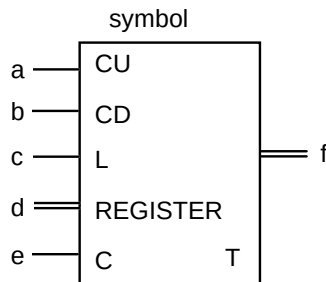
(1) If y=Max, f=1 else f=0
(2) If y=0, h=1 else h=0
(3) If g=0, h=1 else h=0
If g=Max, f=1 else f=0
(4) y<0 or y>Max =Trip

Valid connection combinations are:

- Set and value, others arbitrary.
- Count up, others arbitrary.
- Count down, others arbitrary.

The boolean inputs CU and CD are edge-sensitive only (rising edge).

- REGISTER



All input and output signal types are specified in the symbol.
The symbol height cannot be varied.

A register can be used for:

- Storage of intermediate results of calculations
- Memory element for numbers
- Counter for large numbers

T = B Byte (8 bits)
T = W Word (16 bits)
T = L Long (32 bits)
T = F Floating point (32 bits)

Function:	a	b	c	d	e	f
Clear	*	*	*	x	1	0
Load	*	*	1	y	0	y
Count down	0	1	0	F	0	f-1 every cycle
Count up	1	0	0	f	0	f+1 every cycle
Unchanged	*	*	0	v	0	x

*:0 or 1

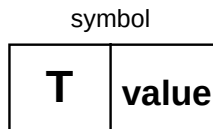
The boolean inputs CU and CD are level-sensitive. This means that, in order to prevent CU/CD every program cycle, a pulse gate should be connected to the corresponding signal.
Count up / Count down is not possible for floating point registers.

If T= Float, Count Up and Count Down may not be connected.

Valid input combinations are:

- Load and value, others arbitrary.
- Count up, Load and value, others arbitrary.
- Count up and Clear, others arbitrary.
- Count down, Load and value, others arbitrary.
- Count down and Clear, others arbitrary.

- CONSTANTS



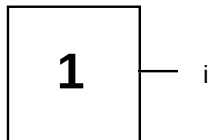
Function: Constant value

T = B Byte (8 bits)
T = W Word (16 bits)
T = L Long (32 bits)
T = F Floating point (32 bits)

The constant is entered in the logics and cannot be changed during program execution but only by modification of the logics.



symbol

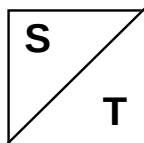


Function: $i:=1$

Constant boolean signal, only one boolean output or an inverter may be connected.

- SIGNAL CONVERSION

symbol



Function: The function checks whether the value of the input symbol is within the range of the type T. If the input value is out of range, the FSC system will shut down due to calculation overflow.
If the source value does not fit into the target (e.g. conversion of -40000 from 'F' to 'W'), the FSC system will shut down.

S = Source signal type (before conversion)

T = Target signal type (after conversion)

Only downward conversions are allowed (see table below).

Source (From)	Target ('To')			
	B	W	L	F
B	--	No	No	No
W	Yes	--	No	No
L	Yes	Yes	--	No
F	Yes	Yes	Yes	--

B = Byte (8 bits)

W = Word (16 bits)

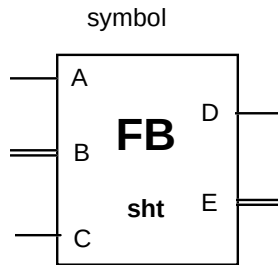
L = Long (32 bits)

F = Floating point (32 bits)

Note:

If, for example, a word-to-byte conversion has been defined and during translation it appears that the destination type must be word, the conversion symbol is corrected to the trivial word-to-word conversion, which is ignored by the 'Translate application' option of FSC Navigator (the compiler). You can use the FLD design editor to delete the conversion.

6.3.2.4 Function block

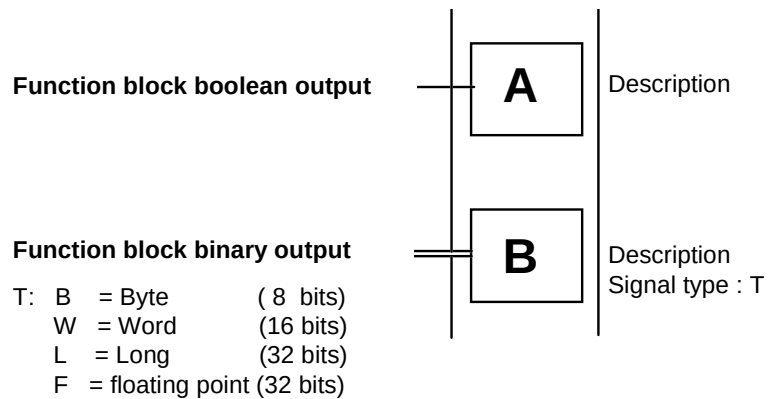
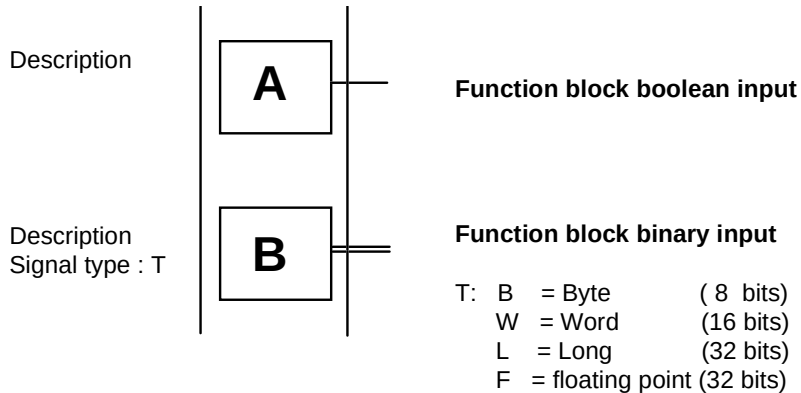


A function block is a subroutine symbol that must be created on another FLD first. This FLD must have a higher number. When placed, the FLD number is characteristic for that function block. A function block has no system outputs and no off-sheet references. It does, however, have function block inputs and outputs to transfer values from and to the calling FLD.

At least one output must be connected. All connected symbols must be of a compatible type.
(I.e. signal of type Byte can be connected to Byte, Word or Float input)

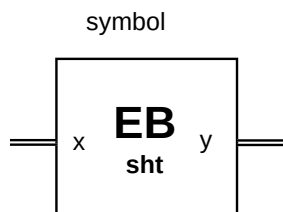
6.3.2.4.1 Function block inputs and outputs

Function block inputs and outputs are symbols that represent the signals entering the function block or leaving the function block. They are essentially different from the FSC system inputs and outputs. The function block I/O can be boolean or binary. A boolean input or output is represented by a single line, a binary input or output is represented by a double line.





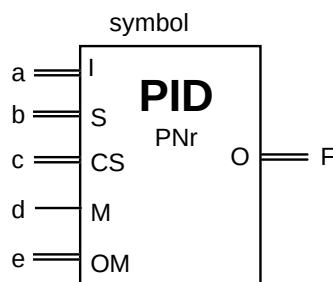
6.3.2.5 Equation block



An equation block is a symbol that must be designed on another FLD first. When placed, the diagram number (Sht) is characteristic for that equation block. Both the binary input and the binary output must be connected. The signal type at the output side is Float, the signal type at the input side is converted internally to type Float if required.

For more information about the equation block usage and definition see subsection 6.5.2.4 in this manual.

6.3.2.6 PID



PID control function

I	= (Analog) Input (Float)
S	= Setpoint (Float)
CS	= Setpoint (Float)
M	= Manual Output condition
OM	= Manual Output value (Float)
O	= (Analog) Output (Float)

The PID function is calculated using the trapezium integration function. You can place a PID using the 'System Configuration' option of FSC Navigator, or define it directly using the FLD design editor. Parameters can be set using 'System Configuration' option of FSC Navigator.

If the PID is in Cascade mode, the CS setpoint will be used; otherwise the S setpoint is used for the PID control function.

At least the input of the setpoint must be connected. The input signals must be of a compatible type. The manual output condition must be both connected or both not connected. The cascade setpoint must be connected to the PID tag number as configured in the variable database, unless it is not defined in the variable database.

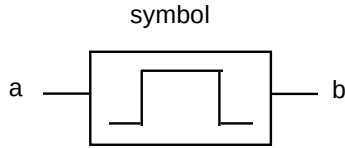
If M = 1, the PID mode will be switched to manual mode and the value OM will be copied to the PID output O.

If M = 0, the PID stays in its current mode and no value of OM will be copied to the PID output O.

For more information on PID controllers refer to Appendix E of the FSC Software Manual.

6.3.2.7 Timers

– Pulse



Function:

N	a	b
1	1	1
>1	1	0

N is the number of the program cycles after 'a' becomes active

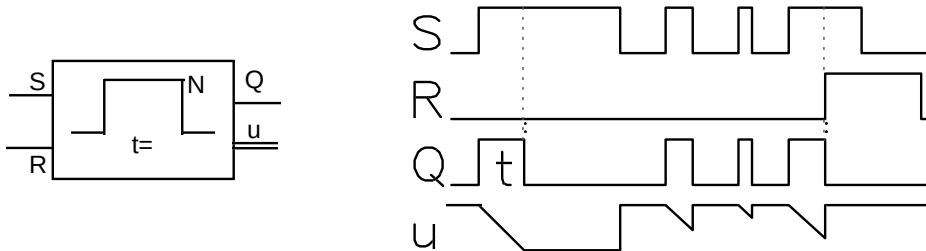
This function can be used to create a pulse with a duration of 1 program cycle e.g. for a CU input of a register. After the generation of a pulse 'a' must become zero first in order to make a new pulse possible.

6.3.2.7.1 Timers with constant timer value

For timer symbols

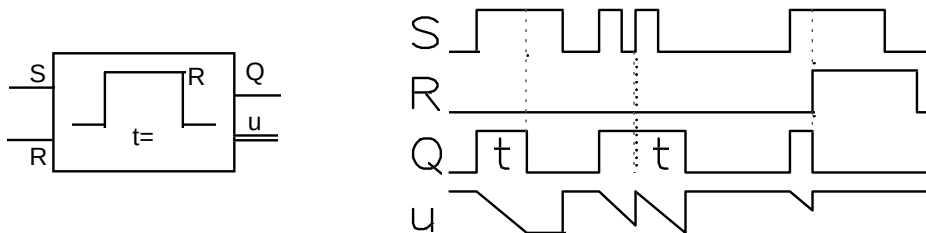
- S = Start timer (1 = start)
- R = Reset timer (1 = reset)
- Q = Timer output
- t = Time value + base (time constant)
- u = Remaining time value

– Pulse



Pulse time function on rising edge of the timer start input. If only one input connected (e.g. in the middle), it is assumed to be the start input.

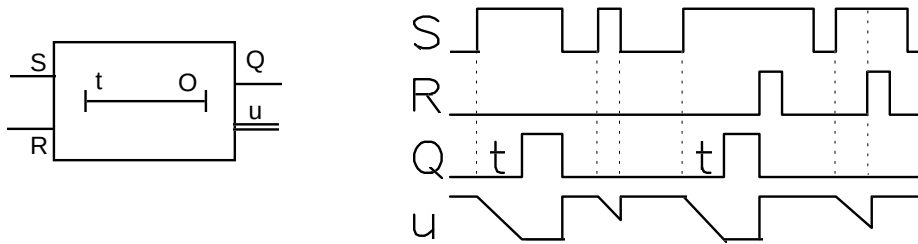
– Pulse retriggerable



Pulse time function on rising edge of the timer start input, the time is restarted after every rising edge. If only one input connected (e.g. in the middle), it is assumed to be the start input.

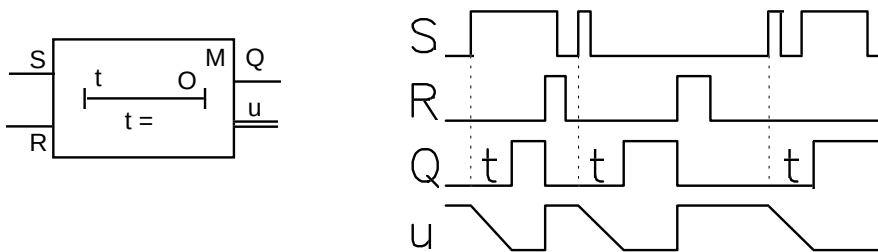


– Delayed ON



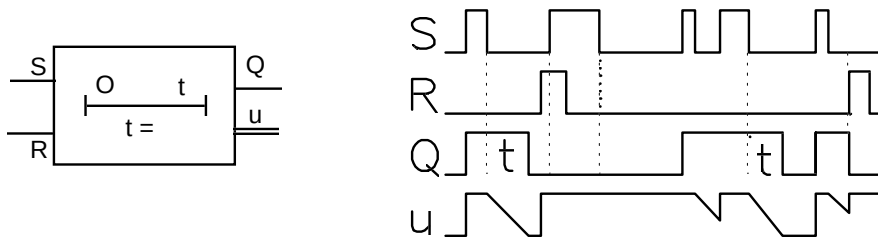
Delayed ON time function on rising edge of the timer start input. If only one input connected (e.g. in the middle), it is assumed to be the start input.

– Delayed ON memorize



Delayed ON time function on rising edge of the timer start input. The timer continues to count even if the timer start input goes back to zero state. Both inputs S and R have to be connected.

– Delayed OFF



Delayed OFF time function on falling edge of the start timer input. If only one input connected (e.g. in the middle), it is assumed to be the start input.

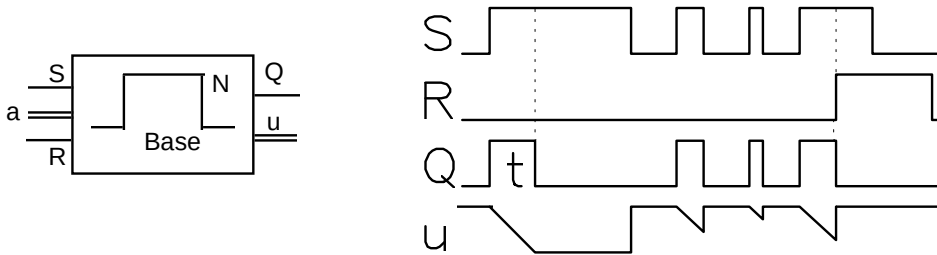
For all timers with constant timer value, the start timer input and the Q output must be connected.

6.3.2.7.2 Timers with variable timer value

For timer symbols

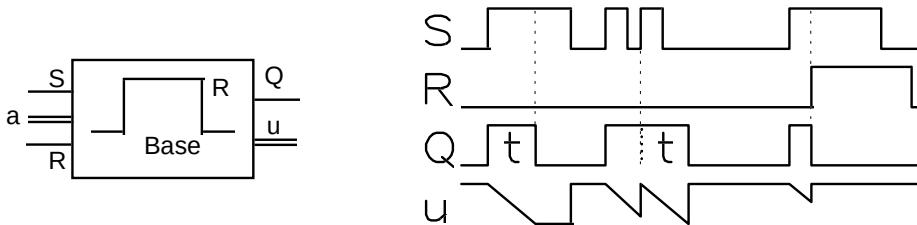
- S = Start timer (1 = start)
- R = Reset timer (1 = reset)
- Q = timer output
- a = The time value that is loaded when the timer is started. Base is the time base of the timer
100 ms, 1 s, 1 m
- u = Remaining time value

– Pulse



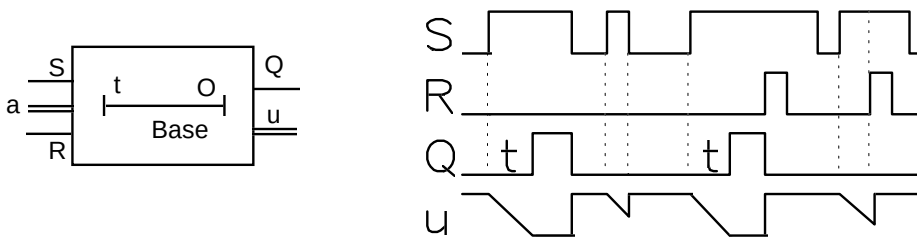
Pulse time function on rising edge of the timer start input.

– Pulse retriggerable



Pulse time function on rising edge of the timer start input, the time is restarted after every rising edge.

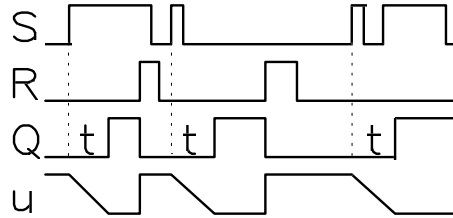
– Delayed ON



Delayed ON time function on rising edge of the timer start input.

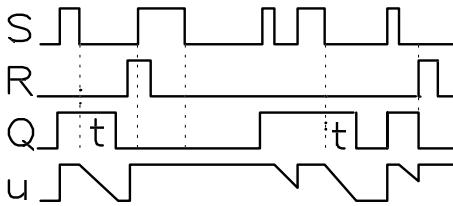
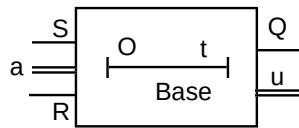


– Delayed ON memorize



Delayed ON time function on rising edge of the timer start input. The timer continues to count even if the timer start input goes back to the zero state. All inputs S, R and 'a' have to be connected.

– Delayed OFF

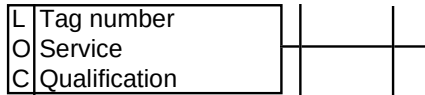


Delayed OFF time function on falling edge of the timer start input.

For all timers with variable timer value: The start timer input, the value input and the Q output must be connected.

6.3.2.8 I/O symbols

Digital I/O signals

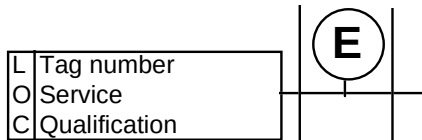
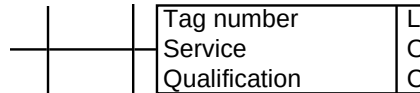


Digital input

Used for boolean signals.
e.g. ON/OFF
START/STOP

Digital output

Used for boolean signals.
e.g. START/STOP
RUNNING/STOPPED



Digital input (with sheet reference)

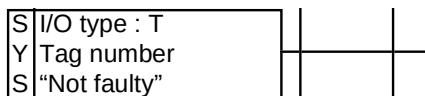
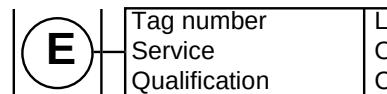
Used for boolean signals that have a direct digital output. The sequence character refers to the connected digital output.

See also digital outputs.

This symbol is always accompanied by the next output symbol. This combination is characterised by a unique character sequence. In this example the character "E" (see subsection 6.5.2.7).

Digital output (with sheet reference)

Used for boolean signals where the digital output is a direct copy from the digital input. The sequence character refers to the connected digital input. This symbol is always accompanied by the previous digital input symbol see (subsection 6.5.2.7).



Diagnostic input

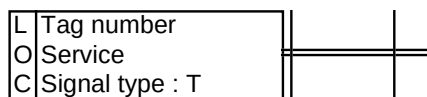
I/O type : I, O, AI, AO, LoopO, SensAI.
Tag number : Tag number of I/O (I, O, AI, AO)
allocated to a fail safe I/O
module

Function:

I, O, AI, AO : The diagnostic input is low when the channel is faulty, else it is high.
loop O : The diagnostic input is low when a loop-monitoring error is detected by the system, else it is high. Only for outputs allocated to a module with loop-monitoring features.
SensAI : The diagnostic input is low when a sensor analog input alarm is generated by the system and the analog input is not forced, else it is high.



Multiple I/O signals



Multiple input (BI)

Used for binary signals

e.g. 37

19

T : B = Byte (8 bits)

W = Word (16 bits)

L = Long (32 bits)

For variables with location COM or FSC the signal type may also be:

F = Floating point (32 bits)

Multiple output (BO)

Used for binary signals

T : B = Byte (8 bits)

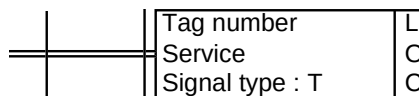
W = Word (16 bits)

L = Long (32 bits)

For variables with location COM or

FSC the signal type may also be:

F = Floating point (32 bits)



Binary Coded Decimal input (BI)

The Binary Coded Decimal (BCD) signal is always converted to a binary signal through a Decimal to Binary converter

T : B = Byte (8 bits)

W = Word (16 bits)

L = Long (32 bits)

F = Floating point (32 bits)

Binary Coded Decimal output (BO)

Binary signals are converted to a

Binary Coded Decimal through a

Binary to Decimal converter.

T : B = Byte (8 bits)

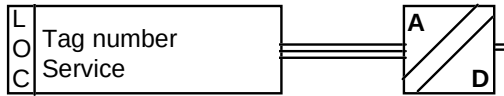
W = Word (16 bits)

L = Long (32 bits)

F = Floating point (32 bits)



Analog I/O signals

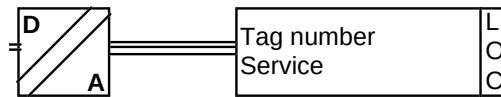


Analog input combined with analog to digital converter

Used to convert analog signals to binary signals. Signal type 'F' (float)
An analog signal is always converted to a binary signal through an analog to digital converter.

Analog output combined with digital to analog converter

Used to convert binary signals to analog signals.
Signal type 'F' (float)
The digital signal is converted to an analog signal through a digital to analog converter.

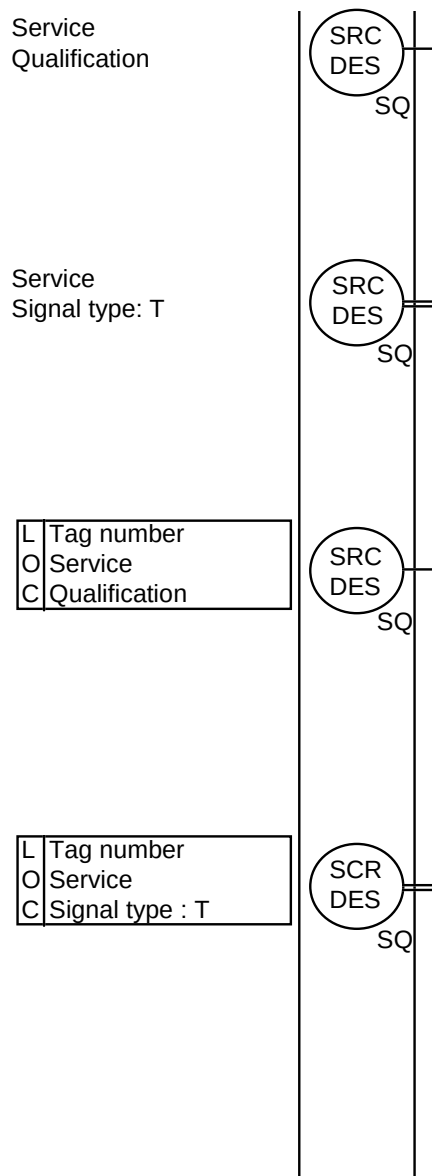




6.3.2.9 Sheet transfers

Sheet transfers are used to connect signals between sheets for those functions that cannot be incorporated on a single sheet. A sheet transfer contains information on the originating sheet (source FLD) and the destination FLD. In order to distinguish more than one sheet transfer between two FLDs, a sheet transfer sequence number is added. For understanding a service (and qualification) can be entered.

6.3.2.9.1 On-sheet transfers



Boolean on sheet transfer

Used for boolean signals.

Binary on sheet reference

Used for binary signals

T : B = Byte (8 bits)

W = Word (16 bits)

L = Long (32 bits)

F = Floating point (32 bits)

On sheet transfer or repeated digital input

This symbol can be used to show that this sheet transfer originates directly from a digital input without intermediate functions, therefore the tag number has to be identical to the tag number of the original input.

On sheet transfer or repeated multiple input

This symbol can be used to show that this sheet transfer originates directly from a multiple input, analog input or BCD input without intermediate functions, therefore the tag number has to be identical to the tag number of the original input.

T : B = Byte (8 bits)

W = Word (16 bits)

L = Long (32 bits)

F = Floating point (32 bits)

6.3.2.9.2 Off-sheet transfers

Boolean off sheet transfer

Used for boolean signals

Binary off sheet transfer

Used for binary signals

T : B = Byte (8 bits)
 W = Word (16 bits)
 L = Long (32 bits)
 F = Floating point (32 bits)

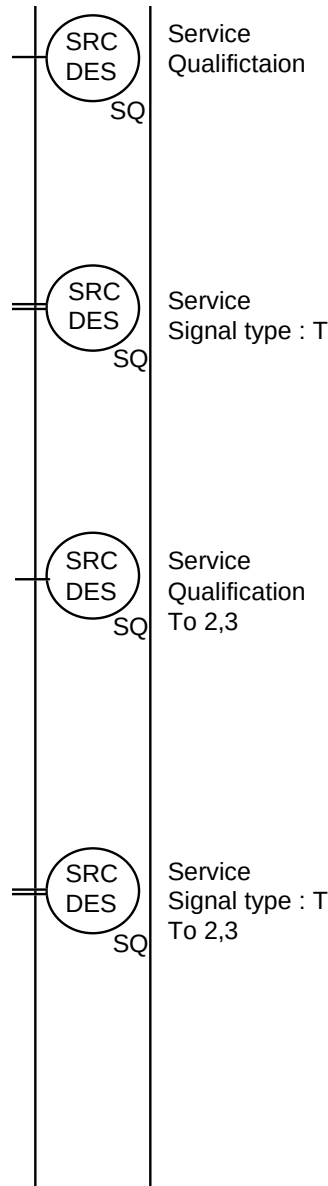
Multiple boolean off sheet transfer

Used for boolean signals to create more than 1 transfer from this sheet originating from the same signal. Destination sheets must be defined. (Max. 5 references)

Multiple binary off sheet transfer

Used for binary signals to create more than 1 transfer from this sheet originating from the same signal. Destination sheets must be defined. (Max. 5 references)

T : B = Byte (8 bits)
 W = Word (16 bits)
 L = Long (32 bits)
 F = Floating point (32 bits)



Remark for sheet transfers:

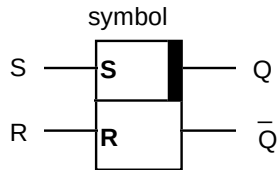
For sheet transfers which are directly connected to an input (I, BI) or an output (O, BO), the service text is used as a tag number of the connected input or output. This is done to create the correct corresponding symbol for the on-sheet transfer on the destination FLD as repeated inputs have a different symbol as normal on-sheet transfer. The tag number is combined with the variable type (I, O, BI, BO) to search in the variable database. If the combination is found, the service (and qualification) of this variable are taken and drawn in the symbol.



6.3.2.10 Flip-flops

- S-R FLIP-FLOP

Set-Reset flip-flop with Set dominant



Function:

S	R	Q	-Q
0	1	0	1
1	0	1	0
0	0	Qt-1	-(Qt-1)
1	1	1	0

State unchanged

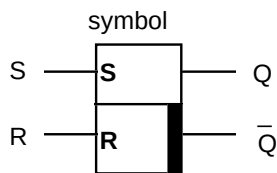
Set is dominant

Qt-1 = state after the previous cycle.

The set input and the reset input must both be connected and at least one of the outputs must be connected.

- R-S FLIP-FLOP

Reset-set flip-flop with Reset dominant



Function:

S	R	Q	-Q
0	1	0	1
1	0	1	0
0	0	Qt-1	-(Qt-1)
1	1	0	0

State unchanged

Reset is dominant

Qt-1 = state after previous cycle.

The set input and the reset input must both be connected and at least one of the outputs must be connected.

6.3.2.11 Texts

Example

This is a text

(size = 1, strokes = 1)

TEXT

(size = 3, strokes = 3)

Function:

Explaining texts and descriptions.

Text is a string of up to 60 alphanumerical characters.

The character size can be selected in the range of 1-10. The number of strokes can be selected in the range of 1 up to the character size to create a bold character by writing the character's multiple.

6.4 The menus of the FLD design editor

6.4.1 MAIN menu

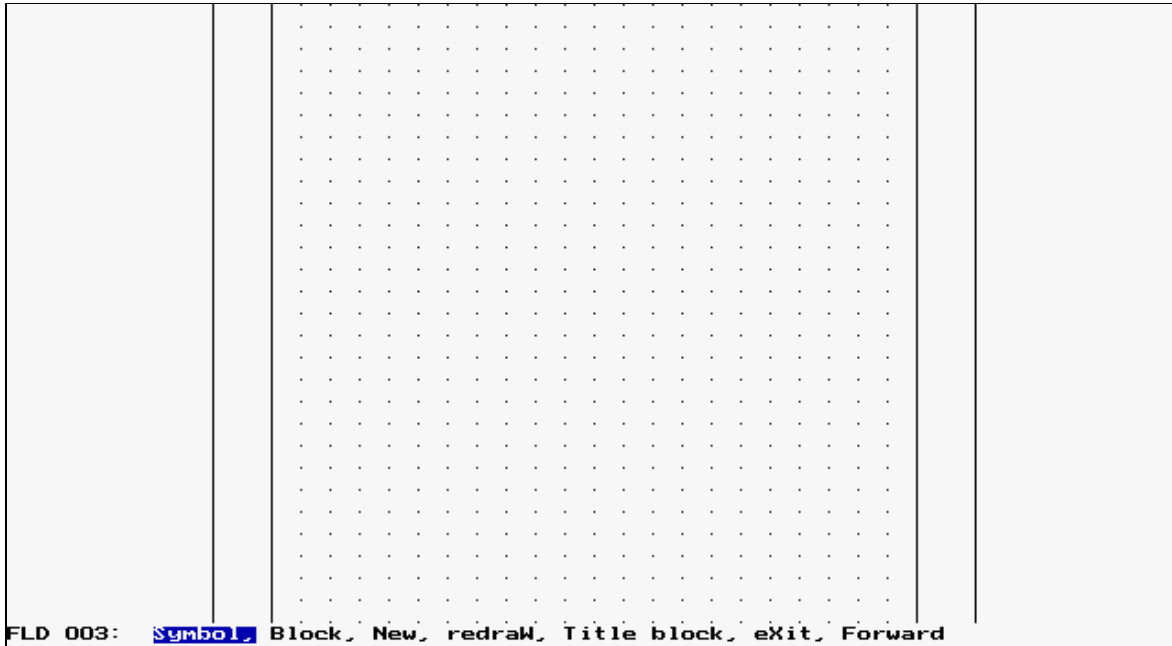


Figure 6-9 FLD design editor: MAIN menu (part 1)

Only the options that can be used will be displayed in the menu. Please note that the FLD number to the left of the menu bar is the number of the functional logic diagram that is being edited.

Brief explanation of the MAIN commands:

Symbol	Leave the MAIN menu and go to the SYMBOL menu.
Block	Leave the MAIN menu and go to the BLOCK menu.
New	After the displayed FLD has been saved to disk, a new FLD number is asked. Enter the FLD number and the corresponding FLD will be displayed. In case of a new FLD, the title block appears on screen first.
saVe	Save the current FLD to disk, but do not leave the FLD.
redraW	Redraw current FLD.
Title block	The FLD data can be changed.
eXit	Save the displayed FLD to disk and return to the FSC Navigator main screen.
Forward	Go from the first menu part to the second menu part.



Additional keyboard functions are:

<PgDn> Go to the next available FLD.

<PgUp> Go to the previous available FLD.

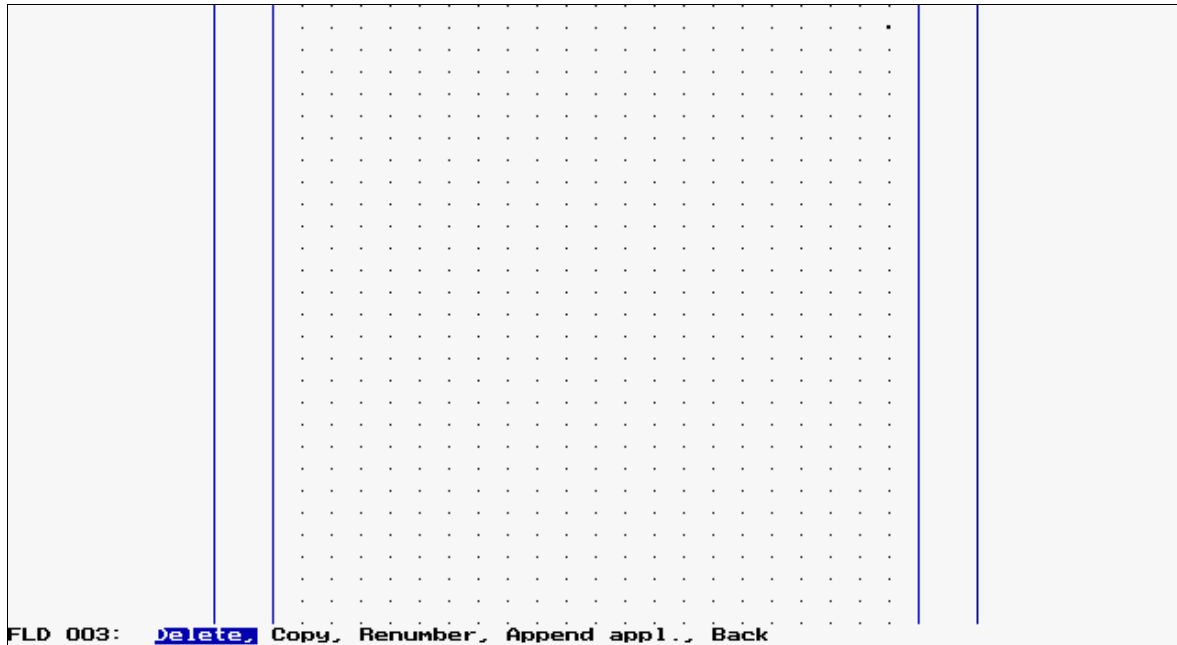


Figure 6-10 FLD design editor: MAIN menu (part 2)

Only the options that can be used will be displayed in the menu. Please note that the FLD number to the left of the menu bar is the number of the functional logic diagram that is being edited.

Brief explanation of the MAIN commands:

Zoom	Is only active if the size of the FLD is too big to be displayed on your screen (library depended e.g. SYM2). With the zoom option the total FLD will be scaled so it can be displayed on your screen.
Delete	The displayed FLD is deleted and undrawn but a backup is made for possible recovery. A confirmation is asked.
Undelete	The recently deleted FLD is reloaded and drawn.
Copy	A complete FLD from the same application or from another application can be copied to the current FLD. This option is not active if the FLD is empty or if on-line modification is active. Make sure the FLD to be copied is drawn with the same symbol library.
Renumber	The FLD number of the displayed FLD can be changed. This option is not active when the FLD is empty or when on-line modification is set to Yes.



Append appl. You can append one or more subapplications to the current application. This option is not active when the FLD is empty or when on-line modification is set to Yes.

Back Go from the second menu part to the first menu part.

Addition keyboard functions are:

<PgDn> Go to the next available FLD.

<PgUp> Go to the previous available FLD.



6.4.2 SYMBOL menu

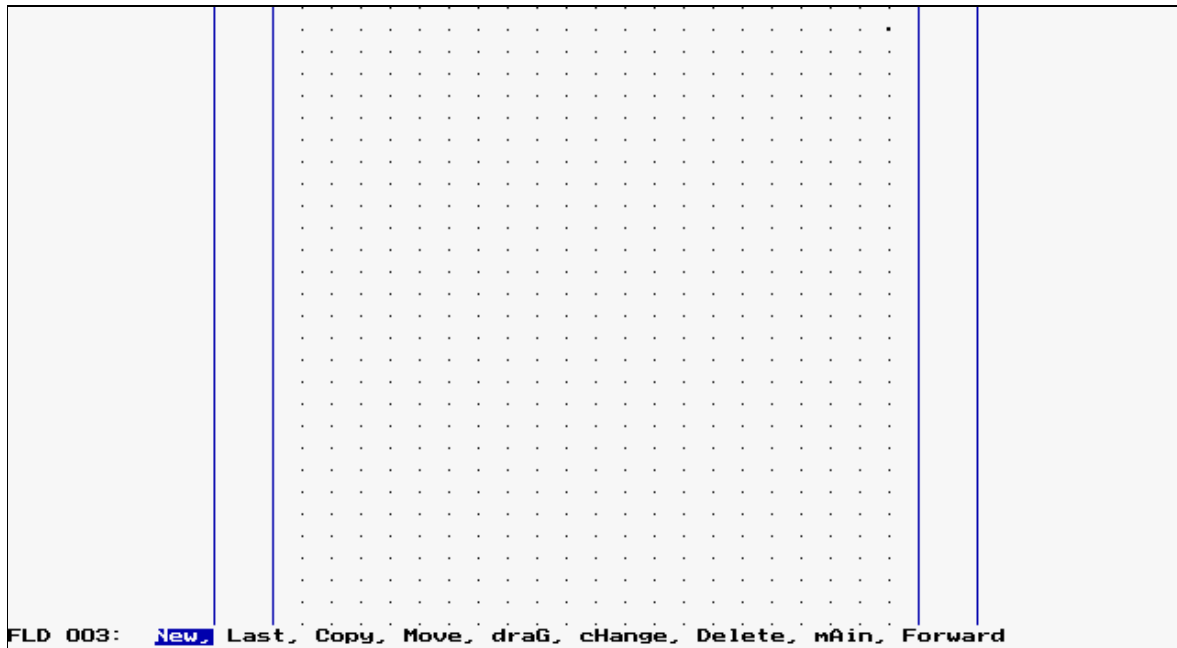


Figure 6-11 FLD design editor: SYMBOL menu (part 1)

Only the options that can be used will be displayed in the menu. Please note that the FLD number to the left of the menu bar is the number of the functional logic diagram that is being edited.

Brief explanation of the SYMBOL commands:

New	The selection sheet is displayed. After a successful symbol choice is made, the design sheet is displayed again and the selected symbol can be placed.
Last	The last symbol selected from the selection sheet is selected for placement. No selection sheet is displayed.
Copy	A symbol can be copied from the design sheet to another location on the FLD. You must select the symbol from the design sheet. Also a new position must be chosen.
Move	A symbol can be moved to another location on the FLD. All connections of this symbol will be broken. You must select a symbol from the design sheet and a new position for that symbol.
draG	A symbol can be moved to another location on the FLD, but only in a way that the connections will be retained. You must select a symbol from the design sheet and a new position for that symbol.
Delete	A symbol can be deleted from the FLD. A copy is made for a possible recovery. You must select the symbol on the design sheet.
Undelete	A recent deleted symbol is recovered.



Change	The variable data of a symbol on the FLD can be changed . The symbol to be changed must be selected from the design sheet.
mAin	The SYMBOL menu is left and program returns to the MAIN menu.
Forward	Go from the first menu part to the second menu part.

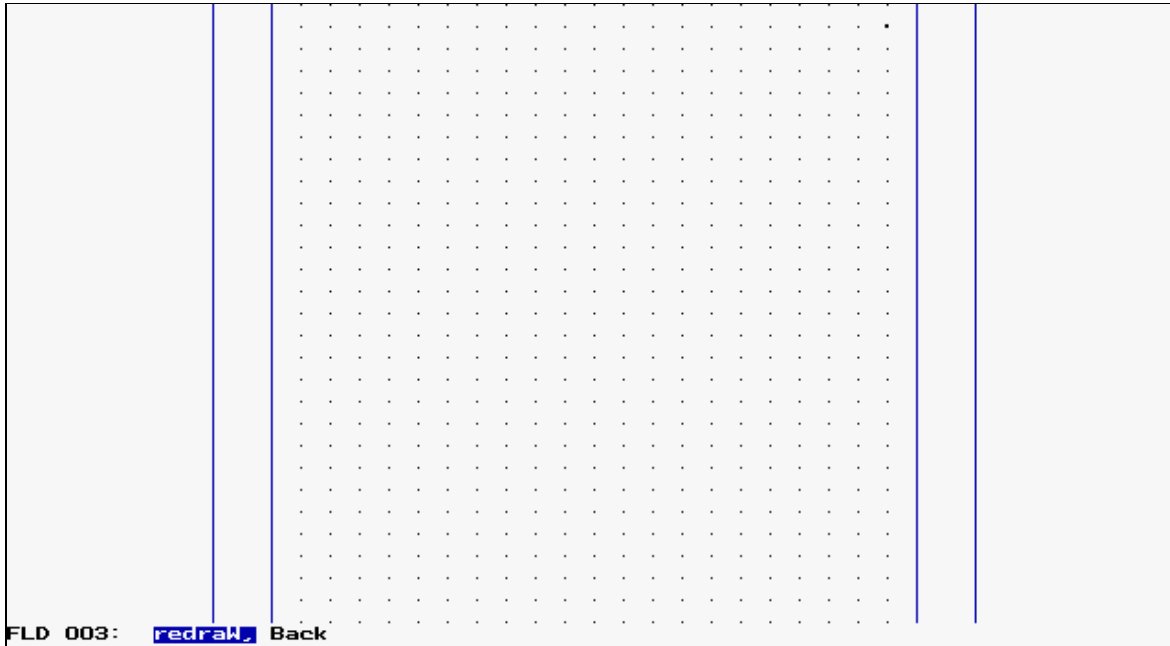


Figure 6-12 FLD design editor: SYMBOL menu (part 2)

Only the options that can be used will be displayed in the menu. Please note that the FLD number to the left of the menu bar is the number of the functional logic diagram that is being edited.

Brief explanation of the SYMBOL commands:

redraW	The displayed FLD is drawn again to correct some point deletions during move or delete.
Symbol set	You can switch between the symbol set for a function block sheet and the symbol set for a program block sheet. This option is only available when you are editing a comment sheet.
Back	Go from the second menu part to the first menu part.



6.4.3 BLOCK menu

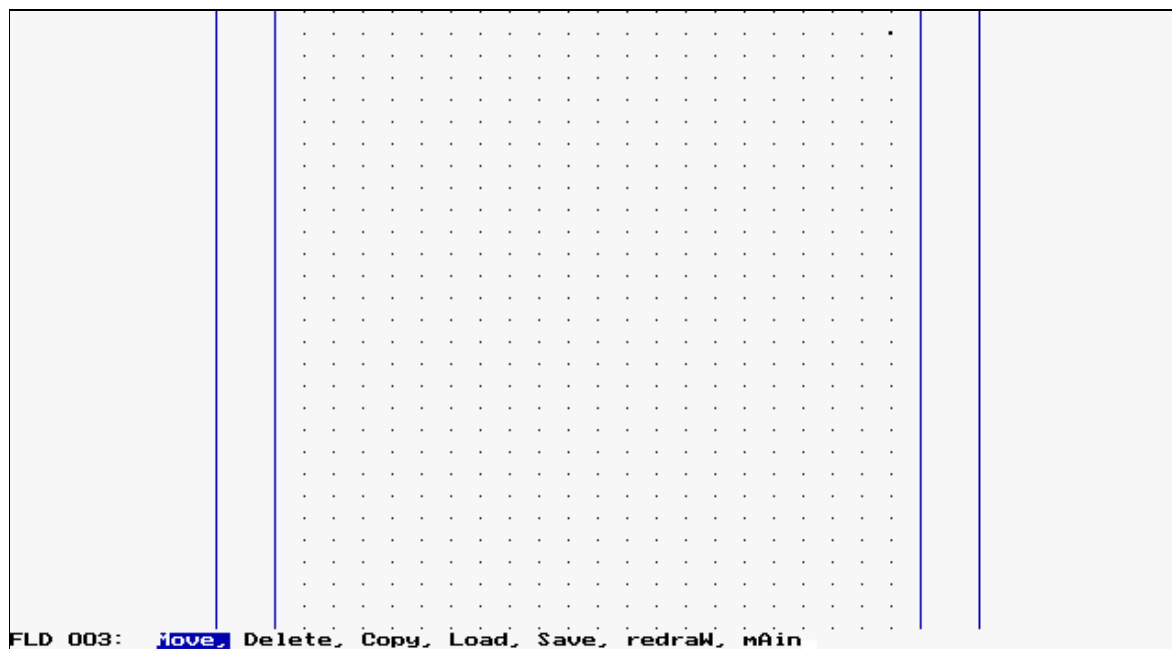


Figure 6-13 FLD design editor: BLOCK menu

Only the options that can be used will be displayed in the menu. Please note that the FLD number to the left of the menu bar is the number of the functional logic diagram that is being edited.

Brief explanation of the BLOCK commands:

Move	You are asked to define a block. The defined block with its contents can be moved to another position on the FLD. A new valid position must be selected.
Delete	You are asked to define a block. The defined block with its contents is deleted. A backup is made of the block contents for a possible recovery. A confirmation is asked.
Copy	You are asked to define a block. The defined block can be copied to another legal position on the FLD. A new legal position must be selected.
Load	A block can be loaded from the disk and after it is drawn on the FLD you can drag it around to a new position.
Save	You are asked to define a block. The defined block will be saved to disk.
Undelete	The recent deleted block will be recovered.
RedraW	The displayed FLD is drawn again to correct some point deletions during move or delete.
Main	The BLOCK menu is left and the program returns to the MAIN menu.



6.5 Designing with the FLD design editor

6.5.1 Designing functional logic diagrams

6.5.1.1 MAIN commands

For each of the menu options the possible path through the program and the actions to be done are explained in this section.

– Symbol

The SYMBOL menu is displayed, which consists of a set of symbol commands. You must choose a symbol option. When the symbol command you entered has been executed, the program returns to the SYMBOL menu.

– Block

The block submenu is displayed, consisting of a set of block commands. You must choose a block option. When the block command you entered has been executed, the program returns to the BLOCK menu.

– Title block

The title block of the current FLD will be displayed. You can change the data you entered before. But also when you entered an FLD number of an empty FLD (New), the title block function is activated by the program. Some data are already filled out for you. They are copied from the last FLD you worked on (e.g. engineer, design date, unit, subunit). This is done to minimize the user input. If you enter <Esc> before entering any data the program will return to the previous situation. The following data must be specified:

FLD number	:	3																					
Block type	:	Program block																					
Unit	:																						
Subunit	:																						
Revision	:	<table border="1"> <thead> <tr> <th>No</th> <th>Date</th> <th>Description</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>11-13-1997</td> <td>FIRST ISSUE</td> </tr> <tr> <td>A</td> <td></td> <td></td> </tr> <tr> <td>B</td> <td></td> <td></td> </tr> <tr> <td>C</td> <td></td> <td></td> </tr> <tr> <td>D</td> <td></td> <td></td> </tr> <tr> <td>E</td> <td></td> <td></td> </tr> </tbody> </table>	No	Date	Description	0	11-13-1997	FIRST ISSUE	A			B			C			D			E		
No	Date	Description																					
0	11-13-1997	FIRST ISSUE																					
A																							
B																							
C																							
D																							
E																							
Engineer	:	PM NL33																					
Design date	:	11-13-1997																					
Text1	:	FUNCTIONAL LOGIC DIAGRAMS																					
Text2	:																						
Text3	:																						
Text4	:																						
Text5	:																						
Text6	:																						
Text7	:																						
Text8	:																						
Text9	:																						

Figure 6-14 Title block



– FLD number

This field cannot be reached or changed in the title block session, but it is only meant as information for the user.

– Block type

The block types are:

- Program block
- Function block
- Equation block
- Comment block
- Sheet index
- Tag number index

You can change the block type by toggling with <Space> or selecting via <?>. The program gets a default block type when an FLD has no block type yet.

If you select a program block, you get the selection sheet containing the symbols which can be used on a program block sheet.

You can use a function block to define your own function and place it as a subroutine on another FLD (with a lower number).

A function block and an equation block must have an FLD number of 500 or higher (except when it is converted from a software version before 2.78). In the demo version of FSC Navigator, function blocks can be defined in FLD 5 or higher.

The function block or the equation block FLD numbers must all be higher than the program block FLD numbers. Between the program blocks and the function blocks, there must be at least one sheet with a block type that is different from a program block or a function block. From these constraints it depends which default block type will be chosen by the program and which block types can be selected.

The index sheets can only be placed once.

The only way to delete an index sheet is to change the block type to a non-index sheet. Next go into this sheet and leave it again eventually after making a design.

– Comment block

This type of sheet can be used as a cover sheet, legend of symbols or just a comment between two normal FLDs. Except for texts, lines and revision clouds, all variable data is now fixed and is taken from the message file of the program. (See selection sheet). This type of sheet is ignored by the 'Translate application' option of FSC Navigator (the compiler).

– Tag number index

This type of sheet contains no logics, but is only used for printing functional logic diagrams. It is used to generate an index of all I/O tag numbers used in the application.

A tag number Index listing contains for each I/O: – I/O type

- Tag number
- Service
- FLD numbers where the I/O is placed

Included in this listing are on-sheet transfers with service equal to a tag number e.g. repeated inputs.

Because this sheet contains no logics, no design sheet is drawn, but a limited main menu is displayed (Title block/New sheet/eXit).

A tag number index sheet is ignored by 'Translate Application' option of FSC Navigator (the compiler).

– Sheet index

This is similar to a tag number index sheet, but now an index is made of all sheets used in the application. For every sheet the last revision date and a sheet description (Text 4 in the title block) will be printed.



Because this sheet contains no logics, no design sheet is drawn, but a limited MAIN menu is displayed (Title block, New, eXit).

The sheet index sheet is ignored by the 'Translate application' option of FSC Navigator (the compiler).

– Unit and subunit

Unit and subunit indicate the relation of the logics, designed on the FLD, with a part of the system to be controlled. These fields are only defined when the block type is a program block. In this case the program displays the text 'Undefined' behind unit and subunit and these field cannot be reached by you.

Note:

A function block can be used in any part of the logics so the unit and subunit can be different. The unit and subunit of all variables in this function block are for this reason determined during placement of the function block.

– Revision

You can add 6 revisions to an FLD. A revision consists of a revision number, a revision date, and a revision description. If an empty FLD is selected, the program generates a zero-revision and prints all default revision numbers. With <Enter> you can move from left to right within a revision field and from the description-field to the next revision field. If you enter an empty description and an empty date this revision is canceled and the program jumps over all revision fields to the engineer field. The revision data is taken from the real time clock of your PC. Be sure the clock has the correct date.

– Engineer

You are asked to enter a name or your initials, maximal 10 characters long.

– Design date

If you enter a new FLD the current date is generated by the program from the real time clock of your PC. If you press <Enter> this date is stored else you can edit it and next press <Enter>. The default date format is DD-MM-YYYY.

– Additional data

You can enter 9 texts. The position where these texts will be printed on the hardcopy depends on your library. For example, in the SYM1 library the text 4 is used for an FLD description.

If you are on the last line of the title block, you can leave it by pressing <Enter>, <↓> or <PgDn>. You can also leave the title block by pressing <Esc>. In that case all listed data is accepted.

If you enter <Esc> directly after entering a new title block you return to the situation before you entered the title block.



– New

The current FLD is saved to disk. You can now enter a new FLD number which must be a number between 1 and 999. The current FLD number is displayed as default value. The program repeats asking for an FLD number until a legal number is entered. Next, the program loads the FLD you have chosen. A sheet file name consists of the application name and the FLD number as extension. If the FLD is empty, the title block is activated and you must enter some data. Next the FLD and its grid are drawn on the screen. The MAIN menu is displayed on the menu line. Now a new MAIN option can be chosen.

You can also press <PgUp> or <PgDn>.

<PgUp>	Go to the previous available FLD.
<PgDn>	Go to the next existing available FLD.
<FLD no.>+<?>	Gives a list of all existing FLDs starting with the FLD number you entered. If you did not enter an FLD number before <?>, all existing FLDs will be listed.

– Save

The current FLD is saved to hard disk, without leaving the FLD. This option must prevent you from losing your data if you leave your system temporarily or if a power dip disorders your PC.

– Delete

If this option is chosen, the complete FLD is deleted from the screen and the sheet database. But before the delete operation is executed a back up is made in order to make a possible recovery after the deletion. Also a confirmation is asked.

– Undelete

The previously deleted FLD is recovered. An undelete function can only be executed directly after a delete action. If you choose one of the options Redraw, Delete, Forward or Back, you can still undelete the deleted FLD. Otherwise the Undelete option cannot be selected.

– Zoom

Is only active if the size of the FLD is too big to be displayed on the screen (library depended e.g. SYM2). With the zoom option the total FLD is scaled so it can be displayed on your screen. While the FLD is scaled, you cannot edit. Press <Enter> to return to normal scaling.

– Redraw

The complete FLD is redrawn. This can be useful when the FLD is damaged because of point deletions e.g. drawing and undrawing of lines or line cuts when a connection is made.

– Copy

You are asked to enter a file name.

This file name must have the format <Folder>_<Sys. no.>. <FLD no.>.

Where:	Folder	Folder of the application from which an functional logic diagram is to be copied.
	Sys. no.	System number of the application from which an functional logic diagram is to be copied.
	FLD no.	The number of the functional logic diagram to be copied (always 3 digits).

e.g. If the second FLD from application TEST, which has system number 1, is to be copied, the file name should be: TEST_1.002



If the file you entered does not exist, a beep is generated and you are asked to enter a new FLD number. Next, all data of the selected file is converted to data which is compatible with the current FLD. All inputs, outputs and PIDs get an illegal tag number: '?'.
This tag number must be changed afterwards. All sheet references are set to zero. You must connect them afterwards with the change option in the SYMBOL menu. If the contents of the copied FLD is not almost as large as the format of the screen, you can move it to a new legal position, else it is placed directly into the new FLD if no overlap is detected by the program.

– **Equation file** (only active on an equation block diagram)

You can enter the name of an equation table file which is located in your project folder. For details on equation blocks refer to subsection 6.3.2.5 and 6.5.2.4.

– **Renumber**

The current FLD number can be changed. All data of this FLD will be moved to the new FLD number. The old FLD will be cleared from the disk and can be used again. This option is not available when on-line modification mode is set to 'Yes'.

– **Append appl.**

With this option you can append applications to the current application under some conditions. These conditions will be checked and possible inconsistencies in both applications will be logged to a file <Application name>.APP. Some inconsistencies with regard to the logics will be corrected: i.e. a double used output or PID will be deleted. When an input is used more than once per CPU, this input will be replaced by a repeated input symbol if possible. More details:

- 1) You can successively append a subapplication to the main application from which you do all appends. You can keep on appending applications until you enter <Esc> or until you appended 255 applications.
- 2) No backup is made from the original main application from which the append is made. All sub applications to be appended will stay unchanged. So it is advised to make a back up from the main application.
- 3) There are two ways to append an application.
 - A The subapplication must be located under the same FSC path, i.e.:
Main application : TEST located in subfolder \FSC\TEST
This is the application from which you do the appends.
First subapplication : TEST1 located in subfolder \FSC\TEST1
Only enter TEST1_1
Second subapplication : TEST2 located in subfolder \FSC\TEST2
Only enter TEST2_1

You only need to enter the name of the subapplications and not the folder path. The path to the application is automatically generated.
 - B You can also append an application from a floppy disk. No FSC path is used in that case. Just enter the name of the drive, ":", and the name of the subapplication.
E.g. A:\TEST1_1
The application files must all be located in the root of the floppy disk.
- 4) FLD numbers in the main application are not allowed to be present in one of the subfolders.
- 5) All function block sheets in the subapplications must have an FLD number which is at least two higher than the largest FLD number of the program block sheets in the main application.



- 6) No check is built in yet for possible double placement of tag number indexes and sheet indexes. This is the responsibility of the user. When it is placed in more than one application, it will be generated more than once during printing/plotting.
It is recommended to place such FLD in the main application after the append action.
- 7) If the configuration type is defined in both applications the complete configuration of both applications must be the same. So it is advised to create a main application first and then make copies from the main application while new names are given to the so called subapplications.
- 8) If the configuration type is defined in both applications the rack positions in each cabinet must be the same in both applications. See advice given under point 7.
- 9) If the configuration type is defined in both applications the fault reset and force enable tag numbers must be equal in both applications. See advice given under point 7.
- 10) If in both applications in a certain rack and position a module is defined, they must be equal as well as the safety relation. If a rack, position in the subapplication is defined while it is not defined in the main application, it is copied to the main application.
- 11) The main application has priority on the subapplications with regard to the use of certain I/O, hardware allocation etc. When possible, I/O is copied else it is deleted or replaced by a repeated input and is placed in the subapplication.
I/O which exists in the data file, but is not placed in the main application will be copied to the main application.
Inputs which are used in both applications but for a different CPU are also copied to the main application.
- 12) PIDs may only be placed once in the logics. So it will be deleted when it is placed in both applications. When a PID from a subapplication does not exist in the main application it is copied to the main application. The PID number may be changed if necessary.
If a PID from a subapplication already exists in the main application, but is not placed in the logics yet, the adjustments of the PID in the subapplication are copied to the main application and the PID number stays unchanged.
- 13) If the hardware allocation of an I/O is not already used in the main application, it is copied to the main application, else it is cleared. The same holds for the COM port assignment of COM or FSC variables.
- 14) Multiplexer numbers or alarm group numbers may be changed during append. If no more multiplexers or alarms can be placed in the main application the variable will be deleted.
- 15) All delete or replace operations and all errors during the append are written to a log file named <Main application name>. APP
You can output this file to a printer afterwards or have it displayed on your screen using the 'View Log' option of FSC Navigator.

Note:

Append application is not active if on-line modification is set to 'Yes'.

Two variables of a certain type are equal if the tag numbers are identical.

– eXit

The FLD design editor will save the current FLD to disk and then returns to the FSC Navigator main screen.



– Forward

The first MAIN menu is left and the second MAIN menu is displayed.

– Back

This option will take you back to the first part of the MAIN menu.

<PgDn> and <PgUp> are keyboard functions, and are not displayed in the menu.

<PgDn> go to the next available FLD.

<PgUp> go to the previous available FLD.

These options are only available in the FLD design editor's main menu.



6.5.1.2 SYMBOL commands

– New

The selection sheet which contains all symbols is displayed on the screen (see chapter 3). You can now choose a symbol by pointing with your cursor at its inside area. The program displays your design sheet on the screen and the selected symbol is positioned in the middle of it. You can drag the symbol to a legal position on the screen. Remember that you can cancel the placement command by entering <Esc>. The way the program goes on depends upon the kind of symbol you have chosen (see chapter 5.2)

– Change

Changing a symbol almost works the same as the 'new' option. When the change symbol option is chosen you must select a symbol from the design sheet. Next all the variable data of the chosen symbol variable data can be changed. The old values are displayed as default input and can be edited with help of the edit functions. For instance, change the tag number of an input or the height of a gate, without having to delete it first and replacing it. If the symbol to be changed is a line you cannot change the type of line but only the length of the line. If you select a line to be changed, the cursor jumps to the nearest end point of the line. When you change the tag number of an I/O or PID, it means that a new I/O or PID is chosen. The old I/O or PID is free for use on that moment. If you choose a new tag number that does not already exist, all data must be entered again.

If you change the tag number using the 'System Configuration' option of FSC Navigator, the variable stays the same, only the tag number will be different. You will then need to change all I/O or PIDs in the FLDs to the new tag number.

– Delete

You are asked to point with the cursor at the inside area of a symbol on the FLD. The indicated symbol is deleted from the screen and the database. But a backup of the symbol record is made for a possible recovery. If you delete an input or an output, it is not deleted from the variable database, but it is now available to be used again. As long as the undelete option is displayed in the menu you can undelete the last deleted symbol.

When you delete an off-sheet transfer symbol, the on-sheet transfer symbol on the destination FLD is deleted when it is not connected to a signal line.

When the on-sheet transfer on the destination sheet is not deleted, the source in that transfer symbol is replaced by zero and all variables in the variable database will also be deleted when it is not connected to a signal line or it is not a multi transfer symbol.

When the off-sheet transfer on the source FLD is not deleted, the destination in that transfer symbol will be replaced by zero and all variables in the variable database belonging to it are changed accordingly. Inverters connected to the selected symbol will be replaced by single lines.

Deleted square root, square or convert signal symbols will be replaced by double lines. A deleted pulse symbol will be replaced by a single line.

– Move, drag and copy

You are asked to point with the cursor at the inside area of a symbol on the FLD. If you choose copy the sequence is the same as with placement of that symbol. You must enter all variable data. If move is chosen, You are asked to select a new position. Next the program checks and corrects the position if possible.

If you selected move or copy all connections to the symbol will be broken. If you selected drag the connections will be maintained, therefore the new position to be selected is limited.

When you select move, inverters connected to the selected symbol will be replaced by single lines, when the symbol is moved to another position.

When you move a square root, square or a convert signal, the symbol is replaced by a double line. When you select a pulse symbol, the symbol is replaced by a single line.



– Undelete

If the delete option was entered previously, the deleted symbol can be undeleted. But if you placed new symbols on the FLD the undelete function cannot be executed anymore. As long as the undelete option is displayed in the menu you can undelete the last deleted symbol.

– Redraw

Redraw can be used to redraw the complete FLD in order to recover damaged symbols, which can be caused after a delete operation (e.g. basic sheet lines when an input is deleted).

– Set safety (not implemented yet)

You can mark the signals and symbols that are safety-related. The lines will get a different line type or a different color. If you switch the safety relation of I/O symbols, other code will be generated by the compiler. Some I/O symbols cannot be switched to safety-related (e.g. variables with location COM).

– Symbol set

You can switch between the symbol set for a function block and the symbol set for a program block. This option is only available when you are editing a comment block.

– Main

This option will return you to the FLD design editor's main menu.



6.5.1.3 BLOCK commands

If you select one of the block options in the BLOCK menu, you have to define a rectangular block area (except for load, undelete and redraw). The block can be defined by entering two block corners. The block will be marked by a dashed line. The program makes the block as small as possible.

– Move, Copy

The block can be moved to a new legal position. Press <Enter> to confirm the position. If an input, output or transfer symbol is included in the block, the x-position of the block (or the y-position if the FLD is vertically oriented) is kept on its old value and only the y-position can be varied.

When you copy an I/O or a PID the tag number is replaced by an illegal non-existing tag number '?'. You must change it afterwards with the change option in the SYMBOL menu.

On-sheet transfers get a source number zero and off-sheet transfers get destination numbers zero. During the move of the block, the background is automatically recovered. If you enter <Esc>, the program returns to the situation before this option was entered. If you press <Enter>, position checks, connection checks and border corrections are executed just as with the 'New symbol' option. When symbols with a sequence number are copied sequence number overflow can occur. If a position error occurs you can try a new position or cancel the operation with <Esc>. If no error occurs, the block is placed on its new position.

– Delete

When delete is chosen, you have to define a block. The block contents is deleted from the screen. Before the deletion is executed, a backup of the old block is made for a possible recovery of the block. Undelete is possible as long as the undelete option is displayed in the menu. A confirmation is asked before the delete operation is executed.

– Undelete

The previous deleted block is recovered. This option can only be executed immediately after you have deleted a block. As long as the undelete option is displayed in the menu you can use it.

– Save

You have to define a block. You are asked to enter a file name without extension. The program adds the extension '.BLK' to the name you entered and saves this block to disk. If that file already exists, you will be asked whether this file may be overwritten. If you answer <N>, the save operation is canceled else the program writes the block contents to that file. By entering <Esc> during input of the file name the save operation is stopped. The saved block can be loaded into any FLD using the 'Load' option.

– Load

You are asked to enter a file name without extension. The program adds the extension '.BLK' and checks whether this file exists. If it does not exist you can try a new file name as long as no <Esc> is entered. If you enter <Esc> the load operation is canceled. If an existing file name is entered, the FLD design editor will load the block from the file. It will check whether this block fits within the screen size and if it does, it will define a new block. In most cases the block is positioned on the place where it was before the save operation (see 'Save' option above). If the sheet borders are exceeded, the program corrects the position if possible.



All data is converted just like when you choose the Copy command. Next the block is drawn on the screen and you can drag it to the place just like copy block (see copy option). If the format of the block is almost as large as the complete screen you cannot drag it to another position. Instead it is placed in the center of the FLD if no block overlap is detected by the program. When the block cannot be placed anywhere, because of symbol overlap, you can abort this operation by entering <Esc>. The block is then deleted from the screen, but not from the disk. If no file with extension .BLK is present in the project folder, this option is not available.

If the block contains symbols that are illegal for the sheet type, these symbols are deleted for this FLD e.g. output symbols in a block into a function block.

– Redraw

See SYMBOL commands for explanation.

– Main

This option returns you to the FLD design editor's main menu.



6.5.2 Placing and connecting symbols

6.5.2.1 Connection lines

If you select a line (boolean or decimal) on the selection sheet you will return to the design sheet. Select a position where you want to start the line and press <Enter>. If you move the cursor a line is drawn in a vertical or horizontal direction depending on the chosen position of the last point. By pressing <Enter> you can select the end point of the line. If you did not make a connection the line will not be cut, but the program will ask you to enter a third position to connect the line. The program will keep on asking for a connection point until you made a connection. If you do not want to connect the line you can use <Esc>. The program will stop drawing the line and undraw the last line to the last connection point. You can make a connection by selecting the end position of a line into a symbol. The line will be cut at the outer borders of the symbol and the connection is made. Lines will also be cut at the sheet borders. If you cross two lines the vertical line will be broken at the cross point, which means no connection is made between the lines. If lines overlap they will be united to one single line. If you make an illegal connection, a beep is generated and the last line is undrawn.

6.5.2.2 Logical functions, calculations and compare functions

For symbols with a variable height (depending on the library definition) you are asked for a height and/or the width (format) expressed in half grid points. In this case a default value is given by the program. This value can be edited. If the symbol exceeds the sheet borders, the program places it at that border. If it coincides with an already placed symbol, a beep is generated. (Symbol overlap) You can keep on selecting a position until the position is legal or until you cancel the option with <Esc>.

6.5.2.3 Function block

First you are asked to enter an FLD number on which the function block has been designed. If that FLD is not a function block, the following message is displayed:

The sheet you entered is not a function block.

You can also use <point of FLD number> + <?> to ask for a function block listing. You get a listing on screen, with all existing function blocks larger or equal to the number you entered before <?>. If you did not enter an FLD number, all function blocks will be listed.

You are asked to enter an FLD number until you entered a number of a legal function block. Next that function block is placed on the screen. All flip-flops, counters, registers and timers are now copied in the FLD and in the variable database. The format of the function block depends upon the number of function block inputs/outputs. All function block I/O is drawn at the function block symbol. Decimal inputs/outputs are drawn as double lines and boolean I/O are drawn as single lines. The function block number of that function block must be higher than the current FLD number. If you enter <Esc>, the placement of a function block is canceled. Once a function block is designed on an FLD, the following FLDs may only be equation blocks or function blocks.

6.5.2.3.1 Changing a function block

If you change a function block, you must change all function block calls on other FLDs. The compiler will also change these function block calls. When you changed the number of function block inputs or outputs the connections will get lost. In this case you must reconnect these signals.

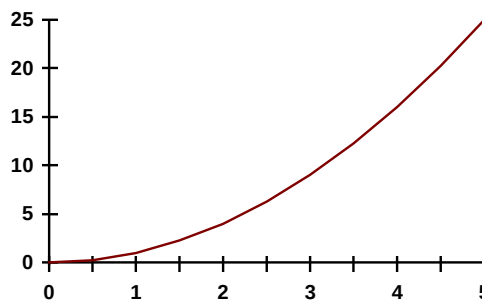
6.5.2.3.2 Function block inputs and outputs

A position must be selected. The symbol is then drawn on the screen. If you choose a decimal input, double lines are drawn at the input side of this symbol else a single line is drawn. For function block outputs the same action is taken on the output side of the symbol. For both types a descriptive text can be entered.

6.5.2.4 Equation blocks

An equation block symbol can be used to approximate complex calculations. Logarithmic, exponential, differential functions, etc. can be approximated using an approximation table. An example of a square approximation table for values between 0 and 5:

X	Y
0	0
0.5	0.25
1	1
1.5	2.25
2	4
2.5	6.25
3	9
3.5	12.25
4	16
4.5	20.25
5	25



Using the equation block symbol and a generated equation table, the equation can be approximated.

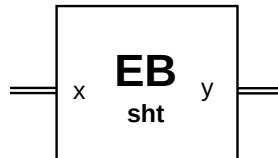


Figure 6-15 Equation block symbol

Equation block definition

Before the equation block symbol can be used on any functional logic diagram, an equation block diagram must be defined. This is accomplished by defining a functional logic diagram as an equation block diagram. The 'Title block', 'Block type' option must be used to select the 'Equation block' as block type. The diagram number of an equation block diagram must be within the range 500 ... 999. Next the equation table for the required calculation must be stored in the logic diagram. Once an equation block is designed on an FLD, the following FLDs may only be equation blocks or function blocks.

Equation table

The equation table can be specified with the 'Equation table' option which is selectable in the main menu of an equation block diagram. This option demands for an ASCII text file in which an equation table is defined. As soon as the file name is entered, the layout of the equation table file is checked and the equation table is stored in the equation block diagram. To remove an equation table from an equation block diagram just enter the name of a non-existing ASCII text file. Once an equation table is stored in the equation block diagram, the original ASCII text file is not needed anymore by the FSC software.



Equation table design

The equation table file is an ASCII file which can be generated with any popular text editor. This equation file consists of two columns. The input (x) values are stored in the first column and the output (y) values are stored in the second column. One row with an 'x' and an 'y' value together form one point pair (x,y).

The following rules apply to the layout of an equation table:

- Each line must contain an input and an output value with at least one empty space between the first and the second value. Empty lines are not allowed.
- The minimum number of rows in an equation table is 3, the maximum number of rows in an equation table is 200.
- The input values must be in an ascending numerical order.
- Both normal and exponential notations may be used for example: 0.05 equals 5.0E-2.
- The range for input values (x) and output values (y) is -10^{38} and $+10^{38}$.

Equation block usage

Once you have defined an equation table in an equation block diagram, it can be used on any functional logic diagram with a lower diagram number. The next figure shows the equation block symbol

Both the input (x) and the output (y) values must be connected. The output (y) signal is of type Float, the input (x) value is internally converted to type Float if required.
'Sht' stands for the FLD number on which the equation block is defined. This FLD number is characteristic for an equation block.

Algorithm used

The FSC CPU software uses linear interpolation to approximate the output value for an input value that is located between two input values:

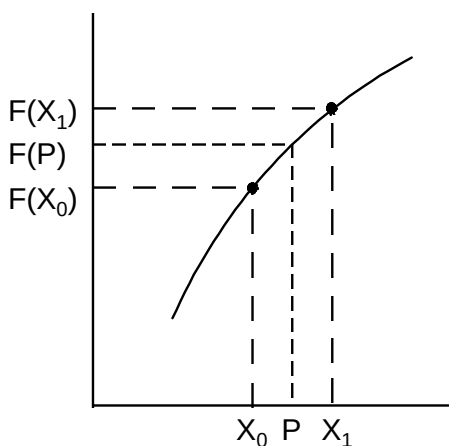


Figure 6-16 Approximating an output value to an input value



The approximation algorithm used is:

$$f(P) = f(X_0) + \frac{f(X_1) - f(X_0)}{X_1 - X_0} * (P - X_0)$$

Remarks:

The input (x) value is clamped on the first and the last values in the equation table: A value smaller than the first input value in the equation table will result in an output value equal to the first output value of the equation table; an input value larger than the last input value in the equation table will result in an output value equal to the last output value of the equation table.

6.5.2.5 Timers

First a position must be chosen. Next, a timer base is asked and for fixed timers also the timer value must be specified. For the variable timer the value must be realized by connecting a binary signal (byte or word) on the input side of the symbol.

With <←> and <→> or <Enter>, you can switch between these fields. The following timer bases can be chosen by toggling with <Space>:

- 100ms 100 milliseconds (0.1 second)
- s 1 second
- m 1 minute

The timer value must be an integer, but for seconds and minutes you can enter a value using a fixed point real. This value will be rounded to the nearest integer, but in some cases a base will be chosen for better accuracy of the timer value. Illegal values are changed to the nearest legal value.

The following base conversions are executed (T = timer value):

- Minutes to seconds : if T < 10 (10 minutes)
- Seconds to minutes : if T ≥ 1200 (20 minutes)
- Seconds to 100 ms : if T < 10 (10 seconds)

The number of timers that you can use in one application is limited:

- 100ms : max. 748
- s : max. 748
- m : max. 446

The unique identification code for the timers is constructed as follows:

'T' + <FLD no.> + <sequence number>

The sequence number is generated by the program. The maximum number of timers per FLD is limited to 32 (this includes timers in function blocks).



6.5.2.6 Inputs and outputs (I/O)

The I/O symbol can be dragged to the correct position and placed by <Enter>.

If the I/O overlaps an error message is displayed: symbol overlap.

For more information about normal I/O (no diagnostic input) see subsection 5.2.5.1, for diagnostic inputs see subsection 5.2.5.2.

6.5.2.6.1 Normal I/O

Next a tag number is asked. With this tag number and the I/O type, the program searches for that I/O in the variable database. If the input is already placed on an FLD of the same CPU the following message appears on screen (except for flasher inputs):

'Input already placed on FLD <FLD no.>'

If the output is already placed on an FLD, the following message will appear on screen:

'Output already placed on FLD <FLD no.>'

If you have deleted an I/O before, you can use it again. If the tag number you entered is found in the variable database all the variable data is taken from the database and no further input needs to be entered unless this data is illegal. The data is drawn on the screen.

You can also type <?> or <string> + <?>. A tag number listing appears on the screen, which contains all valid tag numbers (all existing but free tag numbers). If you specified a string before <?>, only tag numbers starting with this string are listed.

If the tag number is not found you have to enter some other data, starting with a description (service). If the I/O is boolean, a text describing the one status of the input (qualification) must be entered.

Location, which is a string of 3 characters which can be an abbreviation of a part of the installation where that I/O is located. Location can also have a special meaning which has influence on the application function of the I/O:

MUX: This Input/Output must be assigned to a multiplexer. This must be done using the 'System Configuration' option of FSC Navigator.

ANN: This Input/Output must be assigned to an alarm function. This must be done using the 'System Configuration' option of FSC Navigator.

COM: This Input/Output must be assigned to a communication port. This must be done using the 'System Configuration' option of FSC Navigator.

FSC: This Input/Output must be assigned to a FSC system. This must be done using the 'System Configuration' option of FSC Navigator.

SYS: This Input is a reserved location which is used for internal system variables. (You cannot use this location!)

You can use <↑> and <↓> to switch between tag numbers, service, qualification, signal type and location fields. The last selected value will be drawn to the I/O symbol.

If the I/O is new, its information is stored in the variable database.

6.5.2.6.2 Diagnostic input

Next you must choose the I/O type. You can do this by toggling with <Space>. The following types can be selected: I, O, AI, AO, LoopO, SensAI



For detection of a channel fault you must choose I, O, AI or AO depending on the module type for which the I/O is allocated. For a loop-monitoring error of a fail-safe output you must select LoopO. For a sensor alarm of an analog input you must select SensAI.

You cannot select SensAI for safety-related analog inputs with non fail-safe sensors. Use type AI instead. Next you must enter a tag number of a I/O which is allocated to a fail-safe I/O module.

If you entered a non-existing tag number or when you entered a tag number which is not legal you get a message:

'Input/output does not exist: <Tag no.>'

You can also select the I/O from a selection list only containing the selectable I/O for diagnostic input. Type <?> or <string> + <?>. A tag number listing appears on the screen, which contains all valid tag numbers (all existing and allocated to a fail-safe I/O module). If you specified a string before <?>, only tag numbers starting with this string are listed.

6.5.2.7 Sheet references

The positioning is the same as inputs and outputs. The way the program continues depends on the type of the marker:

– MARKER INPUT, MARKER OUTPUT

Now first enter the tag number of the input and secondly enter the tag number of the output.

You can also type <?> or <String> + <?>. A tag number listing will be displayed on your screen, which contains all possible tag numbers starting with the specified string. If no string is specified all tag numbers will be listed.

If no space is available for the related transfer symbol, an error message is displayed on the screen, and the placement is canceled else the transfer and its related symbol are drawn on the screen. A marker input is always placed together with a marker output. With respect to the I/O it is handled the same as I/O. But beside the I/Os a third symbol is placed in the variable database: a marker which connects the input to the output.

The number of marker I/O symbols per FLD is limited to 26. In the variable database the service field of the marker contains the tag number of the input, the qualification field contains the tag number of the output. If the input is already placed on an FLD, the following message appears on screen:

'Input already placed on FLD: <FLD no.>'

If the output is already placed on an FLD, the following message will appear on screen:

'Output already placed on FLD: <FLD no.>'

The unique search code for the marker is:

<FLD no.> + '_' + <Sequence letter>

6.5.2.8 Flip-flops + pulse

A flip-flop and a pulse are placed in the variable database. Every flip-flop gets a sequence number which is needed to create a unique code for searching in the variable database. The tag number of a flip-flop and a pulse are constructed as follows:

<FLD no.> + '_' + <Sequence number>.



Only a limited number of sequence numbers is available. When all sequence numbers are in use, an error message is displayed and no flip-flop or pulse can be placed anymore.

6.5.2.9 Sheet transfers

6.5.2.9.1 Characteristics

A sheet transfer is characterized by two FLD numbers and a sequence number. The first FLD is the source FLD. This is the FLD from which the signal is transferred. The second FLD is the destination FLD. This is the FLD to which the signal is transferred to. The sequence number is a number which is unique on the source FLD in order to discriminate between two signal transfers with the same source and destination FLD.

Note:

Care must be taken in sequence of FLDs. The destination FLD number should be greater than the source FLD number. Failing to do so will increase the response of the FSC system by one program cycle.

Two different types of reference symbols are used for the source FLD and the destination FLD.

– Off-sheet transfer symbol

This symbol is placed on the source FLD at the output side.

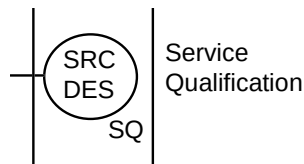


Figure 6-17 Off-sheet transfer (boolean signal)

There are four different off-sheet transfer symbols:

- a) Single boolean off-sheet transfer.
- b) Multi boolean off-sheet transfer: a maximum of 5 transfers to different FLDs.
- c) Single binary off-sheet transfer.
- d) Multi binary off-sheet transfer: a maximum of 5 transfers of binary signals to 5 different destination FLDs.

– On-sheet reference symbol

This symbol is placed on the destination FLD at the input side.

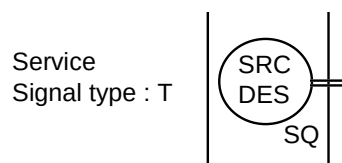


Figure 6-18 On-sheet transfer symbol (binary signal)



There are four different on-sheet transfer symbols:

- a) Boolean on-sheet transfer.
- b) Repeated input transfer: this signal is connected to a digital input (I) on the source FLD.
- c) Binary on-sheet transfer.
- d) Binary repeated input transfer: this signal is connected to a multiple input (BI) on the source FLD.

The unique identification (tag number) of a reference symbol is constructed as follows:

<Source FLD> + '_' + <Destination FLD> + '_' + <Sequence number>

If the type is 'M' (Boolean sheet transfer), a marker is placed in the variable database for each reference you make.

If the type is 'R' (Binary sheet transfer), a register is placed in the variable database for each reference you make.

6.5.2.9.2 How can you make a sheet transfer?

There are two ways to make a sheet transfer:

- A From the source FLD by selecting an off-sheet transfer symbol.
- B On the destination FLD by selecting an on-sheet transfer symbol.

There is a minor difference between linking FLDs from the source FLD or from the destination FLD. There are also two different ways to realize a connection from the destination FLD or from the source FLD:

- A Only make use of the FLD number and a sequence number. This method needs knowledge about the sheet transfer mechanism.
- B Use a list of existing sheet transfers which contains the transfers which can be linked to the current FLD. This method does not require much knowledge about the sheet transfer mechanism but instead makes use of a tag number, service and qualification.

For the tag number listing see chapter 6.5.2.9.5.

6.5.2.9.3 Sheet transfer from the source FLD

First select an off-sheet transfer symbol. This can be a single or a multi sheet transfer symbol. The program always converts this symbol to a multi sheet transfer symbol, which means that you can create maximal 5 transfers to 5 different FLDs. You must first select a legal position. Next you must enter the FLD number of the destination FLD, which must be a number within the range of 0 ... 999.

If you do not know to which FLD you want to transfer a signal to, enter 0. This means that the signal is not connected to a destination FLD yet. You can only transfer to an existing FLD. If you enter a number of a non-existing FLD, the program replaces the number with '0'. You can connect an unconnected transfer symbol afterwards with change or by selecting it from the destination FLD (see sheet transfer from destination FLD).

In order to identify a sheet transfer symbol, each on-sheet transfer symbol gets a unique sequence number which is assigned automatically by the program.

If you enter the number of an existing FLD, you are asked to give the sequence number of a sheet transfer on the destination FLD (range 0 ... 36).



You now have two options:

1. You want to connect the signal to an already existing not connected (source FLD = '0') on-sheet transfer symbol on the destination FLD. In this case you must enter the sequence number of that on-sheet transfer symbol. The program searches for this transfer and connects it when it is really not yet connected and when it has the same type as the source signal. The sequence number of that on-sheet transfer symbol is changed into the sequence number of the source symbol (off-sheet transfer). Also the identification code (tag number) is changed according to the prescribed structure (source sheet '0' is replaced by source FLD = current FLD and old sequence number is replaced by the new sequence number). If the transfer is not found, the sequence number is changed into '0' by the logic editor (see 2).
2. You do not want to connect it to an existing on-sheet transfer symbol on the destination FLD (or the sequence number you entered did not result in a link (see 1)). In this case enter '0' for the sequence number. Now a new on-sheet transfer symbol will be placed on the destination FLD. If there is no space on the destination FLD, the following error message appears on the screen:

No space on referenced sheet.

The sequence number of this new symbol will be copied from the source symbol which is generated by the program.

Note:

Note that the sequence number you enter is only meant to point out which connection you want to make or whether you want to make a connection and that it is not a way to assign a number to the transfer symbol you want to place.

Next you must enter some additional data:

- 1) Service or I/O tag number.
- 2) Qualification (only for boolean sheet references)
- 3) Signal type (only for binary sheet references)

1. Service

You can enter a string with a maximum length of 19 characters. The service text can have two meanings:

- A A descriptive text.
- B An I/O tag number of an I/O which is connected directly to the off-sheet transfer symbol. In this case the service and qualification and signal type are copied from the I/O and else you must enter them yourself (see 2 and 3).
The program searches with the service text (first 16 characters) for an I/O on the FLD the off sheet reference symbol is placed. The type is determined by the program in the following order of priority.

For boolean I/O	: I, O
For binary I/O	: AI, AO, BI, BO

If the type is I, AI or BI the transfer symbol on the destination FLD is converted to a repeated input symbol.

Note:

If an I/O tag number is entered and the connection is not direct, the 'Translate application' option of FSC Navigator (the compiler) will give an error message. If it is not directly connected, you can better use a service text which is slightly different from the tag number.



2. Qualification

This text can only be given for boolean sheet transfers.

You can enter a text with a maximum length of 16 characters. This text describes the 1 status of the signal. Thus, "ALARM ACTIVE" means that the alarm is active when the value of the reference marker is 1.

Note:

The interpretation of the qualification is your responsibility. It is meant as description of the signal status.

3. Signal type

This type can only be specified for binary sheet transfers. You must select a signal type by toggling with <Space>.

Possible types are :

B : Byte
W : Word
L : Long
F : Float

If you did not enter a destination FLD '0', you can make 4 additional transfers to different destination FLDs. For these transfers the service, qualification and signal type are the same as you entered for the first transfer. If you use an already used destination in this sheet transfer symbol, the following error message appears on the screen:

Duplicate reference not allowed.

In this case you must use a new off-sheet transfer symbol which gets a new sequence number from the logical editor.

All on-sheet transfer symbols on the specified destination FLD get the same sequence number, which is the one that is assigned by the program to the on-sheet transfer symbol on the source FLD.

If you enter <Esc> all completed destination FLDs are accepted and the program stops asking for more destination FLDs.

If you enter a destination FLD '0', the logical editor stops asking for new destinations. So it is the same as entering <Esc>.

If you only specified one destination FLD, the program converts the multi off-sheet transfer symbol to a single off-sheet transfer symbol.

6.5.2.9.3.1 Changing an off-sheet transfer

If you want to change an off-sheet transfer the procedure is the same. But now you get the old values as default. You can now enter the same or different destination FLDs. Pressing <Esc> will abort the input of destination FLD numbers. Modified destinations are accepted; other previous existing destinations are not modified.

6.5.2.9.4 Sheet transfer from the destination FLD

First select an on-sheet transfer symbol and a legal position. Making a sheet transfer from the destination FLD is almost identical to making a sheet transfer from the source FLD. One difference is that you can only make one transfer: Two different source signals cannot be transferred to one destination.

Another difference is that you can connect an on-sheet transfer to an off-sheet transfer which does not yet contain the maximum number (5). If you enter the source FLD and the sequence number of an off-sheet



transfer which is single, it will be converted to a multi sheet transfer. In this case that symbol will be extended with the new destination. You cannot make a connection to an off-sheet reference which already contains a transfer to the current FLD. In that case again the following error message appears on the screen:

Duplicate reference not allowed.

If you connect the on-sheet transfer symbol to an off-sheet transfer symbol which has a service equal to the tag number of an I/O on the source FLD, the on-sheet transfer symbol will be converted to a repeated on-sheet transfer symbol. So it does not really matter which on-sheet transfer symbol you choose. If you entered the sequence number of an existing off-sheet transfer symbol on the source FLD, the sequence number of that transfer symbol will be copied to the destination symbol and a new marker or register is stored in the variable database. You can also enter a source '0', which means that you do not want to make a connection to the source FLD yet. You can connect this signal afterward with change or by connecting it from the source FLD (see chapter 5.2.8.3). If you entered the sequence number of an off-sheet reference symbol which already contains 5 transfers or when no off-sheet reference exists with that sequence number or when you enter sequence number '0', a new off-sheet reference symbol will be placed on the source FLD. If there is no space on the source FLD the following error message appears on the screen:

No space on reference sheet.

If you connected the sheet transfer to an off-sheet transfer on the source FLD, which already contains another transfer, you cannot change the service and qualification / signal type. You have to do this on the source FLD.

If a new off-sheet reference symbol is created on the source FLD, the logical editor assigns a new sequence number to it, which will be copied to the on-sheet transfer symbol on the destination FLD. So the sequence number you entered before was just to point out which link you want to make or whether you want to make a link. You do not have any influence on the final choice of this number.

6.5.2.9.4.1 Changing an on-sheet transfer

If you want to change an on-sheet transfer the procedure is the same. But now you get the old values as defaults. You can now enter the same or another source FLD. When you enter <Esc> or '0' the old source FLD is replaced by '0'. In this case the off-sheet transfer symbol on the source FLD is deleted or replaced by an unlinked on-sheet transfer symbol (see delete paragraph 5.1.2) and all variables in the variable database belonging to it are erased or changed accordingly.

When you change an on-sheet transfer which is connected to a multi off-sheet transfer you cannot change the service or qualification or signal type, because they are also related to the other sheet transfers which are stored in that off-sheet transfer symbol. It is possible to change the signal type when the source FLD reference is equal to 0. If you want to change it nevertheless you must go to the source symbol on the source FLD and replace it there.

6.5.2.9.5 Selecting a sheet transfer symbol using a selection list

Often you do not know to which FLD you want to transfer a particular signal. In this case you deliver the signal in an off-sheet transfer symbol with destination FLD '0' (unlinked). Afterwards you can make a link of FLDs by means of an on-sheet transfer symbol on a certain destination FLD. In this case you can ask for all existing unlinked sheet transfers by pressing <?> or the middle key of the mouse. You can do this on two places during the defining of sheet transfers.

1. When you are asked to enter the destination or source FLD number.
Suppose you are placing an on-sheet transfer. The case of placing an off-sheet is equivalent.
Enter <Numeric string>+<?>. A list of all unlinked off-sheet transfers on all FLDs starting with



the numeric string appears on the screen. If you did not specify a string , but only the <?> all unlinked off-sheet transfers are recorded in the list. If no such sheet transfers exist, the following error message appears on the screen :

No such tag number found.

In this list the tag number, service and qualification are recorded. Move the cursor to the sheet transfer you want to select and press <Enter> to select it. Now the list disappears from the screen. Because you already pointed out which transfer you want to make, the program does not ask for a sequence number anymore. If you do not want to select a transfer from the list press <Esc>. Now you return to the situation before you entered <?>. You can go on defining a source FLD.

2. You did not select a transfer from the list yet. Instead you entered a source or destination FLD number S.
You are asked to enter a sequence number for pointing out a sheet transfer on sheet S.
You can still select a transfer from the list. Press <?> and the list described in point 1 appears on the screen, but now only the unlinked transfers from sheet S are recorded in the list. The selection is as described in point 1.

In both cases service and qualification / signal type are copied.

6.5.2.10 Texts

The text must be entered. The text is not allowed to be longer than 60 characters. You can drag the text to the desired position and press <Enter>. Text can be placed anywhere and no position check is done. This means that you can even place texts in symbols.

6.5.2.11 Counters, registers and constants

6.5.2.11.1 Counters

You must first choose a legal position . Next you must specify the maximum counter value which must be a number in the range of 1 - 8191. When a counter reaches this value after a count up it is restarted to count at zero. The number of counters per sheet is limited to 32. The total number of counters per application is limited to 510.

6.5.2.11.2 Registers

You must first choose a legal position. Next you must enter the register type. The different types are:

- 'B' Byte register which can contain 8-bit integers.
- 'W' Word register which can contain 16-bit integers
- 'L' Long register which can contain 32-bit integers
- 'F' Floating-point register which can contain floating-point reals.

The number of registers per sheet is limited to 32.



6.5.2.11.3 Constants

You must first choose a legal position. Next you must enter the register type. For the constant types see registers. Next you must enter the constant value. The range of this value depends upon the signal type you have specified:

B : -128...+127
W : -32,768...+32,767
L : -2,147,483,648...+2,147,483,647
F : -10³⁸...+10³⁸

6.5.2.12 PID

Select a legal position. Next you must enter the tag number of the PID. If it is already placed, the following error message appears on the screen :

PID already placed on sheet <FLD no.>

If the tag number is a new PID, it is placed in the variable database and the PID settings are set to their defaults. You can change them using the 'System Configuration' option of FSC Navigator.

You can also choose a PID tag number by pressing <?> or <string> + <?>. If you specified a string, all selectable tag numbers that start with that string will be listed. If you did not enter a string, all selectable tag numbers will be displayed.

6.5.3 Output generated by FLD design editor

The functional logics are designed on FLDs which have a name <Application name> and a number <FLD no.>. The name of the file on disk in which the drawing is stored is a combination of this name and number: <Application name>.<FLD no.>. If an FLD number has less than 3 digits, zeros are used as leading digits in the file name extension.

All non-empty FLDs are part of the output. In the installation file all used FLDs are registered and also the type of the block (Function block, Program block, Step block, Comment block, Tag number index, Sheet index).

If an existing FLD is changed, the compiler status is set to false. So the installation file is a part of the output too. The FSC variable database is also a part of the output, variables may have been changed or inserted.

If a data error occurs the error is logged to a file named <Application name>.CRC.

If you saved a block to disk with the save option the block file is part of the output.

If you activate the option Append application in the second part of the main menu, the file <Application name>.APP is generated which contains all actions and changes on the appended application.



6.6 FLD formats

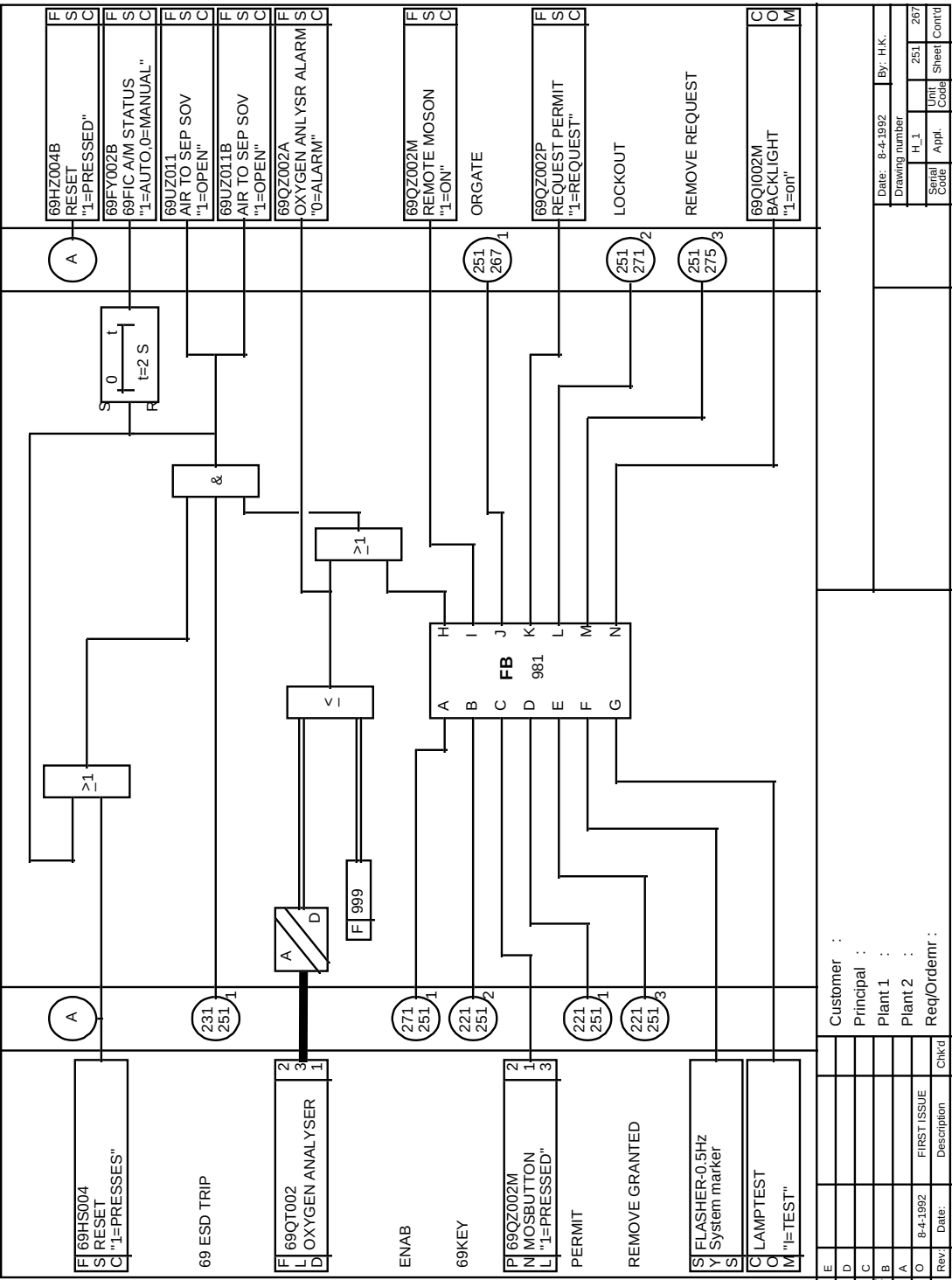


Figure 6-19 Example of an FLD designed with SYM1

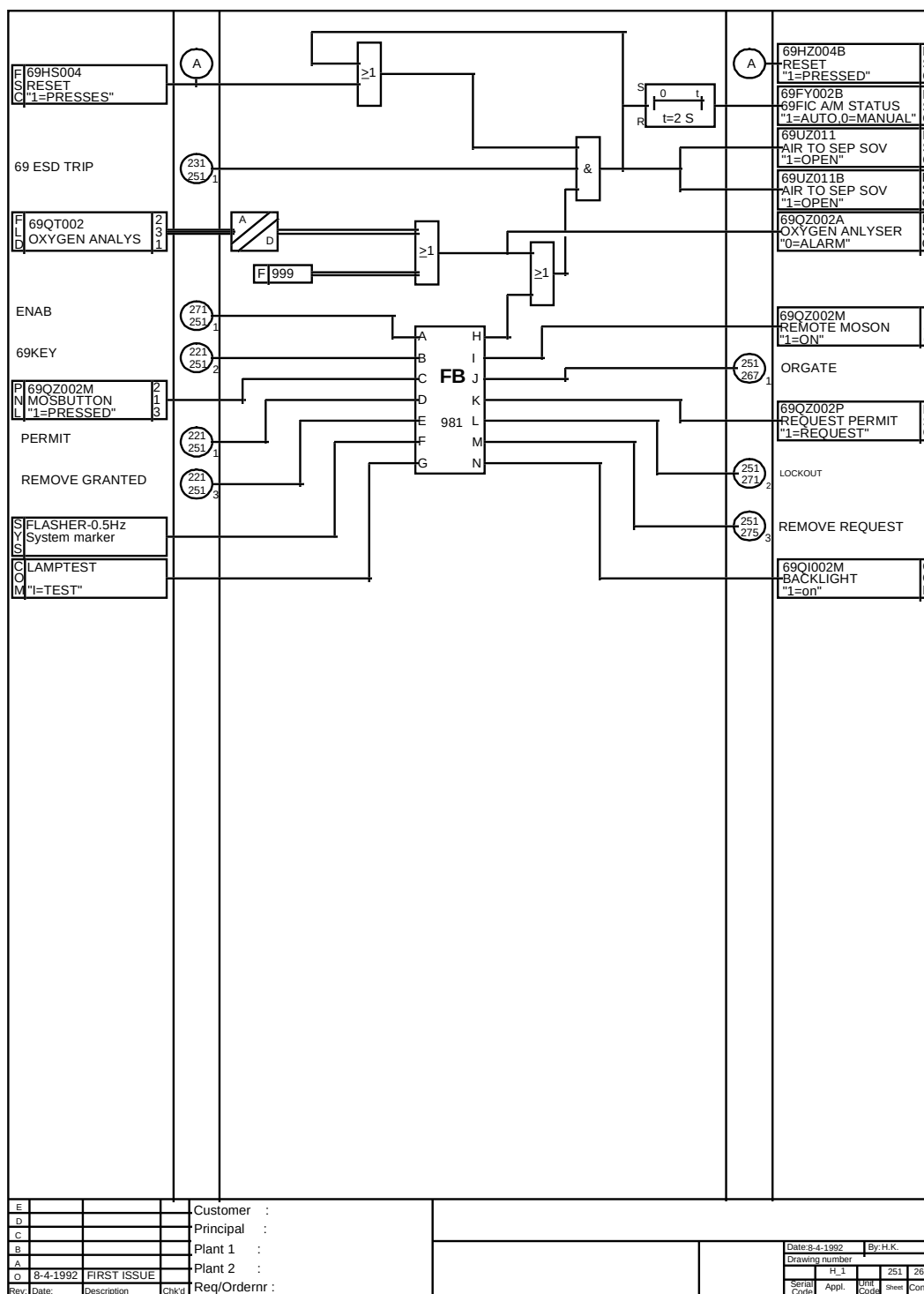


Figure 6-20 Example of an FLD designed with SYM2

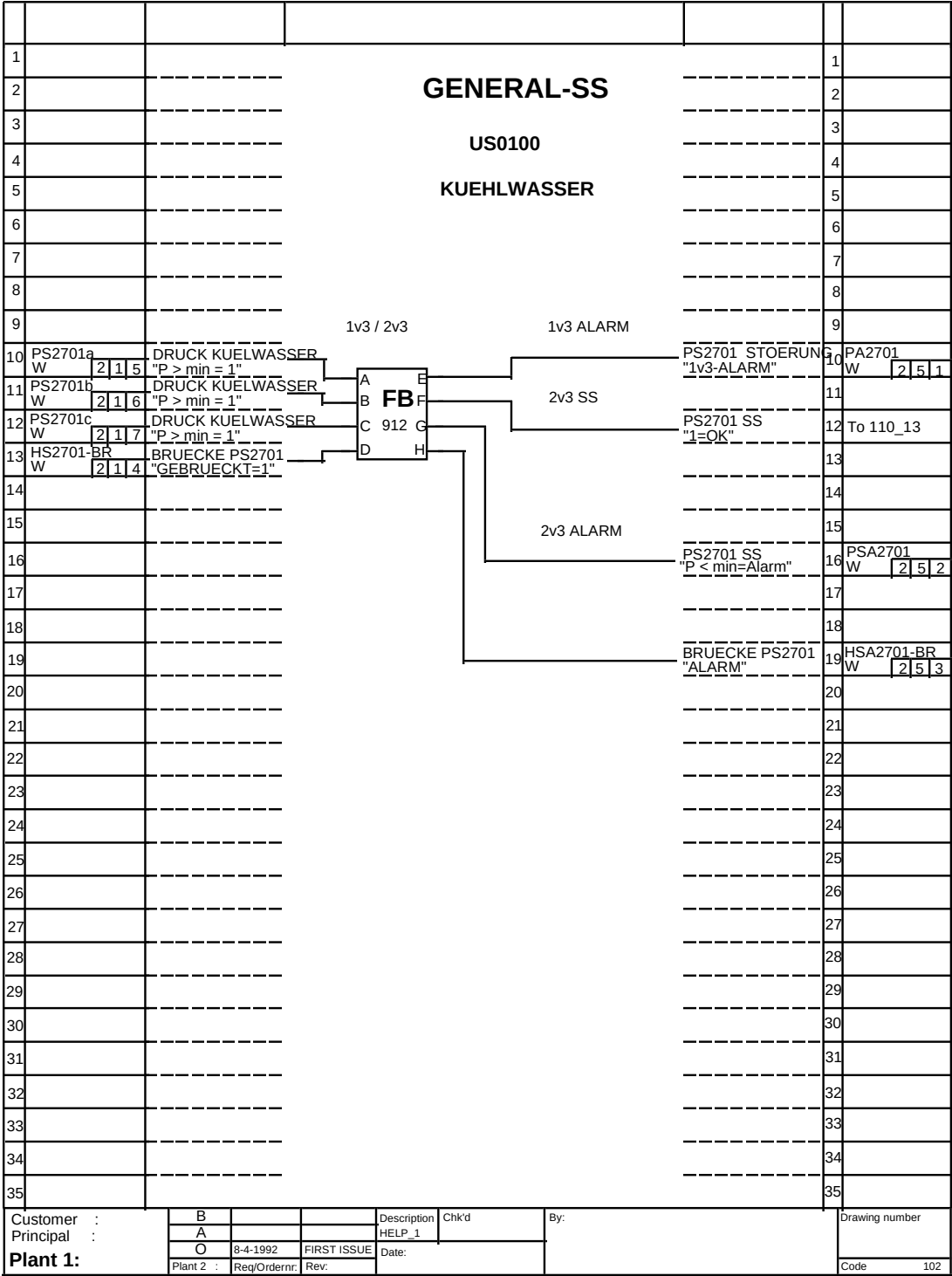


Figure 6-21 Example of an FLD designed with SYM3

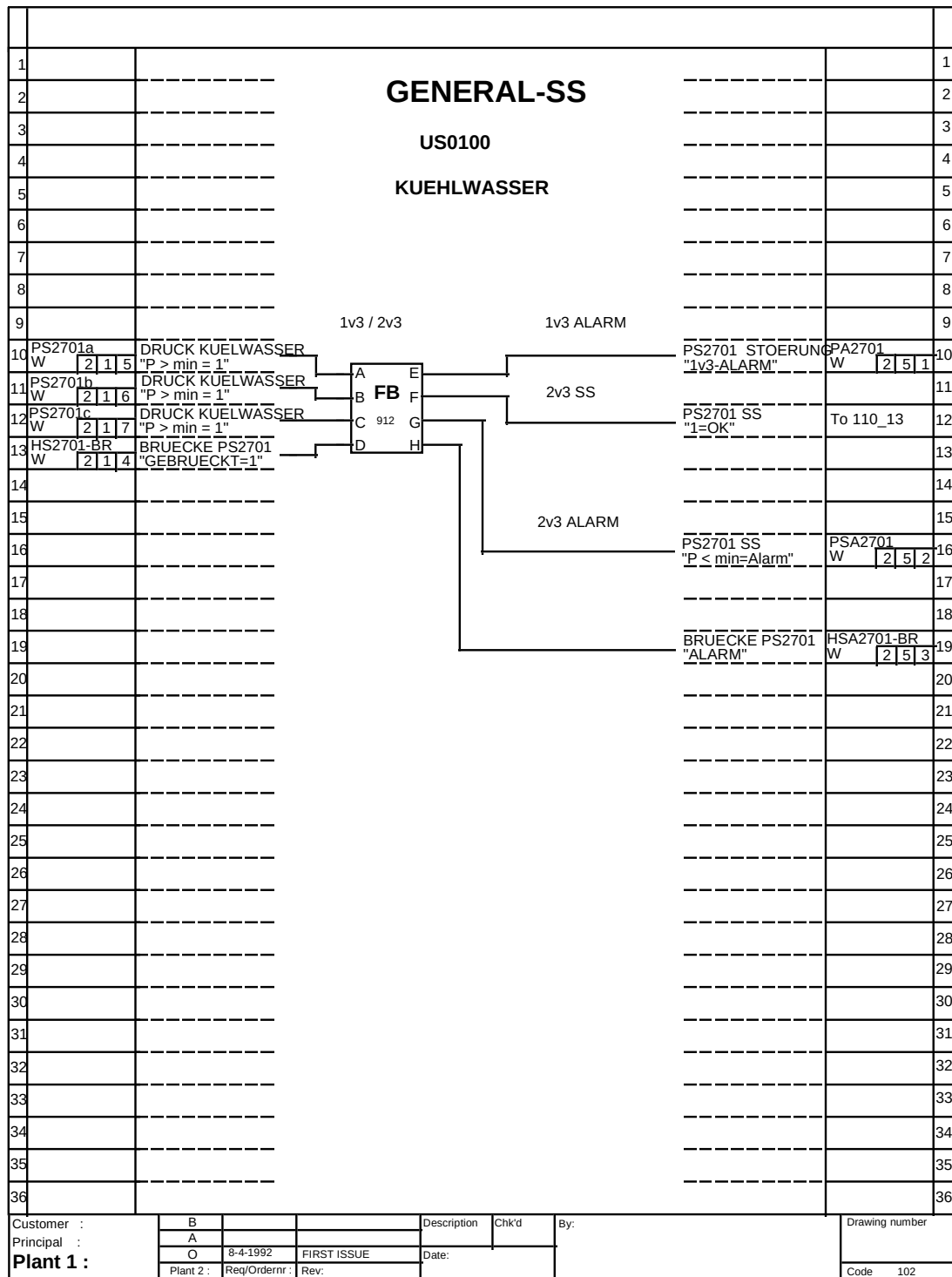


Figure 6-22 Example of an FLD designed with SYM4



6.7 Error messages

Error situations may result in an error message being displayed on screen. This subsection lists the error messages, and also provides information on how to solve the problem indicated by the error message. If the program gives an error message, it warns the user with a beep. The message is displayed for several seconds. If, however, a key is pressed or the mouse is moved, the message will disappear at once.

Alarm function different. Variable not appended: <type + tag number>

The alarm function in the appended application has the same name but other parameters. All alarm I/O assigned to this alarm are not appended.

Block does not exist:

1. You entered a block name which is not present in your project folder or which is not found in the root of the floppy.
2. You entered a name with extension.

Block does not fit on screen.

The block you want to copy is too big.

Cannot be converted (drawn with other type of library).

You can only copy an FLD if the application where the FLD is copied from was drawn with the same type of symbol library.

COM/FSC assignment changed: <type + tag number>

Inconsistent assignment. Reassign.

Communication buffer memory overflow.

No more communication variables can be allocated.

Configuration or safety times not equal.

Must be equal. The append action is canceled.

Constant will not fit in the internal area.

When the constant symbol is changed to a larger value, the symbol will exceed the I/O-borders.

Corrupted block. Deleted from disk:

The block you want to select, contains corrupted data. It has been deleted from the disk.

Corrupted data detected in I/O data file at record: <record no.>

A record in the variable database is corrupted. The FLD design editor will abort immediately. To rebuild the data files, start one of the other options of FSC Navigator.

Corrupted data detected in installation file at record: <record no.>

The FLD design editor will abort immediately. To rebuild the data files, start one of the other options of FSC Navigator. Some FLDs may be considered empty by the program. You can copy the FLD from your backup disks.



Database of application has different version: <version no.>

The application you try to use for copying or appending an FLD has another software version. Conversion is required. The operation is aborted.

Data error in installation file at record: <file name>

A record of the data file contains corrupted data.

Data error in FLD: <FLD no.>

The FLD indicated contains corrupted data. If possible, restore the application from a floppy or delete the FLD and design it again.

Data file too large. Try 'Rebuild' option to optimize file.

Use the Rebuild option of the 'System Configuration' option of FSC Navigator. This may create some new space.

Different number of bits for multiplexer. Variable not appended: <Type + Tag number>

The multiplexer functions in the appended application and the main application differ because of a different number of bits. So the multiplexer I/O assigned to this multiplexer variable are not appended to the main application.

Directory path not found.

Your system is corrupted. Exit FSC Navigator. Reboot your system (hardware reset) and restart FSC Navigator. If the problem persists, consult the dealer of your system.

DISK READ ERROR.

The program cannot read the complete contents of a file.

DISK WRITE ERROR.

The disk is full or corrupted. The data has not been saved. Try to free space on the disk and retry with a previous back-up of your application.

Drawn with invalid symbol library. FLD, Version: <version no.>

You have drawn this FLD with another version of the symbol editor. (Not active yet.)

Drawn with old editor version: <Version number>

The program will ask if you want to convert the FLD.

Duplicate reference not allowed.

You are not allowed to make two sheet transfers to the same destination FLD using one sheet transfer symbol. If you want to create two sheet transfers to the same FLD, make a line splitting and connect this signal to a second sheet transfer to that destination FLD.

Empty block.

You defined a block which does not contain a symbol that can be manipulated.

Empty diagram cannot be copied.

You entered the name of an FLD which is not defined.

**Equation file not found:** <File name>

The file name you specified could not be found in the current project folder.

Fault reset and/or force enable tag numbers not identical.

Must be equal. The append action is canceled.

File creation failed: <file name>

The program tried to create a file, but creation failed. (i.e. disk or folder full).

File opening failed: <file name>

The FLD design editor attempted to open a file, but the file could not be found on the disk.

FLD already exists.

You must enter the number of an FLD which is not defined yet.

<FLD name> not found on disk. Erase from application? Press <Space>: Y/N

An FLD that you created earlier with the FLD design editor could not be found in the project folder. If you answer 'yes', all variables relating to this FLD will be deleted from the variable database, and the FLD is considered to be empty. You may have deleted the FLD "outside" FSC Navigator. In that case, it is better to use the Delete option of the FLD design editor.

FLD number: <FLD no.>. **Illegal signal type(s):** R <FLD no.><tag no.> <conversion source> <conversion target>.

Example: "FLD number: 13. Illegal signal type(s): R 13_0_1 W L".

An invalid conversion was made for a multiple output or an off-sheet reference. Use the signal conversion symbol (see page 7-25) to create a conversion. Only downward conversions are allowed, e.g. from word to byte or from float to word. Make sure that only downward conversions have been defined.

FLD is corrupted: <FLD no.>

Some data which must be present in the sheet database are not found. You must delete this FLD manually (e.g. using Windows Explorer) and redefine this FLD.

The program will ask you whether you want to erase the data belonging to this FLD from the FSC database. You must enter Y.

FLD used twice. Application cannot be appended. Renumber FLD: <FLD name>

The application to be appended must only use FLDs which do not exist in the main application.

For printing logics with a laser printer, select functional logic plotting.

Make sure that you select a laser printer as your plotter, and use the 'Print/Functional Logic Diagrams' command on the File menu of FSC Navigator.

FSC system memory overflow.

The FSC system reached its maximum for this type of variable, so you cannot place it anymore.

You can try to create new variables by deleting variables of this type which you do not use anymore.

If the number of variables of this type is not as big, it is possible that your database is corrupted. Use the 'System Configuration' option of FSC Navigator to rebuild the database (with address reallocation).

Function block cannot be referenced.

You cannot make a sheet transfer to the output side of a function block sheet, as a function block cannot have an off-sheet transfer.

**Function block diagram in program block area: <FLD no.>**

During append some program blocks of the appended application are located in the function block area of the main application or some function blocks are located in the program block area of the main application. The append action will not be executed.

Hardware allocation changed: <type + tag no.>

Inconsistent allocation. Reallocate.

Hardware allocation cleared: <type + tag no.>

Inconsistent hardware allocation and no allocation possible.

Illegal application name or block name, or file does not exist.

You entered an illegal file name or the name of a non-existing file.

Illegal connection.

The connection you tried to make is not allowed or the signal type is illegal for this symbol.

Illegal FLD number

The FLD number you entered is out of range.

Illegal function block input exchange:

Two function block input symbols on a function block are exchanged. Place the function block input symbols as before or remove the function block call from the FLD.

This check is only active if the 'On-line modification' attribute has been set to 'Yes'.

Illegal function block output exchange:

Two function block output symbols on a function block are exchanged. Place the function block output symbols as before or remove the function block call from the FLD.

This check is only active when 'On-line modification' is set to 'Yes'.

Illegal input/output.

The I/O you entered for the diagnostic input is not legal. It must be allocated to a fail-safe module. You will also get this message if the data record is corrupted.

Illegal location for this type of I/O.

This type of input or output cannot have this special location.

Illegal operation in I/O data file at record: <Record number>

Your index file is probably corrupted. Try the 'Rebuild' option of the 'System Configuration' option of FSC Navigator.

Infinite signal cycle.

You made a cycle of connection lines which is not allowed.

Input already placed on FLD: <FLD no.>

You are not allowed to make two inputs with the same type and tag number per CPU.

Input/output does not exist: <type + tag number>

The input or output you want to use for the diagnostic input does not exist.

**Insufficient memory.**

You do not have enough internal memory to run this program. Try to make room by removing resident programs and drivers from your system. Before you restart the program do a hardware reset or switch off your system and switch it on again.

Invalid library version. You need version <version>.

The library contains a version number which is not compatible with the software version you are using.

Invalid numeric order in equation table.

The input value must be in an ascending numerical order.

Invalid software version. You need version <version no.>

The FLDs are drawn with a later software version. Use the correct software version.

Invalid value in equation table.

A value defined in the equation file is out of range. The legal range is 1E-38 to 1E+38.

Inverter must be connected to non-line symbol.

An inverter may only be connected to one of the following symbols:

- at the front of logical functions and pulses.
- at the back of a timer, pulses and Constbool.
- at the set and reset input of a timer and a flip-flop.
- at the top and the bottom of rotated logical functions

Lack of memory: <Number of bytes>

The window needed to shift a block of logics cannot be stored in the memory. Try to free the amount of specified memory by removing memory resident programs or drivers.

Link specification(s) of communication are not compatible.

Must be equal. The append operation is canceled.

Location <location> is reserved for internal FSC variables.

E.g. location 'SYS'.

Locations not compatible: <type + tag no.>

Two I/O variables with the same type and tag number during an append application action, have both a special location but are different. The location of the I/O of the main application is maintained.

Missing data file: <file name>

If you just started a new application, go back to FSC Navigator's main screen and open the project again. A new data file will be created if you answer yes. In case of an existing application, restore a recent backup of the complete application.

Missing index file(s). Use 'System Configuration' to rebuild.

Unless the data file is corrupted, it is enough to rebuild without address reallocation.

**Missing installation file: <file name>**

If you just started a new application, go back to the 'Select application' option in the FSC main menu and a new installation file will be made if you answer yes. In case of an existing application restore a recent backup of the complete application.

Modbus - RKE protocol specification not equal.

Must be equal. The append action is canceled.

Modules not identical.

Must be identical or undefined in one of the applications. The append action is canceled.

MUX number or alarm group or PID number changed: <type + tag no.>

Inconsistency. Another group or number assigned.

No more non fail-safe, safety-related inputs: <type + tag no.>

The input is erased from the appended FLD.

No more PID functions available: <type + tag no.>

The maximum number of PIDs allowed in the system has been reached. Delete all PIDs which you do not use in the logics to make room for new PIDs ('System Configuration' option of FSC Navigator: 'Signal specs' option).

No more square root functions.

The number of square root functions per FLD is limited to 10. Another limitation is the execution time of one FLD which effectively reduces the number of square roots to 1 or 2.

No more symbols on this FLD or on FLD to be linked.

The FLD is full, which means that no more symbols can be placed. If you are placing a sheet transfer symbol, it is also possible that the program cannot place the related symbol on the FLD which is transferred to. You can save space by decreasing the number of unnecessary line splittings.

No space on referenced FLD.

You want to make an FLD transfer to another FLD, but there is no room in the I/O area on the referenced FLD to place the related symbol. You might create new space by moving the I/O closer to each other.

No FLD with legal block type found.

You are asked for a list of FLD to select from, but the system does not contain an FLD with a legal block type.

No such tag numbers found.

You asked for a list of tag numbers starting with a string you entered. The system does not contain such tag numbers. If you did not enter a string before the question mark, the system does not contain any tag number of this type of variable.

Not enough entries in equation table.

The minimum number of entries in an equation table file is 3.

Not enough free disk space. Program will be aborted.

At startup of the FLD design editor, at least 10 Kbytes of free disk space must be available.

**Number of input bits must be a multiple of 4 for decimal I/O.**

A BI or BO with hexadecimal data representation must have a number of data bits which is a multiple of four. Use the 'System Configuration' option of FSC Navigator to change the number of data bits.

One or more illegal symbols deleted.

After a block type change some symbols are incompatible with the new block type or they have an illegal position. These symbols are deleted.

Output already placed on FLD: <FLD no.>

You are not allowed to make two outputs with the same type and tag number.

Passwords not identical.

Must be identical. The append action is canceled.

PID already placed on FLD: <FLD no.>

You are not allowed to place a PID with the same tag number more than once.

Rack positions in cabinet(s) not compatible.

Must be equal in one of the applications. The append action is canceled.

Reference tag number blocked during on-line modification.

Only in case of a SYM3 or SYM4 symbol library and on-line modification on. The sheet reference you want to make is not allowed because it is deleted but is still in use in the system. Try another position (or sequence number) on the source or the destination FLD.

Reference to mnemonic sheet or non-logic sheet is not allowed.

You cannot transfer a signal to a non-logic sheet or a mnemonic sheet.

Reference to other CPU not allowed.

You cannot make a sheet transfer to an FLD which is executed by another CPU.

Safety values of marker input and marker output cannot be made equal.

The safety values of the input and the output cannot be made identical because of a special location or different I/O modules.

Selection sheet corrupted. Selection sheet will be rebuilt.

Automatic recovery.

SER channel specification not equal.

Must be equal. The append action is canceled.

Symbol is too large to move.

Symbols which are larger than the screen cannot be moved.

Symbol library corrupted.

The symbol library is corrupted and needs to be reinstalled.

**Symbol overlap.**

You are trying to place a symbol on a location which is already occupied by another symbol. Try a new position.

Syntax error in equation table.

The equation table contains an illegal entry or text.

Text too long.

The text is too long to be placed on the screen. Change the scale and/or the number of strokes or you must change the text.

The I/O exists but is not placed on the correct FLD.

The I/O tag number you entered during making a sheet reference exists, but the I/O is not placed on the source FLD of the sheet transfer.

This FLD has wrong block type.

The FLD number you entered is illegal because it has not the wanted block type. i.e. you want to place a function block but you specified the number of an empty FLD or of a program block sheet.

Too many connection lines.

You made a path of connection lines which contains too many lines.

Too many entries in equation table.

The maximum number of entries in an equation table file is 200. Remove items from the equation table until the number of entries is lower than 200.

Too many multiplexers or alarms. Variable not appended: <type + tag no.>

The maximum number of multiplexers or alarms is reached so the variable is not appended.

Variable allocated.

Variable has been allocated before.

Variable not found in I/O data file:

The variable is found on the current FLD, but it does not exist in the I/O data file. The program will try to regenerate the variable. If this is not successful, the variable will be erased from the FLD.

Wrong block type.

The FLD you want to copy has a different block type than the FLD you are editing. This is not allowed.



Left blank intentionally.



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The Netherlands

Helping You Control Your World



Fail Safe Control

Section 7: Printing

Version 500

Revision 01 (01/98)



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Version 500
Revision 01 (01/98)

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7.1 Introduction

FSC Navigator allows you to create hardcopy of two main items:

- project configuration (data stored in the project database), and
- functional logic diagrams (FLDs).

This section explains how to use the 'Print' option of FSC Navigator and describes the commands available to you.

The 'Print' option of FSC Navigator will only be available if:

- a printer has been properly defined (see Section 3 of the FSC Software Manual or the Help file), and
- a project has been opened.

If no project is open, you must either create a new project or open an existing one (for details refer to Section 3 of the FSC Software Manual or the Help file).

The 'Print' option can be called in three ways:

- **Menu bar:** File / Print
- **Buttons bar:** Print
- **Navigation area:** FSC Safety Compliance / Print

If you select the 'Print' option, a submenu will appear, which allows you to choose between printing the project configuration (see subsection 7.2) or functional logic diagrams (see subsection 7.3).

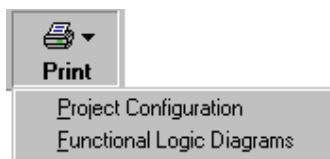


Figure 7-1 Print button with submenu

7.2 Printing the project configuration

If you choose the 'Print\Project Configuration' option of FSC Navigator and you have already defined the FSC configuration (using 'System Configuration'), all print options will be available to you, and the following screen will be displayed:

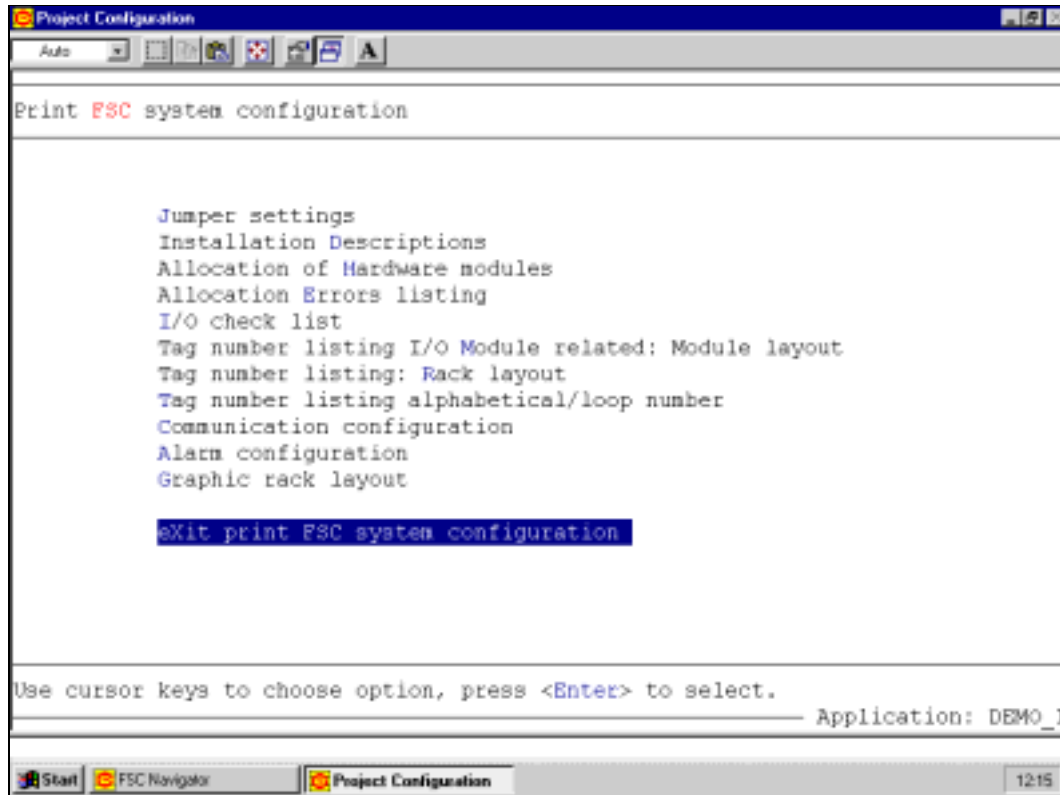


Figure 7-2 Printing the project configuration: main menu (with all options available)

The menu will only show the options that are relevant. If there is nothing to print for a particular option, it will not be shown in the menu.

If you did not define the FSC configuration yet, only the following options will be available:

- Installation descriptions, and
- Tab number listing alphabetical/loop number.

7.2.1 Print options

You can choose one of the options by moving the cursor bar to the desired option and press enter, or by pressing the highlighted shortcut key. Some options will start printing or plotting immediately; others may require you to answer one or more questions first. You cannot use an option if another one is being executed. This section discusses each of the options in the main menu.

The first page of every printout contains information about the application (see Figure 7-3).



Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 11:19 Page: 1

Requirement class : 5
FSC system type specification : Redundant

Tag number fault reset input : RESET
Tag number force enable input : ENABLE
Tag number clock sync. input : CLOCK-SYNC

CPU chip size : 512 KBit
COM chip size : Variable

Prim. SER channel configuration: Yes
Primary SER system number : 1
Primary SER Central Part number: 1
Primary SER com. module : 1
Primary SER channel : B
Primary SER base : 200
Primary SER maximum : 216
Sec. SER channel configuration : Not used

Process safety time : 1 seconds
Interval time between faults : 4320 minutes
FSC-FSC timeout : 1.20 seconds

Power-on mode : Simulation 23126/1/1, cold start
On-line modification wanted : Yes
Application program on : CPU

Central Part 1 Rack number : 1
Central Part 2 Rack number : 2
Slots Central Part bus(es) : 12

Cabinet	Rack 1	Rack 2	Rack 3	Rack 4	Rack 5	Rack 6	Rack 7	Rack 8	Rack 9
1	1	2	3	4	5	-	-	-	-
2	11	12	13	-	-	-	-	-	19

Figure 7-3 Sheet with application information

7.2.2 Jumper settings

This option allows you to print all jumper settings of the hardware modules. If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes (see Figure 7-4). After you have selected the paper size, printing will start.

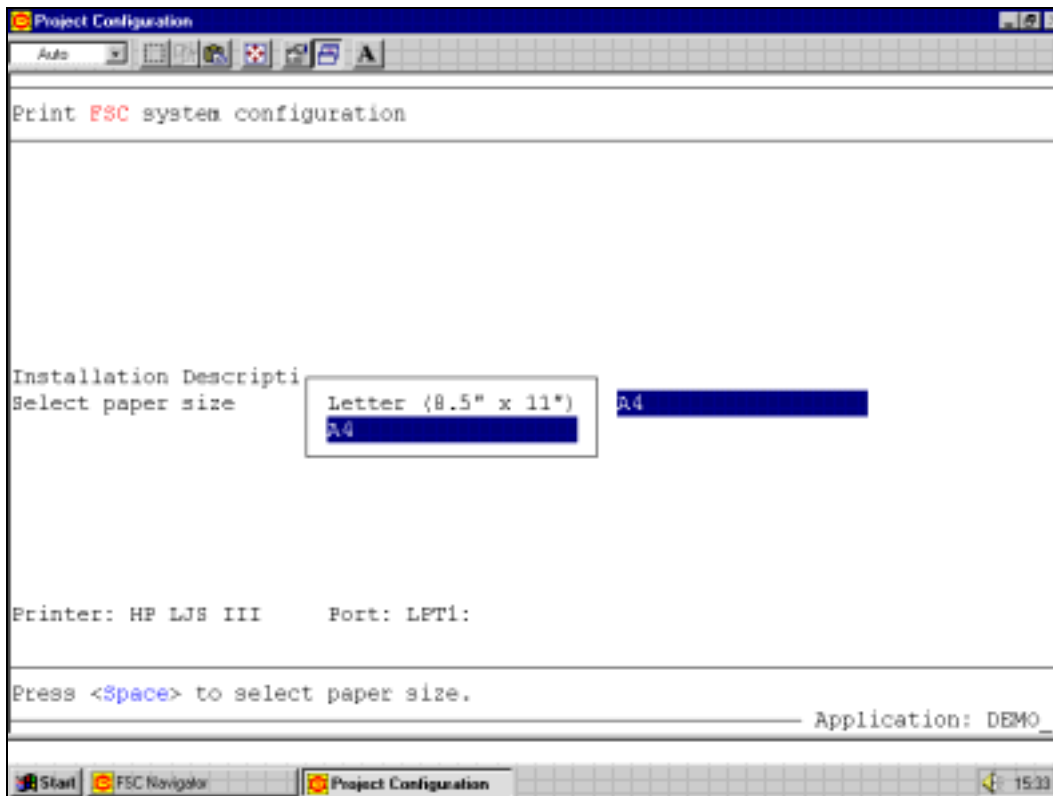


Figure 7-4 Selecting the paper size

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 11:19 Page: 2

Jumper settings (see module layouts in FSC Hardware Manual)

Central Part: 1

Module Rack Position

CPU	1	1	6	J1: 1 J9: 0	J2: 1	J3: 1	J4: 0	J5: 0	J6: 1	J7: 1	J8: 1
WD	1	1	14	J1: 0	J2: 0	J3: 0					
VBD	1	1	16	J1: 1	J2: 0	J3: 0	J4: 0	J5: 1	J6: 0		
VBD	2	1	15	J1: 0	J2: 1	J3: 0	J4: 0	J5: 1	J6: 0		
DBM		1	17	J1: On							
COM	1	1	12	J1: EPROM J9: Other	J2: 0	J3: 0	J4: 1	J5: 1	J6: EPROM	J7: EPROM	J8: Other
COM	2	1	10	J1: EPROM J9: Other	J2: 1	J3: 0	J4: 1	J5: 1	J6: EPROM	J7: EPROM	J8: -
COM	3	1	8	J1: 0	J2: 1						

Compile first; not all jumper settings are known before compilation.

Figure 7-5 Jumper settings (CP 1)



Note:

If the application is not compiled before using this option, not all information is available (e.g. the size of the EPROM). In that case the following message is printed:

Compile first; not all jumper settings are known before compilation.

7.2.3 Installation descriptions

This option allows you to print the installation descriptions that you entered using the 'System Configuration' option of FSC Navigator (Install\Description). If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:30 Page: 2

Installation Descriptions

Number	Description
1	Customer :
2	Principal :
3	Plant :
4	
5	Req/Ordernr :
6	Rev
7	Date
8	Description
9	Chk'd
10	By:
11	Drawing number:
12	Code
13	
14	
15	Honeywell SMS BV
16	
17	Honeywell NL33
18	HSMS Product Marketing
19	Branderijstraat 6
20	5223 AS 's-Hertogenbosch
21	
22	Serial
23	Project
24	Unit
25	Sheet
26	Cnt'd
27	
28	Tel +31 73-6273273
29	Fax +31 73-6219125
30	P.O. Box 116
31	5201 AC
32	's-Hertogenbosch

Figure 7-6 Installation descriptions (SYM1 and SYM2)



7.2.4 Allocation of hardware modules

This option allows you to create listings of the allocation of the hardware modules per Central Part. Each rack that belongs to a Central Part is printed. If this option is selected and more than one Central Part exist, you are asked to specify whether to print all modules or to print the module configuration of each Central Part separately.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:33 Page: 2

Allocation of hardware modules

Cabinet: 1

Position	Rack: 1	SR	Tested	Voting	Rack: 2	SR	Tested	Voting	Rack: 3	SR	Tested	Voting	Rack: 4	SR	Tested	Voting
1									10101/1/1	Yes	Yes	1002D	10101/1/1	No	Yes	1001D
2									10101/1/1	Yes	Yes	1002D	10101/1/1	No	Yes	1001D
3									10101/1/1	Yes	Yes	1002D	10209/1/1	No	No	1001
4									10101/1/1	Yes	Yes	1002D	10209/1/1	No	No	1001
5									10102/1/2	Yes	Yes	1002D	10209/1/1	No	No	1001
6	10002/1/2	Yes	Yes		10002/1/2	Yes	Yes		10102/1/2	Yes	Yes	1002D	10209/1/1	No	No	1001
7									10105/2/1	Yes	Yes	1002D	10101/1/1	No	Yes	1001D
8	10008/2/U	Yes	Yes		10008/2/U	Yes	Yes		10105/2/1	Yes	Yes	1002D	10102/1/1	No	Yes	1001D
9	*****	***	***		*****	***	***		10101/1/1	Yes	Yes	1002D	10209/1/1	No	No	1001
10	10004/F/F	Yes	Yes		10004/F/.	Yes	Yes		10101/1/1	Yes	Yes	1002D	10101/1/1	No	Yes	1001D
11	*****	***	***		*****	***	***		10201/2/1	Yes	Yes	1002D	10205/1/1	No	Yes	1001D
12	10004/H/I	Yes	Yes		10004/H/I	Yes	Yes		10201/2/1	Yes	Yes	1002D	10215/1/1	No	Yes	1001D
13	*****	***	***		*****	***	***		10209/1/1	No	No	2002	10216/1/1	No	Yes	1001D
14	10005/1/1	Yes	Yes		10005/1/1	Yes	Yes		10209/1/1	No	No	2002	10209/1/1	No	No	1001
15	10001/R/1	Yes	Yes		10001/R/1	Yes	Yes		10209/1/1	No	No	2002	10209/1/1	No	No	1001
16	10001/R/1	Yes	Yes		10001/R/1	Yes	Yes		10209/1/1	No	No	2002	10205/1/1	No	Yes	1001D
17	10006/2/1	Yes	Yes		10006/2/1	Yes	Yes		10209/1/1	No	No	2002	10209/1/1	No	No	1001
18	*****	***	***		*****	***	***		10209/1/1	No	No	2002	10302/1/1	No	Yes	
19	10303/1/1	Yes	Yes		10303/1/1	Yes	Yes									
20	10300/1/1	Yes	Yes		10300/1/1	Yes	Yes		10100/2/1	Yes	Yes					
21	*****	***	***		*****	***	***		10100/2/1	Yes	Yes		10100/2/1	No	Yes	

***** = Reserved for the second position of the module on the previous position.

Figure 7-7 Allocation of hardware modules (all Central Parts)

7.2.5 Allocation errors listing

This option allows you to make a printout of all I/O tag numbers that were not allocated. This means that:

- for variables with location 'FSC', the system number is invalid.
- for variables with location 'COM', the COM channel has not been specified (CP, COM module channel).
- for variables with location 'ANN', the alarm group or the alarm type has not been specified.
- for all other variables, the combination of rack, position, channel has not been fully specified.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:35 Page: 2

Allocation Errors listing

Type	Tag number	Service	Qualification	Rack	Position	Channel	Sheet number	Module type
0	SEC.SWITCH-OFF			0	0	0	No sheet	-
XI	KE-576-D1	DATA SELECT		4	0	0	No sheet	-

Figure 7-8 Allocation errors listing



7.2.6 I/O check list

This option allows you to print a checklist which contains all channels of the I/O modules that can be used during testing of the system wiring.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:36 Page: 2

I/O check list Test date:

By:

Cabinet: 1 Rack: 3 Position: 1 Module type: Digital input 24 Vdc FS (10101/1/1) Safety-related: Yes Tested: Yes Voting: 1002D

Channel	Tag number	Tested	Remark	Retested
1	53HS-101			
2	Test_pushButton			
3	Ack_PushButton			
4	RESET-ALARM			
5	EARTH-LEAKAGE			
6	PSU-2			
7	PSU-1			
8	LAMPTTEST			
9	FIRSTUP-RESET			
10	SHUTDOWN			
11	SENSOR-CP2			
12	SENSOR-CP1			
13	SENSOR-B1			
14	SENSOR-B3			
15	ENABLE			
16	RESET			

Cabinet: 1 Rack: 3 Position: 3 Module type: Digital input 24 Vdc FS (10101/1/1) Safty-related: Yes Tested: Yes Voting: 1002D

Channel	Tag number	Tested	Remark	Retested
1	Transm.-Value			
2	Transm.-Value			
3	Max-Discrepancy			
4	Max-Discrepancy			
5	SENSOR-B2			
6	SENSOR-A2			
7	SENSOR-A1			
8	SENSOR2			
9	SENSOR3			
10	SENSOR1			
11	SENSOR_2			
12	SENSOR-1			
13	RESET-PUSHBUTTON			
14	TEST-PUSHBUTTON			
15	ACK-PUSHBUTTON			
16	ALARM-2			

Cabinet: 1 Rack: 3 Position: 5 Module type: Analog input FS (10102/1/2) Safety-relaead: Yes Tested: Yes Voting: 1002D

Channel	Tag number	Tested	Remark	Retested
1	53PT-920			
2	53TT-900			
3	PRESSURE			
4	MAINLINE			

Figure 7-9 I/O check list



7.2.7 Tag number listings I/O module related: module layout

This option allows you to create a listing of all tag numbers which are allocated to I/O modules. An allocation error list is also printed after all other modules.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:37 Page: 2

Tag number listing I/O Module related: Module layout

Cabinet: 1 Rack: 3 Position: 1 Module type: Digital input 24 Vdc FS (10101/1/1) Safety-related: Yes Tested: Yes Voting: 1002D

Channel	Tag number
1	53HS-101
2	Test_pushButton
3	Ack_PushButton
4	RESET-ALARM
5	EARTH-LEAKAGE
6	PSU-2
7	PSU-1
8	LAMPTTEST
9	FIRSTUP-RESET
10	SHUTDOWN
11	SENSOR-CP2
12	SENSOR-CP1
13	SENSOR-B1
14	SENSOR-B3
15	ENABLE
16	RESET

Cabinet: 1 Rack: 3 Position: 2 Module type: Digital input 24 Vdc FS (10101/1/1) Safety-related: Yes Tested: Yes Voting: 1002D

Channel	Tag number
1	53HS-101
2	Test_pushButton
3	Ack_PushButton
4	RESET-ALARM
5	EARTH-LEAKAGE
6	PSU-2
7	PSU-1
8	LAMPTTEST
9	FIRSTUP-RESET
10	SHUTDOWN
11	SENSOR-CP2
12	SENSOR-CP1
13	SENSOR-B1
14	SENSOR-B3
15	ENABLE
16	RESET

Figure 7-10 Tag number listing I/O module related: module layout

7.2.8 Tag number listings: rack layout

This option allows you to print a tag number list of a number of racks. All modules will be printed which are placed in one rack. If you select this option, and more than one rack is specified, you will be asked to enter the first and the last rack number you want to print, starting with the first rack number. The default rack number displayed is the lowest existing rack number. You can only enter existing rack numbers.

Enter first rack number to print: <lowest rack no.>

After entering the first rack number, you will be asked to enter the last rack number to print. The default is the highest existing rack number.

Enter last rack number to print: <highest rack no.>



The rack number is a positive integer in the range 1 to 99. If the rack does not exist, a message is displayed. This option can be used to create a listing of all tag numbers of I/O types in the system with their operating parameters such as power-on values, rack, position, channel (I/O tag numbers), etc. This option will list only existing tag numbers. This means that if, for example, no analog inputs have been defined, no analog inputs will be printed. If you select this option, printing will start immediately without any further questions.

On digital input modules, variables of type I, BI and XI can be allocated.

On digital output modules, variables of type BI (Location = 'MUX'), O, BO, XO can be allocated.

On analog input modules, variables of type AI can be allocated.

On analog output modules, variables of type AO can be allocated.

For the other modules, the name of the module is printed, followed by the Central Part number and the module number. The Central Part number, the VBD number and the module number are printed for HBDs and VBDs.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:38 Page: 2

Tag number listing: Rack layout

Cabinet: 1 Rack: 1

Channel	1	2	3	4	5	6 10002/1/2	7
1						CPU 1. 1	
2						*****	
3						*****	
4						*****	
5						*****	
6						*****	
7						*****	
8						*****	
9						*****	
10						*****	
11						*****	
12						*****	
13						*****	
14						*****	
15						*****	
16						*****	

Channel	8 10008/2/U	9 10008/2/U	10 10004/F/F	11 10004/F/F	12 10004/H/I	13 10004/H/I	14 10005/1/1
1	COM 1. 3A	COM 1. 3B	COM 1. 2A	COM 1. 2B	COM 1. 1A	COM 1. 1B	WD 1. 1
2	*****	*****	*****	*****	*****	*****	*****
3	UCN	Undefined	Modbus RTU	Modbus RTU	FSC-FSC	Development sys.	*****
4	10008/U/1	*****	10004/F/1	10004/F/1	10004/H/1	10004/I/1	*****
5	*****	*****	(RS232 multidrop	(RS232 multidrop	(RS422 readback)	(RS485) No encod	RESET
6	*****	*****	19K2, n, 8, 1, n	19K2, n, 8, 1, n	2M, n, 8, 1, n	19K2, n, 8, 2, n	*****
7	*****	*****	*****	*****	*****	*****	*****
8	*****	*****	*****	*****	*****	*****	*****
9	*****	*****	*****	*****	*****	*****	*****
10	*****	*****	*****	*****	*****	*****	*****
11	*****	*****	*****	*****	*****	*****	*****
12	*****	*****	*****	*****	*****	*****	*****
13	*****	*****	*****	*****	*****	*****	*****
14	*****	*****	*****	*****	*****	*****	*****
15	*****	*****	*****	*****	*****	*****	*****
16	*****	*****	*****	*****	*****	*****	*****

Channel	15 10001/R/1	16 10001/R/1	17 10006/2/1	18 10006/2/1	19 10303/1/1	20 10300/1/1	21 10300/1/1
1	VBD 1. 2	VBD 1. 1	DBM 1. 1	DBM 1. 1	PSD 1. 1	PSU 1. 1	PSU 1. 1
2	*****	*****	*****	*****	*****	*****	*****
3	*****	*****	*****	*****	*****	*****	*****
4	*****	*****	*****	*****	*****	*****	*****
5	*****	*****	*****	*****	*****	*****	*****
6	*****	*****	*****	*****	*****	*****	*****
7	*****	*****	*****	*****	*****	*****	*****
8	*****	*****	*****	*****	*****	*****	*****
9	*****	*****	*****	*****	*****	*****	*****
10	*****	*****	*****	*****	*****	*****	*****
11	*****	*****	*****	*****	*****	*****	*****
12	*****	*****	*****	*****	*****	*****	*****
13	*****	*****	*****	*****	*****	*****	*****
14	*****	*****	*****	*****	*****	*****	*****
15	*****	*****	*****	*****	*****	*****	*****
16	*****	*****	*****	*****	*****	*****	*****

Figure 7-11 Tag number listing I/O module related: rack layout

***** = This channel is no I/O channel (only printed on positions which contain a module).

7.2.9 Tag number listings alphabetical/loop number

This option allows you to create a listing of all tag numbers of all variable types in the system with their operating parameters such as power-on values, rack, position, channel (I/O tag numbers), etc. This option will list only existing tag numbers. This means that if, for example, no analog inputs have been defined, no analog inputs will be printed (and no page for analog inputs will be reserved). If you choose this option, you must specify which tag numbers will be printed:

- All To print all variables in the database.
- I/O To print all digital input (DI), digital output (DO), multiple input (BI), multiple output (BO), analog input (AI) and analog output (AO) variables.
- Specific To print a specific type of variables (see below).

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

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Input signal specification

Type	Tag number	Service	Qualification	Location	Unit	Subunit	Sheet	Safety	Force	En.	Write	En.	SER	En.	SER	seq.	no.
I	53HS-101	LAMPTTEST	TEST	MCP			102	Yes	Yes	No	No	No	No	No	No	-	-
I	53_HS_101	LAMPTTEST	"TEST"	MCP			104	Yes	Yes	No	No	No	No	No	No	-	-
I	91XA-651A	Door switch	Close	AH	5000	91UZ-650	0	Yes	No	No	No	No	No	No	No	-	-
I	ACK-PUSHBUTTON			PNL			107	Yes	Yes	No	No	No	No	No	No	-	-
I	ACKNOWLEDGE			DCS			106	Yes	Yes	No	No	No	No	No	No	-	-
I	AF_Audible			ANN			105	No	No	No	No	No	No	No	No	-	-
I	AF_Common_Alarm			ANN			105	No	No	No	No	No	No	No	No	-	-
I	ALARM-1		ALARM STATUS	DCS			107	Yes	Yes	No	No	No	No	No	No	-	-
I	ALARM-2		ALARM STATUS	DCS			107	Yes	Yes	No	No	No	No	No	No	-	-
I	AUDIBLE			ANN			107	No	No	No	No	No	No	No	No	-	-
I	Ack_PushButton			PNL			105	Yes	Yes	No	No	No	No	No	No	-	-
I	CENTR.PART-FAULT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	CLOCK-SYNC	FSC-CLOCK-SYNCHRON.	CLOCK-SYNC	SYS			0	No	No	No	No	No	No	No	No	-	-
I	COMMON			ANN			107	No	No	No	No	No	No	No	No	-	-
I	DEVICE-COM.FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	EARTH-LEAKAGE	EARTH LEAKAGE PSU'S	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-	-
I	ENABLE	FORCE-ENABLE	ENABLE	SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	EXT. COMMUNIC. FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	FIRSTUP-ALARM-1	SUBLOCATION-FSC	FIRSTUP FLAG	DCS			107	Yes	Yes	No	No	No	No	No	No	-	-
I	FIRSTUP-ALARM-2	SUBLOCATION-FSC	FIRSTUP FLAG	DCS			107	Yes	Yes	No	No	No	No	No	No	-	-
I	FIRSTUP-RESET			DCS			106	Yes	Yes	No	No	No	No	No	No	-	-
I	FLASHER-0.5Hz	System marker		SYS			107	No	No	No	No	No	No	No	No	-	-
I	FLASHER-1Hz	System marker		SYS			107	No	No	No	No	No	No	No	No	-	-
I	FLASHER-2Hz	System marker		SYS			105	No	No	No	No	No	No	No	No	-	-
I	FSC-SYSTEM-FAULT	System marker		SYS			123	Yes	No	No	No	No	No	No	No	-	-
I	INPUT-FAILURE	System marker		SYS			122	Yes	No	No	No	No	No	No	No	-	-
I	INT. COMMUNIC. FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	IO-COMPARE	System marker		SYS			120	Yes	No	No	No	No	No	No	No	-	-
I	IO-FORCED	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	LAMPTTEST	LAMPTTEST	TEST	PNL			123	Yes	Yes	No	No	No	No	No	No	-	-
I	OUTPUT-FAILURE	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	PSU-1	PSU-1 24VDC	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-	-
I	PSU-2	PSU-2 24VDC	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-	-
I	RED. INPUT-FAULT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-	-
I	RESET	FSC-FAULT-RESET	RESET	SYS			121	Yes	No	No	No	No	No	No	No	-	-
I	RESET-ALARM	RESET ALARM	RESET	CAB			123	Yes	Yes	No	No	No	No	No	No	-	-
I	RESET-PUSHBUTTON			PNL			107	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-1						109	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-A1						111	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-A2						111	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-B1						112	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-B2						112	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-B3						112	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-CP1						113	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR-CP2						113	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR1						110	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR2						110	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR3						110	Yes	Yes	No	No	No	No	No	No	-	-
I	SENSOR_2						109	Yes	Yes	No	No	No	No	No	No	-	-

Figure 7-12 Tag number listing alphabetical/loop number (all) (signal specification)



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Input hardware information

Type	Tag number	Rack	Pos	Chan	Sensor	Sensconf	Max. on.	Max. discr.	Rack2	Pos2	Chan2	Rack3	Pos3	Chan3	Bit no.	Modbus	PLC address
I	53HS-101	3	1	1	FS										8	-	-
I	53_HS_101	3	9	3	FS										42	-	-
I	91XA-651A	3	9	13	Red	R 202	480 M	10 S		3	9	14			36	-	2002
I	ACK-PUSHBUTTON	3	3	15	FS										22	-	-
I	ACKNOWLEDGE	3	9	11	FS										34	-	-
I	AF_Audible	-	-	-	-										5471	-	-
I	AF_Common_Alarm	-	-	-	-										5468	-	-
I	ALARM-1	3	9	9	FS										32	-	1020
I	ALARM-2	3	3	16	FS										23	-	-
I	AUDIBLE	-	-	-	-										5551	-	-
I	Ack_PushButton	3	1	3	Red	R 202	480 M	10 S		3	9	12			10	-	-
I	CENTR.PART-FAULT	-	-	-	-										0	-	-
I	CLOCK-SYNC	3	9	15	FS										38	-	-
I	COMMON	-	-	-	-										5548	-	-
I	DEVICE-COM.FLT	-	-	-	-										9	-	-
I	EARTH-LEAKAGE	3	1	5	FS										12	-	-
I	ENABLE	3	1	15	FS										6	-	-
I	EXT.COMMUNIC.FLT	-	-	-	-										13	-	3
I	FIRSTUP-ALARM-1	3	9	10	FS										33	-	-
I	FIRSTUP-ALARM-2	3	9	8	FS										47	-	-
I	FIRSTUP-RESET	3	1	9	FS										0	-	-
I	FLASHER-0.5Hz	-	-	-	-										4	-	-
I	FLASHER-1Hz	-	-	-	-										3	-	-
I	FLASHER-2Hz	-	-	-	-										2	-	-
I	FSC-SYSTEM-FAULT	-	-	-	-										15	-	-
I	INPUT-FAILURE	-	-	-	-										6	-	-
I	INT.COMMUNIC.FLT	-	-	-	-										11	-	-
I	IO-COMPARE	-	-	-	-										14	-	-
I	IO-FORCED	-	-	-	-										8	-	-
I	LAMPTTEST	3	1	8	FS										15	-	-
I	OUTPUT-FAILURE	-	-	-	-										7	-	-
I	PSU-1	3	1	7	FS										14	-	-
I	PSU-2	3	1	6	FS										13	-	-
I	RED.INPUT-FAULT	-	-	-	-										10	-	-
I	RESET	3	1	16	FS										7	-	40
I	RESET-ALARM	3	1	4	FS										11	-	-
I	RESET-PUSHBUTTON	3	3	13	FS										20	-	-
I	SENSOR-1	3	3	12	FS										19	-	-
I	SENSOR-A1	3	3	7	FS										30	-	-
I	SENSOR-A2	3	3	6	FS										29	-	-
I	SENSOR-B1	3	1	13	FS										4	-	-
I	SENSOR-B2	3	3	5	FS										28	-	-
I	SENSOR-B3	3	1	14	FS										5	-	-
I	SENSOR-CP1	3	1	12	FS										3	-	-
I	SENSOR-CP2	3	1	11	FS										2	-	-
I	SENSOR1	3	3	10	FS										17	-	-
I	SENSOR2	3	3	8	FS										31	-	-
I	SENSOR3	3	3	9	FS										16	-	-
I	SENSOR_2	3	3	11	FS										18	-	-

Figure 7-13 Tag number listing alphabetical/loop number (all) (hardware information)

The addresses may have the following values:

<value>: Address value

'-' (dash): An address is possible but none has been assigned.

' ' (space): No address possible.

'Specific' option

The 'Specific' option can be used to print a single variable type with its operating parameters such as power-on values, rack, position, channel (I/O tag numbers), etc. If no tag numbers of this type exist, no list will be created. You will be prompted to select the type and to filter the variables that belong to that type.

The first question will be:

Variable type:

The expected answer is a string of one or two characters which represents the variable type (I, AI, BI, XI, O, AO, BO, XO, M, C, T, R, P, A) followed by <Enter>, e.g.:

I <Enter> (print only inputs)



The following abbreviations are valid (see also Section 4, "System Configuration"):

I	digital inputs	O	outputs	M	markers
AI	analog inputs	AO	analog outputs	C	counters
BI	multiple inputs	BO	multiple outputs	T	timers
XI	input multiplexers	XO	output multiplexers	R	registers
				P	PIDs
				A	alarm groups

If you have entered M, C, T or R as the variable type, the next question will be:

Tag number:

Here you can enter the first part of the tag number. If you enter an empty tag number and then press <Enter>, all tag numbers of the selected type will be printed.

If you have entered I, AI, BI, XI, O, AO, BO, XO, P or A as the variable types, the next question will be:

Search prefix, description or loop number ? Prefix

Here you can choose to print the tag number listings sorted by prefix, description or loop number, which are a part of the tag number. A tag number consists of three parts:

Tag number = <Prefix> '_' <Description> '_' <Loop number>

The underscore ('_') may also be replaced by a dash ('-').

Use <Space> to toggle between the available options: 'prefix', 'description' and 'loop number'. You can also press <?>, which will open a small window that contains the available options. After you have made your choice, press <Enter> to confirm.

Enter string to search for

This option allows you to enter the first part of the string to search for. Enter the string and press <Enter>.

After you have answered these questions, printing will start. While printing, messages will be displayed on screen to show the program status.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.



7.2.10 Communication configuration

This option allows you to print the communication configuration of an application. It lists the configuration of all used channels (channels which are **not** undefined), and generates a list of all variables allocated to the communication channel.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

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Channel configuration of central part 1, module 2, channel A.

Protocol : Modbus RTU
Redundant : Yes
Network : Multidrop
Link status : Primary
Connected system : None
Timeout : 25 (100 ms)
Interface : 10004/F/1 (RS232 multidrop)
Baud rate : 19K2
Handshake : None
Number of stop bits : 1
Number of bits/char : 8
Parity : None

Communication address : 4
Link specification
Marker bytes : bit start address : 2256
Marker bytes : to FSC 1 from device: 200
Marker bytes : from FSC 1 to device: 200

Register bytes: byte start address : 2336
Register bytes: to FSC 1 from device: 200
Register bytes: from FSC 1 to device: 200

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Channel configuration of central part 1, module 2, channel A.

Link specification

Rel. bit no.	Type	Tag number	Service	Qualification	Unit	Subunit	Sheet	Safety	SER	En.	SER	seq. no.	Power-on
1	I	XK7F30A	Gasleak test	Start	F3300	Burner 1	10	Yes	Yes			1105	Off

Figure 7-14 Communication configuration listing



7.2.11 Alarm configuration

This option allows you to print the alarm configuration of an application. It lists all alarm groups and lists all allocated variables of each alarm group.

If more than one paper size is supported, FSC Navigator prompts to select the desired paper size. Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes. After you have selected the paper size, printing will start.

```
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Alarm configuration

Alarm group       : A AF_AlarmPoint2
Alarm group number: 2
Alarm sequence    : AF
Alarm group size   : 16
Start address     : 5456

Rel. bit no. Type Tag number      Service      Qualification  Unit    Subunit  Sheet Safety SER En. SER seq. no. Alarm type
12 I AF_Common_Alarm              105 No      No              - Common alarm
15 I AF_Audible                    105 No      No              - Audible
1 0 AF_Acknowledge2                101 No      No              - Acknowledge
4 0 AF_Flasher_1                   105 No      No              - Flasher 1

Type Tag number      Rack Pos Chan
0 AF_AlarmPoint2      3 11 3
```

Figure 7-15 Alarm configuration listing

7.2.12 Graphic rack layout

This option allows you to create a tag number plotting of a single rack. All modules will be plotted which are placed in one rack. If you select this option, you will be asked to enter the first rack number you want to plot. The default rack number displayed is the lowest existing rack number. You can only enter existing rack numbers.

Enter first rack number to print : 1

After entering the start rack number, you will be asked to enter the last rack number to plot. The default is the highest existing rack number

Enter last rack number to print : 1

The rack number is a positive integer in the range 1 to 99. If the rack does not exist, a message is displayed. After entering the range to plot, you will be asked for the size of the plotting.

Select paper size: A4

Use <Space> to toggle between available paper sizes. You can also press <?>, which will open a small window that contains the available paper sizes.

For the HP7475 plotter, the paper size should be A3 (paper fits) or 297 x 500 mm (cabinet sized).

Fit cabinet on selected paper size (Y/N) : No

Use <Space> to toggle between 'Yes' and 'No'. You can also press <?>, which will open a small window that contains the available options.



If you select the HP 7550 plotter, HP LaserJet series II or III, the Cannon or Brother laser printer, you will be asked whether the paper will be changed automatically or manually.

Print to (printer/file): Printer

Pause for paper change between sheets? <Y/N>: No

Use <Space> to toggle between the available options. You can also press <?>, which will open a small window that contains the available options.

If you select manual paper change or the HP7475 plotter, the program will prompt after every completed sheet:

Change paper and press <Enter> to continue.

[illegible]

Figure 7-16 Rack layout

7.2.13 Interrupting the printing process

After any of the printing options on the main menu has been selected, FSC Navigator will gather and format the data for printing. This is prompted with the message:

Generate print

or

Generate graphic print

When the program is ready to send the data to the plotter or the printer, you can interrupt the program. If interrupting is possible, a message on the screen will be displayed.

Press any key to interrupt.

You can press any key to interrupt the program. When the printing (or plotting) has stopped, you are prompted:

Continue (Y/N): Y

If you decide to continue, press <Enter>. You are then asked to press <Enter> once more, and printing (or plotting) will continue.

If you really want to stop the printer or plotter, use <Space> to toggle the Y/N prompt to N and press <Enter> to confirm. FSC Navigator will then stop sending data and return to the main print menu (see Figure 7-2).

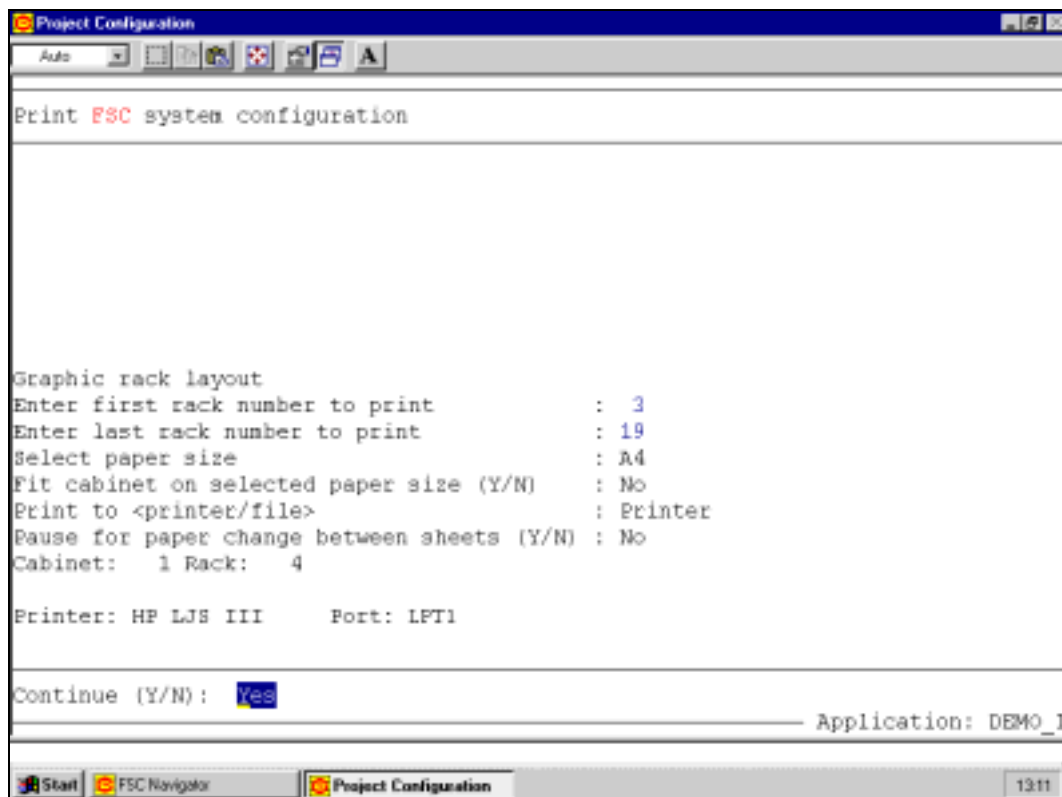


Figure 7-17 Interrupting the printing process



7.2.14 Error messages

Aborted manually.

When a print option is aborted, an abortion message will be printed.

FSC_ENG.MSG missing.

This error will occur if the file FSC_ENG.MSG is not in the FSC Navigator application folder (C:\FSC by default).

Not a legal FSC file.

This error will occur if the program has not been properly installed.

Version of message file does not match.

This error will occur if you try to call the 'Print' option of FSC Navigator using a message file that belongs to another version of the application.



7.3 Printing functional logic diagrams

The 'Print\Functional Logic Diagrams' option of FSC Navigator allows you to create hardcopy of the functional logic diagrams (FLDs) that were created using the 'Design FLDs' option of FSC Navigator.

Before you can start plotting/printing, you must have installed the output device correctly (see Section 3 of the FSC Software Manual or the Help file). The program supports the following output devices:

Plotters: HP 7475 HP 7550
Laser printers: HP LaserJet series II (HP LJS II)
HP LaserJet series III (HP LJS III)
Brother HL-8e (BR HL-8e)
Canon LBP 811 (CA LBP-811)

and all other plotters and laser printers that are compatible to the above devices.

Notes:

1. You cannot print functional logic diagrams on matrix printers.
 2. If you intend to use the HP LaserJet series III (HPLJSIII) with full graphics, make sure that the printer has at least 2 Mbytes of memory (and preferably more) to avoid memory overflow problems when printing complex functional logic diagrams.
Also, enable page protection for the paper size that you expect to use most (using the printer's front panel menu). If page protection is set, additional memory is reserved for page image processing, which allows the printer to create the entire page image in memory before outputting to paper.
If no extra memory is available, select HPLJSII (HP LaserJet series II) as the printer. The sheet will then be printed as a bitmap.
 3. If you choose HPLJSIII (HP LaserJet series III), the sheet will be printed in HPGL instructions.
 4. If you choose HPLJSII (HP LaserJet series II), the sheet will be printed as a bitmap.
-



7.3.1 Options

If you choose the 'Print\Functional Logic Diagrams' option of FSC Navigator, the following screen will appear:

```
Application: DEMO_1

Hardcopy of
functional logic diagrams

Print to graphical device/file           : Graphical device
Select FLD(s) <?> for list              : 101
Blank printing paper <Y/N>               : Y
Paper size                               : A4
Pause for paper change between sheets <Y/N> : N
X translation:      0   Y translation:    0

When graphical device ready and connected, press <Enter>.

Graphical device: HP LJS III      Port: Lpt1:
```

Figure 7-18 Printing functional logic diagrams: options menu

This screen allows you to define the printer/plotter settings for the functional logic diagram(s) to be printed. The selections that you can make for the various options depend on the configuration of the output device.

Print to graphical device/file

If the output device you have selected is HP7475, HP7550 or HP LaserJet III, you can output your drawings to paper or write them to a plot file. Otherwise the option 'File' is not available. You can use <Space> to toggle between the available options. You can also press <?>, which will open a small window that contains the available options. There are two options:

Graphical device : Outputs the drawings to paper.
File : Writes the drawings to an HPGL file. You can use this file to print to a plotter/laser printer that can handle HPGL instructions (or you must use an HPGL interface or HPGL emulation program).

The default selection is 'Graphical device'.

Select FLD(s)

You can enter one FLD number to plot or press <?> to open a selection window (see Figure 7-19), which allows you to enter five ranges of sheets you want to print. Only existing FLDs will be printed.

Application: DEMO_1

Hardcopy of functional logic diagrams

Print to graphical device/file
Select FLD(s) <?> for list
Blank printing paper <Y/N>
Paper size
Pause for paper change between sheets <Y/N>
X translation: 0 Y translation: 0

Press <Esc> to leave FLD selection list.

Graphical device: HP LJS III Port: Lpt1:

: Graphical device
: 0
: from..to
: 100..105
: 120..145
: 900..907
: 907..0
: 0..0

Figure 7-19 Selecting an FLD range to print

Blank printing paper <Y/N>

Here you specify whether you use blank paper or pre-printed paper. If you use pre-printed paper, no title block and sheet borders will be drawn. This option is only available if you have installed an HP7475 or HP7550 plotter. For all other graphical devices no pre-printed paper can be used.

You can use <Space> to toggle between the available options. There are two options:

Y: use blank paper

N: use pre-printed paper

Paper size

Here you specify the size of the paper that you use. You can use <Space> to toggle between the available options. You can also press <?>, which will open a small window that contains the available paper sizes. The following paper sizes may be available, depending on the output device selected: A3, A4 and Letter (8.5"x11").

Pause for paper change between sheets ? <Y/N>

Here you can select if your graphical device should stop after completion of an FLD. This is helpful when your graphical device does not have automatic paper feed. You can use <Space> to toggle between 'Yes' and 'No'. You can also press <?>, which will open a small window that contains the available options.

If you answer 'Yes', printing will start. After FSC Navigator has finished printing a sheet, the following message will be displayed:

Pausing for paper change. Press <Enter> to continue.



Place a new paper sheet in the graphical device and press <Enter>. The graphical device will now continue printing/plotting the next FLD.

X Translation: 0 Y Translation: 0

You are asked to enter two coordinates (X and Y) which are used to translate the drawings. This can be necessary for some graphical devices in order to move the drawing into the printable area. The X and the Y coordinate may be positive or negative. The value must be entered as points (28 pts = approx. 1 cm; 72 pts = approx. 1 inch).

After you have answered the last question, the following message will be displayed:

When graphical device connected and ready, press <Enter>.

Make sure your graphical device is ready (turned on and on-line), and make sure that your paper is positioned correctly. After you press <Enter>, the graphical device will start. For the laser printers (HPLJS II, BR HL-8e and CA LBP8-II), the program will first load a number of special fonts.

When the graphical device starts printing, the following message will be displayed:

Press any key to interrupt.

If you press <Esc>, printing will stop immediately. If you press any other key on your keyboard, printing will be interrupted and the following question will be asked:

Continue? Press <Space> to alter.

If you select <N>, printing will stop and the program returns to FSC Navigator's main screen.

If you select <Y>, the graphical device will resume printing.



Left blank intentionally.



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Helping You Control Your World



Fail Safe Control

Section 8: Translating an Application

Version 500

Revision 01 (01/98)



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8.1 Introduction

After the FSC system has been configured and the functional logic diagrams (FLDs) have been designed, the application must be translated into code which can be used by the FSC processor(s). This translation is done by the compiler. The compiler also checks if errors were made during the configuration and design stage.

Subsection 8.2 ("Getting started") specifies the files that are required for translation, and explains how the compiler is started.

Subsection 8.3 ("Compiler tasks") explains the tasks of the compiler and provides a number of conditions that must be satisfied for correct translation. It also describes how the compiler is operated.

Subsection 8.4 ("Compiler output") describes the compiler output, especially the messages that are generated during translation, with possible solutions to any errors.

8.2 Getting started

The 'Translate Application' option can be called in three ways:

- **Menu bar:** Project / Translate Application
- **Buttons bar:** Translate
- **Navigation area:** FSC Project Configuration / Translate Application

An application can only be translated if the project folder contains the following files:

- Installation file : <Application name>.INS
- Data file: <Application name>.DAT
- Index file: <Application name>.IXT
- The functional logic diagrams (FLDs) to be translated: <Application name>.<FLD number>
- The SER format file (optional): <Application name>.FMT
- The report files that were generated by the 'SER Format' option of FSC Navigator (optional): <Application name>.R<report number>

Note:

If the application is part of an FSC network, the installation, data and index files of all connected applications must be present in their project folders.

The following files must be present in the FSC Navigator application folder (C:\FSC by default):

- The library which is used by the 'Design FLDs' option of FSC Navigator: <Library name>.SYM
- The device driver file: DEVICE.DRV
- RKE3964R.BIN (optional)
- MB_HB.BIN (optional)
- FSMM.BIN (optional)
- PBUS.BIN (optional)
- MB_RTU.BIN.



8.3 Compiler tasks

The 'Translate Application' option of FSC Navigator translates all the functional logic diagrams (FLDs) that were designed using the 'Design FLDs' option of FSC Navigator and that are present in the project folder. Before a correct translation can be made, the FLDs must be syntactically correct (for example, all required connections must have been made). The application compiler checks whether the FLD can be translated, and will generate error messages if any errors are detected. It will also give a warning when the syntax is correct but probably does not match the intention of the programming engineer, e.g. if a function block input has not been connected.

The compiler tries to find as many errors as possible, which means that the translation of an FLD is continued after an error has been detected. This is the reason that some errors are reported more than once (e.g. if more than one signal has been connected at one node). The compiler also continues to translate all subsequent FLDs. All messages are written to a log file (see subsection 8.4) and displayed on screen.

The compiler generates one or more output files from all FLDs that have been found to be syntactically correct. These files contain the FSC program code for the various processors in the Central Part(s). These files are used as input for the 'Program EPROMs' option and the 'Load Application' option of FSC Navigator (the latter in the case of RAM mode).

During translation, compiler registers or markers may be introduced. These special registers and markers are needed to store intermediate results or to store feedback results which are required in the next program cycle. The compiler markers and registers can be recognized by their tag number, which always starts with the character 'C' (Compiler). The different types of compiler markers and registers can be distinguished visually by their service text. The marker and register types are the following:

Compiler marker : storage of an intermediate boolean result.
Feedback marker : storage of a boolean feedback result.
Compiler register : storage of an intermediate binary result.
Feedback register : storage of a binary feedback result.

If an FLD is changed and translated again, the compiler markers and compiler registers that are no longer required are erased from the variable database.

8.3.1 Compiler conditions

A number of conditions must be met for correct translation. The FLDs are translated by FLD number (in descending order). This means that the FLD with the highest number is translated first and the FLD with the lowest number last. If an FLD was designed using the 'Design FLDs' option of FSC Navigator but is not present in the project folder, it will be ignored. If one or more FLDs are missing, an error is generated with the FLD number of the first detected missing FLD.

8.3.1.1 Function blocks

Function blocks must have higher FLD numbers than program blocks and step blocks in order to ensure that they are translated first. The first function block must have an FLD number which is at least 2 higher than the program block (or step block) with the highest FLD number. Also, the nesting of the function blocks must be such that the function block which is nested deepest is translated first (i.e. has a higher FLD number).



8.3.2 Compiler operation

The compiler displays on-screen status information during the translation. Warnings and messages are displayed in green (low video on monochrome monitors) and all errors are displayed in red (highlighted on monochrome monitors).

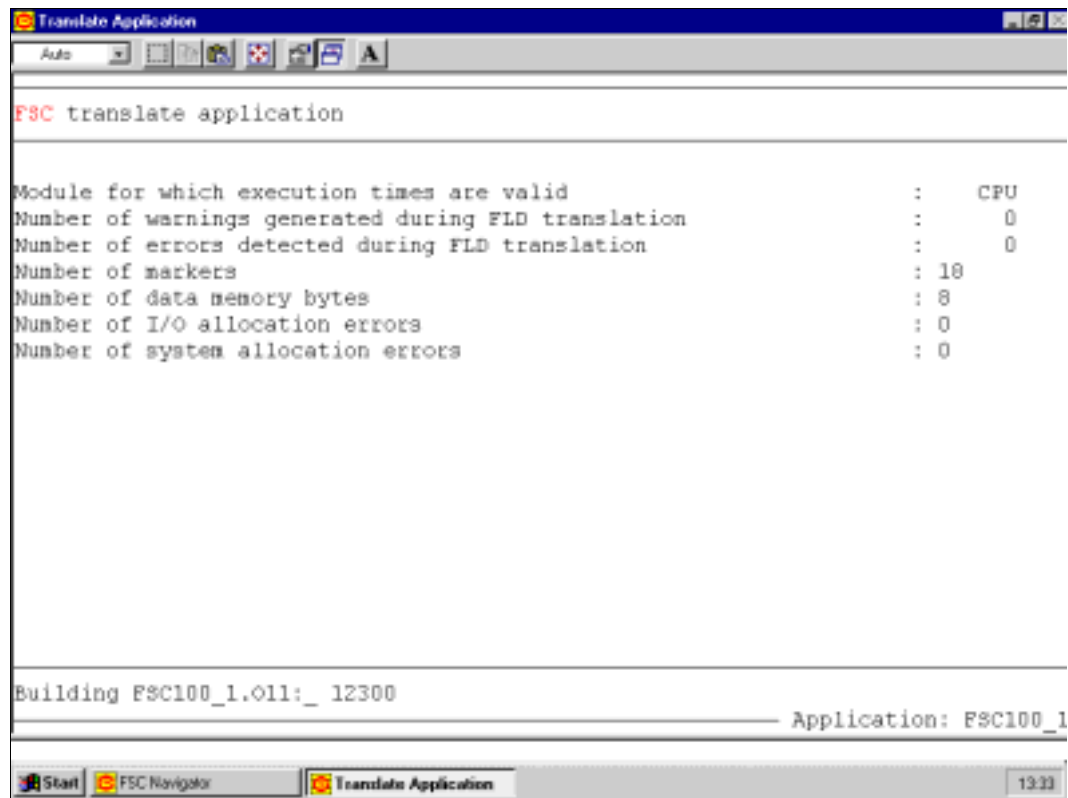


Figure 8-1 Warnings and messages screen

If you want to abort the translation process for some reason, just press **<Esc>**. The compiler will report a manual interrupt and writes the time and date of interruption in the log file. After the translation has ended, the number of errors and warnings is written to the log file. If severe errors occur, the translation is aborted automatically and no additional data is reported.

Note:

If the compiler detects an error, no output files are generated except the log file.



8.3.3 Compiler modes

The compiler can translate the project for the CPU or the MEM modules. Use the 'System Configuration' option of FSC Navigator (Install \ Modules \ Central Parts) to specify the location of the application program. At the start of the translation process, the compiler reports for which module the execution times are calculated.

8.3.3.1 Simulation mode

Simulation mode can be used to test the application software of large systems on an FSC simulation unit 23125/1/1, 23125/R/1, 23126/1/1, 23250/1/1, 23260/1/1 or 23251/1/1. The number of variables that can be forced or written regardless of the force enable status depends on the configured EPROM size (see Table 8-1).

Table 8-1 EPROM sizes

EPROM size	Force / Write variables
512 K	2080
1 M	8192
2 M	20800
4 M	44800

If the number of variables of the same type in the original application exceeds the maximum number, the remaining variables of that type cannot be forced or written. The compiler will then give a warning.

The application software is located in RAM on the memory module. To be able to reset the tested system, the RESET input is located on a default address, although it may be anywhere else in the original FSC system. The default address of the reset input is Rack 2, Position 8, Channel 2.

The input and output modules will get the address, which is the same as in the non-simulated system. This means that the inputs which have the same address as the Force enable and the reset input in the simulated system cannot be debugged normally but should be changed to location 'COM'.

The FSC configuration of the simulated application can only be one of the following configurations:

23125/1/1	single configuration
23126/1/1	single configuration in one rack
23125/R/1	redundant configuration
23250/1/1	redundant configuration
23260/1/1	single configuration in one rack
23251/1/1	redundant configuration

The hardware configuration of the test systems is fixed. The tables below provide an overview of the hardware contained in each test system.



Table 8-2 Hardware allocation of 23125/1/1 training unit

Rack	Position	Module
1	8	10002/1/. (CPU)
1	13,14	10004/I/F (COM)
1	15	1003/1/1 (MEM)*
1	16	10005/1/1 (WD)
1	17	10001/1/1 (VBD)
1	18,19	10006/1/1 (DBM)
1	20	10300/1/1 (PSU)
2	8,9,10	10101/1/1
2	11	10102/1/1
2	14,15	10201/1/1
2	17	10206/1/1
2	18	10205/1/1

***Note:**

If the location of the application program is CPU, no MEMory module (10003/1/1) is required.

Table 8-3 Hardware allocation of 23126/1/1 training unit

Rack	Position	Module
1	1	10101/1/1
1	3	10102/1/1
1	4	10205/1/1
1	6	10201/1/1
1	7	10206/1/1
1	8	10209/1/1
1	12,13	10004/I/F (COM)
1	14	10002/1/. (CPU)
1	15	10003/1/1 (MEM)*
1	16	10007/1/1 (SBD)
1	17	10005/1/1 (WD)
1	18,19	10006/1/1 (DBM)

Table 8-4 Hardware allocation of 23125/R/1 training unit

Rack	Position		Module
	CP 1	CP 2	
1	2	12	10002/1/. (CPU)
1	3,4		10004/H/I (COM)
1		13,14	10004/H/F (COM)
1	5	15	10003/1/1 (MEM)*
1	6	16	10005/1/1 (WD)
1	7	17	10001/1/1 (VBD)
1	8,9	18,19	10006/1/1 (DBM)
1	10,11	20,21	10300/1/1 (PSU)
2	9,10		10101/1/1
2	15,16		10201/1/1

Table 8-5 Hardware allocation of 23250/1/1 training unit

Rack	Position	Module
1	8	10002/1/ (CPU)
1	13,14	10004/I/F (COM)
1	15	10003/1/1 (MEM)*
1	16	10005/1/1 (WD)
1	17	10001/1/1 (VBD)
1	18,19	10006/1/1 (DBM)
1	20	10300/1/1 (PSU)
2	8,9,10	10101/2/1
2	11	10102/2/1
2	14,15	10201/2/1
2	17	10206/2/1
2	18	10205/2/1

***Note:**

If the location of the application program is CPU, no MEMory module (10003/1/1) is required.

Table 8-6 Hardware allocation of 23260/1/1 training unit

Rack	Position	Module
1	1	10101/2/1
1	3	10102/2/1
1	4	10205/2/1
1	6	10201/2/1
1	7	10206/2/1
1	8	10209/2/1
1	12,13	10004/I/F (COM)
1	14	10002/1/ (CPU)
1	15	10003/1/1 (MEM)*
1	16	10007/1/1 (SBD)
1	17	10005/1/1 (WD)
1	18,19	10006/1/1 (DBM)

Table 8-7 Hardware allocation of 23251/1/1 training unit

Rack	Position		Module
	CP 1	CP 2	
1	2	12	10002/1/ (CPU)
1	3,4		10004/H/I (COM)
1		13,14	10004/H/F (COM)
1	5	15	10003/1/1 (MEM)*
1	6	16	10005/1/1 (WD)
1	7	17	10001/1/1 (VBD)
1	8,9	18,19	10006/1/1 (DBM)
1	10,11	20,21	10300/1/1 (PSU)
2	9,10		10101/2/1
2	15,16		10201/2/1



***Note:**

If the location of the application program is CPU, no MEMory module (10003/1/1) is required.

COM channels in simulation mode

The communication (COM) channels of a simulated project are assigned by the compiler. At least one channel will be a Development System channel. The compiler indicates which COM channel of the original application is used by the simulated project.

If no Development System channel was available in the original application or if the COM channel is occupied by another protocol, the compiler will use the 10004/I/F COM module by default:

I = RS485, 19k2, no encoding

F = RS232, 19k2, no encoding

CPU module type in simulation mode

The CPU module type which is required in simulated project will be identical to the CPU module type in the original application. The location of the application software is also adapted from the original application.

8.3.4 Default SER format file

The compiler will create a default SER format file if no user-specified SER format file was found. The default format file is identical to the default SER file which is created if the SER format is specified for the first time using the 'SER format' option of FSC Navigator.



8.4 Compiler output

8.4.1 Output files

The compiler generates three types of files:

- **A log file**

The log file (<Application name>.LOG) contains all messages, errors and warnings that were generated by the compiler. This file can be printed or displayed on screen using the 'View Log' option of FSC Navigator.

- **Application code files**

One application code file (<Project name>.O<CP no.>1) will be generated for the CPU of each Central Part. The first digit in the file extension refers to the Central Part number; the second digit is always '1'. For example, TEST_1.O21 is the file for Central Part 2. The code files contain FSC code and can be used as input for the 'Program EPROMs' option of FSC Navigator.

- **Communication files (if COM modules are used)**

One communication file (<Application name>.C<number>) will be generated for each COM module of each Central Part. The number in the file extension consists of the Central Part number (first digit) and the COM module sequence number (second digit). For example, TEST_1.C13 is the communication file for Central Part 1, COM module 3.

If the compiler detects an error, no CPU or COM application files are generated.

8.4.2 Error messages and warnings

All errors found in the FLDs and databases during translation are written to the log file. Also, warnings are generated in situations where the code is syntactically correct but will probably not perform the function intended by the design engineer. Finally, some translation data is reported in the log file.

The following expression is printed behind several error messages: Node(x, y).

Node (x, y) means x grid points from the left side of the FLD and y grid points from the upper FLD border. The nodes are also shown on screen when editing the functional logic diagram, i.e. choose 'Move symbol' to start the coordinate counter.

If a message contains an FLD number, the variable concerned is located on this FLD. If no FLD number is printed, this means that the variable (tag number) is not used on an FLD.

This subsection lists all possible messages and warnings in alphabetical order.

4-Megabit EPROMs are required for COM EPROMs.

The file created for the communication module needs 4 Megabit EPROMs. This size is not supported yet. Do not use service and qualification for markers in the SER messages.

A functional logic diagram must contain at least one output symbol.

At least one output symbol (output, BO, off-sheet ref., etc.) is required on each functional logic diagram.

**Alarm group number undefined:** <type> <tag no.>

The alarm group number of the variable (location 'ANN') has not been specified.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to specify the alarm group number.

Alarm group size undefined: <type> <tag no.>

The group size of the alarm group has not been specified.

Use the 'System Configuration' option of FSC Navigator to specify the group size or delete the variable.

Alarm type undefined: <type> <tag no.>

The alarm type of the variable (location 'ANN') has not been specified.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to specify the alarm type.

Alarm variable double declared: <type> <tag no.>

The alarm type of the variable (Location 'ANN') has already been specified for this alarm group, but should be unique for each alarm group.

Alarm variable is allocated to hardware: <type> <tag no.>

The variable <type> <tag no.> should be allocated to a communication channel, but is now configured as a hardware-related variable.

Use the 'System Configuration' option of FSC Navigator to change the sub-allocation.

Alarm variable is not allocated to hardware: <type> <tag no.>

The variable <type> <tag no.> should be allocated to a digital output module, but is now configured as a communication variable.

Use the 'System Configuration' option of FSC Navigator to change the sub-allocation.

Alarm variable not used on sheet: <type> <tag no.>

The specified variable exists in the variable database, but is not used in the application program.

Use the 'Design FLDs' option of FSC Navigator to change the program or delete the variable (if the variable is not necessary for the alarm group).

Analog I/O allocation error: <type> <tag no.>

This means that an analog variable has not been properly allocated to the hardware (rack, channel, position is not specified).

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to allocate the variable.

If the FLD number of the variable is zero, the variable is not used in the FLDs and may be deleted. You can use the 'Print' option of FSC Navigator to print all variables in the variable database that were not properly allocated.

Average application time (ms): <time>

After the translation is complete, the average application time is listed.

BI/BO with DEC/BIN conversion must have BCD data representation: <type> <tag no.>

The variable <type> <tag no.> is converted from/to BCD data representation in the FLD, but the variable is specified as a binary input or output.

Use the 'System Configuration' option of FSC Navigator to change the data representation to BCD, or replace the BCD input or output symbol in the FLD with a binary input or output signal.

Called block is no or incorrect equation block: <equation block>

The referenced diagram is no equation block diagram.

**Called block is no function block:** <function block>

A call is made to an FLD which is not a function block FLD.

Delete the call in the calling FLD.

Cascade PID not found: <type> <tag no.>

PID <type> <tag no.> has been specified as a cascade PID, but was not found in the database.

Use the 'System Configuration' option of FSC Navigator to change the cascade PID tag number (Hardware specs) or enter a new PID with this tag number.

Code file too large to fit on RAM set(s).

The generated files will be too large for the current configuration.

Use the 'System Configuration' option of FSC Navigator to change the application program to memory modules.

Communication module <CP><module><chan.> uses protocol specified at <CP><module><chan.>

This warning is generated while translating in simulation mode and specifies which protocols are used for the channels of the communication module(s).

Compiler out of space: no more compiler sequence numbers.

Not enough space available to store more compiler sequence numbers (on-line modification set to 'Yes').

Use the 'System Configuration' option of FSC Navigator (rebuild option) to create new space.

Configured process safety time (s): <time>

The process safety time as specified in the database.

Connected FSC application name not configured: <appl. no.>

An FSC-FSC link has been specified for system <appl. no.>, but no valid application name has been specified.

Use the 'System Configuration' option of FSC Navigator (Install / Configuration / FSC network) to specify the name of the application with application number <appl. no.>.

Converting sheets to new 'Design FLDs' version...

Before translation, all FLDs will be converted to the correct compiler version.

Corrupted data: <type> <tag no.>

The data of the variable is corrupted, but the CRC is OK.

Use the 'System Configuration' option of FSC Navigator (Signal Specs option) to delete the variable, and enter the data for the variable again.

Corrupted data detected.

The current installation FLD contains a corrupted record.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to regenerate the configuration.

Corrupted data detected in variable data file at record: <record no.>

The data file contains a corrupted record.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to rebuild the data file.

Corrupted file: <file name>

A (temporary) file which was created by the compiler is corrupted.

Use a disk utility to remove any bad sectors from your disk and restart the translation.

The report or SER format file is corrupted.

Use the 'SER Format' option of FSC Navigator to correct the corrupted file(s).

**Count-up or count-down not allowed for a float register.**

The count-up or count-down operation is only valid for integer-type registers (byte, word or long).
Use the 'Design FLDs' option of FSC Navigator to modify the FLDs.

Counter allocation error: <type> <tag no.> Byte number: <byte no.>

The counter has an address <byte no.> which is used by one or more other counters.
Use the 'System Configuration' option of FSC Navigator (rebuild option) to check the addresses of all variables.

CRC-32 of application software in Central Part <CP> : <CRC>

Displays the CRC over the application software of the specified Central Part to validate the verification.

Current application not linked in: <linked appl. name>

The linked application was found, but no link has been specified in this database to the current application (no FSC-FSC link to the current application has been specified).

Use the 'System Configuration' option of FSC Navigator (Install/Modules/Central Parts/COM) to specify a link to the current application name in the linked application. Remove the link to the other application. For example, the current application is TEST_1 and has a link to TEST_2. In application TEST_2, no FSC-FSC link has been specified to application TEST_1.

Current application version: <version no.>

If the translation is complete, the version of the software is listed.

Data error in sheet.

The current translated FLD contains a corrupted record.

Use the 'Design FLDs' option of FSC Navigator to see if all symbols are still present and correct.

Data representation in sheet does not match variable data file.

If input or output representation is of type BCD (configured using the 'System Configuration' option of FSC Navigator), there must be a conversion between input/output and the other sheet symbols.

Database version mismatch detected in file:

During the network check by the compiler, an attempt was made to read data from an application with a different version than the current application.

Use the 'Open Project' option of FSC Navigator to convert the application to the current version.

Default Development System protocol used at: <CP No.> <Com No.> <Chan>

The application is translated in a simulation mode. No Development System protocol exists in the original application. The compiler uses the specified communication channel as the Development System channel.

**Digital I/O allocation error: <type> <tag no.>**

This means that a variable has not been properly allocated to the hardware (rack, channel, position has not been specified).

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to allocate the variable. If the FLD number of the variable is zero, the variable is not used in the FLDs and may be deleted. You can use the 'Print' option of FSC Navigator to print all variables in the variable database that were not properly allocated.

This error message (with "O SEC.SWITCH-OFF" as the type and tag number) will also appear in the following two cases:

- If the FSC system has been configured for requirement class AK6 and the secondary switch-off has not been allocated to a hardware output.
- If one or more 10216/./ output modules have been configured with 1002D voting and the secondary switch-off has not been allocated to a hardware output.

(For FSC systems with requirement class AK5, the secondary switch-off does not need to be allocated, and the above message will only be issued as a translation warning.)

DISK READ ERROR: <file name>

An unsuccessful attempt was made to read data from the specified file.

Use a disk utility to remove any bad sectors from your disk and restart the translation.

DISK WRITE ERROR: <file name>

An unsuccessful attempt was made to write data to the specified file.

Use a disk utility to remove any bad sectors from your disk and restart the translation.

Disk full.

This can be solved by removing unused files from your disk or using a disk with a larger disk capacity.

Duplicate off-sheet reference on sheet: <FLD no.>

An off-sheet reference with the specified tag number has been placed on one FLD twice.

Use the 'Design FLDs' option of FSC Navigator to remove one reference.

Duplicate on-sheet reference on sheet: <FLD no.>

An on-sheet reference with the specified tag number has been placed on one FLD twice.

Use the 'Design FLDs' option of FSC Navigator to remove one reference.

Duplicate sheet transfer in index file: <tag no.>

The index file is corrupted.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to rebuild the index file.

Duplicate tag number on sheet: <tag no.>

A variable with the specified tag number has been placed on one FLD twice.

Use the 'Design FLDs' option of FSC Navigator to remove one tag number.

Error in the COM module specifications.

A CRC error has occurred.

Use the 'System Configuration' option of FSC Navigator to verify the communication module specifications.

Error in link configuration to system: <system>

A CRC error has occurred.

Use the 'System Configuration' option of FSC Navigator to verify the link specification to the specified system.

**Error in sheet:**

The current translated FLD contains a corrupted record.
Use the 'Design FLDs' option of FSC Navigator to check the FLD.

Error in the application description field

A CRC error has occurred in the description fields on the specified field.
Use the 'System Configuration' option of FSC Navigator to verify the specified rack application description field.

Error in address allocation.

A CRC error has occurred.
Use the 'System Configuration' option of FSC Navigator to verify the address allocation.

Error in the block definitions of the RKE3964R protocols.

A CRC error has occurred.
Use the 'System Configuration' option of FSC Navigator to verify the block definitions.

Error in the configuration definition.

A CRC error has occurred.
Use the 'System Configuration' option of FSC Navigator to verify the configuration definition.

Error in the module in rack: <Rack> at position: <Pos.>

A CRC error has occurred.
Use the 'System Configuration' option of FSC Navigator to verify the module at the specified rack and position.

Error in password definitions.

A CRC error has occurred.
Use the 'System Configuration' option of FSC Navigator to verify the password definitions.

Error in the rack definition of cabinet

A CRC error has occurred.
Use the 'System Configuration' option of FSC Navigator to verify the specified rack definition of cabinet.

Error in the SER channel definition.

A CRC error has occurred.
Use the 'System Configuration' option of FSC Navigator to verify the SER channel definition.

Execution time too large.

The total calculated maximum execution time exceeds the maximum time (498 ms).
Change the timing of multiplexers and/or PIDs. Rebuild the database without saving the addresses of the variables. Increase the process safety time or split the application over two FSC systems.

Fatal error. Translation aborted: <time> <date>

This message is printed if for some reason the translation process cannot continue. Possible reasons are disk full or insufficient memory.

**File creation failed:** <file name>.<file extension>

The file could not be created. For example, the disk may be full or the number of open files has been exceeded.

Clean your disk, i.e. delete unused files or save currently unused files to a diskette and delete them from your hard disk.

Use a system with a larger disk capacity.

Extend number of files in your CONFIG.SYS file: Files = <number of files>. The recommended number of files is 30.

File opening failed: <file name>

An attempt was made to open the specified file, but it was not found or could not be opened. If the file was the <application name>.FMT, a default format file will be used by the compiler and it will be reported that the default format file is used.

FSC system out of memory for this type of variable: <type> <tag no.>

The compiler cannot place a compiler register or compiler marker in the variable database.

It is possible that new space will be created after the translation is complete. Therefore, recompile the whole set. Erase unused variables (FLD is 0). Change your design so as to decrease the number of required markers and/or registers.

Function block input not connected. Function block input sequence number: <seq. no.>

An input of a function block has not been connected to a signal in the FLD. This is allowed but a warning is nevertheless given, because it might be not intended. The input value will be zero during execution.

Function block sheet in program block area: <no.>

The FLD number of a program block must be at least two less than the FLD number of the first function block.

Use the 'Design FLDs' option of FSC Navigator (renumber option) to shift down the program block or to shift up the function block. Check all calls to this function block.

Function block sheet missing: <FLD no.>

The function block which is called at FLD <FLD no.> is not available in the project folder (file is not found). Delete the function block on the calling FLD or restore the function block sheet.

Function block sheet must contain at least one function block output.

A function block must have a function block output, otherwise it is of no use in the functional logic diagrams.

Place a function block output or delete the FLD.

Function block symbol corrupted.

Use the 'Design FLDs' option of FSC Navigator to delete and replace the corrupted function block.

Function change failed.

The FLD contains too many variables of the specified type (markers, counters, timers or registers).

Use the 'Design FLDs' option of FSC Navigator to edit the FLD (e.g. split into more FLDs).

Identical source and destination sheet are not allowed.

The source and the destination FLD in a sheet transfer must be different.

Illegal address: <type> <tag no.>

The database contains a variable which has an illegal address.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to check the new addresses of all variables.

**Illegal bit number:** <bit no.>

The bit number is corrupted.
Translate again.

Illegal connection: Node(x, y)

A signal line has been connected to a symbol at a point where it is not allowed, or the signal line is not a valid signal type. Function blocks and registers have predefined input and output positions.
Run the 'Design FLDs' option of FSC Navigator to change the connection.

Illegal constant: <type> <tag no.>

The value of the constant is not allowed (e.g. a negative constant is loaded to a counter).
Use the 'Design FLDs' option of FSC Navigator to change the constant.

Illegal counter value: Node(x, y)

The value which will be loaded into the counter is illegal.
Use the 'Design FLDs' option of FSC Navigator to change the constant value or change the range of the counter.

Illegal function block call: <function block>

The function block called has a larger or an equal FLD number than the current FLD.
Remove the function block from the current FLD. Renumber the function block to an FLD with a number larger than the current FLD number. Place the function block again on the current FLD.

Illegal function block input exchange.

Two function block inputs have been exchanged illegally.
Use the 'Design FLDs' option of FSC Navigator to place the symbols as before.

Note:

Only active if on-line modification is set to Yes.

Illegal function block output exchange.

Two function block outputs have been exchanged illegally.
Use the 'Design FLDs' option of FSC Navigator to place the symbols as before.

Note:

Only active if on-line modification is set to 'Yes'.

Illegal input signal type of function block: <function block> Node(x, y)

The signal type of the register on the FLD cannot be converted to the signal type of the function block input. The calling FLD long register cannot be connected to a word register.

Illegal modules used for multiplexer I/O.

The multiplexer configuration contains I/O modules which are not valid.
Use the 'System Configuration' option of FSC Navigator to change the allocation of the multiplexers.

Illegal name of application stored in database: <incorrect appl. name>

The name of the current application has not been specified for the current system number. For example, the current application is TEST_1. The application name for system 1 in the current database is, for example, DEMO. This will happen if an application has been renamed "outside" FSC Navigator (e.g. using Windows Explorer).
Use the 'System Configuration' option of FSC Navigator (Configuration option) to update the application name.

**Illegal printer reference:** <file name>

The printer specified in a report file or in the SER format file is not known in the specified file.
Use the 'SER Format' option of FSC Navigator to install another printer. The specified file might be an old version of the file DEVICE.DRV.

Illegal protocol specified: <Central Part module> <COM no.> <Chan.>

The protocol specified in the database is not valid.
Use the 'System Configuration' option of FSC Navigator (Install option) to select a valid protocol.
The channel has been specified as the SER channel, but has an invalid protocol.
Use the 'System Configuration' option of FSC Navigator to specify the primary or secondary link, or define the channel as a primary link.

Illegal record number: <no.> (<file name>)

The compiler attempted to read a record which does not exist.
Use the 'System Configuration' option of FSC Navigator to rebuild the index files.

Illegal record type: <type> <tag no.>

The variable has been defined in a report, but cannot be used as a condition to generate a report.
Use the 'SER Format' of FSC Navigator to remove the variable from the condition of a report.

Illegal sec. link specified for COM chan. (CP, Mod): <CP> <Module> <Chan.>

The protocol of the secondary channel does not match the protocol of the primary link.
Use the 'System Configuration' option of FSC Navigator to change the protocol or change the secondary link.

Illegal secondary SER channel specification.

The secondary SER channel has been allocated to an invalid channel.
Use the 'System Configuration' option of FSC Navigator to change the secondary SER channel specification.

Illegal SER printer specification (CP no., COM module, chan): <CP> <Module> <Chan.>

The SER channel has been configured in this system, but the Central Part, COM module and channel have not been fully specified.
Use the 'System Configuration' option of FSC Navigator to specify the SER channel.

Illegal SER printer specification (system number): <system no.>

The SER printer has been configured to another system, but no link with this system is available.
Use the 'System Configuration' option of FSC Navigator (Install/Configuration option) to specify the SER channel.

Illegal signal type(s): Node(x, y)

The signal type is illegal for the symbol to which it is connected. For example, a signal of type Long cannot be connected to a counter or timer.
The types of the signal coming from the FLD to an internal off-sheet symbol (i.e. decimal marker) do not match and conversion is not allowed for internal symbols.
Use the 'Design FLDs' option of FSC Navigator to change the signal type.

Illegal symbol for this block type:

The FLD contains a symbol which should not be present in an FLD with the specified block type.
Use the 'Design FLDs' option of FSC Navigator to delete the symbol.

Illegal timer base

Base must be 100 ms, 1 second or 1 minute.
Use the 'Design FLDs' option of FSC Navigator (Change option) to change the symbol.

**Illegal value: <type> <tag no.>**

The value of the variable (timer or counter) is less than or equal to zero, or greater than the maximum value (counter maximum = 8191; timer maximum = 2047).

Use the 'Design FLDs' option of FSC Navigator to change the variable.

Illegal value specified in report condition.

The value of the variable is not within the valid range.

Use the 'SER Format' option of FSC Navigator to edit the value of the report condition.

Illegal variable type specified for <type> <tag no.>

The database contains a variable ('I' or 'O' with location 'ANN') which is connected to this alarm with an illegal alarm type.

Use the 'System Configuration' option of FSC Navigator to disconnect the illegal variable.

Impossible scan time: <type> <tag no.>

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to change the scan time.

Inaccurate scaling specified for analog input:

The difference between the bottom and top scale value is too small.

Use the 'System Configuration' option of FSC Navigator to change the scaling values of the specified variable.

Incompatible version of file: <file name>

The specified file is not compatible with the current FSC Navigator version.

Use the 'SER Format' option of FSC Navigator to convert the files or remove the file.

Index files are corrupted.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to create new index files. If a timer or register symbol was found on an FLD which is not found in the I/O data file, the FLD design editor will first try to place this symbol in the I/O data file. If this fails, the symbol is deleted from the FLD.

Index files will be rebuilt as soon as another FSC program is started.

A CRC error has occurred and the index files have been removed from disk.

As soon as another main option of FSC Navigator is started, the index files will automatically be rebuilt.

Infinite signal cycle: Node(x, y)

A signal line has been connected to itself (perhaps via several other FLDs), creating a loop.

Use the 'Design FLDs' option of FSC Navigator to delete the loop.

Input/output/PID already placed on sheet:

An input, output or PID has been placed on different FLDs twice.

Use the 'Design FLDs' option of FSC Navigator to remove the symbol on one of the FLDs.

Invalid data exchange with com. server system: <CP No.> <Com No.> <Chan.>

The specified channel contains a number of markers and/or registers to read and/or write to a communication server channel.

Use the 'System Configuration' option of FSC Navigator to delete the number of bytes to read or write.

Use the 'System Configuration' option of FSC Navigator to change the communication server to a master channel.

Invalid I/O for diagnostic status.

Diagnostic input must be allocated to a fail-safe module.

**Invalid library version. You need version <version>**

The library has a different version than the compiler, but cannot be converted.
Use an FSC Navigator software set of the specified version.

Invalid master channel configured (CP,COM,Ch):

The channel specified has been configured as a redundant master channel.
Use the 'System Configuration' option of FSC Navigator to change the master channel to a redundant slave channel or to a single master channel.

Invalid network configuration detected in appl:

The configured network contains errors:

- The master of the current slave channel was not found.
- The current application has not been specified in the connected slave systems.
- Invalid data was detected.

Use the 'System Configuration' option of FSC Navigator to correct the network configuration.

Invalid numeric order in equation table.

The input value defined in the equation table must be in ascending numerical order.
Invalid PID sample time configured.
The sample time of the PID is too small.
Use the 'System Configuration' option of FSC Navigator to change the PID sample time to a larger value (it should be higher than the application time).

Invalid redundancy type configured: <type> <tag no.>

The specified input variable contains non-fail-safe sensors, but the redundancy type of the sensor has not been specified (yet).
Use the 'System Configuration' option of FSC Navigator to change the sensor to fail-safe, or to specify the sensor redundancy type.

Invalid redundant link definition (CP,COM,Ch):

The specified channel has been configured as a redundant link, but the configuration of both channels does not match.
Use the 'System Configuration' option of FSC Navigator to change the redundant link to single, or redefine the redundant link after a rebuild.

Invalid requirement class configured: <requirement class>

The requirement class of the application is not valid.
Use the 'System Configuration' option of FSC Navigator to specify a valid requirement class.

Invalid SER sequence number area specified.

The SER sequence number area is too small or is not in the valid range of SER sequence numbers.
Use the 'System Configuration' option of FSC Navigator (Install/Configuration/SER channel) to change the SER sequence base and/or SER sequence maximum.

Invalid SER sequence number specified: <type> <tag no.>

The SER sequence number specified for <type> <tag no.> is out of the specified SER number range or is already in use.
Use the 'System Configuration' option of FSC Navigator (Rebuild option) to change the SER sequence number.

Invalid sheet reference: <tag no.>

The sheet reference with <tag no.> is not correct (not found in the variable data file or found twice on FLDs).
Use the 'Design FLDs' option of FSC Navigator to remove the reference.

**Invalid simulation mode specified.**

The specified simulation mode is not known by compiler.

Use the 'System Configuration' option of FSC Navigator to change the power-on mode.

Invalid software version. You need version <version>

The database is of a different version as the compiler, but cannot be converted. Use an FSC Navigator software set of the specified version.

Invalid system link configured at (CP,COM,Ch): <CP>, <COM Module>, <Chan.>

The specified channel contains systems which cannot be used for FSC-FSC communication.

Use the 'System Configuration' option of FSC Navigator to remove the invalid systems, or specify the path for the invalid systems in the complete network.

Invalid/unknown I/O module configured:

The I/O module configured is not supported by the system.

Use the 'System Configuration' option of FSC Navigator to remove the invalid I/O module.

Invalid value in equation table.

A value defined in the equation table is out of range. The legal range is 1E-38 to 1E+38.

I/O buffer full: <i,o,I,O>

The buffer used for digital or analog I/O is full (i : Analog input, o : analog output, I : digital input, O : digital output).

Use the 'System Configuration' option of FSC Navigator to delete some I/O modules of the specified type.

I/O module address(es) have been changed.

This warning will occur if the compiler detects invalid addresses of I/O modules in the database. I/O variables which are allocated to these modules will have other addresses.

Isolated symbol: Node (x,y)

Symbol without any direct or indirect output connection to an off-sheet symbol.

Connect the symbol or delete it.

Larger size of application EPROMs required: <EPROM size>

A specific type of EPROM/RAM has been configured for the application module (CPU only), but the application does not fit in the configured type. The compiler shows which type should be configured to make the application fit.

Use the 'System Configuration' option of FSC Navigator (Install/Configuration/Memory chip size) to select the suggested EPROM type (or a larger type), or specify variable sizing. The generated files can be used, but should be programmed with a larger type of EPROM.

Larger size of communication EPROMs required: <EPROM size>

A specific type of EPROM has been configured for the communication module, but the application does not fit in the configured type. The compiler shows which type should be configured to make the communication file fit.

Use the 'System Configuration' option of FSC Navigator (Install/Configuration/Memory chip size) to select the suggested EPROM type (or a larger type), or specify variable sizing. The generated files can be used, but should be programmed with a larger EPROM type.

Library corrupted: <library name>

The specified library file is corrupted.

Reinstall the library from the CD-ROM.

**Library not found on disk: <library name>.SYM**

The library used to create the functional logic diagrams cannot be found in the FSC Navigator application folder (C:\FSC by default).

Make sure that the required library file is in the FSC Navigator application folder, or use the Project/Properties menu item of FSC Navigator to select another library and translate again. Make sure that the library is compatible with the drawings.

Link area of connected appl. does not match: <linked appl. name>

The number of bytes to read and/or write markers and/or registers of the current application does not match the number of bytes to read and/or write in the linked application.

Use the 'System Configuration' option of FSC Navigator to specify (or copy) the number of bytes used for external FSC-FSC protocols.

Link specification contains errors.

The compiler has detected an error for the link specification of the COM channel specification.

Use the 'System Configuration' option of FSC Navigator to correct the error (rebuild option).

Link appl. name of connected appl. not OK: <linked appl. name>

The linked application is found, but in this database the application name to the current application number is not the current application name.

Use the 'System Configuration' option of FSC Navigator (Install/Configuration) to specify the current application name in the linked application. For example, the current application is TEST_1 and has a link to TEST_2. In application TEST_2, the application name for system number 1 is not TEST, but (for example) TEST1.

Load and value must both be connected or not connected: <type> <tag no.>

If you have a load function, you need a value to be loaded into the register. If you do not have a load function, no value is needed.

Logic sequence too complex: Node (x,y)

The compiler has memory problems when translating the sequence to the specified node point.

Use the 'Design FLDs' option of FSC Navigator to split the functional logic diagrams into simpler or shorter sequences.

Marker allocation error: <type> <tag no.> Bit number: <bit no.>

The variable in the marker area has an address <bit no.>, which is used by one or more other variables in the marker area.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to check the addresses of all variables.

Maximum delay time on communication link (CP no., COM, Chan.): <value> ms

This message is given for each multidrop communication channel in a communication server FSC application. It shows the maximum response time (in ms) of that channel. This information helps the user decide how to balance overloaded communication channels.

Maximum number of report entries exceeded.

The number of conditions used for report generation is too large. The maximum number is 16368.

Use the 'SER Format' option of FSC Navigator to reduce the number of conditions or the number of reports.

Missing alarm variable(s) for <type> <tag no.> <alarm type>

The alarm type mentioned is not specified with the Alarm group.

Use the 'System Configuration' option of FSC Navigator to append variables to this alarm group with the missing alarm type.

**Missing sheet:** <appl. name>.<FLD no.>

The file that contains the FLD cannot be found on disk.

Copy the FLD file to the project folder. If the FLD is really lost, run the 'Design FLDs' option of FSC Navigator. Enter the FLD number of the lost FLD and answer 'Y' when asked whether you want to erase the FLD from the database.

Missing signal or unconnected signal line: Node(x, y)

A signal line which should be connected has not connected at node point (x, y).

Use the 'Design FLDs' option of FSC Navigator to connect or delete the line.

Missing slave system detected at (CP, COM, Ch): <CP No.> <Com No.> <Chan.>

Not all systems connected have been configured at the specified channel.

Use the 'System Configuration' option of FSC Navigator to add the system to the specified external application.

Missing variable data file : <data file name>

The variable data file not found.

Use the 'System Configuration' option of FSC Navigator to create a new data file.

Module for which execution times are valid : <module type>

The module type as specified using the 'System Configuration' option of FSC Navigator.

More than one signal at a node: Node(x, y)

Two independent signals may not be connected.

Disconnect them.

Multiple I/O allocation error: <type> <tag no.>

This means that a variable has not been properly allocated to the hardware for all necessary I/O bits (rack, channel, position is not specified).

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to allocate the variable. If the FLD number of the variable is zero, the variable is not used in the FLDs and may be deleted. You can use the 'Print' option of FSC Navigator to print all variables in the variable database that were improperly allocated.

Multiplexer contains no variable: <type> <tag no.>

The data bus of the multiplexer contains no variables. In case of an XI, no BIs with the number of multiplexer XI do exist. In case of an XO, no BOs with mux number of the XO exist.

Use the 'System Configuration' option of FSC Navigator to delete the XI or XO, or append variables to this XI or XO.

Multiplexer RAM out of space.

Too much RAM is required for all multiplexed I/O (variables with location 'MUX') that have been configured.

Try to minimize the size of variables (bytes instead of words or longs, words instead of longs) or change the location of some variables.

Multiplexer tag number not defined: <type> <tag no.>

The multiplexer tag number of the variable should be specified to connect this variable to a XI or XO.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to specify the multiplexer tag number.



No application connected for COM chan. (CP no., COM module, chan.): <CP> <Seq.> <Chan.>
Channel <Chan.> from the communication module in Central Part <CP> with sequence number <Seq.> has been defined as an external FSC-FSC protocol, but no other application has been specified. Use the 'System Configuration' option of FSC Navigator (Install/Central Part/COM) to specify the communication link.

No communication channel found to load application into RAM.

The application is translated for test mode, but no FSC user station (or printer) is available. Set the power-on mode to RUN mode or specify a Development System channel.

No conditions specified in report file.

A report has been specified, but the report is never printed. Use the 'SER Format' option of FSC Navigator to edit the conditions or frequency to generate the report, or to delete the report.

No DBM module present, so real-time functions not available: <type> <tag no.>

The application program uses system variable <type> <tag no.>, which can be used only if a DBM module has been configured.

Use the 'System Configuration' option of FSC Navigator to specify a DBM module, or remove the real-time variables (YEAR, DATE, DAY, MONTH, HOURS, MINUTES or SECONDS) from the functional logic diagrams.

No hardware specified.

The database has not been configured; no hardware has been specified.

Use the 'System Configuration' option of FSC Navigator (Install option) to specify the hardware of the application.

No linked variable specified for: <type> <tag no.>

The variable <type> <tag no.> has been allocated to a TFS_COM1 alarm group and should be linked to another variable.

Use the 'System Configuration' option of FSC Navigator to specify a linked variable, allocate the variable to another alarm group (with a different alarm group sequence), or change the alarm type of the variable.

No load signal, so a count signal and clear signal must be connected: <type> <tag no.>

If you do not load a register, you must at least use a count-up or a count-down in combination with a clear, else the register is of no use.

No master system detected for (CP,COM,Ch): <CP No.> <Com No.> <Chan.>

The slave channel specified does not communicate with an external (master) system.

Use the 'System Configuration' option of FSC Navigator to add a master application to the specified channel.

No primary link specified for COM chan. (CP no., COM module, chan.) : <CP> <COM> <Chan.>

The specified COM channel is a secondary channel, but the primary link has not been fully specified.

Use the 'System Configuration' option of FSC Navigator to specify the primary link, or define the channel as a primary link.

No protocols specified for COM module (Central Part, COM module): <CP> <Seq.>

The protocol of channel A and channel B from the communication module in Central Part <CP> with sequence number <Seq.> have not been specified (Protocol = undefined).

Use the 'System Configuration' option of FSC Navigator (Install/Central Part/COM option) to specify a protocol, or remove this COM module.



Not enough disk space found. Required disk space: <MB>

The compiler requires <MB> megabytes of hard disk space.
Free disk space by deleting files that are no longer needed.

Not enough memory modules configured. Expected number: <No. of MEM modules>

The amount of memory on the configured number of modules is not sufficient for the application.
Use the 'System Configuration' option of FSC Navigator to change the number of memory modules to the specified amount.
If the power-on mode is not RUN mode, Use the 'System Configuration' option of FSC Navigator to change the power-on mode to RUN mode.

Not enough memory to execute required translation function(s).

The compiler needs more memory to execute.

Not enough register memory on COM modules: <Number of bytes>

The number of bytes used on COM modules exceeds its maximum limit.
Minimize the number of register bytes used for each communication channel.

Not enough watchdog repeaters placed.

The number of watchdog repeaters in the system is too small. The number of watchdog repeaters should be at least the number of PSUs (10300/1/1).
Use the 'System Configuration' option of FSC Navigator to add one or more watchdog repeaters (10302/1/1).

Not enough watchdog repeaters placed in single I/O part.

At least one watchdog repeater should be placed in the single I/O part of the system.
Use the 'System Configuration' option of FSC Navigator to add a watchdog repeater (10302/1/1).

Number of alarm groups without alarm type <Alarm type> connected <Number>

The application database contains a number of alarm groups with no <Alarm type> used.

Number of data memory bytes: <number>

Total number of register bytes in the application program.

Number of errors detected during safety calculation:

Not implemented yet.

Number of I/O allocation errors: <number>

Number of unallocated I/O, or not completely allocated I/O.

Number of markers: <number>

Total number of markers used in the application program.

Number of MEM modules to expand data memory: <number>

The number of bytes used for registers does not fit in the RAM of the CPU module.
Expand the RAM with an extra memory module.

Number of memory modules used for application: <no. of modules>

The application is translated for the memory module, but not enough memory modules have been configured.
Use the 'System Configuration' option of FSC Navigator to specify enough memory modules, or change the location of the application program to the CPU.

**Number of system allocation errors: <number>**

Number of double allocated registers, markers, counters or timers.

Number of warnings generated during safety calculation: <Number>

Not implemented yet.

Numerical SER and ASCII SER specified on single channel:

An SER channel in the FSC network is used as a numerical SER channel, but it is also used as an ASCII SER channel.

Use the 'System Configuration' option of FSC Navigator to change all SER channels to numerical SER or ASCII SER, or select different channels for ASCII SER and numerical SER.

Numerical SER range overlap detected.

The numerical SER ranges in the applications of the network overlap.

Use the 'System Configuration' option of FSC Navigator to change the numerical SER ranges in the various applications.

One or more sheets missing. First sheet: <Number>

Not all FLD files were found in the project folder, although they are specified in the databases.

Use the 'Design FLDs' option of FSC Navigator to delete the FLDs from the database. The displayed FLD number is the first FLD that is found missing and will be the FLD with the highest number.

Original system contains too many <variable type>

The application is translated in simulation mode, but the original application contains more variables of the specified type. You can use the generated files, but not all.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to change all addresses.

Output low clamp and output high clamp are identical: <type> <tag no.>

The output clamps of the PID are equal.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to specify the output clamp.

PID cascade in logics differs from cascade configuration: P<tag no.> P<Cascade PID> P<tag no. of connected PID>

Make sure that you also made a cascade connection at this PID in the functional logic diagrams.

- Cascade:
- You did not configure a PID cascade tag number using the 'System Configuration' option of FSC Navigator but you connected the cascade setpoint signal nevertheless.
 - You configured a PID tag number using the 'System Configuration' option of FSC Navigator but you did not connect the cascade input signal or you connected it to the wrong PID or to another symbol.

PID register not found: <type> <tag no.>

The register <type> <tag no.> which should exist in the database was not found.

Use the 'System Configuration' option of FSC Navigator to rebuild the index files, or delete the PID and insert it to create the register again.

PID timer not found: <type> <tag no.>

The timer <Type> <Tag number> which should exist in the database was not found.

Use the 'System Configuration' option of FSC Navigator to rebuild the index files, or delete the PID and insert it to create the timer again.

**Power-on value out of range.** <type> <tag no.>

The power-on value of the variable is invalid. The number of bits is too small to represent the power-on value.

Use the 'System Configuration' option of FSC Navigator to change the power-on value.

Press <Enter> to return to FSC main screen.

The compiler waits until <Enter> is pressed, because the compiler has generated errors or warnings.

Previous error may be caused by multiplexer scan time being too high.

This warning is generated when the compiler detects errors.

Program aborted. Application configuration not defined.

Before using the compiler, the FSC application configuration should be known.

Use the 'System Configuration' option of FSC Navigator (Install option) to define the FSC application.

Program communication EPROMs.

The application has been translated, but the COM EPROMs should be programmed for the application to function correctly.

Program interrupted manually.

The translation process was interrupted because the <Esc> key was pressed.

Protocol mismatch detected: <Chan.>

The channel specified contains a protocol definition which does not match the protocol of connected systems.

Use the 'System Configuration' option of FSC Navigator to change the protocol of the specified channel.

Register allocation error: <type> <tag no.> **Byte number:** <byte no.>

The variable in the register area has address <byte no.>, which is used (or partially used) by one or more other variables in the register area.

Use the 'System Configuration' option of FSC Navigator to check the addresses of all variables.

Register type mismatch with current appl.: <type> <tag no.>

The register type in the current application is different from the register type in the connected external application.

Use the 'System Configuration' option of FSC Navigator to copy the allocated variables or to change the register type of the specified variable.

Report files configured, but no valid report communication channel available.

No COM channel has been configured which may be used for report printing.

Use the 'System Configuration' option of FSC Navigator to specify a COM channel for report printing, or delete the report file.

Reset signal must be connected: <type> <tag no.>

A memorize timer must always have a reset signal because you cannot get it low after you started it.

SER format file configured, but no valid communication channel available.

No COM channel has been configured which may be used for SER printing.

Use the 'System Configuration' option of FSC Navigator to specify a COM channel for SER printing, or delete the SER format file.

**SER/report communication port must be set at 8 bits:**

The COM channel which is used for SER and/or reports is configured at 7 bits/char, but the report and/or SER file contain 8-bit characters.

Use the 'System Configuration' option of FSC Navigator to select 8 bits/char, or use the 'SER Format' option of FSC Navigator to remove the 8-bit characters.

Serial number:

The serial number of the FSC Navigator software.

Set and value signals must both be connected or not connected: <type> <tag no.>

If you have a set function, you need a value to be loaded into the counter. If you do not have a set function, no value is needed.

Set, count-up or count-down signal must be connected: <type> <tag no.>

You must connect one of these functions, else the counter is useless. See the symbol description in Section 6 of the FSC Software Manual ('Functional Logic Diagrams').

Set signal must be connected.

The set signal of a timer has not been connected.

Use the 'Design FLDs' option of FSC Navigator to create a connection.

Setpoint low clamp and setpoint high clamp are identical: <type> <tag no.>

The setpoints of the PID are equal.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to specify the setpoint clamp.

Sheet contains different 'Design FLDs' version: <version>

The FLD was drawn using an older version of the FLD design editor.

Convert the FLD by editing the FLD in the current FLD design editor.

Sheet contains different symbol editor version: <version>

The FLD was drawn with a symbol library which is specified with an old symbol library editor.

Convert the FLD by editing the FLD in the current FLD design editor.

Sheet contains too many <symbol type>

Too many symbols of <symbol type> have been placed on an FLD.

Use the 'Design FLDs' of FSC Navigator to remove some symbols.

Sheet is corrupted: <FLD no.>

The FLD contains corrupted data.

Use the 'Design FLDs' option of FSC Navigator to load the FLD, and then delete it.

Sheet reference not found on both sheets: <tag no.>

The sheet reference with <tag no.> is found on one FLD but not on a referenced FLD.

Use the 'Design FLDs' option of FSC Navigator to replace the reference.

Sheet reference to non-existing sheet: <FLD no.>

The specified FLD is undefined or cannot be found on disk.

**Sheet too big or nesting too deep. Sheet time =**

The execution time of the FLD is too large because function blocks have been nested too deep or there are too many time-consuming functions.

Use the 'Design FLDs' option of FSC Navigator and change the FLDs so that fewer functions or other functions are needed. Change the nesting of the function blocks, since too many function block calls can cause the execution time to become too large. Split the FLD into two or more FLDs. Particularly time-consuming functions are square root and PID.

Sheet transfer expected to be directly connected to I/O symbol. <type> <tag no.>

You defined a service text equal to the tag number of an I/O symbol, but no direct connection exists. You must change the service test or make a direct connection.

Signal conversion not allowed (type mismatch): Node(x, y)

The result of the calculation is not float, but one of the signals at the input side of the calculation symbol has type float. This means that a conversion from float to long would be necessary. But this conversion means a loss of information, which is not allowed.

Use the 'Design FLDs' option of FSC Navigator and change signal type(s).

Signal type does not match number of bits: <type> <tag no.>

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) and change signal type of the variable.

Signal type undefined. Signal cannot be saved or loaded: <type> <tag no.>

The signal type of the variable is undefined.

Use the 'System Configuration' option of FSC Navigator to specify the type of the variable.

Specified communication area(s) and/or alarm area(s) show overlap.

The compiler has detected an error in the database for the reserved areas of alarm groups or communication channels.

Use the 'System Configuration' option of FSC Navigator (rebuild option).

Single fault in single I/O part may cause system shutdown.

The system contains a single I/O part. If a severe error is detected, the possibility of a system trip exists.

Specified communication area(s) and/or alarm area(s) have illegal range.

The compiler has detected an error in the database for the reserved area specification of alarm groups or COM channels.

Use the rebuild feature of the 'System Configuration' option of FSC Navigator.

Specified SER format ignored, default format assumed.

The configured SER format file is not allowed.

Use the 'System Configuration' option of FSC Navigator to specify a proper SER channel.

Use the 'SER Format' option of FSC Navigator to specify a proper printer configuration.

System contains too many <symbol type>

Too many symbols of <symbol type> have been used in the application.

Use the 'Design FLDs' option of FSC Navigator to remove some symbols.

Timer allocation error <type> <tag no.> Byte number: <byte no.>

The timer has an address <Byte number> which is used by one or more other timers.

Use the 'System Configuration' option of FSC Navigator (rebuild option) to check the addresses of all variables.

**Too many appl. connected for COM channel (Central Part, COM module, Chan):**

<CP> <Seq.> <Chan.>

Channel <Chan.> from the communication module in Central Part <CP> with sequence number <Seq.> has been specified as an external point-to-point FSC FSC protocol, but too many external applications have been specified.

Use the 'System Configuration' option of FSC Navigator to specify only one communication link, or change the point-to-point link to a multidrop link (Install \ Central Part \ COM).

Too many linked variables specified for: <type> <tag no.>

The variable <type> <tag no.> has been allocated to a TFS_COM1 alarm group and should have only one link variable.

Use the 'System Configuration' option of FSC Navigator to rebuild the database.

Too many links configured between two systems: <system no.>

More than one physical path exists between the current and the specified external application.

Use the 'System Configuration' option of FSC Navigator to remove links to the specified external application.

Too many masters specified for (CP,COM,Ch): <CP No.> <Com No.> <Chan.>

The specified communication channel has been linked to several applications, but too many masters have been specified.

Use the 'System Configuration' option of FSC Navigator to remove master applications from the specified channel.

Too many registers specified for on-line modification.

The application is translated using on-line modification, but the database contains too many register bytes.

Use the 'System Configuration' option of FSC Navigator to change on line modification to 'No', or to rebuild with address reallocation.

Too many signal lines connected to one symbol: Node(x, y)

The number of signal lines to one symbol is limited by the compiler. See symbol description in Section 6 of the FSC Software Manual ("Functional Logic Diagrams").

Change the functional logic diagrams by splitting the symbol into two symbols.

Too much information to fit on EPROM set(s).

The generated files will be too large for the current configuration.

Use the 'System Configuration' option of FSC Navigator to expand the number of MEM modules, or change the location of the application program to memory modules.

Top and bottom values are identical: <type> <tag no.>

The range of the variable <type> <tag no.> has not been defined.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to specify the top and bottom value of the variable.

Total number of warnings generated during translation: <number>**Total number of errors detected during translation: <number>****'Translate application' program out of space.**

The number of mnemonics or the number of found symbols in the FLD cannot be stored internally.

Try to optimize the FLD with regard to the number of symbols needed for particular functions.

Try to change the partition of your total functional logics design in such a way that fewer functions are needed per FLD.

**Translating for simulation mode 23125/1/1.**

This message is used to show that the application is translated as a simulated application to debug the application program (see subsection 8.3.3).

Translating for simulation mode 23125/R/1.

This message is used to show that the application is translated as a simulated application to debug the application program (see subsection 8.3.3).

Translating for simulation mode 23126/1/1.

This message is used to show that the application is translated as a simulated application to debug the application program (see subsection 8.3.3).

Translating for simulation mode 23250/1/1.

This message is used to show that the application is translated as a simulated application to debug the application program (see subsection 8.3.3).

Translating for simulation mode 23260/1/1.

This message is used to show that the application is translated as a simulated application to debug the application program (see subsection 8.3.3).

Translating for simulation mode 23251/1/1.

This message is used to show that the application is translated as a simulated application to debug the application program (see subsection 8.3.3).

Translation completed.

The compiler has checked all FLDs and all data in the databases without encountering serious errors.

Translation log file: <appl. name> <date> <time>

Heading of log file.

Unable to create all dummy sheets. Number of unplaced dummy sheets: <no. of FLDs>.

In order to extend the total execution time to a certain minimum, some delay sheets must be appended to the application program. However, no space can be found between the program blocks and the function blocks.

Move the function blocks to higher FLD numbers. Renumber the first function block to create more unused sheet numbers lower than the lowest function block number.

Undefined sheet reference: <type> <tag no.>

Reference to sheet 0 or from sheet 0, which has not been defined.

Use the 'Design FLDs' option of FSC Navigator and define sheet reference.

Unexpected I/O module at (rack, position, channel): (<Rack>, <Pos.>, <Chan.>) <type> <tag no.>

The variable has been allocated to an invalid module type (e.g. a digital input variable has been allocated to an analog output module).

Use the 'System Configuration' option of FSC Navigator to reallocate the variable.

Unknown printer reference.

The printer specified in a report file or in the SER format file is not known by the compiler.

Use the 'SER Format' option of FSC Navigator to install another printer.

Unnecessary signal conversion. Node(x,y)

A signal conversion is performed but the source signal is equal to the destination signal.

Use the 'Design FLDs' option of FSC Navigator to remove this signal conversion.

**Value (or base) in sheet and variable data file are not equal: <type> <tag no.>**

The value of the variable (timer or counter) has been changed in the database but not in the FLD.
Use the 'Design FLDs' option of FSC Navigator to change the variable.

Value signal must be connected.

A binary input for this type of timers must be connected.
Use the 'Design FLDs' option of FSC Navigator to create a connection.

Value signal must not be connected.

A binary input for this type of timers may not be connected.
Use the 'Design FLDs' option of FSC Navigator to remove the binary input.

Variable address is not assigned: <type> <tag no.>

The database contains a variable which has not been assigned to a communications channel, and has an illegal address.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to assign the variables with location 'COM' or location 'FSC' to a communications channel.

Variable address mismatch with current appl.: <type> <tag no.> (<Linked appl. name>)

The relative address of the variable specified does not match the relative address in the current application. This will occur if, for example, I FSCvar has relative address 3 in this application, but O FSCvar does not have relative address 3 in the application specified.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs option) to change the address, or use Configuration\Module\Central Part\COM\Copy allocation to make all addresses used in FSC networks equal.

Variable address not within communication area: <type> <tag no.>

The variable type tag number has an address which is not within the range of the COM channel to which it has been allocated.

Use the 'System Configuration' option of FSC Navigator (Hardware Specs or Rebuild option) to recover.

Variable data file too large. Try 'Rebuild' option to optimize file.

The data file cannot be extended anymore.
Rebuilding may create some space.

Variable not found in variable data file: <type> <tag no.>

The variable found in the FLD is not found in the data file. The compiler will try to regenerate the variable. The result of that action is logged.

Use the 'Design FLDs' option of FSC Navigator (Symbol \ Change) to enter the required data. The variable will then be placed in the database. If the variable is a repeated input, you must first place the actual input. It is possible that the data was deleted when using the 'System Configuration' option of FSC Navigator. Use this option to (re-)enter the missing variable.

Variable is not SER enabled: <type> <tag no.>

The variable is used for a report, but is not SER-enabled.

Use the 'System Configuration' option of FSC Navigator to enable the SER, or use the 'SER Format' option of FSC Navigator to remove the variable from the condition of a report.

Variable is SER enabled, but no DBM available: <type> <tag no.>

Use the 'System Configuration' option of FSC Navigator to define a DBM in the system, or reset SER enable of the specified variable.



Variable is SER enabled, but no valid SER communication channel found: <type> <tag no.>

No COM channel has been specified to be used for SER.

Use the 'System Configuration' option of FSC Navigator to specify the SER channel (Install\Configuration).

Variable not linked to current application: <type> <tag no.> (<linked appl. name>)

The specified variable in the other application database does not exist, but is linked to another FSC system. For example, TEST_1 in the current application is a variable I Link1, which is linked to TEST_2. There is a variable O Link1 in application TEST_2, but this variable is linked to system 3.

Wrong number of input signals: Node(x, y)

The number of signals at input side of the symbol does not match the expected number of signals.

Run the 'Design FLDs' option of FSC Navigator to change.

Wrong number of output signals: Node (x,y)

See the symbol description in Section 6 of the FSC Software Manual ("Functional Logic Diagrams").



Left blank intentionally.



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Section 9: Log Files

Version 500

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Total Plant



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9.1 Introduction

Several options of FSC Navigator create a log file to report the results of specific functions. The following log files exist:

- | | |
|--|--|
| – Translation log file (*.LOG): | This file contains all messages, errors and/or warnings that were generated during translation of the application. |
| – Verification log file (*.VRF): | This file contains all messages, errors and/or warnings that were generated during verification of the application. |
| – Append log file (*.APP): | This file contains all changes and/or errors that were generated when two or more projects were appended to the currently open project. The append log file is generated for the currently open project. |
| – Conversion log file (*.FAD): | This file contains information about required changes and errors that were detected during conversion of the application to the current release. |
| – Database integrity log file (*.CRC): | This file contains information about CRC errors that are detected by FSC Navigator, and is used to verify the integrity of the FSC databases. |
| – On-line rebuild log file (*.OLR): | This file contains all messages that were generated during an on-line rebuild. |
| – Menu selection log file (CRC.LOG): | This file lists any errors that occurred (e.g. CRC errors) when the various FSC Navigator application files were called from the main screen. |

All log files are project-related and are located in the corresponding project folder. An exception is the menu selection log file (CRC.LOG), which is project-independent and can be found in the FSC Navigator application folder (C:\FSC by default).

The contents of the log files can be viewed, printed or saved to disk. The 'View Log' option of FSC Navigator can be opened in three ways:

- **Menu bar:** File / View Log
- **Button bar:** View Log
- **Navigation area:** FSC Safety Compliance / View Log

9.2 Selecting a log file

If you select the 'View Log' option, a submenu will appear, which allows you to choose which log file you wish to view:

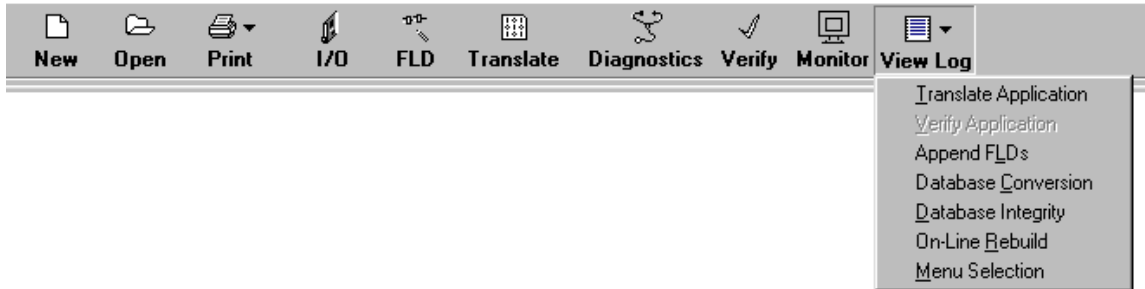


Figure 9-1 Button bar, with 'View Log' submenu

Only the log files that are actually available for viewing will be valid menu options. The submenu as shown in Figure 9-1, for example, shows the 'Verify Application' option disabled. This means that the verification log file is not available for viewing, either because verification has not yet taken place or because verification did take place but no errors were found (and logged).

Note:

The log files will always relate to the project that is currently open. If you wish to view the log files of a different project, you must first close the current project and open the project whose log files you wish to view.



9.3 Log file viewer

After a log file has been selected, the log file viewer will open and show the requested log file. Figure 9-2 shows the log file viewer with an example of the verification log file.

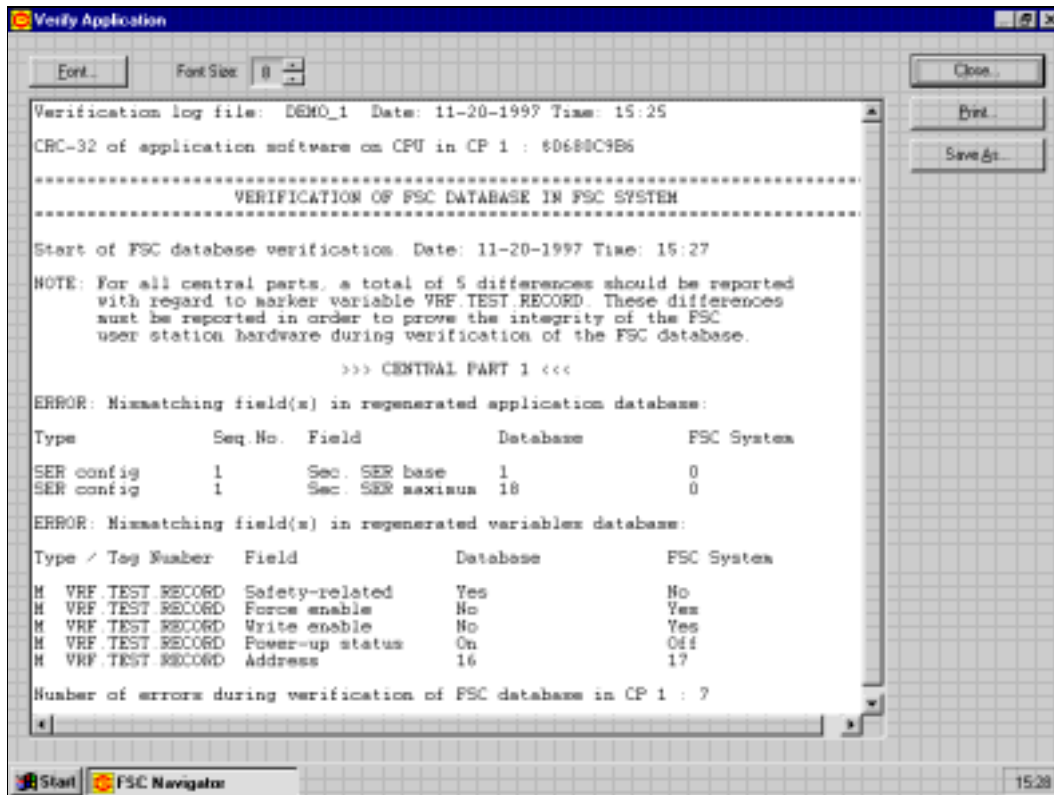


Figure 9-2 Log file viewer with verification log file

The log file viewer has a number of options that allow you to modify the font settings (see subsection 9.3.1), as well as buttons for printing or saving the log file (see subsections 9.3.2 and 9.3.3).

If the data contained in the log file does not fit in the viewer window, horizontal and/or vertical scroll bars are provided which enable you to view the entire log file.

Please note that the log file viewer window is resizable, which means that you can use the mouse to "drag" the window edges to change the window size.

9.3.1 Font type and size

By default, the log files will be shown in the Courier New font with font size 11 pt. If you want, you can modify these font settings. FSC Navigator "remembers" the font settings, which means that the log file viewer will always open with the font settings that were used during the last session.

To change the font type, choose the Font button in the upper left corner of the log file viewer. This will open the window as shown in Figure 9-3. Here you can choose the font type and style that you wish to use for viewing and printing the log files.

Notes:

- In the font selection list, only non-proportional fonts will be displayed. This means that all letters will have the same width, regardless of their shape ('i' vs. 'm').
- The font type and style that you choose will be used for both on-screen viewing and printing.

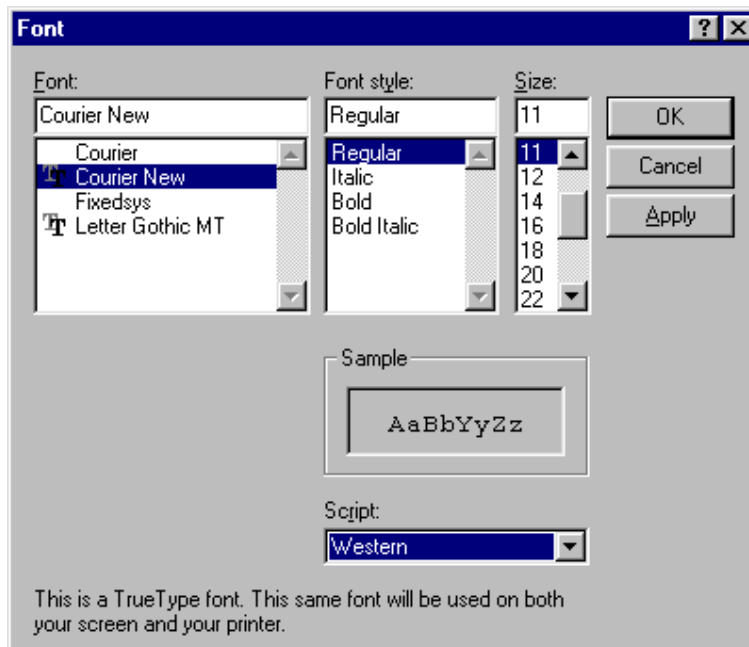


Figure 9-3 Changing the font type and style

To change the font size, you can use the Size selection list in the Font window (see Figure 9-3), or you can use the Font Size spin button in the log file viewer (see Figure 9-4).

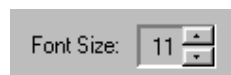


Figure 9-4 Font size spin button

The default font size is 11 pt, but you can choose any value between 8 pt and 24 pt. Please note that the font size that you choose will be used for both on-screen viewing and printing.



9.3.2 Printing a log file

The Print button in the log file viewer allows you to print the log file currently being displayed. If you choose the Print button, the standard Windows print dialog will open:

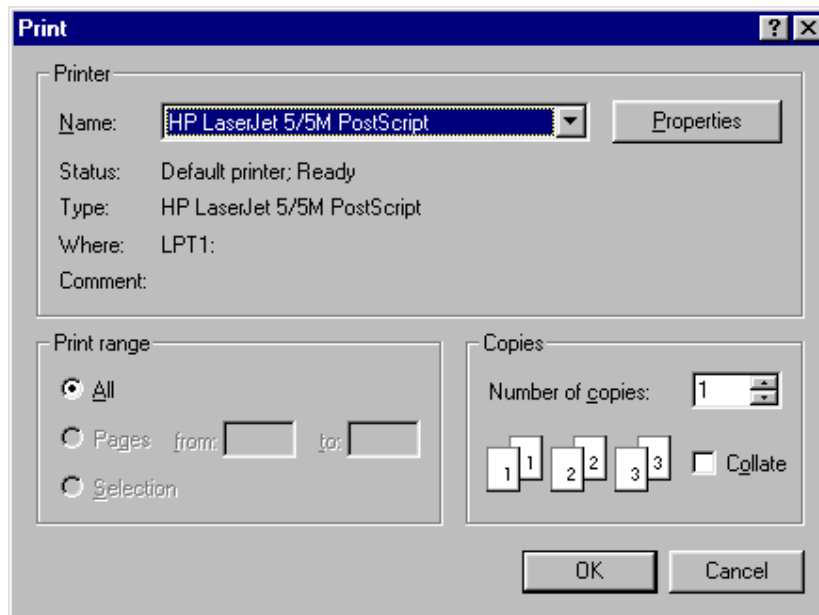


Figure 9-5 Printing log files

Note:

The Print dialog may look slightly differently than shown in Figure 9-5, depending on the exact Windows version that you are running.

The printer shown will be the printer that has been defined as the default printer in Control Panel. Make sure that you choose a printer from the drop-down list that is compatible with the printer settings as defined in FSC Navigator (using the Printers option of the File menu). If it is not, printing may produce garbled output, or printing may not be possible at all.

For details on setting the printer options in FSC Navigator, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the FSC Navigator Help file.

If you want to change the paper size that should be used for the print job, you can use the Properties button to modify the printer settings.



9.3.3 Saving a log file to disk

The Save As button in the log file viewer allows you to save the log file currently being displayed to disk. If you choose this button, the standard Windows Save As dialog will open (see Figure 9-6).

To save the log file to disk, move to the folder you wish to save the file to, and enter a valid file name in the File Name text box. The default file extension is 'LOG'. This extension is assumed, which means that you do not have to enter it. However, if you add a different extension to your file name, this will be the extension that the file will have. Thus, if you enter MYFILE as the file name, the log file will be saved as MYFILE.LOG. If you enter MYFILE.TXT, it will be saved as MYFILE.TXT.

Log files are saved to disk as plain ASCII files, which means that they can be opened in any word processor or any other application that has text-processing capabilities.

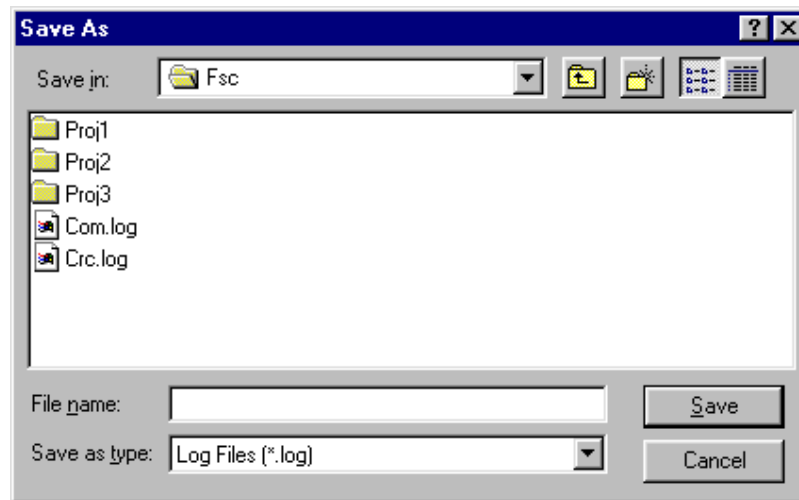


Figure 9-6 Saving a log file to disk



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Section 10: Programming EPROMs

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10.1 Introduction

The application logic that was created using the functional logic diagrams (FLDs) must be programmed into EPROMs before it can be used in the FSC system.

The 'Program EPROMs' option of FSC Navigator is used to program, copy or read EPROMs, using the EPROM programmer (07170/.I.) and the PC serial interface + key module (07177/1/.).

Apart from programming of the FSC application EPROMs for the CPU, MEM and COM modules, the program also supports programming of the CPU system program EPROMs.

The 'Program EPROMs' option can be called in two ways:

- **Menu bar:** Project / Program EPROMs
- **Navigation area:** FSC Project Configuration / Load Application / Program EPROMs

On start-up, the program will read the FSCPROM.DAT file in the FSC Navigator application folder. This file contains the default programming configuration. If this file is not present, it will be created automatically. The program will also check whether the EPROM programmer (07170/.I.) has been connected to the FSC user station. The hexadecimal I/O address \$300 is assumed.

If the programmer cannot be found – for example because the address is incorrect or the programmer has not been properly connected – you are asked to select another address or connect the programmer properly (see Figure 10-1). If you change the address setting, it will be stored in the FSC.INI initialization file, which is located in the FSC Navigator application folder (C:\FSC by default).

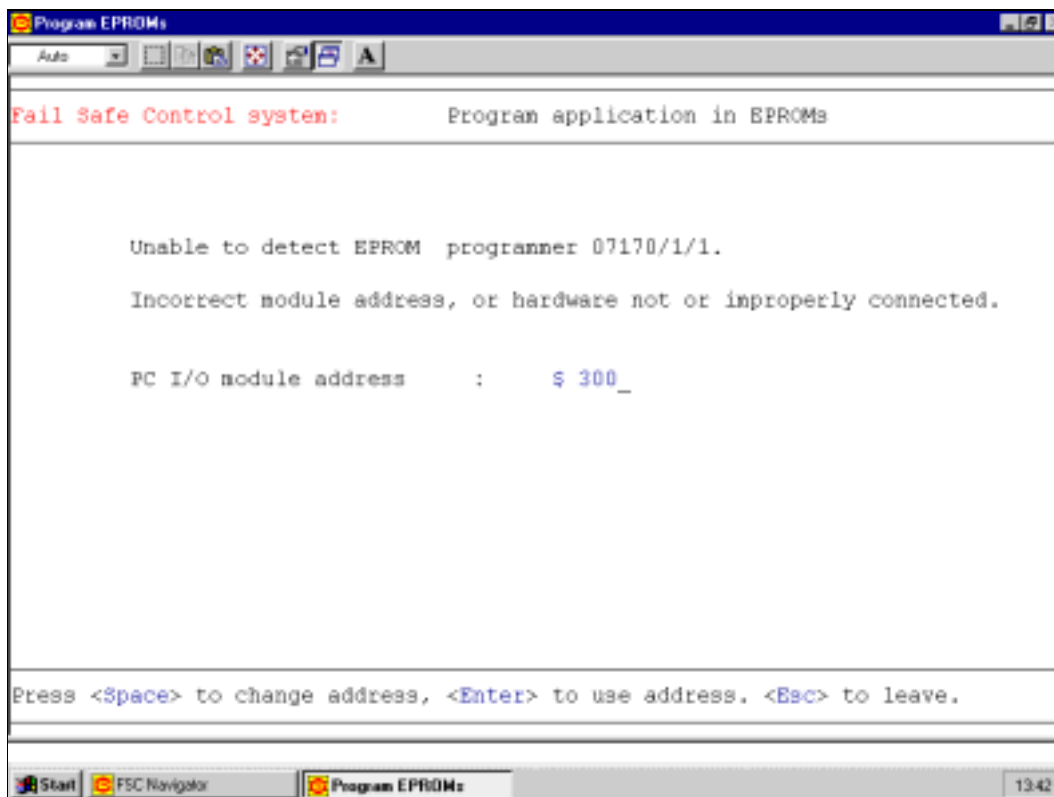


Figure 10-1 Setting the programmer's I/O address



Notes:

1. Switch off the power supply to your PC before connecting or disconnecting the 07170/1/1 module.
 2. Do not place the EPROMs until the 'Program EPROMs' option of FSC Navigator has completed its startup procedure.
-



10.2 Programming EPROMs

10.2.1 Main menu

The 'Program EPROMs' option of FSC Navigator can only be used if the EPROM programmer has been properly connected. If that is the case, the main menu will be displayed on screen, which allows you to program, read, copy and blank-check EPROMs.

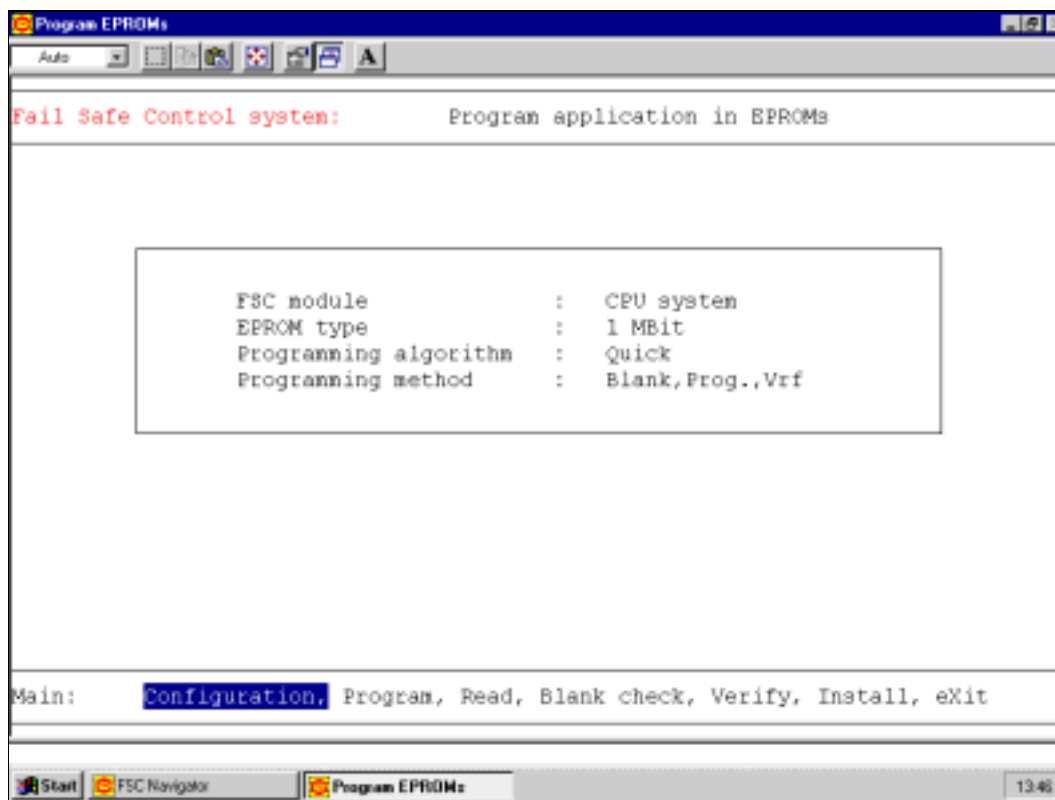


Figure 10-2 Program EPROMs option: main menu

The main menu has seven options:

- Configuration** : Modify the configuration of the 'Program EPROMs' option of FSC Navigator.
- Program** : Program or copy the application into one or more EPROMs.
This option can only be selected if the EPROM type specified matches the size of the file to be programmed.
- Read** : Read one or more EPROMs and write the contents to a file.
- Blank check** : Check whether one or more EPROMs are empty.

- Verify** : Check whether one or more EPROMs contain the same data as a given file.
This option can only be selected if the EPROM type specified matches the size of the file to be programmed.
- Install** : Calibrate the 07170/1/1 module.
- eXit** : Return to the main screen of FSC Navigator.

You can always return to the main menu from anywhere in the program by (repeatedly) pressing <Esc>.

10.2.2 'Configuration' option in main menu

After selecting the **Configuration** option in the main menu, you can modify the 'Program EPROMs' configuration. The following screen will be displayed:

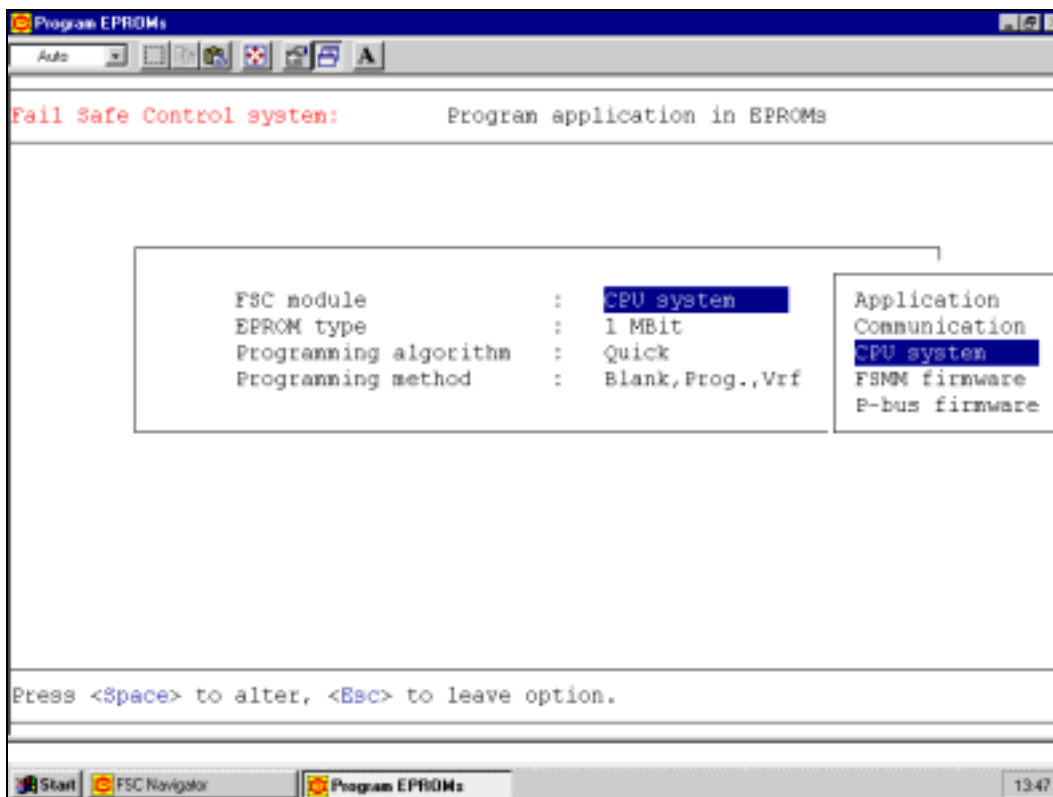


Figure 10-3 Configuring 'Program EPROMs'

You can use <Space> to alter the following options:

- FSC module** : The selection list will display up to five options:
- Application: Work with the application software which is either located on the memory module (Location = MEM) or on the CPU module (Location = CPU).
 - Communication: Work with the communication EPROMs of the application.
 - CPU system: Work with the CPU system software.
 - FSMM firmware: Work with the FSMM firmware.
 - P-Bus: Work with the Hartmann & Braun P-bus.



Notes:

1. Application software, communication EPROMs and CPU system software always require four EPROMs of 0.5, 1 or 2 MB. FSMM firmware requires one EPROM of 4 MB, which must be placed on position 1 of the EPROM programmer.
 2. CPU, COM and MEM can only be selected if valid files exist for these modules.
-

EPROM type	:	Specifies the size of the EPROMs used. Three or four different sizes are supported by the program.
Programming algorithm	:	Can be set to Intel Intelligent and Intel Quick Pulse algorithm (see subsection 10.3).
Programming method	:	Specifies the way the program command will act. Selectable options are: 'Program': used to program EPROMs. 'Blank, Prog., Vrf': used to perform a blank check before programming and a verify after programming automatically.

The program prevents you from creating an invalid configuration. If you alter one of the options, the other options are automatically updated to show only valid configurations. It may also be that some of the options are blocked once a certain configuration has been chosen. Therefore, always start at the top if you want to alter the configuration.

10.2.3 'Program' option in main menu

If the FSC module type has been set to 'Communication' or 'Application', FSC Navigator will search for all possible files to select. If more than one file is available, you will be prompted to toggle the Central Part number and/or the module sequence number (e.g. COM1, COM2, CPU1, MEM1, etc.). If the FSC module type is 'Application' and the location of the application is 'MEM', more than one set of EPROMs may be required. In that case you are prompted to specify the first set of the EPROMs to program. The default size which has been defined is used to program or verify the current application software. The size can be changed if you want to read a set of EPROMs. If the default is changed, the 'Program' and 'Verify' options are deleted from the main menu.

If the FSC module type is 'Communication', FSC Navigator will check if the COMxxx.BIN (where 'xxx' is the FSC software release number), CTV.BIN and MTV.BIN files are present in the FSC directory. If they are not, an error message is displayed. If they are present, FSC Navigator will compose one file from these files together with the application file. The resulting file will be programmed to the EPROMs.

If the FSC module type is 'Application' and the location of the application is MEM, FSC Navigator will calculate the number of EPROM sets (one set is 4 EPROMs) that are required to hold the amount of data. If one set of EPROMs is filled with data, you place the next set of EPROMs until all data has been programmed. Every time you need to place a new set of EPROMs, a message is displayed on your screen which tells you to do so (see Figure 10-4). In this way all data can be programmed.

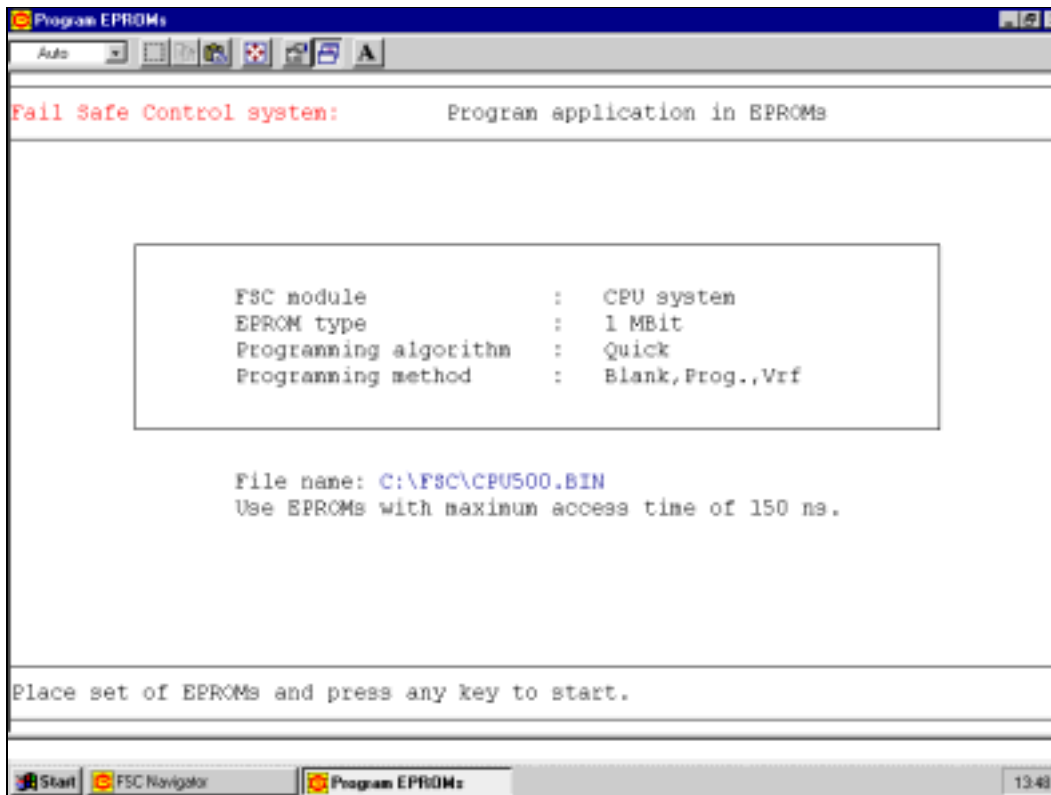


Figure 10-4 Placing a set of EPROMs

Before programming or verifying is started, the checksum (CRC) of the file is calculated and verified to detect any data corruption.

Before FSC Navigator starts programming the EPROMs, a check is made whether the EPROMs have been correctly inserted and whether they are of the correct type (27512, 27010, 27C2001 or 27C4001) and manufacturer (see subsection 10.4). You must insert the EPROMs correctly before you can start programming (green LEDs on). If the EPROMs are not of the correct type or manufacturer, a message is displayed on screen. (For a list of supported EPROM types refer to subsection 10.4.) During programming, the screen will show the address currently being programmed. This will give you an idea how many bytes have been programmed.

Programming can stop in three ways:

1. Programming was successful.
2. Programming crashed.
3. You pressed <Esc> during programming.

The way in which programming stopped will be displayed on screen. If programming crashed (i.e. unable to program the desired data), the address where the programming crashed is also displayed. If your file format is 'FSC' and programming was successful, the address of the last bytes which were programmed in your last EPROM set is also given.



10.2.4 'Read' option in main menu

The 'Read' option allows you to read a set of EPROMs and put the contents in a file.

Before FSC Navigator starts programming the EPROMs, a check is made whether the EPROMs have been correctly inserted and whether they are of the correct type (27512, 27010, 27C2001 or 27C4001) and manufacturer (see subsection 10.4). You must insert the EPROMs correctly before you can start programming (green LEDs on). If the EPROMs are not of the correct type or manufacturer, a message is displayed on screen.

You must first enter a valid file name before FSC Navigator can start writing the EPROM contents to a file (see Figure 10-5).

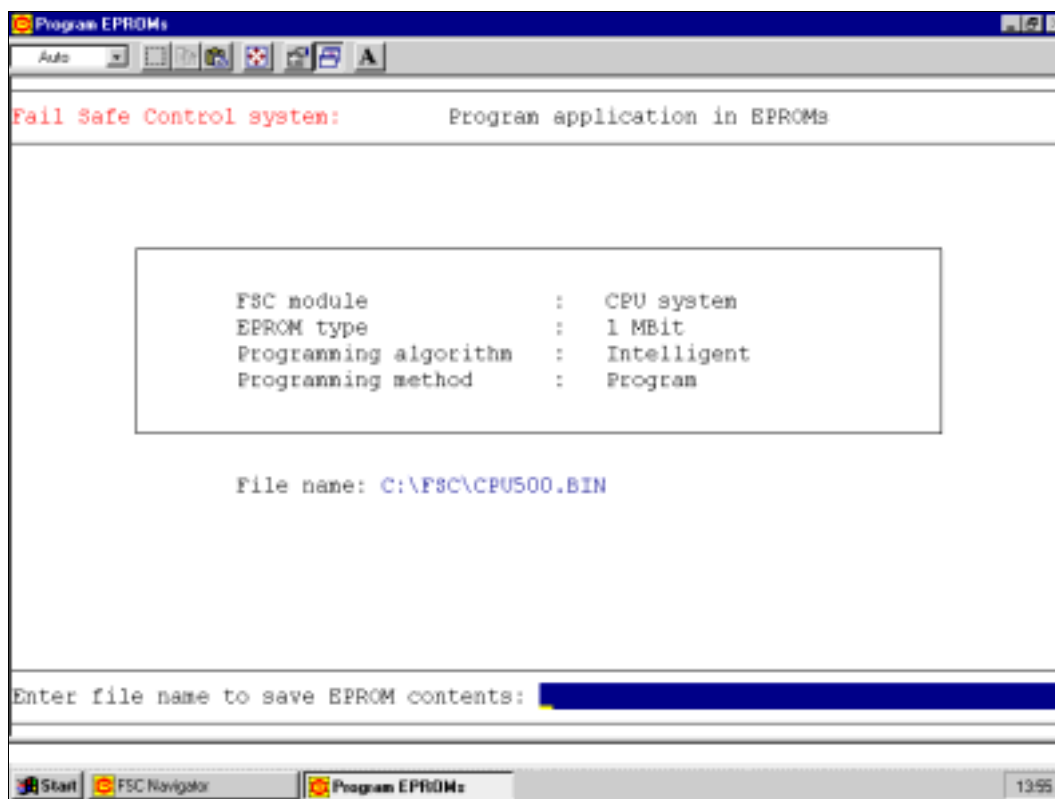


Figure 10-5 Saving EPROM contents to file

10.2.5 'Blank check' option in main menu

The 'Blank check' option allows you to perform a test to check whether the EPROMs are empty. If an error occurs, FSC Navigator will tell you in which EPROM the error occurred and at what address.

10.2.6 'Verify' option in main menu

The 'Verify' option allows you to check whether the data in the EPROMs is the same as the data in a file. Before verification starts, you must first enter a file name. The data contained in the EPROMs is compared with the data in that file. If the data is not the same, the address in the EPROM, the data contained in the EPROM and the data in the file are displayed on screen (see Figure 10-6).

Program EPROMs

Auto File Edit Format Help A

Fail Safe Control system: Program application in EPROMs

Address EPROM	File	Address EPROM	File
0900D 47 40 47 40	35 50 35 50	09092 05 46 05 46	00 F3 00 F3
0900E 47 40 47 40	35 70 35 70	09093 34 9B 34 9B	7F 6A 7F 6A
09011 47 40 47 40	35 8A 35 8A	09101 ED C8 ED C8	5D AA 5D AA
09013 47 40 47 40	35 A0 35 A0	09103 EE 0E EE 0E	5D F0 5D F0
09015 47 40 47 40	35 AA 35 AA	09105 EE 52 EE 52	5E 34 5E 34
09017 47 40 47 40	35 C0 35 C0	09107 EE 98 EE 98	5E 7A 5E 7A
09019 47 40 47 40	36 8E 36 8E	09109 EE 98 EE 98	5E 7A 5E 7A
0901B 47 40 47 40	36 96 36 96	0910B EE 98 EE 98	5E 7A 5E 7A
0901D 47 40 47 40	36 9E 36 9E	0910D EE 98 EE 98	5E 7A 5E 7A
0901F 47 40 47 40	36 A6 36 A6	0910F EE 98 EE 98	5E 7A 5E 7A
09084 00 00 00 00	11 00 11 00	09111 EE 98 EE 98	5E 7A 5E 7A
09086 BB D8 BB D8	5B BA 5B BA	09113 EE 98 EE 98	5E 7A 5E 7A
09089 04 7C 04 7C	03 4D 03 4D	09115 EE 98 EE 98	5E 7A 5E 7A
0908A 68 0C 68 0C	E5 3F E5 3F	09117 EE 98 EE 98	5E 7A 5E 7A
0908F D4 F2 D4 F2	22 3D 22 3D	09119 EE 98 EE 98	5E 7A 5E 7A

Press any key.

Press <ESC> Or Continue

Start FSC Navigator Program EPROMs 13:59

Figure 10-6 Data comparison

10.2.7 'eXit' option in main menu

The 'eXit' option returns you to the FSC Navigator main screen.



10.3 Programming algorithms

There are two available programming algorithms:

- Intel Intelligent, and
- Intel Quick Pulse.

10.3.1 Intel Intelligent algorithm

The duration of an intelligent programming pulse is 1.0 ms. This will be repeated until the desired data is settled in the EPROM. After this, an 'overprogram' pulse will follow. The number of these pulses is three times the number of program pulses needed. For example, if two pulses were needed before a byte was settled, the algorithm will give an overpulse of $3 \times 2 = 6$ ms.

10.3.2 Intel Quick Pulse algorithm

The Quick Pulse algorithm is different from the Intelligent algorithm in two ways:

- the duration of the programming pulse is not 1.0 ms but 0.1 ms, and
- there is no overprogram pulse.

10.3.3 Algorithm implementation

The 'Program Application' option of FSC Navigator does not execute these algorithms exactly as they are described above.

- The overprogram pulse of the Intel Intelligent algorithm is replaced by a fixed number of three (separate) pulses of 1 ms.
- The Intel Quick Pulse algorithm uses no overprogram pulse. FSC Navigator will, however, nevertheless give three overprogram pulses of 0.1 ms each.

10.4 Supported EPROMs

Table 10-1 below lists the EPROMs that are supported by the 'Program Application' option of FSC Navigator.

Table 10-1 EPROMs supported by 'Program Application' option of FSC Navigator.

Manufacturer	Type	Size	Remarks
Intel	27512	512 K	
	27010	1 M	
NEC	27C512	512 K	
	27C1001	1 M	
	27C2001	2 M	Only programmable with Intel Quick Pulse.
	27C4001	4 M	Only for programmers produced after June 1, 1990. The 'Program Application' option of FSC Navigator detects if this EPROM type can be programmed with the programmer used. Only programmable with Intel Quick Pulse.
AMD	27512	512 K	
	27C010	1 M	
	27C020	2 M	
	27C040	4 M	
Toshiba	571000D	1 M	
	574000D	4 M	Only for programmers produced after June 1, 1990. The 'Program Application' option of FSC Navigator detects if this EPROM type can be programmed with the programmer used. Only programmable with Intel Quick Pulse.
WSI	27010	1 M	Only programmable with intelligent mode.
Honeywell Safety Management Systems B.V.	27C010 on 10002/A/1 module	1 M	Only programmable with 07170/A/1 adapter module and Intel Intelligent programming mode.



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Helping You Control Your World



Fail Safe Control

Section 11: Verifying an Application

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Revision 01 (01/98)



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11.1 Introduction

The 'Verify Application' option of FSC Navigator verifies if the FSC application software as present in the FSC system is consistent with the FSC database and the functional logic diagrams (FLDs) as stored on the FSC user station. The 'Verify Application' option can be called in three ways:

- **Menu bar:** On-Line / Verify Application
- **Buttons bar:** Verify
- **Navigation area:** FSC Safety Compliance / Verify Application

Differences between the application at the FSC user station and the application software in the FSC system may be the result of:

1. changes made to the application at the FSC user station without transferring the new application to the FSC system,
2. corruption of the FSC database or the functional logic diagrams (FLDs), or
3. errors occurring in the complete path from configuring the application to programming the FSC system EPROMs or downloading the application.

In the first case, the 'Verify Application' option of FSC Navigator may be used as a revision control tool in order to register the modifications. In the other two cases, it is used for safety reasons to ensure the correctness of the FSC application software with respect to the FSC application at the FSC user station.

All differences will be reported and stored in a log file for later reference. The log file can be viewed and printed using the 'View Log' option of FSC Navigator.

The FSC application software may be configured for 'RUN' or 'RAM' mode. The 'Verify Application' option of FSC Navigator cannot verify applications that have been configured for simulation mode, for use in one of the FSC simulation units.



11.2 Conditions for verification

To verify the FSC application software, the following conditions must be met:

- All functional logic diagrams of your application must have been translated, i.e. the 'Translate Application' option of the FSC Navigator must have been performed successfully.
- The FSC user station must be able to communicate with the FSC system.
- An application must be present in the FSC system (programmed in EPROM or loaded into RAM).
- The power-on mode of the FSC system and of the application at the FSC user station must either be 'RUN' mode or 'RAM' mode. Simulation modes are not supported.
- The configuration of the FSC system must match the FSC configuration as defined in the FSC database. In other words, you cannot use a single (i.e. non-redundant) application to verify a redundant FSC system and vice versa.
- There must be sufficient free disk space to store intermediate results of the verification process. The required amount of disk space is approximately the disk space needed to store the compiler output (.Oxx and Cxx files).
- **The version of the application on the CPU/MEM module must match the version of the application on the COM module, i.e. the EPROMs of the COM module must have been programmed at all times.**



11.3 Verification process

To verify an application, the FSC application software as present in the FSC system is uploaded to the FSC user station. The FSC application software is temporarily stored in the project folder and will be deleted before the program terminates.

The time required to upload the FSC application software depends on the configuration, the size of the application and the baud rate of the configured Development System protocol.

During communication, an asterisk (*) is displayed in the upper right corner of the screen. The progress of the upload function is visualized by displaying the percentage of each file that is being uploaded to the FSC user station.

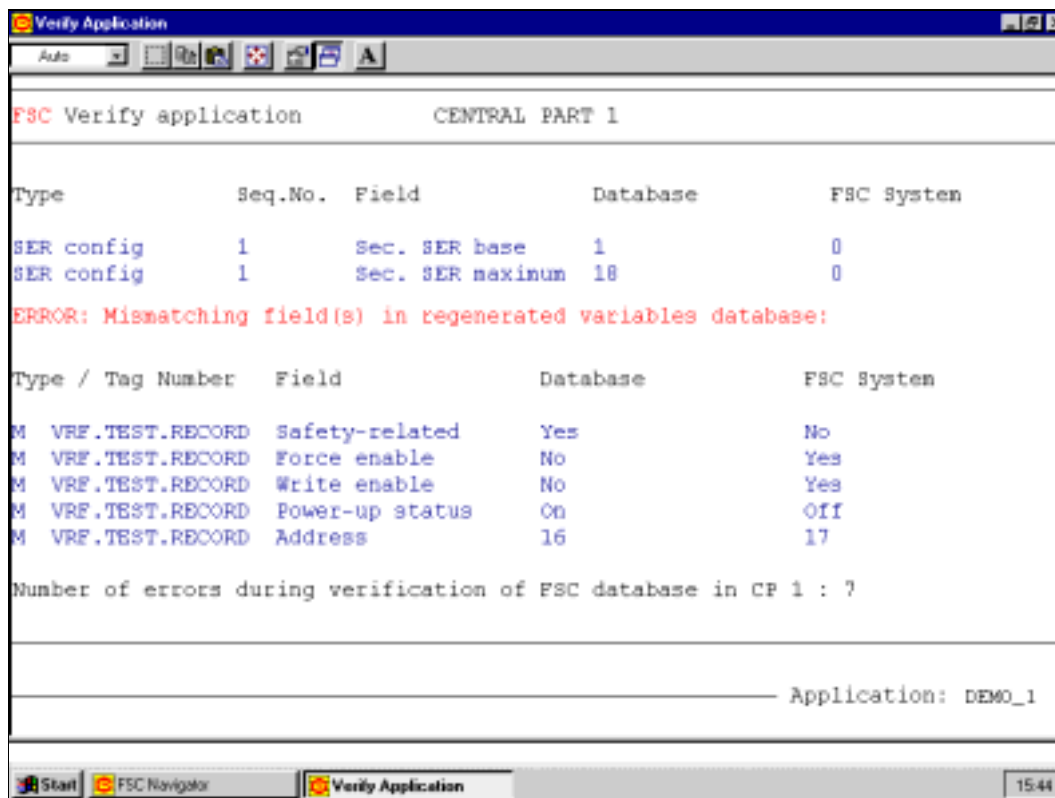
As soon as all FSC application software has been uploaded, the program will start verifying the application. Verification of an application is divided into two main functions:

1. Verification of the FSC application software with regard to the information in the FSC database.
2. Verification of the FSC application software with regard to the functional logic diagrams (FLDs).

All differences that are found between the application stored at the FSC user station and the FSC application software are reported.

Figure 11-1 shows an example of the output to the screen during verification of the FSC database. In this case, a difference was found in the numerical SER configuration.

The following subsections describe the two main functions of the verification program in more detail.



The screenshot shows a window titled 'Verify Application' with a menu bar (Auto, File, Edit, View, Options, Help) and a toolbar. The main content area displays the following text:

```
FSC Verify application          CENTRAL PART 1
```

Type	Seq.No.	Field	Database	FSC System
SER config	1	Sec. SER base	1	0
SER config	1	Sec. SER maximum	10	0

ERROR: Mismatching field(s) in regenerated variables database:

Type / Tag Number	Field	Database	FSC System
M VRF.TEST.RECORD	Safety-related	Yes	No
M VRF.TEST.RECORD	Force enable	No	Yes
M VRF.TEST.RECORD	Write enable	No	Yes
M VRF.TEST.RECORD	Power-up status	On	Off
M VRF.TEST.RECORD	Address	16	17

Number of errors during verification of FSC database in CP 1 : 7

Application: DEMO_1

The taskbar at the bottom shows 'Start', 'FSC Navigator', and 'Verify Application' (active), with a clock showing 15:44.

Figure 11-1 Example of output to screen



Verification of an application can be terminated at any time by pressing <Esc>. The program will report a manual interrupt, and will write the time and date of interruption to the log file. If one or more differences were found or if an error occurs which does not allow further program execution, you are asked to press <Enter> to return to the FSC Navigator main screen. If no differences were found, the program will automatically return to the the FSC Navigator main screen.

11.3.1 Verification of FSC application using the FSC database

This part verifies the information in the FSC database with the application software in the FSC system.

Differences between the FSC database and the FSC application software are preceded by an error message in red if it is the first difference found within the current Central Part. In addition, a descriptive header in black is generated. The actual difference is displayed in blue.

If a difference is detected in a field that affects related information, only this field is reported. For this reason, when you decide to correct the difference and verify the application for the second time, additional differences may be reported. For example, if differences are detected in the characteristics of a specific communication channel (protocol, interface, baud rate, etc.), then only the protocol is reported. If the protocol is subsequently made identical to the protocol as defined within the FSC system, the interface and baud rate may be reported after a second verification.

Verification of the FSC database is performed once for every Central Part of the FSC configuration.

11.3.1.1 Restrictions

This subsection provides an overview of the data in the FSC database that will not be verified.

Application database

- If the interface of a communication channel is RS485, the program does not verify the network configuration in the FSC database which can be 'multidrop' or 'point-to-point.'
- The program does not distinguish between Development System protocol and Modem protocol.
- The program does not verify which rack numbers are controlled by which HBD.
- The program does not verify any information regarding PSU or PSD modules.
- The program does not verify user-defined rack numbers.
- Descriptive installation texts are not verified.
- The maximum number of cabinets and racks are not verified.
- The program does not verify the number of slots on the Central Part bus.
- CPU and COM EPROM types are not verified.
- The program does not verify the maximum numerical SER number.
- Minimum and maximum FLD execution times are not verified.
- The program does not verify application names of the other FSC systems connected to a network.

Note:

If a communication channel has been configured as a master channel, but no data is exchanged with its slave systems (which means that the system actually serves as a communication server), the program will report this as a difference.



Variables database

- The program does not verify service fields of register variables, multiplexer variables and multiple input and output variables.
- Service and qualification fields of input and output variables and marker variables will only be verified if the appropriate information is present in the FSC application software.
- User-defined locations of variables (non-reserved) will not be verified.
- The signal type of variables BI and BO (BCD or normal) will not be verified.
- The program does not verify the signal type of analog input variables.
- Strings specifying the Engineering Units of several parameters are not verified.
- The program does not verify top and bottom values of PID variables.

Note:

The 'On-Line Environment' option of FSC Navigator can be used to change the parameters of PID variables, the power-up value of variables with location COM, etc. These new values are stored in the FSC database. As long as the new values are not transferred to the FSC system, the differences will be reported by the 'Verify Application' option of FSC Navigator.

11.3.2 Verification of the FSC application software using the functional logic diagrams

After verification of the contents of the FSC database, the functional logic diagrams (FLDs) are verified.

All differences between the FLDs and the FSC application software are displayed in red. If the connections on a functional logic diagram do not match, the connection is further specified by a description in blue. Any differences found will be output to the screen and to the log file.

11.3.2.1 Restrictions

This section provides an overview of the data that will not be verified and of some logic sequences that are not supported.

- The program does not verify the signal type of binary signals (byte, word, long or float).

11.3.3 Revision control

The 'Verify Application' option of FSC Navigator is very useful as a revision control tool to register the differences between various versions of an application during application development or in case of subsequent application updates.

This feature is available for both single-channel configurations and redundant configurations:

- If application software for use on any FSC system configuration is modified, the 'Verify Application' option of FSC Navigator can be used to register and verify the changes that were made. This is accomplished by running the 'Verify Application' option of FSC Navigator while the previous version of the application software is still present in the FSC system.

- For redundant configurations, additional benefit is achieved when the 'Verify Application' option of FSC Navigator is used during on-line modification (OLM). During OLM, running the program only once is sufficient to both verify the new application in one Central Part and to register the differences with respect to the previous version of the application.
- When Central Part 1 has been loaded with a new version of the application software while Central Part 2 is still running on the previous version of the application software, the 'Verify Application' option of FSC Navigator will perform the following functions:

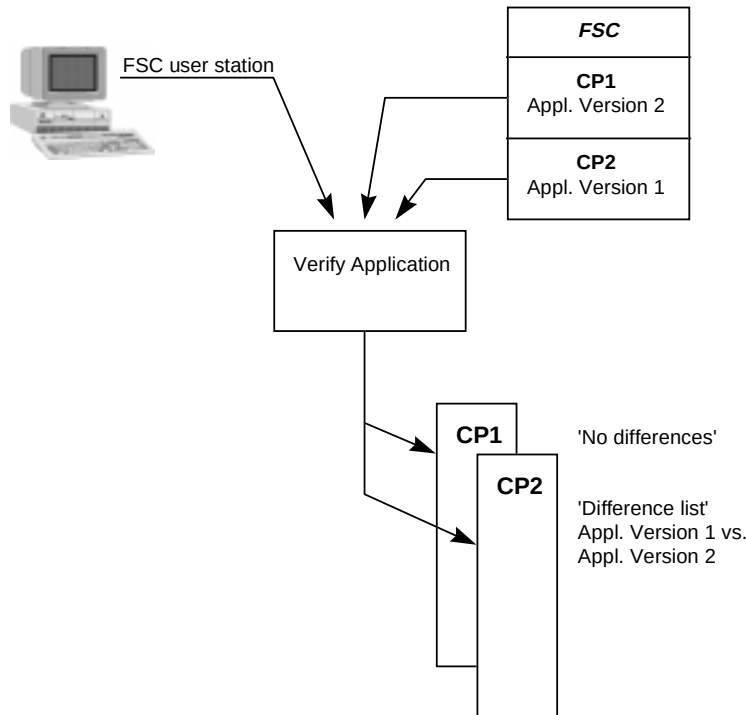


Figure 11-2 Verifying an application during OLM

- The program will verify that there are no differences between the application software at the FSC user station and the application software in Central Part 1. Therefore the program basically serves as a verification program for the application software in Central Part 1. The sections in the log file that refer to Central Part 1 should contain no differences, except for the differences that pertain to the test data for integrity purposes. (For details on test data, refer to the FSC Safety Manual).
- The program will register the differences between the latest version of the application software at the FSC user station and the previous version of the application software in Central Part 2. Therefore, for Central Part 2, the program operates as a revision control tool to register differences between two versions of the application software. The sections in the log file that refer to Central Part 2 should contain all changes that were made with respect to the previous version of the application software.



11.4 Verification results

11.4.1 Log file

The log file <Application name>.VRF contains all messages and errors that were generated during the verification process. The log file can be printed or displayed on screen using the 'View Log' option of FSC Navigator.

11.4.2 Error messages

The following sections contain the error messages that may be reported during the verification process. All messages (except for the manual interrupt message) are preceded by the word 'ERROR: '.

Unless the 'Verify Application' option of FSC Navigator is used as a revision control tool (see subsection 11.3.3), no differences should be reported except for the differences as listed in the FSC Safety Manual. If any differences are reported, the application in the FSC system is not consistent with the application at the FSC user station. In that case, it is necessary to correct the differences at the FSC user station, translate the application again, and reload it into the FSC system. The 'Verify Application' option should then report the proper differences (see FSC Safety Manual).

11.4.2.1 Common error messages

Application not translated. Translate application.

The application must have been translated correctly before it can be verified.

Corrupted data detected. Translate application.

The FSC application software in the FSC system contains data that does not allow further program execution.

Could not write to file <file name>

An unsuccessful attempt was made to write data to the specified file.

Could not create file <file name>

An unsuccessful attempt was made to create the specified file.

Could not open file <file name>

An unsuccessful attempt was made to open the specified file.

Could not read from file <file name>

An unsuccessful attempt was made to read data from the specified file.

Could not close file <file name>

An unsuccessful attempt was made to close the specified file.

Could not open log file.

An unsuccessful attempt was made to open the log file.

Could not write to log file.

An unsuccessful attempt was made to write data to the log file.

**Could not close log file.**

An unsuccessful attempt was made to close the log file.

CRC error in file <file name>

An attempt was made to read from the specified file, but the file was found to be corrupted.

Fatal error occurred during communication.

Communication with FSC system is not possible. Try again.

Note:

All errors that may be reported during the communication with the FSC system are specified in Section 11 of the FSC Software Manual ("On-Line Environment").

File of illegal format: <file name>

An attempt was made to read data from the specified file, but the format of the file was not correct.

File of incorrect version: <file name>

The program cannot continue since the version of the specified file is not the same as the current program version.

FSC configuration in database and FSC system do not match.

It is not possible to verify an application in an FSC system whose configuration is not compatible with the database as stored on the FSC user station.

Incorrect version of application software on CP, CPU : <Central Part>, <CPU module>.

The version of the FSC Navigator software that was used to generate the FSC application software does not match the current program version.

Incorrect version of communication EPROMs belonging to CP, COM : <Central Part>, <COM module>.

The version of the FSC Navigator software that was used to generate and program the communication EPROMs does not match the current program version.

No application present in FSC system.

Verification of the application is not possible. The application has not been downloaded yet.

Not enough disk space. Required additional space : <###> kB.

There is not enough disk space to store the temporary files that are required to successfully complete the verification process.

Not enough memory.

The program has not enough memory to proceed.

Program interrupted manually.

You have interrupted the program by pressing <Esc>.

Simulation mode not supported or illegal power-on mode.

The application in the FSC system or the application at the FSC user station has been configured for simulation mode, or the power-on mode of the FSC system or the FSC user station is illegal.

Variable not found in variables database: <variable>.

A variable that was expected to be present in the variables database could not be found.



11.4.2.2 Verification of FSC application using the FSC database

Application database contains <no.> additional block definition record(s).

The application database contains more block definition records than could be generated from the FSC application software in the FSC system.

Application database record(s) that were not regenerated for CP <CP no.>

The application database contains more records than could be generated from the FSC application software in the FSC system. The following lines will provide a list of not regenerated application database records.

Illegal record number found in FSC application software.

The generated record cannot be compared, since the record number is out of range (the record is not present in the FSC database).

Mismatching field(s) in block definition records:

Differences were found between the RKE block definition records in the application database and the FSC system. The lines that follow specify where the differences were found, and will contain the mismatching values.

Mismatching field(s) in regenerated application database:

Differences were found between the application database and the FSC system. The lines that follow specify which differences were detected, and will contain the mismatching values.

Mismatching field(s) in regenerated variables database:

Differences were found between the variables database and the FSC system. The lines that follow specify where the differences were found, and will contain the mismatching values.

Missing block definition record in application database.

Use the 'System Configuration' option of FSC Navigator to modify the application or rebuild your databases.

Missing RKE block definition table.

The records belonging to an RKE communication channel cannot be regenerated since the specified table is not present in the FSC application software.

Variables database record(s) that were not regenerated for CP <CP no.>

The variables database contains more records than could be generated from the FSC application software in the FSC system. The following lines will provide a list of variables database records that were not generated.



11.4.2.3 Verification of the FSC application software using the functional logic diagrams

Connection between symbols not found on FLD <FLD no.>:

From : <symbol>, output <#>

To : <symbol>, input <#>

The specified connection generated from the FSC application software is not present on the FLD.

Connection between symbols on FLD <FLD no.> has not been regenerated:

From : <symbol>, output <#>

To : <symbol>, input <#>

The specified connection on the FLD was not generated from the FSC application software.

CRC error in FLD <FLD no.>

The specified FLD in the project folder is corrupted.

Expected symbol not found. FLD number: <FLD no.>

Internal storage error. This error should never occur.

FLD in application directory has not been regenerated. FLD number: <FLD no.>

The project folder contains more FLDs than could be regenerated from the FSC application software in the FSC system.

Function block I/O data not found on original FLD <FLD no.>

The original FLD is corrupted. Use the 'Design FLDs' option of FSC Navigator to remedy the problem.

Function block I/O data not found on regenerated FLD <FLD no.>

Internal storage error. This error should never occur.

Incorrect initialization of function block <FB> on FLD: <FLD no.>

The FSC application software to initialize the specified function block, which is called on the specified FLD, is missing, incomplete or incorrect.

Illegal instruction detected. FLD: <FLD no.>

The FSC application software in this FLD contains an illegal 68000 instruction.

Illegal bit number detected. FLD: <FLD no.>

The bit address of a variable in the FSC application software is illegal.

Illegal constant value detected. FLD: <FLD no.>

A constant value in the FSC application software at the specified FLD is out of range.

Illegal displacement detected. FLD: <FLD no.>

The address of a variable in the FSC application software is out of range.

Illegal equation table address detected: <FLD no.>

The absolute address in the equation block FSC application software does not point to an equation block table, or the format of the equation table is not correct.

Illegal 32-bit address detected. FLD: <FLD no.>

The absolute address of a variable in the FSC application software is out of range.

**Illegal timer value detected. FLD: <FLD no.>**

The value belonging to a timer variable in the FSC application software is out of range.

Illegal counter value detected. FLD: <FLD no.>

The value belonging to a counter variable in the FSC application software is out of range.

Illegal LPEC detected. FLD: <FLD no.>

An illegal program execution in the FSC application software has been detected in the specified FLD.

Incorrect type(s) of I/O to and from symbol <symbol>

The signal types of the connections of a symbol do not match. In most cases, the symbol will be a function block symbol.

Mismatching equation table entries on FLD: <FLD no.>

The equation table FLD as found in the project folder contains entries that are different from the equation table in the FSC application software in the FSC system.

Missing logic sequence at start of FLD: <FLD no.>

The specified FLD does not start with the expected program code.

Number of entries in equation table does not match for FLD: <FLD no.>

The equation table FLD as found in the project folder does not contain the same number of entries as the equation table in the FSC application software in the FSC system.

Regenerated FLD not found in application directory. FLD number: <FLD no.>

The project folder does not contain the FLD that was regenerated from the FSC application software in the FSC system.

Regenerated symbol <symbol> not found on FLD <FLD no.>

The specified symbol generated from the FSC application software is not present on the FLD. (See note below.)

Symbol <symbol> on FLD <FLD no.> has not been regenerated

The specified symbol on the FLD was not generated from the FSC application software. (See note below.)

Note:

The preceding two messages are usually reported simultaneously as a result of only one difference. If, for example, a constant value symbol with value 10 is generated for FLD 100 while the value of the constant on the FLD is 20, the following two messages will be reported:

- Regenerated symbol CONSTANT VALUE 10 not found on FLD 100.
 - Symbol CONSTANT VALUE 20 on FLD 100 has not been regenerated.
-

Signal type of connection on FLD <FLD no.> does not match:

From : <symbol>, **output** <#>

To : <symbol>, **input** <#>

The specified connection on the FLD is binary while the connection generated from the FSC application software is digital or vice versa.



Type of symbol could not be determined. FLD number: <FLD no.>

68000 code in FSC system cannot be converted to a logic sequence of symbols.

This message means that there is a problem with the application loaded into the FSC system. The specified FLD in the project folder is most likely corrupted. This situation may occur if you copy an FLD "outside" FSC Navigator (e.g. using Windows Explorer) instead of using the 'Copy' command in the functional logic editor (the 'Design FLDs' option of FSC Navigator). To solve this problem, delete the FLD and redraw it from scratch, or use the correct copy command.

Unexpected mnemonic found. FLD: <FLD no.>

The program code of the specified FLD contains an unexpected sequence of 68000 code.

Untranslatable code detected. FLD: <FLD no.>

The FSC application software contains a sequence of code that cannot be translated.

The following differences and error messages address the attributes of the different types of symbols. These messages speak for themselves.

Counter value of symbol <symbol> does not match.

Incorrect number of inputs to symbol <symbol>

Incorrect number of outputs from symbol <symbol>

Regenerated symbol <symbol> on FLD <FLD no.> has no output signals.

Regenerated symbol <symbol> on FLD <FLD no.> has no input signals.

Register type of symbol <symbol> does not match.

Signal conversion of symbol <symbol> does not match.

Signal type of symbol <symbol> does not match.

Symbol <symbol> on FLD <FLD no.> has no output signals.

Symbol <symbol> on FLD <FLD no.> has no input signals.

Timer base of symbol <symbol> does not match.

Timer value of symbol <symbol> does not match.

Note:

If the specified symbol in one of the preceding messages is followed by (FB <FLD no.> <sequence>), the symbol is an internal variable belonging to function block <FLD no.> which is used to initialize the function block.



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Helping You Control Your World



Fail Safe Control

Section 12: On-Line Environment

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12.1 Introduction

The 'On-Line Environment' option of FSC Navigator is a powerful tool for interaction between the FSC user station and the FSC system. It allows you to obtain information on the status of your process and the FSC system. It is also a useful tool in case of problems in your process and/or the FSC system. Many functions have been implemented which allow you to obtain a clear overview of the FSC system and process status. They include:

- display the status of the application within the context of the functional logic diagrams (FLDs),
- display the status of any process variable,
- force the status of any FSC variable (if enabled),
- modify PID parameters on-line,
- show a graphic overview of the PID control function,
- display and store the diagnostics of the system, and
- load the application into RAM.

The main menu of the 'On-Line Environment' option of FSC Navigator can be called in two ways:

- **Menu bar:** On-Line / Monitor System
- **Button bar:** Monitor

The **FSC On-Line Environment** screen provides a number of shortcuts to specific on-line functions:

- **Process Status Monitoring / Monitor FLDs**
Use this option to view the actual status of the variables in the FLDs of the current FSC project. (For details refer to subsection 12.5.1.)
- **Process Status Monitoring / Variable Status**
Use this option to view the actual status of the variables in the FLDs of the current FSC project. (For details refer to subsection 12.5.2.)
- **Process Status Monitoring / List Forces**
Use this option to view a list of all forced variables in the FSC system. (For details refer to subsection 12.5.3.)
- **I/O Signal Status**
Use this option to view the status of the input/output loops. This menu option has a submenu, which allows you to choose which loops you wish to view. (For details refer to subsections 12.4.2 and 12.4.3.)
- **Diagnostics / Extended Diagnostics**
Use this option to view and store all diagnostics of the Central Part(s) in the FSC system. (For details refer to subsection 12.4.4.)
- **Diagnostics / Diagnostics to Disk**
Use this option to back up the extended diagnostic database to hard disk or retrieve a previously stored database. (For details refer to subsection 12.4.5.)
- **Diagnostics / I/O Loop Diagnostics**
Use this option to view all diagnostics of the I/O loops in the FSC system. (For details refer to subsection 12.5.5.)
- **Diagnostics / System Information**
Use this option to view important system parameters of the FSC system. (For details refer to subsection 12.4.1.)

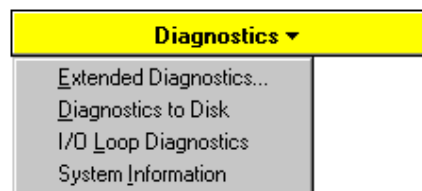


Figure 12-1 Button in FSC On-Line Environment screen (with submenu)

If an on-line function of FSC Navigator is called, the program will search for the file <application>.SCR. This file is used to store screens (see subsection 12.5.2.2). If this file is not found, it will be created automatically. This may take some time, depending on the size of the application.

After FSC Navigator has finished reading the data files, you will see the 'On-Line Environment' main menu (see subsection 12.3). To select an option, just move to the desired option using the left and right arrow keys and press <Enter>, or press the highlighted shortcut key.



12.2 Off-line menu

As soon as any of the on-line functions of FSC Navigator is called, the program will try to establish a communication link with the FSC system. If this fails, the following message will be displayed:

Check connection with FSC system. Press <Enter> for communication retry
or 'C' to enter the off-line menu.

If you press <C>, the off-line menu will appear (see Figure 12-2). The options in this menu allow you to perform a number of off-line tasks.

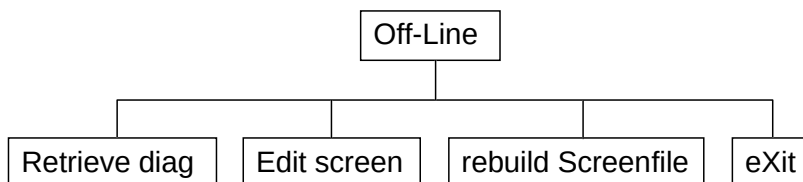


Figure 12-2 Off-line menu

A brief description of each option is given below:

Retrieve diag

This option allows you to retrieve and display extended diagnostic data that is stored on disk. For details refer to subsection 12.4.5.4.

Edit screen

This option allows you to edit screens that were defined earlier. Screens can be composed by the user and can contain up to 32 FSC variables or 8 PID controllers whose values are continuously updated on screen. For details refer to subsection 12.5.2.2.1.

Rebuild screen file

This option allows you to rebuild your screen file according to the current database in case it has changed. For details refer to subsection 12.5.2.6.

12.3 On-line main menu

The various options of the main menu are shown in Figure 12-3.

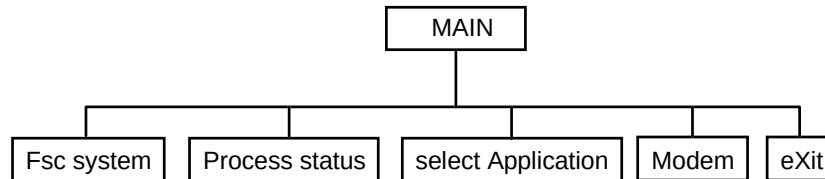


Figure 12-3 On-line main menu

A brief description of each option is given below.

FSC system

This option provides information which is related to the FSC system status, such as:

- diagnostics of the system,
- DBM status,
- setting of real-time clock,
- hardware I/O status, and
- loading the application in RAM.

Process status

This option provides information which is related to the process the FSC system is controlling, e.g.:

- the status within the context of functional logic diagrams,
- the status of all variables (per type or per functional logic diagram),
- analog input sensor alarms, and
- sequence of events.

You can change the status of the variables by performing force or write commands to the variables.

Select application

Use this option to switch to another FSC system, which is also a part of the configured FSC network or FSC user station network. This allows you to select another application to communicate with, without leaving the program.

Modem

This option allows you to establish a connection with the FSC system via a modem.

eXit

This option returns you to the FSC Navigator main screen.



12.4 FSC system

The options of the 'FSC system' menu are shown in Figure 12-4.

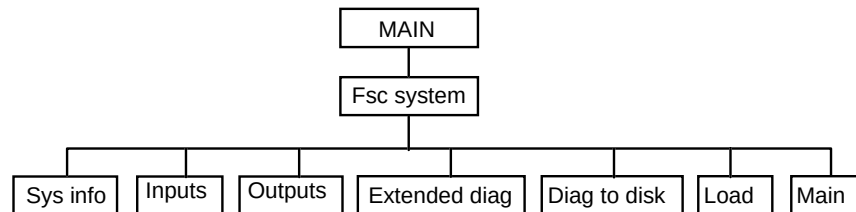


Figure 12-4 Structure of 'FSC system' menu

A brief description of each option is given in the next subsections.

12.4.1 'Sys info' menu

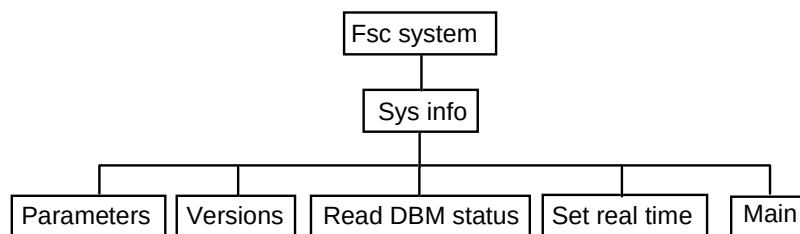


Figure 12-5 Structure of 'Sys info' menu

There is a shortcut to the 'Sys Info' menu. In the FSC On-Line Environment screen, choose the Diagnostics button and then the System Information option (see Figure 12-6).

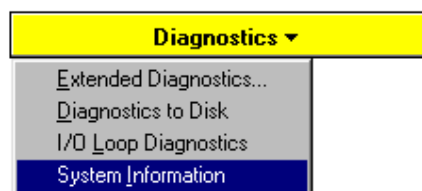


Figure 12-6 Diagnostics button with System Information option

12.4.1.1 Parameters

This option shows the value of configuration parameters of the running FSC system:

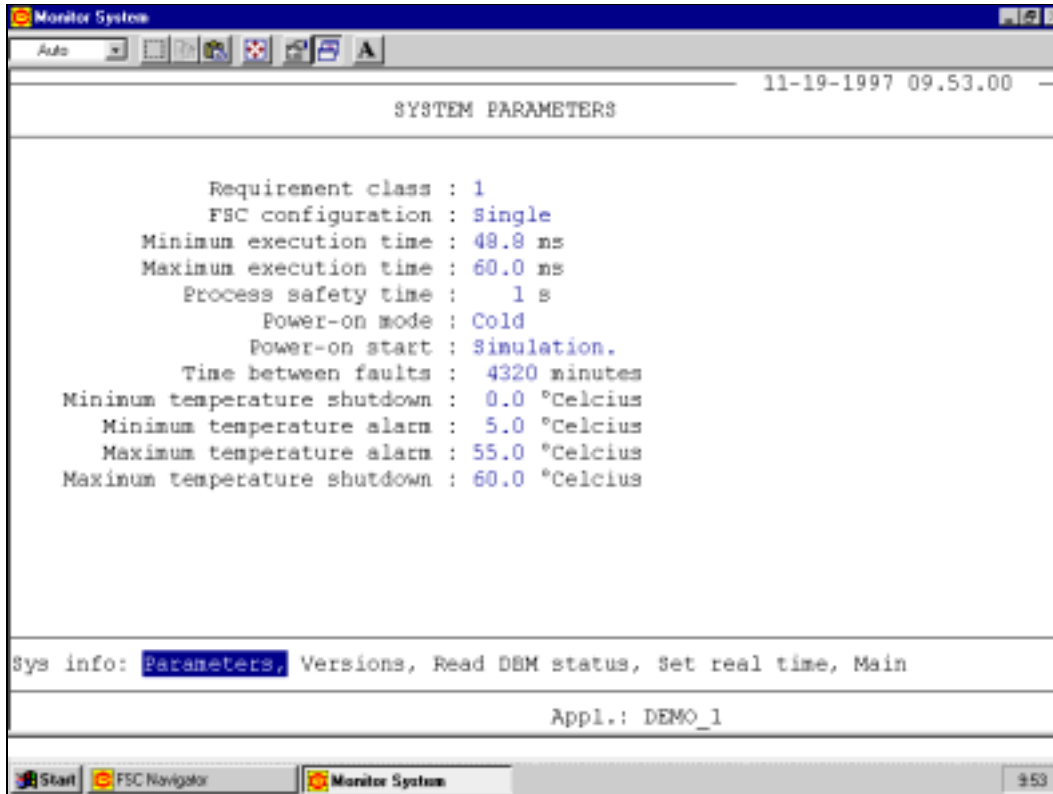


Figure 12-7 System parameters



12.4.1.2 Versions

This option allows you to view the versions of the embedded software in the FSC system components:

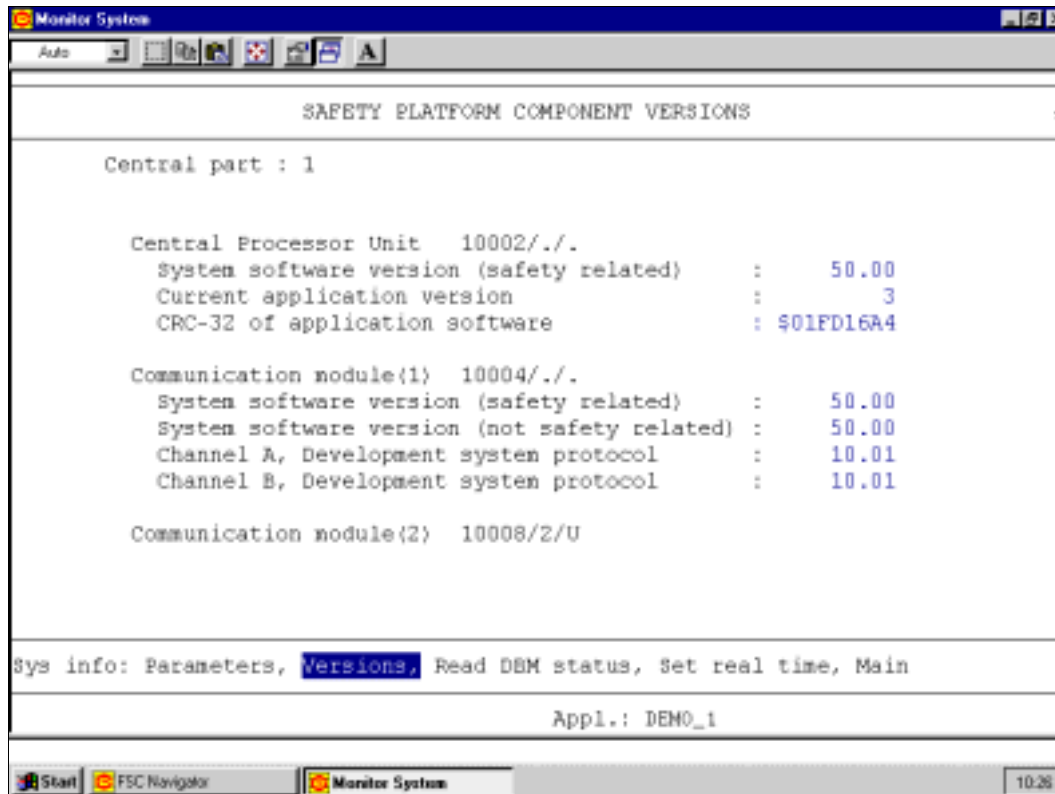


Figure 12-8 FSC component versions

Note:

The version of the FSC Safety Manager Module (FSC-SMM, 10008/2/U) is only displayed if the Safety Manager firmware is v2.1 or higher. If the firmware version is older, the window will show that an FSC Safety Manager Module is present, but will not indicate its firmware version. The FSC-SMM firmware version can also be retrieved from the TPS Universal Station (US).

12.4.1.3 Read DBM status

This option is only available if a Diagnostic and Battery Module (10006/.I.) has been placed in the FSC system. If you choose this option, the following screen will appear:

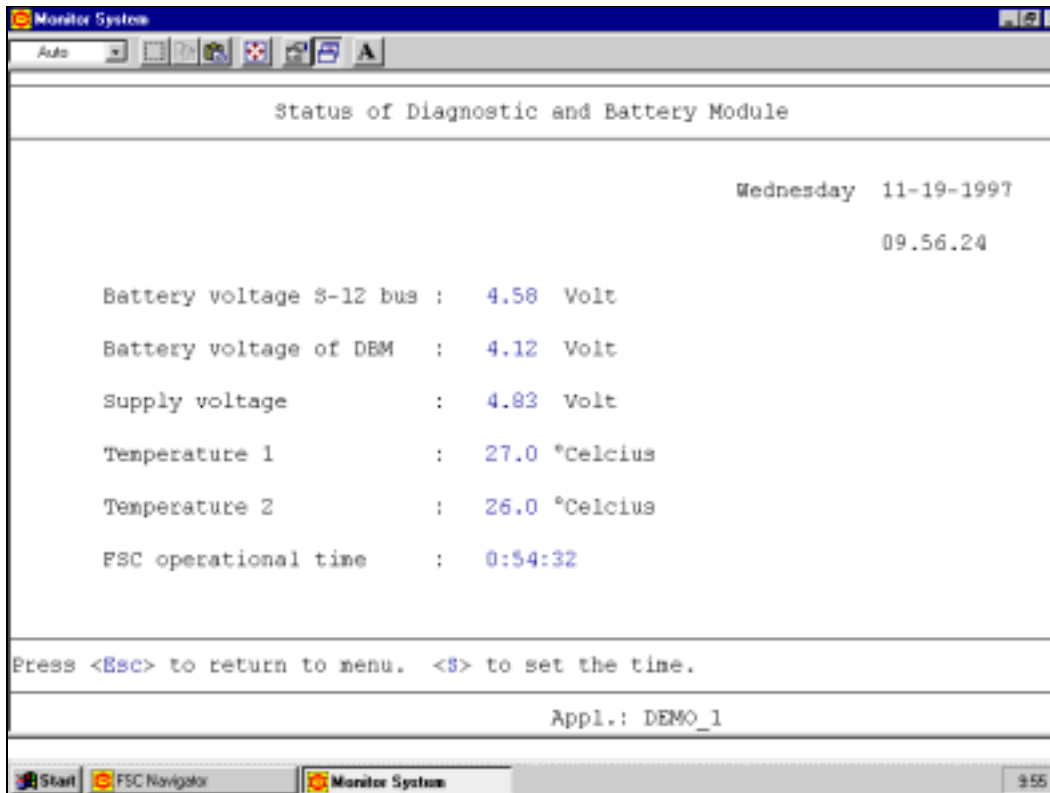


Figure 12-9 Status of diagnostic and battery module (DBM)

You can set the time by pressing <S>. After the time has been set, you will return to this option. The time and date appear in the upper right corner. The screen is continuously updated. The FSC operational time is the time that the FSC system has been controlling and safeguarding your process.

Note:

After each power-up of the FSC system, the time setting should be checked and corrected, if necessary.

12.4.1.4 Set real time

Use this option to set the FSC system's real-time clock. If the time and date have been set, you will automatically go to the 'Read DBM status' option (see subsection 12.4.1.3).

12.4.1.5 Main

Use this option to return to the main menu of the 'On-Line Environment' option of FSC Navigator.



12.4.2 'Inputs' menu

This menu contains options concerning digital and analog inputs. These options are hardware-related, which means that the inputs are sorted by rack, position and channel.

The structure of the 'Inputs' menu is shown Figure 12-10.

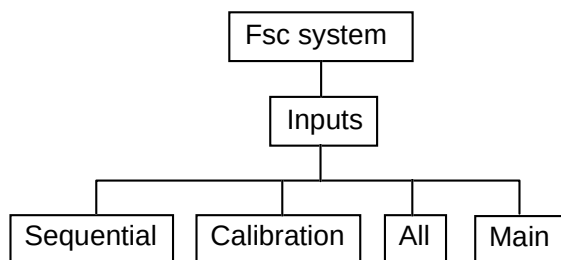


Figure 12-10 Structure of 'Inputs' menu

There is a shortcut to the 'Inputs' menu. In the FSC On-Line Environment screen, choose the I/O Signal Status button and then the Inputs option (see Figure 12-11).

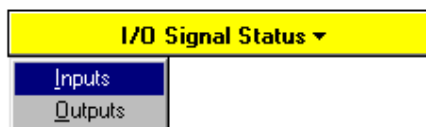


Figure 12-11 I/O Signal Status button with Input option

You must first choose whether to operate on all inputs at the same time ('All'), to operate on a single input ('Sequential'), to calibrate an analog input module ('Calibration') or to go back to the 'On-Line Environment' main menu ('Main').

12.4.2.1 Sequential inputs

The 'Sequential' option of the Inputs menu is used to present on-screen information about a specific hardware input.

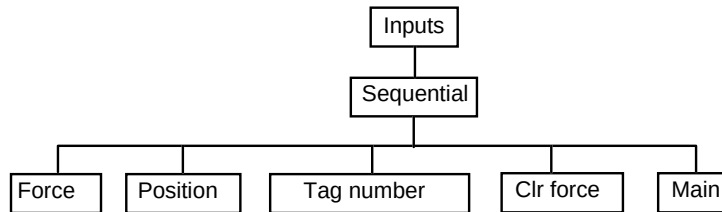


Figure 12-12 Structure of 'Sequential inputs' menu

If you choose the 'Sequential' option, the following screen will appear:

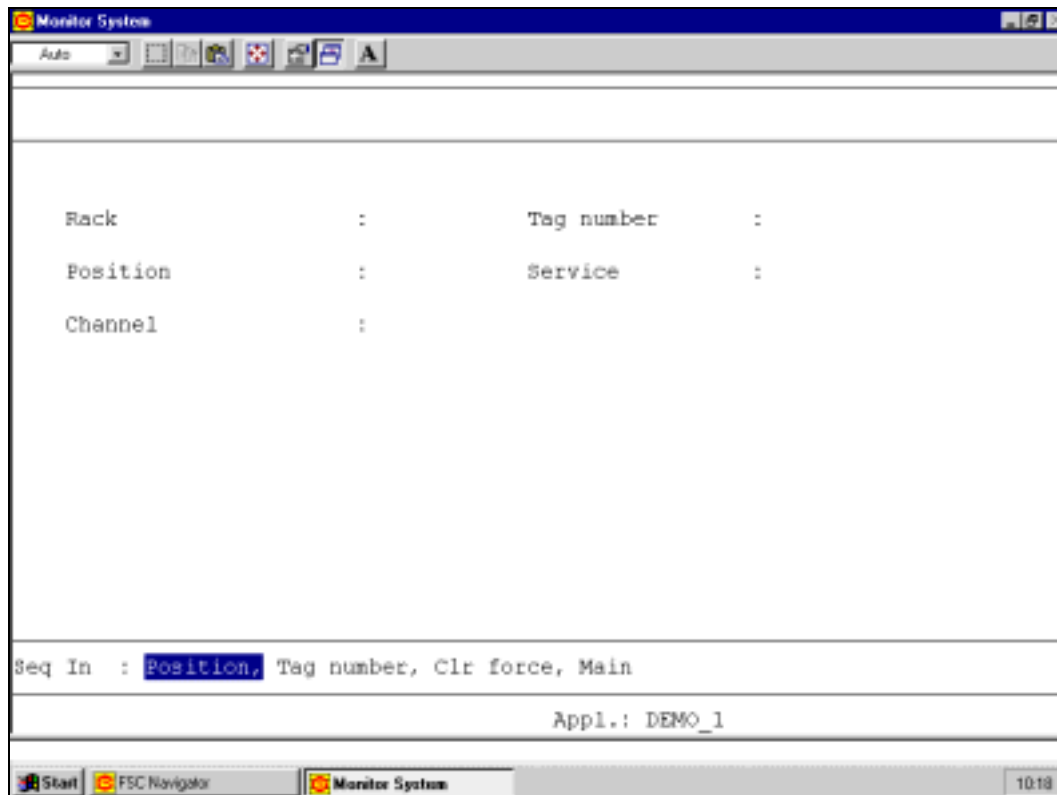


Figure 12-13 Sequential inputs

To select a specific input, use the following keys:

<PgDn> If used for the first time, the first input channel in the hardware configuration is displayed. After this the next channel in the hardware configuration is displayed. If the current channel is the last hardware configured channel and <PgDn> is pressed, the first configured hardware channel will be displayed.

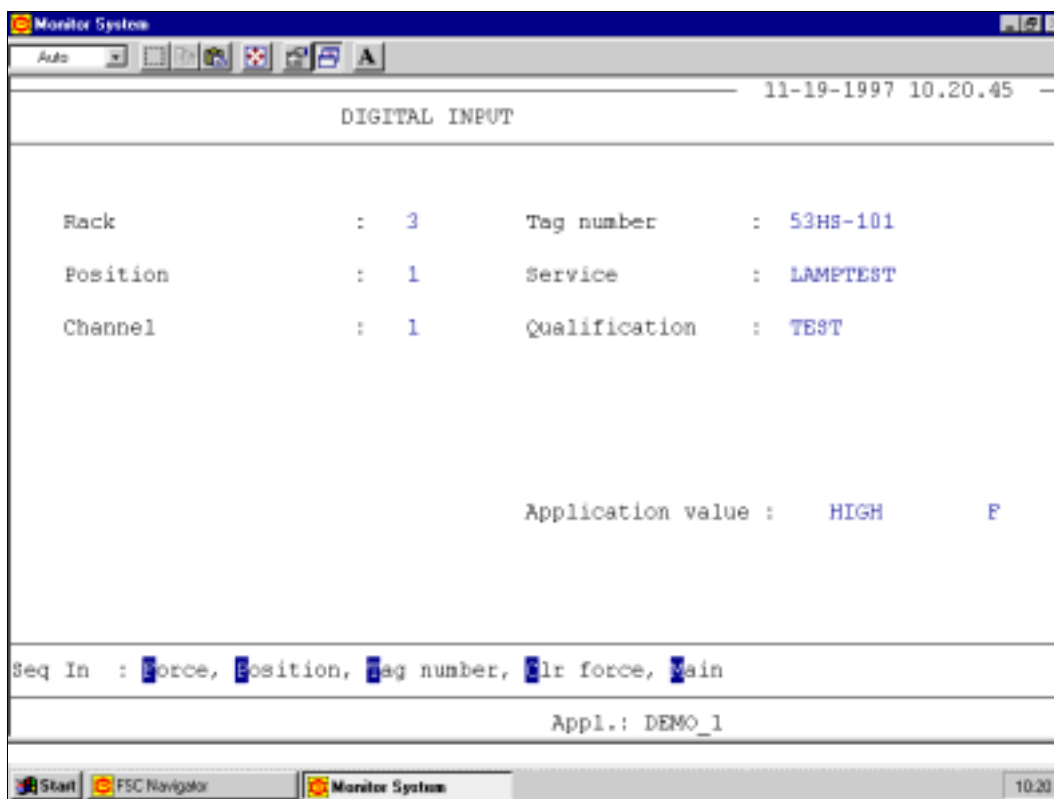


Figure 12-14 Example of sequential inputs

If the application has a redundant configuration and the next channel of a module is controlled by the other Central Part, the Central Part will be switched. If the other Central Part is in shutdown, all channels of that module will be omitted.

<PgUp>

If used the first time, the last input channel in the hardware configuration is displayed. After this, the previous channel in the hardware configuration is displayed. If the current channel is the first hardware configured channel and <PgUp> is pressed, the last hardware configured channel is displayed.

If the application has a redundant configuration and the previous channel of a module is controlled by the other Central Part, the other Central Part will be selected. If the other Central Part is in shutdown, all channels of that module will be omitted.

P

Enter the hardware configuration for a specific input.

If the application has a redundant configuration and a channel is selected that is controlled by a Central Part that is in shutdown, the corresponding channel of the other Central Part is displayed.

T

Find a specific input by entering the type (I, AI) and tag number.



The following information is presented on screen:

At the bottom line:

Appl.:	The name of the application
Central Part:	The number of the Central Part (only in case of a redundant Central Part configuration).

On screen:

Rack:	Number of the rack where the input module is placed.
Position:	Number of the position in the rack where the input module is placed.
Tag number:	The tag number which is assigned to the input channel.
Service:	Description of the tag number.
Qualification:	Information about the status of the variable. Only displayed for digital inputs.
Bottom scale:	Lowest possible value. Only for analog inputs.
Top scale:	Highest possible value. Only for analog inputs.
Signal type:	The type of the signal. Only for analog inputs.
Scan value:	Current value of the input signal (in engineering units).
Application value:	Value of the signal as used by the system for the application calculations (i.e. scan value, diagnostic status and force status included).
Input current:	The current detected current or voltage. Only for analog inputs.
Loop status:	Indicates the loop status. Only for analog inputs.
Diagnostic data:	Diagnostic status of the input channel. The indication will be 'OK' or 'Defect' if the channel is valid.



12.4.2.1.1 Force

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

The input status displayed on screen can be forced to a desired value, if forcing of this input is allowed. If the current input is digital, the status is selected by toggling with <Space>. If it is an analog input, the desired value can be entered via the keyboard. The value entered must be between the bottom scale and the top scale, which is presented on screen.

Note:

The force option is only present in the menu if the force enable system input is high and the input is set to force-enabled in the database.

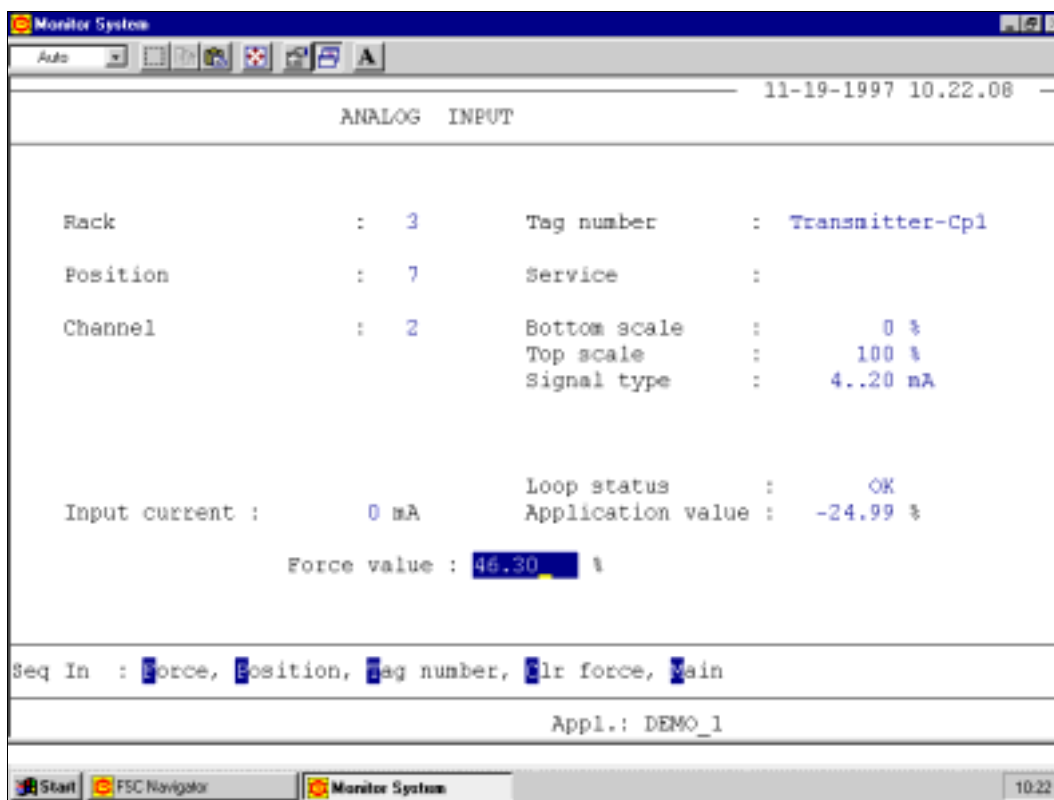


Figure 12-15 Example of force

If an input is forced, this is indicated by the letter 'F' behind the application value.

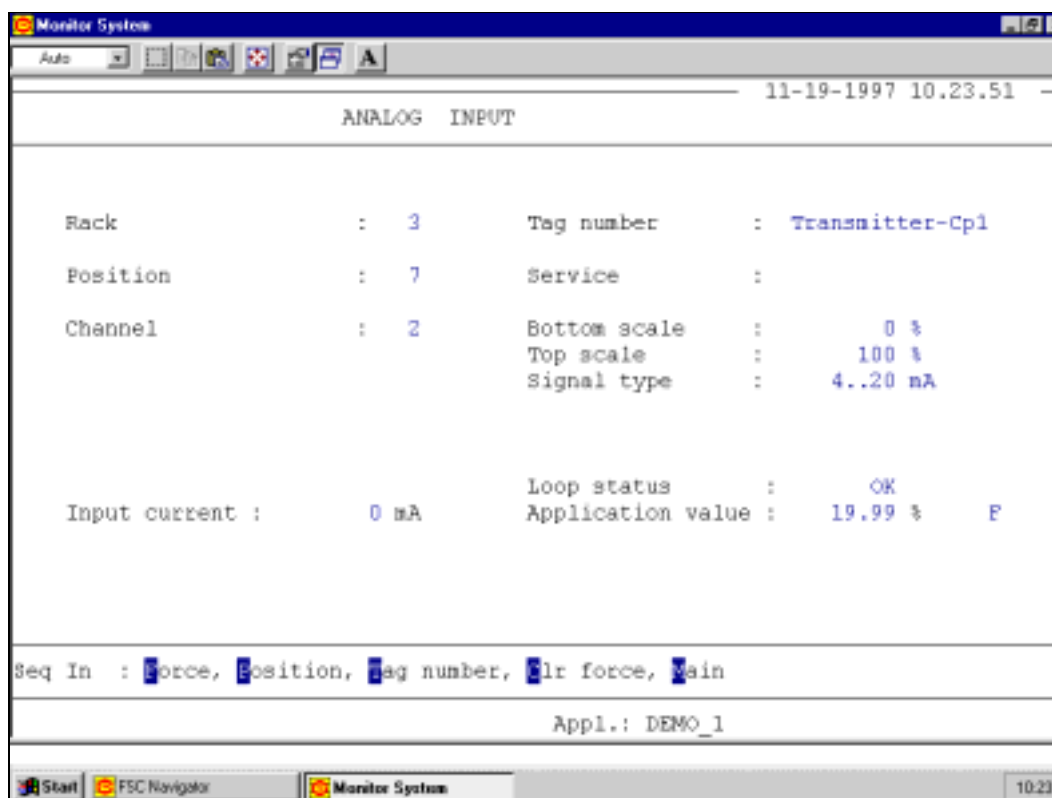


Figure 12-16 Example of a previously forced input



12.4.2.1.2 Position

After selecting the 'Position' option, you can enter a hardware location of an input. You must enter the rack, position number (1 to 18) and channel number at the bottom left corner of the screen.

Monitor System

Auto

11-21-1997 15.14.50

DIGITAL INPUT

Rack	:	1	Tag number	:	59qz002M
Position	:	1	Service	:	MOSBUTTON
Channel	:	1	Qualification	:	1-PRESSED

Scan value : LOW

Diagnostic data : OK

Application value : LOW

Rack : 1

Position : 1

Channel : 9

Seq In : Position, Tag number, Bit force, Gain

Appl.: DEMO_1

Start FSC Navigator Monitor System 15:14

Figure 12-17 Example of position

The result of these entries is shown below.

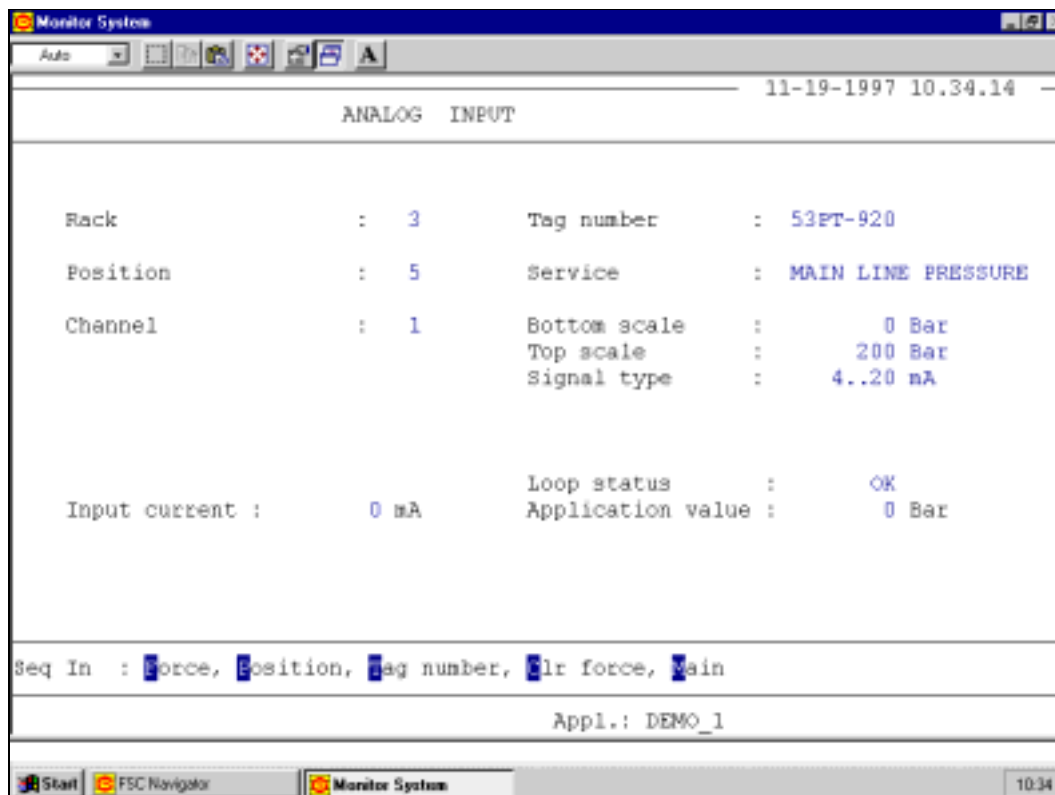


Figure 12-18 Result of the position option



12.4.2.1.3 Tag number

Using this option, a specific input can be allocated through its type (I, AI) and tag number. After selecting the tag number option, you will see the following screen:

The screenshot shows the 'Monitor System' window with a menu bar containing 'Auto' and several icons. The main area displays a configuration form with the following fields:

Rack	:	Tag number	:
Position	:	Service	:
Channel	:		

Below these fields, the 'Variable type' is set to 'AI' and the 'Tag number' is set to a blue box. The application name 'Appl.: DEMO_1' is displayed at the bottom of the main area. The taskbar at the bottom shows 'Start', 'FSC Navigator', and 'Monitor System' with a clock showing 10:35.

Figure 12-19 Tag number

You can now enter the tag number of the desired input, e.g. PRESSURE.

The result of this entry is shown below.

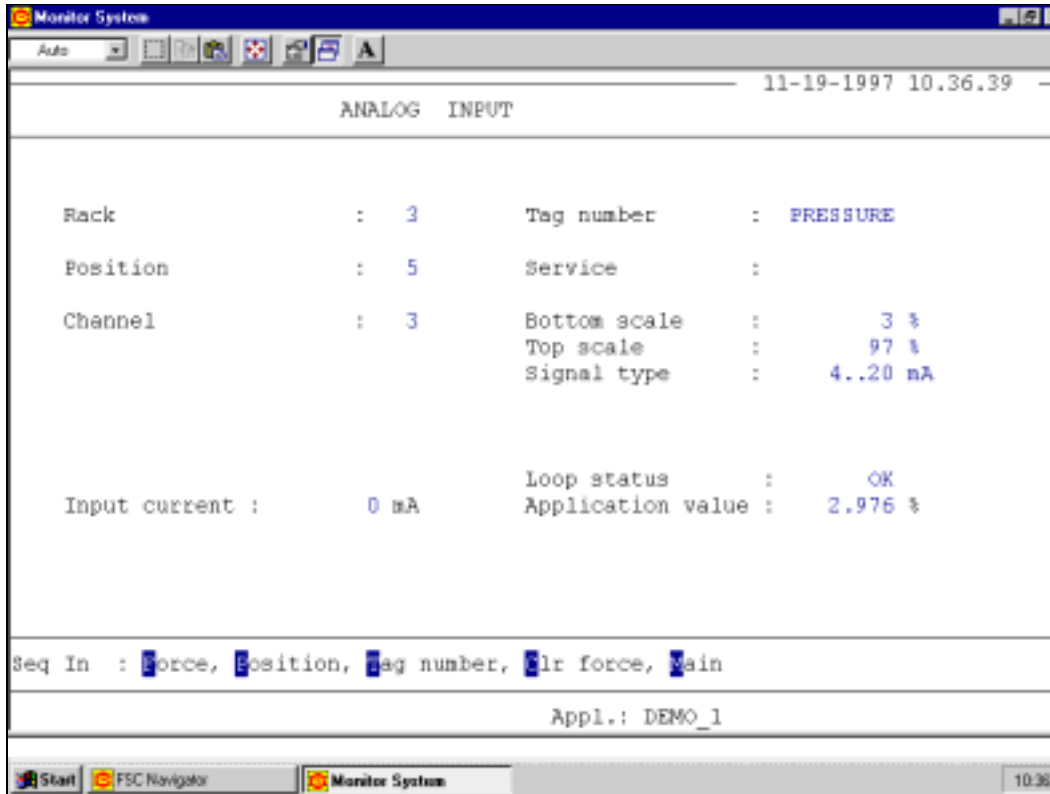


Figure 12-20 Result of the tag number option

12.4.2.1.4 Clr Force

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. The forced input status which is displayed on screen is cleared and the input gets its current process status.

12.4.2.1.5 Main

This option returns you to the main menu.



12.4.2.2 Calibration

The calibration option is a useful tool when calibrating the analog input modules 10102/1/1, 10102/1/2 or 10102/2/1. It gives you information about how to manipulate the potentiometers at the module to achieve the proper calibration. Figure 12-21 below shows the location of the potentiometers at the 10102/./ modules.

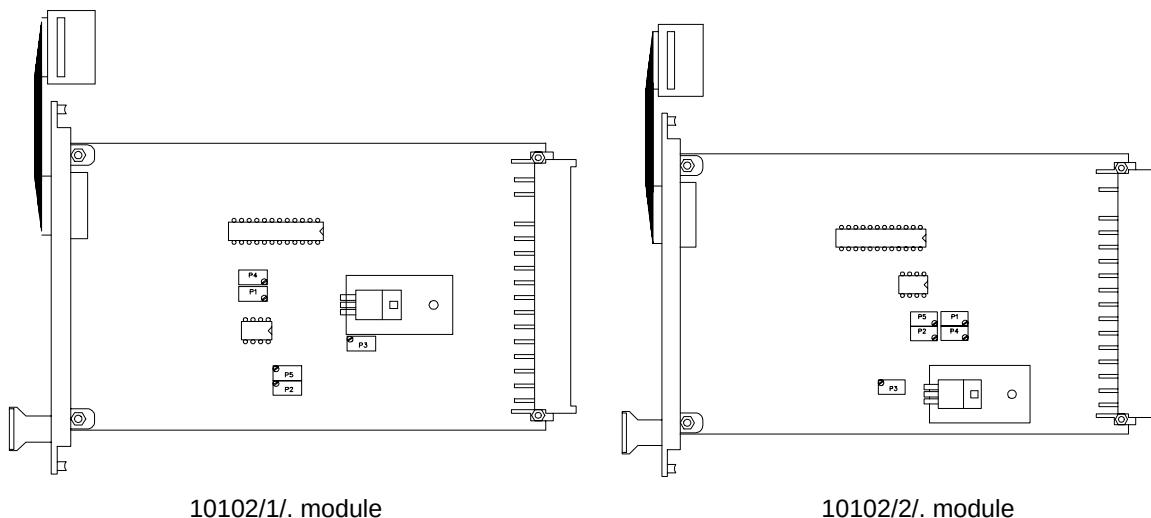


Figure 12-21 Location of potentiometers on the 10102/./ module

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

If you choose this option, the rack and the position of the module to calibrate must be entered. If this has been done correctly, you must enter the calibration value for channel 1 (Span). There are six standard calibration values: 4, 8, 10, 12, 16 and 20 mA.

Connect a mA source to channel 1 (voltage source for redundant configurations).

Place the 10102/./ module in the extender card and connect the flatcable via the extender flatcable to the horizontal bus.

Press the space bar to select a value and press <Enter> to confirm.

Note:

After this, the calibration value of channel 2 (Zero) is set to 0 mA.

Short-circuit the input terminals for channel 2 (d22 and z22 on 10102/1/. connector).

In redundant configurations, the calibration is preferably performed with only one Central Part running.

The screen will now look like this:

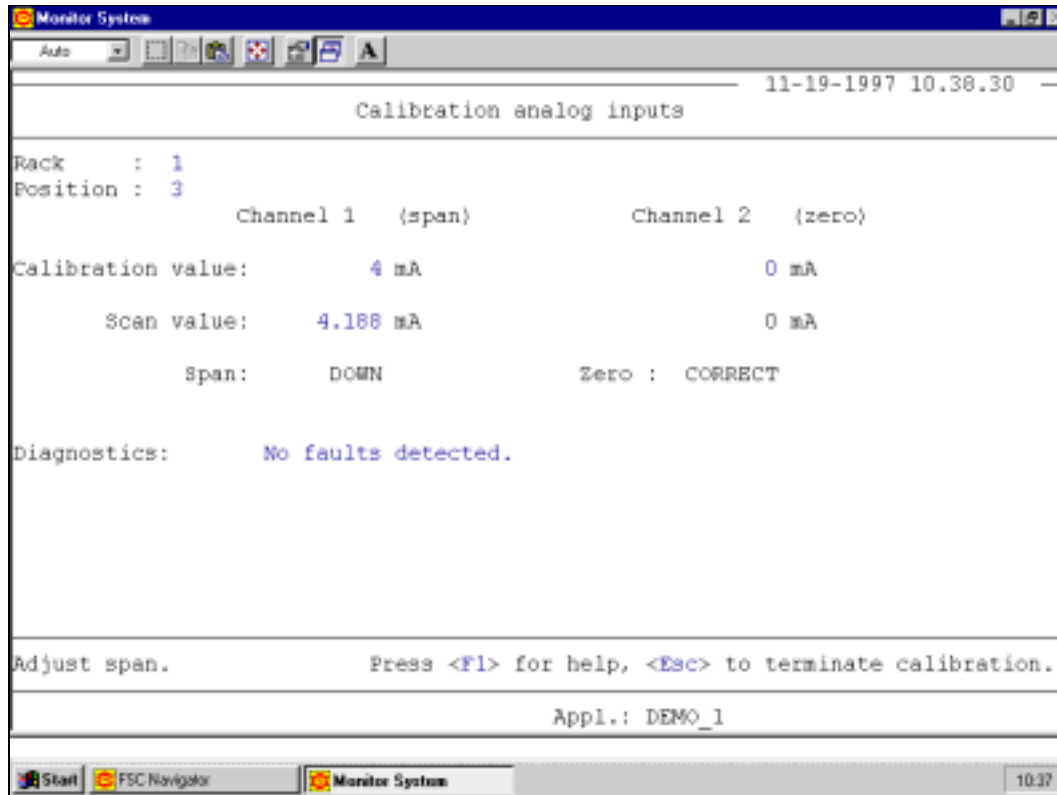


Figure 12-22 Input calibration (zero + span)

First you must calibrate the zero. The action to take is displayed in the menu line.

- | | | |
|-------|-----------|---|
| Zero: | – DOWN | – Turn potentiometer P4 (see Figure 12-21) counter-clockwise. |
| | – UP | – Turn potentiometer P4 (see Figure 12-21) clockwise. |
| | – HOLD | – P4 (see Figure 12-21) is in the correct position, hold for stabilizing. |
| | – CORRECT | – P4 (see Figure 12-21) is correctly set. |

The zero value must change between 0 mA and 0.006 mA (Steady 0 mA means that the zero is too low).

If Zero indicates CORRECT, you must calibrate the Span. The action to take is displayed in the menu line.

- | | | |
|-------|-----------|---|
| Span: | – DOWN | – Turn potentiometer P1 (see Figure 12-21) counter-clockwise. |
| | – UP | – Turn potentiometer P1 (see Figure 12-21) clockwise. |
| | – HOLD | – P1 (see Figure 12-21) is in the correct position, hold for stabilizing. |
| | – CORRECT | – P1 (see Figure 12-21) is correctly set. |

Note:

When adjusting the Span, the Zero result may change. Correct this by adjusting P4 (see Figure 12-21). Keep adjusting potentiometers P1 and P4 (see Figure 12-21) until both Span and Zero are CORRECT.

If Span and Zero results are CORRECT, the test value calibration will be performed.



Now the screen will look like this:

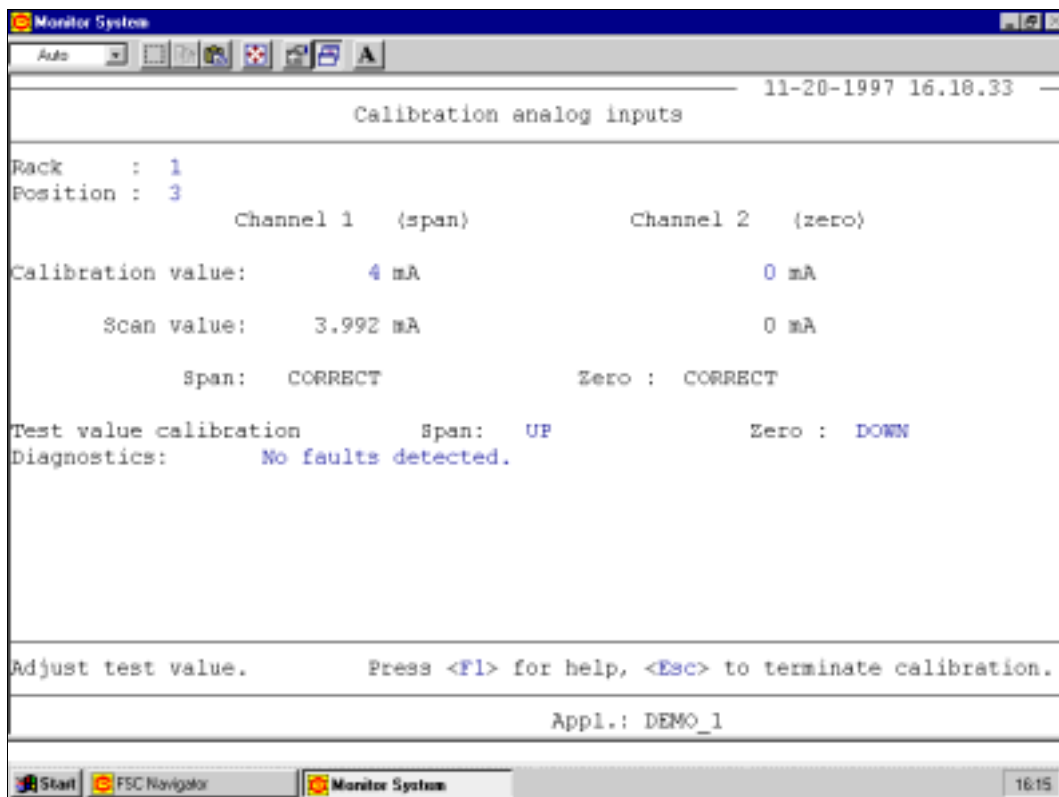


Figure 12-23 Input calibration (test value)

First you must calibrate the Span roughly. This is done by adjusting potentiometer P2 (see Figure 12-21) according to the Span result displayed at the lower part of the screen.

- | | | |
|-------|-----------|---|
| Span: | – DOWN | – Turn potentiometer P2 (see Figure 12-21) counter-clockwise. |
| | – UP | – Turn potentiometer P2 (see Figure 12-21) clockwise. |
| | – HOLD | – P2 (see Figure 12-21) is in the correct position, hold for stabilizing. |
| | – CORRECT | – P2 (see Figure 12-21) is correctly set. |

The Span is roughly calibrated when the result is HOLD or switches between Up and DOWN. After this, you must calibrate the Zero. This is done by adjusting potentiometer P5 (see Figure 12-21) according to the Zero result displayed at the lower part of the screen

- | | | |
|-------|-----------|---|
| Zero: | – DOWN | – Turn potentiometer P5 (see Figure 12-21) counter-clockwise. |
| | – UP | – Turn potentiometer P5 (see Figure 12-21) clockwise. |
| | – HOLD | – P5 (see Figure 12-21) is in the correct position, hold for stabilizing. |
| | – CORRECT | – P5 (see Figure 12-21) is correctly set. |

If the Zero result is CORRECT, the Span must be calibrated again. This is done by adjusting potentiometer P2 (see Figure 12-21) as described above until the Span result is CORRECT.



Note:

When adjusting the Span of the Test Value Calibration, the Zero result may change. Correct this by adjusting P5 (See Figure 12-21). If both values are on HOLD or CORRECT, press the fault reset key switch to reset the diagnostics. Keep adjusting the potentiometers (P2 and P5) until both Zero and Span are CORRECT.

During calibration, the diagnostics will be continuously updated at the bottom of the screen.



12.4.2.3 All inputs

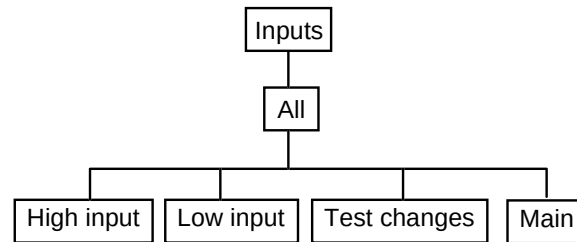


Figure 12-24 Structure of 'All inputs' menu

This option displays all inputs that meet a certain condition. The condition is specified by selecting an option from the menu line.

High input:	All inputs with status 'not low' are presented on screen.
Low input:	All inputs with status 'low' are presented on screen.
Test changes:	All inputs are displayed on screen when a change of their status occurs.

After selecting one of the options **H**igh Input, **L**ow Input, or **T**est changes, you will see a menu on screen, which allows you to choose digital inputs or analog inputs.

12.4.2.3.1 High input

All inputs with status 'not low' are displayed on screen and represented by their rack, position, channel and tag number (if implemented). This option can be used to test if all inputs can be set high independently (for wiring check).

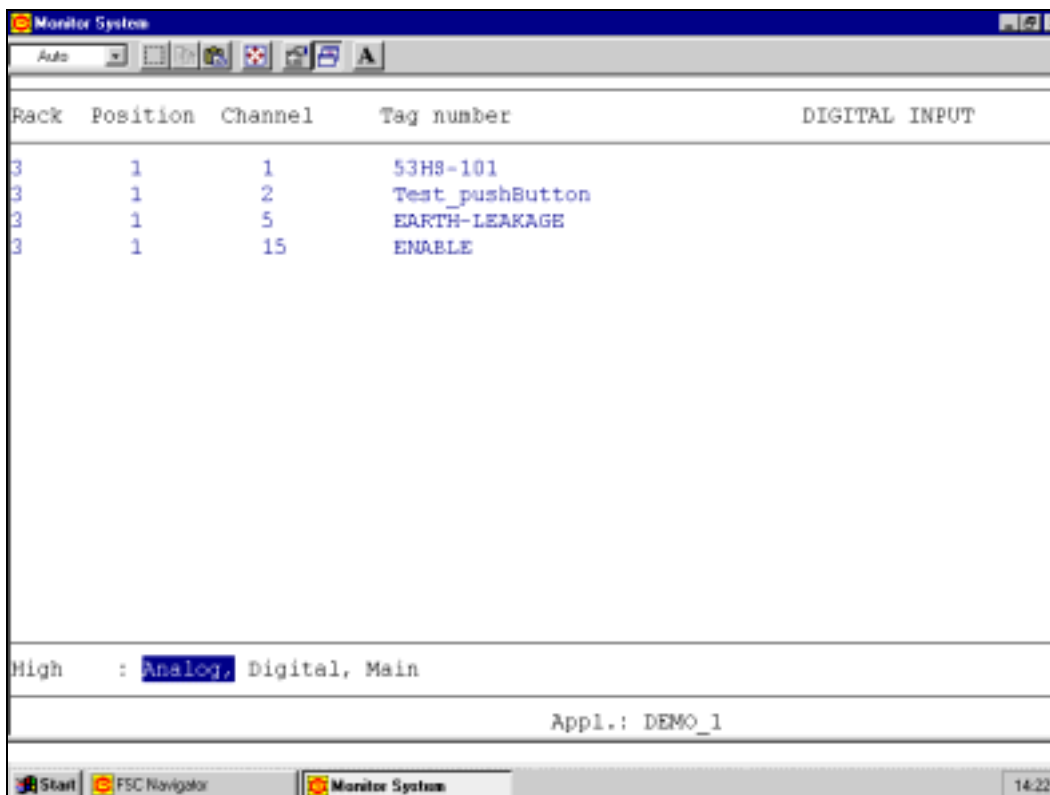


Figure 12-25 High input

12.4.2.3.2 Low input

All inputs with status 'low' are displayed on screen and represented by their rack, position, channel and tag number (if implemented). This option can be used to test if all inputs can be set low independently (for wiring check).

12.4.2.3.3 Test changes

If you use this option, all inputs are displayed on screen when a change in their status occurs. The screen is continuously updated.

12.4.2.3.4 Main

This option returns you to the main menu.



12.4.3 'Outputs' menu

This menu contains options concerning digital and analog outputs. These options are hardware-related, which means that the outputs are sorted by rack, position and channel.

The structure of the 'Outputs' menu is shown in Figure 12-26.

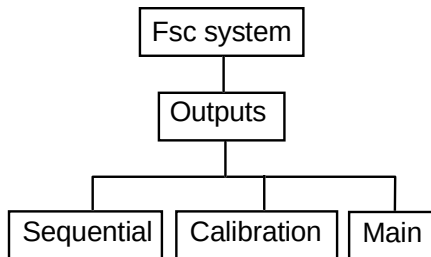


Figure 12-26 Structure of 'Outputs' menu

There is a shortcut to the 'Outputs' menu. In the FSC On-Line Environment screen, choose the I/O Signal Status button and then the Outputs option (see Figure 12-27).

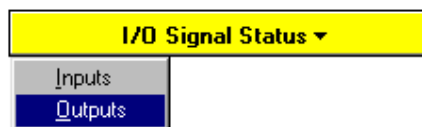


Figure 12-27 I/O Signal Status button with Input option

The 'Sequential' option is used to present information about a specific hardware output on screen. The 'Calibration' option is used to calibrate an analog output channel.

12.4.3.1 Sequential outputs

The 'Sequential' option of the Outputs menu is used to present on-screen information about a specific hardware output.

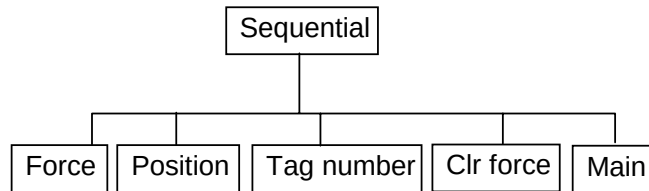


Figure 12-28 'Sequential outputs' menu

If you choose the 'Sequential' option, the following screen will appear:

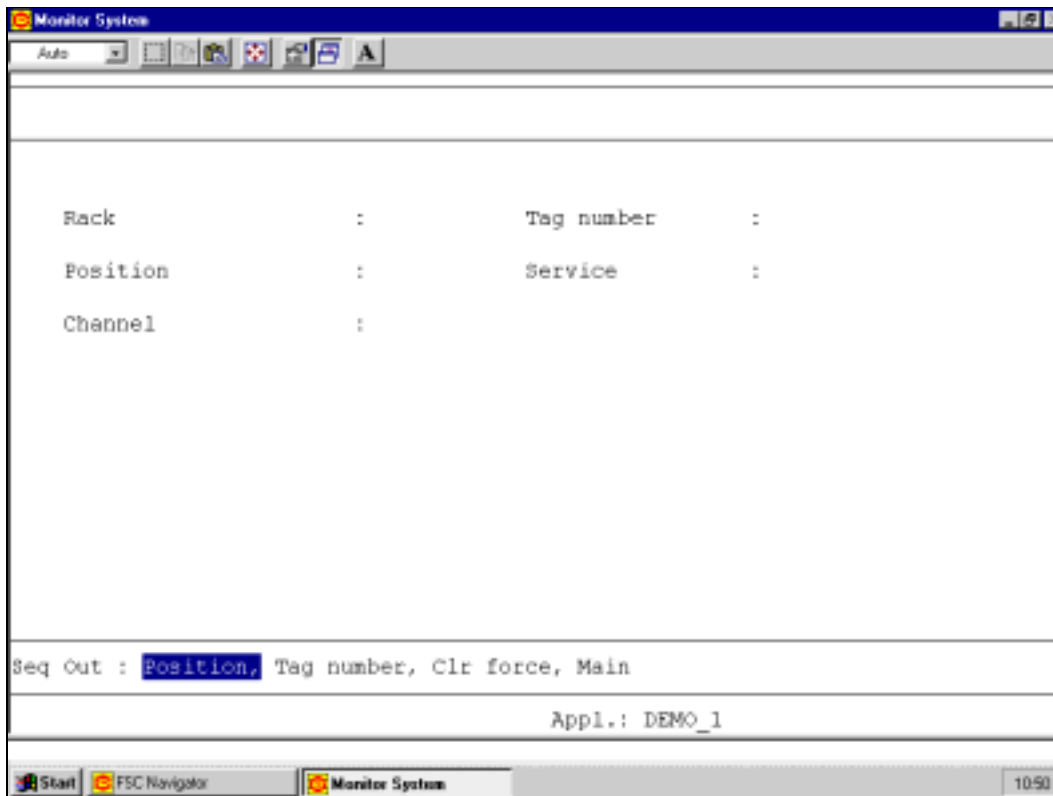


Figure 12-29 'Sequential outputs' screen

To select a specific output, use the following keys:

- <PgDn>** If used for the first time, the first output channel in the hardware configuration is displayed. After this the next channel in the hardware configuration is displayed. If the current channel is the last hardware configured channel and <PgDn> is pressed, the first configured hardware channel will be displayed.

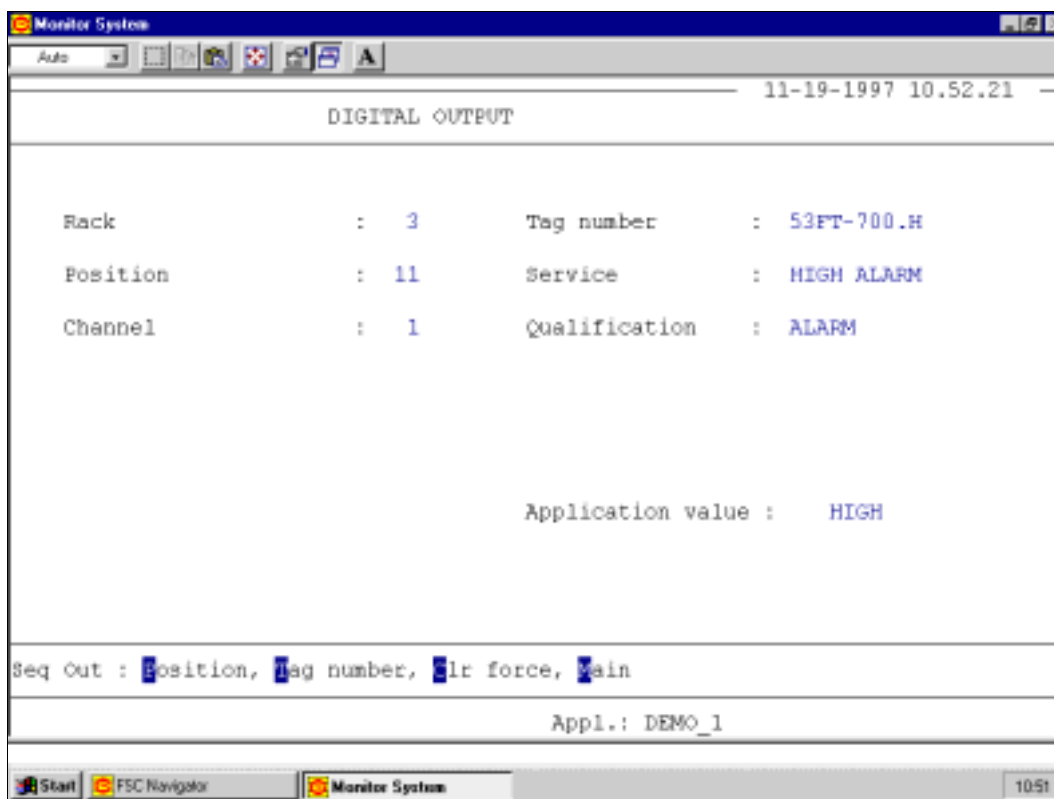


Figure 12-30 Example of sequential outputs

If the application has a redundant configuration and the next channel of a module is controlled by the other Central Part, the Central Part will be switched. If the other Central Part is in shutdown, all channels of that module will be omitted.

<PgUp>

If used the first time, the last output channel in the hardware configuration is displayed. After this, the previous channel in the hardware configuration is displayed. If the current channel is the first hardware configured channel and <PgUp> is pressed, the last hardware configured channel is displayed.

If the application has a redundant configuration and the previous channel of a module is controlled by the other Central Part, the Central Part will be switched. If the other Central Part is in shutdown, all channels of that module will be omitted.

P

Enter the hardware configuration for a specific output.

If the application has a redundant configuration and a channel is selected that is controlled by a Central Part that is in shutdown, the corresponding channel of the other Central Part is displayed.

T

Find a specific output by entering the variable type (O, AO) and tag number.



The information presented on screen is:

At the bottom line:

Appl.: The name of the application.

Central Part: The number of the Central Part (only in case of a redundant Central Part configuration).

On screen:

Rack: Number of the rack where the output module is placed.

Position: Number of the position in the rack where the output module is placed.

Tag number: The tag number which is assigned to the output channel.

Service: Description of the tag number.

Qualification: Information about the status of the variable. Only displayed for digital outputs.

Bottom scale: Lowest possible value. Only for analog outputs.

Top scale: Highest possible value. Only for analog outputs.

Signal type: The type of the signal. Only for analog outputs.

Scan value: Current value of the output signal in engineering units.

Loop status: Indicated the loop status. Only for a channel of a 10214/1/2 or 10216/1/1 module.

Diagnostic data: Diagnostic status of the output channel. The indication will be 'OK' or 'Defect'.



12.4.3.1.1 Force

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

The output status displayed on screen can be forced to a desired value, if forcing of this output is allowed. If the current output is digital, the status is selected by toggling with <Space>. If it is an analog output, the desired value can be entered via the keyboard. The value entered must be between the bottom scale and the top scale, which is presented on screen.

Note:

The force option is only present in the menu if the force enable system output is high and the output is set to force-enabled in the database.

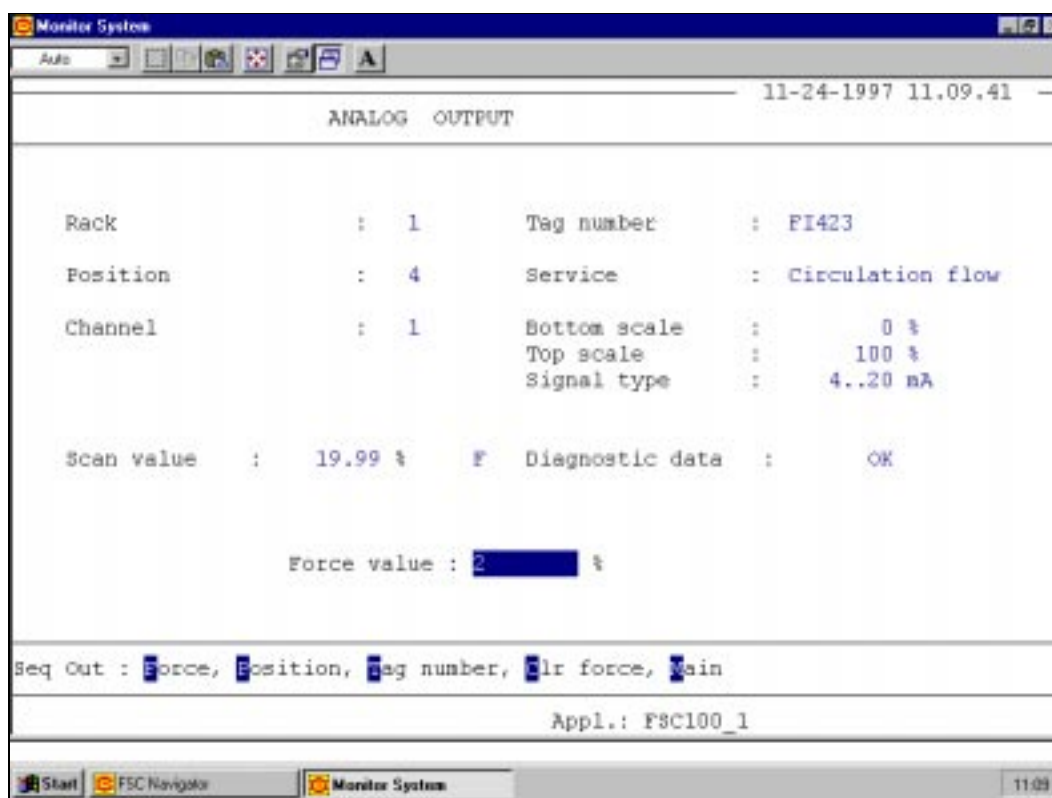


Figure 12-31 Example of force

If an output was forced previously, this is indicated by the letter 'F' behind the current status.

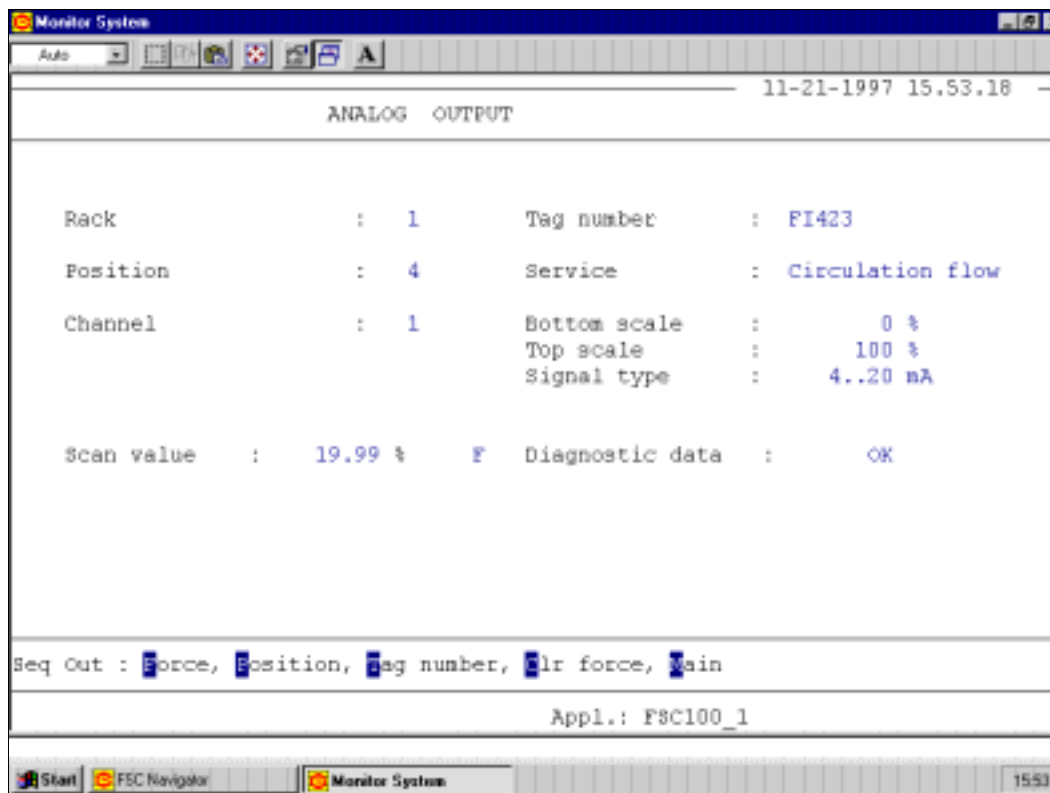


Figure 12-32 Example of a previously forced output



12.4.3.1.2 Position

After selecting the option 'Position', you can enter a hardware location of an output. You must enter rack, position number (1 to 18) and channel number at the bottom left corner of the screen.

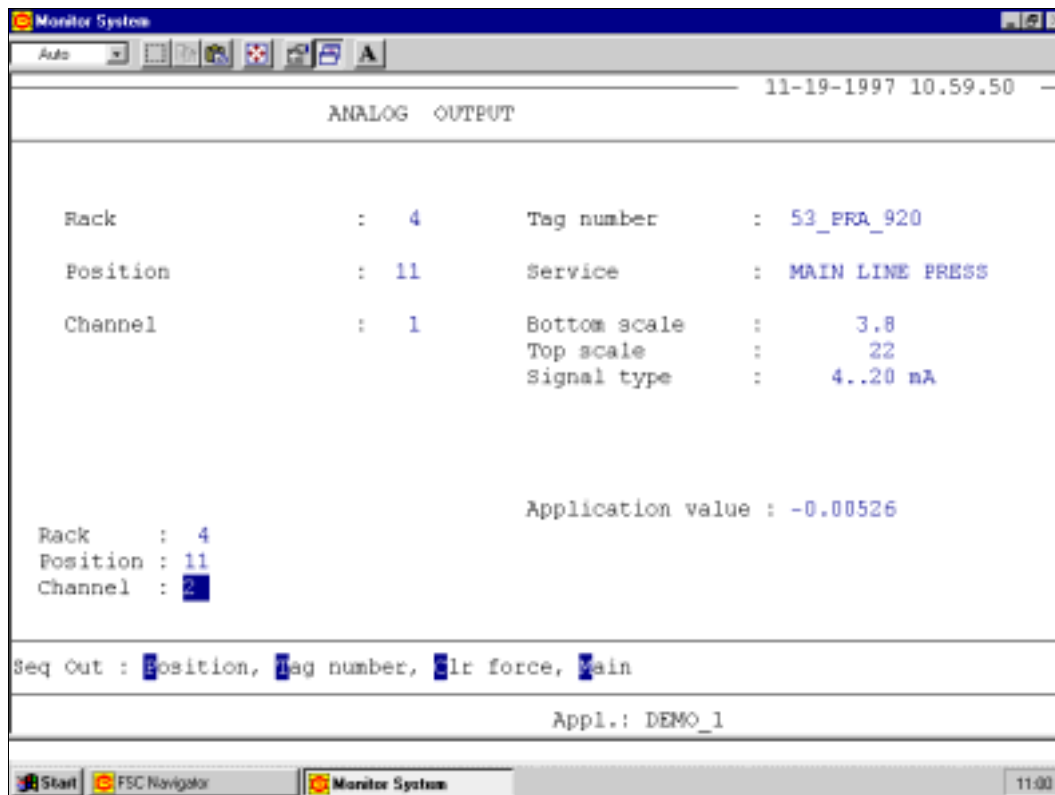


Figure 12-33 Sequential outputs menu

The hardware allocation is entered by rack and channel of this output.

The result of these entries is shown below.

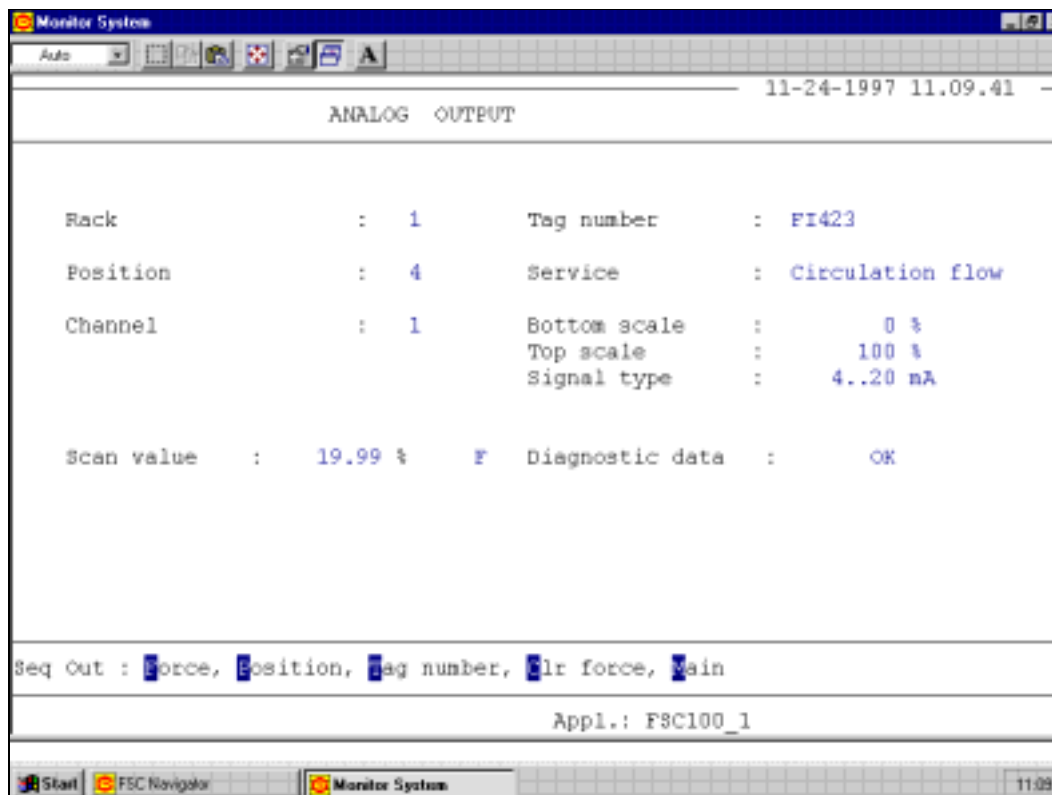


Figure 12-34 Result of the position option



12.4.3.1.3 Tag number

Using this option, a specific output can be allocated through its variable type (O, AO) and tag number. After selecting the tag number option, you will see the following screen:

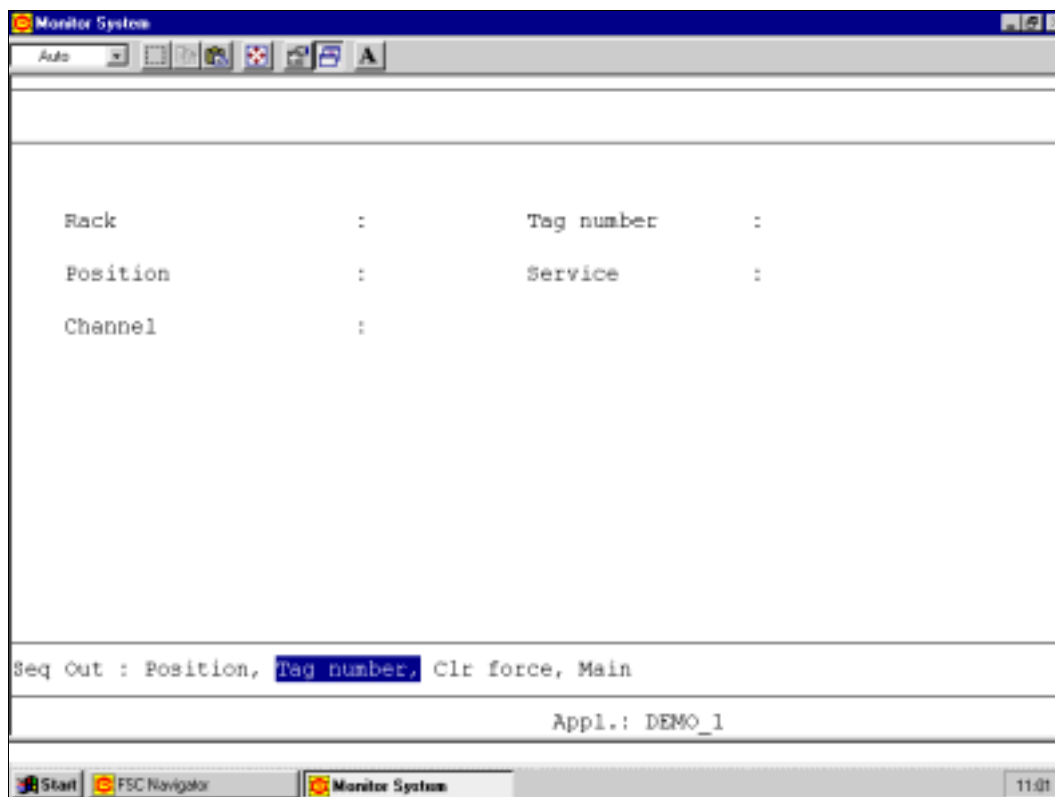


Figure 12-35 Tag number

You can now enter the tag number of the desired output, e.g. 53_PRA_920.

The result of this entry is shown below:

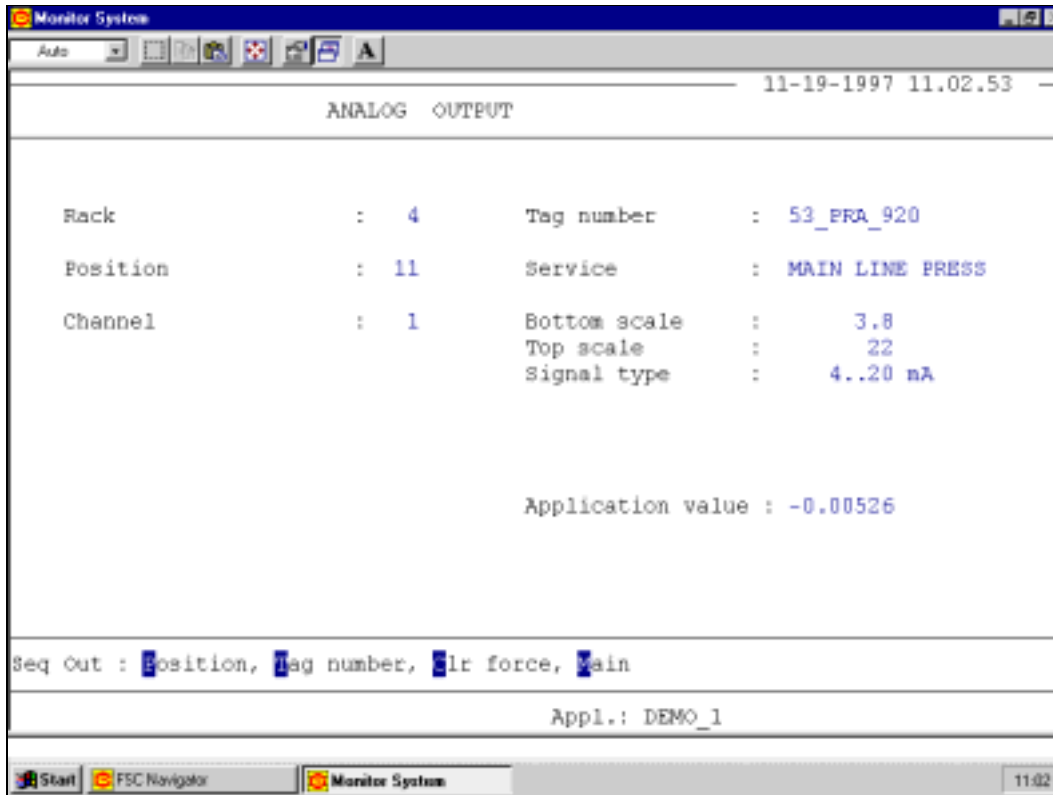


Figure 12-36 Result of the tag number option

12.4.3.1.4 Clr Force

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. The forced output status which is displayed on screen is cleared and the output gets its current process status.

12.4.3.1.5 Main

This option returns you to the main menu.



12.4.3.2 Calibration

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

If you choose this option, the rack, position and channel of the module to be calibrated must be entered. If this has been done correctly, you must enter the calibration value. There are seven standard calibration values: 4, 8, 10, 12, 16, 20 and 24 mA.

Just press the space bar to select a value and press <Enter> to confirm. The output current must be measured with, for example, a multimeter. The span can be adjusted with the potentiometer for the channel specified. The output has been correctly calibrated if the meter indicates the value set. The diagnostics of the module will be continuously updated at the bottom of the screen.

Note:

If faults are mentioned in these diagnostics a fault reset has to be given in order to retest the channel(s).

The screen will now look like this:

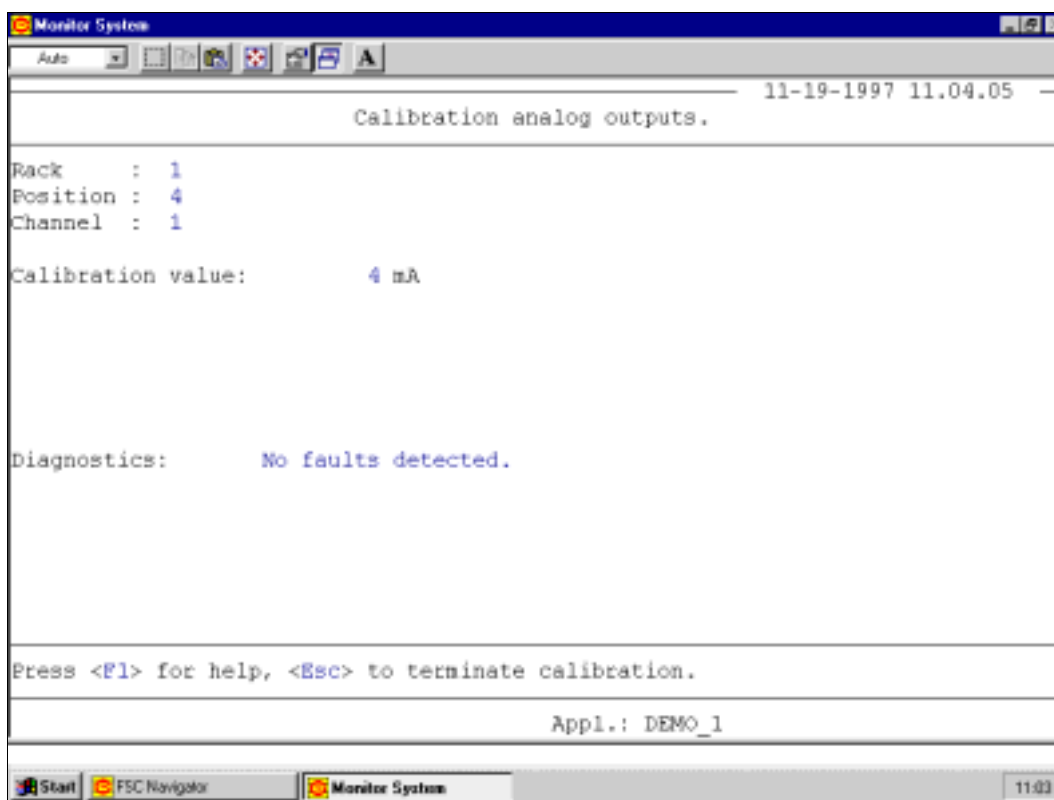


Figure 12-37 Output calibration

Press <Esc> to stop the calibration procedure. A menu will then appear which allows you to select another module, change the calibration value or return to the main menu. Use <PgDn> or <PgUp> to select the next or previous channel.

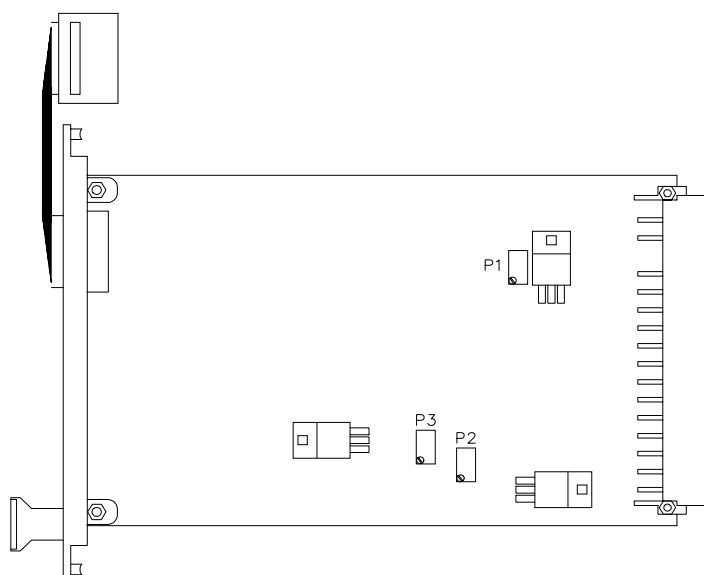


Figure 12-38 Location of potentiometers on the 10205/1/1 module



12.4.4 'Extended diag' option

Important!

Always read diagnostic data prior to a system fault reset. Otherwise diagnostic information will be lost.

The 'Extended diag' option shows all the diagnostics of the Central Part(s) of the FSC system. The retrieved diagnostic data will be stored in a data file on hard disk. The name of the file is the current application name with the extension '.EXD'. The file is located in the application directory.

The following information will be displayed:

- module name,
- short description of the event,
- rack and position of the module for hardware modules, and
- time and date of event.

The 'Extended diag' option has a number of shortcuts:

- **Menu bar:** On-Line / Extended Diagnostics
- **Button bar:** Diagnostics
- **Navigation area:** FSC On-Line Environment / Diagnostics / Extended Diagnostics (see Figure 12-39)

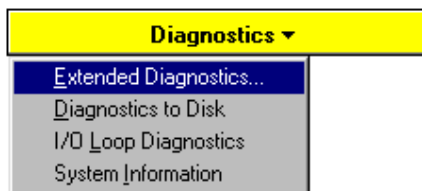


Figure 12-39 Diagnostics button with Extended Diagnostics option

The screen may look like this:

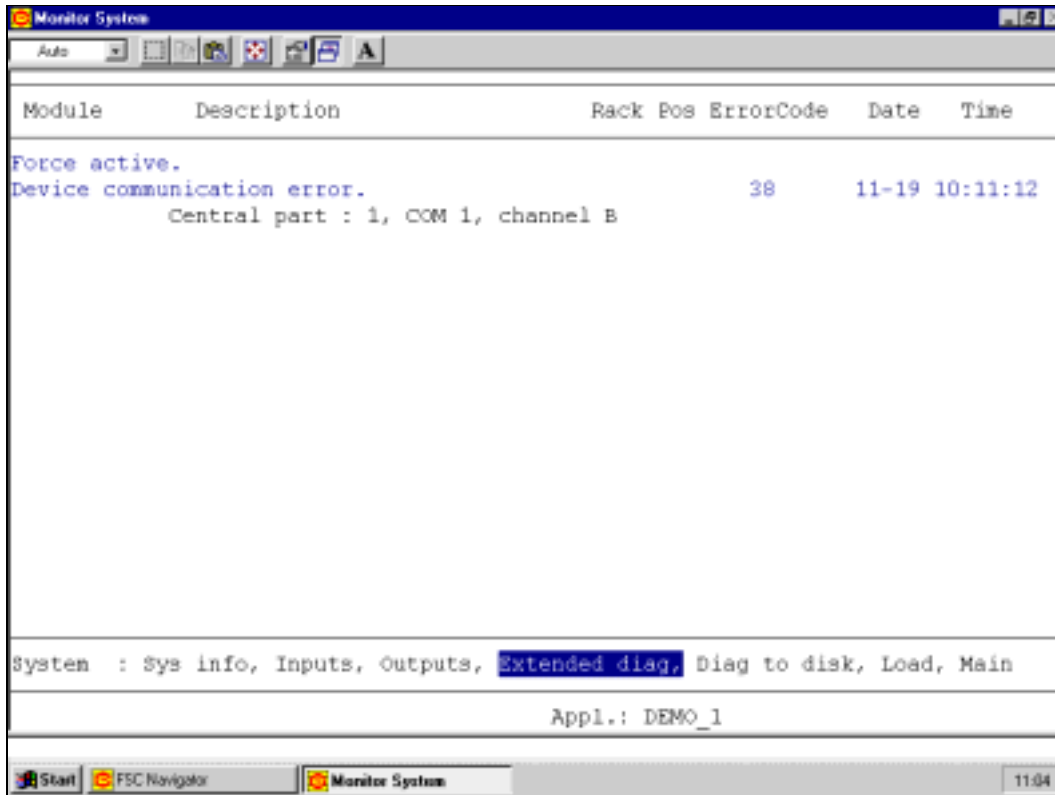


Figure 12-40 Presentation of diagnostic extended

Refer to subsection 12.10 ("Extended diagnostic descriptions") for possible descriptions.

If the 'Extended diag' option is selected, the diagnostic information retrieved from the application is stored in the extended diagnostics data file. If the FSC system has a redundant configuration, diagnostic information is retrieved and stored from all Central Parts.

The diagnostic state of the selected Central Part of the application is decoded and displayed.

After the extended diagnostics data is displayed, you will return to the 'Fsc system' menu.



12.4.5 'Diag to disk' menu

The menu structure of this option is shown below.

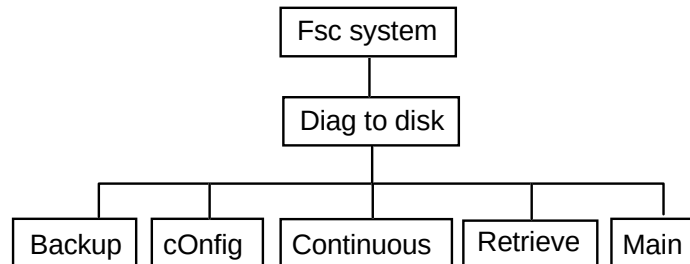


Figure 12-41 Structure of 'Diag to disk' menu

The 'Diag to disk' menu has the following options:

- Backup Back up stored extended diagnostic data.
- cOnfig Select FSC systems to be polled for diagnostic data.
- Continuous Continuous diagnostic interrogations of selected FSC systems.
- Retrieve Retrieve and display stored extended diagnostic data.

The 'Diag to disk' option has a number of shortcuts:

- **Menu bar:** On-Line / Diagnostics To Disk
- **Navigation area:** FSC On-Line Environment / Diagnostics / Diagnostics To Disk (see Figure 12-42)

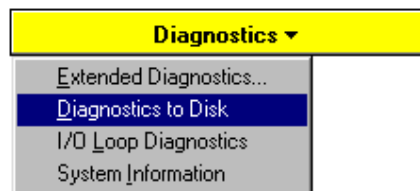


Figure 12-42 Diagnostics button with Diagnostics To Disk option

12.4.5.1 Backup

This option allows you to store the extended diagnostic database (or a part of it) into another database file. First you must specify the name of the backup file. The name must have the format <Disk>:\<Path>\<Filename>.

The extension of the backup file is fixed with the extension '.EXD', and must not be specified.

If the file already exists, a message appears asking whether the file should be overwritten. If you choose to overwrite, the backup file is deleted and created again.

Note:

The current application name can never be used as a backup file name.

After you have specified the file name, the range of the records to be stored in the backup file must be specified. The selection of the records is done by specifying the first and the last time of the data records. The time has the format: DD-MM-YY,HH.MM.SS.

By default, the first time is set to the time of the first record stored in the database. You must at least enter the day, month and year. The last time is default set to the specified first time.

Note:

Incomplete time stamps are allowed. For example, '24-11-97,10' means the first extended diagnostic data block stored from November 24, 1997, 10 a.m.

Before records are stored in the backup file, you are asked whether to delete the records stored from the current extended diagnostic database.

If you confirm to delete records from the extended diagnostic database, the extended diagnostic database will be rebuilt after all selected records have been stored in the backup file. This may take some time, depending on the size of the diagnostic database.

12.4.5.2 Config

If you select this option, a window appears which contains all possible FSC systems that can communicate with the FSC user station. The maximum size of the extended diagnostics database is displayed (see Figure 12-43). All applications which must be continuously interrogated can be (de)selected. After (de)selection of the applications, the size of the extended database must be specified. The information is stored in the file DIAGEXT.CFG.

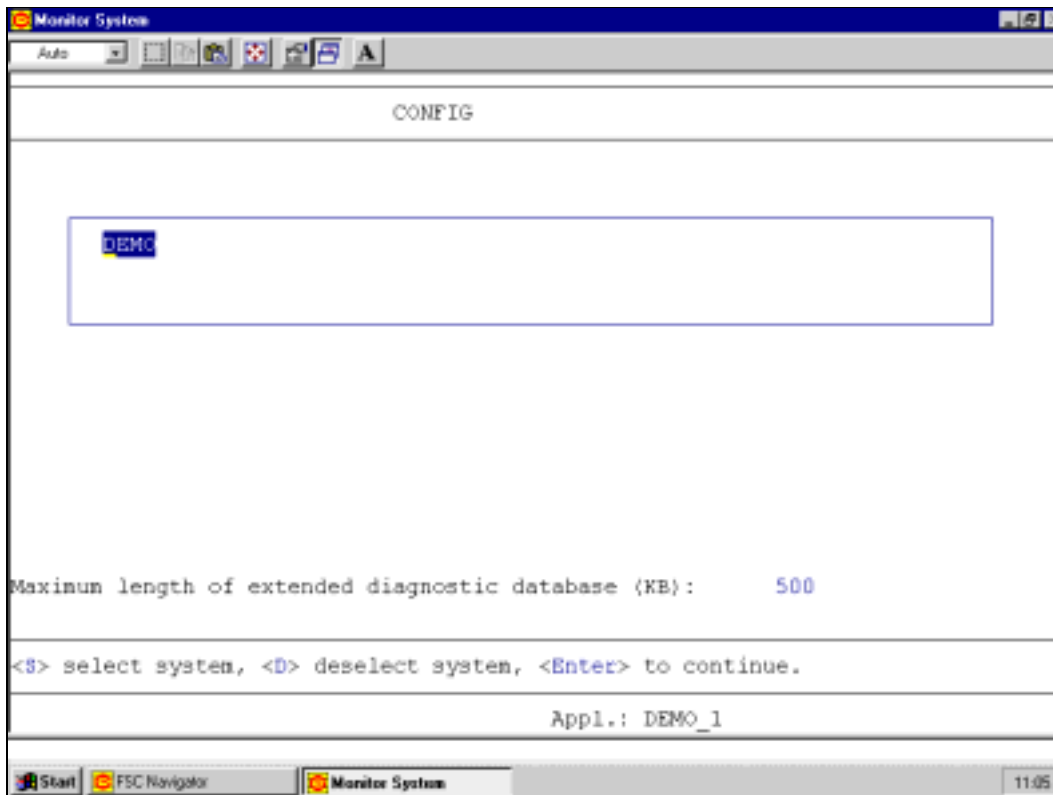


Figure 12-43 Config



12.4.5.2.1 Select application(s)

If the option 'cOnfig' is selected, a window appears with all applications the FSC user station can communicate with. To select an application press <S>, to de-select an application press <D>. The selected applications are highlighted. If an application name with more than one system number is present, a separate window appears, containing all the valid system numbers.

If you choose 'cOnfig' for the first time, the current application is the default selected application. If you have used 'cOnfig' before, the information of the last time is retrieved from the DIAGEXT.CFG file and presented on screen.

12.4.5.2.2 Define maximum size of extended diagnostic file

The size of the extended diagnostics data file must be specified. The default size is 500 Kbytes. The minimum size is 100 Kbytes or, if the current size is greater than 100 Kbytes, the minimum size is 110% of the current size.

The maximum size is 30 Mbytes or the remaining disk space plus the current size of the extended diagnostic data file (whichever of the two is smallest).

12.4.5.3 Continuous

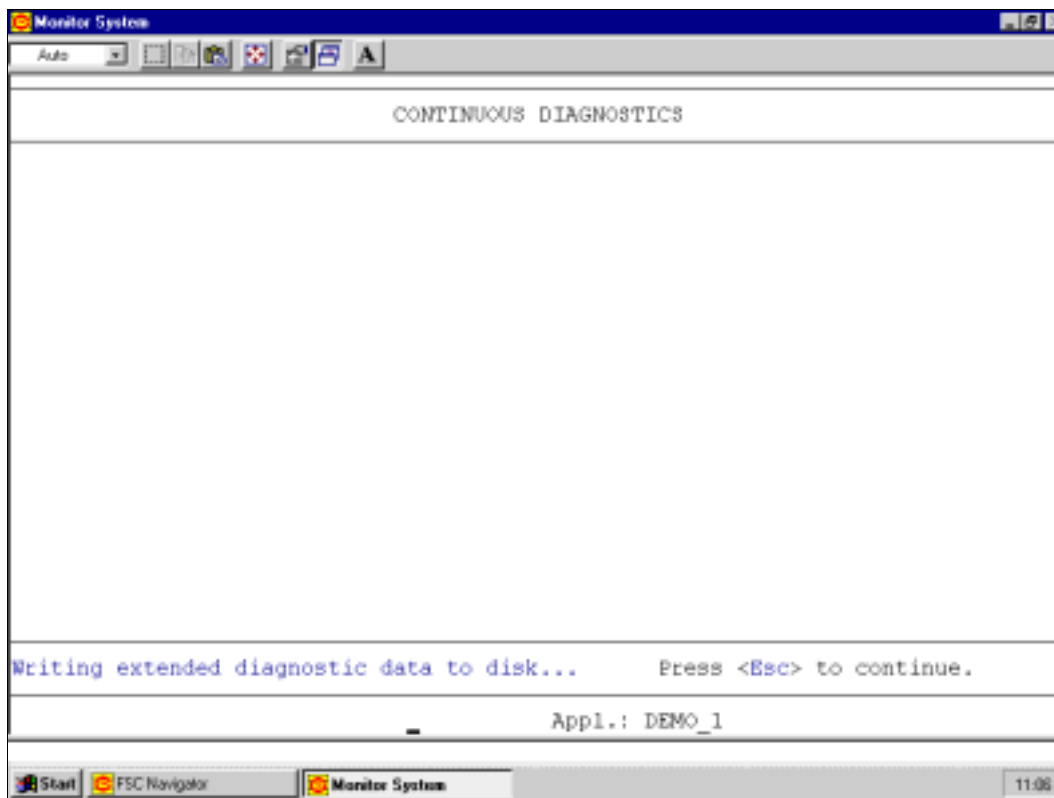


Figure 12-44 Continuous

If the continuous option is selected, all diagnostic data of all selected applications will be retrieved and stored in the extended diagnostics file continuously.



The bottom line shows you the application that is currently being interrogated.

Before extended diagnostics data of a Central Part is stored, the size of the extended diagnostic database will be checked.

If the size of the extended diagnostic data file exceeds 90% of the maximum size, a message will appear denoting that the database is almost full. You can backup a part of the database.

If the current size exceeds 95% of the maximum size, the size will be reduced to 85% of the maximum size by automatically deleting the oldest information stored in the database. This to create more space for new data. A message will appear to denote that data is being deleted.

If extended diagnostic data is deleted from the database, it can never be restored.

The database will be rebuilt after the deletion of the extended diagnostic data. This may take some time.

To stop continuous interrogation, press <Esc>.

When you stop continuous interrogation, the communication to all the Central Parts of the application which is currently interrogated will be finished and the retrieved data will be stored in the extended diagnostic database. Then you return to the 'Diag to disk' menu.

Note:

If the time of the FSC system has changed, there may be inconsistencies between the time of the diagnostic messages and the save time.

12.4.5.4 Retrieve

The retrieve option decodes stored diagnostic data, and displays it on screen (see Figure 12-45).

First you must specify the name of the data file to be retrieved. The name must have the format <Disk>:\<Path>\<File name>. The extension of the file to be displayed is fixed with '.EXD', and must not be specified.

The name of the application is the default extended diagnostic data file name.

If the stored data was retrieved from an application, which can communicate with the current application, then the data will be decoded. Otherwise a message appears which indicates the diagnostic data cannot be displayed.

The last extended diagnostic data which is stored in the diagnostic data file is first displayed when selecting 'retrieve'.

To browse through the stored extended diagnostic data you have to press <PgUp>, <PgDn> to select the previous/next stored diagnostic extended data.

If you search specific diagnostic data you have to enter the date/time of that diagnostic data using the format DD-MM-YY,HH.MM.SS. The screen will show the first extended diagnostic data from that specific time. If from that time on no data is stored, a message will appear.



Figure 12-45 Retrieve

12.4.6 'Load' option

This option allows you to load your application program from your hard disk into the FSC system. If the password for 'Loading application in RAM' is active, you must enter the correct password. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

The 'Load' option has a number of shortcuts:

- **Menu bar:** Project / Download Application
- **Navigation area:** FSC System Configuration / Load Application / Download Application (see Figure 12-46)

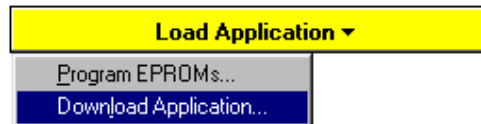


Figure 12-46 Load Application button with Download Application option

To use this option, the application must be in RAM or simulation mode. To configure RAM or simulation mode, use the 'System Configuration' option of FSC Navigator.

Notes:

This option is disabled if you have a connection with the FSC system via modem.
This option will only be enabled if the Force input key is present and has been allocated.

If you choose this option, you can select the Central Part to be loaded by toggling with <Space>. Press <Enter> to confirm. If your system is non-redundant, you do not need to specify a Central Part.

If the FSC system is running, loading will not be performed and a warning is displayed on screen:

System is running !! Stop execution (Y/N)

Type 'Y' to stop execution and perform the load or type 'N' to return to the system menu.
If you continue with the load, the text 'Central part stopped' appears in the bottom right corner of the screen. The following message will be displayed:

Waiting for acknowledge from FSC system to load application in RAM...



If acknowledgement is received from the FSC system, the load is performed. During this operation the screen will look like this:

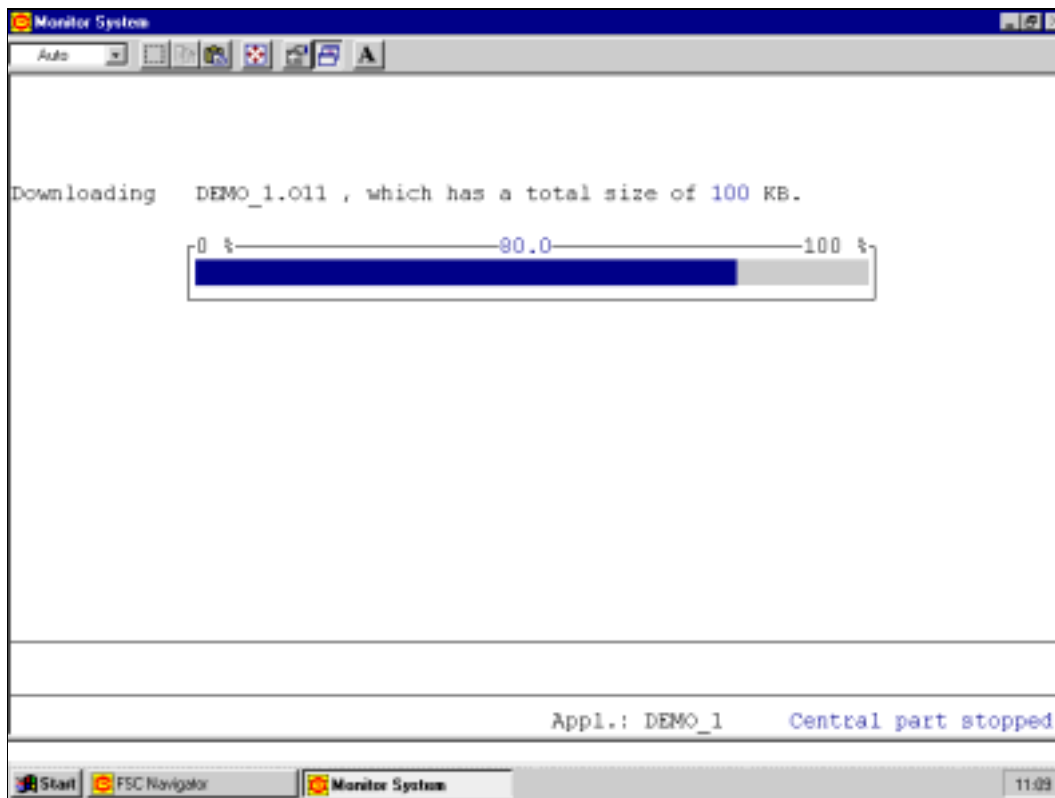


Figure 12-47 Loading an application

The name of file that is being loaded and its size are displayed.
The progress bar indicates how much of the file has been downloaded.

If the load was successful, FSC Navigator will report that no errors were detected. The message bar will contain one of the following messages:

If the download of the application is performed and the system was not running (off-line), the message will be:

Download successful. Please VERIFY extended diagnostics.

After the download of an application that is being activated while the system is running (e.g. during on-line modification), the message will be:

Waiting for OLM check and FSC-FSC COM check...

This message indicates that the system is busy comparing the application of the Central Part and checking the FSC-FSC communication. This will take some time, depending on the size of the application.

If the other Central Part has been downloaded and the result of FSC-FSC communication check is correct, the message will be:

Load successful. Use the RESET key switch to start up the central part(s).



After completing the OLM check and the FSC-FSC communication check, diagnostics of the FSC system will be retrieved and displayed.

If no faults were detected during downloading (see Appendix D, "On-Line Modification"), you must press the watchdog reset to start the downloaded application and the message 'Central stopped' will disappear. If any faults were detected during download, messages will be displayed. Refer to subsection 12.9 for an explanation of these messages.

After download has been completed, you will return to the 'Fsc system' menu.



12.5 Process status

The various options for the process status menu are shown in Figure 12-48:

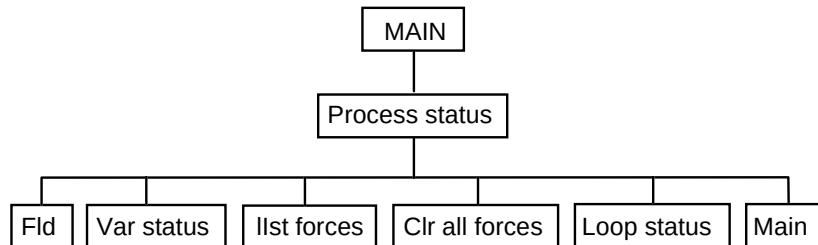


Figure 12-48 Structure of 'Process status' menu

A description of each option is given in the next subsections.

12.5.1 FLD status

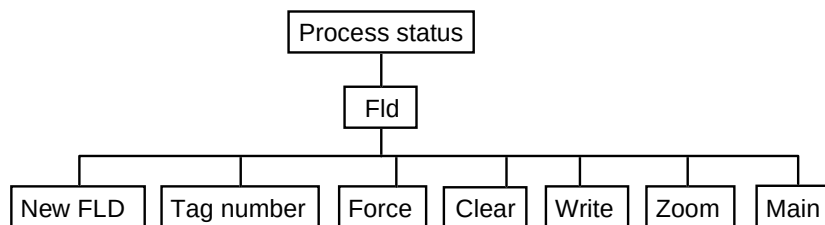


Figure 12-49 The 'Fld' menu

This option shows the status of all signals present on a selected functional logic diagram, which has been created using the 'Design FLDs' option of FSC Navigator. The status of the signal is continuously updated. Figure 12-51 shows an example of an FLD status screen.

The 'FLD status' option has a number of shortcuts:

- **Menu bar:** On-Line / Monitor FLDs
- **Navigation area:** FSC On-Line Environment / Process Status Monitoring / Monitor FLDs (see Figure 12-50)

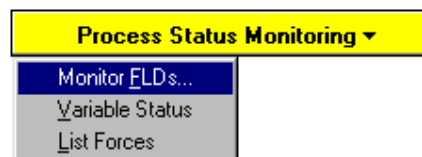


Figure 12-50 Process Status Monitoring button with Monitor FLDs option

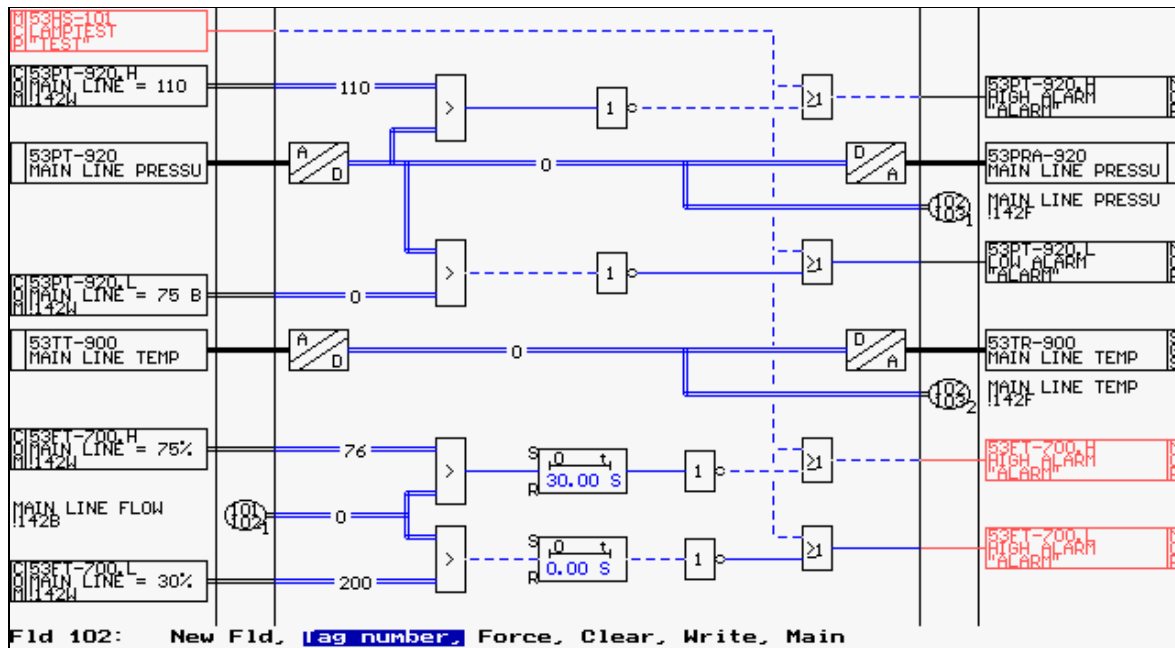


Figure 12-51 FLD status

You can change the status of variables using the 'Write', 'Force' and 'Clear' options in the menu.

Signal value representation:

- **Boolean signals (I, O)**
A straight line indicates the signal is high. A dotted line indicates the signal is low.
- **Binary signals (BI, BO) and analog signals (AI, AO)**
The values are displayed across the double lines (for AI/AO in engineering units).

Note:

If the length of a binary value exceeds the width of the double line, no value will be displayed. The only solution is to change the size of the double line in your functional logic diagram.

- **Timer, counter and register values (T, C, R)**
The current value is displayed in the symbol.
- **Annunciator outputs**
The current status of the annunciator output is indicated between the two vertical lines before the output.
If this output is high this is indicated by 'A' else no indication is displayed.
Note that if the annunciator output is flashing the indication might not be consistent in frequency to the current output, because of the refreshing time of the screen display.

12.5.1.1 New FLD

This option allows you to enter the number of another functional logic diagram to view its status.



12.5.1.2 Tag number

This option allows you to enter a tag number to view the status of the corresponding functional logic diagram.

12.5.1.3 Force

If the password for "Forcing variables" is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

This option allows you to force a specific variable. To select a variable (only inputs or outputs), move the cursor with the mouse or the arrow keys, and press <Enter> or the left mouse button.

Note:

The force option is only available if the force enable system input is active!

The screen will look like this:

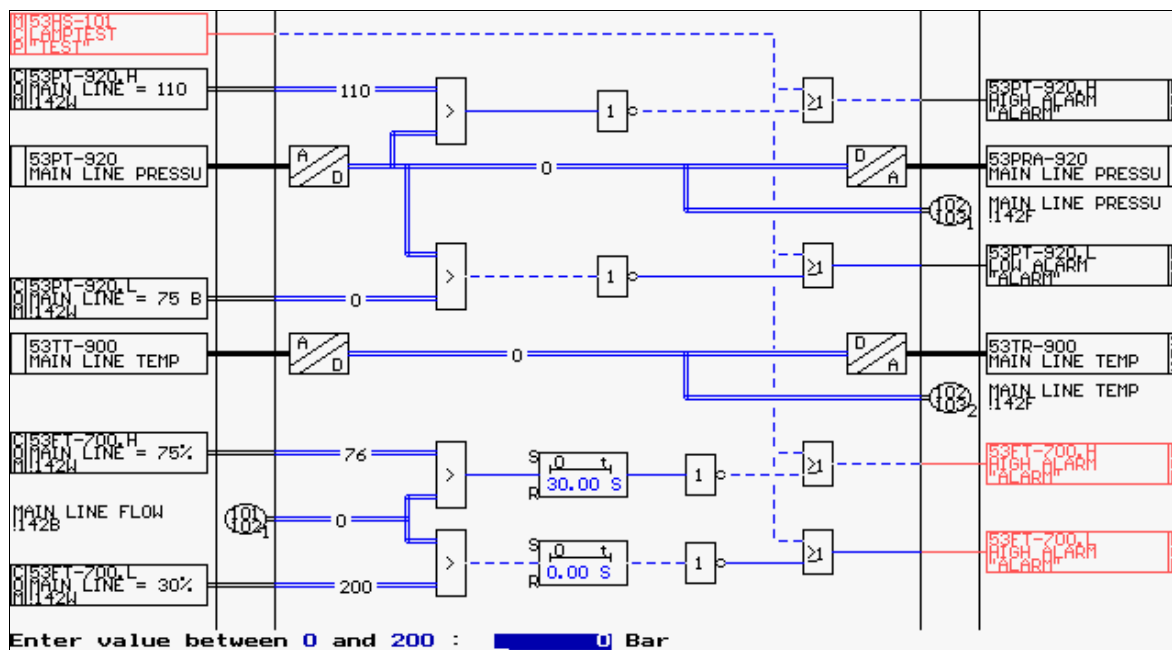


Figure 12-52 Forcing a value

If the variable is a digital value, toggle the <Space> bar and press <Enter> to confirm the desired force value. If the variable is a binary value, enter the desired force value and press enter to confirm.

If an input is forced, the current scan value will be displayed between the two vertical lines behind this input. If an output is forced the force value will be displayed between the vertical lines before the output.

The screen will look like this:

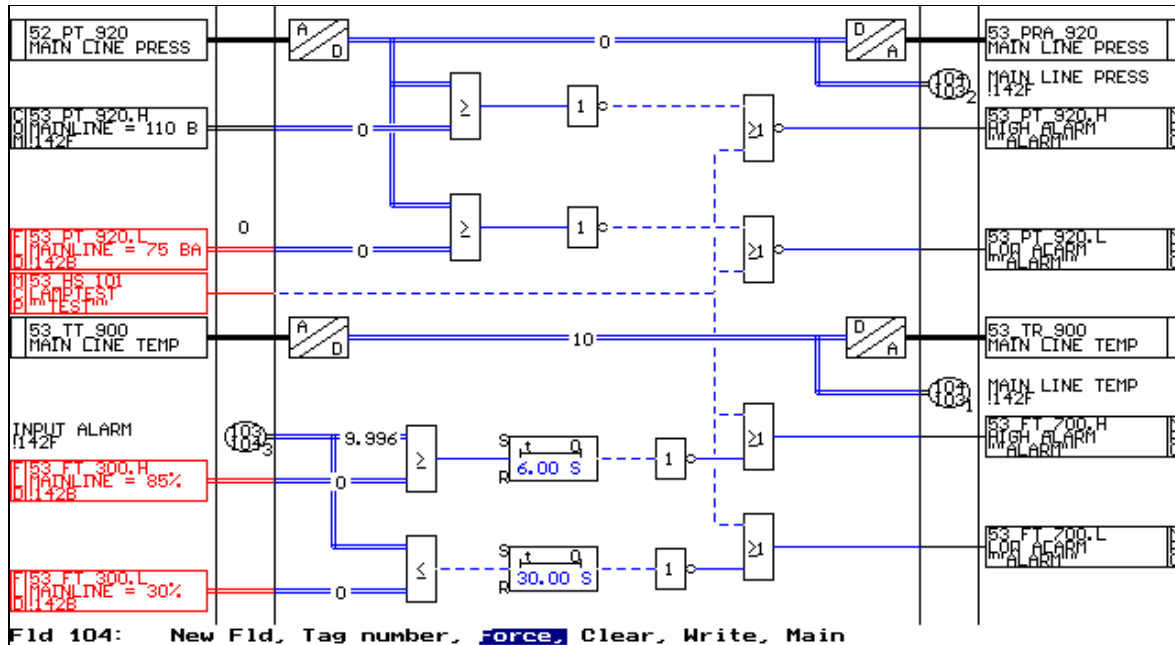


Figure 12-53 Forcing a value

Note:

Forcing is only accepted by the system:

- if the force enable (key) switch is ON.
- if the system variable has been configured as 'force enable' in the configuration file.

12.5.1.4 Clear

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. If you choose this option, you will be able to clear the forced status of a specific variable. To select a variable (only inputs or outputs), move the cursor with the mouse or the arrow keys, and press <Enter> or the left mouse button.

12.5.1.5 Write

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. If you choose this option, you will be able to write to a specific variable. To select a variable (only inputs or outputs), move the cursor with the mouse or the arrow keys, and press <Enter> or the left mouse button.

If the variable is a digital value, toggle <Space> and press <Enter> to confirm the desired write value. If the variable is a binary value, enter the desired write value and press <Enter> to confirm.



Note:

Writing is only accepted by the system:

– If the system variable is configured as 'write enable' in the configuration file.

12.5.1.6 Additional options

PgUp: This option will step to the previous sheet of the application.

PgDn: This option will step to the next sheet of the application.

If you used the SYM2, SYM3 or SYM4 symbol library for drawing the functional logic diagrams, the functional logic diagrams consist of more than one display. Therefore the next options are available:

View: Displays the entire sheet on screen. This means that the sheet is compressed and some text may not be readable anymore (depending on the graphic possibilities of the PC). This option is only to obtain an overview of the entire functional logic diagram. The 'View' option is only available in zoom mode.

Zoom: This option allows you to go back to the normal display when you are in view mode. The 'Zoom' option is only available in view mode.

Up: This option allows you to scroll the sheet half a screen upwards. The 'Up' option is only available in zoom mode.

Down: This option allows you to scroll the sheet half a screen downwards. The 'Down' option is only available in zoom mode.

Note:

If the FSC system is part of an FSC network and not directly connected to the FSC user station, communication will take more time. In that case mouse response may be slower when selecting variables or menu options. Keep the mouse button pressed until the option has been selected.

12.5.2 Variable status

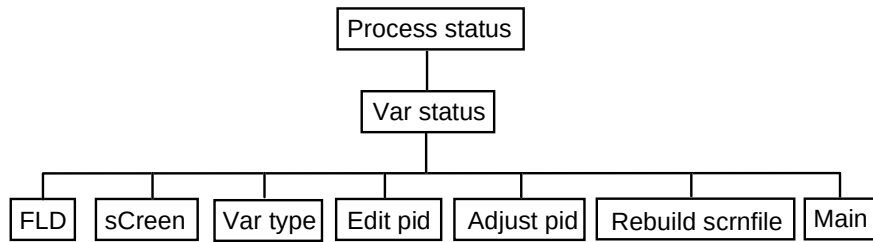


Figure 12-54 The 'Var status' menu

The 'Var status' option has a shortcut:

- **Navigation area:** FSC On-Line Environment / Process Status Monitoring / Variable Status (see Figure 12-55)

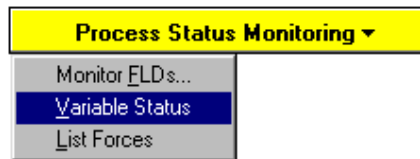


Figure 12-55 Process Status Monitoring button with Variable Status option

12.5.2.1 FLD

This option displays a selected functional logic diagram (as designed with the functional logic editor) and updates the FSC variables placed in this FLD. If you choose this option, you will be asked to enter an FLD number is asked. Valid entries are S1-S999. After you have selected the FLD, it will be displayed. You have the option of selecting another FLD by entering <PgDn> for the next FLD, <PgUp> for the previous FLD and 'S' for selecting a specific sheet. Figure 12-56 shows an example of variable status screen.



Type	Tag number	Status	Type	Tag number	Status
I	53HS-101	HIGH	O	53FT-700.H	HIGH
AI	53PT-920	0 Bar	O	53FT-700.L	HIGH
AI	53TT-900	0	O	53PT-920.H	HIGH
T	102_1	0 S	O	53PT-920.L	HIGH
T	102_2	0 S	AO	53PRA-920	0 %
R	101_102_1	0	AO	53TR-900	-0.0206 m3/s
BI	53FT-700.H	0	R	102_103_1	0
BI	53FT-700.L	0	R	102_103_2	0
BI	53PT-920.H	0			
BI	53PT-920.L	0			

Screen : Search screen, Force, Clear force, Write, Edit

Appl.: DEMO_1

Figure 12-56 Variable status screen

If <PgDn> or <PgUp> has been entered, the program searches the next or previous FLD that has been defined as a status sheet.

If one or more variable can be forced or written to, the force and/or write option will be displayed in the menu line. Just press the option you wish and select a variable with the mouse or the arrow keys, (the selected variable is highlighted) and press <Enter> to confirm.

Enter the desired value (or toggle space bar for digital values) followed by <Enter>.

If the variable is forced, this is indicated with 'F'.

If the variable is regarded faulty, this is indicated with 'D'.

Note:

When you are selecting a variable (with the mouse or the arrow keys), only the variables which are allowed to be forced or written to will be highlighted.

12.5.2.2 Screen

Screens can be composed by the user and can contain up to 32 variables. These variables can, for example, be the inputs and the outputs of a process unit.

This option displays a selected screen and updates the FSC variables placed in this screen. When entering this option a screen number is asked. Valid entries are 1-99. After you have selected the screen, it will be displayed. You have the option of selecting another screen by entering <PgDn> for the next screen, <PgUp> for the previous screen and 'S' for selecting a specific screen. Example:

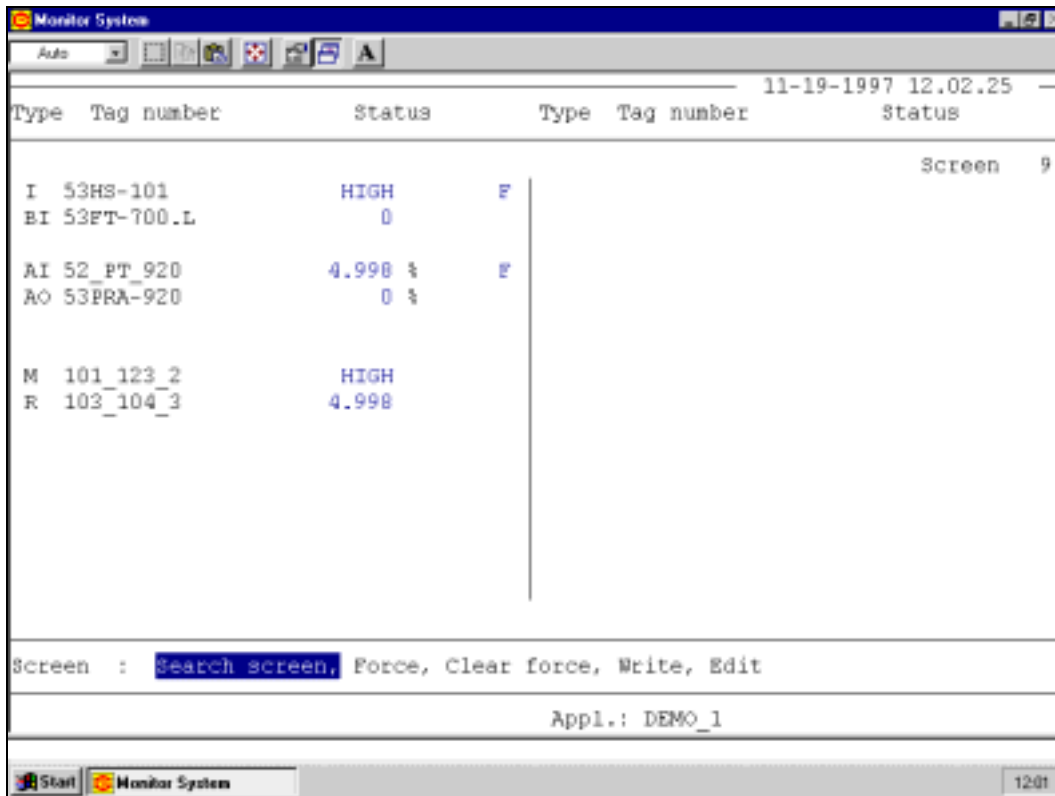


Figure 12-57 Display screen

If <PgDn> or <PgUp> has been entered, the program searches the next or previous screen that has been defined as a status screen.

If one or more variable can be forced or written to, it will be displayed at the menu line. If the selected option is password protected, you must enter the correct password to start the option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. Just press the option you wish and select a variable with the arrow keys (the selected variable is highlighted), and press <Enter> to confirm. Enter the desired value (or toggle space bar for digital values) followed by <Enter>.

If the variable is forced, this is indicated with 'F'.

If the variable is regarded faulty, this is indicated with 'D'.

Note:

When you are selecting a variable (with the mouse or the arrow keys), only the variables which are allowed to be forced or written to will be highlighted.



12.5.2.2.1 Edit screen

This option allows you to select a screen for editing.

A screen consists of a maximum of 32 FSC variables (type and tag number) or a maximum of 8 PID controllers. Each screen has a sequence number. This option is available in the 'Sheet' and 'Screen' menu. The numbers 1-99 are selectable for PIDs or any other FSC variable.

The sheet numbers S1-S999 are assigned to corresponding functional logic diagrams 1-999. These sheets are created automatically when the 'On-Line Environment' option of FSC Navigator is called, but they can be changed if necessary. A PID cannot be assigned to sheet S1-S999

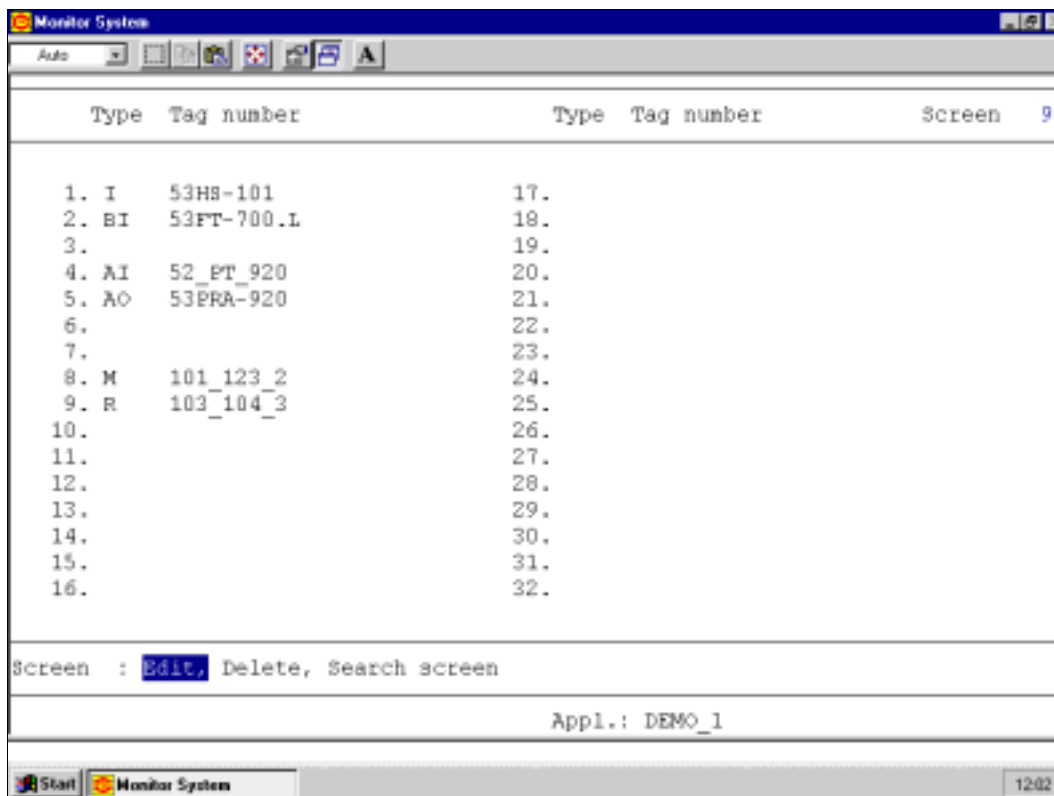


Figure 12-58 Status screen



If you have entered a number in the range of 1-99 and this screen number has been defined as a PID screen, you will, for example, see the following for screen 10:

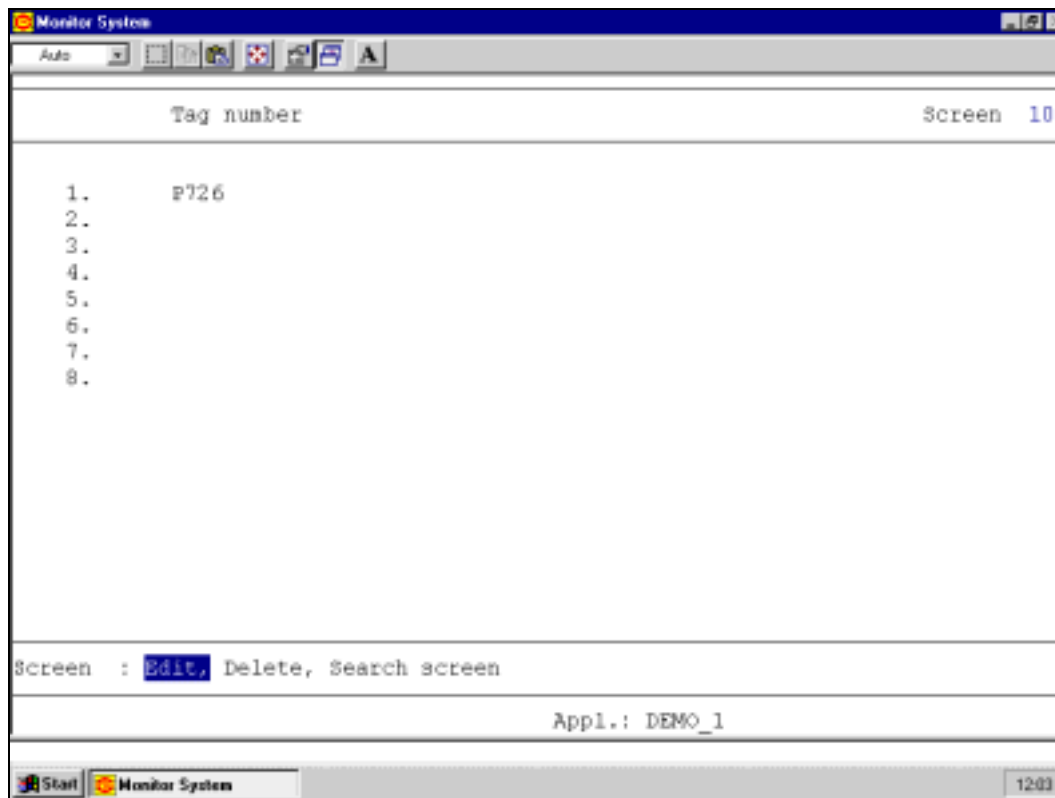


Figure 12-59 PID screen



If you have entered a number in the range of 1-99 and this screen number has been defined as a status screen, you will, for example, see the following for screen 9:

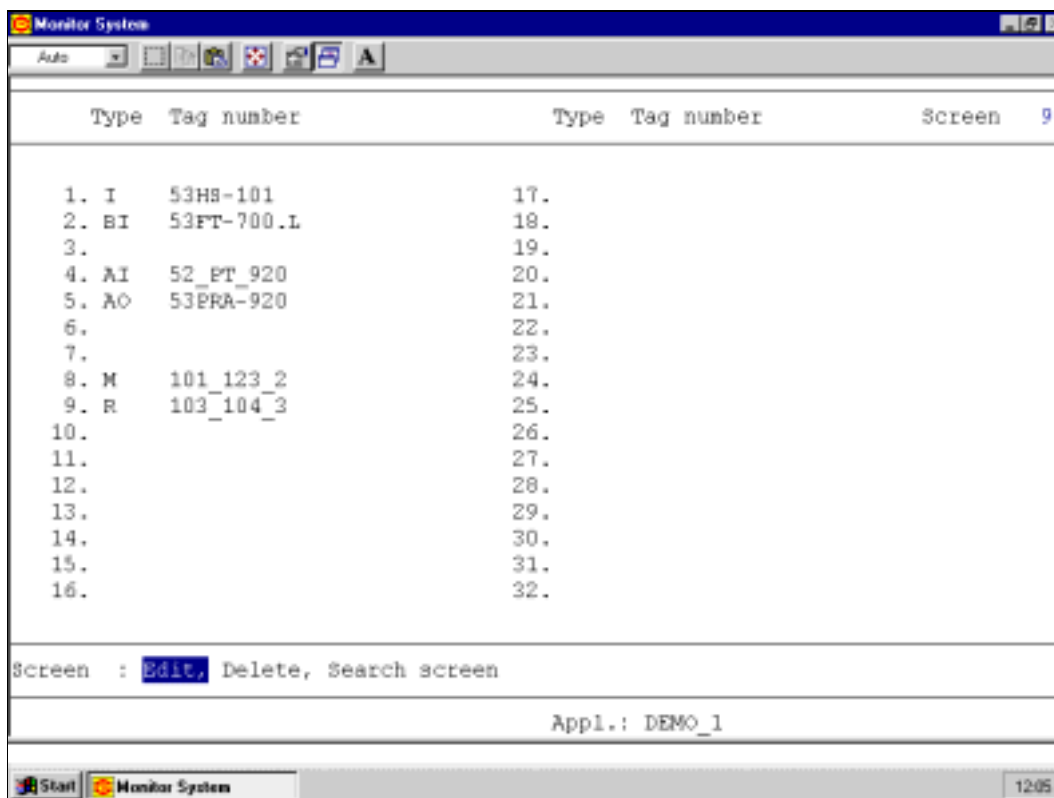


Figure 12-60 Status screen

If the selected screen has already been defined, the variables will be displayed on screen. Otherwise a selection must be made (only in screen option) if the screen must be defined as PID screen or a status screen. The following options are available:

- Edit:** When this selection is accepted, the screen can be edited. If you press <Esc>, the screen is saved to disk. After saving, the screen will be displayed, which allows you to check it.
- Delete:** If you choose this option, the current screen will be deleted. You will be asked to confirm the deletion. If you do not confirm it, you will return to the 'Edit screen' menu and the current screen will still be available.
- Search screen:** You will be asked to enter a screen number. If this screen has been defined as a PID screen or status screen, it will be displayed. Figure 12-61 shows an example of what the screen might look like.

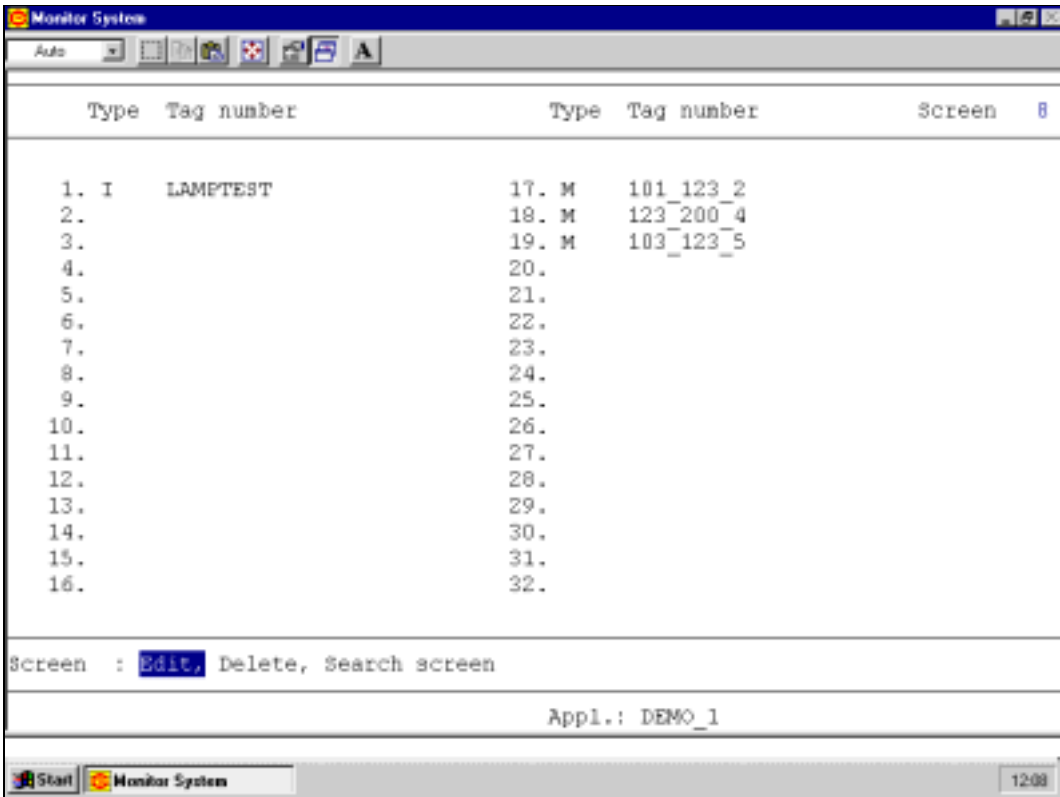
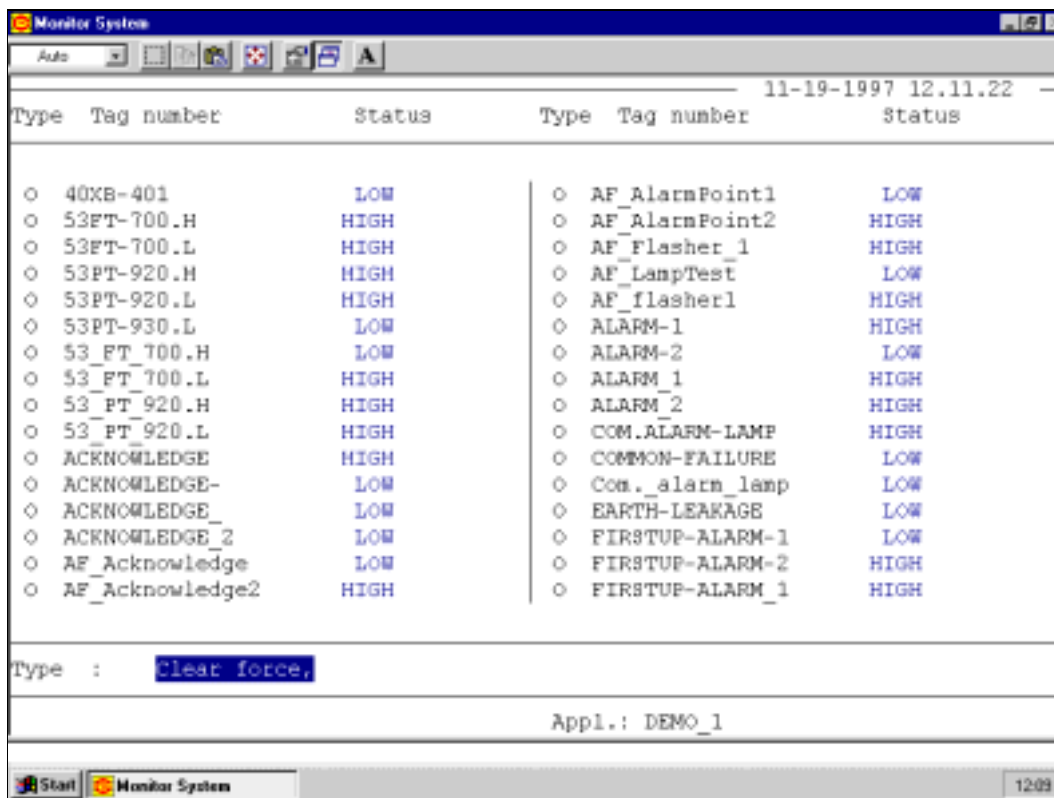


Figure 12-61 Status screen



12.5.2.3 Variable type

If you choose this option, you are asked to enter an FSC variable (type and tag number). The status of this FSC variable is displayed and continuously updated. If you have entered an incomplete tag number, all FSC variables with a tag number that starts with the incomplete tag number are displayed and updated. The maximum number of FSC variables displayed is 32. Figure 12-62 shows an example of what the screen might look like.



The screenshot shows a window titled "Monitor System" with a menu bar (Auto, icons, A) and a status bar (11-19-1997 12.11.22). The main area displays a table of variables with columns: Type, Tag number, Status, Type, Tag number, Status. The variables are listed in two columns. The first column includes variables like 40XB-401, 53FT-700.H, 53FT-700.L, 53PT-920.H, 53PT-920.L, 53PT-930.L, 53_FT_700.H, 53_FT_700.L, 53_PT_920.H, 53_PT_920.L, ACKNOWLEDGE, ACKNOWLEDGE-, ACKNOWLEDGE_, ACKNOWLEDGE_2, AF_Acknowledge, and AF_Acknowledge2. The second column includes variables like AF_AlarmPoint1, AF_AlarmPoint2, AF_Flasher_1, AF_LampTest, AF_flasher1, ALARM-1, ALARM-2, ALARM_1, ALARM_2, COM.ALARM-LAMP, COMMON-FAILURE, Com._alarm_lamp, EARTH-LEAKAGE, FIRSTUP-ALARM-1, FIRSTUP-ALARM-2, and FIRSTUP-ALARM_1. The status of each variable is displayed in blue text. At the bottom, there is a "Type :" field with the text "Clear force," and an "Appl.: DEMO_1" field. The taskbar at the bottom shows the Start button, the Monitor System icon, and the time 12:03.

Type	Tag number	Status	Type	Tag number	Status
○	40XB-401	LOW	○	AF_AlarmPoint1	LOW
○	53FT-700.H	HIGH	○	AF_AlarmPoint2	HIGH
○	53FT-700.L	HIGH	○	AF_Flasher_1	HIGH
○	53PT-920.H	HIGH	○	AF_LampTest	LOW
○	53PT-920.L	HIGH	○	AF_flasher1	HIGH
○	53PT-930.L	LOW	○	ALARM-1	HIGH
○	53_FT_700.H	LOW	○	ALARM-2	LOW
○	53_FT_700.L	HIGH	○	ALARM_1	HIGH
○	53_PT_920.H	HIGH	○	ALARM_2	HIGH
○	53_PT_920.L	HIGH	○	COM.ALARM-LAMP	HIGH
○	ACKNOWLEDGE	HIGH	○	COMMON-FAILURE	LOW
○	ACKNOWLEDGE-	LOW	○	Com._alarm_lamp	LOW
○	ACKNOWLEDGE_	LOW	○	EARTH-LEAKAGE	LOW
○	ACKNOWLEDGE_2	LOW	○	FIRSTUP-ALARM-1	LOW
○	AF_Acknowledge	LOW	○	FIRSTUP-ALARM-2	HIGH
○	AF_Acknowledge2	HIGH	○	FIRSTUP-ALARM_1	HIGH

Type : Clear force,

Appl.: DEMO_1

Figure 12-62 Variable type screen

If one or more variable can be forced or written to, it will be displayed at the menu line. If the force option is password-protected, you must enter the correct password to start the option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. Just press the option you wish and select a variable with the mouse or the arrow keys (the selected variable is highlighted), and press <Enter> to confirm. Enter the desired value (or toggle space bar for digital values) followed by <Enter>.

If the variable is forced, this will be indicated with 'F'.

If the variable is regarded faulty this will be indicated with 'D'.

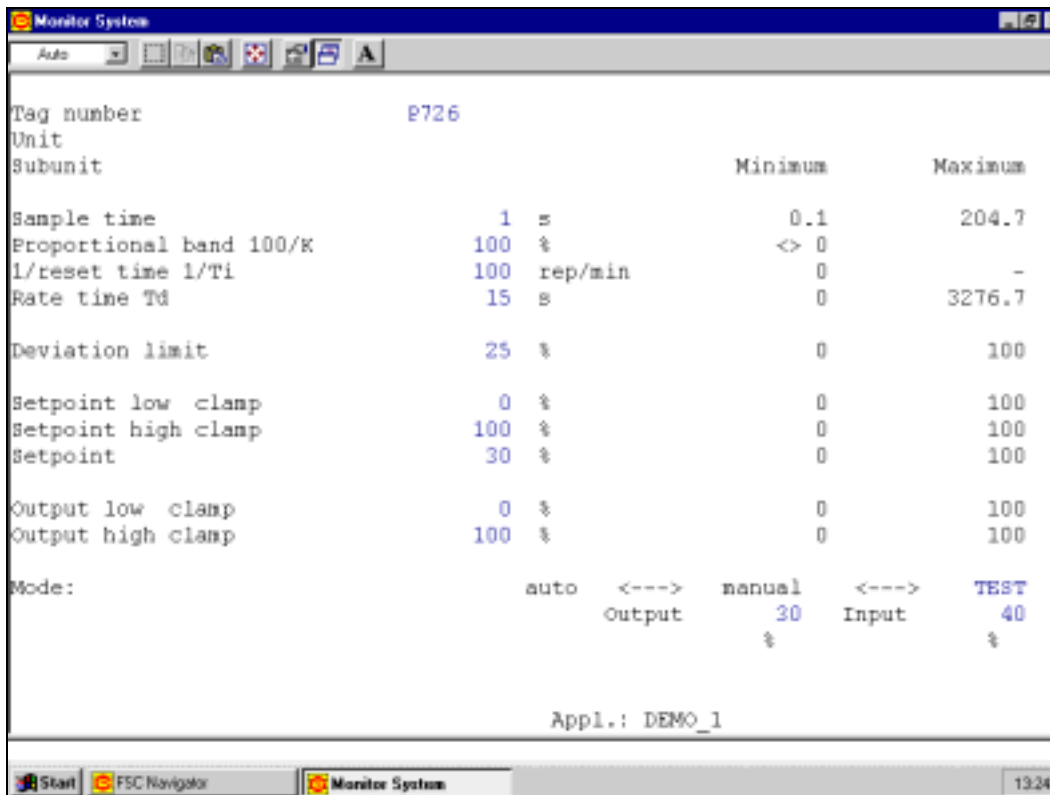
Note:

When you are selecting a variable (with the mouse or the arrow keys), only the variables which are allowed to be forced or written to will be highlighted.

If you press <PgDn>, the program will search the next variables that correspond to the entered type and tag number, and displays them. You can always go back to the previous screen by pressing <PgUp>.

12.5.2.4 Edit PID

If you choose this option, you will be asked to enter a tag number to search for. Enter the tag number for the PID controller you want to see or edit. If the tag number you entered is found in the database, the following screen will appear:



The screenshot shows a window titled "Monitor System" with a menu bar (Auto, View, Print, Help) and a toolbar. The main display area shows the following parameters for tag P726:

Parameter	Value	Unit	Minimum	Maximum
Tag number	P726			
Unit				
Subunit				
Sample time	1	s	0.1	204.7
Proportional band 100/K	100	%	<> 0	
1/reset time 1/Ti	100	rep/min	0	-
Rate time Td	15	s	0	3276.7
Deviation limit	25	%	0	100
Setpoint low clamp	0	%	0	100
Setpoint high clamp	100	%	0	100
Setpoint	30	%	0	100
Output low clamp	0	%	0	100
Output high clamp	100	%	0	100
Mode:	auto		manual	
	Output		Input	
			30	40
			%	%

At the bottom of the window, it says "Appl.: DEMO_1". The taskbar at the bottom shows "Start", "FSC Navigator", "Monitor System", and the time "13:24".

Figure 12-63 Edit parameters

The input screen contains the following information and input options.

- Tag number: Displays the tag number of the PID which is on screen. (Cannot be changed)
- Unit: Displays the process Unit in which the PID is configured. (Cannot be changed)
- Sub unit: Displays the process Sub-Unit in which the PID is configured. (Cannot be changed)
- Sample time: The sample time for the PID controller. Since this is a digital PID controller, this is the time between two samples. The range is 0.1...204.7 sec.
- Proportional Band 100/K: This indicates the percentage change in the error (setpoint input) for doubling the output. Can be changed over full range, but cannot be 0.
- 1/Reset time: This indicates the times per minute in which the integration is calculated. E.g. if 1/Reset Time is 2 Rep/Min, the time is integrated in 1/2 minute. Can be changed in the full positive range. If 0, the integrating part is out of the controller.



Rate time:	This is the time section over which the differentiations are discreted. The range is 0.0...3276.7 Sec. If 0, the differentiating part is out of the controller.
Deviation limit:	This is the high clamp for the error (setpoint - input). Take care: If this parameter is zero, there will be no correct controlling. Any value in engineering input units can be assigned to this parameter.
Setpoint Low Clamp:	This is the low clamp for the setpoint. The setpoint will never become smaller then this value. Any value in engineering units can be assigned to this parameter.
Setpoint High Clamp:	This is the high clamp for the setpoint. The setpoint will never exceed this value. Any value in engineering units can be assigned to this parameter.
Setpoint:	The setpoint cannot be entered in auto mode (In this mode it is generated by the application program from, for example, BCD switches or a constant.). In test mode, a test setpoint can be entered. If you enter a test input in this mode too, you can simulate a process. Any value in engineering units can be assigned to this parameter.
Output Low Clamp:	This is the low clamp for the output. The output will never become smaller then this value. Any value in engineering units can be assigned to this parameter.
Output High Clamp:	This is the high clamp for the output. The output will never exceed this value. Any value in engineering units can be assigned to this parameter.
Mode:	If the password for 'Writing variables' is active, you have to enter the correct password to start-up this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. Only if the cursor position is at this field the input, output and setpoint are continuously updated. The other parameters are less frequently updated. Cascade: Acts as slave, controlled by a master. Can only be switched in this mode if the master is in Auto mode. The output of the master will control the (cascade) setpoint of this PID. Auto: This is the automatic mode, the output is calculated each sample time out of the error between input and setpoint and the parameters. The input is an FSC input, the setpoint is generated by the application program and the output is calculated, so the setpoint and the output cannot be changed. Manual: The output can be entered. This output is directly copied to an FSC output and is not calculated. Test: In this mode the input (a virtual input) and the setpoint (a virtual setpoint) can be entered. The output is calculated using these values. So the PID function can be simulated. Output: This is the FSC output and is updated if the cursor position is at 'mode'. In manual mode this value can be changed. Input: In Auto and Manual mode this is the FSC input and is updated if the cursor position is at 'mode'. In Test mode this value is a virtual test input.

After you have edited one parameter, you can select another in the same PID by using the mouse, the up and down arrow keys or the enter key. Other key commands are the following:

<ESC>	: Exit edit screen without saving the entered parameters
<PgUp>	: Checks if all inputs are in the correct range, saves and selects the previous PID in the database for editing.

<PgDn> : Checks if all inputs are in the correct range, saves and selects the next PID in the database for editing.

12.5.2.5 Adjust PID

This option will display only one PID. You can change the mode. If the password for 'Writing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file. In test mode, the (test) setpoint and (test) input can be changed. In manual mode, the output can be changed.

You will be asked to enter the tag number to search for. Enter the tag number for the PID controller that you want to adjust.

If you have entered a correct tag number, the picture for this PID is drawn on screen. This picture contains 3 bars. The left bar (PV) represents the setpoint, the middle (M) the input and the right (O) the output. These bars are filled between accessory bottom and top scaling. For example, a fully filled bar has its maximum value; a value in the middle of the bottom and top scaling will fill the bar by 50 percent. The value in engineering units is placed under each bar.

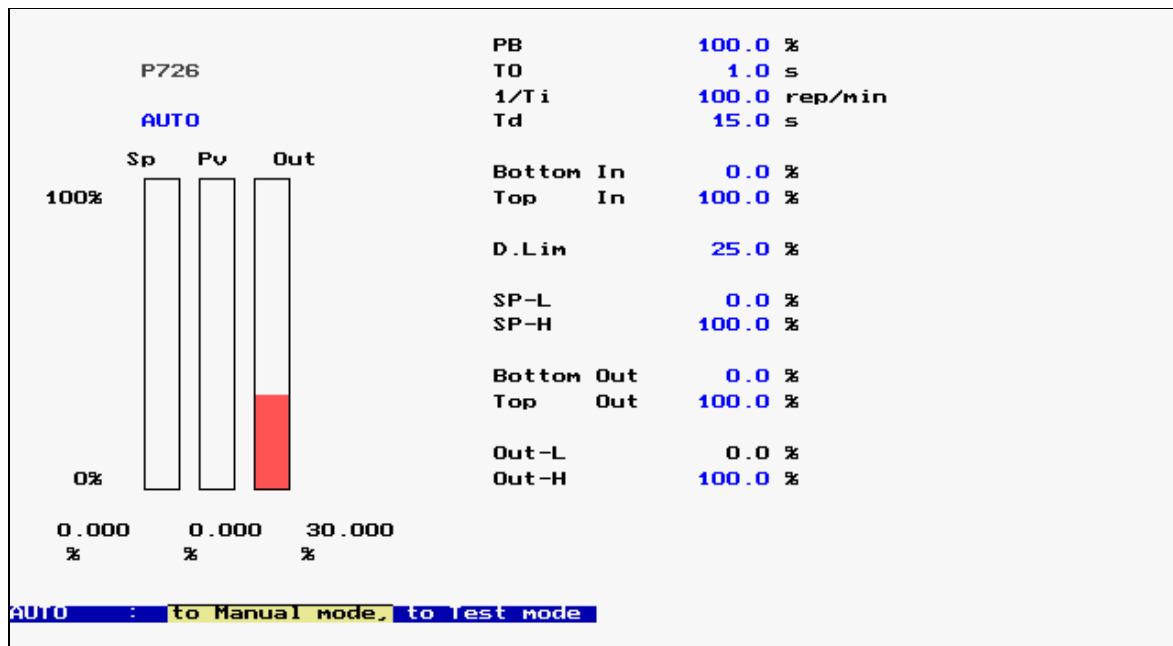


Figure 12-64 Adjust PID

At the right of the picture, the parameters of the PID are displayed. These values cannot be changed in this option.

The bars and the accessory values are continuously updated. The following key commands are available:

- <Esc> Return to the PID menu
- + , <PgDn> Display the next PID in the database.
- , <PgUp> Display the previous PID in the database.



Depending of the mode of the PID, a menu will be displayed:

Auto : to **Manual**-mode, to **Test**-mode, (optional) **Cascade** mode
Manual : to **Auto**-mode, to **Test**-mode, set **Output**
Test : to **Auto**-mode, to **Manual**-mode, set **Setpoint**, set **Input**
Cascade : to **Auto**-mode, to **Manual**-mode, to **Test**-mode

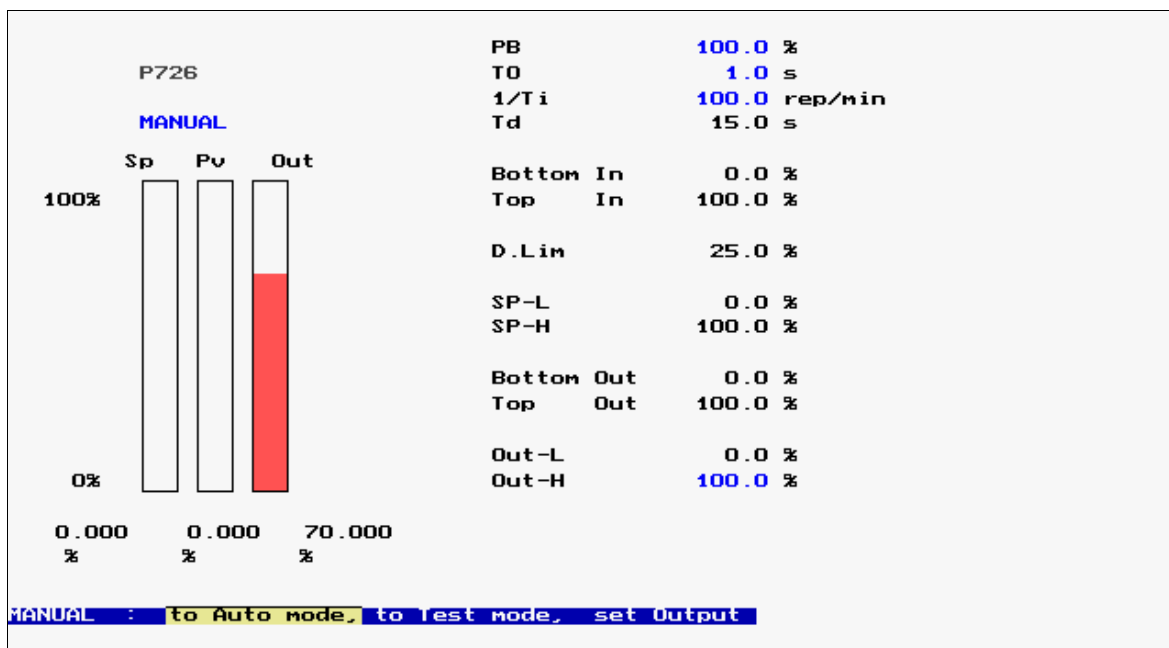


Figure 12-65 PID in manual mode

The meaning of these options are:

to **Auto** mode : Switch to auto mode and return to updating.
to **Manual** mode : Switch to manual mode and return to updating.
to **Test** mode : Switch to test mode and return to updating.
to **Cascade** mode : Switch to cascade mode and return to updating.
set **Output** : Set the output to a value which must be entered and return to updating.
set **Input** : Set the input to a value which must be entered and return to updating.
set **Setpoint** : Set the setpoint to a value which must be entered and return to updating.



12.5.2.6 Rebuild screen file

This option is used whenever you want to update your screen file according to the current database in case it is changed or modified. The screens (PID or STATUS) which you have created (no. 1-99) will still be available when this function is completed.

If you choose this option, you will be asked if the present screen file must be saved or deleted. Just press <Space> to select Yes or No, and press <Enter> to confirm. If you select Yes, the current file will be saved on disk.

The file name is identical to the project name and the extension will be changed in OLD. So whenever you wish to reuse your old screen file, change the extension OLD into SCR (e.g. using Windows Explorer).

The rebuilding of the screen file may take some time, depending on the size of the current database.



12.5.3 List forces

This option shows all the variables which are currently forced (this means inputs, outputs and markers). The variables will be represented on screen by their rack, position, channel, type and tag number. If no variable is forced, this will be mentioned at the top of the screen. Figure 12-66 shows an example of what the screen might look like.

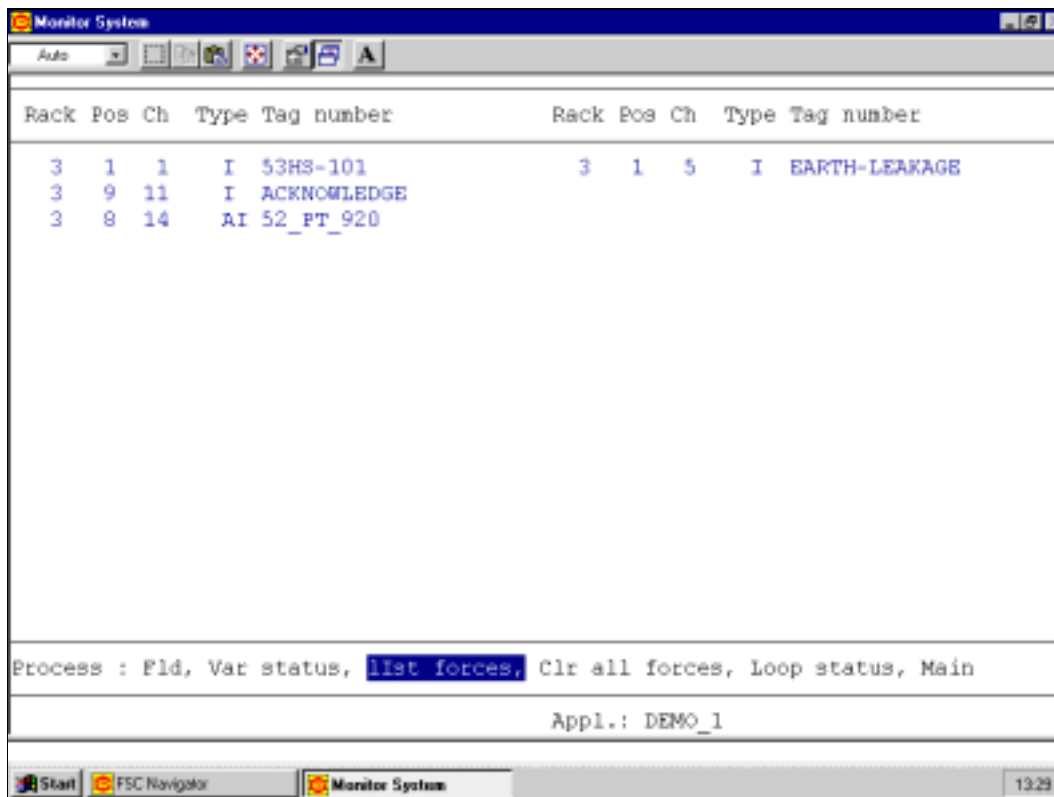


Figure 12-66 List forces

The 'List forces' option has a shortcut:

- **Navigation area:** FSC On-Line Environment / Process Status Monitoring / List Forces (see Figure 12-67)

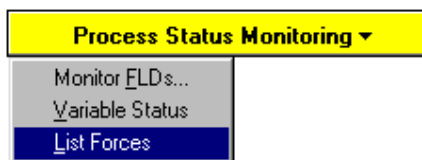


Figure 12-67 Process Status Monitoring button with List Forces option

If the list does not fit on the screen, you can use the up and down arrow keys to scroll through the list.

12.5.4 Clr all forces

If the password for 'Forcing variables' is active, you must enter the correct password to start this option. For more information on passwords, refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

The program asks you if you really want to continue. If you answer 'Yes', all forced FSC variables will be cleared. The FSC variables will get their current status.

12.5.5 Loop status

This option will show the specification of the analog input channels whose values are out of range (e.g. $< 3.8 \text{ mA}$ or $> 20.2 \text{ mA}$ for 4 - 20 mA; $> 20.2 \text{ mA}$ for 0 - 20 mA, etc.) and variables allocated to the 10214/1/2 and the 10216/1/1 module, which outputs have an open loop or short circuit. The 10214/1/2 module is a high-voltage digital output module. The 10216/1/1 module is a 24 Vdc digital output module with line-monitoring. For details on these modules, refer to the FSC Hardware Manual.

The 'Loop status' option has a shortcut:

- **Navigation area:** FSC On-Line Environment / Diagnostics / I/O Loop Diagnostics (see Figure 12-68)

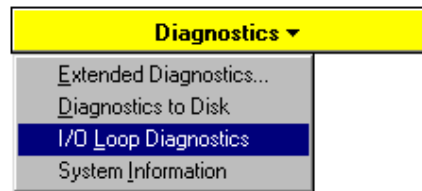


Figure 12-68 Diagnostics button with I/O Loop Diagnostics option

Figure 12-69 shows an example of what the screen might look like.

If the list does not fit on the screen, you can use the up and down arrow keys to scroll through the list.



Type	Tag number	Cabinet	Rack	Position	Channel
O	DO4	1	7	9	4
AI	AI2	1	3	1	2
AI	AI3	1	3	1	3

Process : FLD, Var status, list forces, Ctr all forces, Loop status, Main

Central part : 1 Appl.: ALF63_63

Figure 12-69 Presentation of loop status

12.6 Select application

You can use this option to switch communication to another FSC system which is also a part of the configured FSC system or Development System network.

If you choose this option, the following message will appear at the bottom of the screen:

Searching for applications on disk. Please wait...

The program creates a list of applications that are present on your hard disk. The screen will look like this:

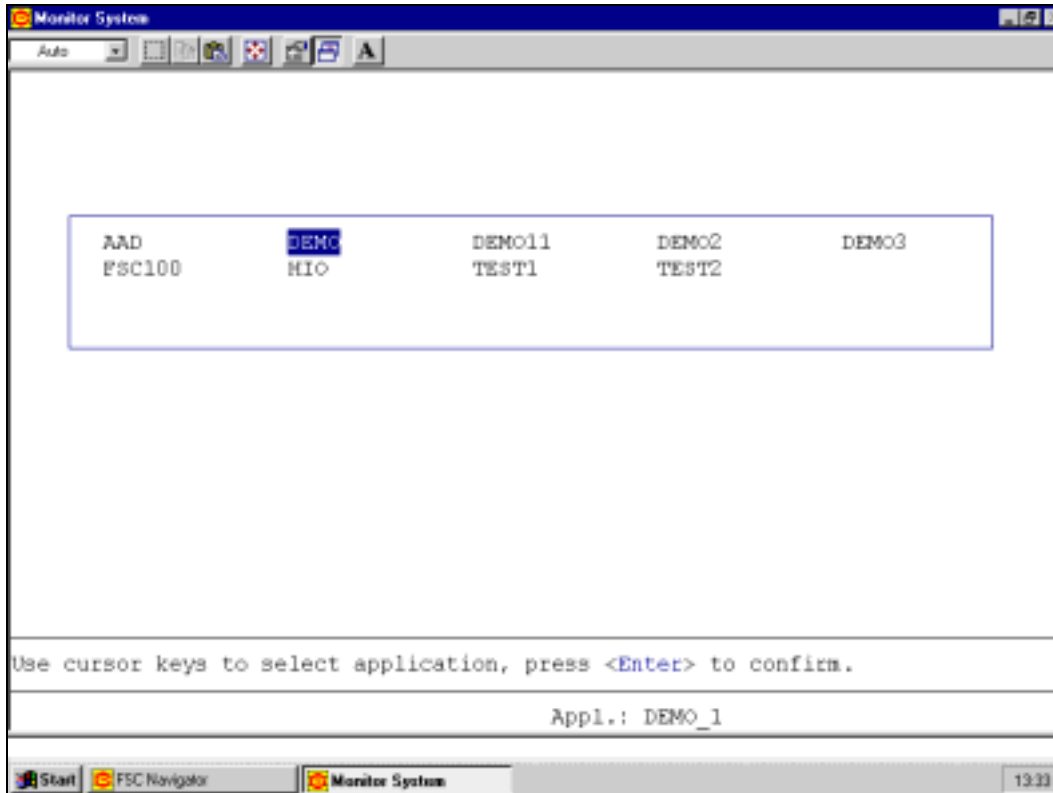


Figure 12-70 Select application screen

To select an application, move the cursor bar to the application you wish to communicate with (using the arrow keys), and press <Enter>. If the application is part of the FSC system or Development System network, the 'On-Line Environment' option of FSC Navigator will restart after switching to the new application.



12.7 Modem

This option allows you to set up communication between the FSC user station and an FSC system via a telephone line.

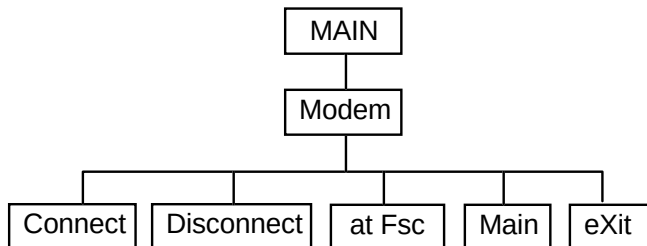


Figure 12-71 Modem menu

To be able to use modem communication, you must have configured a modem channel in your application. The channel can be configured for a modem using the 'System Configuration' option of FSC Navigator.

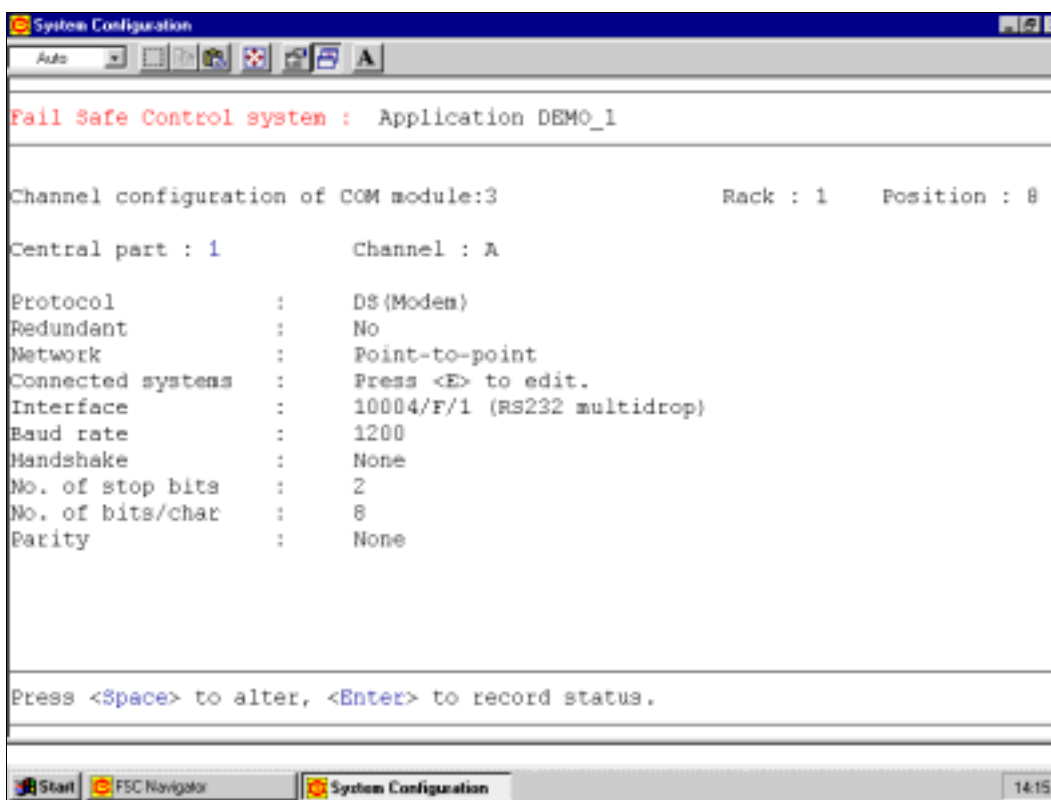


Figure 12-72 Configuration of modem (using 'System Configuration' option of FSC Navigator)



Setting up the modem link is done using the 'On-Line' Environment' option of FSC Navigator, and consists of the following steps:

- Connect the FSC modem to the FSC user station's COM port and select option 'at FSC'.
- Connect the FSC modem to the FSC system.
- Connect the DS modem to the FSC user station's COM port and select option 'Connect'.
- After the FSC user station has dialed the phone number entered, the modem link is established.
- To terminate the modem link, select the 'Disconnect' option.

After the modem link has been set up, all options of the 'On-Line Environment' option of FSC Navigator can be selected except the 'Load' option.

All options of the 'On-Line Environment' option of FSC Navigator can be selected via modem communication except 'Load application'.

12.7.1 At FSC

This option allows you to configure the FSC modem which is connected to the FSC system.

To use this option effectively, you must first connect the FSC modem to the FSC user station's COM port. This is done via a PC modem cable, which is delivered with the modem.

The FSC user station's COM port is selected using the 'Communication Port' option of the On-Line menu. For details on setting the communication port refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

The configuration is done by sending commands to the FSC modem.

To enable the use of any Hayes-compatible modem, the configuration commands are stored in the FSC Navigator initialization file FSC.INI.

After the modem has been configured, you can connect it to the FSC modem channel and a telephone line.

12.7.2 Connect

If you choose this option, a telephone number can be entered. The maximum length of the telephone number is 20 characters. After you have entered the phone number, the modem link will be set up.

After connection has been established, data can be retrieved from the FSC system(s) by selecting the desired function. If there is no response (after 5 retries), the following message appears:

No connection with <tel. no.>

In that case, check:

- the modem connections,
- the phone number, and
- the configuration commands in the modem definition file.

12.7.3 Disconnect

If you choose this option, the established connection will be terminated. If the connection cannot be terminated, the following message appears:

Disconnect not successful. Press any key to retry or <Esc> to quit.

If the connection is terminated, pressing <Esc> will return you to the 'Modem' menu.



12.7.4 Main

This options returns you to the 'On-Line Environment' main menu.

12.7.5 Exit

This options leaves the 'On-Line Environment' and returns you to the FSC Navigator main screen.



12.8 Passwords

Passwords are used to protect unauthorized execution of specific functions. For details on passwords refer to Section 3 of the FSC Software Manual ("Using FSC Navigator") or the Help file.

The following functions in the 'On-Line Environment' are password-protected:

- Forcing of variables
 - Force
 - Clear force
 - Calibrate
 - Clear all forces
- Writing of variables
 - Write
 - Edit PID parameters
 - Adjust PID
- Loading application into RAM

If a password is entered and the function (force, write or load) is executed, the program allows the user to execute the same function again for 10 minutes without re-entering the password. After every execution of the function, the period of 10 minutes is started again. If these 10 minutes have elapsed, the password must be entered again before the function can be executed.



12.9 Error messages

Address allocation of I/O variables not correct.

You have added new I/O, but the application is not translated. Translate again and place new EPROMs or perform on-line rebuild.

Application files not translated for download.

Change the mode of operation of the FSC system to RAM mode (or simulation mode) and translate again.

Application versions in database and FSC system do not match.

You have translated your application without changing the EPROMs in the FSC system. The application in the database may differ with the application in the FSC system.

Cannot be forced.

You want to force a variable which is not force-enabled according to the database.

Cannot be written

You want to write to a variable which is not write-enable according to the database.

Cannot find tag number.

You have entered a valid type, but the tag number is not found in the database. Type a correct and existing tag number.

Cannot receive PID update values.

This happens when the communication with the FSC system was not successful. Correct this by trying it again, or checking the hardware of the connection between FSC system and PC.

Cannot update diagnostic configuration file.

The DIAGEXT.CFG file cannot be created or written to. Check disk space and attributes of this file.

Command not allowed.

You have entered a force command, but the FSC system does not accept it. It is no use to try again.

Communication module (07177/1/1) not present or incorrect module address.

You have selected RS485 communication for the development system, but the RS485 communication module for the development system has not been detected.

Communication with too many retries.

- a) Check if the communication cable is terminated correctly, has no broken wires, etc.
- b) Decrease the communication speed as the length of the communication cable may be too long for the selected communication speed.

Corrupt extended diagnostic data file(s).

Delete the diagnostic index files (.EXR and .EXT)

Error in communication between user station and FSC system.

Check all modules on the system bus for correct EPROMs and jumper placements. In RAM and simulation modes, check if the RAMs have been inserted correctly.

**Error updating extended diagnostic data file(s).**

Check surface of your hard disk. Delete <application name>.EXT. The database will be rebuilt.

Force enable input not activated.

You want to force a variable, but the force enable input is not activated, so the FSC system does not accept a force command. Forces are only allowed when this input is activated.

Please note that application download is only allowed when the system is running and the force key is present and active.

FSCVIEW.MSG missing

The message file FSCVIEW.MSG is not present in the FSC path directory of the disk. Copy FSCVIEW.MSG to the correct directory.

Illegal variable type.

You have entered an invalid variable type. Valid variable types are I, O, AI, AO, BI, BO, C, T, M, R.

Invalid destination address or system number not correct.

You try to communicate with an FSC system that is loaded with a different application and/or a different system number.

Load aborted due to disk file error. Press any key.

Check the surface of your hard disk and translate again.

Load NOT successful.

Turn power OFF and remove memory module from the FSC system for 60 seconds to discharge capacitors on the MEM module.

Then turn power ON and reset CPU. Press any key to restart download.

Load NOT successful. Press any key.

Downloading of the FSC program has not been properly ended due to a communication error. Try again.

Maximum output clamp must be higher than minimum output clamp.

This can happen in Edit parameters, if you enter the mode selection field or try to save the parameters (Esc, PgUp, PgDn). Output high clamp is smaller then output low clamp. Correct this by entering a smaller output low clamp or a larger output high clamp.

MEM - COM incompatible versions

You try to download a file from a different application than the CPU/MEM EPROMs/RAMs, or the COM EPROMs are from another application.

No application files found. Translate first before choosing this option.

You try to run the download program although no application files (e.g. <Application name>.O11) are found on disk.

No Development System communication port configured for

You try to communicate via a communication port in the FSC user station that has not been configured in the FSC system. For example, FSC user station = RS485 and FSC only has RS232 port or vice versa.

**No EPROMs programmed yet for COM module.**

You try to communicate with an FSC system for which no communication EPROMs have been programmed yet.

NO INPUT CHANNEL

The selected rack, position contains no input module. Please enter proper values for rack and position.

NO OUTPUT CHANNEL

The selected rack, position contains no output module. Enter proper values for rack and position.

No PID screens have been created.

This happens if you want to enter the 'Graphic view' option and have not created any PID screen in 'Edit screen'. Correct this by entering the 'Edit screen' option in the status main menu. Create a PID screen and select at least one PID. Then try the 'PID screen' option again.

No records of this type in database.

You have entered a valid type, but there are no FSC variables of that type in the current database.

No response from FSC system.

- a) Check if the cable is plugged in the correct communication port of the development system and/or the FSC system.
- b) Check if the EPROMs on the COM module are placed correctly. If the EPROMs are placed correctly, verify the EPROMs.
- c) Check all modules on the system bus for correct EPROMs and jumper placements.

Power-on modes in database and FSC system do not match.

Check if the power on mode in the database is the same as in the FSC system (RAM or RUN mode).

Proportional band NOT in the range.

This can happen in Edit parameters, if you enter the mode selection field or try to save the parameters (Esc, PgUp, PgDn). You have entered a proportional band, which is not in the range -327.67 .. 327.67 or 0. Correct this by entering a value in the correct range.

Rate time NOT in the range.

This can happen in Edit parameters, if you enter the mode selection or try to save the parameters (Esc, PgUp, PgDn). You have entered a rate time, which is not in the range 0.0 .. 3276.7. Correct this by entering a value in the correct range.

Reset time NOT in the range.

This can happen in Edit parameters, if you enter the mode selection field or try to save the parameters (Esc, PgUp, PgDn). You have entered a 1/Reset time, which is not in the range 0.00 .. 327.67. Correct this by entering a value in the correct range.

Sample time NOT in the range.

This can happen in Edit Parameters, if you enter the mode selection field or try to save the parameters (Esc, PgUp, PgDn). You have entered a sample time, which is not in the range 0.1 .. 3276.7. Correct this by entering a value in the correct range.

**Screen cannot be selected for PID overview.**

You have entered a sheet number in the range S1 - S999, and these sheets cannot be selected for PIDs (these screens are assigned to sheets). Please enter a screen number in the range of 1 - 99.

Screen has NOT been selected for PID display.

This can happen in the 'Screen' option. You have entered a screen number, which is not defined as a PID screen. Correct this by entering a defined PID screen number or select this number for a PID screen. This can be done by the 'Edit screen' option in the status main menu. Enter the desired screen number and create a PID screen.

Screen has NOT been selected for status display.

You have entered a screen number, which was not defined as a status screen. The selected screen is a PID screen or not defined at all.

Setpoint Low must be smaller then setpoint High.

This can happen in Edit parameters, if you enter the mode selection field or try to save the parameters (Esc, PgUp, PgDn). Setpoint high clamp is lower than setpoint low clamp. Correct this by entering a smaller setpoint low clamp or a larger setpoint high clamp.

Version of message file and program do not match

The message file FSCVIEW.MSG is found in the FSC path directory, but the version number does not match with the program FSCVIEW.



12.10 Extended diagnostic descriptions

This section lists the diagnostic error messages per module type. As a reference, error codes are listed and the type of error message:

- E Problem which caused the FSC system to shut down or prevent its start-up.
- W Warning. A problem has been detected that allows the FSC system to continue operation (for a limited time period)
- M Message giving information on the status of the FSC system.

Each diagnostic error message includes a description which provides information about the cause of the problem and the way to remedy it. If this does not work for hardware-related problems, try to resolve the problem by exchanging modules in the circuitry, e.g. the modules of the Central Part in case of a problem report related to a Central Part module or in case of I/O modules check the VBD, vertical bus, HBD and horizontal bus connections as well as the power supply and watchdog wiring.

Diagnostic messages other than those listed below are possible. If these occur, document the message, the message number and circumstances as completely as possible, and contact Honeywell Safety Management Systems B.V. in the Netherlands.

12.10.1 General

Error in communication between user station and FSC system.

1. Check if the cable has been plugged in the correct communication port of the FSC user station and/or the FSC system.
2. Check if the FSC COM port has been set correctly.
3. Check if the EPROMs on the COM module have been placed correctly. If they are, verify the EPROMs.
4. Check all modules on the system bus for correct EPROMs and jumper settings.
5. Decrease the communication speed (the length of the communication cable may be too long for the communication speed).

12.10.2 Diagnostic messages not related to specific hardware modules

Analog output scaling error. (E-88)

Check the calculation of the analog output value in the functional logic diagrams as calculation overflow occurred on scaling the value from engineering units to the 12-bit value for the A/D converter.

Analog input sensor break alarm. (W-24)

An analog input value is out of its specified range. Use 'Loop status' option to determine the tag number. Check field wiring and transmitter output.

Application not yet loaded into RAM. (M-45)

The FSC system does not have an application program loaded yet and is waiting for the download. Use the 'Download Application' option of FSC Navigator.

Calculation overflow (E-70)

Check the calculation in the logics at the indicated sheet.

**Command status buffer overflow. (E-51)**

Check the functioning of the COM module or replace the COM module.

Communication failure during OLM (CPU). (E-149)

During the on-line modification check between the Central Parts a communication error has occurred. Check the communication cable between the Central Parts.

Corrupted installation file. (E-117)

Rebuild the application files, check the hardware configuration, re-compile and re-load the application.

CPU module failure. (E-30, 31)

The 10 ms timerbase is faulty. Replace the CPU module.

Cycle CPU STOP-RUN key switch. (E-27)

This message will always be displayed in combination with an external communication failure message (E-27).

Device communication error. (W-38)

An external device (SER PC, DCS printer or MMI PC) has stopped communicating to the FSC system. Check the external device and the communication table.

Divide by zero (E-71)

Check the calculations in the logics at the indicated sheet.

External communication error (E-27)

The communication between the system and the specified FSC system (SystemNr, Communication module, Channel) is not correct. Check if the cable between the systems is correctly connected.

External communication failure. (E-27)

Error detected in FSC network check. The on-line modification check between FSC systems in an FSC network was not successful.

Illegal argument e.g. square root of -1 (E-72)

Check the calculations in the logics at the indicated sheet.

Illegal counter value (0 .. 8191). (E-74)

Check the calculations in the logics at the indicated sheet.

Illegally placed COM module. (E-104..107)

The number of communication modules or jumper settings on one or more communication modules is not correct (see FSC hardware manual).

Illegally placed MEM module. (E-100..103)

The number of MEM modules or the jumper settings on one or more MEM modules is not correct.

Illegally placed VBD module. (E-116, 118..131)

The number of VBD modules or the jumper setting on one or more VBD module is not correct.

**Illegal timer value (0 .. 2047) (E-73)**

Check the calculations in the logics at the indicated sheet.

Incorrect version of 'System Configuration...'. (E-134)

Use the correct version of the 'System Configuration' option of FSC Navigator (FSC_IO).

Incorrect version of 'Design FLDs...'. (E-135)

Use the correct version of the 'Design FLDs' option of FSC Navigator (FSCAD).

Incorrect version of 'Translate application...'. (E-137)

Use the correct version of the 'Translate application' option of FSC Navigator (FSCCODE, FSCTABLE).

Internal communication failure. (E-26, 150..155, 158..164, 167..169, 171)

One Central Part has shut down. If the Central Part does not start up after several retries, check the following:

- the communication cable has been properly connected.
- lead breakage in the communication cable.
- EPROMs on the COM or CPU module of the Central Part that does not start up.

Jumper setting COM module on system bus not correct. (E-245)

Check the jumper settings on the system bus and 10004/./ communication module, and put them in the correct positions (see FSC Hardware Manual).

Maximum number of I/O faults exceeded. (E-200)

There are too many I/O faults which cause the FSC system to overload and shut down. This situation may, for example, occur if there are serious problems with a horizontal bus or if the power supply to all modules in a rack is lost.

No differences detected in application program. (M-37)

A new application version has been loaded, but no differences in the functional logic diagrams and hardware configuration were detected.

Redundant input fault (E-25)

The maximum on time or the maximum discrepancy time of the displayed redundant inputs is expired.

Second fault timer expired. (M-21)

The fault that caused the start of the second fault timer has not been repaired within the 'Interval time between faults' time period. This resulted in the system shutdown.

Second fault timer started (W-28)

Warning: An error has occurred and the second fault timer has been started. Repair the error before the timer runs out (otherwise a shutdown of the Central Part will take place).

Shutdown due to I/O error on single I/O section (E-156, 157)

An I/O module has been detected faulty by the other Central Part in the single I/O section, which has resulted in a shutdown of both Central Parts.

**Shutdown due to on-line modification (M-35)**

Shutdown of the Central Part (running with the unmodified software) due to exchange of software in the other Central Part which runs with the modified software.

Software shutdown. (M-8)

Always reported in case the system shuts down as a result of a severe error detected during the self-test. The reported error listed above this one in the extended diagnostics is the real cause.

Spurious watchdog interrupt. (E-9)

This shutdown can be caused by:

- Memory error
- 5 Vdc out of range
- Deactivation of ESD input (removed watchdog flat cable)
- Program runtime error

The watchdog status is displayed automatically.

System stopped via FSC Navigator. (M-99)

A system stop command was given in the 'On-Line Environment' option of FSC Navigator.

Temperature alarm. (W-23)

Warning: Temperature of the FSC system is outside of the configured range.

12.10.3 Module types VBD (10001/1/1 or 10001/R/1)

Address CP not set correctly (J5/J6 on VBD/SBD) (E-146)

Jumpers for addressing the Central Part are not correctly set. Check and set the jumpers in the correct position for the specific Central Part (see FSC Hardware Manual).

Configuration error (E-117)

Module configured but not placed in the FSC system.

CP address not set correctly (J5/6 on VBD/SBD) or module faulty (E145)

The logic on the VBD module cannot be activated. This can be caused by incorrect jumper setting or due to a faulty module.

Illegal VBD function. (E-22)

Check placement of EPROMs, or translate and load the application program again.

Module faulty (E-0..53, 58..144)

Check the vertical bus flat cable and the vertical bus terminator. If not successful, replace the VBD module.

VBD correlation test. (E-13)

Check the HBDs, the vertical bus flat cable and the vertical bus terminator. If not successful, replace the VBD module.

VBD, vertical bus or HBD faulty. (E-54..57)

Check the HBDs, the vertical bus flat cable and the vertical bus terminator. If not successful, replace the VBD module.



12.10.4 Module types CPU (10002/1/1 or 10002/1/2)

Application checksum error. (E-44)

The FSC system detected a CRC error in the application program during the start-up check. Reload the application program or exchange the memory circuits.

Application program corrupted. (E-41)

Replace the EPROMs on the CPU module of the Central Part that reports the fault or check for bent EPROM pins or reload the application program.

Bus error (E-240)

The CPU module tried to access the module indicated above but did not get access. Replace the indicated module above or the CPU module. If this does not cure the problem, contact Safety Management Systems in The Netherlands.

Communication failure during OLM (COM). (E-2)

Check communication cable between Central Parts.

Configuration coprocessor. (E-133)

The coprocessor has been placed and not configured, or it has not been placed and configured.

CPU address setting (SO 2,3). (E-12)

Set the address of the CPU (10002/1/1) correctly (see FSC Hardware Manual).

CPU address setting (J1,2,3). (E-12)

Set the address of the CPU (10002/1/2) correctly (see FSC Hardware Manual).

CPU, MEM or COM RAMs/EPROMs faulty. (E-39)

Check the contents of the EPROMs and make sure that all EPROMs and RAMs on the CPU/MEM and COM modules have been inserted correctly.

CPU module error. (E-230, 231)

Replace the CPU module.

CPU test error. (E-247..255)

The microprocessor is faulty. Replace the CPU module.

Difference found at FLD(s): (M-3)

Warning: during the on-line modification check, the system has found differences at the specified sheets. Check if the sheet numbers correspond with the changes you have made.

Error in hardware comparator test. (E-1)

Replace the CPU module.

Memory error. (E-10, 14..19)

Replace the EPROMs on the indicated module. If this is not successful, replace the module.

**Module error. (E-241)**

No access possible to the system bus. Replace CPU module

No start-up due to different application. (E-5)

Central Parts have different application programs. Reload the Central Part that has the incorrect application program.

Shutdown due to faulty temperature sensors. (E-60)

Both temperature sensors on the DBM module are regarded faulty. Exchange the DBM module.

Start-up not possible. Modify off-line. (M-4)

Changes have been made in the application software, which cannot be modified on-line (only off-line).

System program corrupted. (E-40)

Replace the CPU EPROMs or the CPU module.

12.10.5 Module type MEM (10003/1/1)

Application program corrupted. (E-41)

Replace the EPROMs on the MEM modules of the Central Part that reports the fault or check for bent EPROMs or reload the application program.

Configuration error (E-43, 100..103)

The jumper settings for the RAM/EPROM memory circuits on one or more memory modules are not correct. Refer to the FSC Hardware Manual or the jumper setting printout (output via 'Print' option of FSC Navigator).

Hardware comparator failure. (E-1)

Replace the MEM module.

Memory error. (E-10,14..19)

Replace the EPROMs on the indicated module. If this is not successful, replace the module.

Module error. (E-240, 241)

No access possible to the system bus. Replace the module.

RAM error. (E-2, 3)

Replace the RAM memory chips.

12.10.6 Module type COM (10004/./., 10008/2/U)

COM module not responding. (E-59)

The FSC-SMM communication module does not respond to initialization commands. Check the jumpers and the EPROMs.

**Communication failure during configuration check. (E-45)**

A time-out occurred during the configuration check between FSC systems. Check system cables.

Communication failure during OLM check. (E-45)

A time-out occurred during the configuration check between FSC systems. Check system cables.

Communication program corrupted. (E-42)

Replace the EPROMs on the COM modules of the Central Part that reports the faults or check for bent EPROMs pins.

COM software version mismatch. (E-20)

Translate the application program again, and replace all application EPROMs on the COM and MEM/CPU modules.

Error in self-test COM module (E-220)

Replace the indicated COM module.

Hardware comparator failure. (E-1)

Replace the COM module.

Incompatible FSC-FSC configuration. (E-44)

The FSC-FSC communication block sizes for two FSC systems do not have equal size. Translate and load the application again, or modify off-line.

Incompatible FSC-FSC variable allocation. (E-44)

Tag numbers have changed within the last modification of the application software. Only adding or deleting of tags is allowed during on-line modification.

Incompatible variable allocation (E-44)

Tag numbers have changed within the last modification of the application software. Only adding or deleting of tags is allowed during on-line modification.

MEM - COM incompatible Central Part number. (E-6)

Check if all EPROMs are placed in the correct Central Part.

MEM - COM incompatible run mode. (E-6)

CPU/MEM and COM EPROMs are not compatible with regard to the power-on mode. Translate the application again, and replace all EPROMs on the CPU/MEM module and the COM modules.

MEM - COM incompatible system number. (E-6)

EPROMs on CPU/MEM and COM module contain different system number. Check if the correct EPROMs are placed on the modules.

Module faulty (E-11, 13)

Replace the COM module

RAM error. (E-25, 27)

Replace the COM module.



Wrong EPROMs or jumpers (J2, J3) on COM module set incorrectly. (E-6)

EPROM size and jumper setting do not match (see the FSC Hardware Manual for jumper settings).

12.10.7 Module type WD (10005/1/1)

Illegally placed watchdog module. (E-108..115)

The number of watchdog modules or the jumper setting on one or more watchdog modules is not correct (see the FSC Hardware Manual).

Maximum runtime test failed. (E-49)

Replace the watchdog module.

Minimum runtime test failed. (E-48)

Replace the watchdog module.

Module faulty (E-0..38)

Replace the watchdog module.

12.10.8 Module type DBM (10006/./.)

Configuration error (E-132)

Module configured but not placed in FSC system

Error in temperature sensor (E-4..7)

The indicated temperature sensor is regarded faulty. Replace DBM module.

Error in voltage reference check (E-1..3)

The reference voltage to check the functionality of the A/D converter for the temperature sensors is faulty.

Illegally placed DBM module. (E-132)

The number of DBM modules or the jumper settings on one or more DBM modules is not correct.

Jumper set incorrectly or fuse blown. (W-8)

Battery voltage is not correct. Check if jumper is set to the 'battery on' position and check fuse.

Sensor temperature difference out of range. (E-9,10)

The temperature difference between the two sensors exceeds 5 °C (9 °F). Replace DBM module.

Temperature out of range. (M-20)

Check the ambient temperature of the Central Part. If the temperature is within the range, replace the DBM module.



12.10.9 Module type SBD (10007/1/1)

Address CP not set correctly (J5/J6 on VBD/SBD). (E-146)

Jumpers for addressing the Central Part are not set correctly. Check and set jumpers in the correct position for the specific Central Part (see FSC Hardware Manual).

Address CP not set correctly (J5/6 on VBD/SBD) or module faulty. (E145)

The logic on the SBD module cannot be activated. This can be caused by incorrect jumper setting or due to a faulty module.

Configuration error (E-117)

Module configured, but not placed in the FSC system.

Illegal placed SBD module (E-116, 118..131)

The number of SBD modules or the jumper setting on one or more SBD modules is not correct.

Module faulty (E-0..49)

Replace SBD module.

12.10.10 Module type HBD (10100/1/1 or 10100/2/1)

HBD addressing error. (E-98)

Check address setting on the HBD connectors of all HBD modules. Make sure that the vertical bus flat cable in VBD and HBDs has been inserted correctly.

HBD addressing error or other HBD fault. (E-99)

Check address setting on HBD connectors of all HBD modules. Make sure that the vertical bus flat cable in VBD, HBDs and bus terminator has been inserted correctly. If this does not help, replace the HBD module.

Module faulty (E-0..47)

Replace the HBD module electronic part.

12.10.11 Module types 10101/1/1, 10101/1/2, 10101/1/3, 10101/2/1, 10101/2/2 or 10101/2/3

Input compare error (E/W-99)

A discrepancy was detected between the inputs scanned via Central Part 1 and Central Part 2. Check the input signal on fast transients or replace both modules.

Module faulty (E/W-0..21)

One or more channels of the input module are faulty. Replace the module.



12.10.12 Module types 10102/1/1, 10102/1/2 or 10102/2/1

Calibration error. (E/W-4..20)

Calibrate the analog input module using the calibration option of FSC Navigator.

Capacitor test. (E/W-0..3)

The leakage current of the input circuitry is too high. Replace the module.

Input compare error (E/W-99)

A discrepancy was detected between the inputs scanned via Central Part 1 and Central Part 2. Check the input signal on fast transients, or replace both modules.

Module faulty (E-21..23, 28)

Check calibration. If it is not possible to calibrate, then replace the 10102/1/1 or 10102/1/2 module.

Voltage output short circuit or module faulty (E/W-24..27)

Make sure that the 26 Vdc output gives approx. 26 V. If it does not, check for short-circuiting, or replace the 10102/1/1 or 10102/1/2 module. If the transmitter current is > 25 mA, change the maximum current setting to < 25 mA.

12.10.13 Module type 10105/2/1

Internal power-down (E-1, 4, 5, 9, 10, 12, 56, 59-61, 88, 91-93, 136)

There is no internal voltage.

External power-down (E-196)

There is no external voltage.

Module faulty (all other codes)

The hardware is defective.

12.10.14 Module type 10201/1/1 or 10201/2/1

External power-down complete module (W-17)

Check fuses / circuit breaker for external power supply / watchdog signal.

External power-down channels 1, 2, 3, 4 (W-15)

Check fuses / circuit breaker for external power supply channels 1-4 / watchdog signal.

External power-down channels 5, 6, 7, 8 (W-16)

Check fuses / circuit breaker for external power supply channels 5-8 / watchdog signal.

**Mismatch between expected and actual output value (E/W-13)**

The actual output status is not identical to the expected value that was calculated by the application program. Replace the module.

Module faulty (E/W-0..4)

A fault is detected in the common part of the output module. Replace the module.

Output channel cannot be switched off (E/W-5..12)

Replace the module.

Output short circuit (W-14)

Check actuator and field wiring for short-circuiting.

SEC. SWITCH-OFF channel defective. (W-18)

The output channel used for the secondary switch-off cannot be switched on. Replace the module.

12.10.15 Module type 10203/1/2

External power-down complete module (W-15)

Check fuses / circuit breaker for external power supply / watchdog signal.

Mismatch between expected and actual output value (E/W-13)

The actual output status is not identical to the expected value that was calculated by the application program. Replace the module.

Module faulty (E-0..4)

A fault was detected in the common part of the output module. Replace the module.

Output channel cannot be switched off (E/W-5..12)

Replace the module.

Output short circuit (W-14)

Check actuator and field wiring for short circuit.

12.10.16 Module type 10205/1/1 or 10205/2/1

Mismatch between expected and actual output value (E/W-10, 11)

Check for open loop on analog output. Check calibration or replace the module.

Module faulty (E-0..9)

A fault was detected in the common part of the output module. Replace the module.



12.10.17 Module type 10212/1/1

External powerdown channels 1, 2, 3, 4 (W-15)

Check fuses / circuit breakers for external power supply channels 1-4 / watchdog signal.

Mismatch between expected and actual output (E/W-13)

The actual output status is not identical to the expected value that was calculated by the application program. Replace the module.

Module faulty (E-0..4)

A fault was detected in the common part of the output module. Replace the module.

Output channel cannot be switched off (E/W-5..12)

Replace the module.

Output short circuit (W-14)

Check actuator and field wiring for short circuiting.

12.10.18 Module types 10213/1/1, 10213/1/2, 10213/1/3, 10213/2/2 or 10213/2/3

External power-down (E-8)

Check fuses / circuit breakers for external power supply / watchdog signal.

Mismatch between expected and actual output value (E/W-7)

The actual output status is not identical to the expected value that was calculated by the application program. Replace the module.

Module faulty (W-0..2)

A fault was detected in the common part of the output module. Replace the module.

Output channel cannot be switched off (E/W-3..6)

Replace the module.

Output short circuit (W-9)

Check actuator and field wiring for short circuiting.

12.10.19 Module type 10214/1/2

Current detected in output loop (E-10)

Current is detected in an output loop although the channel is switched off. Check for short circuit between channels in the field.

External power-down (E-11)

Check fuses / circuit breakers for external power supply / watchdog signal.

**Line-monitor circuit faulty (E-7..9)**

The circuitry sensing the output loop current is faulty. Replace the module.

Module faulty (E-0..2)

A fault is detected in the common part of the output module. Replace the module.

Open loop or line monitor circuit faulty (W-13)

Check the field wiring for lead breakage or the lead breakage current setting (see the FSC Hardware Manual). If no problem can be found, replace the module.

Output channel cannot be switched off (E/W-3..5)

Replace the module.

Output short circuit (W-12)

Check actuator and field wiring for short circuiting.

Output short circuit or output cannot be switched on (W-6)

Check actuator and field wiring for a short circuit. If no problem can be found, replace the module.

12.10.20 Module type 10215/1/1 or 10215/2/1

External power-down (W-12)

Check fuses / circuit breakers for external power supply / watchdog signal.

External power-down channels 1, 2 (W-10)

Check fuses / circuit breaker for external power supply for channel 1-2 / watchdog signal.

External power-down channels 3, 4 (W-11)

Check fuses / circuit breaker for external power supply for channel 3-4 / watchdog signal.

Mismatch between expected and actual output value (E/W-8)

The actual output status is not identical to the expected value that was calculated by the application program. Replace the module.

Module faulty (E-0..3)

A fault is detected in the common part of the output module. Replace the module.

Output channel cannot be switched off (E/W-4..7)

Replace the module.

Output short circuit (W-9)

Check actuator and field wiring for short circuiting.



12.10.21 Module type 10216/1/1 or 10216/2/1

Correlation between channels (E-3, 4, 30)

The output channel cannot be switched on or off independent of the other channel(s). Replace the module.

Current detected in output loop (W-5, 6, 29, 31, 33, 34)

Current is detected in an output loop, although the channel is switched off. Check for short-circuiting between output loops in the field.

External power-down complete module. (W-23, 24)

Check fuses/circuit breakers for external power supply/watchdog signal.

Line-monitor circuit faulty (E-8, W-17..19)

The circuitry that senses the output loop current is faulty. Replace the module.

Module faulty (E-1..2, 10..12, 25)

A fault is detected in the common part of the output module. Replace the module.

Open loop or line-monitor circuit faulty (W-20..22, 26)

Check field wiring for lead breakage or the lead breakage current setting (see FSC Hardware Manual). If no problem can be found, replace the module.

Output channel cannot be switched on (E-14)

The output is de-energized and cannot be energized if required. Replace the module.

Output channel cannot be switched off (E-13)

The output is energized and cannot be de-energized if required. Replace the module.

Output is incorrectly switched on (E-7)

The actual output status is On, although it should be Off according to the application program. Replace the module.

Output short circuit (W-9, 15, 16)

Check actuator and field wiring for short circuiting.

12.10.22 Module type WDR (10302/1/1 or 10302/2/1)

Module faulty (E-0..17)

Check the 24 Vdc, 5 Vdc and the incoming watchdog. If no problem can be found, replace the module.



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Helping You Control Your World



Fail Safe Control

Appendix A: Alarm Functions

Version 500

Revision 01 (01/98)



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Revision 01 (01/98)

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A.1 Alarm description

The FSC system software has several standard alarm functions, which can easily be configured by the user. An alarm function group is defined by the variable type 'A'. A predefined standard sequence and the number of alarm points can be configured for an alarm group. Once this has been done, the alarm function is included in the application program via inputs and outputs with location 'ANN', which are allocated to the alarm function. The function of the alarm sequence of each input and output is defined by the alarm signal type. The inputs with location 'ANN' are virtual inputs for the application. The status of these inputs is generated by the standard alarm sequence function in the system software. The outputs with location 'ANN' are virtual outputs for the application. The status of these outputs is used as input in the standard alarm sequence function in the system software.

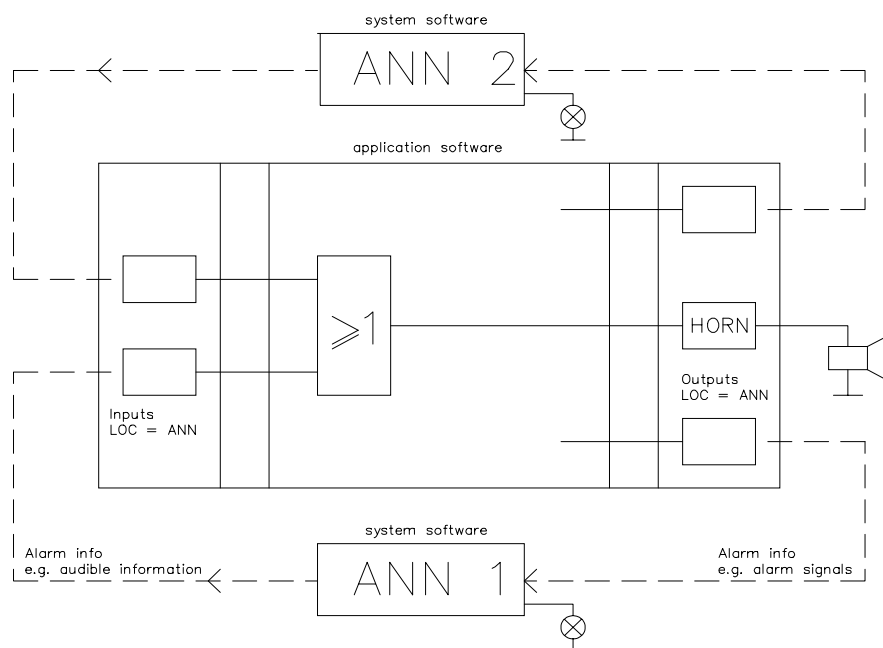


Figure A-1 Alarm functions

Variables are allocated to an alarm group by specifying the alarm group tag number. This tag number must already be defined as a variable with type 'A'. The number of variables with alarm type 'alarm' are specified by the user. This number should be a multiple of 8. All alarm types of a specific group must be configured, except for alarm type 'test' and alarm type 'common'.

A.1.1 Alarm types

Table A-1 shows the inputs and outputs used for each alarm sequence.

Table A-1 Inputs and outputs per alarm sequence

Alarm sequence										
Type	Alarm type	AF	AM	FR	FRM	AR	ARR	TFS	TFS COM1	TFS COM2
O	alarm(s)	X	X	X	X	X	X	X	X	X
O	acknowledge	X	X	X	X	X	X	X	X	X
O	flasher 1	2 Hz	2 Hz	2 Hz	2 Hz	0.5 Hz	0.5 Hz	0.5 Hz		0.5 Hz
O	flasher 2					2 Hz	2 Hz	2 Hz		2 Hz
O	lamp test	X	X	X	X	X	X	X		X
O	lamp reset		X		X					
O	reset					X	X			
O	flasher reset			X	X					
O	first up reset							X	X	X
O	first up status									X
I	audible	X	X	X	X	X		X		X
I	common	X	X	X	X	X	X	X	X	X
I	audible reset						X			
I	audible set						X			

The alarm types of an alarm function group are:

- alarm(s): This is an output variable which is used to activate the alert function of the alarm group. The variable should be allocated to a digital output module, which contains the current status of the specific alarm. The number of alarm variables of an alarm group are specified within the alarm function group variable ('A' <tag number>). At least one output variable with alarm type 'alarm' should be allocated to each alarm group.
- acknowledge: The acknowledge signal (output variable) is used to confirm the alarm, e.g. by the system operator.
- flasher 1: Flasher 1 (output variable) is the signal which is used for the flash functions of the alarm type 'alarm'. In case two flashers are needed for the alarm type, this flasher is used for the slow flash function. Table A-1 indicates the standard frequency for the flash functions for each alarm function.
- flasher 2: Flasher 2 (output variable) is only used in case two flash functions are needed for the alarm type 'alarm'. Flasher 2 is used for the fast flash function. Table A-1 indicates the standard frequency for the flash test functions for each alarm function.
- lamp test: Lamp test (output variable) is used to check the hardware of the digital output module which are used by an alarm type 'alarm' and the alarm indicator. The configuration of this variable is optional.
- lamp reset: Lamp reset (output variable) is used to reset the alarm type 'alarm' to the normal (low) state when the condition of the 'alarm' is returned to normal.



- reset: Reset (output variable) is used to reset the alarm type 'alarm' to the normal (low) state and to reset the audible signal when the condition of the 'alarm' is returned to normal.
- flasher reset: Flasher reset (output variable) is used to reset the alarm type 'alarm' to the steady (high) state. This signal is only accepted when the alarm was already confirmed by an acknowledge.
- first up reset: First up reset (output variable) is used to reset the alarm type 'alarm' to the steady (high) state. This signal is only accepted when the alarm was already confirmed by an acknowledge.
- first up status: The status of the first up reset in the receiver alarm status. This output must be linked to the output of type alarm of the same alarm group to which it belongs.
- audible: Audible (input variable) is the audible result of the alarm group. The signal is set when an alarm is detected or when the state of the alarm changes. The signal is reset by the acknowledge signal or by the reset signal of the alarm group.
- common: This input variable is high if any of the alarm types 'alarm' of the alarm group is in the alarm state. The configuration of this variable is optional.
- audible reset: Audible reset (input variable) is the audible result of the alarm group. The signal is set when an alarm returns to the normal state. The signal is reset by the reset signal of the alarm group.
- audible set: Audible set (input variable) is the audible result of the alarm group. The signal is set when an alarm is detected. The signal is reset by the acknowledge signal of the alarm group.

A.1.2 Alarm sequences

The alarm sequences supported by FSC are a subset of the annunciator sequences specified by the ISA S18.1 standard.

A.1.2.1 Symbols

The symbols used to describe the alarm sequences are in accordance with DIN 19 235 (October 1980).

Dark	
Steady light	
Slow flashing	
Flashing	
Fast flashing	
Intermittent fast flashing	

Visual symbols

No sound	
Sound	

Audible symbols

Figure A-2 Symbols used in description of alarm sequences

A.1.2.2 Alarm sequence AF, basic flashing

Alarm signal						
Alarm output						
Audible						
Acknowledge						

Figure A-3 Basic flashing

When an alarm becomes active, the output of the alarm flashes, and the audible is set. When acknowledge becomes high, the output of the alarm is steady, and the audible is reset. The output will be cleared if the alarm returns to its normal (high) state. Common will be high when one or more alarms of the alarm group are in the active state.



A.1.2.3 Alarm sequence AM, manual lamp reset

Alarm signal								
Alarm output								
Audible								
Acknowledge								
Lamp reset								

Figure A-4 Manual lamp reset

If an alarm becomes active, the output of the alarm flashes, and audible is set. When acknowledge becomes high, the output of the alarm is steady, and audible is reset. The output continues steady when the alarm returns to its normal high state. The output will be cleared, when 'lamp reset' is set. Common will be high when one or more alarms of the alarm group are in the active state.

A.1.2.4 Alarm sequence FR, flasher reset

Alarm signal								
Alarm output								
Audible								
Acknowledge								
Flasher reset								

Figure A-5 Flasher reset

When an alarm becomes active, the output of the alarm flashes, and audible is set. When acknowledge becomes high, audible is reset. The output becomes steady, when 'flasher reset' is set. The output will be cleared, when the alarm returns to its normal (high) state. Common will be high when one or more alarms of the alarm group are in the active state.

A.1.2.5 Alarm sequence FRM, flasher / lamp reset

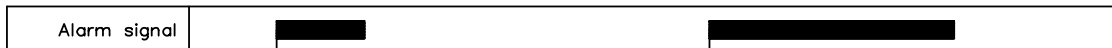
Alarm signal								
Alarm output								
Audible								
Acknowledge								
Flasher reset								
Lamp reset								

Figure A-6 Flasher/lamp reset

When an alarm becomes active, the output of the alarm flashes, and audible is set. When acknowledge becomes high, audible is reset. The output will be steady, when 'flasher reset' is high. The output will be cleared, when the alarm returns to its normal (high) state. Common will be high when one or more alarms of the alarm group are in the active state.

A.1.2.6 Alarm sequence AR, ringback

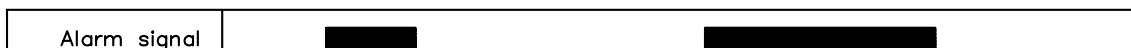
Alarm signal								
Alarm output								
Audible								
Acknowledge								
Reset								

Figure A-7 Ringback

When an alarm becomes active, the output of the alarm flashes fast, and audible is set. When acknowledge becomes high, the output of the alarm is steady, and audible is reset. The output will flash slow, when the alarm returns to its normal (high) state. The output will be cleared if 'lamp reset' is set. Common will be high when one or more alarms of the alarm group are in the active state.



A.1.2.7 Alarm sequence ARR, double audible ringback

Alarm signal									
Alarm output									
Audible set									
Audible reset									
Acknowledge									
Reset									

Figure A-8 Double audible ringback

When an alarm becomes active, the output of the alarm flashes fast, and audible_set is set. When acknowledge becomes high, the output of the alarm is steady, and audible_set is reset. When the alarm returns to its normal (high) state, the output will flash slow, and audible_reset is set. When 'reset' is set, the output will be cleared and audible_reset is reset. Common will be high when one or more alarms of the alarm group are in the active state.

A.1.2.8 Alarm sequences TFS, TFS_COM1, TFS_COM2, first up indication

Alarm signal 1								
Alarm output 1								
Alarm signal 2								
Alarm output 2								
Audible								
Acknowledge								
First up reset								

Figure A-9 First up indication (fast flash, slow flash)

When the first alarm becomes active, the output of this alarm has an intermittent flash, and audible is set. When other alarms become active, the output of these alarms have steady fast flashes, and audible is set. When acknowledge becomes high, the output of the first alarm flashes slowly, the output of the other active alarms will become steady and audible is cleared. The output of the first alarm will be steady, when 'first out reset' is set. When the alarm returns to its normal (high) state, the outputs will be cleared. Common will be high when one or more alarms of the alarm group are in the active state.

Alarm signal 1								
Alarm output 1								
Alarm signal 2								
Alarm output 2								
Audible								
Acknowledge								
First up reset								

Figure A-10 First up indication (single flash frequency)

When the first alarm becomes active, the output of this alarm has a steady fast flash, and audible is set. When other alarms become active, the output of these alarms also have fast flashes, and audible is set. When acknowledge becomes high, the output of the first alarm remains flashing, the output of the other active alarms will become steady and audible is cleared. The output of the first alarm will be steady, when 'first out reset' is set. When the alarm returns to its normal (high) state, the outputs will be cleared. Common will be high when one or more alarms of the alarm group are in the active state.



A.2 Alarm configuration examples

This section discusses two examples of how alarm functions are used.

The first example shows the basic flashing sequence (AF). The second example discusses the configuration of the first up alarm sequence with separate alarm detection and indication which can be used in communication networks.

A.2.1 AF alarm configuration

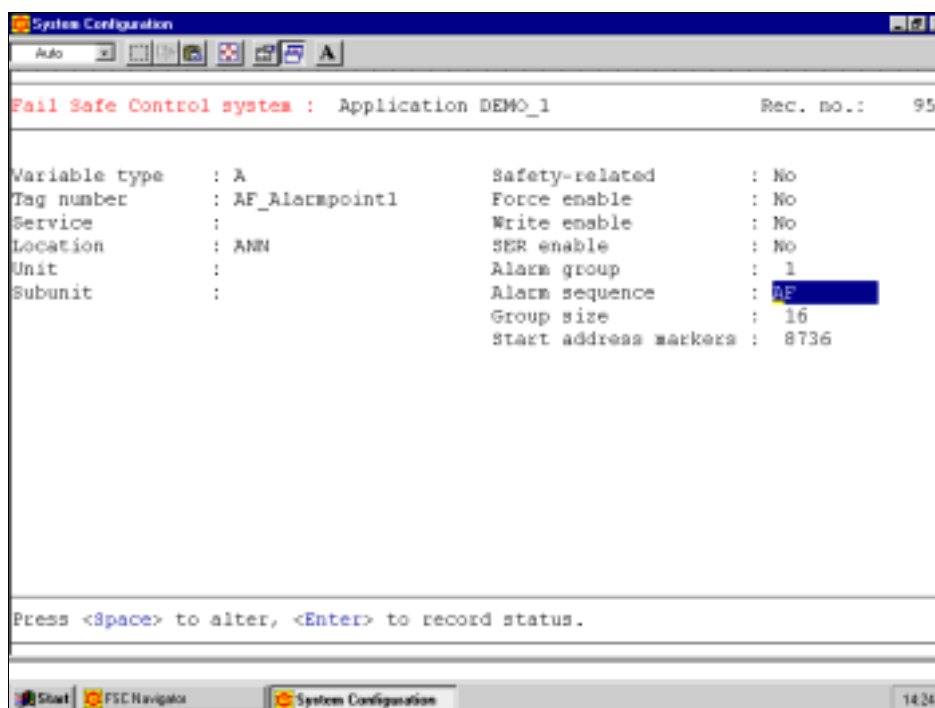


Figure A-11 AF alarm configuration

Use the 'System Configuration' option of FSC Navigator ('Signal Spec' menu item) to add an alarm group. The type of an alarm group is always 'A'. Use the 'Hardware Spec' option to assign this variable to an alarm sequence and to define the number of maximum alarm points for this group, e.g. 8 points for the alarm sequence AF in Figure A-11.

Variables with location 'ANN' can now be assigned to this alarm group in accordance with Table A-1. For the sequence AF, you can assign inputs to the signal type audible and common (Figure A-12 and Figure A-13).

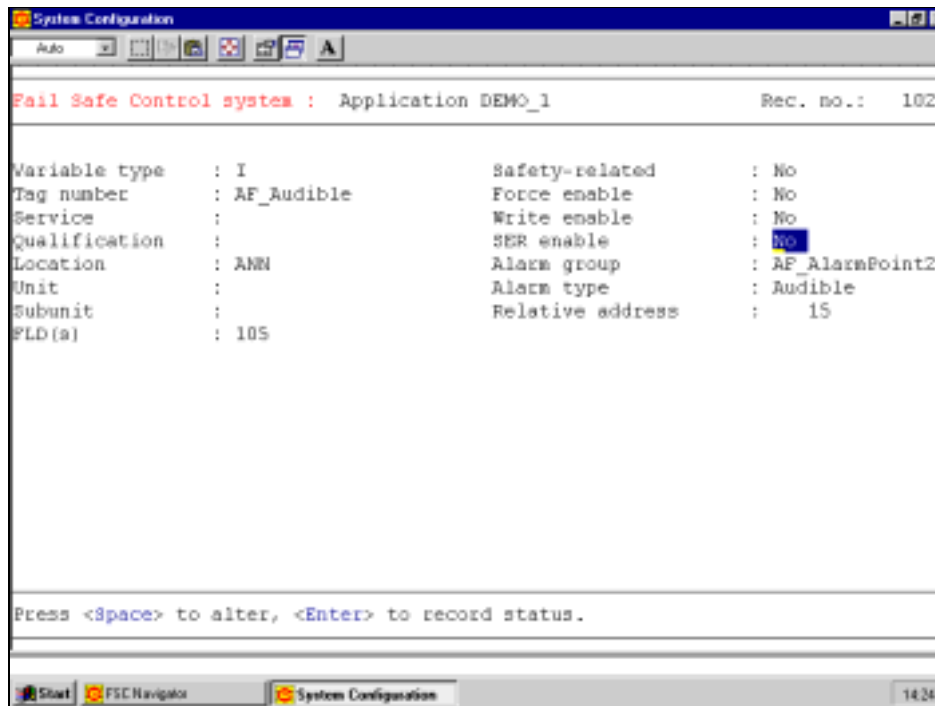


Figure A-12 AF audible

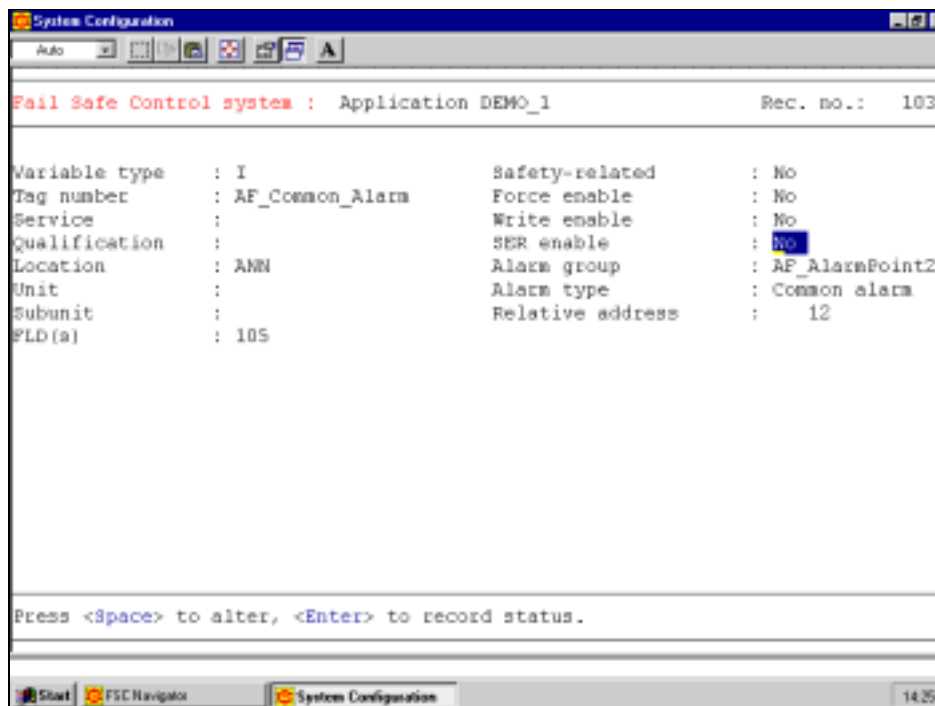


Figure A-13 AF common alarm

The alarm types flasher 1, lamp test and acknowledge can be allocated in the same manner.



The actual alarm points can be allocated to an alarm lamp driver (output module) (Figure A-14).

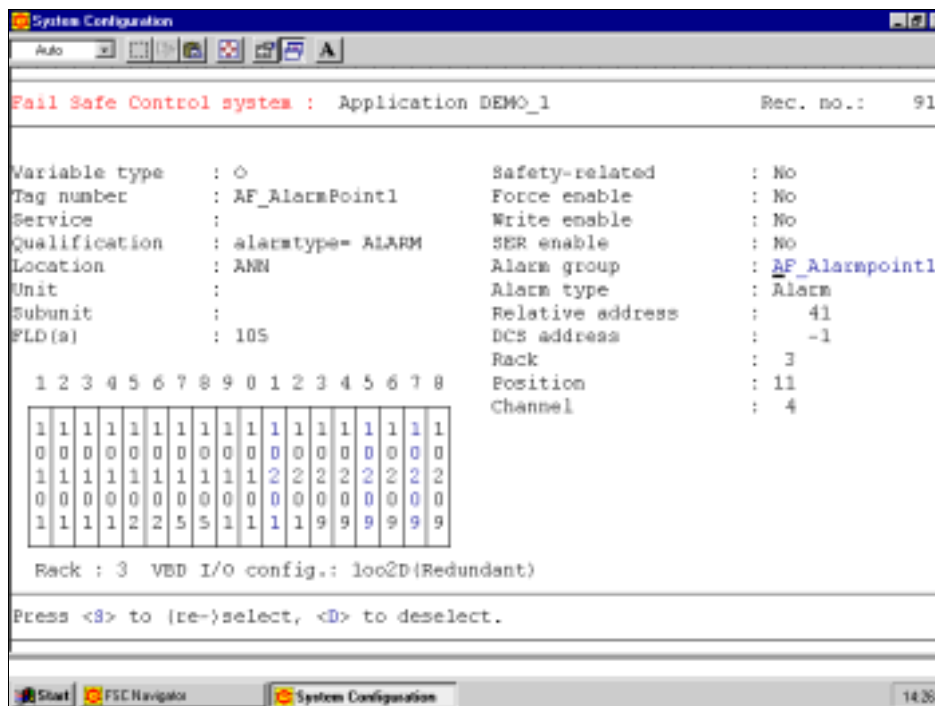


Figure A-14 Allocation of alarm point to alarm lamp driver



The alarm function can now be included in the application program as shown in Figure A-15.

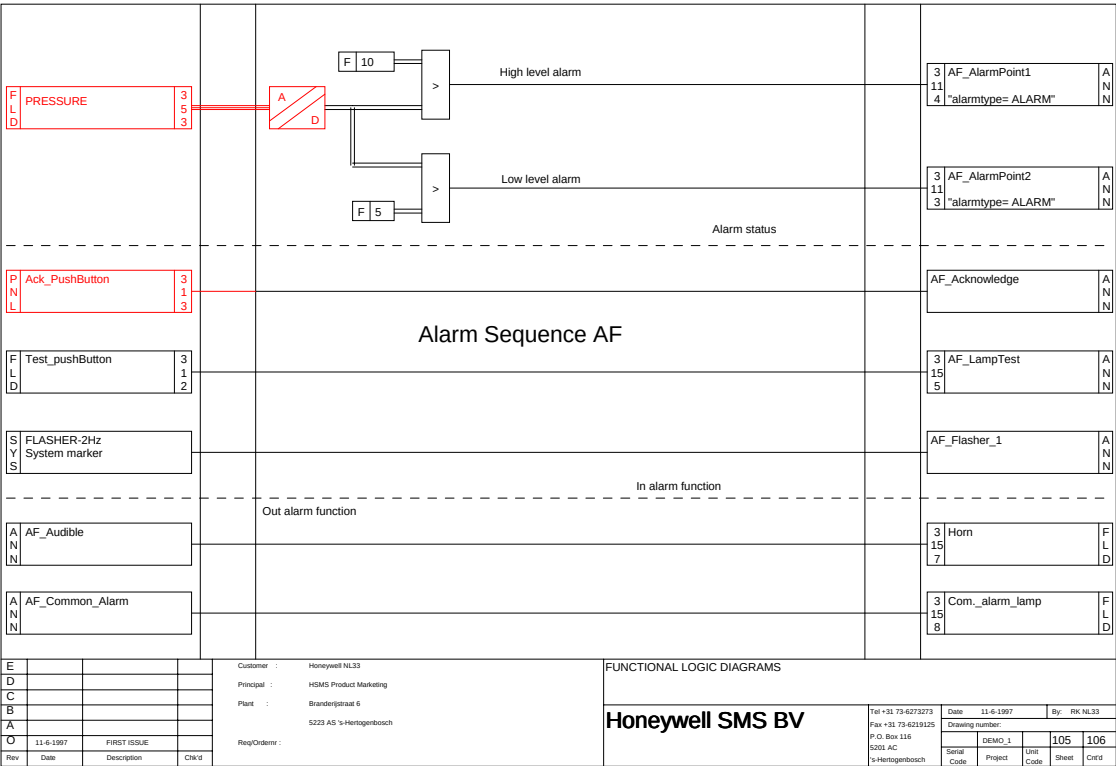


Figure A-15 Alarm function included in application program

A high and low level pressure alarm will bring up alarms called AF_AlarmPoint1 and AF_AlarmPoint2. A horn and a common alarm lamp are driven by the inputs AF_Audible and AF_Common_Alarm. The 2 Hz. system flasher controls the flashing speed of this group. The acknowledge of the alarm sequence is realized via an input button which is connected to AF_Acknowledge. The lamp test button is connected to AF_LampTest.



A.2.2 First up alarm configuration via communication links

Alarm groups of the sequence TFS COM_1 and TFS COM_2 are related. The sequence is nearly the same as TFS. The alarm detection and the visual display and manipulation of the alarm sequence is done in different FSC systems or an FSC system in cooperation with a Distributed Control System (DCS). An example of these sequences is given in Figure A-16.

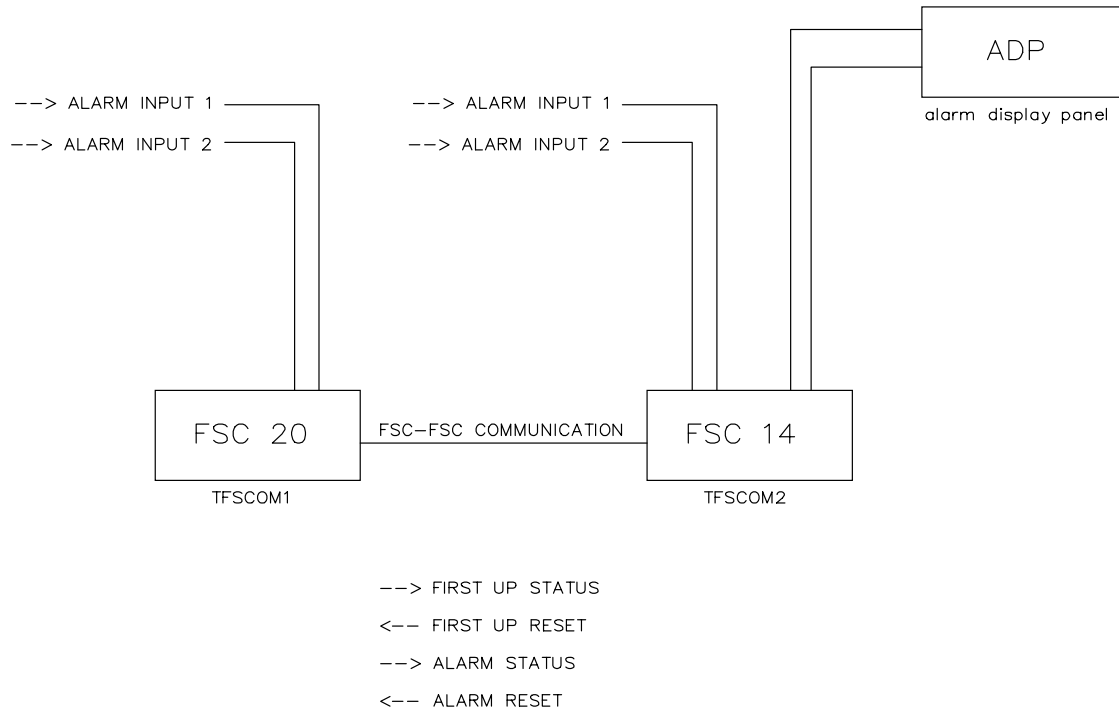


Figure A-16 Example of alarm sequence

In FSC 20, an alarm group with sequence TFS COM1 is configured. In FSC 14, an alarm group with sequence TFS COM2 is configured. The actual alarm is detected in FSC 20. Since it is a TFS sequence and the first in the alarm group must be detected, two signals for each alarm point are transferred to the other device (here FSC 14). The signals are the alarm status and information whether it was the first alarm in the group or not (the first up status). In FSC 14 these alarms are displayed and can be acknowledged. The acknowledge signal is transferred back to FSC 20. If the alarm must be reset, this is initiated in FSC 14 and copied to FSC 20.

F I D		3 7 13			<table border="1"> <tr> <td>3</td> <td>FIRSTUP-ALARM-1</td> </tr> <tr> <td>15</td> <td>SUBLOCATION-FSC</td> </tr> <tr> <td>4</td> <td>"FIRSTUP-FLAG"</td> </tr> <tr> <td>3</td> <td>ALARM-1</td> </tr> <tr> <td>17</td> <td></td> </tr> <tr> <td>1</td> <td>"ALARM STATUS"</td> </tr> </table>	3	FIRSTUP-ALARM-1	15	SUBLOCATION-FSC	4	"FIRSTUP-FLAG"	3	ALARM-1	17		1	"ALARM STATUS"										
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D C S		3 1 9	FIRSTUP-RESET		<table border="1"> <tr> <td>3</td> <td>FIRSTUP-RESET</td> </tr> <tr> <td>15</td> <td></td> </tr> <tr> <td>2</td> <td></td> </tr> </table>	3	FIRSTUP-RESET	15		2																	
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Figure A-17 Functional logic for alarm sequence

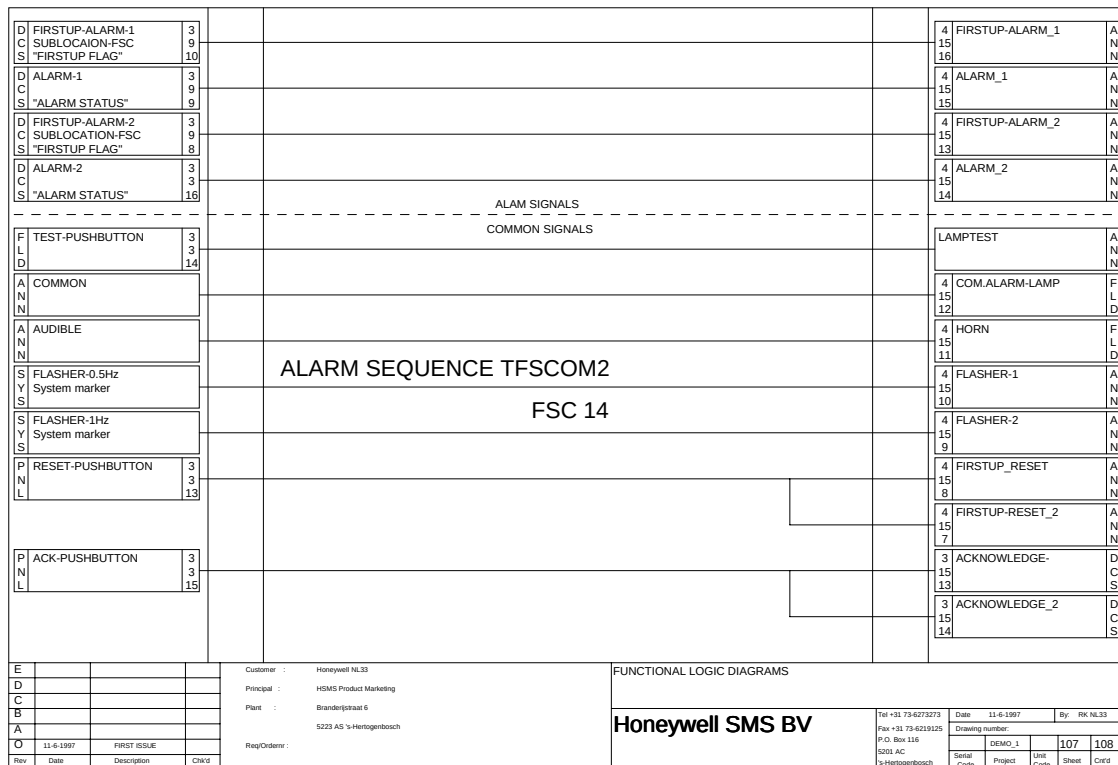


Figure A-18 Functional logic for alarm sequence

A high-level alarm is detected in FSC 20 and connected to two signals ALARM-1 and FIRSTUP-ALARM-1. ALARM-1 has location ANN because it must detect whether it was the first alarm in the group. Since this signal has to be sent to the other system, it has a sub-allocation FSC as well. The other signal FIRSTUP-ALARM-1 contains the status of the alarm and must only be sent to the display unit. This signal has location FSC (in case a DCS system is used as alarm display driver the (sub)location is COM).

These signals arrive in FSC 14 as an input and are directly connected to the alarm group FSC_COM_2 via outputs with location ANN ALARM-1 and FIRSTUP-ALARM-1 (signal type alarm and first up status). If FSC 14 receives an active alarm status, the alarm group will activate the inputs COMMON and AUDIBLE. In the functional logic diagrams (FLDs), these are connected to outputs which drive the common alarm lamp and horn.

Two flashers control the flashing sequence in FSC 14.

The input ACK-PUSHBUTTON in FSC 14 is hardwired. The FLDs connect these inputs to outputs ACKNOWLEDGE and ACKNOWLEDGE-2. One of these signals is used for TFS COM2 and one will be used for TFS COM1. Since TFS_COM_1 is in another system, the signal must first be send via FSC-FSC communication to FSC 20.

The hardwired input to reset the alarm group RESET-PUSHBUTTON works identically.

Another hardwired input is used to activate a lamp test. This input is called TEST-PUSHBUTTON and is connected to the output LAMPTEST in FSC14.



A.2.3 Flasher for alarm functions

The FSC system supports built-in alarm functions. These alarm functions include flashing of the alarm lights. The flasher frequency for the alarm lights can be generated from the FSC system flashers (originating from the real-time clock data) or built up using the cycle pulse function (output coupled to the input via an inverter with optional divider created via counter). The use of FSC system flashers result in variation of the on/off time of the flashing lights. The flashers based on the cycle time of the application program do not have these disadvantages (the on/off time of the flashing lights is constant). Figure A-19 shows an example of flashers based on the cycle time.

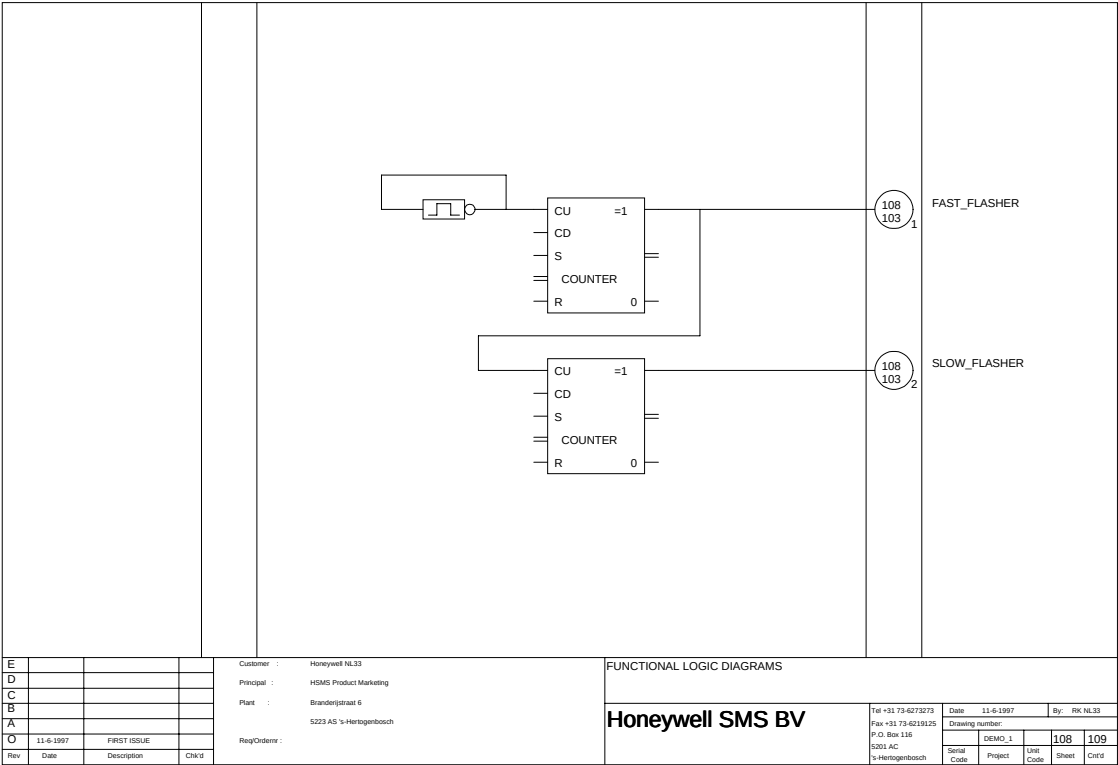


Figure A-19 Flasher for alarm functions



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Appendix B: Sequence of Event Recording

Version 500

Revision 01 (01/98)



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Version 500
Revision 01 (01/98)

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B.1 Introduction

In an operational plant, events may occur which indicate or cause deviations from normal process behavior. Examples of such events are:

- change of state of a valve limit switch,
- steam pressure becoming too high,
- maintenance override effected by an operator,
- faults in the field (e.g. output short circuit), and
- faults within the safeguarding system.

The sequence-of-event recording (SER) function of the FSC system outputs the events to a printer or to FSCSOE. FSCSOE is a separate Windows application which allows you to view, filter and log all reported events. This allows you to evaluate abnormal situations that have occurred.

The SER function reports events if:

- an event occurs for a process variable which is SER enabled,
- a process variable is forced (maintenance override),
- an active force on a process variable is removed,
- a fault is detected in the FSC system.

B.2 Event specification

This section describes the events recorded by the SER function of the FSC system.

B.2.1 Variable events

A variable event is recorded if an event occurs for a process variable that is SER-enabled. Variables can be enabled for sequence-of-event recording using the 'System Configuration' option of FSC Navigator.

Variable type I

For variable type I, an event occurs if the application value of the variable as applied to the application logic changes, i.e. Low $\rightarrow$ High or High $\rightarrow$ Low. The event report contains the variable identification, the current variable value and the date and time that the event has occurred.

Variable type O

For variable type O, an event occurs if the scan value of the variable, as applied to the process, changes i.e. Low $\rightarrow$ High or High $\rightarrow$ Low. The event report contains the variable identification, the current variable value and the date and time the event has occurred.

Variable types AI, AO

For event recording of variable types AI and AO, three operational areas are defined (Figure B-1):

1. the normal operational area, between the SER setpoint low and the SER setpoint high,
2. the low level alarm area, below the SER setpoint low, and
3. the high level alarm area, above the SER setpoint high.

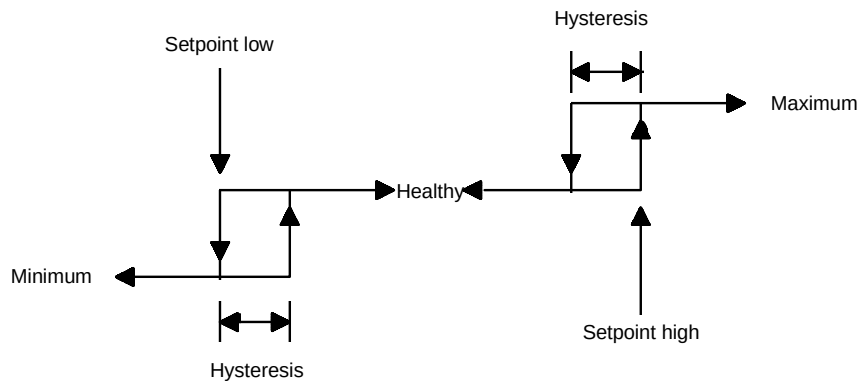


Figure B-1 Event definition for variable types AI and AO



For variable type AI, an event occurs if the variable application value, as applied to the application logic:

- becomes less than the SER setpoint low value (minimum event),
- exceeds the SER setpoint high value (maximum event),
- re-enters the normal operational area, between the SER setpoint low and SER setpoint high values, compensated with a hysteresis (healthy event).

The event report contains the variable identification, the current operational area (high, low, healthy) and the date and time that the event occurred.

For variable type AO, an event occurs if the variable scan value, as applied to the process:

- becomes less than the SER setpoint low value (minimum event),
- exceeds the SER setpoint high value (maximum event),
- re-enters the normal operational area, between the SER setpoint low and SER setpoint high values, compensated with a hysteresis (healthy event).

The event report contains the variable identification and the date and time the event has occurred.

The hysteresis is 0.5% of the full scale of the analog value.

The absolute value depends on the configuration of the variable and can be calculated as follows:

- for channels with range 0-20 mA, 0-5 V or 0-10 V;

$$\text{hysteresis} = \frac{10 * (\text{Top} - \text{Bottom})}{1638} \quad [\text{engineering units}]$$

- for channels with range 4-20 mA, 1-5 V or 2-10 V;

$$\text{hysteresis} = \frac{20 * (\text{Top} - \text{Bottom})}{2621} \quad [\text{engineering units}]$$

Variable type M

For variable type M, an event occurs if the value of the variable changes i.e. Low → High or High → Low. The event report contains the variable identification, the current variable value and the date and time the event has occurred.

Variable type T

For variable type T, an event occurs if a running timer runs out.

The event report contains the variable identification and the date and time the event has occurred.

Variable type C

For variable type C, an event occurs if the counter value changes from non zero to zero.

The event report contains the variable identification and the date and time the event has occurred.



Variable type BO

For variable type BO, an event occurs if a trigger output, associated with the BO changes from Low to High. The BO event can be regarded as the ability to sample the value of the BO variable at user definable criteria. The event report contains the variable identification, the current value of the BO variable and the date and time the event has occurred.

The associated trigger output is automatically generated when a BO is enabled for SER. The tag number of the trigger output will be equal to the tag number of the BO variable. The location of the trigger output is 'SYS'. Thus, it can be included in the application logic to generate the event and needs no further allocation to hardware.

Only BO variables with location 'COM' or 'FSC' can be SER-enabled.

B.2.2 Force events

A force event occurs if a process variable is forced or a force is removed.

The event report contains the variable identification (tag number), an indication if the force was activated or removed, and the date and time that the event occurred.

If numerical SER has been configured and the forced variable is SER enabled, the variable is identified by its numerical sequence number. If the forced variable is not SER enabled, a generic (non-specific) force event message is used.

If the force concerns a variable that is SER enabled, a variable event may occur as well.

An exceptional force event is generated if all forces are removed simultaneously.

The event report contains the 'All forces cleared' event specification and the date and time that the event occurred.

B.2.3 Fault events

A fault event occurs if a fault is detected by the self diagnostic of the FSC system.

The event report contains the system number, rack and position of the module the fault was detected at, and the date and time the event has occurred.

If the fault affects a variable that is SER enabled, a variable event may occur as well.



B.3 SER resolution

The resolution of the SER channel is equal to the execution time of the application. The application execution time is reported in the compiler logging file, which can be viewed using the 'View Log' option of FSC Navigator (see Section 9 of the FSC Software Manual). The time stamping of the events is in 10 ms resolution.

B.4 SER buffer capacity

The FSC system stores all events that occur until reported to the outside world, via the SER communication channel. The exact size of the SER buffer depends on the system configuration but the buffer can hold at least 448 events. When the maximum number of events has been reached, the events in the buffer are preserved for at least four hours. New events that occur within this time period are lost. After four hours, the stored events are overwritten if new events occur. The overwritten events are lost.

If a buffer overflow occurs, a specific event report is generated, specifying the 'SER buffer full' event, which contains the date and time the event has occurred.

B.5 SER channel specification

The SER channel specifies the communication channel through which events are reported to the outside world. The SER channel is configured using the 'System Configuration' option of FSC Navigator (see Section 4 of the FSC Software Manual).

In FSC communication networks, the SER channel may be located in another system. This allows events of multiple systems to be reported at a central point in the network.

If the SER function is used in combination with the FSCSOE program, the SER channel must use the Development System (DS) communication protocol. Otherwise the protocol must be printer.

Figure B-2 to Figure B-5 and Table B-1 show various possibilities for assigning SER channels for event reporting.

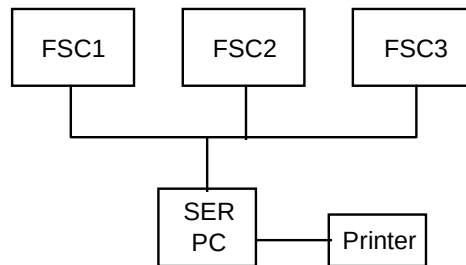


Figure B-2 Event collection via a SER communication network, with each system having its own local SER channel (only in combination with FSCSOE application)

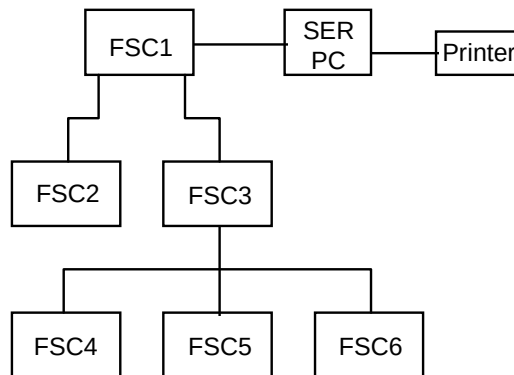


Figure B-3 Event collection via a single, global SER channel in the FSC communication network



B.5.1 Redundant SER channels

To improve availability of the SER function, the SER channel may be configured for redundant communication (see Figure B-4). SER events are reported as long as one of the redundant communication channels is operational. Refer to Appendix F ("FSC Communication Manual") for further details on redundant communication.

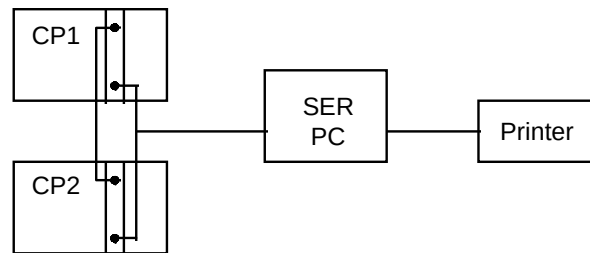


Figure B-4 Redundant SER communication channel

B.5.2 Dual SER channels

For optimum availability of the SER function, two SER channels may be used (see Figure B-5). As long as both channels are operational, all events are reported at both channels. If one of the channels fails, events will continue to be reported by the other channel.

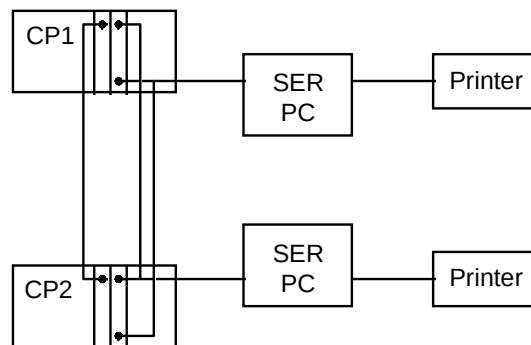


Figure B-5 Dual, redundant SER channels

B.5.3 SER channel diagnostics

The operation of the SER channel is monitored by the FSC system. If the channel is not able to report any events for a period of one minute (e.g. because the SER collecting device is switched off), a device communication error will be reported for that channel. This error will persist until the channel is able to report events again.



B.6 SER format

The layout of SER reports can be customized using the 'SER format' option of FSC Navigator. If the SER function is used in combination with the FSCSOE program, the standard "Numerical SER" format must be configured. Any custom SER format will be ignored.

In FSC networks, the SER format file is used from the application that contains the SER communication part, e.g. the printer or user station port.

B.6.1 Numerical SER format

The numerical SER format allows event recording with minimal communication overhead. Numerical SER format can be configured for two devices:

- the HSMS proprietary FSCSOE program, or
- a printer port (for third-party devices).

When using the numerical SER format, a unique sequence number is assigned to variables that are SER enabled. If an event occurs for the variable, its sequence number is contained in the event report. The format requires an intelligent device to collect the event reports and to relate the sequence numbers to the original process variable.

B.6.1.1 Configuring numerical SER format

To configure the numerical SER format, proceed as follows:

1. Specify the appropriate communication protocol using the 'System Configuration' option of FSC Navigator (Install / Modules / Central parts / COM):
 - 'DS' for sequence-of-event recording to FSCSOE, or
 - 'Printer' for sequence-of-event recording to a printer.
2. Specify the numerical SER range using the 'System Configuration' option of FSC Navigator (Install / Configuration). Specify the minimum and maximum value of the SER range in the 'SER base' and 'SER maximum' fields, respectively.

Note:

Variables must be SER-enabled and have a sequence number.

3. Specify the appropriate printer type using the 'SER format' option of FSC Navigator (SER format file / Printer config):
 - 'Numerical SER format' for sequence-of-event recording to FSCSOE, or
 - 'Printer' for sequence-of-event recording to a printer.

The SER function is now ready for use.



B.6.2 Numerical SER to FSCSOE

The following event types are supported when recording sequence of events to FSCSOE:

- I, O, AI and AO
- BO.

The events are sent to the FSCSOE program, which processes the events and presents them to the user in a variety of user-definable formats.

B.6.3 Numerical SER to printer

The following event types are supported when recording sequence of events to a printer:

- I, O, AI and AO
- M, T and C.

Note:

The printer must be set to HEX mode. Consult the printer manual for instructions on how to set the printer mode.

Events are reported in records. The length of the record is 16 bytes.

Table B-1 Event record structure

Byte	Field	Data type	Range
1,2	Record start indication	Binary	\$ffff
3	Event type	Binary	\$40 - \$4a, \$c1 - \$c3, \$c9, \$ca
4	Event status / Message specification	Binary	0, \$0f, \$ff
5	Reserved		0
6	System number	Binary	1 - 63
7,8	Sequence number	Binary	0 - 65534
9	Reserved		0
10	Weekday	BCD	1 - 7
11	Date	BCD	1- 31
12	Month	BCD	1 - 12
13	Hours	BCD	0 - 23
14	Minutes	BCD	0 - 59
15	Seconds	BCD	0 - 59
16	Hundredths of seconds	BCD	0 - 99

The following subsections discuss the fields of the event record in more detail.



B.6.3.1 Record start indication

Field : Record start indication
 Position : byte no. 1
 Length : 2 bytes
 Data type : binary
 Value : \$ffff
 Description : The record start indication field marks the start of the event record. The fields can be used for synchronization of external devices.

B.6.3.2 Event type

Field : Event type
 Position : byte no. 3
 Length : 1 byte
 Data type : binary
 Range : \$40 - \$4a, \$c1 - \$c3, \$c9, \$ca
 Description : The event type field specifies the type of event.

Value specification:

System messages:

\$40 system message

Variable events:

\$41 digital input event
 \$42 digital output event
 \$43 marker event
 \$44 10 millisecond timer event
 \$45 100 millisecond timer event
 \$46 1 second timer event
 \$47 1 minute timer event
 \$48 counter event
 \$49 analog input event
 \$4a analog output event

Variable force events:

\$c1 digital input force event
 \$c2 digital output force event
 \$c3 marker force event
 \$c9 analog input force event
 \$ca analog output force event

B.6.3.3 Event status/message specification

Field : Event status
 Position : byte no. 4
 Length : 1 byte
 Data type : binary
 Range : 0, \$0f, \$ff
 Description : The event status field contains information about the kind of event. The information depends on the type of event.



B.6.3.3.1 System messages

For system messages, the event status field specifies if the event has occurred or the system has recovered from the event:

0 = event occur
\$ff = event recover

The sequence number of the system messages are calculated by the FSC system, based on the configured base sequence number. The following system messages have been defined:

Table B-2 System messages

Message	Sequence number
All forces cleared	Base
FSC system fault	Base + 1
SER buffer full	Base + 2
SER buffer empty	Base + 3
Force event	Base + 5

B.6.3.3.2 Variable events

For variable events, the status field specifies the current status of the variable.

Variable type I, O, M

For variables of type I, O and M, the status field indicates the current status of the variable:

0 = variable status is Low
\$ff = variable status is High

Variable types T, C

For all timer events and the counter event, the event status has no meaning. The field value is always 0.

Variable types AI, AO

For the variable types AI and AO, the status field indicates in what range the analog value currently is:

0 = value is larger than SER high value as configured.
\$0f = value is less than SER low value as configured.
\$ff = value within healthy range.

Note:

Sequence-of-event recording to a printer does not support the variable type BO.



B.6.3.3.3 Variable force events

For variable force events, the event status field specifies if a force was activated or deactivated.

0 = force activated
\$ff = force deactivated

B.6.3.4 System number

Field : System number
Position : byte no. 6
Length : 1 byte
Data type : binary
Range : 1 - 63
Description : The system number field specifies the number of the FSC system that generated the event report. The system number can be combined with the variable sequence number to create a unique key to recognize an event.

B.6.3.5 Sequence number

Field : Sequence number
Position : Byte no. 7
Length : 2 bytes
Data type : binary
Range : 0 - 65534
Description : The variable type sequence number field contains the sequence number of the variable. The variable sequence number can be combined with the system number to create a unique key to recognize an event.

B.6.3.6 Weekday

Field : Weekday
Position : Byte no. 10
Length : 1 byte
Data type : BCD
Range : 1 - 7
Description : The variable sequence number field contains the day of the week, the event was generated.

Value specification:

1 = Sunday	5 = Thursday
2 = Monday	6 = Friday
3 = Tuesday	7 = Saturday
4 = Wednesday	



B.6.3.7 Date

Field : Date
Position : byte no. 11
Length : 1 byte
Data type : BCD
Range : 1 - 31
Description : The date field specifies the date of the month, the event was generated. The value of this field can only contain valid dates for the current month.

B.6.3.8 Month

Field : Month
Position : byte no. 12
Length : 1 byte
Data type : BCD
Range : 1 - 12
Description : The month field specifies the month, the event was generated.

Value specification:

1 = January	7 = July
2 = February	8 = August
3 = March	9 = September
4 = April	10 = October
5 = May	11 = November
6 = June	12 = December

B.6.3.9 Hours

Field : Hours
Position : byte no. 13
Length : 1 byte
Data type : BCD
Range : 0 - 23
Description : The hours field specifies the hour, the event was generated.

B.6.3.10 Minutes

Field : Minutes
Position : byte no. 14
Length : 1 byte
Data type : BCD
Range : 0 - 59
Description : The minutes field specifies the minute, the event was generated.



B.6.3.11 Seconds

Field : Seconds
Position : byte no. 15
Length : 1 byte
Data type : BCD
Range : 0 - 59
Description : The seconds field specifies the second, the event was generated.

B.6.3.12 Hundredths of seconds

Field : Hundredths of seconds
Position : byte no. 16
Length : 1 byte
Data type : BCD
Range : 0 - 99
Description : The hundredths of seconds field specifies the hundredths of seconds that the event was generated.

B.7 SER configuration requirements

If you wish to use the sequence-of-event function, the FSC configuration must contain:

1. a DBM module (except for the FSC slaves),
2. a communication channel that has been configured for SER reporting, using the FSC-FSC protocol, Development System protocol or printer communication protocol.

If these requirements are not met, you will not be able to set any variable to 'SER-enabled' and translate successfully.



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Fail Safe Control

Appendix C: Safety-Related Inputs with Non-Fail-Safe Sensors

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C.1 Introduction

Safety-related inputs require the use of fail-safe input modules (e.g. 10101/1/1, 10101/1/2 or 10101/1/3 for digital inputs, and 10102/1/1, 10102/1/2 or 10105/2/1 for analog inputs). It is also required that fail-safe input devices (e.g. sensors, switches and transmitters) are used. If the input device is not fail-safe, then redundant sensors (transmitters) are necessary which are each connected to its own input. The sensor configuration depends on the number of sensors and the function of the VBD that controls the rack which the input is connected to. The various redundancy types are shown in Table C-1.

Figure C-1 shows an example of redundancy type 201 for digital inputs, which can be used for VBD functions with single I/O.

To check the safety of the digital sensors, they must switch in a certain time interval. Also, the maximum on time is monitored, which can be set in the range between 1 second and 2047 minutes (deactivation of the maximum on time is also possible). If the maximum on time is exceeded, the resulting sensor status is switched off. To detect if all inputs execute the switch function, an extra timer is added: the maximum discrepancy timer (range 1-2047 seconds). If the maximum on timer or the maximum discrepancy timer expires, a redundant input fault (system marker) and a sensor fault are generated.

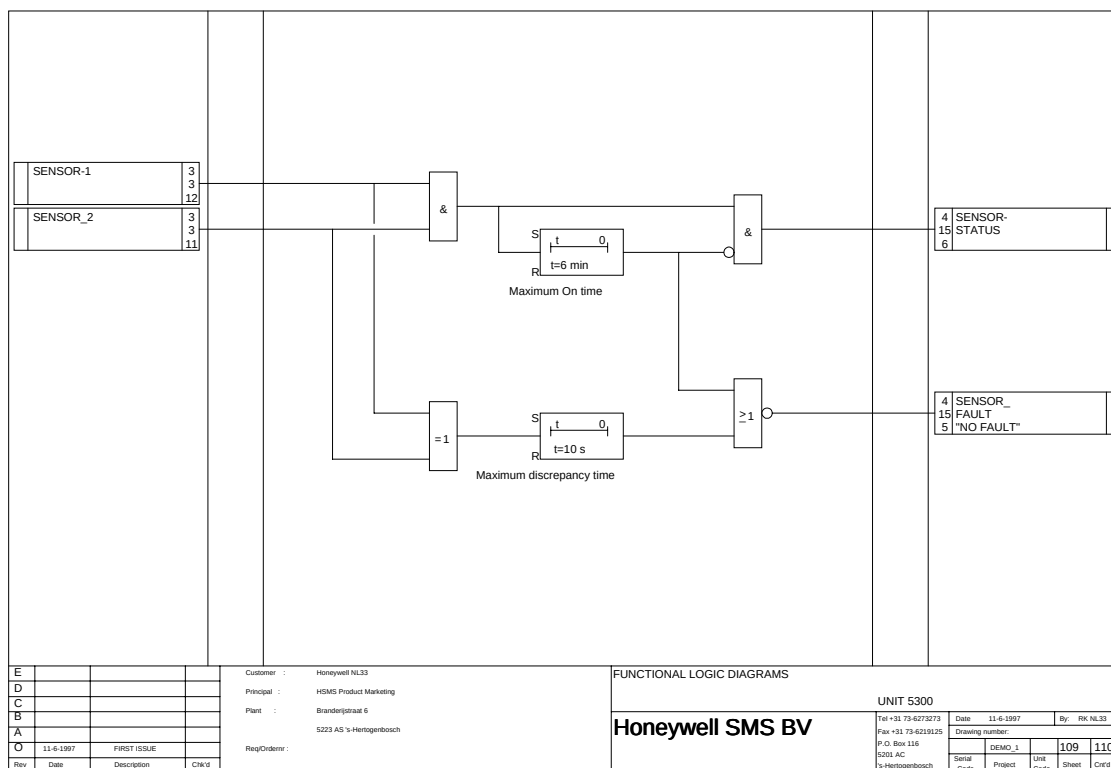


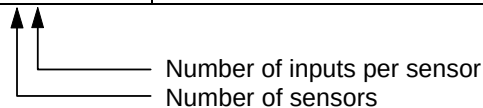
Figure C-1 Example of a safety-related input function with dual sensors

Note:

The maximum on time may also be deactivated. In that case, an organizational procedure must exist which ensures periodical testing of the sensors.

Table C-1 Possible redundancy types for the different VBD functions

VBD function	Redundancy types	Description
1oo1D (single)	201	Two input sensors, each connected to one input channel (2-out-of-2 function).
	301	Three input sensors, each connected to one input channel (2-out-of-3 function).
1oo2D (redundant)	202	Two input sensors, two input channels per Central Part (2-out-of-2 function per Central Part).
	302	Three input sensors, each connected to one input channel per Central Part (2-out-of-3 function per Central Part).



The redundancy type can be selected using the 'System Configuration' option of FSC Navigator. Depending on the redundancy type, extra inputs may have to be allocated. Extra inputs can only be allocated if the previous inputs have been allocated correctly. If the allocation of the main input is changed, the redundant inputs must be reallocated as well. The redundant inputs must be allocated at an input module, which is controlled by the same VBD as the input module to which the main input is allocated.



C.2 Safety-related digital inputs with non-fail-safe sensors

C.2.1 FSC systems with single Central Part

For single-channel FSC configurations it is necessary to use two or three input sensors connected to a fail-safe digital input module:

C.2.1.1 2 sensors: redundancy type 201

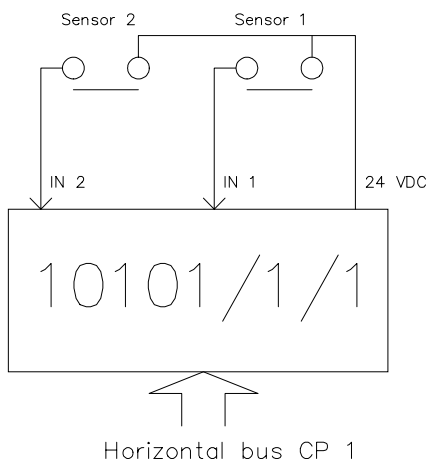


Figure C-2 Redundancy type 201

Both input sensors are connected to the FSC system (this may be the same or different input modules). In the system software, an AND function is performed for the resulting sensor status.

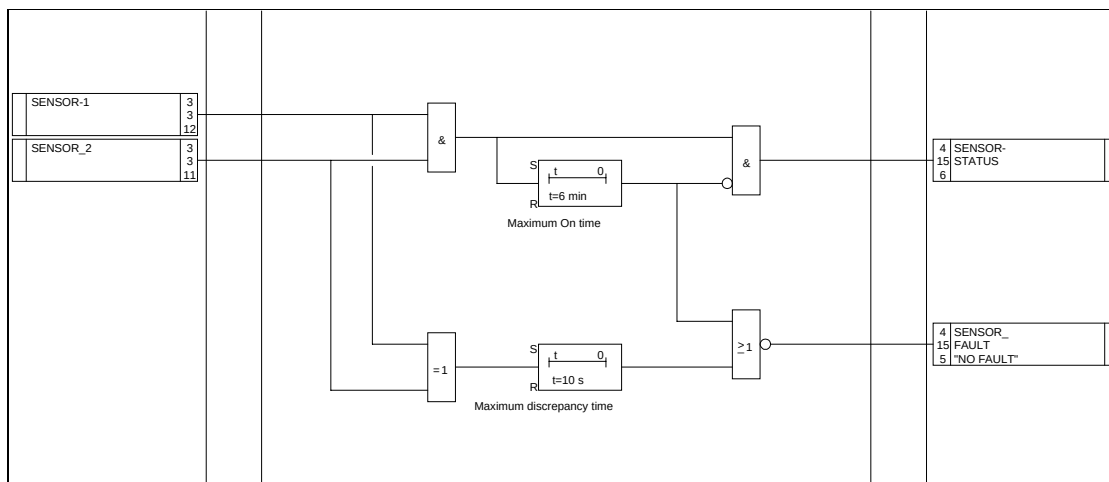


Figure C-3 Redundancy type 201 for FSC systems with single CP



C.2.1.2 3 sensors: Redundancy type 301

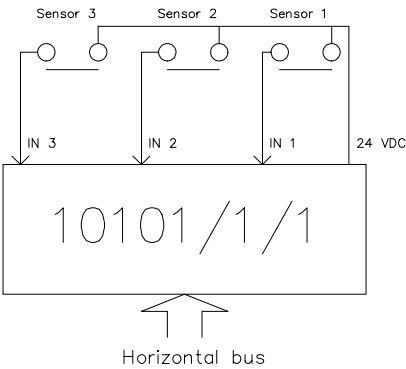


Figure C-4 Redundancy type 301

All three input sensors are connected to the FSC system (this may be to the same or different input modules). In the system software, a 2-out-of-3 function is performed for the resulting sensor status.

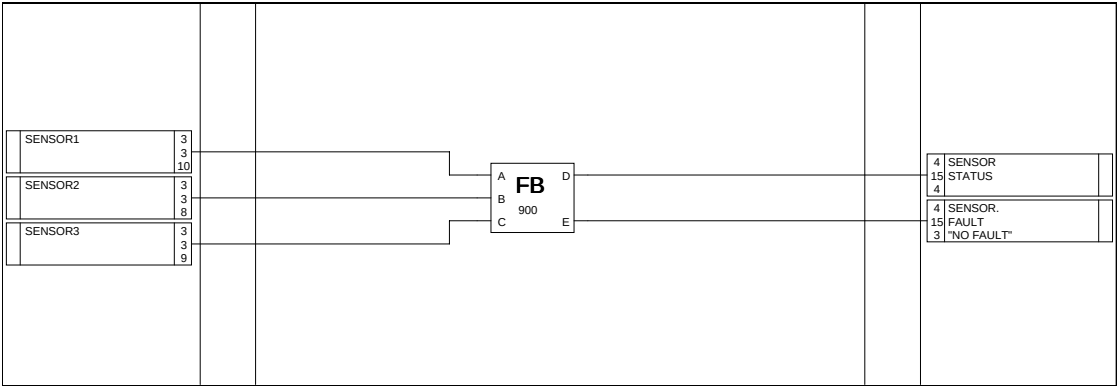


Figure C-5 Redundancy type 301 for system with single CP

For the 2-out-of-3 function block FB-900 refer to Figure C-10.



C.2.2 FSC systems with redundant Central Parts

For redundant FSC system configurations, the same sensor configurations apply as for single-channel configurations as one part of the redundant system must be able to control the process. For these configurations, too, it is necessary to use two or three input sensors.

C.2.2.1 2 sensors: redundancy type 202

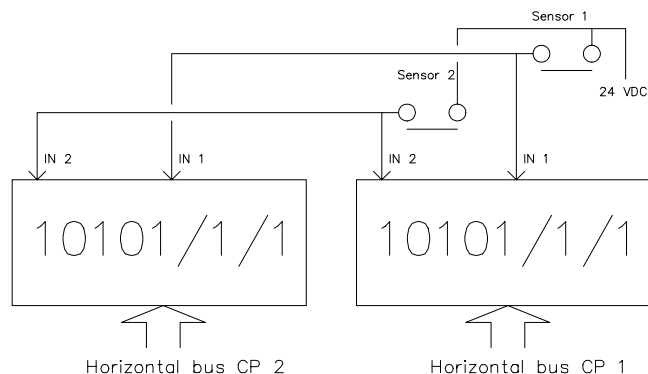


Figure C-6 Redundancy type 202

Both input sensors are connected to both Central Parts of the FSC system. In the system software, an OR function is performed for the synchronization in cases where faults can be located, and an AND function in all other cases.

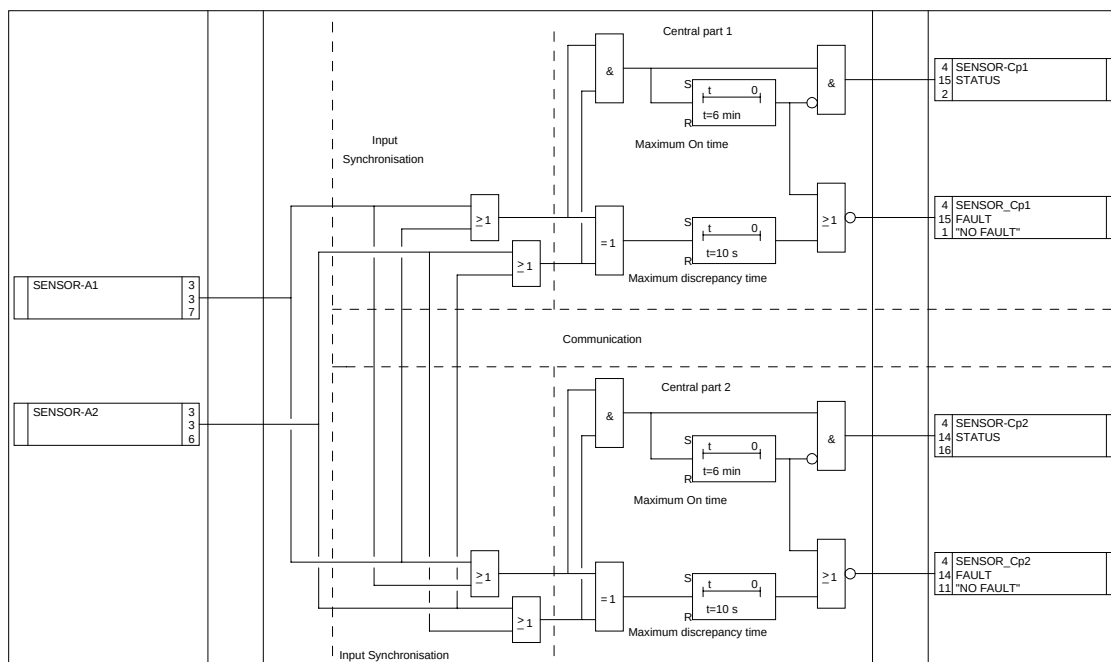


Figure C-7 Redundancy type 202 for FSC systems with redundant CPs

During the communication the status of the inputs is exchanged. The input synchronization function for the redundant systems is the OR function in cases where faults can be located, and an AND function in all other cases.

C.2.2.2 3 sensors: redundancy type 302

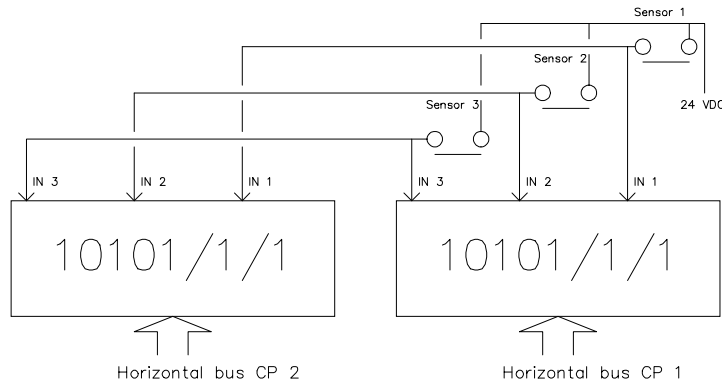


Figure C-8 Redundancy type 302

All three input sensor are connected to both Central Parts of the FSC system. For the synchronization in the system software, an OR function is performed in cases where faults can be located, and an AND function in all other cases. A 2-out-of-3 function is performed for the resulting sensor status.

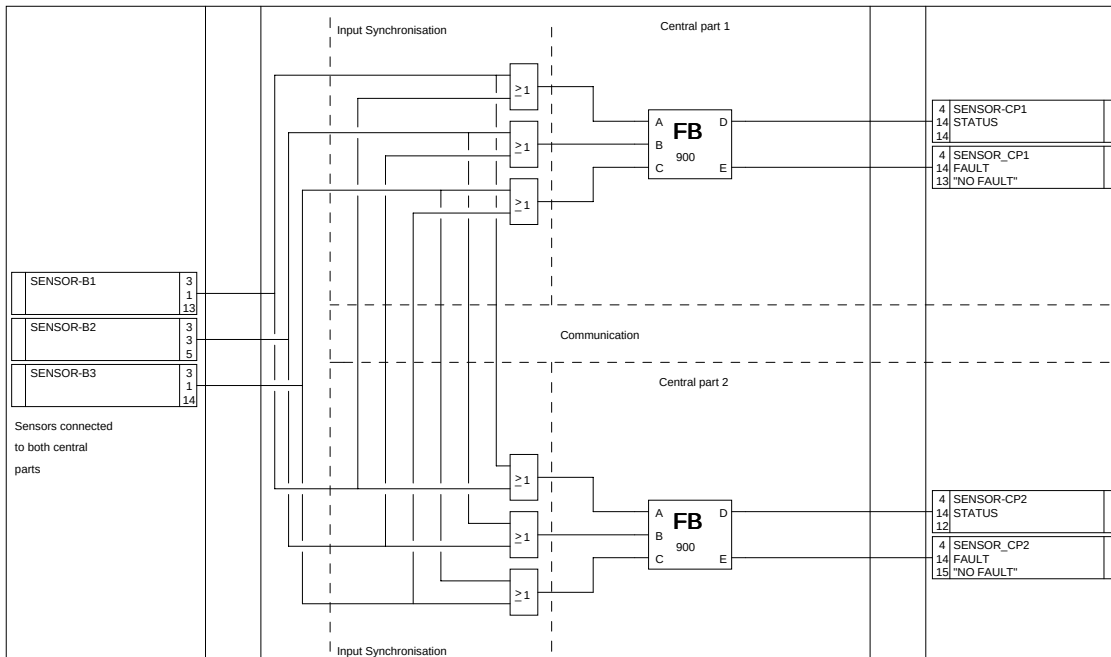


Figure C-9 Redundancy type 302 for FSC systems with redundant CPs



During the communication the status of the inputs is exchanged. The input synchronization function for redundant systems is the OR function in cases where faults can be located, and an AND function in all other cases. For the 2-out-of-3 function block FB-900 refer to Figure C-10.

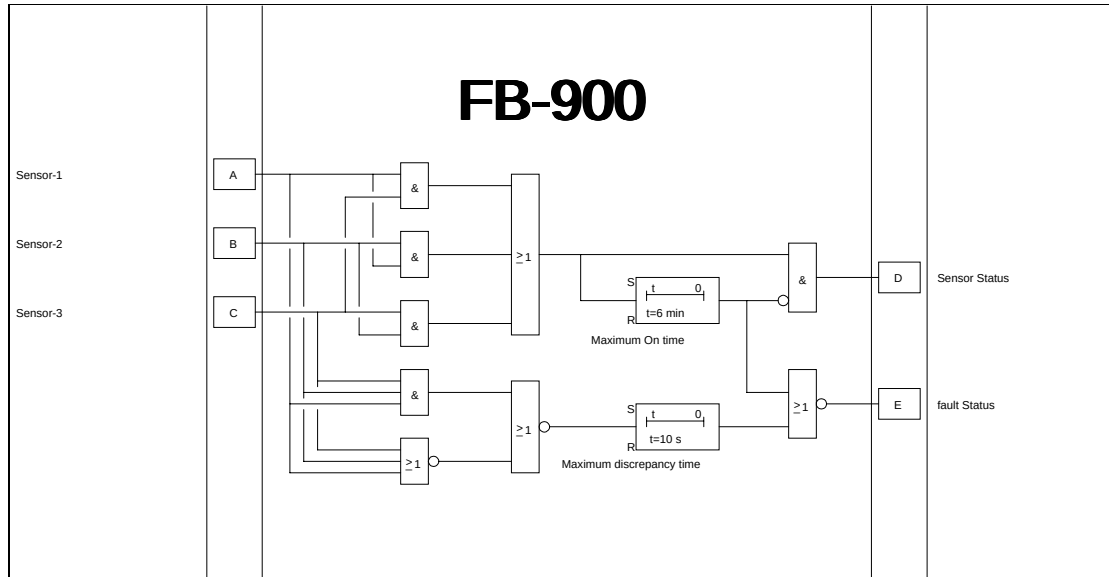


Figure C-10 2-out-of-3 function block FB-900



C.3 Safety-related analog inputs with non-fail-safe transmitters

To check the functioning of redundant transmitters, they must be within a certain bandwidth: the "maximum discrepancy value". This maximum value is monitored, and can be set during the configuration stage. If the maximum discrepancy value is exceeded, the resulting transmitter status is switched to the bottom scale of the analog input. If the maximum discrepancy value is exceeded, also a redundant input fault (system marker) and a transmitter fault are generated.

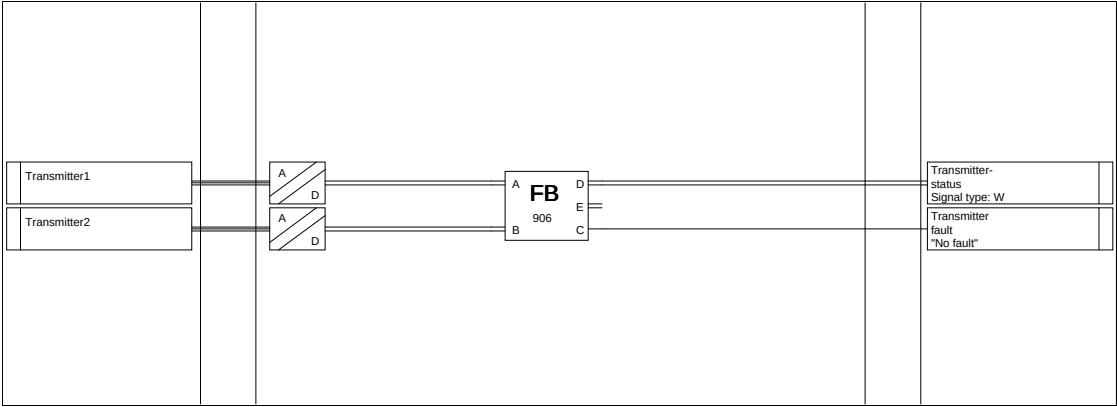


Figure C-11 Example of a safety-related analog input function with dual sensors



C.3.1 FSC systems with single Central Part

For single-channel FSC configurations it is necessary to use two or three transmitters connected to a fail-safe analog input module:

C.3.1.1 2 transmitters: redundancy type 201

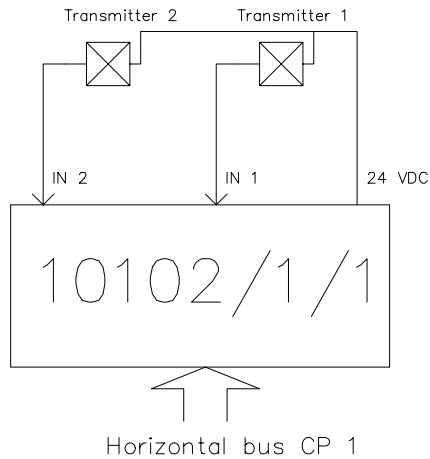


Figure C-12 Redundancy type 201

Both transmitters are connected to the FSC system (this may be the same or different 10102/1/1, 10102/1/2 modules). In the system software, a mean function is performed for the resulting transmitter status.

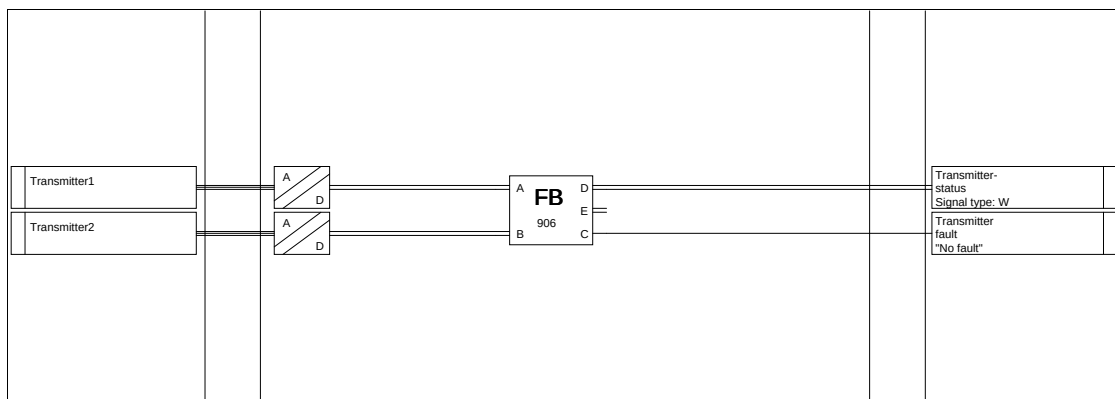


Figure C-13 Redundancy type 201 for FSC systems with single CP

C.3.1.2 3 transmitters: redundancy type 301

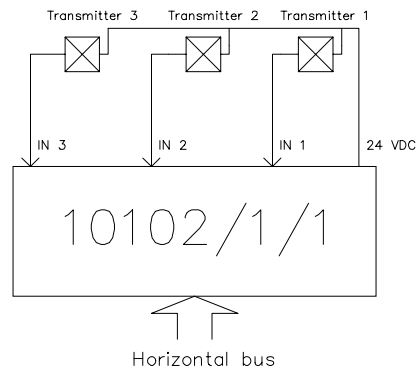


Figure C-14 Redundancy type 301

All three input transmitters are connected to the FSC system (This may be to the same or different 10102/1/1, 10102/1/2 module). The system software calculates the resulting transmitter status using the following algorithm:

1. If the difference between the values of each pair of transmitters (1-2, 1-3, 2-3) is less than the maximum discrepancy value, the resulting transmitter status is the mean value of the three transmitters.
2. If the difference between the values of only 2 transmitters is less than the maximum discrepancy value, the resulting transmitter status is the mean value of these 2 transmitters. The third transmitter is regarded faulty. Its value is ignored.
3. If the difference between the value of one transmitter (e.g. transmitter 1) and both other transmitters (2 and 3) is less than the discrepancy value and the difference between the values of transmitters 2 and 3 is more than the discrepancy value, then two calculation algorithms are possible:
 - If the differences between 1 and 2 and the difference between 1 and 3 are identical, the resulting transmitter status is the mean value of the three transmitters.
 - If the differences between 1-2 and 1-3 are not identical, the resulting transmitter status is the mean value of the 2 transmitters with the smallest difference. The third transmitter is regarded faulty. Its value is ignored.

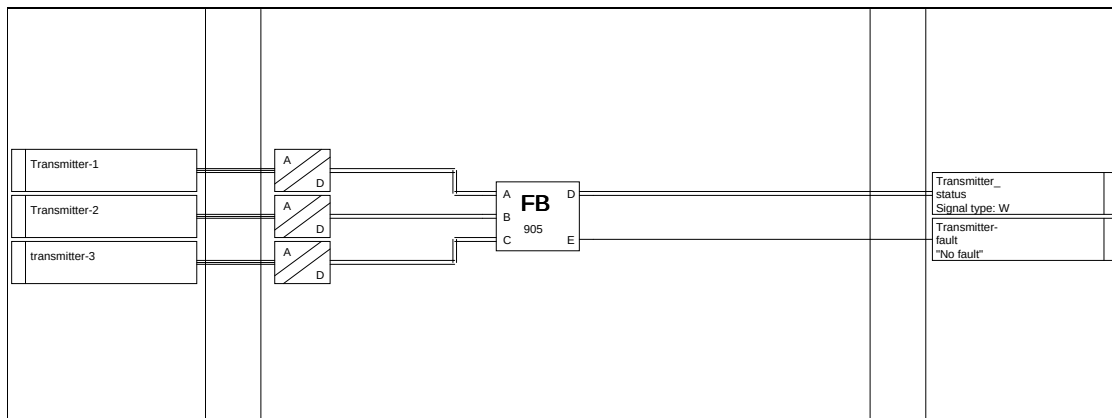


Figure C-15 Redundancy type 301 for FSC systems with single CP

For the function block FB-905 refer to Figure C-22.



C.3.2 FSC systems with redundant Central Parts

For the redundant FSC system configurations the same transmitter configurations apply as for the single-channel configuration as one part of the redundant system must be able to control the process. For these configurations, too, it is necessary to use two or three transmitters.

C.3.2.1 2 transmitters: redundancy type 202

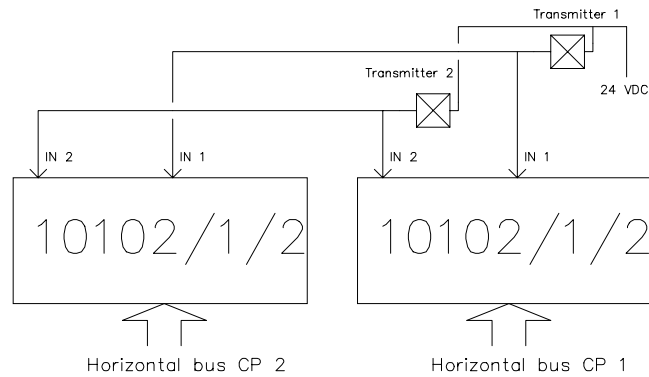


Figure C-16 Redundancy type 202

Both transmitters are connected to both Central Parts of the FSC system. In the system software, a mean function is performed for the resulting transmitter status.

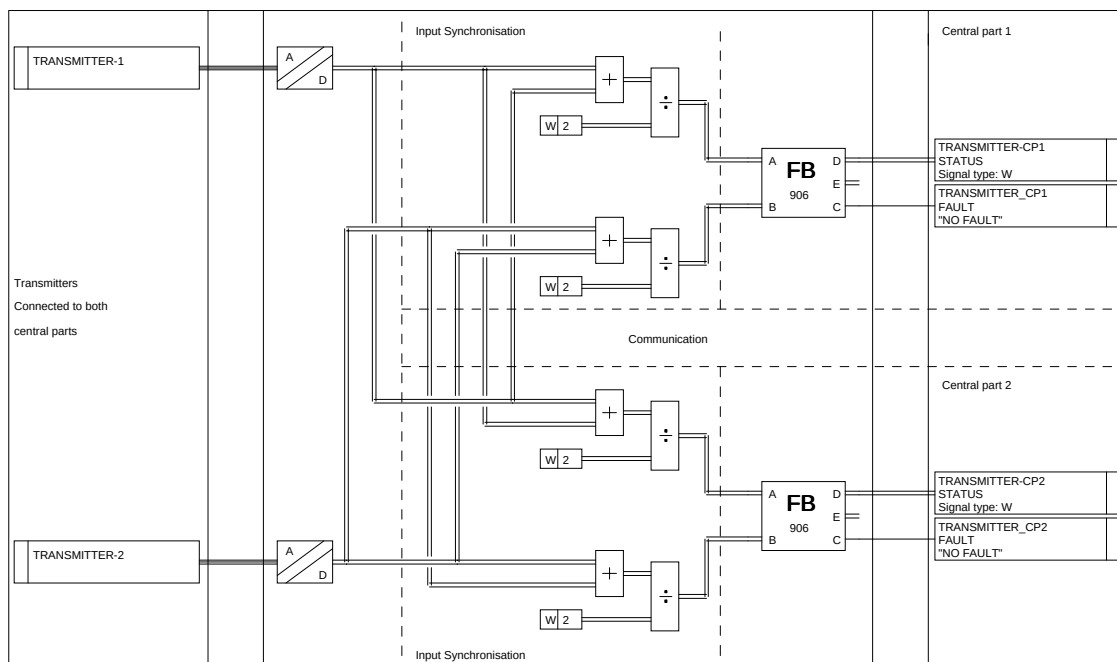


Figure C-17 Redundancy type 202 for FSC systems with redundant CPs

During the communication the status of the inputs is exchanged. The input synchronization function for the redundant systems is the mean function. For the function block FB-906 (mean function) refer to Figure C-22.

C.3.2.2 3 transmitters: redundancy type 302

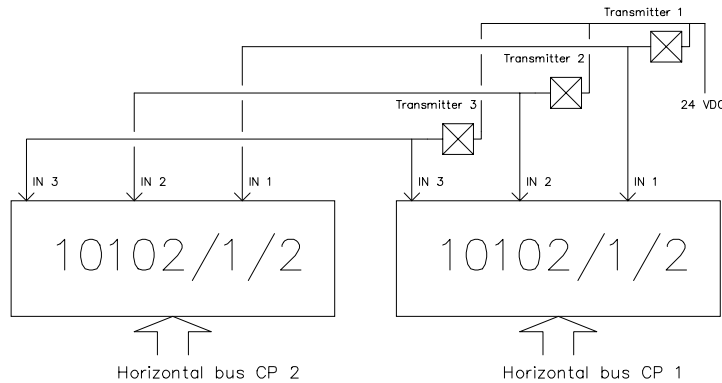


Figure C-18 Redundancy type 302

All three transmitters are connected to both Central Parts of the FSC system. The system software calculates the resulting transmitter status. The algorithm used is described in subsection C.1.2.1

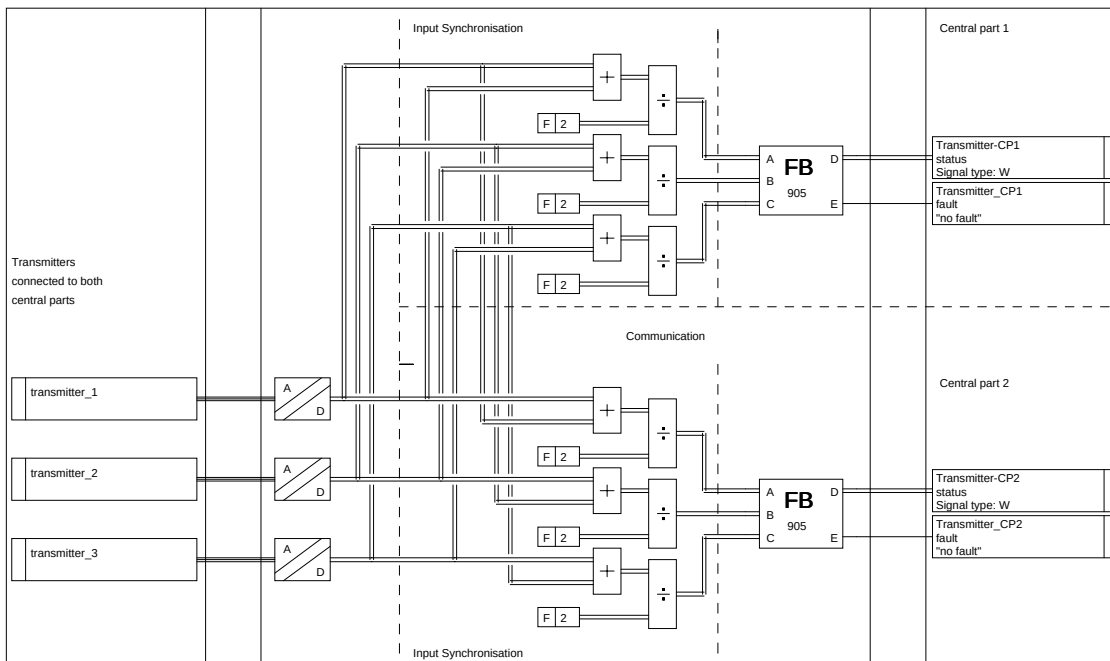


Figure C-19 Redundancy type 302 for FSC systems with redundant CPs

During the communication the status of the inputs is exchanged. The input synchronization function for the redundant systems is the mean function (see function block 906, Figure C-22). For the function block FB-905 (2-out-of-3 function) refer to Figure C-21.



C.4 Function blocks

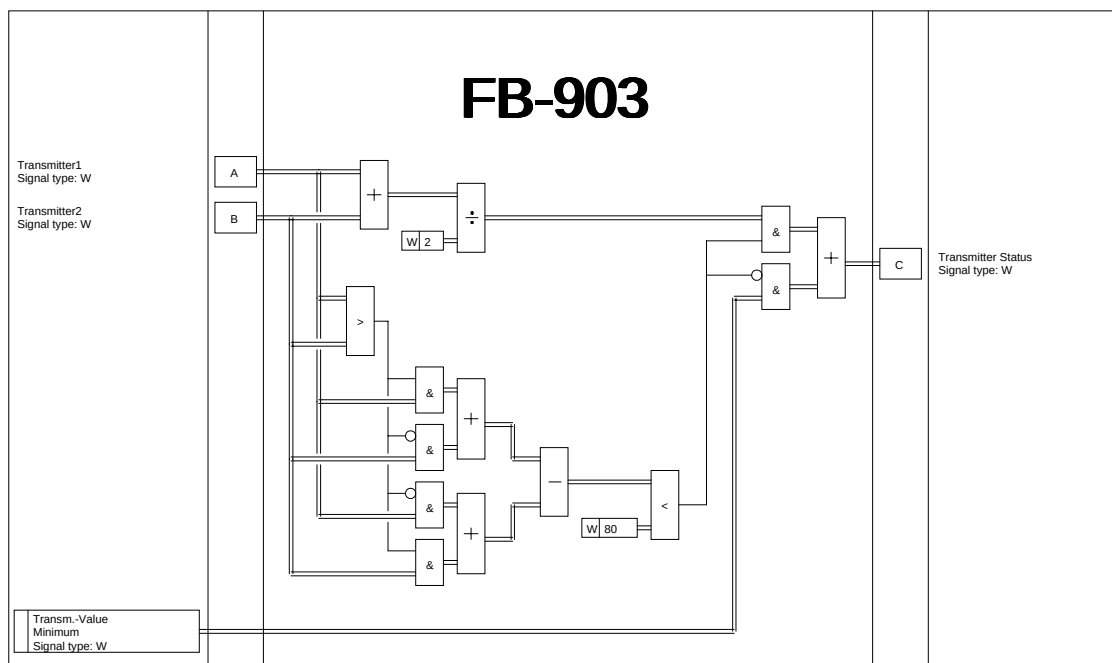


Figure C-20 Function block FB-903 (mean function if discrepancy < 3%, else 0)

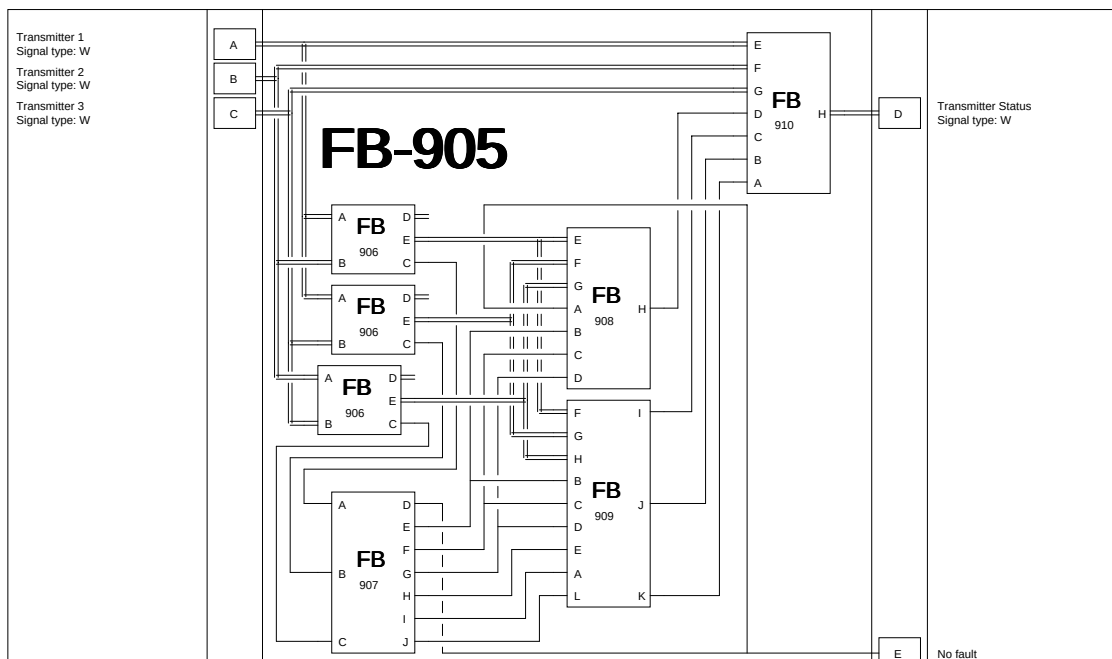


Figure C-21 Function block FB-905 (2-out-of-3 function)

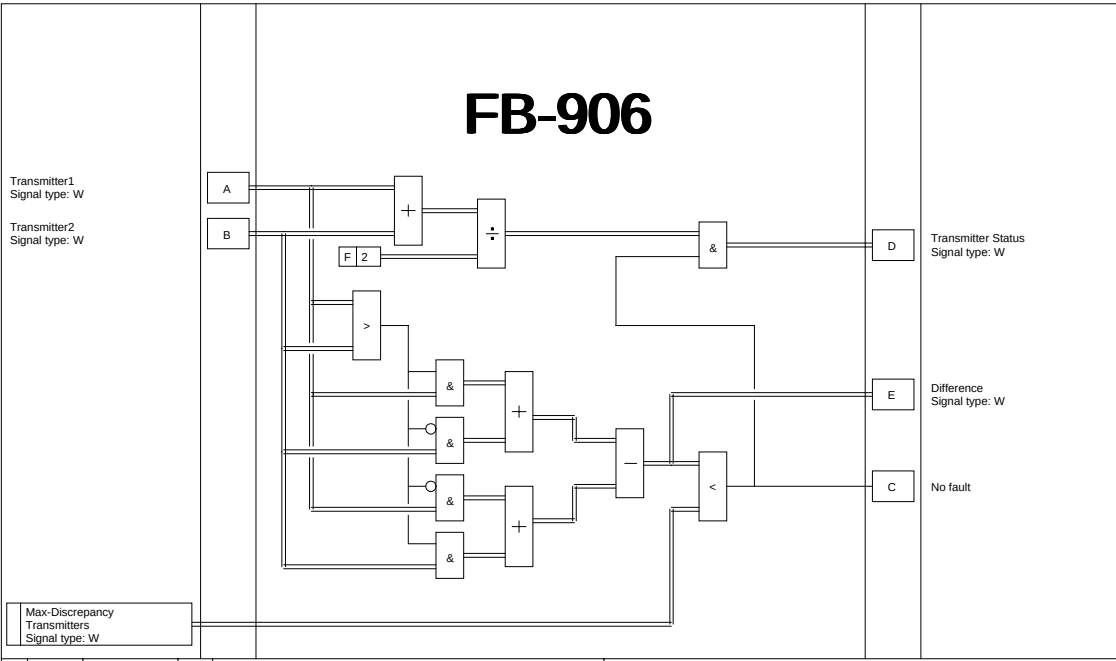


Figure C-22 Function block FB-906 (mean function)

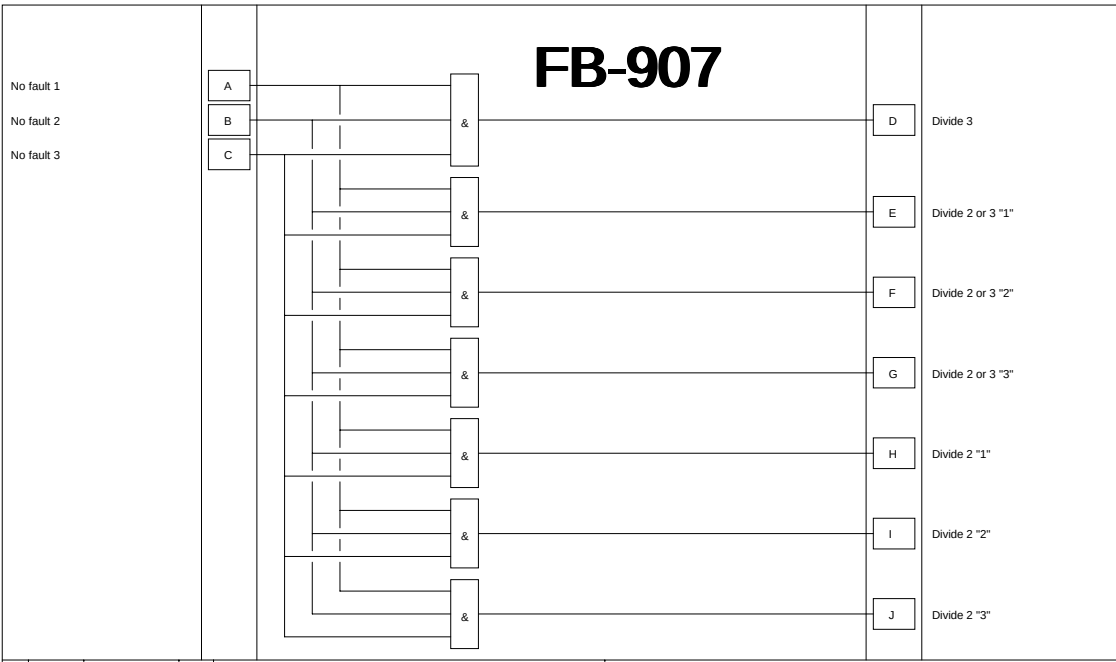


Figure C-23 Function block FB-907 (part of function block FB-905)

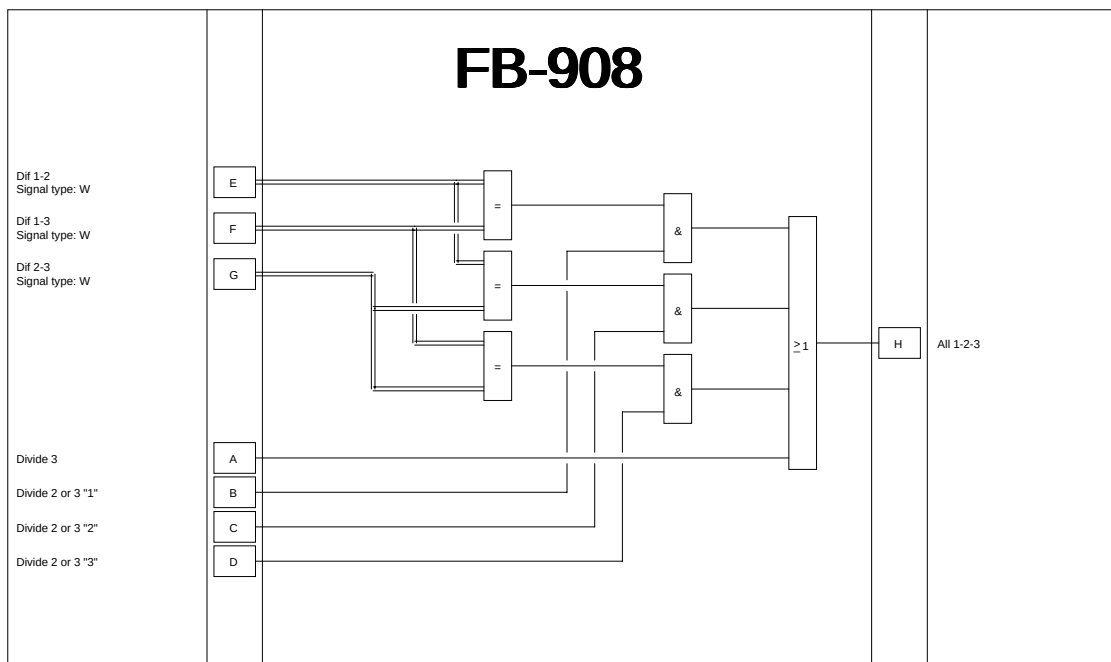


Figure C-24 Function block FB-908 (part of function block FB-905)

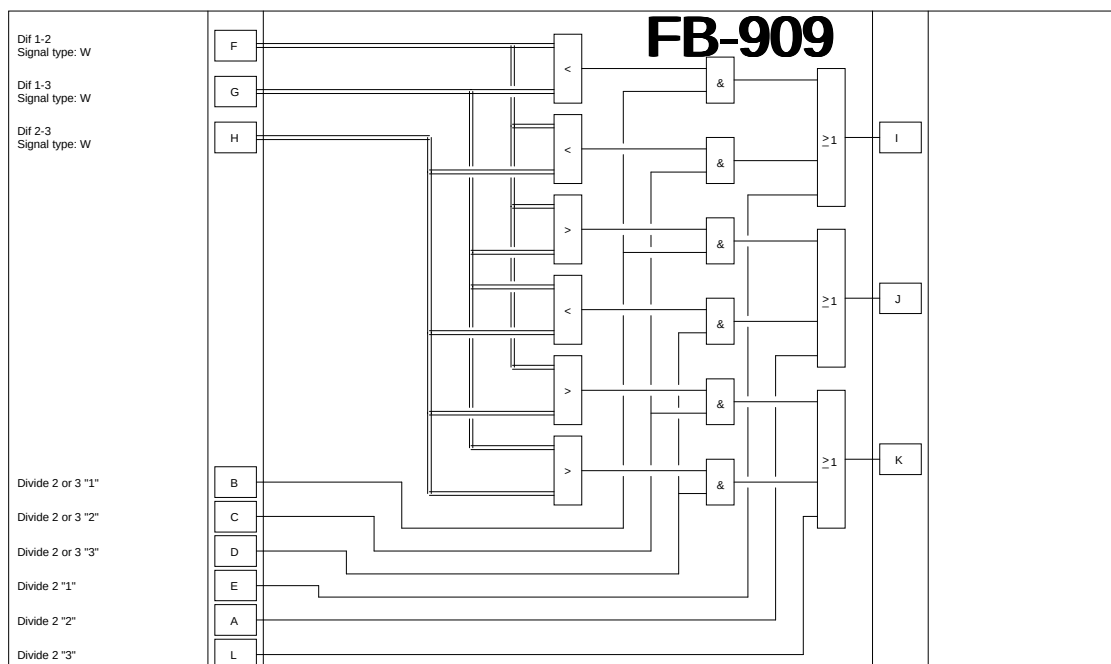


Figure C-25 Function block FB-909 (part of function block FB-905)

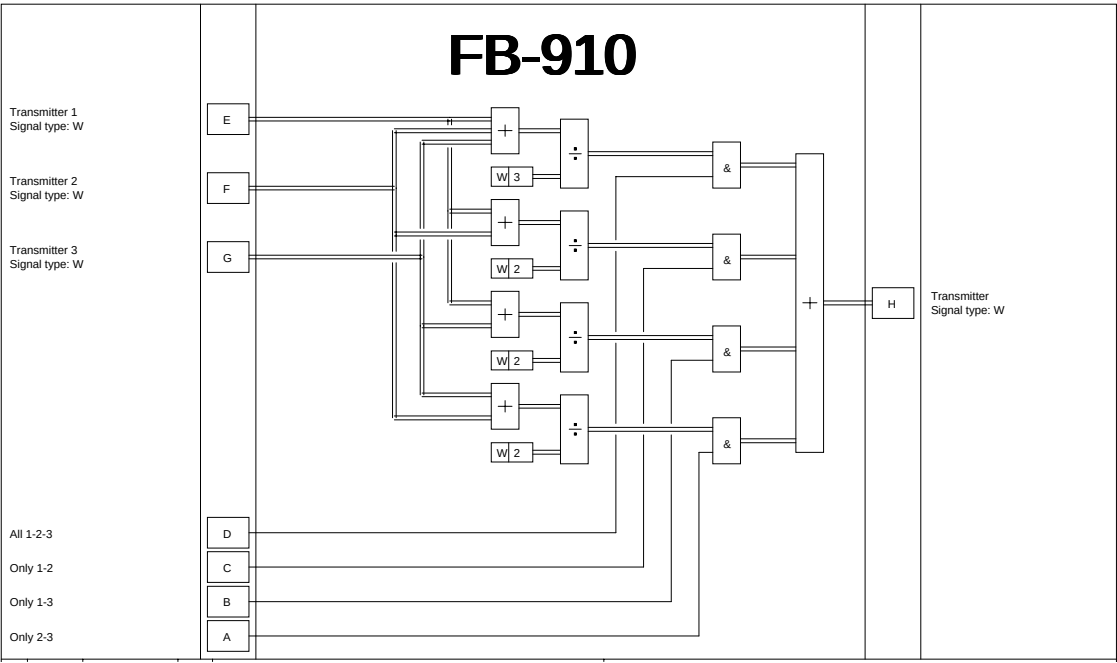


Figure C-26 Function block FB-910 (part of function block FB-905)



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Helping You Control Your World



Fail Safe Control

Appendix D: On-Line Modification

Version 500

Revision 01 (01/98)



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Version 500
Revision 01 (01/98)

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D.1 Introduction

On-line modification is an FSC system option which allows you to modify the application software, system software and the FSC hardware configuration of redundant systems, while the system remains operational. During on-line modification, the changes are upgraded in one Central Part at a time. Meanwhile, the other Central Part continues to monitor the process.

During the upgrade, the FSC system performs a compatibility check across the application-related data, in order to guarantee a safe changeover from the old software to the new software. The system reports the numbers of the functional logic diagrams (FLDs) that have changed. This allows easy verification of the implemented modification within the application. Additional verification of the changes can be obtained using the 'Verify Application' option of FSC Navigator.

The on-line modification option of the FSC system allows changes in the functional logic diagrams, the FSC system configuration and the system software to be implemented in the system without the need of a plant shutdown.

Note:

The amount of human interaction during the modification process is considerable. On-line modification should therefore be carried out with the utmost care, and by authorized and qualified persons only. In case of problems during the on-line modification, the FSC system can go to a safe state resulting in a process shutdown.

D.2 Description

This section describes the stages in the on-line modification process on a functional basis. The detailed procedure for on-line modification is described in section D.7.

D.2.1 Activating the on-line modification option

On-line modification (OLM) is only possible in systems with redundant Central Parts. Before on-line modification can be performed, the on-line modification option must first be activated. This is done using the 'System Configuration' option of FSC Navigator (Install \ Configuration \ On-line modification wanted) (see Figure D-1).

Note:

For systems that do not have redundant Central Parts, the on-line modification option will be disabled.

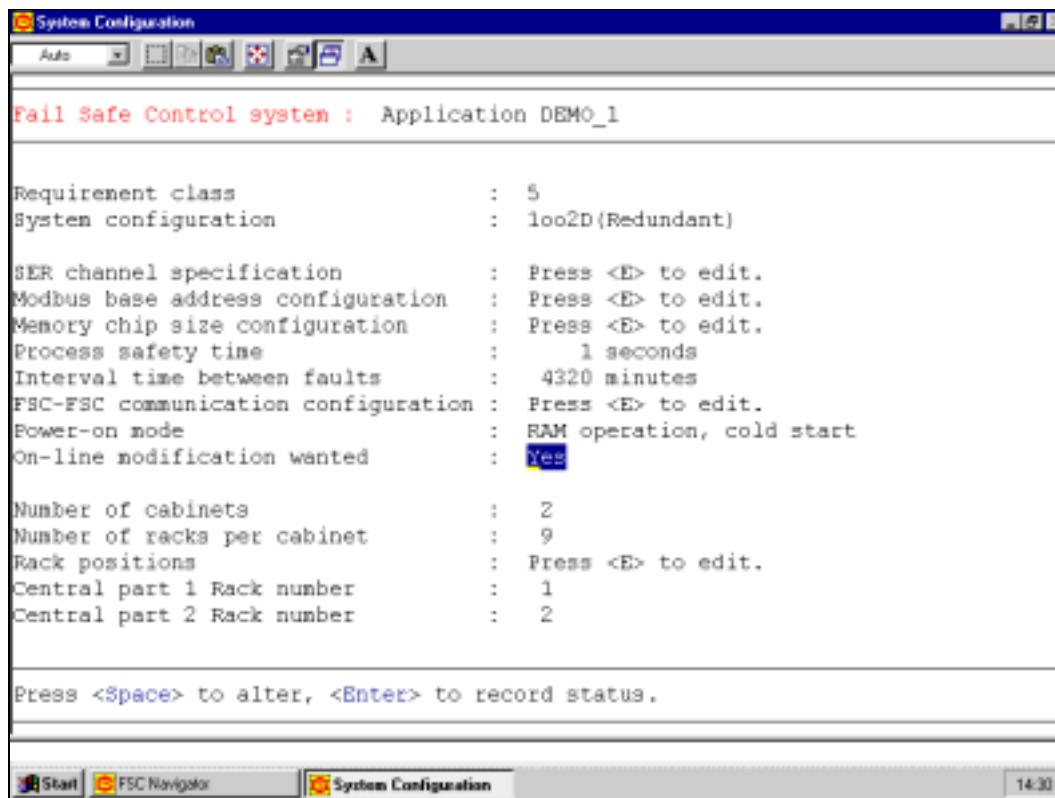


Figure D-1 Activating the on-line modification option

Setting the on-line modification option to 'Yes' will not affect the behavior of the FSC system. The system will always perform a safety compatibility check across the application-related data when new software is loaded into the system.



As the creation of application-related data is controlled by certain options of the FSC Navigator software, these options must be prevented from performing any functions that could lead to incompatibilities between the old and new application. This is done by setting the on-line modification to active. Section D.5 contains a list of the FSC Navigator functions that are affected when on-line modification has been activated.

D.2.2 Modifying the application

The steps taken in this stage are identical to those taken during application development. Use the FSC Navigator functions to modify the application, translate it, and load it into EPROM/RAM.

If there are any changes in the application software, it is strongly recommended that you test the application in an FSC simulation unit (23125/./., 23126/./., 23250/./., 23251/./., or 23260/./.) prior to upgrading the actual system. In this way, any serious errors that could lead to unsafe situations or a total system shutdown (e.g. calculation errors) are detected without consequences.

D.2.3 Upgrading the FSC system

During this stage, the modifications are implemented in the FSC system, i.e. EPROMs are exchanged and, if the system operates in RAM mode, the application is downloaded.

The software is upgraded in one Central Part at a time. If both the system software and the application software are upgraded, both are upgraded simultaneously.

After the upgrade of the software in the first Central Part, the FSC system carries out a compatibility check to verify if a safe changeover can be made from the old software to the new software. This check may last several minutes. While the check is being executed, the time at the display of the diagnostic and battery module (DBM, 10006/./.) is not updated.

As soon as the time at the DBM display starts running again, the check is finished. You can now use the 'On-Line Environment' option of FSC Navigator to read the results. The extended diagnostics screen will show the numbers of the functional logic diagrams that have been changed (see Figure D-2). This allows final verification of the modifications.

Notes:

If a function block has changed, a difference will be reported for all functional logic diagrams that use this function block.

If the force enable status of the inputs with location 'COM' or 'FSC' has changed, a sheet difference will be reported for the sheets that contain these variables.

During on-line modification, the 'Verify Application' option may be used to log all revision information. Refer to Section 11 of the FSC Software Manual ("Verifying an Application") for further details.

If no sheet differences exist, this is also reported.

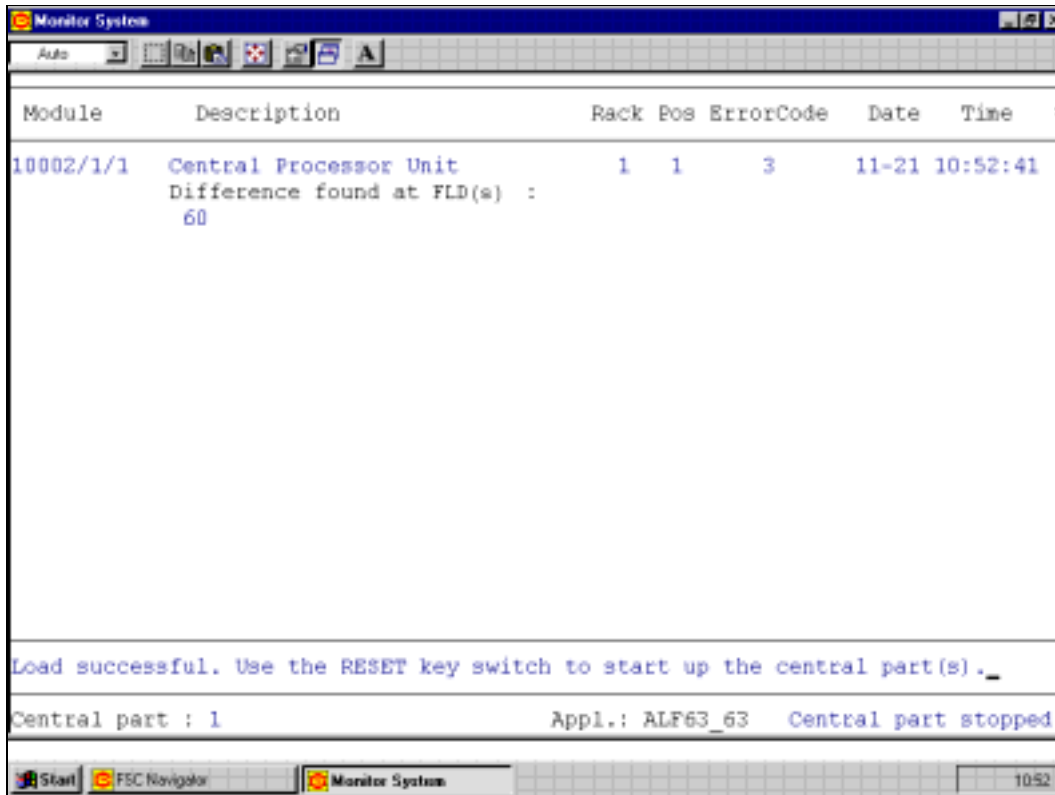


Figure D-2 Sheet difference during on-line modification

Apart from the report regarding sheet differences, no messages should be present. Any other message indicates an incompatibility of the current application with the previous version. As a result, either on-line modification is not possible, or the new software will run with limited functionality. The possible messages and their consequences are listed in subsection D.3.

If the first Central Part reports no errors, it may be started. At the same time, the other Central Part will shut down and software can be exchanged in the other Central Part.

D.2.4 On-line modification in FSC networks

If a system is part of an FSC communication network, it performs a compatibility check for all connected systems. If inconsistencies are detected or if the check for a specific system cannot be completed for any other reason, an error message is generated in the extended diagnostics. In case of such an error, no data will be exchanged with the system after start-up. The communication can only be re-established after successful completion of the compatibility check by any of the systems.

Note:

The FSC system performs the compatibility check for connected systems each time it is started, regardless if modifications were made in the system or not. The extended diagnostics should therefore always be examined prior to starting the system, to verify if all communication links are intact.



D.3 Error diagnostics during on-line modification

Apart from the sheet differences report, the following messages may appear when reading extended diagnostics during on-line modification:

D.3.1 Diagnostics relating to internal on-line modification check

Communication failure during OLM.

This message indicates that the compatibility check was aborted because communication with the running Central Part failed. Additional information is included to indicate during which part of the check the error was detected.

Corrective action: Check the communication cable between the Central Parts.

Incompatible variable allocation.

This message indicates that the variable allocation in the new software collides with the variable allocation in the old software. If this message appears, on-line modification is not possible. The type and tag number of the improperly allocated variable are included in the diagnostic report.

Corrective action: Check if the on-line modification option has been activated (see subsection D.2.1).
Perform an on-line rebuild to obtain an application database which is compatible with the application as present in the system (refer to Section 4 of the FSC Software Manual).
Reallocate any unallocated variables.

Proceed in accordance with the normal application development procedures.

No start-up due to different application.

This message appears if you try to start up both Central Parts simultaneously, with incompatible software loaded in the Central Parts. Refer to Section 12 of the FSC Software Manual ("On-Line Environment") for details on FSC system diagnostics.

Corrective action: Load consistent software in both Central Parts before system start-up.

Shutdown due to on-line modification.

This message will always appear during the upgrade stage. The message is reported by the Central Part that contains the old software if it has shut down just after the changeover to the Central Part that contains the new software.

Note:

If either the 'Communication failure during OLM' message or 'Incompatible variable allocation' message is reported, it is not possible to start up the Central Part that contains the new software. Trying to start up anyhow may result in a total system shutdown.



D.3.2 Diagnostics relating to FSC network configuration check

Communication failure during configuration check for system <No.>.

The message indicates that the configuration check was aborted because communication with the target system failed. The number of the target system is included in the diagnostic report.

Corrective action: Check the communication connections between the systems.

Incompatible FSC-FSC configuration for system <No.>.

The message indicates that the configuration check was aborted due to different sizes of the marker and/or register areas for data exchange with the target system. The number of the target system is included in the diagnostic report.

Corrective action: Adjust the size of the marker and/or register areas.

Incompatible FSC-FSC variable allocation.

The message indicates that the configuration check was aborted because the variable allocation in the system collides with the variable allocation in the target system. The type and tag number of the improperly allocated variable are included in the diagnostic report.

Corrective action: Check if the on-line modification option has been activated (see subsection D.2.1).
Perform an on-line rebuild to obtain an application database which is compatible with the application as present in the system.
Reallocate any unallocated variables.

Note:

If the system is started while one of these faults is present, no communication with the target system will be accomplished.



D.4 On-line rebuild

When an application is created, variables are allocated to locations in system memory (see Figure D-3). If the application is then changed, variable allocations will change as well. This is partly because of user interaction (e.g. removal of a variable allocated to a hardware channel) and partly because of actions by the FSC Navigator software (e.g. change of the number of compiler markers required to realize a functional logic diagram).

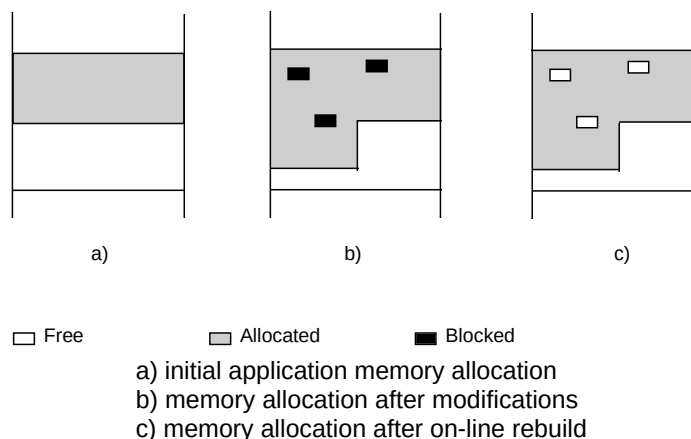


Figure D-3 Allocation in variables to locations in system memory

If the application has been configured for on-line modification, every time a variable is deleted from the application, its allocation is blocked for further use (Figure D-3) in order to guarantee application compatibility with the previous version. As a consequence of this mechanism, the required amount of system memory will increase, while the allocated memory becomes more and more fragmented.

The only way to release blocked allocations is via the 'On-Line Rebuild' option of FSC Navigator. This option must be selected after the application has been compiled and upgraded into the FSC system. During the on-line rebuild, the variable allocation is read from the FSC system and the unused memory locations are unlocked.

For details on the on-line rebuild function refer to Section 4 of the FSC Software Manual ("System Configuration").



D.5 Consequences for functionality of FSC Navigator options

If on-line modification is active, the following functions in FSC Navigator are not available:

- System Configuration:
 - change the name of allocated tag numbers,
 - change configured hardware modules.
- Design FLDs:
 - change the name of used tag numbers,
 - renumber FLDs,
 - append an application.

D.6 Limitations

The following modifications cannot be made using the on-line modification option:

- I/O racks with function 1001D (single) cannot be added/deleted.
- Upgrade of system software if the system is running an FSC release older than 2.80b1.

On-line modification and start-up of redundant systems configured in "Warm Start" mode may give spurious diagnostic messages.

During an on-line modification it is possible to add and delete all types of variables, including timers, inputs and outputs. If an output is deleted which was high at the time of the on-line modification, the output will remain high after removal from the application program.



D.7 On-line modification procedure

1. Verify TPS compatibility with FSC system software (only for FSC-SM).

Prior to proceeding with an on-line upgrade of the FSC system software, verify that the correct TPS software release has been loaded.

Note:

If an upgrade of the TPS software is required (see Table D-1), load the TPS software according to the procedure mentioned in the TPS documentation without loading the Personality Image (PI) software of the FSC Safety Manager. The PI software of the FSC Safety Manager must be loaded during the on-line modification procedure mentioned below.

Table D-1 Compatibility matrix FSC-SM vs. TPS

FSC-SM	TPS
Release 400.2, 400.3	Release 510.7, 510.8, Release 520.1 and 520.2
Release 420.x	Release 530.x and 531.x
Release 421.x	Release 530.x and 531.x
Release 500.x	Release 530.x and 531.x

2. Make sure the application in the FSC system is identical to the application that is stored on the user station (FSC Navigator).

The FSC Navigator software offers two ways to accomplish this:

- a) Compare the version of the application in the FSC system with the version of the application that is stored on the user station (FSC Navigator).
 - The version of the application in the FSC system can be determined using the 'Monitor System' option of FSC Navigator (FSC System \ Sys Info \ Versions).
 - The version of the application on the FSC user station can be determined by viewing the log file of the 'Translate Application' option of FSC Navigator.

Note:

This method will only ensure that the applications in the FSC system and on the FSC user station are identical if no modifications were made to the application at the FSC user station without running the 'Translate Application' option of FSC Navigator afterwards.

- b) Use the 'Verify Application' option of FSC Navigator to verify the application in the FSC system against the application on the FSC user station. For details refer to Section 11 of the FSC Software Manual ("Verifying an Application").

3. Make a backup of the application files in your project folders.



4. Activate the on-line modification option.

In order to be able to carry out on-line modification, you must first activate the on-line modification option. To activate this option, do the following:

1. Start FSC Navigator.
2. Open the application you wish to modify.
3. Choose the 'System Configuration' option from the Project menu or the FSC Project Configuration screen.
4. From the Maintain menu, choose 'Install'.
5. From the Install menu, choose 'Configuration'.
6. Make sure that the 'On-line modification wanted' option is set to 'Yes'. If it is not, move the cursor to this line and use <Space> to toggle the option to 'Yes'.

Note:

For systems that do not have redundant Central Parts, the on-line modification option will be disabled.

5. Make the required modifications to the application.

6. Translate the application.

After translating, use the 'View Log' option of FSC Navigator to read the 'Translate Application' log file. If any errors were detected, correct these errors and translate the application again.

Repeat these steps until the compiler reports no errors.

7. Program the EPROMs.

Use the 'Program EPROMs' option of FSC Navigator to program the EPROMs for the **MEM**ory modules (EPROM mode only) and **COM**munication modules for both Central Parts.

The following sequence is recommended:

- Blank check
- Program
- Verify.

If you want to perform an on-line upgrade of the system software, program EPROMs for the **CPU** module as well.

For FSC-SM systems, the new firmware EPROM of the 10008/2/U communication module must also be programmed.

Provide all EPROMs with a unique identification with regard to module type, module number, Central Part number, date and version number (in case of more than one upgrade on one day).



The following steps must be carried out per system. If you have a master/slave configuration, start with the master system.

8. Check the system before exchanging the EPROMs.

Before exchanging the EPROMs, use the 'Extended Diagnostics' option of FSC Navigator to verify that the FSC system is OK. If FSC Navigator reports any errors, resolve these errors before exchanging the EPROMs. (Please note that the error messages 'Analog input sensor break alarm' and 'Analog output value invalid' may be ignored here.)

Note:

If you fail to repair all hardware failures before on-line modification, the process may shut down during the on-line modification process.

9. Switch off Central Part 1.

- a) Trip Central Part 1 by pulling the flatcable from the watchdog module (type 10005/1/1).
- b) After Central Part 1 has tripped, you may reconnect the flatcable of the watchdog.
- c) Switch the CPU RUN/STOP key switch (type 10002/1/1 or 10002/1/2) from Central Part 1 to the 'STOP' position (horizontal).
- d) Remove all communication cables from the COM modules (type 10004/./.) which are placed in Central Part 1.
- e) Switch off the 24 Vdc power supply by switching off circuit breaker from the Central Part 1 (optional).

10. Exchange the memory EPROMs in Central Part 1 (only if the system operates in EPROM mode).

If the application is located in EPROMs at module 10002/A/1 on the CPU module (10002/1/2) instead of EPROMs on the memory module (10003/1/1), steps 10a to 10g below should read: 'EPROMs on the 10002/A/1 module on the CPU module'.

- a) Loosen the screws of the memory modules.
- b) Carefully pull the first memory module from Central Part 1.
- c) Remove the EPROMs from the memory module.
- d) Install the new EPROMs on the memory module.
- e) Check the EPROMs visually and check if the correct set of EPROMs has been installed and in the correct sequence.
- f) Put the memory module back into place.

Note:

If more memory modules are installed in the system, repeat steps 10b to 10f for all other memory modules.

- g) Fasten all the screws of the memory modules.



11. Exchange the communication EPROMs in Central Part 1.

- a) Loosen the screws of the communication modules.
- b) Carefully pull the first communication module from Central Part 1.
- c) Remove the EPROMs from the communication module.
- d) Install the new EPROMs on the communication module.
- e) Check the EPROMs visually and check if the correct set of EPROMs has been installed and in the correct sequence.
- f) If the EPROM size has changed, check the jumper settings on the communication module.
- g) Put the communication module back into place.

Note:

If more communication modules are installed in the system, repeat steps 11b to 11g for the other communication modules.

- h) Fasten all the screws of the communication modules.

12. Exchange the CPU EPROMs (only if the system software is upgraded).

- a) Loosen the screws of the CPU module (type 10002/1/1 or 10002/1/2).
- b) Carefully pull the CPU module from Central Part 1.
- c) Remove the EPROMs from the CPU module (10002/A/1 for CPU 10002/1/2).
- d) Install the new EPROMs on the CPU module.
- e) Check the EPROMs visually and check if the correct set of EPROMs has been installed and in the correct sequence.
- f) Make sure that the CPU RUN/STOP key switch is in the 'STOP' position (horizontal), and put the CPU module back into place.
- g) Fasten the screws of the CPU module.

13. Exchange the FSMM EPROM (only for FSC-SM if the firmware software is upgraded).

- a) Make sure that the TPS software is compatible with the FSC system software (see Table D-1).
- b) Disconnect the UCN drop cables from the FSC-SMM module (type 10008/2/U) of Central Part 1.
- c) Loosen the screws of the FSC-SMM module.
- d) Remove the EPROM from the FSC-SMM module.
- e) Install the new EPROM on the FSC-SMM module.
- f) Check the EPROM visually.
- g) Put the FSC-SMM module back into place.
- h) Fasten the screws of the FSC-SMM module.
- i) Reconnect the UCN drop cables to the FSC-SMM module.



14. Start up Central Part 1.

- a) Reconnect the communication cables to the communication modules. Make sure that the cables are connected to the correct module and channel, and verify that the cables have been properly inserted.
- b) Switch the CPU RUN/STOP key switch of Central Part 1 to the 'RUN' position (vertical).
- c) Wait for the vertical bus drivers (VBDs) to "click".
- d) If the VBDs do not "click", switch the CPU RUN/STOP key switch to the 'STOP' position (horizontal), and disconnect the 24 Vdc power supply by switching off its circuit breaker (if required). Then check all the modules whose EPROMs have been replaced. Make sure that they have been placed in the correct position of the Central Part rack.

Steps 14f and 14g are only carried out if the system operates in RAM mode and if downloading takes place via a Development System link with the FSC user station.

- f) Use the 'Download Application' option of FSC Navigator to download the application software for Central Part 1. Make sure that the Force Enable key switch is in the 'ON' position (vertical). If it is not, downloading to FSC will not be possible.
- g) Wait for the message 'Load successful. Use the RESET key switch to start up the central part(s)' to appear.
If another message is displayed, check the communication link between the FSC user station and the FSC system, and retry.

Steps 14h to 14j are only carried out if the system operates in RAM mode and if downloading takes place via an FSC-FSC link for multidrop (MD) slaves.

- h) Disconnect the multidrop cable from the FSC-FSC link port on the communication module in Central Part 1, and connect the PC-FSC cable.
- i) Use the 'Download Application' option of FSC Navigator to download the application software for Central Part 1.
- j) Wait for the message 'Waiting for OLM check and FSC-FSC COM check...' to appear. Then switch the CPU RUN/STOP key switch in Central Part 1 to the 'STOP' position (horizontal). Disconnect the PC-FSC cable and reconnect the multidrop cable. Switch the CPU RUN/STOP key switch on Central Part 1 to the 'RUN' position (vertical).
- k) Go to the FSC system and wait until the diagnostic and battery module (DBM, 10006/./.) starts "running" again. The DBM was stopped when downloading of the application software was started. As long as the DBM time is "frozen", the system is performing the internal check between Central Parts and systems.
- l) When the DBM starts "running" again, use the 'Extended Diagnostics' option of FSC Navigator to see if any errors have occurred via the FSC user station that is connected to the FSC (master) system.
- m) FSC Navigator will display the FLD differences between the two application versions. The error "Diagnostic data invalid" may also be displayed. This will be the case if the CPU has been removed and no sheet differences exist. The message "No faults detected" on systems with FSC-FSC communication is also possible.
If any other messages are displayed, it is not allowed to reset the FSC system. A reset can cause a trip on both Central Parts causing a shutdown of the plant.



Cycle the CPU RUN/STOP keyswitch and repeat steps 14d to 14m, until all faults have been traced and restored which are mentioned in this step as 'any other messages'.

Steps 14n to 14p are only carried out for an FSC-SM system.

- n) Wait for the secondary Safety Manager to become 'ALIVE' on the TPS operator station (US, U^XS, GUS).
- o) Load the Personality Image (PI) into the secondary Safety Manager Module (SMM).
- p) Wait for the message 'LOADED' to appear. The status will go to 'BKUP_PF'.

- q) If the diagnostics show no messages other than "Diagnostic data invalid", "No faults detected", "Device communication error" or "Sheet differences", you may restart the Central Part using the watchdog reset key, or you may want to run the 'Verify Application' option of FSC Navigator at this point. This option will verify that there are no differences between the application in Central Part 1 and the application on the FSC user station. In addition, this option will generate a list of differences between the current version of the application and the 'old' version, which is still present in Central Part 2 (revision control).

Note:

The log file will be overwritten if you run the 'Verify Application' option again. You should therefore make a backup of the log file (by saving it under a different name or by printing it).

- r) If a reset is given, Central Part 1 will start "running" and Central Part 2 will shut down.
- s) Use the 'Monitor System' option of FSC Navigator to see if Central Part 1 is OK.

15. Switch off Central Part 2.

- a) Due to the on-line modification, Central Part 2 will already be shut down.
- b) Switch the CPU RUN/STOP key switch (type 10002/1/1 or 10002/1/2) from Central Part 1 to the 'STOP' position (horizontal).
- c) Remove all communication cables from the COM modules (type 10004/./.) which are placed in Central Part 2.
- d) Switch off the 24 Vdc power supply by switching off the circuit breaker from Central Part 2 (optional).

16. Exchange the memory EPROMs in Central Part 2 (only if the system operates in EPROM mode).

If the application is located in EPROMs at module 10002/A/1 on the CPU module (10002/1/2) instead of EPROMs on the memory module (10003/1/1), steps 16a through 16g below should be read: 'EPROMs on the 10002/A/1 module on the CPU module'.

- a) Loosen the screws of the memory modules.
- b) Carefully pull the first memory module from Central Part 2.
- c) Remove the EPROMs from the memory module.
- d) Install the new EPROMs on the memory module.



- e) Check the EPROMs visually and check if the correct set of EPROMs has been installed and in the correct sequence.
- f) Put the memory module back into place.

Note:

If more memory modules are installed in the system, repeat steps 16b to 16f for all other memory modules.

- g) Fasten all the screws of the memory modules.

17. Exchange the communication EPROMs in Central Part 2.

- a) Loosen the screws of the communication modules.
- b) Carefully pull the first communication module from Central Part 2.
- c) Remove the EPROMs from the communication module.
- d) Install the new EPROMs on the communication module.
- e) Check the EPROMs visually and check if the correct set of EPROMs has been installed and in the correct sequence.
- f) If the EPROM size has changed, check the jumper settings on the communication module.
- g) Put the communication module back into place.

Note:

If more communication modules are installed in the system, repeat steps 17b to 17g for the other communication modules.

- h) Fasten all the screws of the communication modules.

18. Exchange the CPU EPROMs (only if the system software is upgraded).

- a) Loosen the screws of the CPU module (type 10002/1/1 or 10002/1/2).
- b) Carefully pull the CPU module from Central Part 2.
- c) Remove the EPROMs from the CPU module (10002/A/1 for CPU 10002/1/2).
- d) Install the new EPROMs on the CPU module.
- e) Check the EPROMs visually and check if the correct set of EPROMs has been installed and in the correct sequence.
- f) Put the CPU module back into place.
- g) Fasten the screws of the CPU module.

19. Exchange the FSMM EPROM (only for FSC-SM if the firmware software is upgraded).

- a) Disconnect the UCN drop cables from the FSC-SMM module (type 10008/2/U) of Central Part 2.
- b) Loosen the screws of the FSC-SMM module.
- c) Remove the EPROM from the FSC-SMM module.
- d) Install the new EPROM on the FSC-SMM module.



- e) Check the EPROM visually.
- f) Put the FSC-SMM module back into place.
- g) Fasten the screws of the FSC-SMM module.
- h) Reconnect the UCN drop cables to the FSC-SMM module.

20. Start up Central Part 2.

- a) Switch on the circuit breaker for the 24 Vdc power supply to Central Part 2 (optional).
- b) Reconnect the communication cables to the communication modules. Make sure that the cables are connected to the correct module and channel, and verify that the cables have been properly inserted.
- c) Switch the CPU RUN/STOP key switch of Central Part 2 to the 'RUN' position (vertical).
- d) Wait for the vertical bus drivers (VBDs) to "click".
- e) If the VBDs do not "click", switch the CPU RUN/STOP key switch to the 'STOP' position (horizontal), and disconnect the 24 Vdc power supply by switching off its circuit breaker (if required). Then check all the modules whose EPROMs have been replaced. Make sure that they have been placed in the correct position of the Central Part rack.

Steps 20f and 20g are only carried out if the system operates in RAM mode and if downloading takes place via a Development System link with the FSC user station.

- f) Use the 'Download Application' option of FSC Navigator to download the application software for Central Part 2. Make sure that the Force Enable key switch is in the 'ON' position (vertical). If it is not, downloading to FSC will not be possible.
- g) Wait for the message 'Load successful. Use the RESET key switch to start up the central part(s)' to appear.
If another message is displayed, check the communication link between the FSC user station and the FSC system, and retry.

Steps 20h to 20j are only carried out if the system operates in RAM mode and if downloading takes place via an FSC-FSC link for multidrop (MD) slaves.

- h) Disconnect the multidrop cable from the FSC-FSC link port on the communication module in Central Part 2, and connect the PC-FSC cable.
- i) Use the 'Download Application' option of FSC Navigator to download the application software for Central Part 2.
- j) Wait for the message 'Waiting for OLM check and FSC-FSC COM check...' to appear. Then switch the CPU RUN/STOP key switch in Central Part 1 to the 'STOP' position (horizontal). Disconnect the PC-FSC cable and reconnect the multidrop cable. Switch the CPU RUN/STOP key switch on Central Part 2 to the 'RUN' position (vertical).
- k) Go to the FSC system and wait until the diagnostic and battery module (DBM, 10006/./.) starts "running" again. The DBM was stopped when downloading of the application software was started. As long as the DBM time is "frozen", the system is performing the internal check between Central Parts and systems.
- l) When the DBM starts "running" again, use the 'Extended Diagnostics' option of FSC Navigator to see if any errors have occurred via the FSC user station that is connected to the FSC (master) system.



- m) FSC Navigator will display the FLD differences between the two application versions. The error "Diagnostic data invalid" may also be displayed. This will be the case if the CPU has been removed and no sheet differences exist. The message "No faults detected" on systems with FSC-FSC communication is also possible.

If any other messages are displayed, it is not allowed to reset the FSC system. A reset can cause a trip on both Central Parts causing a shutdown of the plant.

Cycle the CPU RUN/STOP keyswitch and repeat steps 20d to 20m, until all faults have been traced and restored which are mentioned in this step as 'any other messages'.

Steps 20n to 20p are only carried out for an FSC-SM system.

- n) Wait for the secondary Safety Manager to become 'ALIVE' on the TPS operator station (US, U^XS, GUS).
- o) Load the Personality Image (PI) into the secondary Safety Manager Module (SMM).
- p) Wait for the message 'LOADED' to appear. The status will go to 'BKUP_PF'.
- q) If no error messages are displayed other than the ones mentioned in step n, you may restart the Central Part using the watchdog reset key.
- r) If a reset is given, Central Part 2 will start "running" together with Central Part 1.
- s) Use the 'Monitor System' option of FSC Navigator to see if the FSC system is OK.
- t) Run the 'Verify Application' option of FSC Navigator to verify that the FSC application software in both Central Parts is consistent with the application as stored on the FSC user station.

Note:

This verification will overwrite the log file generated during the previous verification. You should therefore make sure that you have a backup of the 'old' log file, if desired (by saving it under a different name or by printing it).

21. Back up the application, e.g. on floppy disk.



D.8 On-line modification of tag numbers

D.8.1 Changing input tag numbers

In order to change input tag numbers in a system on-line, the following steps must be carried out:

1. Delete the tag numbers that are to be modified. Modify the FLD relating to these tags in such a way such the FLDs remain in a steady state. Translate and download the application software as per OLM procedure in section D.7.
2. Carry out the on-line rebuild.
3. Add the modified tag numbers in the FSC database and add the tag numbers back into the corresponding FLD. Translate and download the application software as per OLM procedure in section D.7.

D.8.1.1 Deleting tag numbers

- a) Start the 'System Configuration' option of FSC Navigator.
From the main menu, choose 'Signal specs'.
Use the 'Search' option to list the related tag number(s).
Press <Ctrl>+<U> to delete the tag numbers.
The message "Variable allocated. Deallocate because of on-line modification? (Y/N)" will be displayed. Enter <Y> to confirm.
- b) To modify the FLD relating to the deleted tag number(s), start the 'Design FLDs' option of FSC Navigator. Delete the I/O symbols of the tag number(s) and replace them with, for example, a constant boolean signal or force-enable input in order to maintain the correctness of the FLD. If a constant boolean signal is used, the process cannot be shut down anymore via that input and closer process monitoring is required to manually trip the application in case of dangerous situations.
- c) Translate the application and upgrade the system in accordance with the OLM procedure as described in section D.7.

D.8.1.2 On-line rebuild

- a) Connect the FSC user station to the FSC system via the Development System link port or FSC-FSC link port (FSC multidrop slave systems).
- b) From the On-Line menu of FSC Navigator, choose the 'On-Line Rebuild' option.
- c) You will be asked to confirm the on-line rebuild. Press <Y> to confirm.
- d) The message "Rebuilding screen file for status display..." will appear.
Shortly after that, the message "Toggle to select the Central Part to be rebuilt: 1" will be displayed.
Use <Space> to select the Central Part number that the PC is connected to. Press <Enter> to continue.
- e) The results of the rebuild are recorded in a log file, which can be viewed using the 'View Log' option of FSC Navigator.
- f) Back up the application, e.g. on floppy disk.



D.8.1.3 Adding changed tag numbers

- a) Start the 'System Configuration' option of FSC Navigator.
From the main menu, choose 'Signal specs'.
Choose the 'Add' option to enter the related changed tag number(s).
- b) From the main menu of 'System Configuration', choose 'Hardware specs'.
Reallocate the tag numbers as per its previous allocation or relative address (COM I/O).
Also set the 'Safety related', 'SER enable', etc., attributes to their previous settings.
- c) Use the 'Design FLDs' option of FSC Navigator to reassign the tag number to the respective FLD(s).
- d) Translate the application and upgrade the system in accordance with the OLM procedure as described in section D.7.

D.8.2 Changing output tag numbers

In order to change output tag numbers in a system on-line, the following steps must be carried out (depending on the operational status of the outputs):

1. If the output may be de-energized, the procedure is identical to the procedure for changing input tag numbers (see subsection D.8.1).
2. If continuous control of the output must be maintained, a temporary output which is allocated to a spare channel is used to control the process while the output tag number is being changed.

D.8.2.1 Changing output tag numbers for outputs that may be de-energized

Use the same procedure as for input tag numbers.

However, instead of a constant symbol, use an off-sheet reference to maintain the integrity of the functional logic diagrams.

If you use this procedure, the output in question will be de-energized as soon as Central Part 1 is started with the intermediate application software. Output control is regained as soon as the final application software (which contains the new output tag number) has been upgraded in the system.

D.8.2.2 Changing output tag numbers, with continuous output control

If continuous output control is required, use the following procedure.

D.8.2.2.1 Deleting tag numbers

- a) Start the 'System Configuration' option of FSC Navigator.
From the main menu, choose 'Signal specs'.
Choose the 'Add' option to add a temporary output tag number for each output tag number being changed.
- b) From the main menu of 'System Configuration', choose 'Hardware specs'.
Allocate the temporary tag number(s) to spare output channel(s).



- c) Use the 'Design FLDs' option of FSC Navigator to modify the FLD relating to the added temporary tag number(s). Add a new output symbol for the temporary tag number in parallel with its corresponding output which is being changed.
- d) Wire the hardware channel(s) of the output(s) being changed to the corresponding temporary output channel(s) in the FSC cabinet.
- e) Translate the application and upgrade the system according the OLM procedure in section D.7.
- f) Select 'Configure FSC system' from the FSC main menu. Select 'Signal specs' and list the related tag number(s) via the 'Search' option. Delete the tag numbers of the outputs being changed by pressing <Ctrl>+<U>. The message 'Variable allocated. De-allocating because of on-line modification (Y/N)' is displayed. Enter <Y> to confirm.

Note:

It is important that the tag numbers that are being changed, are deleted from the FSC database. If the tag number is only deleted from the functional logic diagrams while its hardware allocation is maintained, the corresponding output channel status is preserved during the on-line modification and cannot be controlled after the on-line modification.

- f) To modify the FLD relating to the deleted tag number(s), go to 'Design functional logic diagrams'. Delete the output symbol for the tag number(s) being changed.
- g) Translate the application and upgrade the system in accordance with the OLM procedure as described in section D.7.

D.8.2.2.2 On-line rebuild

Carry out an on-line rebuild as described in section D.2.

D.8.2.2.3 Adding tag numbers

- a) Start the 'System Configuration' option of FSC Navigator.
From the main menu, choose 'Signal specs'.
Choose the 'Add' option to add the new output tag number(s).
- b) From the main menu of 'System Configuration', choose 'Hardware specs'.
Allocate the tag number(s) to the original output channel(s).
- c) Use the 'Design FLDs' option of FSC Navigator to modify the FLD relating to the added tag number(s). Add a new output symbol for the tag number in parallel with its corresponding temporary output.
- d) Translate the application and upgrade the system in accordance with the OLM procedure as described in section D.7.
- e) Start the 'System Configuration' option of FSC Navigator.
From the main menu, choose 'Signal specs'.
Use the 'Search' option to list the temporary tag number(s).
Press <Ctrl>+<U> to delete the tag numbers of the temporary outputs.
The message "Variable allocated. Deallocate because of on-line modification? (Y/N)" will be displayed. Enter <Y> to confirm.



- f) Use the 'Design FLDs' option of FSC Navigator to modify the FLD relating to the deleted tag number(s). Delete the output symbol for the temporary tag number(s).
- g) Remove the wiring to the temporary outputs from the FSC cabinet.
- h) Translate the application and upgrade the system in accordance with the OLM procedure as described in section D.7.

D.8.2.2.4 On-line rebuild

Carry out an on-line rebuild as described in subsection D.8.1.2.



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Appendix E: PID Controllers

Version 500

Revision 01 (01/98)

Total Plant



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Version 500
Revision 01 (01/98)

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E.1 Introduction

The most common way to introduce feedback in a control loop is via PID algorithms. The process input is controlled in such a way that the process output will approximate a predefined setpoint.

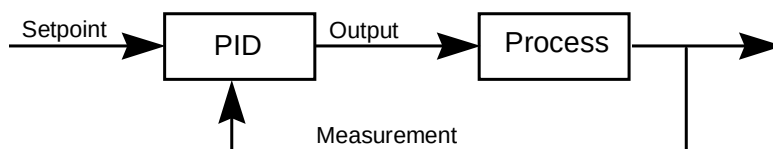


Figure E-1 Feedback in control loop using PID algorithms

The characteristics of the PID controller – such as gain, damping, overshoot, etc. – are adjusted by several parameters, with the proportional, integrational and differential (PID) being the most important ones.

A standard software PID algorithm has been implemented in the FSC system software. The user can easily use this algorithm via a symbol in the functional logic diagrams (see Figure E-2).

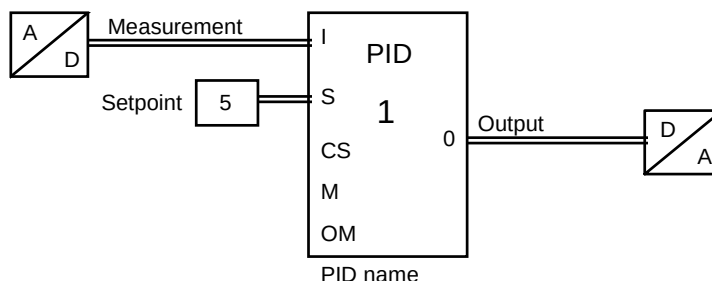


Figure E-2 Standard PID algorithm in FSC system software

Control parameters can be altered on-line using the 'Monitor System' option of FSC Navigator (Process status \ Var status \ Edit PID).

The controller inputs and outputs are shown in graphic bars in the PID menu of the 'Monitor System' option of FSC Navigator.

Several operating modes are supported, including Auto, Manual and Cascade.

E.2 Theoretical approach

To control several analog I/O, the FSC system can use a software controller. The most commonly used controller is the PID (Proportional, Integrational and Differential controller). The PID will control a process variable of the process controlled by the FSC system. This process (output) value is compared to a predefined (setpoint) value. The PID will control the process variable (output) in such a way that the process state (input) value will approximate the setpoint. Figure E-3 shows the model used for the FSC PID function.

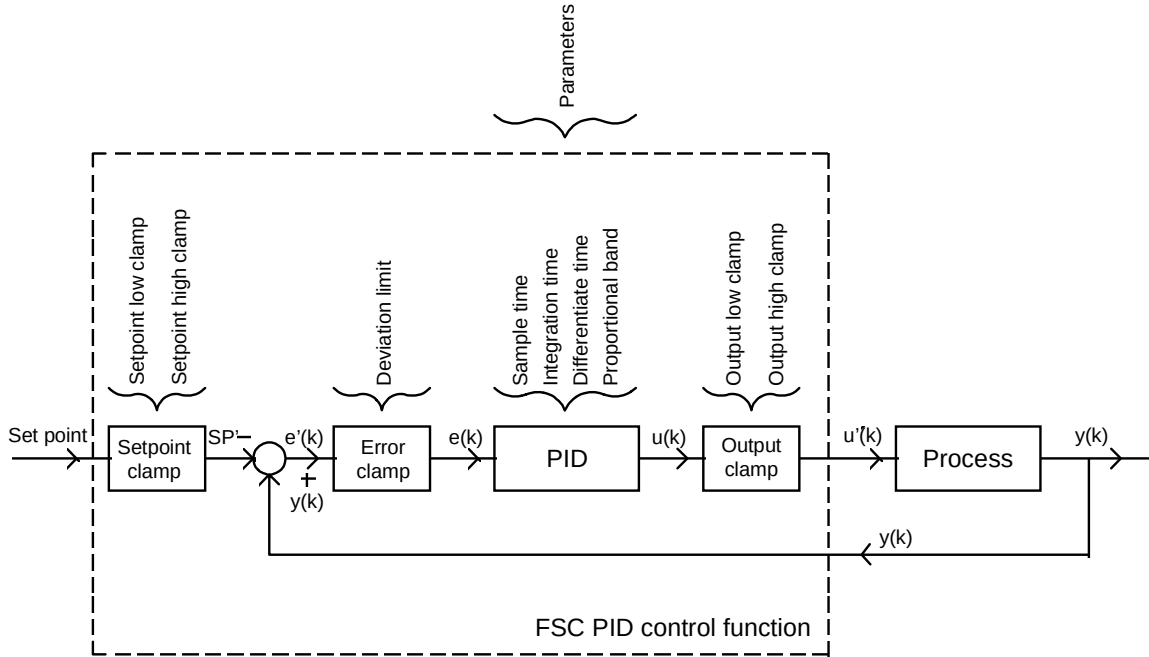


Figure E-3 FSC PID control function

The ideal equation for a PID controller is:

$$u(t) = K * \left[e(t) + 1 / T_i \int e(t) dt + T_d \frac{de(t)}{dt} \right]$$

or

$$u(t) = K * \left(1 + \frac{1}{T_i * S} + T_d * S \right) e(t)$$



where:

$u(t)$ = PID output
 $u'(t)$ = process input (clamped PID output)
 $y(t)$ = process output
 $e'(t)$ = error function = $y(t) - SP'$
 $e(t)$ = PID input (clamped error function)
 SP = setpoint
 SP' = clamped setpoint
 K = gain factor
 T_i = integration time (reset time)
 T_d = differentiate time (rate time) (or derivative time)
 PB = proportional band ($100 / K$)
 S = derivative operator.

$y(k)$ is sampled by the FSC system with a sample time T_0 .

The differential quotient can be converted to a first-order difference and the integral to a sum (using trapezium approximation).

The continue equation can be made discrete by:

$$u(k) = K \left[e(k) + \frac{T_0}{T_i} \sum_{i=1}^k \frac{e(i-1) + e(i)}{2} + \frac{T_d}{T_0} (e(k) - e(k-1)) \right]$$

The following recursive relation can be derived from the above formula:

$$u(k) = u(k-1) + q_0 * e(k) + q_1 * e(k-1) + q_2 * e(k-2)$$

The following constants are used to implement a number of clamp functions (see below):

$$\begin{aligned} q_0 &= K * (1 + 0.5 * T_0 / T_d + T_d / T_0) \\ q_1 &= -K * (1 + 2 * T_d / T_0 - 0.5 * T_0 / T_i) \\ q_2 &= K * (T_d / T_0) \end{aligned}$$

The following clamp functions are implemented:

Setpoint clamp

If setpoint < setpoint low clamp then the setpoint (SP) is set to the value of setpoint low clamp.

If setpoint > setpoint high clamp then the setpoint (SP) is set to the value of setpoint high clamp.

Output clamp

If output < output low clamp then the output ($u'(k)$) is set to the value of output low clamp.

If output > output high clamp then the output ($u'(k)$) is set to the value of output high clamp.

Error clamp

$e(k)$ becomes

$$\{ \text{sign}(e'(k)) \} * \text{MIN} \{ \text{abs}(e'(k)), \text{abs}(\text{deviation limit}) \}$$

In general the FSC PID control function (the function in the dashed lines) can be seen as a block with the process output $y(k)$ and the setpoint as input. The output is the process input $y'(k)$. By changing the parameters, the function within the block can be controlled.



E.3 How to configure, implement and use a PID controller in the FSC system

The project database will contain a variable with type 'P' if a PID controller:

- has been added to the database (using the 'System Configuration' option of FSC Navigator), or
- has been placed on a functional logic diagram (using the 'Design FLDs' option of FSC Navigator).

The configuration of the parameters, mode and scaling can be done using the 'System Configuration' option of FSC Navigator (via 'I/O\Hardware Specs').

There are four PID modes:

- Auto: Normal operation: the output is calculated by the input, setpoint and all the configured parameters.
- Manual: The output (set by the 'Monitor System' option of FSC Navigator or via the OM input in the logic) is copied straight into the logic. To have a bumpless transfer back to Auto mode, the manual output is also copied to the calculations.
- Test: In this mode, the input and the setpoint can be entered in the 'Monitor System' option of FSC Navigator. The input and setpoint connected by logic are overwritten by these values. This option can be used to test the PID function.
- Cascade: If the setpoint of the PID is variable and is directly connected to the output of another PID, this PID is in cascade mode. Cascade mode is used to control outputs that depend on more than one input variable.

The PID can be implemented in the logic by a standard PID symbol. For normal operation (Auto mode), only the setpoint (S) and input (I) must be connected.

If the PID is in manual mode, the output (O) can be set via the 'Monitor System' option of FSC Navigator. If $M=1$, the PID mode will be switched to manual mode and the value OM will be copied to the output O. If $M=0$, the PID mode stays in the current mode. Since the input M and OM are linked together, either both or none must be connected.

In cascade mode, the cascade input (CS) must be connected to the output (O) of another PID. In the configuration of the cascade PID, the controller which is connected to the cascade input (CS) must be configured. This is done in the I/O hardware specs by toggling the cascade input to YES and enter the tag number of the PID which is connected to the CS input.

If the PID is in cascade mode, the connected PID must be in master mode. If more than two PIDs are configured, these can be connected in the chain of controllers. This is done by switching the first PID in Auto mode and all the other PIDs in cascade mode. The PID in Auto mode is called master and all others are slaves.

All connected binary inputs and outputs must be of signal type Float (F).



Example

In the following configuration two PID controllers have been configured:

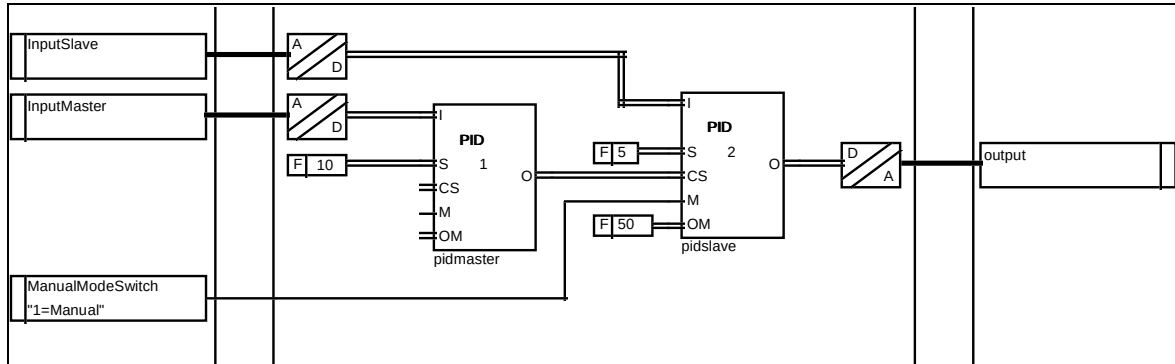


Figure E-4 Implementation example of a cascade PID

The cascade input of the PID 'pidslave' is used, so the cascade input must be configured to YES and the master PID tag number must be configured as 'pidmaster'.

If PID 'pidslave' is in Auto mode, the output is calculated based on the AI 'Input slave' and the setpoint S (5). The digital input 'ManualModeSwitch' (M) must be low.

If 'ManualModeSwitch' is high, the value of OM (50) is copied directly to the value of the output O 'Output'. In none of the above situations, the operation mode of PID 'pidmaster' is important.

If PID 'pidmaster' is in Auto mode, the PID 'pidslave' can be switched over to cascade mode. The master PID 'pidmaster' will calculate its output based on the input 'InputMaster' and the setpoint S (10). The cascade PID 'pidslave' will calculate its output based on the input 'InputSlave' and its cascade setpoint (CS), which is the output of the 'pidmaster'.

A practical example is the temperature control of an 'open' vessel filled with a certain gas as shown in the figure below:

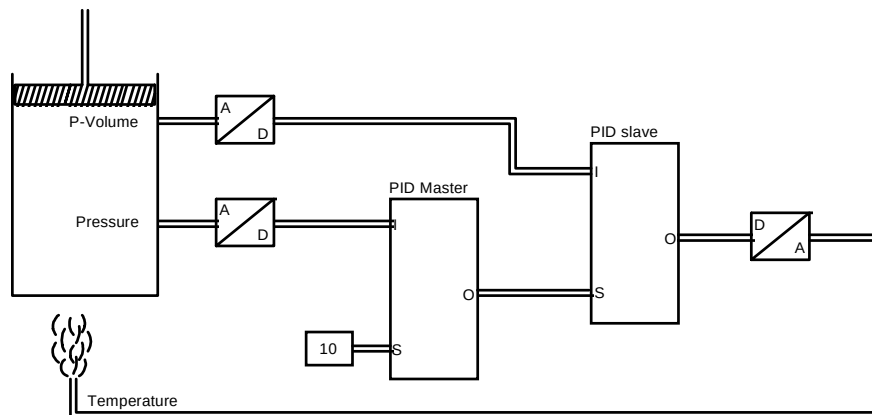


Figure E-5 Temperature control using PID

Volume and pressure are measured. The temperature is controlled by the master/slave PID algorithm.



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Fail Safe Control

Appendix F: Communication Manual

Version 500

Revision 01 (01/98)

Total Plant



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Revision 01 (01/98)

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F.1 Introduction

Nowadays, process control and safeguarding functions in chemical and petrochemical plants are highly automated via computer systems. Apart from the processing power offered by these computer systems, they are also capable of gathering valuable information on the actual status of the process part they are monitoring. In order to make optimum use of this information and to be able to provide adequate information to plant operators, both the process control and the safeguarding systems must have communication capabilities to exchange process information.

The FSC system offers the following communication capabilities:

1. Communication with Distributed Control Systems or visualization programs running on PC, to make optimum use of the available process information.
2. Communication to printer devices, to produce status reports and sequence of event records.
3. Communication with FSC Navigator or visualization programs running on PC, to monitor the process and FSC system status.
4. Communication between FSC systems, to decentralize process safeguarding in combination with central monitoring and control capabilities.

For optimum availability of the communication, the redundant FSC system configurations allow the use of redundant communication links as well.

The following subsections describe each of the communication capabilities. First, several important definitions are explained.

Note:

The Modbus and the Siemens RKE3964R protocols can accept write commands to the FSC system. FSC Navigator can send force commands to the FSC system. The number of write / force commands to the FSC system per application program cycle is limited in redundant Central Part configurations. An approximation for the number of write / force commands per application is: $(\text{application program cycle time in ms} - 160 \text{ ms}) / 10$. This is the total number of write / force commands of all communication protocols to one FSC system.

F.2 Glossary

Logical link

The logical link expresses the capability of two systems to exchange information via the communication network. A logical link may include several physical links (e.g. via Communication servers, see Figure F-1).

Physical link

The physical link defines the interconnection of systems within the communication network. Multiple logical links may be realized across a physical link (see Figure F-1).

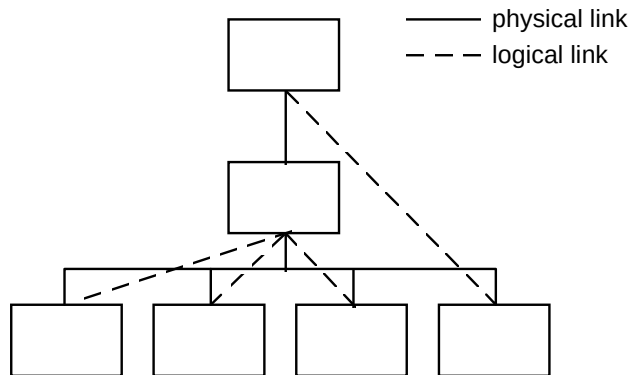


Figure F-1 Physical and logical links

Point to point

A point-to-point link is a physical link which interconnects two systems, with one system connected to each end of the link (see Figure F-2).

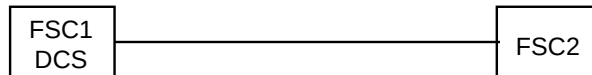


Figure F-2 Point-to-point link

Multidrop link

A multidrop link is a physical link which interconnects multiple systems (see Figure F-3). Within the context of the FSC communication, a multidrop link is the connection of a single master system with multiple slave systems.

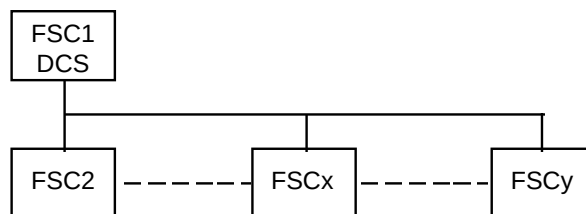


Figure F-3 Multidrop link



Redundant link

A redundant link is a logical link which is connected to both Central Parts of a redundant FSC system (see Figure F-4 to Figure F-6). For FSC-FSC communication, a redundant link is realized via two independent physical links, each connected to a single Central Part. For communication with non-FSC systems, a redundant link is realized via a single physical link which is connected to both Central Parts.

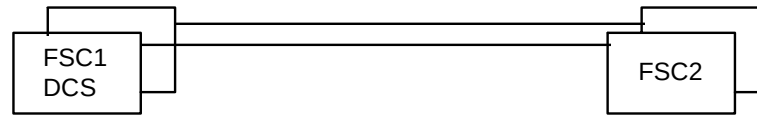


Figure F-4 Redundant point-to-point link

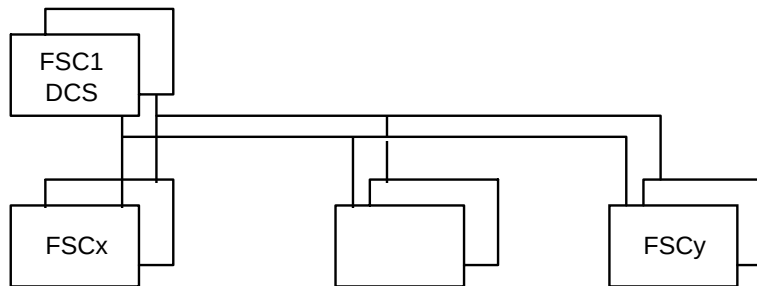


Figure F-5 Redundant multidrop link



Figure F-6 Redundant device link

Communication master

A communication master is a system which is able to initiate an exchange of data with other systems without being requested by another system.

Communication slave

A communication slave is a system which only exchanges data with another system only when requested by the other system.

Communication server

A communication server is a system which is not an end user of the exchanged data, but serves as a routing device between master and slave systems.

F.3 Communication with Distributed Control Systems

The FSC system is able to communicate with Distributed Control Systems (DCS) via three different communication protocols:

1. TPS network protocol
2. Modbus RTU
3. RKE3964R

This section describes the capabilities of the FSC system with respect to the communication via these protocols.

Note:

There can be a maximum of one write command to the same address per application cycle.

F.3.1 Communication with the TPS system via UCN

The FSC Safety Manager is directly connected to the Universal Control Network (UCN) of the TotalPlant Solution (TPS) system as shown in Figure F-7. Its interface to the UCN enables the FSC-SM to readily share data with its peers on the UCN. For more detailed information on communication between the FSC-SM and the TPS system please refer to the FSC-SM documentation.

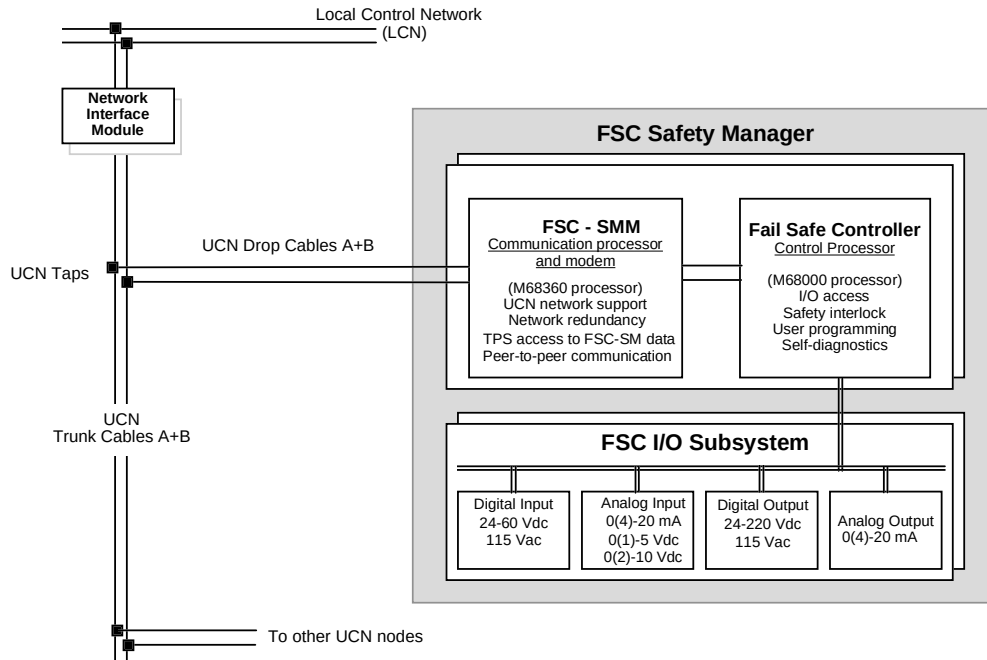


Figure F-7 FSC Safety Manager architecture



F.3.2 Communication via the Modbus RTU protocol

The Modbus protocol as implemented in the FSC system conforms to the description in the Gould Modbus protocol reference guide; PI-MBUS-300 Rev B, 1985.

The FSC system supports the Modbus RTU protocol. A special "dialect" of the protocol, for communication with Hartmann & Braun Contronic E and Contronic P system, is available as an option. The protocol details are described in chapter F.3.2.5.

Within the Modbus protocol, information exchange is accomplished in frames. Each frame contains the following information: system address, command type (function code), function parameters, checksum. The length of a frame depends on the function code and the function parameters. The checksum is used to verify the correctness of the transferred information.

During the configuration of the application, the communication of the FSC system to the DCS is also configured, by assigning the Modbus protocol to a communication channel, and defining the protocol parameters.

F.3.2.1 Supported networks

The Modbus communication protocol is supported on both point-to-point and multidrop networks. The communication link may be redundant (see Figure F-2 to Figure F-6).

A point-to-point redundant link can be configured by assigning a primary point-to-point Modbus link to Central Part 1 and a secondary point-to-point Modbus link to Central Part 2. The timeout must have a value that is larger than the maximum read cycle on both links.

F.3.2.2 Functional operation

During the communication process, the FSC system performs a slave function, which means that data exchange is initiated by the DCS.

Please note that Modbus communication is only running if the FSC system is running. This means that there will be no Modbus communication if the FSC system has shut down or is waiting for a reset.

If a redundant device link is used and both Central Parts are healthy, then communication is established via both Central Parts alternately. Every 10 seconds a changeover to the other Central Part is made. If one Central Part fails, communication is realized via the healthy Central Part.

The data exchange between FSC and DCS system is realized via configured marker and register areas (see Figure F-8). The link of the communication to the FSC system application is made via application variables (I, O, BI, BO with location COM). These are allocated to the FSC communication link with the DCS.

The FSC system is able to handle a maximum of 250 data bytes per read/write transfer, i.e. a maximum number of 2000 coils or 125 holding registers.

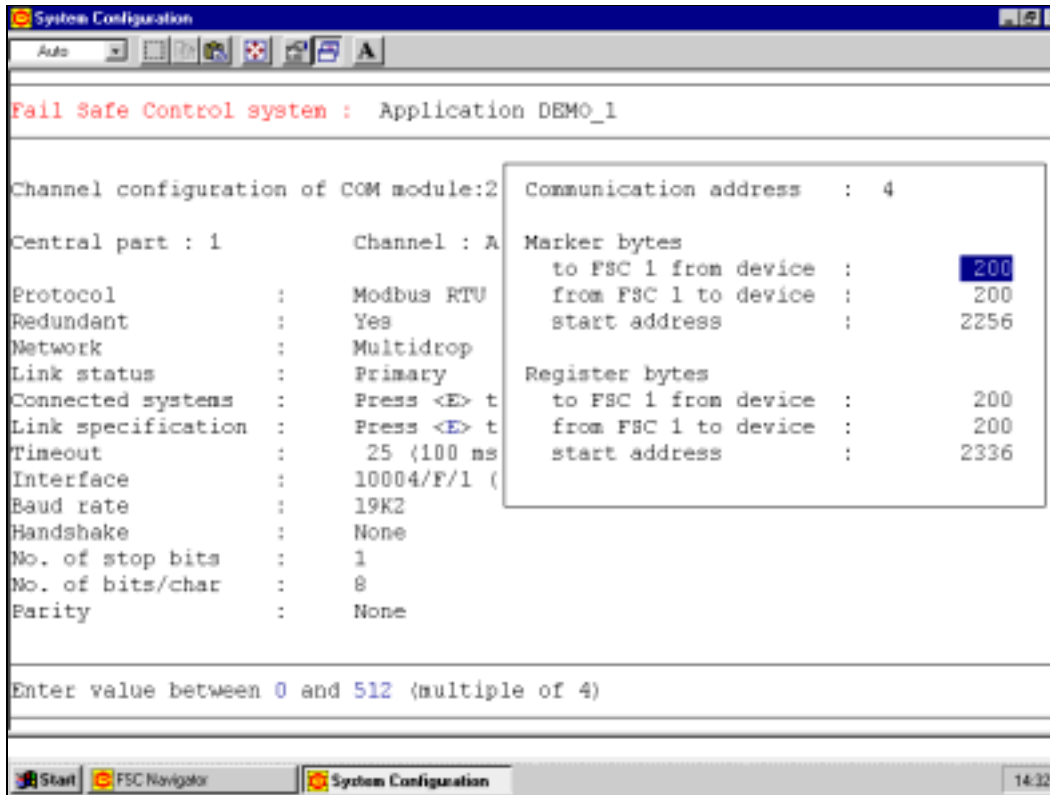


Figure F-8 Modbus data area configuration

F.3.2.2.1 Supported function and error codes

Table F-1 and Table F-2 show the Modbus function and error codes supported by the FSC system.

Table F-1 Supported function codes

Code	Modbus description
1	read coil status
2	read input status
3	read holding register
4	read input register
5	force coil
6	load register
8	loop back test
15	force multiple coils
16	load multiple registers

Notes: Function code 2 is handled the same as function code 1.
Function code 4 is handled the same as function code 3.
If function code 15 is used to force more than 32 coils at the same time,
the first coil and the number of coils must both be an integer multiple of 8.



Table F-2 Supported error codes

Code	Modbus description	Cause
1	Illegal function code	An unsupported function code is applied.
2	Illegal data address	The applied variable address exceeds the configured communication area boundaries.
3	Illegal data value	Incorrect data format (e.g. invalid time stamp).
4	Device failure	FSC system shut down.
6	Busy	Data cannot be accepted at this time.

For a detailed description of the supported function codes, incl. function parameters and frame formats, see the Gould Modbus Protocol reference guide: PI-MBUS-300, Rev B, 1985.

Real-time clock synchronization

The real-time clock of the FSC system can be set by the DCS system via the Modbus register write function codes 6, 16 with specific register addressing. This feature can be used to synchronize the time of the FSC and the DCS system.

If function code 16 is used, the following format is required:

Function code:	16	
Register address:	49996	
Number of registers:	4	
Register contents:	49996 – Year	: Decimal value, e.g. 1997
	49997 – Month / Date	: Decimal value, e.g. 0812 means August 12.
	49998 – Hour / Minute	: Decimal value, e.g. 2233 means 22.33 h.
	49999 – Sec / Not used	: Decimal value, e.g. 5900 means 59 sec.

If function code 6 is used, the four registers can be written separately. Upon reception of the first three registers (Year, Month/Date, Hour/minute) the FSC system only stores the received data. Upon reception of the last register (Sec), the FSC system checks the entire time stamp. It will accept the data if it represents a valid time.

Due to variable delays in the data transfer of the DCS to the FSC system (scan delay + duration of the transfer), an additional hardwired clock synchronization input must be provided in the FSC system. At assertion of the clock synchronization input (0 → 1 edge), a previously transferred time stamp is copied to the FSC real-time clock.

Via the clock synchronization input, easy adjustment of the FSC clock to daylight saving time is possible. If the input is asserted while no time stamp was transmitted by the DCS, the FSC time will be set at 3.00 am.

Read FSC event buffer

Via the function code 3, the FSC event buffer contents can be read. At each request, the FSC system sends a response message containing 16 events. If the FSC event buffer holds less than 16 events, the remaining entries in the response message will be filled with an SER buffer empty event.

Each event contains 8 bytes as specified in the table below.

Table F-3 Event structure

Seq. no. High	Seq. no. Low	
Value	Milliseconds	
Tenths of seconds	Seconds	
Minutes	SZ	Hours

Seq. no. H, Seq. no. L. : [0..FFFE_H]

Value : [0, 0F_H, F0_H, FF_H]

SZ¹ (Hours.b₇) : 0 = winter time
1 = summer time

All other fields in hexadecimal representation, in 24-hour time scale.

¹ Summer/winter time only available if the 10006/2/2 diagnostic and battery module module is used.

Note:

For correct operation of the Modbus protocol, the number of registers in the event request must be 64.

When reading FSC event data, the DCS shall use the register addresses 49994 and 49995 alternately. The FSC system uses the mechanism for confirmation of the last event response message that was sent. While receiving alternate register addresses, the FSC system sends the next events in sequence. When the same address is received, the FSC system re-sends the last event message. The event data read option is only valid when the FSC Numerical SER format is configured.

F.3.2.2.2 FSC addressing

System addressing

Within the communication, the address of the target FSC system that the DCS wants to communicate with must be specified. The communication address of the FSC system is the system number * 4.

Variable addressing

The representation of the addresses of variables used for FSC-DCS communication differs between the FSC and DCS system. The following algorithms can be used to calculate the Modbus addresses from the FSC addresses. These are generated automatically when configuring the communication (see also Figure F-8).

Table F-4 Calculation of Modbus addresses

Type	Modbus address
I	FSC address + 1
O	FSC address + 5001
BI	(FSC address / 2) + (4)0001
AI	(FSC address / 2) + (4)3001
BO	(FSC address / 2) + (4)5001



Notes:

- No addressing information is contained in function code 8.
 - The Modbus addresses for the variable types I, O, BI and BO can be output to hardcopy using the 'Print / Project Configuration' option of FSC Navigator.
 - The FSC addresses for variable type AI can be output to hardcopy using the 'Print / Project Configuration' option of FSC Navigator (column 'Byte').
 - For some DCS systems, the offset of 40.000 for addresses of BI and BO variables cannot be configured. Refer to the documentation of your DCS vendor for further details.
-

The FSC system allows access of variable type I and O in multiples of 16 via the Modbus register function codes 3, 4, 6 and 16. Table F-5 shows an overview of the supported Modbus function codes, the applicable address ranges and the FSC variable types.

Table F-5 Address ranges per function code

Function code	Address range	Remarks
1, 2, 5, 15	1 - 4096	FSC type I
1, 2	5001 - 9096	FSC type O
3, 4, 6, 16	(4)0001 - (4)2048	FSC type BI
	(4)8001 - (4)8256	FSC type I
3, 4	(4)3001 - (4)3256	FSC type AI
	(4)5001 - (4)7048	FSC type BO
	(4)9001 - (4)9256	FSC type O
3	(4)9994 / (4)9995	Event data
6, 16	(4)9996 - (4)9999	RTC data

Note:

The address ranges in release $\geq 2.9x$ differ from the ranges in older versions. However, release $\geq 2.9x$ still supports these ranges. Therefore no adaptation of the DCS address configuration is required when upgrading the FSC system to release $\geq 2.9x$. It is recommended to use the above mentioned address ranges for new applications and to modify old applications when appropriate.

F.3.2.2.3 Response times

Communication response time

The response time of the FSC system is defined as the time period between reception of the last character of the DCS message to the transmission of the first character of the response message. The response time has a typical value of 10 ms (maximum value 25 ms).

Process control response time

The process control response time is defined as the time between two specific events:

- The first event is the alteration of a control variable in the DCS.
- The second event is the detection by the DCS of the FSC system response upon the first event.

The maximum process control response time is calculated by adding the following parameters:

- 1) internal delay in the DCS system to detect the control operation
- 2) the communication scan time of the DCS system
- 3) the transmission time of the Modbus write command frame
- 4) the application program time of the FSC system
- 5) the communication scan time of the DCS system
- 6) the communication response time of the FSC system
- 7) the transmission time of the Modbus read command
- 8) the transmission time of the Modbus response frame
- 9) the internal delay in the DCS system to process the received information.

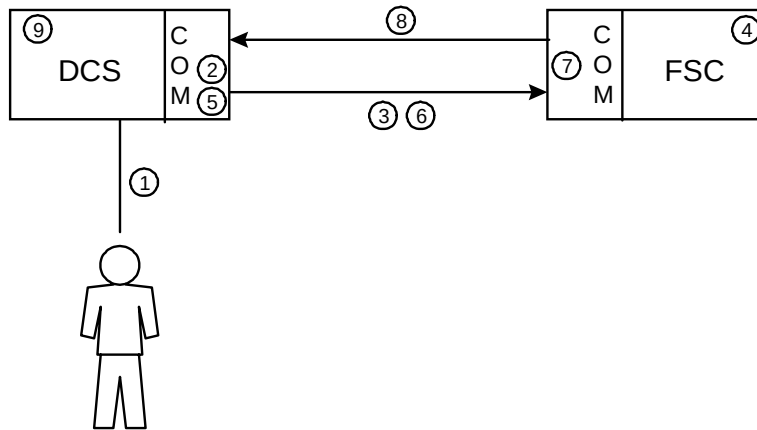


Figure F-9 Response time parameters



F.3.2.3 Fault handling

Communication channels that are configured for the Modbus protocol are expected to be communicating continuously. The correct operation of these communication channels is monitored by the FSC via timeouts, which can be defined by the user when configuring of the communication channel.

A channel is regarded faulty if no valid Modbus frame was received within the configured timeout period. A fault is then reported via the extended diagnostics and the DEVICE-COM.FLT alarm marker.

The system automatically recovers from this fault as soon as communication resumes.

F.3.2.4 Configuration

Table F-6 Modbus protocol

Modbus protocol ¹	Point to point		Multidrop	
	----- Interfaces types	----- Baud rates ²	----- Interface types	----- Baud rates ²
Interface ³			RS485	
			----- No encoding	4800 - 19K2
	RS422			
	----- No encoding	4800 - 19K2		
	RS232	1200 - 38K4	RS232	1200 - 38K4
	Current loop	1200 - 19K2		

Notes:

- 1:
 - Configuration is applicable both to link status Primary and Secondary.
 - The link can either be redundant or not redundant. However, a redundant link is implicitly considered to be a multidrop link.
 - Handshake type (none, hardware), number of stop bits (1, 2) bits per character (7, 8) and parity type (odd, even, none) are configurable.
- 2:
 - Only 1 channel on a COM module can be set at a baud rate of 2M.
 - If a COM module is used with /F/G or /F/I interface, the maximum baud rate of channel A is 19K2 instead of 38K4.
- 3: Maximum allowable cable length:

– RS 422 / 485	– RS232 PtoP / RS232C MD	– Current loop
2 km	38K4 baud: 10 m	19K2 baud: 5 m
	Other: 15 m	9600 baud: 10 m
		4800 baud: 25 m
		2400 baud: 50 m
		1200 baud: 100 m

F.3.2.5 Modbus for communication with Contronic E/P

The FSC modbus protocol driver (MB_HB.BIN) for communication with Hartmann & Braun systems Contronic E and Contronic P conforms to the description of the Gould Modbus protocol reference guide; PI-MBUS-300.REV12, 1985.

The functionality of the driver is identical to the functionality of the general FSC Modbus RTU driver, with the following exceptions:

- 1) Real-time clock synchronization (chapter F.3.1.2) is not supported.
- 2) Event specification (chapter F.3.1.2.1)
 - two specific event sequence numbers are defined, independent of the configured numerical SER range
 - EEEE_H = Event buffer empty
 - FFFF_H = Event buffer overflow
 - The event value range is [0, 1]
- 3) Address ranges (chapter F.3.1.2.2)

Table F-7 Address ranges per function code

Function code	Address range	Remarks
1, 2, 5, 15	0 - 3999	FSC type I
1, 2	5000 - 8999	FSC type O
3, 4, 6, 16	1000 - 1499	FSC type BI
	2500 - 2749	FSC type I
3, 4	1500 - 1999	FSC type BO
	2000 - 2255	FSC type AI
3	3072 / 3076	event data



F.3.3 Communication via the RKE3964R protocol

The RKE3964R protocol, as implemented in the FSC system, conforms to the Siemens RKE3964/RKE3964R protocol specification. The protocol can be used for data exchange between an FSC system and, for example, the TELEPERM M AS 230.

The RKE3964R protocol is block-oriented. All data to be exchanged is allocated within a pre-defined communication block. Throughout data exchange, either complete blocks or partial blocks can be transferred.

Data exchange is accomplished in messages. A message consists of a command (read/write), data type, block number, data offset address within the block, a checksum, and, if the message is a data write command, data. The checksum is used to verify the correctness of received information. A message contains a maximum of 128 data bytes.

The FSC system supports master and slave operating modes. In slave mode the FSC system will only send data upon a specific request from the connected DCS. In master mode the FSC system may be configured to initiate data transfer itself, either periodically or at occurrence of a pre-defined event.

F.3.3.1 Supported networks

The RKE3964R protocol can be used for point-to-point links, redundant device links and redundant point-to-point links (see Figure F-2, Figure F-4 and Figure F-6).

A point-to-point redundant link can be configured by assigning a primary point-to-point RKE3964R link to Central Part 1 and a secondary point-to-point RKE3964R link to Central Part 2.

F.3.3.2 Functional operation

Data exchange between the FSC system and DCS is realized via the configured marker and register areas (Figure F-10). The link of the communication to the FSC system application is made through application variables (I, O, BI, BO) with location COM (see Figure F-10).

Slave mode

If slave mode is configured for the RKE3964R communication protocol, then initiation of data transfer is always done by the DCS.

If a redundant device link is used and both Central Parts are healthy, then communication is established via both Central Parts alternately. Every 10 seconds a change over to the other Central Part is made. If one Central Part fails, communication is realized via the healthy Central Part.

Master mode

If master mode is configured for the RKE3964R communication protocol, then initiation of data transfer can be done by the FSC system as well as by the DCS. The FSC system may be configured for each block to transmit data under different kinds of conditions. Possible conditions are:

- upon a specific request by the DCS only,
- each application program cycle,
- timed (each $N * 0.5$ seconds),
- at occurrence of a specific output activation, or
- combined timed and output activation.

If a redundant device link is used, communication will only be accomplished via one of the connected Central Parts. Normally this is the link via Central Part 1. The communication link connected to the redundant Central Part serves as a hot stand-by. It takes over communication if the first link fails.

Note:

If both redundant links fail, the FSC system will continuously try to re-establish Central Part communication via both links. Such a situation could occur when the connected system is taken off line. At restart of the system, incorrect data could be received by the DCS due to collision of data sent by both Central Parts within the FSC system. This situation normally lasts only for a short period of time.

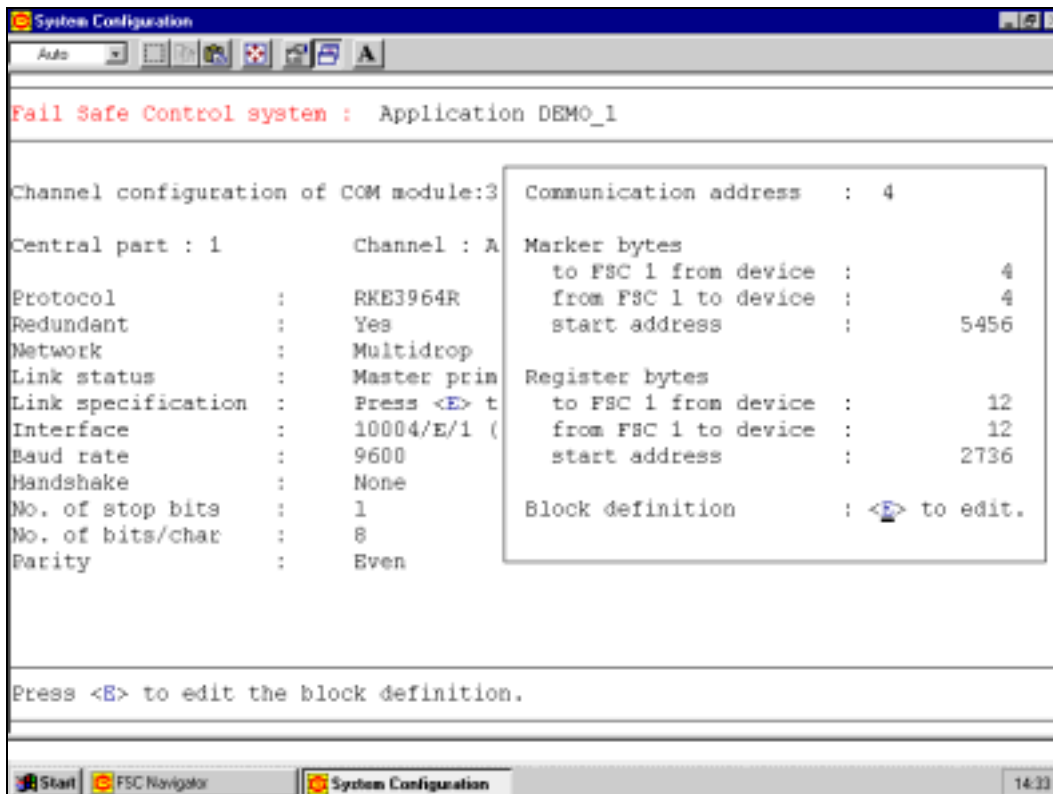


Figure F-10 RKE3964R block definition in slave mode



F.3.3.2.1 Supported commands and error messages

Table F-8 and Table F-9 show the supported RKE3964R commands and error codes.

Table F-8 Supported RKE3964R commands

Command	FKE3964R description	FSC equivalent
AD	send data unit(s)	write marker or register area (I, BI)
ED	read data unit(s)	read from marker or register area (I, O, BI, BO)
ES	test message	loopback test

Table F-9 Supported RKE3964R error codes

Code	RKE3964R description	Cause
0F	communication is blocked	Central Part shutdown

Real-time clock synchronization

The real-time clock of the FSC system can be set by the DCS via a dedicated block write command. This feature can be used to synchronize the time of the FSC system and DCS. The frame format of the command is as follows:

Command:	"A","D"	: write command
Block number:	63	
Start address:	00	
Data word count:	00 04	
Data format:	FF FF	
Data:	YY YY	: Four digit BCD
	MM DD	: Four digit BCD
	HH mm SS	: hours, minutes and seconds in BCD (24 hour clock is used)
	xx	: not used

For example, August 12, 1997 22:33:59 would become:

19 97 (year)
08 12 (month and date)
22 33 59 (hours, minutes, seconds)

Due to variable delays in data transfer from DCS to FSC, an additional hardwired clock synchronization input must be provided in the FSC system. At assertion of this clock synchronization input (0 → 1 edge), a previously transferred time stamp is copied to the FSC real-time clock.

Via the clock synchronization input easy adjustment of the FSC clock to daylight saving time is possible. If the clock synchronization input was asserted while no time stamp was present within the FSC system, then the FSC real-time clock will be set to 03.00 am.

F.3.3.2.2 Block definition

RKE3964R protocol block definitions are made when defining data exchange areas using the 'System Configuration' option of FSC Navigator. See Figure F-11 for block definition fields in master mode and Figure F-12 for block definition fields in slave mode.

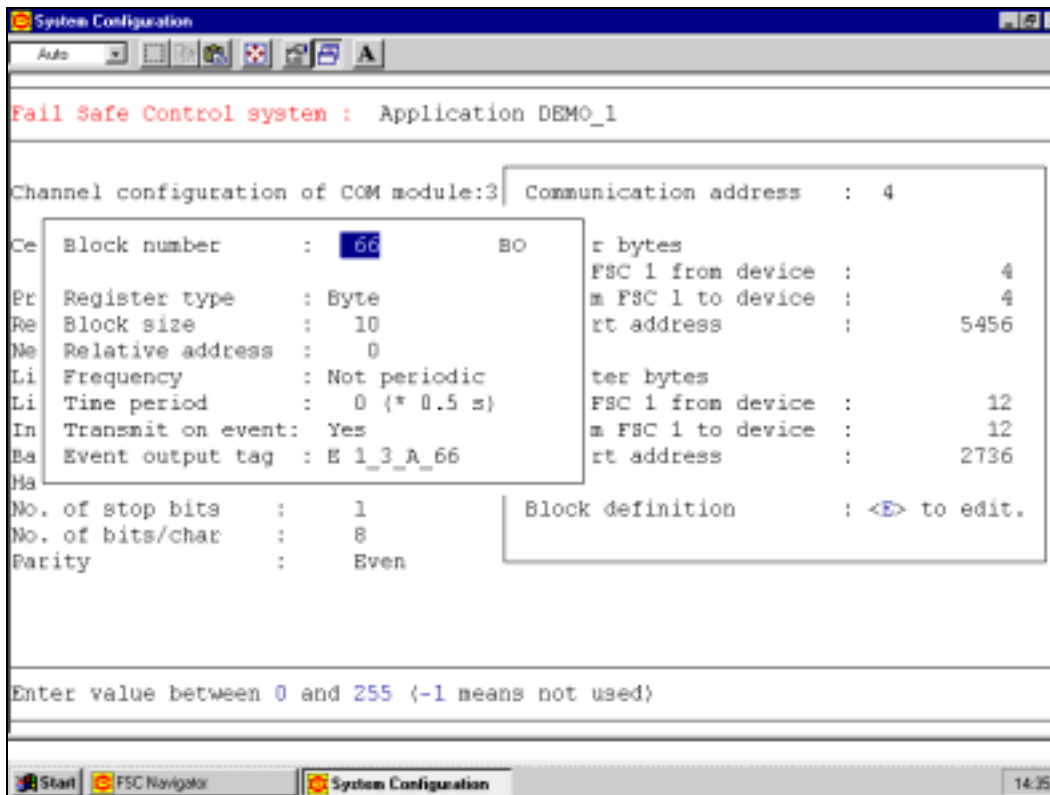


Figure F-11 RKE 3964R block definition in master mode

Fields that can be defined as well in slave mode as in master mode are:

Block number

You can enter a value between 0 and 255. Block numbers are divided into four groups, each with its specific type of data:

0 - 62	: binary inputs
64 - 127	: binary outputs
128 - 191	: digital inputs
192 - 255	: digital outputs

Block number 63 is reserved for the real-time clock set command.

Block size

This field defines the number of data bytes in the block. The maximum number of bytes in a block depends on the number of marker or register bytes configured for input or output, and on the size of already defined blocks in the same variable type.



Register type

This field is available for block numbers 0 - 127, i.e. for input or output registers. It determines the register type that is used within the block. Only registers of the defined type can be allocated to this block. Possible types are: Byte, Word, or Float.

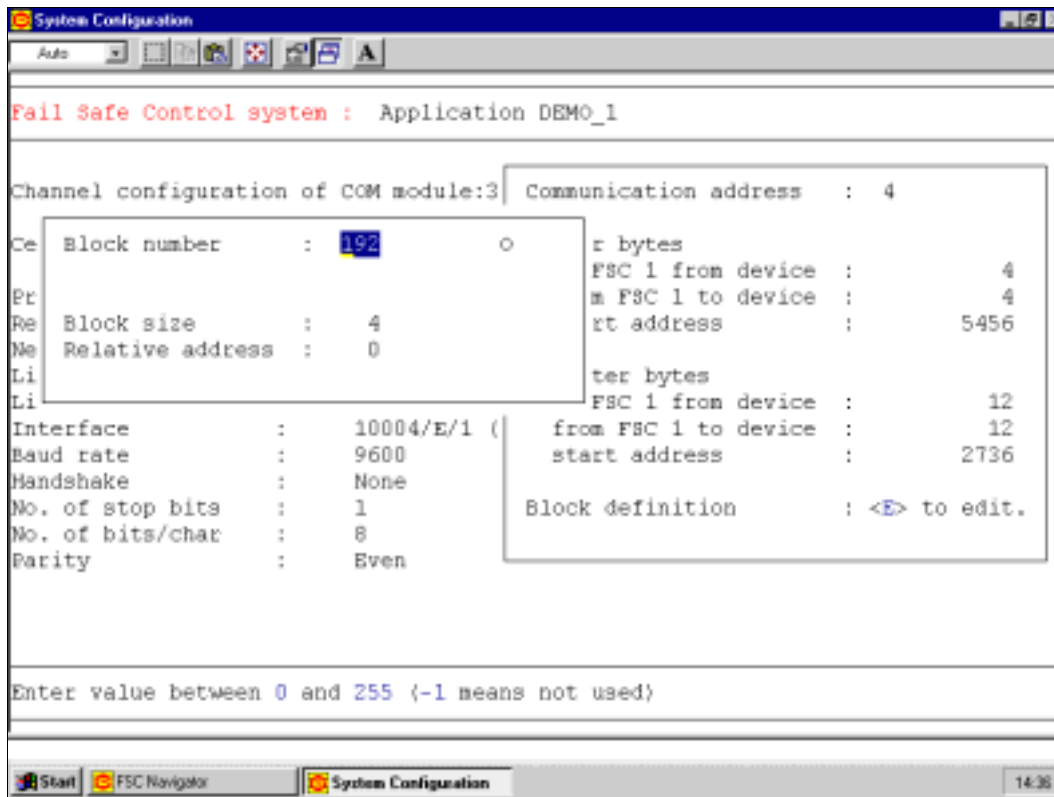


Figure F-12 Block definition in slave mode

The following fields can only be accessed when in master mode.

Frequency

This field is only available in master mode. With this field a selection can be made of the condition that must be met for sending data by the FSC system:

- Undefined : No periodic master function
- Not periodic : No periodic master function
Send data on event possible
- Time periodic : Send data time periodic
- Each program cycle : Send data each program cycle

Note:

All blocks can be configured to be transmitted by the FSC system without specifically being requested by the DCS. This applies also for input data blocks. The reason for this is that data transmitted by the DCS via RKE3964R cannot be viewed local by the DCS itself. Therefore also the FSC input areas can be read by the DCS to allow a kind of echoing of received data.

**Time period**

This field is only available in master mode and when selecting "Time periodic" at the "Frequency" field. This field allows you to define the time between data exchanges in steps of 0.5 seconds. Possible values are: 0 - 120, which implies a maximum time period of 1 minute.

Transmit on event

This field is only available in master mode and when selecting "Not periodic" or "Time periodic" at the "Frequency" field. This field allows you to add a digital output (location SYS) that is related to start of data exchange. Whenever this output goes active, data exchange is started for the related block number. For example see Figure F-11: at BO block number 66 field "Frequency" is set to "Not periodic", while field "Transmit on event" is set to "Yes". This results in an automatic generation of a digital output with location SYS and its tag number composed of Central Part number, communication module number, communication channel and block number: "E 1_2_A_66". If this output becomes active, block number 66 of Central Part 1, communication module 2, channel A will be sent to the connected DCS.

F.3.3.2.3 Response times

Communication response time

The response time of the FSC system is defined as the time period between reception of the last character of the DCS message and the transmission of the first character of the response message. The response time has a typical value of 10 ms (maximum value 25 ms).

Process control response time

The process control response time is defined as the time between two specific events:

- The first event is the alteration of a control variable in the DCS.
- The second event is the detection by the DCS of the FSC system response upon the first event.



The maximum process control response time is calculated by adding the following parameters (see Figure F-13):

- 1) internal delay in the DCS to detect the control operation
- 2) the communication scan time of the DCS
- 3) the transmission time of the RKE3964R write command frame
- 4) the application program time of the FSC system
- 5) the communication scan time of the DCS
- 6) the communication response time of the FSC system
- 7) the transmission time of the RKE3964R read command frame
- 8) the transmission time of the RKE3964R response frame
- 9) the internal delay in the DCS to process the received information

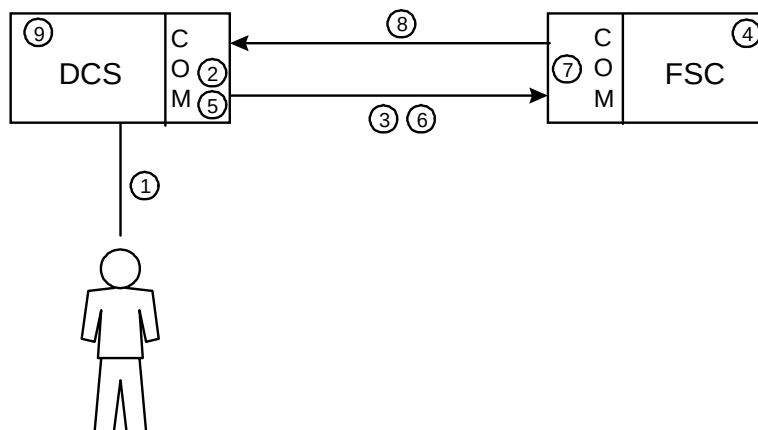


Figure F-13 Response time parameters

Note:

The Siemens RKE3964R protocol in master mode has a capacity to transmit messages on event. The number of events that can initiate a communication message is limited when the RKE3964R protocol is used on a communication module that also supports FSC-FSC communication (including the communication between the Central Parts). When the RKE3964R protocol is used in combination with FSC-FSC communication, the maximum number of events is 15 per program cycle.

F.3.3.3 Fault handling

Communication channels that are configured for the RKE3964R protocol are expected to be communicating continuously. The correct operation of these communication channels is monitored by the FSC system via a fixed timeout of 3 seconds.

A channel is regarded faulty if no valid RKE3964R frame was received within the timeout period. A fault is then reported via the extended diagnostics and the DEVICE-COM.FLT alarm marker.

The system automatically recovers from this fault as soon as communication resumes.



F.3.3.4 Configuration

Table F-10 RKE3964R protocol

RKE3964R protocol ¹	Point to point		Redundant device to link	
	----- Interface types	----- Baud rates	----- Interface types	----- Baud rates
Interface ² :	RS232 MD	≤ 19K2	RD232 MD	≤ 19K2
	Current loop	≤ 19K2	Current loop	≤ 19K2
	RS232C PtoP	≤ 19K2		

Notes:

- 1:
 - Configuration is applicable to link types.
 - The link either be a redundant device link or a point-to-point link.
- 2: Maximum allowed cable length:
 - RS232C PtoP / RS232 MD, all baud rates: 15 m
 - Current loop

19K2 baud:	5m
9600 baud:	10 m
4800 baud:	25 m
2400 baud:	50 m
1200 baud:	100 m



F.4 Hardcopy via the Printer protocol

The Printer protocol within the FSC system is used for creating hardcopy of certain application output. There are two different types of applications that create output for the Printer protocol:

- Report generation, and
- SER generation.

The format for these output types can be defined using the 'SER format' option of FSC Navigator. Reports and SER generation are briefly discussed subsection F.4.2.

F.4.1 Supported networks

The Printer protocol can only be used for point-to-point links and redundant device links.

F.4.2 Functional operation

The communication between FSC system and printer is done simplex, i.e. the FSC system sends ASCII characters and the printer prints them on paper.

There is, however, a possibility at configuration, to allocate binary output variables (location COM) to the printer channel. This is done to enable generation of BO SER messages for the printer (see Figure F-14).

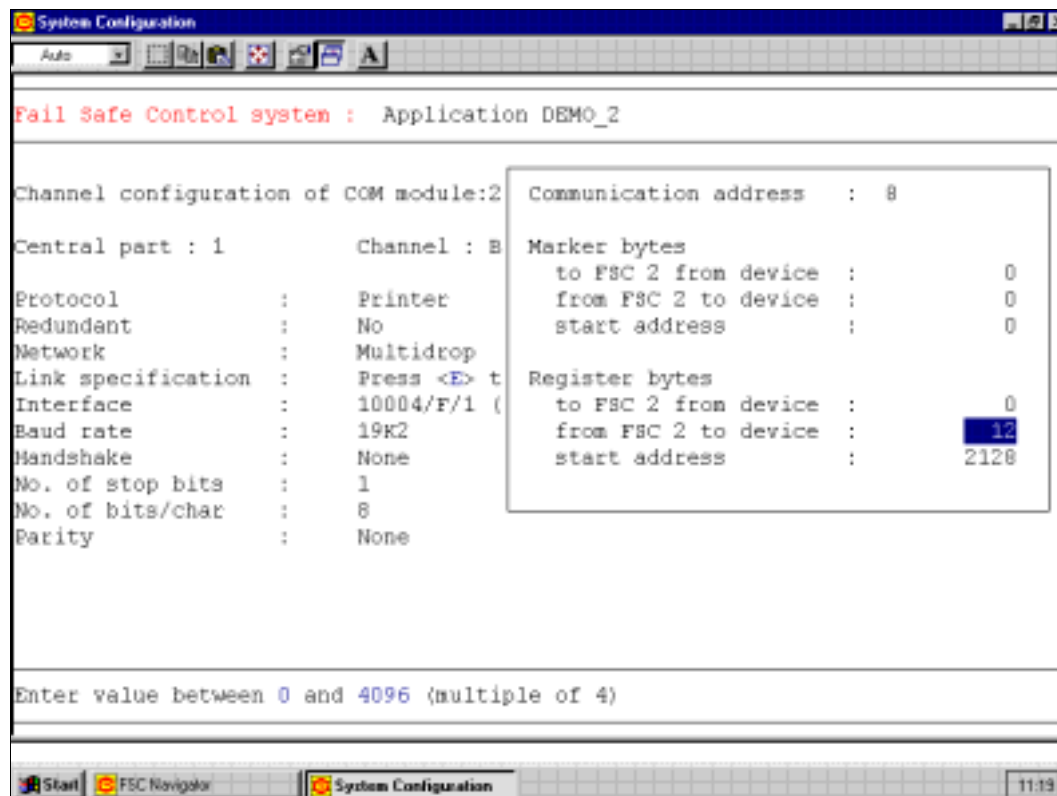


Figure F-14 Printer data area configuration



Two different types of applications use the printer as an output device:

- SER printing, and
- reports printing.

The next two subsections briefly discuss both applications.

F.4.2.1 SER printing

In an operational plant the need may exist to evaluate occurrences of abnormal situations. The FSC system provides sequence-of-event recording (SER) features. The SER data provided by the FSC system can be output to a printer, or it can be used by FSCSOE, a separate Windows application which allows you to log, view and interpret process events. For details on how to configure SER refer to Section 4 of the FSC Software Manual ("System Configuration").

Sequence-of-event recording (SER) reports changes in the state of user-defined application variables, input and output forcings, and FSC system markers. These events are output to an SER logging device. For details about SER events and message generation refer to appendix B ("Sequence of Event Recording").

If the SER logging device is a printer, then the 'SER formats' option of FSC Navigator enables you to define an output format for each variable type. This means that you can indicate yourself which data are of interest to you when printing SER and which data are not. For details about editing SER message formats refer to Section 5 of the FSC Software Manual ("Editing Reports and SER format").

There is one special feature for generating SER messages that are sent to a printer device: numerical SER format. This means that an intelligent device is used to process short hexadecimal SER records instead of complete ASCII format messages. For more information refer to appendix B ("Sequence of Events Recording").

F.4.2.2 Reports printing

Not only abnormal situations may be of interest at a running plant. For example, the wish may exist to print out a daily report containing important information about a number of process variables for evaluation of process efficiency.

The 'SER format' option of FSC Navigator allows you to define up to 64 report formats. Based on such a format definition, a report will be generated and sent to a printer device when at least one of up to three conditions for that report becomes true. Conditions that can be defined are:

- SER event : a specific SER event occurs,
- Counter value : a specific counter value is reached,
- Register value : a specific register value is reached.

For details on how to configure and edit report formats refer to Section 5 of the FSC Software Manual ("Editing Reports and SER Format").



F.4.3 Fault handling

The on-line status of communication channels that are configured for the Printer protocol are monitored by the FSC system via a fixed link timeout of 1 minute and by the configured handshake protocol (see Table F-11).

If no handshake was configured (option "None" at the field "Handshake"), then the FSC system has no means of detecting that a printer is off-line or that it is being disconnected. In this case the link timeout is disabled, and this implies that SER messages and Reports can be lost when the printer goes off-line.

If XON/XOFF handshaking was configured (option "XON/XOFF" at the field "Handshake"), then the FSC system expects the printer to repetitiously send a XON character when it is on-line and to send a XOFF character when it is off-line. In this case the time needed for the FSC system to detect that the printer is off-line, is the time between two succeeding XON/XOFF characters sent by the printer. Further, if the printer would be completely disconnected from the FSC system, then no characters at all are received by the FSC system, and it takes as long as 1 minute (the link timeout) for the FSC system to detect the link to the printer as being faulty. As a result of this still SER messages may be lost. For example there may be some time between a printer going off-line and it sending a XOFF character to the FSC system. All SER messages, that are sent from the FSC system to the printer during this time, will be lost.

If hardware handshaking was configured (option "Hardware" at the field "Handshake"), then the chance is negligible that SER is lost. Because, in this last case, when the printer goes off-line, it is immediately detected by the FSC system. Also disconnecting the printer from the FSC system is immediately detected. This allows the FSC system to buffer SER messages until the printer goes on-line again or until the link is detected faulty, in which case the FSC system will try to re-route SER message via the redundant link.

It is therefore strongly recommended that, if you do not want to lose SER messages, you choose the hardware handshaking option when configuring the printer protocol using the 'System Configuration' option of FSC Navigator.

F.4.4 Configuration

Table F-11 Printer protocol

Printer protocol ¹	Point to point		Multidrop	
	----- Interface types	----- Baud rates	----- Interface types	----- Baud rates
Interface ² :	RS232 MD	≤ 19K2	RS232 MD	≤ 19K2
	Current loop	≤ 19K2		
	RS232C PtoP	≤ 19K2		

Notes:

- 1:
 - * Configuration is applicable to link types.
 - * The link can either be a redundant device link or a point-to-point link.
- 2:
 - Maximum allowed cable length:
 - o RS232C PtoP / RS232 MD, all baud rates : 15 m
 - o Current loop
 - 19K2 baud: 5 m
 - 9600 baud: 10 m
 - 4800 baud: 25 m
 - 2400 baud: 50 m
 - 1200 baud: 100 m

F.5 Communication with the FSC user station

There may be communication between the FSC user station (with FSC Navigator and/or FSCSOE running on it) and the FSC system for a number of functions:

- monitor process behavior,
- monitor the FSC system status,
- rebuild the FSC databases on-line,
- verify the application as present in the FSC system, and
- collect events logged by the FSC sequence-of-event (SER) function.

The controlled process and FSC system can be monitored using the 'On-Line Environment' function of FSC Navigator. This function also includes the capability of changing application variables on-line, e.g. setpoints.

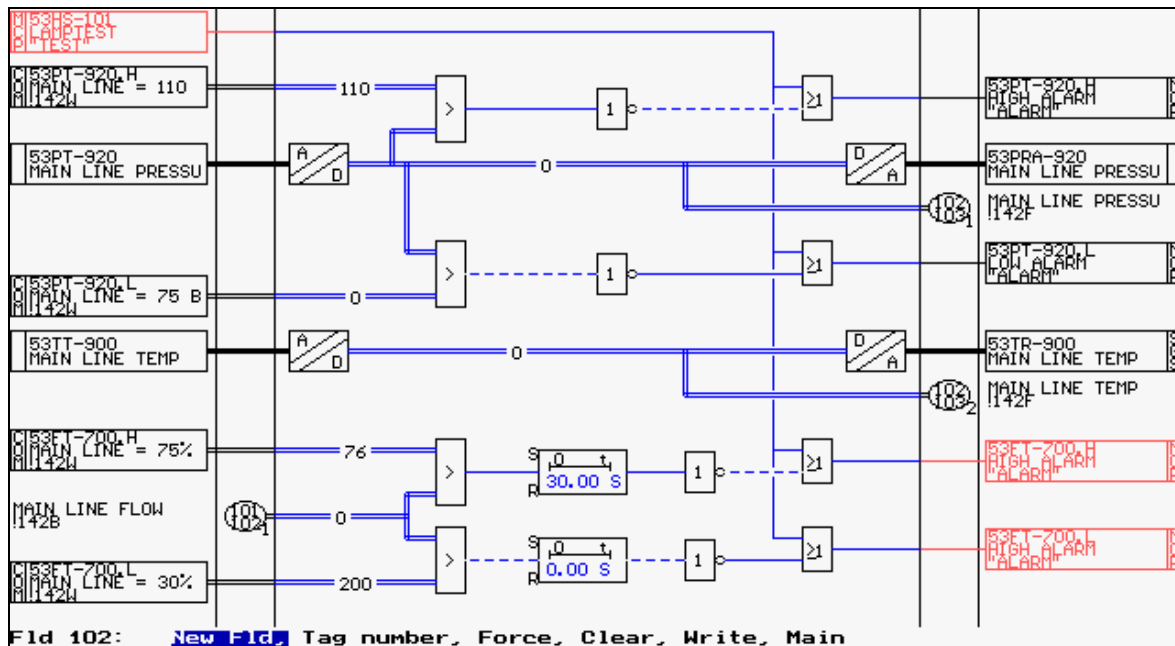


Figure F-15 Example FLD status

The on-line rebuild of the FSC system database is part of the 'System Configuration' option of FSC Navigator. The function is used in combination with the on-line modification function of the FSC system (see appendix D). The on-line rebuild extracts information from the FSC system and rebuilds the databases so they can be used for further project development allowing new modifications to be implemented on-line.

The 'Verify Application' option of FSC Navigator extracts information from the FSC system and validates its correctness against the FSC databases. Events can be collected via the sequence-of-event recording feature. Logged events are stored in a database which can be inspected at any time.



The protocol used is the Development System communication protocol of Honeywell Safety Management Systems. For more information on the above-mentioned options of FSC Navigator, refer to the applicable parts of the FSC Software Manual.

F.5.1 Supported networks

The communication between the FSC system and the FSC user station supports both point-to-point and multidrop networks connected to one or all Central Parts (redundant) of an FSC system.

F.5.2 Functional operation

In the communication with the FSC user station, the FSC systems perform a slave function. Data is only sent at request from the FSC user station.

The target system(s) for the FSC user station to communicate with is/are determined either by the currently selected FSC Navigator option (e.g. 'Monitor System' option) or by configuration within the program option itself.

The communication may be specific to a Central Part, e.g. to read diagnostic information, or not Central Part specific (process status, event data read). In the latter case, when a redundant communication link is being used, the communication will be established via both links alternately. A change over to the other link is made every 10 seconds. If communication via one of the links fails, all communication is realized via the healthy link.

F.5.3 Fault handling

If a communication channel is configured for communication with the FSC user station and it is configured as the SER reference, then the FSC system monitors the operation of the communication channel. If no valid communication has been accomplished during a period of 1 minute, the channel is reported faulty via the extended diagnostics and the DEVICE-COM.FLT alarm marker.

The system automatically recovers from this fault as soon as communication is re-established.

F.5.4 Configuration

Communication with the FSC user station is fully configured via the communication channel configuration screen of the 'System Configuration' option of FSC Navigator. Possible configuration combinations are presented in Table F-12.

Table F-12 Development system protocol

DS protocol ¹	Point to point ----- Interface types	Baud rates ²	Multidrop ----- Interface types	Baud rates ²
Interface ³ :	RS485 ----- No encoding Manchester FM0 FM1	300 - 125K 1M, 2M 1M, 2M 1M, 2M	RS485 ----- No encoding Manchester FM0 FM1	300 - 125K 1M, 2M 1M, 2M 1M, 2M
	RS232C MD	≤ 38K4 ⁴	RS232C MD	≤ 38K4 ⁴
	RS232C PtoP	≤ 38K4 ⁴		

Notes:

- 1:
 - * Configuration is applicable to link status Primary and Secondary.
 - * The link can either be redundant or not redundant.
- 2:
 - * Only 1 channel on a COM module can be set at a baud rate of 2M.
 - * If a COM module is used with /F/G or /F/I interface the maximum supported baud rate of channel A is 19K2 instead of 38K4.
- 3: Maximum allowed cable length:

– RS 422 / 485 - 2M baud: 60 m - 1M baud: 120 m - 125K baud: 1 km - 50K baud: 2 km	– RS232C - 38K4 baud: 10 m - Other: 15 m
--	--
- 4: On PCs with clock frequency < 12 MHz. and RS-232 via the 07177/1/1 module the maximum baud rate is 19K2.

Note:

If the FSC user station is used for collection of event data, the channel must be assigned as the SER reference channel (using the 'Install Configuration' screen of the 'System Configuration' option of FSC Navigator).



F.6 Communication between FSC systems

Within a plant several production units may exist which operate independent of one another, but often have common utilities as e.g. steam and air. In order to minimize the consequences for the entire plant in case of emergencies in one production unit, separate safeguarding systems can be used to surveil the independent production units.

A disadvantage of this decentralization strategy is that process information is only locally available whereas it is often required in a central control room.

Communication between FSC systems provides the means to decentralize process safeguarding in combination with central process monitoring and control capabilities.

The protocol used for communication between FSC systems is the FSC communication protocol of Safety Management Systems. The protocol includes a high level of error detection and recovery which makes it suitable for exchange of safety related information while maintaining optimal availability.

The FSC communication network is configured, during the configuration of the application, by defining the interconnections between the FSC systems and the parameters of the FSC protocol. The exchange of information is realized via input (I, BI) and output (O, BO) variables which are allocated to the connected FSC systems. Both safety related and not safety related information may be exchanged.

The inputs and outputs are included in the functional logic diagrams to realize the required application task of the systems.

F.6.1 Supported networks

The FSC-FSC communication supports both point-to-point and multidrop networks. If a master system has a redundant configuration, the communication links to all slaves are redundant as well. A maximum number of 10 slaves may be connected to one master via one multidrop link. The actual number of slaves depends on the baud rate and amount of exchanged data.

F.6.2 Functional description

The communication between FSC systems is based on a master slave concept. Within this concept, the master system is responsible for all communication activities.

When both redundant links are healthy, communication is established alternately via both links. If one link fails, communication is established via the healthy link.

Apart from master and slave systems, the FSC system also supports communication server activities. Communication servers are FSC systems interconnected between the communicating master and slave system. Their task is to route the data which is exchanged between master and slave. Figure F-16 shows an example of a network containing all three system types.

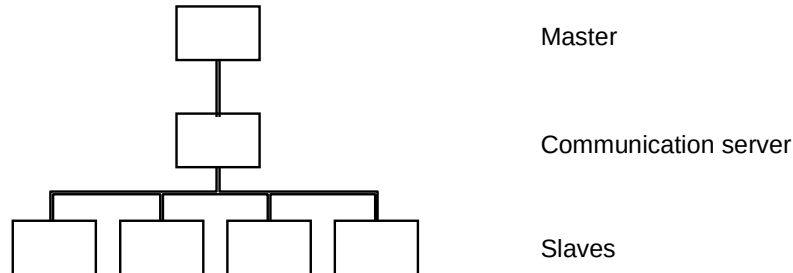


Figure F-16 FSC communication network

Note:

Only one level of communication server systems is allowed.

F.6.2.1 Data exchange

The master periodically sends data to the slave systems and initiates a request for data from the slaves. For point-to-point links, the period is one application program cycle of the master system. For multidrop links, a new data transmission and request for a slave are initiated after reception of slave data in reply to the previous request.

Data is exchanged between the master and the slave through pre-defined marker and register areas. These areas contain variables (I, O, BI and BO with location FSC) which are allocated for communication with the other system. The configuration of the data areas and contained variables is accomplished via the 'System Configuration' option of FSC Navigator.

Note:

The configuration of the master and slave systems must be consistent, i.e. the size of the marker/register output area in one system must be equal to the size of the marker/register input area in another system and the tag numbers of the output variables (O, BO) in one system must be equal to the tag numbers of the corresponding input variables (I, BI) in the other system.

F.6.2.2 Real-time clock synchronization

The real-time clock of all FSC systems interconnected in a communication network are synchronized. This ensures proper sequencing of events across the systems, e.g. in case of a plant upset. The synchronization accuracy between FSC systems is < 100 ms.



F.6.2.3 Low baud rate communication

Low baud rate communication is used in those applications where high baud rates are not possible. This may have a number of reasons:

- The type of cable that is used for communication is not suited for high baud rate communication.
- The communication takes place over large distances.
- There is intermediate equipment that does not support high baud rate communication.
- Multiple communication channels are sent via one communication line (using multiplexing), which means that the baud rate of the channels is smaller than that of the communication line.

To accommodate for large communication distances, the FSC-FSC communication timeout (i.e. the maximum communication time to exchange all data between the FSC master and all its slaves) is user-configurable. It can be set between 1.2 and 30 seconds (in steps of 0.1 second) using the 'System Configuration' option of FSC Navigator (Install \ Configuration \ FSC-FSC communication configuration \ FSC-FSC communication timeout).

The response time must be flexible due to the delay time of signals in the communication medium (cable, radiographic, etc.). The exact delay time depends on the distance to be traveled, and can be calculated by the user taking into consideration the time delay for the distance (300,000 km/sec.) and delays in the intermediate equipment such as repeaters, modems, etc.

The 'Translate Application' option of FSC Navigator (the compiler) will calculate the maximum delay time based on the system and communication configuration in the log file. If the communication distance is short and there is no intermediate equipment, the compiler will provide a warning if the user defined a communication timeout which results in a response time of less than 15 ms. If the calculated response time is lower than 10 ms, an error message will be generated. It is the user's responsibility to check the actual required response time.



F.6.3 Fault handling

The FSC system monitors the operation of its physical communication links by means of a timeout. The timeout depends on the system function and the type of communication link (see Table F-13).

Table F-13 FSC-FSC communication timeout

Link type	System	Timeout
Point-to-point	Master	Response of slave system is expected within the same application program cycle
	Slave	1 second
Multidrop	Master	User-configurable between 1.2 and 30 seconds (in steps of 0.1 second). The default value is 1.2 seconds. See also Section 4 ("System Configuration") of the FSC Software Manual.
	Slave	2x configured communication timeout. See also Section 4 ("System Configuration") of the FSC Software Manual.

Failure of a physical link is reported via the extended diagnostics and the EXT.COMMUNIC.FLT alarm marker. The extended diagnostics report includes the numbers of all systems for which a logical link was defined across the physical link.

Within redundant configurations, the communication is preserved as long as one of the logical links is healthy.

When communication via both links fails, i.e. communication to a connected system fails, the value of safety related input variables (I, BI) allocated to that system are forced to the safe, '0' state. The value of the non safety-related input variables allocated to the system are frozen at the last received value.

F.6.4 Configuration

This section describes the sequence of configuration items which are of interest for FSC-FSC communication, in general terms. For details refer to Section 4 of the FSC Software Manual ("System Configuration").

F.6.4.1 System identification

The first step in the definition of the FSC communication network is the identification of the interconnected systems by their application name.

Application names are coupled to system numbers via the 'FSC-FSC communication configuration' input screen.

The information can be used to simplify successive configuration steps. It is also used by the 'Translate Application' option of FSC Navigator (the compiler) to verify the correctness of the overall network configuration.



F.6.4.2 Communication channel configuration

The next step is the configuration of the communication channels. This includes:

- protocol definition
- network definition
- link status
- connected systems
- definition of the size of the marker and register areas

Table F-14 presents the valid combinations for the FSC protocol.

Table F-14 Configuration combinations for the FSC protocol

FSC protocol ¹	Point to point		Multidrop	
	Interface types	Baud rates ²	Interface types	Baud rates ²
Interface ³ :	Fiber optics			
	Manchester	1M, 2M		
	FM0	1M, 2M		
	FM1	1M, 2M		
			RS485	
			No encoding	50K, 125K
			Manchester	1M, 2M
			FM0	1M, 2M
			FM1	1M, 2M
	RS422			
	Manchester	1M, 2M		
	FM0	1M, 2M		
	FM1	1M, 2M		

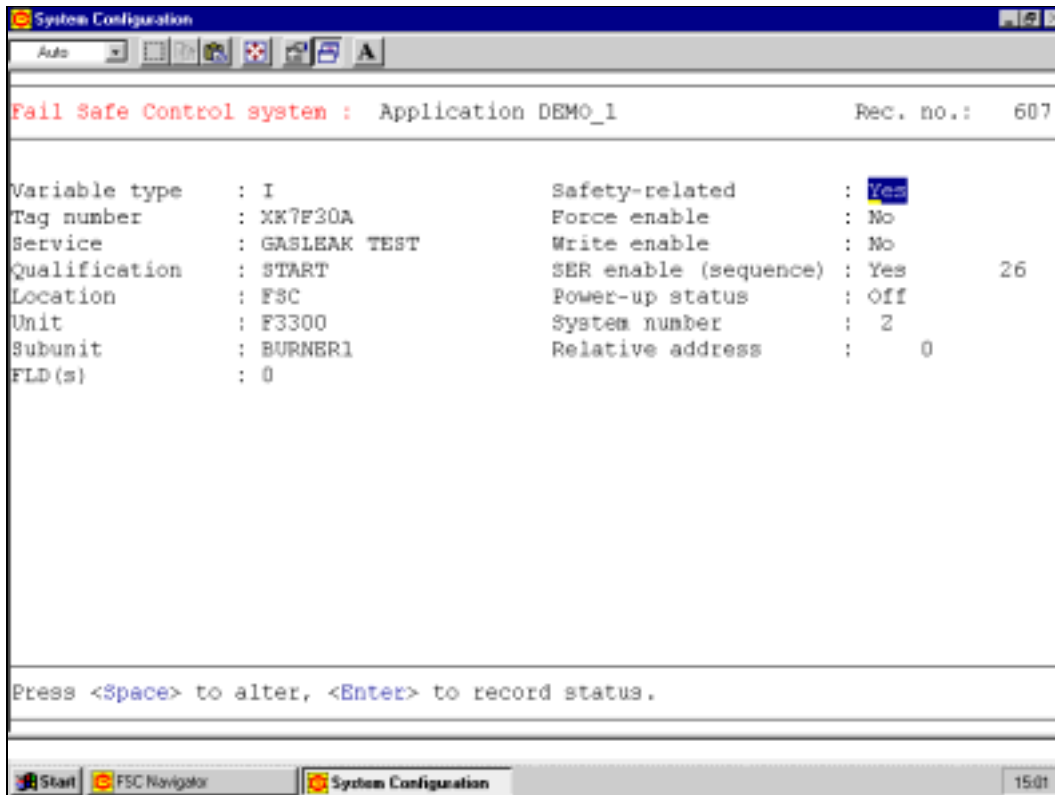
Notes:

- 1: Configuration is applicable to link status Master, Slave and Communication server.
- 2: Only 1 channel on a COM module can be set at a baud rate of 2M.
- 3: Maximum allowed cable length:
 - Fiber optics: 4.5 km
 - RS 422 / 485
 - 2M baud: 60 m
 - 1M baud: 120 m
 - 125K baud: 1 km
 - 50K baud: 2 km

F.6.4.3 Variable allocation

The variables that are intended for exchange of information between FSC systems (I, BI, O, BO with location FSC) are defined using the 'System Configuration' option of FSC Navigator (via 'Signal Specs') or during creation of the functional logic diagrams (FLDs).

The 'Hardware Specs' option of System Configuration allows the variables to be allocated to a connected FSC system.



The screenshot shows the 'System Configuration' window with the following details:

- Fail Safe Control system :** Application DEMO_1
- Rec. no.:** 607
- Variable type :** I
- Tag number :** XK7F30A
- Service :** GASLEAK TEST
- Qualification :** START
- Location :** FSC
- Unit :** F3300
- Subunit :** BURNER1
- FLD(s) :** 0
- Safety-related :** Yes
- Force enable :** No
- Write enable :** No
- SER enable (sequence) :** Yes
- Power-up status :** Off
- System number :** 2
- Relative address :** 0

Additional information: 26

Press <Space> to alter, <Enter> to record status.

The taskbar at the bottom shows 'Start', 'FSC Navigator', and 'System Configuration'.

Figure F-17 Hardware specification



F.6.4.4 Configuration strategy

As mentioned before, the configuration of communicating FSC systems must be consistent with respect to the size of the marker and register areas and the tag numbers of the allocated variables.

To guarantee correct configuration, the 'System Configuration' option of FSC Navigator offers a 'Copy Allocations' feature, which extracts this information from the database of the connected systems. The option is contained in the 'Link specification' item of the communication channel configuration screen (see Figure F-18).

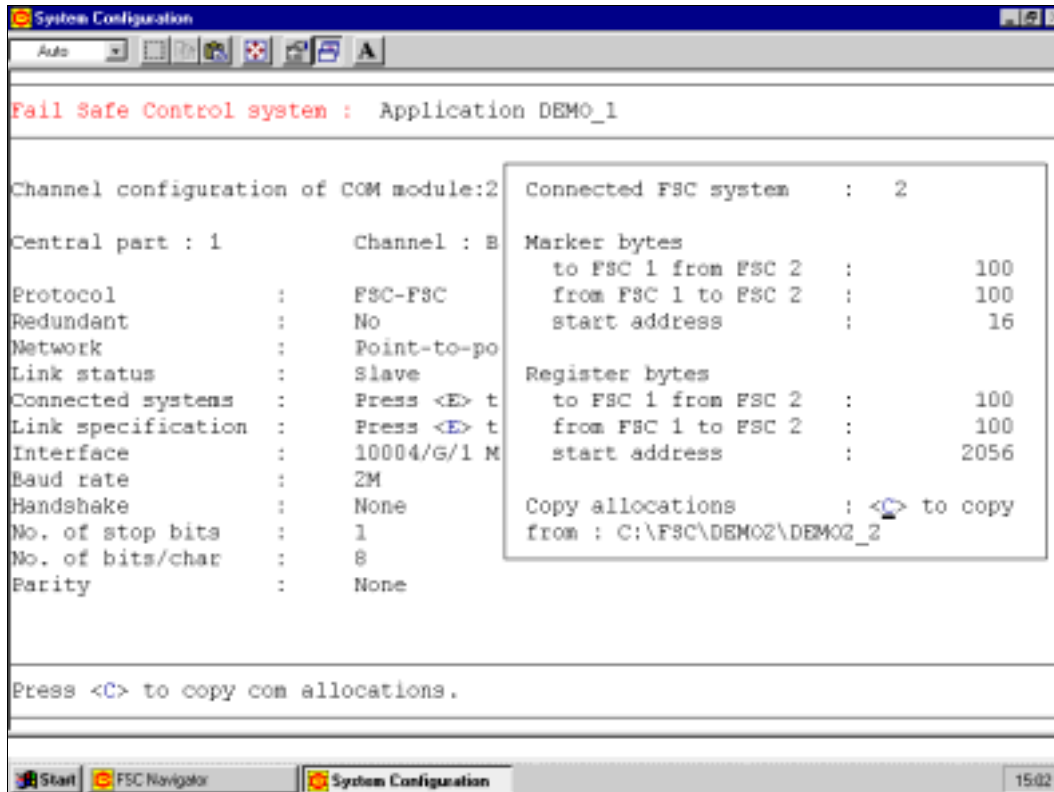


Figure F-18 Copy allocations

Using this option, valuable time during project engineering can be saved. The configuration strategy is to completely define the data exchange in one system, either master or slave, and then copy the information to the connected system.

F.7 Real-time clock set

The FSC network accepts several external sources to synchronize its real-time clock: the German atomic clock (DCF-77), a connected DCS system and the FSC user station (with FSC Navigator running on it). To ensure optimum time synchronization, a priority algorithm is used. This means that clock synchronization data from sources of lower priority will be ignored as long as data from sources of higher priority is valid. Thus, synchronization data from only one source is used at any time. The priorities are defined as follows:

- 1) German atomic clock (DCF-77)
- 2) DCS system connected via UCN
- 3) DCS system connected via P-Bus
- 4) DCS system connected via Modbus or RKE3964, with or without a hardware strobe input (CLOCK-SYNC)
- 5) FSC user station (with FSC Navigator and/or FSCSOE).

As soon as the device from the current priority level is no longer available, the FSC network will use the time synchronization data from the device with the next-highest priority.

The FSC network can use the radio signal of the German atomic clock (DCF-77) as the primary source for time synchronization, providing it is located within the receiving range of the DCF-77 transmitter near Frankfurt, Germany (see Figure F-19). The (gray) inner circle in Figure F-19 shows the region of Europe where reception of the DCF radio signal is usually good. The outer circle in Figure F-19 shows the region where reception of the DCF signal is usually possible.



Figure F-19 Receiving range of the DCF-77 radio transmitter near Frankfurt

As long as the DCF-77 signal is valid (which is determined by special hardware), the FSC network will use this signal every 30 seconds to set its real-time clock. If, for whatever reason, the DCF-77 signal is no longer found to be valid (e.g. due to interference), a three-minute timeout will be initiated. This will prevent transient errors having a greater impact than they should. If the time signal recovers before the timeout expires, the FSC network will continue to use the DCF-77 signal as the source for time synchronization. Otherwise it will switch to the source with next-highest priority: a DCS system or – if none has been connected to the FSC network or the connected DCS system is not used to issue time-set commands – to the FSC user station. A DCS system connected via UCN has a higher priority than a DCS system connected via Modbus or RKE3964. Thus, if both are present, a UCN-based DCS system will take precedence.

A connected DCS system can be configured to send time-set commands automatically in fixed (configurable) intervals, or the user can send them manually. Every time a time-set command from the DCS system is executed, a 30-hour timeout is initiated. The DCS system is regarded faulty for time synchronization if no new time-set command is received within these 30 hours. During this 30-hour time



period, the FSC network will remain synchronized with the DCS system, and will ignore any time synchronization data from other sources of lower priority: a 'lower-level' DCS system or the FSC user station. As soon as the timeout expires without a new time-set command having been received, the FSC network will accept time-set commands from a lower-level source. Only if no DCS system (connected via UCN or Modbus/RKE3964) is available any more will the system accept time-set commands from the FSC user station (e.g. the FSC sequence-of-event recording option). Time-set commands from FSC Navigator can be issued by the user via the 'Monitor System' option, or they are issued automatically by the FSC sequence-of-event recorder.

A DCS system connected via Modbus or RKE3964 may be equipped with a hardware strobe input (CLOCK-SYNC). If that is the case, any time-set command sent by the DCS system is executed only after the CLOCK-SYNC input has been asserted. If no CLOCK-SYNC input is present, the time-set command is executed immediately.

In some configurations, the hardware strobe input may also be asserted without a time-set command being sent by the DCS system (e.g. if it can be actuated manually by turning a switch). In that case, the FSC real-time clock will be set to 03:00:00 (3 a.m.).

As soon as a higher-level source of time synchronization becomes available again, the FSC network will immediately and automatically switch back to the source with the highest priority that is available.

Figure F-20 illustrates the time synchronization process in the FSC network:

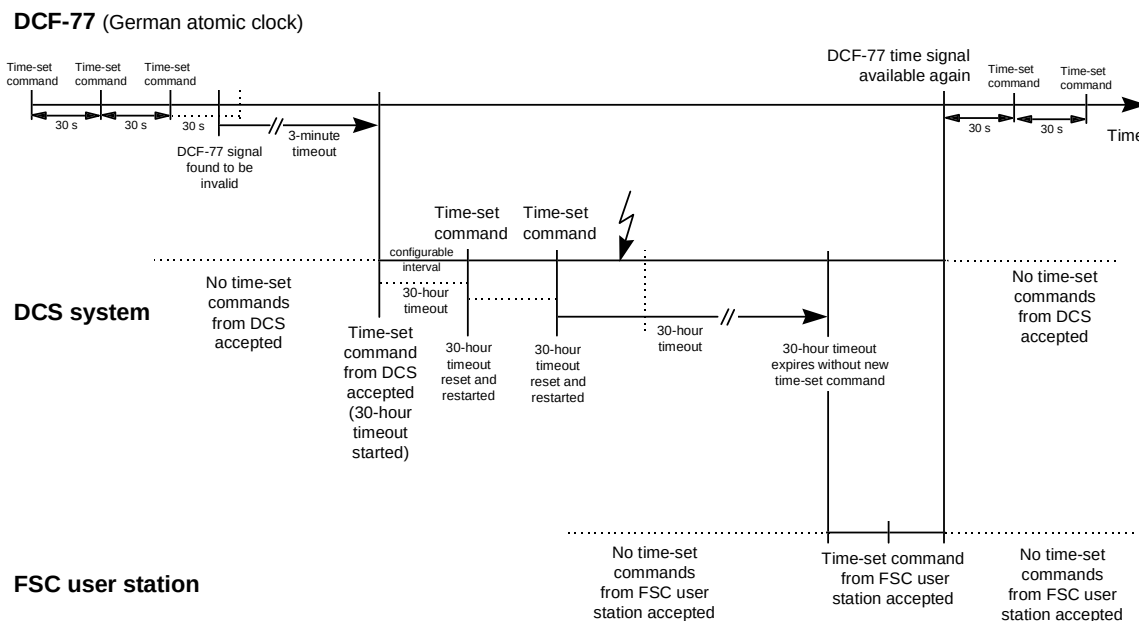


Figure F-20 Real-time clock set process in FSC network



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Appendix G: FSC Multiplexers

Version 500

Revision 01 (01/98)



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G.1 Introduction

Multiplexers are supported by the FSC software to reduce the number of input and output channels to and from the FSC system. The FSC system does not use hardware multiplexers; the principle of a multiplexer is simulated by the FSC software.

Each multiplexer (de)multiplexes the register data of a number of multiple input/output variables. A maximum of 16 multiplexers can be configured.

Terminology for input multiplexers:

Scan time:	The time in which all multiple inputs of a multiplexer are read.
Sample:	The data of a multiple input is read in parts. One part is a sample.
Debounced data:	The data of the multiplexer is read if the data on the data bus is stable. The data on the data bus is stable if two consecutive samples on the input data bus of the multiplexer return the same data result.
Selection outputs:	The status of the selection outputs determines which sample of the data will be read. Only one of the selection outputs can be active at a time.
Selection output active time:	The time which the selection output should be active before the data of the multiplexer bus is valid.
Data hold time:	The time in which the data stabilizes on the multiplexer data bus.
Selection output active status:	Selection of the sample can be either active high or active low. On = active high, Off = active low.
Multiplexer data bus status:	The status of the multiplexer data specifies if the data is either represented normally or inverted ('normal' or 'invert'). The data is inverted after the data is read from the multiplexer data bus if the status is 'invert'.

Terminology for output multiplexers:

Scan time:	The time in which all multiple outputs of a multiplexer are written.
Sample:	The data of a multiple output is written in parts. One part is a sample.
Selection outputs:	The status of the selection outputs determines which sample of the data will be written. Only one of the selection outputs can be active at a time.
Selection output active time:	The time during which the selection output must be active.
Data hold time:	The time during which the data on the multiplexer data bus must stay active after the selection output is deactivated.



Data setup time:	The time during which the data on the multiplexer bus has the correct data status before the selection output will be activated.
Selection output active status:	Selection of the sample can be either active high or active low. On = active high, Off = active low.
Multiplexer data bus status:	The status of the multiplexer data specifies if the data is either represented normally or inverted ('normal' or 'invert'). The data is inverted before it is written on the multiplexer data bus if the status is 'invert'.



G.2 Output multiplexers

Output multiplexers may be used, for example, to provide a numerical value at a seven-segment display.

G.2.1 Attributes of an output multiplexer

An output multiplexers has the following attributes:

- Number of output multiplexer channels. This is the number of channels used to write the multiplexed data of all connected multiple outputs.
- Scan time. The value must be in the range of 100 ms to 60,000 ms.
- Data setup time. The value must be in the range of 0 ms to 100 ms.
- Selection output active time. The value must be in the range of 1 ms to 100 ms.
- Data hold time. The value must be in the range of 1 ms to 100 ms.
- Selection output active status.
- Multiplexer data bus status.

G.2.2 Timing calculation for output multiplexers

The calculation of the output multiplexer timing concerns all output multiplexers. The compiler calculates a worst case timing that will be used by the FSC system software for every output multiplexer. Worst case timing calculation means:

- use the multiplexer with the lowest scan time.
- use the multiplexer with the highest data setup time.
- use the multiplexer with the highest selection output active time.
- use the multiplexer with the highest data hold time.
- use the multiplexer that needs the highest number of samples.

The time necessary to scan all multiple outputs connected to every multiplexer configured cannot exceed the scan time.

The algorithm for the calculation of the time to scan all mentioned outputs is as follows:

- Sample Time = Data Setup Time + Selection Output Active Time + Data Hold Time
- Number of samples necessary for a multiple output =
Number of channels of the multiple output divided by the number of channels per sample of the output multiplexer it is assigned to, rounded to the next whole value:
$$\text{Samples per multiple output} = (\text{Num. chan. mult.o.} - 1) \text{ DIV } (\text{Num. chan. o.mux.}) + 1.$$
- Total number of samples necessary to scan every multiple output of an output multiplexer is found by adding the number of samples of every connected multiple output.
- The total time to scan = Total number of samples * sample time.

If the total time to scan is larger than the lowest scan time, the FSC system is not able to scan the output multiplexers within the required time. This requires adjustments to one of the calculation parameters by the user.

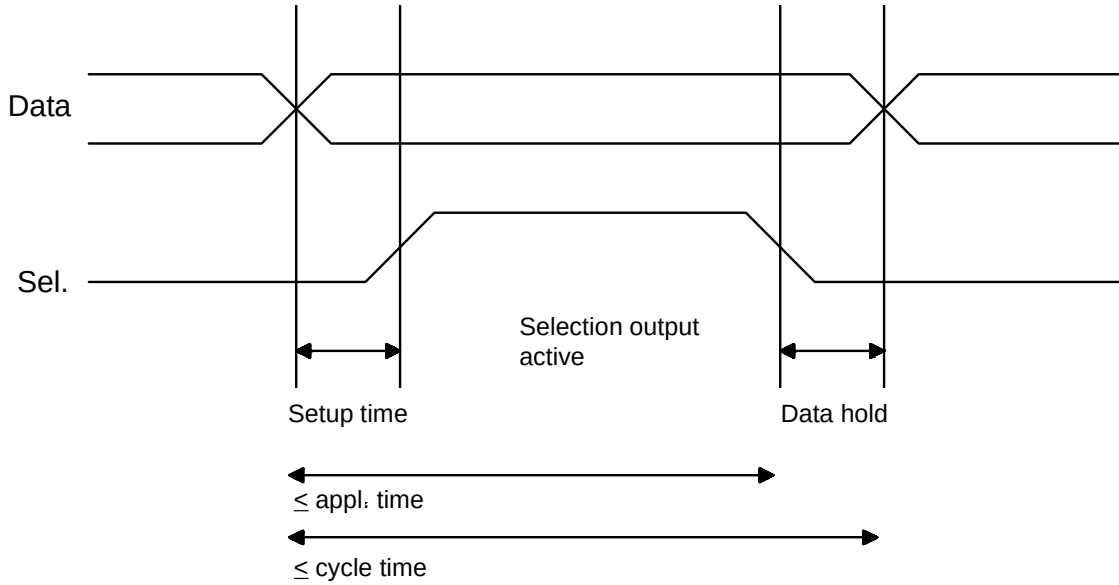


Figure G-1 Output multiplexer timing



G.3 Input multiplexers

Input multiplexers may be used e.g. to read a number of thumbwheel switches constituting a multiple digit numerical value.

G.3.1 Attributes of an input multiplexer

An input multiplexers has the following attributes:

- Number of input multiplexer channels. This is the number of channels used to read the multiplexed data of all connected multiple inputs.
- Scan time. The maximum value is 60,000 ms. The minimum time is at least as large as the application cycle time.
- Selection output active time. The value must be in the range 2 to 100 ms. The minimum value depends on the delay time of the inputs of the input module used for the multiplexer channels. The minimum value is at least twice the delay time of the inputs.
- Data hold time. The value must be in the range of 0 ms to 100 ms.
- Selection output active status.
- Multiplexer data bus status.

G.3.2 Timing calculation for input multiplexers

The calculation of the input multiplexer timing concerns all input multiplexers. The compiler calculates a worst-case timing that will be used by the FSC system software for every input multiplexer. Worst-case timing calculation means:

- use the multiplexer with the lowest scan time,
- use the multiplexer with the highest selection output active time,
- use the multiplexer with the highest data hold time, and
- use the multiplexer that needs the highest number of samples.

The time necessary to scan all multiple inputs connected to every multiplexer configured cannot expire the scan time.

The algorithm for the calculation of the time to scan all mentioned inputs is as follows:

- Sample Time = Selection Output Active Time + Data Hold Time.
- Number of samples necessary for a multiple input =
Number of channels of the multiple input divided by the number of channels per sample of the input multiplexer it is assigned to, rounded to the next whole value:
$$\text{Samples per multiple input} = (\text{Num. chan. mult.i.} - 1) \text{ DIV } (\text{Num. chan. i.mux.}) + 1.$$
- Total number of samples necessary to scan every multiple input of an input multiplexer is found by adding the number of samples of every connected multiple input.
- The total time to scan = Total number of samples * sample time.



If the total time to scan is larger than the lowest scan time, the FSC system is not able to scan the input multiplexers in the required time. This requires adjustments to one of the calculation parameters by the user.

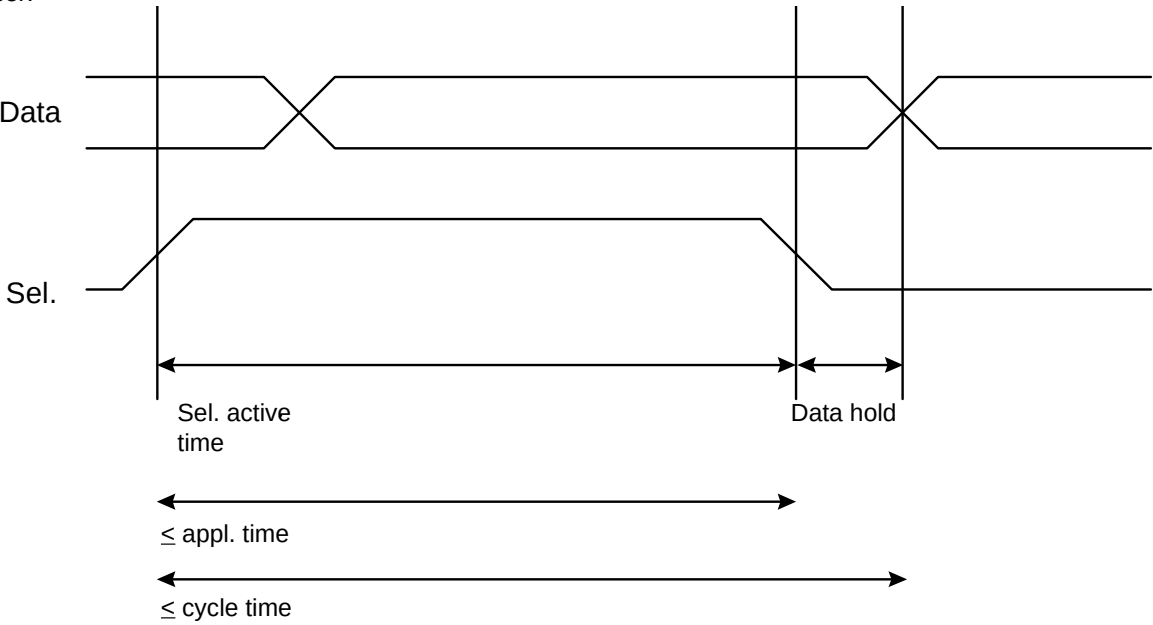


Figure G-2 Input multiplexer timing



G.4 Multiple I/O that uses the multiplexer

Multiple input/output variables that use multiplexers (multiplexed I/O) to exchange data do not need the 'normal' number of channels an ordinary multiple input/output uses. It uses of course the channels of the multiplexer to exchange data. The data is exchanged in a number of samples (see timing calculation). The application controls the sample that is read/written: it uses selection outputs to select the sample. Each sample has its own selection line.

Note:

Both BIs and BOs whose data is exchanged via a multiplexer are allocated to output channels (the selection lines)!



G.5 Configuring and using multiplexers

G.5.1 Configuring multiplexers

Multiplexers are added to the application by selecting the XI (multiplexer input) or the XO (multiplexer output) variable in the signal specification menu of the 'System Configuration' option of FSC Navigator. Up to 16 multiplexers can be configured.

The attributes of the multiplexers can be set in the hardware specification menu of the 'System Configuration' option of FSC Navigator.

G.5.2 Configuring multiplexed I/O

After specifying the attributes of the multiplexers that you will be using, all multiplexed I/O is configured. These are BI and BO variables with special location MUX. Use the signal specification menu to configure these variables.

Multiplexed inputs must be configured to an input multiplexer, multiplexed outputs must be configured to an output multiplexer.

In the hardware specification menu, specify the multiplexer the multiplexed I/O is using. Multiplexed I/O can be configured for a multiplexer as long as all connected multiplexed I/O can be scanned (see timing calculation) within the configured scan time of the multiplexer.

After specifying the number of channels of the multiplexed I/O variable, the application program calculates the number of samples it needs to exchange the variable data. For each sample a selection output must be configured. You must allocate the selection outputs for the variable. Example:

XI 4 channels.
BI Loc Mux 20 channels.

The number of samples is $20/4 = 5$ samples. An equal number of selection outputs must be allocated (5).

Configuration example:

XO:	scan time	= 1000 ms
	data setup time	= 10 ms
	data hold time	= 15 ms
	selection output active time	= 25 ms
	number of channels	= 4

BO 1: 20 channels (= $20/4 = 5$ samples)

BO 2: 32 channels (= $32/4 = 8$ samples)

BO 3: 16 channels (= $16/4 = 4$ samples)

Sample time = $10 + 15 + 25 = 50$ ms

Total number of samples = $5 + 8 + 4 = 17$.

Total scanning time needed by the FSC software is $17 * 50 = 850$ ms.

It is possible to configure an additional multiplexed BO using a maximum of 12 channels (leads to a scanning time of 1000 ms). It is not possible to configure a multiplexed BO with more than 12 channels.



G.5.3 Using output multiplexers

To control a 7-segment display, multiplexed I/O can be very efficient with respect to channel usage. If, for example, the application controls a 5-digit 7-segment display, the application would normally need $5 * 4 = 20$ channels if no multiplexer is used. However, if a multiplexer is used, 9 channels are necessary: 5 (20/4) for the selection outputs of the BO and 4 channels of the XO. The BO contains the segment information (bit 0-3 = segment 1, etc.).

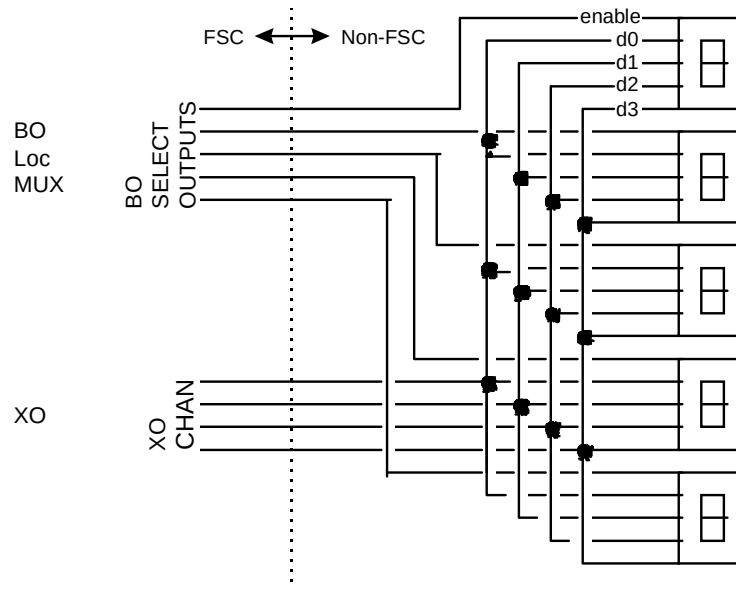


Figure G-3 Multiplexers and 7-segment displays

G.5.4 Using input multiplexers

To monitor four hardware registers (8 bit), multiplexed I/O can be very efficient with respect to channel usage. The application would normally need $8 * 4 = 32$ channels if no multiplexer is used. However, if a multiplexer is used, 12 channels are necessary: 1 (8/8) for each selection output of the BI and 8 channels of the XI. Each BI contains the value of one of the hardware registers.

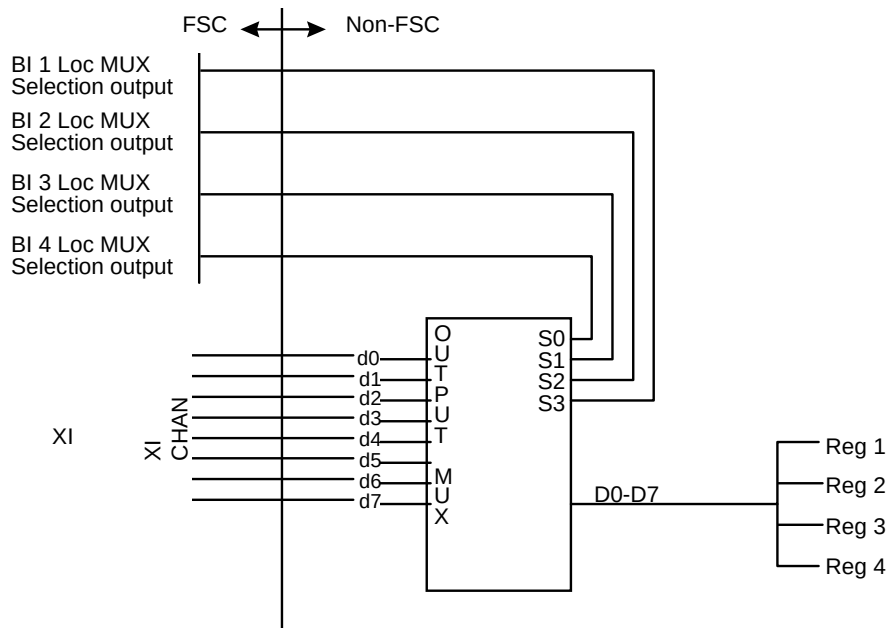


Figure G-4 Multiplexed register read-out

For another application example refer to the data sheet of the 10207/1/1 module in the FSC Hardware Manual.



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